SD-22

Diminishing Manufacturing Sources and Material Shortages

A Guidebook of Best Practices for Implementing a Robust DMSMS Management Program



DEFENSE STANDARDIZATION PROGRAM OFFICE

December 2025

STDZ

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Foreword

Diminishing manufacturing sources and material shortages (DMSMS) issues apply to every acquisition pathway and every phase of a system's life cycle. These issues can contribute to cost increases, readiness and availability declines, and schedule slips. Mitigating such ill-effects requires robust, risk-based DMSMS management to identify issues as early as possible and to implement cost effective resolutions that may only be viable when obsolescence is first discovered.

Robust DMSMS management does not only pertain to Department of Defense (DoD) assets. It also applies to foreign military sales (FMS) equipment with U.S. allies and partners who our warfighters depend on. This revision to Standardization-related Document 22 (SD-22) adds best practices and considerations associated with performing DMSMS management on DoD systems sold by DoD in a security cooperation context consistent with the terms and conditions of the sale. The guidance in this revision supports the speed in planning and execution of FMS as directed in the Acquisition Transformation Strategy.[1]

Successful DMSMS management requires adequate funding be available via programming and budgeting processes. Every program office should allocate funding for both DMSMS management operations and the implementation of resolutions to avoid delays in issue identification and resolution that result in the above negative outcomes.

An important first step in securing adequate funding is the development of a DMSMS management plan (DMP) that is approved by program office leadership. This update of the SD-22 discusses a better integration of the DMP with program capability levels that evaluate DMSMS management processes and identify improvement targets and approaches where needed.

Recommended changes to this document should be addressed to the Defense Standardization Program Office, 8725 John J. Kingman Road, Stop 5100, Fort Belvoir, VA 22060-6220, or via email to DSPO@dla.mil.



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[1] Acquisition Transformation Strategy: Rebuilding the Arsenal of Freedom, November 2025, available at: https://media.defense.gov/2025/Nov/10/2003819441/-1/-1/1/ACQUISITION-TRANSFORMATION-STRATEGY.PDF

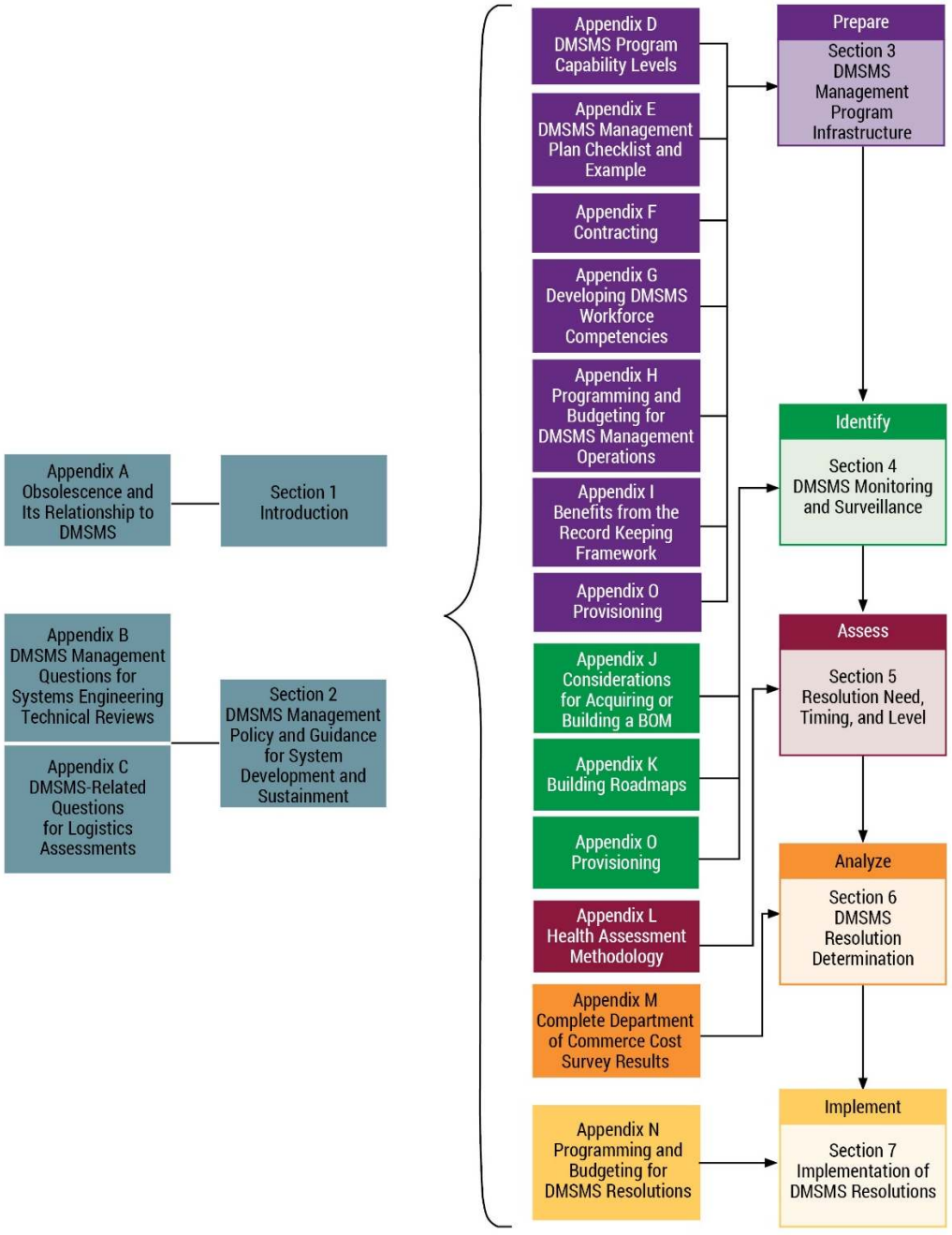
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Section 1. Introduction

1.1 Organization of This Document

This document is organized around the five steps of the management process for diminishing manufacturing sources and material shortages (DMSMS)—prepare, identify, assess, analyze, and implement. Figure 1 shows the high-level interrelationships of the five steps, the corresponding sections of this document, and the supporting appendixes.

Figure 1. Mapping the DMSMS Processes into This Document's Sections



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The following paragraphs describe the content of this document and highlight the major changes to the March 2024 version of SD-22.

What's new in Section 1

No major change.

Section 1 provides an overview of DMSMS as both a concept and a multidisciplinary process. Specifically, it describes the DMSMS scope and objective, the mechanisms that drive DMSMS, the importance of establishing a DMSMS management program, and an overview of the DMSMS management process steps.

What's new in Section 2

No major change.

Section 2 discusses policy and guidance pertaining to DMSMS management. This section provides important input to and context for the rest of the document.

What's new in Section 3

- Adds a new section on communication processes for DMSMS management on FMS equipment and adjusts FMS related aspects of the DMSMS management team and DMSMS management plan (DMP).
- Revises the description of DMP content and better integrates that content with the program capability levels found in Appendix D. References a new Appendix E that contains a DMP checklist and example.

Section 3 addresses the *Prepare* step of DMSMS management. Specifically, it describes best practices for establishing a strong infrastructure—data, people, processes, management reports, and financial resources—for successful DMSMS management.

What's new in Section 4

- Adds FMS considerations for DMSMS management in the Identify step.

Section 4 focuses on the *Identify* step, which encompasses DMSMS monitoring and surveillance throughout the life cycle and includes best practices for determining where to focus DMSMS management efforts.

What's new in Section 5

- Reinforces data needs for DMSMS management on FMS equipment.

Section 5 discusses the *Assess* step of DMSMS management. It begins with a discussion of the monitoring and surveillance data collected as well as other supporting programmatic and logistics data. The Section 5 assessment describes how to determine whether a case should be opened and, if so, how to prioritize and whether a resolution should be considered at a higher level of assembly.

What's new in Section 6

- Adds FMS considerations for DMSMS management in the Analyze step.

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Section 6 focuses on the *Analyze* step, which deals with analyzing alternative approaches for resolving DMSMS issues and identifying the preferred alternative. This chapter lists DMSMS resolution options and provides a basis for estimating their cost. It also identifies risk factors associated with these options.

What's new in Section 7

- Adds FMS considerations for DMSMS management in the Implement step.

Section 7 addresses the *Implement* step, which covers the implementation of the preferred resolution option. It discusses potential sources of implementation funding, the roles of the DMT during implementation, and some considerations associated with common implementation issues.

What's new in the Appendixes

- Refines the program capability levels in Appendix D and adds FMS capability level processes to address communication with FMS partners.
- Adds a new Appendix E showing a DMP checklist and example. Renumbers all subsequent appendices.

Appendix A through Appendix O contain supporting detail about DMSMS activities, such as questions for systems engineering technical reviews (SETRs) and independent logistics assessments (ILAs), sustainment reviews (SRs), DMSMS management plans, DMSMS contract language, DMSMS management program capability levels, DMSMS workforce competencies, the capabilities of a robust DMSMS management program, DMSMS implications of counterfeit parts, how to construct a bill of materials (BOM)[2] if it was not received on a contract, building roadmaps, working capital funds and other sources of funding for DMSMS resolutions, programming and budgeting for DMSMS management operations, budget execution considerations, DMSMS record keeping benefits, health assessments, and provisioning. Appendix P provides the abbreviations, Appendix Q provides references used in this document, and Appendix R contains definitions of important DMSMS management terms.

1.2 Scope and Objective

This guidebook provides best practices for implementing an effective DMSMS management program throughout the entire life cycle of a system. All systems are susceptible to DMSMS issues. A DMSMS issue is the loss, or impending loss, of manufacturers or suppliers of items, raw materials, or software.[3] The Department of Defense (DoD) loses a manufacturer or supplier when that manufacturer or supplier discontinues production and/or support of needed items, raw materials, or software or when the supply of raw material is no longer available.

DMSMS issues affect materiel readiness and operational availability, which, in turn, affect both combat operations and safety. A robust risk-based DMSMS management program, using practices outlined in this guidebook, can minimize the negative impacts of DMSMS on program costs, schedule, and system performance—and ultimately warfighter readiness and lethality.

[2] A BOM is a list of the original equipment manufacturer (OEM)–assigned part numbers within a higher level of assembly.

[3] The term *software* encompasses commercial off-the-shelf (COTS), custom, or any combination thereof of firmware, middleware, wrappers, gateways, firewalls, application programs, and operating systems.

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Consequently, robust risk-based DMSMS management is needed. DMSMS management is a multidisciplinary process to identify issues resulting from obsolescence,[4] loss of manufacturing sources, or material shortages; to assess the potential for negative impacts on schedule and/or readiness; to analyze potential mitigation strategies; and then to implement the most cost-effective strategy.

This guidebook replaces the March 2024 version and is intended primarily for the DMSMS management practitioner community. However, successful DMSMS management benefits others. Therefore, program office managers (PMs), engineers, product support managers (PSMs), life cycle logisticians (including supply chain managers, inventory managers, and maintainers) and their associated policymakers are also part of the intended audience.

The objectives of this document are as follows:

- Create awareness of the extent and impact of DMSMS issues on DoD systems;
- Provide best practices to PMs for implementing a robust, risk-based DMSMS management process, and building a cost-effective DMSMS management program;
- Encourage DMSMS resilience by using a modular, open system design approach along with other supportability-related design considerations in conjunction with part selection procedures that choose items with significant time left in their life cycle and with viable replacement options whenever possible to reduce the likelihood that a design will experience near-term DMSMS issues and increase the probability of a quick recovery when issues do occur;
- Define DMSMS support metrics to measure the effectiveness, efficiency, and return on investment (ROI) of a robust DMSMS management program;
- Promote affordable and efficient program office support through rapid and cost-effective DMSMS management best practices and resolutions that consider equipment life cycles, technology changes, and planned obsolescence;[5] and
- Promote the exercise of best practices to address obsolescence risks throughout the life cycle.

1.3 DMSMS Mechanisms

DMSMS issues are inevitable and can occur at any point in a system's life cycle. They affect short- and long-lived systems; reparables and consumables; and space-based, air-based, ground-based, and sea-based equipment (including support and test equipment). DMSMS issues are not confined to piece parts or devices; obsolescence may occur at the part, module, component, equipment, or system level. DMSMS issues are also not limited to defense-unique items; COTS items represent a significant obsolescence problem, because such items are most susceptible to market forces. While traditionally thought of as applying to electronic items, DMSMS can affect any item within a system, including software, and non-electronic components—materials and structural, mechanical, and electrical (MaSME)

[4] The term *obsolescence* is similar, yet different, from DMSMS from a dictionary perspective. The differences are small—DMSMS encompasses 1) items that are not obsolete but where there are shortages and 2) obsolete items that are out of production and there is demand. Because there is no difference between *DMSMS management* and *obsolescence management* from a process perspective, both terms are used interchangeably throughout this document with a distinction being made only where needed for clarity. For more information on the relationship between obsolescence and DMSMS, see Appendix A.

[5] In general, DMSMS management is performed by organizations which include acquisition program offices and other organizations responsible for performing DMSMS management activities in support of those program offices. The general terminology used is: DMSMS management performing organization. This document uses the term *program office* throughout to imply all types of DMSMS management performing organizations.

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items. Consequently, DMSMS management should address materials, mechanical items, and software, in addition to traditional electronic items.[6]

DMSMS issues can be caused by many factors—such as low-volume market demand, new or evolving science or technology, changes to detection limits, toxicity values, and regulations related to chemicals and materials—that significantly affect the DoD supply chain and industrial base. Another aspect of DMSMS is when an item, although still available commercially, no longer functions as intended because of hardware, software, and/or requirements changes to the system. This is often referred to as functional obsolescence. Any of these situations may endanger an ongoing production capability, the life cycle support of a weapon system, or training, support, and test equipment already in the field.[7]

Whether dealing with hardware items or software, obsolescence is ultimately driven by one of a number of DMSMS mechanisms, influenced by the short life cycle of technology as compared to the long life cycle of defense systems.[8] Figure 2 illustrates the symmetrical nature of both first-order and lower-order DMSMS mechanisms for hardware and software. Although the figure may give an impression of a sequential progression of events, that is not the case. Hardware and software obsolescence may occur simultaneously. To help drive home the point that both hardware and software are part of the same process, the figure shows “hardware and software monitoring” and “DMSMS resolution identification” as single boxes, rather than hardware- and software-specific boxes. Both hardware and software may also become functionally obsolete, as a secondary impact of some other change to the system.

There are two broad DMSMS mechanisms: first-order obsolescence and lower-order, or derived, obsolescence. First-order obsolescence is directly driven by market factors and regulation changes that affect DoD’s ability to buy, license, obtain support for, or use hardware and software.[9] Lower-order obsolescence is functional obsolescence caused by hardware or software changes made to resolve first-order obsolescence issues, perform a proactive upgrade, or respond to a requirement change. Hardware and software monitoring identifies instances of first-order obsolescence. This monitoring also identifies functional obsolescence, resulting from proactive upgrades to hardware and software and driving the need to make proactive upgrades.

[6] Generally, the use of the word *item* in this document is intended to be inclusive of parts, assemblies, software, and material.

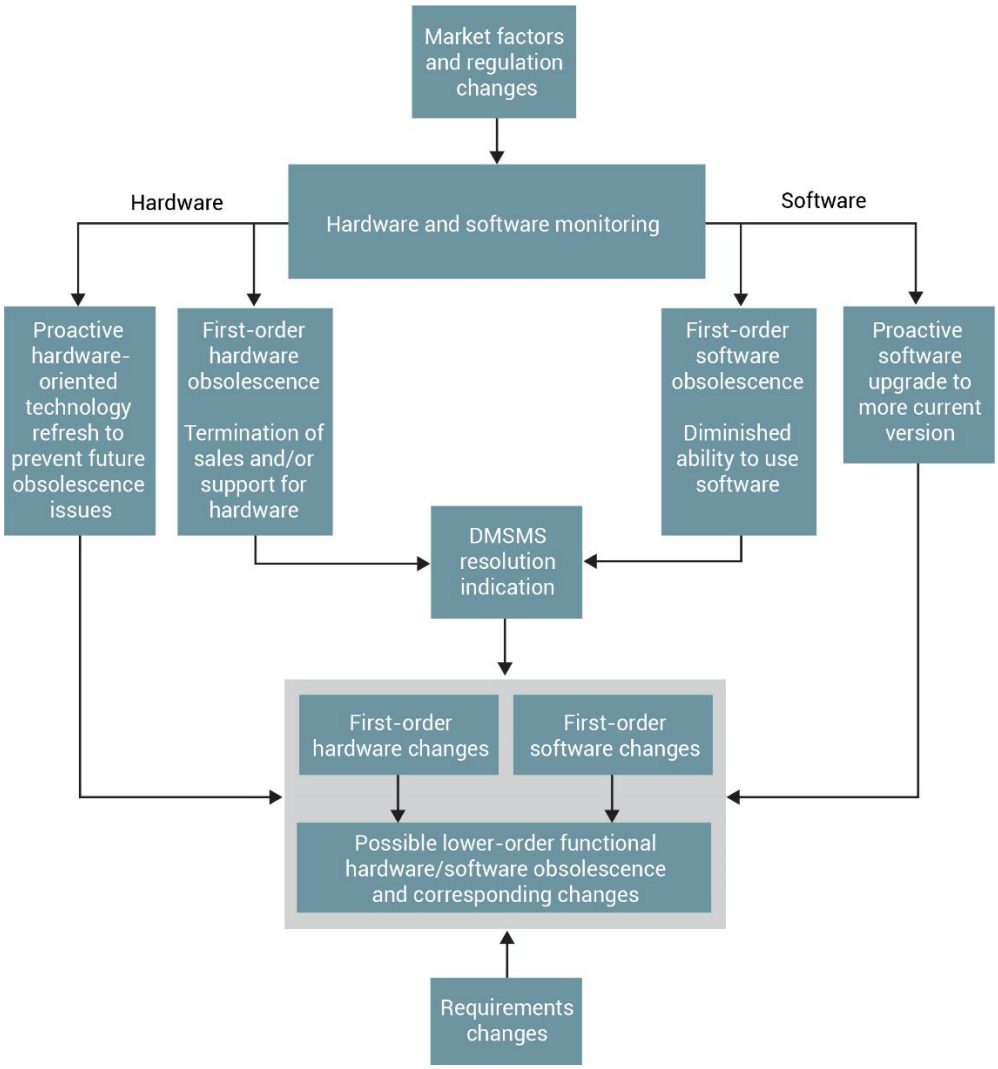
[7] The word *system*, as used in this document, encompasses weapon systems and training, support, and test equipment.

[8] Hereafter, except if otherwise specified, if *hardware* is used alone, it refers to both electronic and MaSME items.

[9] Once an item is identified as becoming obsolete, an additional market factor further limiting DoD’s ability to buy is the potential for increased demand to acquire the remaining supply matter.

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Figure 2. Mechanisms for Hardware and Software Obsolescence



First-order obsolescence situations should be identifiable from hardware and software monitoring of the effects of market forces. Leading causes of first-order obsolescence are as follows:

- ***Hardware—Electronic Items***. Sales and support may be terminated because of some combination of low demand, demand for new technologies and capabilities, lack of profitability, vulnerabilities, loss of production capability because of modernization or disposal of manufacturing equipment or mergers/acquisitions, unavailability of items, unavailability of materials, loss of repair support expertise, and so forth. For example, manufacturing tooling may be disposed of after production is completed.
- ***Software***. Competitive market forces can also lead to software obsolescence. Companies discontinue software sales and support for various reasons:
  - Lack of sufficient profitability;
  - A preference that customers upgrade to the most recent products;
  - Development of a new version that overcomes fundamental security or quality issues, inherent flaws, weaknesses, and/or vulnerabilities found in prior versions;
  - Reliance on inadequately maintained open source software components;
  - Mergers and acquisitions;

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- Loss of the skilled personnel needed to support older versions of the software; and
- Unavailability of software development tools for building, testing, and integrating the software (e.g., compilers, databases, regression testing capability, and configuration management (CM) software).[10]

A program office's ability to use software is inextricably linked to obtaining a license and receiving support. Software support includes product enhancements to increase capability or to decrease vulnerability to malicious attacks, error correction, and general support for its application in particular environments. If support is no longer obtainable, updated security patches will not be available, bugs cannot be fixed, routine maintenance cannot occur, and modifications can no longer be made.

- ***Hardware—MaSME Items***. Market forces also lead to DMSMS issues for hardware—MaSME items, but not quite in the same way as for electronic items and software. Unlike electronic items, MaSME items are typically on the market for long periods of time and, depending on the situation or support posture, are often repairable. Manufacturing processes for structural, mechanical, and electrical items have also remained relatively stable over time. Principal factors for DMSMS issues pertaining to materials (including critical materials resident within the supply chain) and structural, mechanical, and electrical items include the following:
  - Hazardous materials are being used. Regulations on hazardous materials may become stricter (e.g., Restriction of Hazardous Substances [RoHS]). Such materials may be banned completely or become difficult and/or prohibitively expensive to use or obtain (e.g., Freon or lead [Pb]).
  - The supplier goes out of business or through a merger and acquisition in which the existing product line no longer fits in the new product portfolio. This is more likely with small and/or financially vulnerable suppliers that supply items to larger companies that use the items to assemble their product. Item lines that are sold from original companies to other businesses are vulnerable to being discontinued.
  - The supplier's business case is no longer viable. This may occur with low-demand products potentially containing exotic materials that are difficult to manufacture or involve major disruptions to more profitable business activities. This may also occur because of regulation changes (e.g., tungsten rhenium wire was affected because of energy-related regulations on the disuse of incandescent light bulbs; Freon).
  - Supply-constrained materials are used. There may be U.S. or foreign regulations or supplier policies that affect availability. For example, a foreign source may limit its exports or not be willing to sell its product for a DoD application. The United States Code, Title 10, Section 4863 (10 U.S.C. § 4863) imposes restrictions on some specialty metals, for example, steel (specific mixes); metal alloys containing nickel, iron-nickel, and cobalt base alloys; titanium and titanium alloys; and zirconium and zirconium base alloys.[11]
  - The tooling is no longer available. Depending on the specific situation, resolving this issue may involve substantial time and cost.

Figure 2 also includes a resolutions block because implementing changes (which are called first-order hardware and software changes) to resolve first-order obsolescence can lead to lower-order or derived obsolescence. Similarly, when a program office takes an action to proactively upgrade or to satisfy new requirements, the changes implemented have the potential to cause lower-order obsolescence. The

[10] Regression testing seeks to ensure that software changes have not introduced new faults.

[11] 10 U.S.C. § 4863, available at https://www.govinfo.gov/link/uscode/10/4863.

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following describes the different factors that influence the need for hardware or software changes and how such changes can cause lower-order obsolescence:

- ***First-order hardware changes***. These changes can result from one of three factors. First, robust DMSMS management seeks to refresh items before they become obsolete. Proactive technology refreshment results in a hardware change. Technology refreshment is a "technology-related, periodic, planned change of items within a system's design to ensure continued access to the items necessary for system's support."[12] Second, hardware changes may be driven by implementing engineering resolutions to address DMSMS issues such as the termination of sales or support of existing hardware. The resulting engineering resolutions range from the use of a simple substitute with an existing replacement item to the redesign of the item and to a redesign at a higher level of assembly. Finally, a new performance requirement may necessitate a hardware change. Regardless of the factors influencing first-order hardware changes, lower-order hardware or software obsolescence may result because the existing hardware or software no longer functions as intended. For example, legacy software may not work, or may not work correctly, on new hardware configurations for a variety of reasons, such as data transmittal, storage, access, processing, display, interface issues, operating system capability issues, or timing concerns. Similarly, legacy hardware may suffer from physical incompatibility.
- ***First-order software changes***. These changes can result from factors that are analogous to the three factors that cause first-order hardware changes. Software upgrades or technology refreshments are not always compatible with other software applications that have been tested on or with earlier versions. For example, an upgrade to a COTS operating system may no longer support the assumptions made by the existing firmware; consequently, the firmware may become functionally obsolete. DMSMS resolutions implemented because of the inability to use software also lead to a first-order software change. Finally, first-order software obsolescence may be driven by changes to performance requirements that necessitate a software change. For example, scalability may be an issue if the transaction volume changes significantly. Greater processing power and memory size may also be required. The above rationale for first-order hardware changes leading to lower-order software or hardware obsolescence is directly applicable to first-order software changes leading to lower-order software or hardware obsolescence.

1.4 The Importance of a DMSMS Management Program

Ultimately, DMSMS management is important for a simple reason: it protects systems. Robust DMSMS management of inevitable obsolescence is the most cost effective and efficient way to minimize the negative effects of DMSMS for all acquisition pathways throughout a system's life cycle. As discussed in Section 2.2, the following objectives of DMSMS management minimize the impact on the cost, schedule, and performance of a system:

- Ensure DMSMS resilience during design,
- Minimize the scope of DMSMS-related out-of-cycle redesigns when they cannot be eliminated or avoided,
- Eliminate DMSMS-caused production schedule impacts, and
- Eliminate readiness degradations caused by DMSMS issues.

Because DMSMS management has a significant effect on many aspects of a program office, it is not a standalone function, nor does it benefit a single stakeholder. DMSMS management is inherently linked with reliability, maintainability, supportability, and availability. Within this context, it is important to plan for,

[12] Jay Mandelbaum and Christina Patterson, "Be Strategic!—Leverage Technology Insertion and Refreshment on DMSMS Issues," *Defense Acquisition Magazine*, Defense Acquisition University (DAU), July–August 2021, p. 38.

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minimize, and manage the risks associated with DMSMS issues, due to their detrimental impact on materiel readiness, operational mission capability, safety of personnel, and affordability.

Materiel readiness is an immediate and urgent concern for the warfighter. Missions are affected if equipment cannot be supported; either the equipment is not available for the mission, or it cannot be sustained throughout the mission. DMSMS issues can negatively affect supportability if the items needed to repair a system are not available or are in scarce supply. It is unacceptable for a system to be non-mission-capable due to a DMSMS issue. To allow a DMSMS situation to progress to the point of affecting a mission (because items are not available) is contrary to DoD policy and is an indication of ineffective DMSMS management as well as poor sustainment planning. In addition, ineffective DMSMS management can cause item costs to rapidly escalate.

A robust DMSMS management program is the most effective and efficient way to minimize readiness risks caused by DMSMS issues, deliver better buying power, and improve overall life cycle management. DMSMS resolutions are based on the most cost-effective approach to managing the problem before operations are affected. The cost avoidance of being proactive can be substantial, as discussed below:

- The Terminal High Altitude Area Defense (THAAD) program office in the Missile Defense Agency (MDA) faced an obsolescence issue with its existing one-color seeker—a redesign was determined to be the only feasible resolution. However, a redesign to upgrade the seeker to a two-color capability was being considered for five years in the future. The THAAD obsolescence team recommended a one-time redesign of the seeker to mitigate the current obsolescence, while providing the ability to upgrade the THAAD system later. This effort added capability benefit without increasing risk to the production line or production timelines. That resolution avoided $100 million in potential future, redundant, seeker redesign costs while accelerating two-color capability. Additionally, these actions enabled timely production cut-in to support both U.S. and FMS obsolescence cases equally, establishing a foundation for future THAAD improvements and overall ballistic missile defense capability across all architectures.
- The Navy effectively resolves over 3,000 DMSMS cases annually, leading to a remarkable cost avoidance exceeding $100 million each year. Notably, one key mitigation approach alone contributes significantly to this figure, accounting for over $930.04 million in cost savings since April of 1996. A primary objective of the Miniature/Microminiature Electronics Repair (2M) and Module Test and Repair (MTR) program is to devise diagnostics and testing protocols for circuit card assemblies, facilitating the sustained operation of electronic assemblies at a substantially reduced cost. By doing so, the program sidesteps the need for expensive redesigns, reverse engineering endeavors, and qualification expenses associated with alterations to the original assembly design, particularly for assemblies no longer supported in the commercial sector. These mitigations can be implemented at the microcircuit level, representing a fraction of the expense incurred by modifying the assembly's design. Furthermore, the program establishes the capacity to repair and troubleshoot assemblies at both organizational and intermediate maintenance facilities, serving ashore and afloat commands alike. On average 60 percent of 2M/MTR program repair sites that report metrics conduct 8000 repairs annually, thereby preventing or correcting 212 instances of equipment degradation that would impact readiness.
- The Black Hawk UH/HH-60M is an Acquisition Category 1C program currently in Multi-Year X production with plans to be sustained until 2075. The Black Hawk DMT, established in 2017, provides analysis and develops recommendations related to DMSMS/obsolescence. The team is responsible for 31 critical/complex line replaceable units (LRUs) in addition to other electronic and hardware component issues that may arise. One notable area of success was creating the data needed to be proactive; the team worked with the program office to use unserviceable LRUs to identify active components on each board, then develop an indentured BOM. Through its proactivity, advanced warning time for DMSMS issues has increased, allowing time to develop optimum resolutions that yield the lowest total ownership

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cost to the program office. All these efforts have not only prevented an impact to the production line, but also closed 900 obsolescence cases and led to more than $156 million in cost avoidance.

In the following example, robust DMSMS management was directly responsible for securing the resources needed to resolve serious DMSMS issues.

- The C-130 Enterprise DMT avoided potential production delays and sustainment impacts. To ensure the proper level of resources were programmed and budgeted, the DMT performed a detailed and complex schedule driven risk-based analysis that identified DMSMS components and forecasted timelines for required availability with projected impact dates. That analysis also developed risk mitigation techniques and was instrumental in their implementation. The actions of the team triggered the program office to actively track those DMSMS components, resulting in Congress appropriating funding to develop and implement solutions, avoiding production disruptions and impacts to aircraft availability. More than $700 million was allocated to developing new parts and the redesigning others in future fiscal years.

Without a robust DMSMS management program, a program office will likely incur unnecessary cost as the following example shows:

- A contract was awarded to produce five low rate initial production (LRIP) lots for an Air Force lead joint program office. Unfortunately, the contract did not clearly require the contractor to proactively monitor and manage obsolescence effecting future production lots. While the contractor was contractually committed to delivering all units, the contractor had no contractual obligation to preserve future production capacity. This resulted in 19 different DMSMS contract actions that threatened the program office's ability to successfully ramp up to full rate production (FRP) quantities and cost $40 million to resolve. Because of this negative and costly experience, the program office now operates and addresses DMSMS in a proactive manner by conducting health assessments every six months and assessing and planning how best to resolve each DMSMS case. In addition, the program office audits supplier's awareness, practices, and procedures regarding DMSMS to reduce the occurrence of pop-up obsolete parts. Finally, the program office initiated a Technology Refresh trade study to identify critical components, sub-assemblies, and sections of the system that have a limited production life.

These examples demonstrate how DMSMS management can result in better value for the taxpayer and the warfighter. DMSMS management helps target affordability and control cost growth in several ways. By accounting for DMSMS issues during design (trades), future operating and support costs will be controlled and possibly reduced. For example, the use of standardized parts and the latest technologies can reduce the impact of DMSMS issues during sustainment by enhancing the interchangeability, reliability, and availability of items.

Robust DMSMS management also includes cultivating long-term relationships with suppliers. Given such relationships, suppliers should be less likely to discontinue an item, and if they decide to discontinue the item for business reasons, the government is more likely to have advanced warning, placing it in a better position to plan an alternative course of action.

Furthermore, benefits extend well beyond these situations. The following excerpt from the *2022 National Defense Strategy* highlights obsolescence as a key challenge for the future force:

> *Transform the Foundation of the Future Force. Building the Joint Force called for by this strategy requires overhauling the Department's force development, design, and business management practices. Our current system is too slow and too focused on acquiring systems not designed to address the most critical challenges we now face. This orientation leaves little incentive to design*

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open systems that can rapidly incorporate cutting-edge technologies, creating longer-term challenges with obsolescence, interoperability, and cost effectiveness.[13]

Finally, DMSMS management results should be a key driver of technology refreshment plans. The ultimate goal is to replace items before they become obsolete. This is enabled by both open systems architecture and data rights in designs which enhance competition by providing a framework for decomposing a system into items and obtaining the necessary technical information for them. Modular open systems architecture is also an important DMSMS-related design consideration, because identifying and standardizing major system interfaces makes substitution of alternative technologies easier. Robust DMSMS management will secure data rights and necessary technical data for items that are highly likely to face DMSMS issues and likely to be good targets for technology refreshment.

1.5 Overview of the DMSMS Management Process

The DMSMS management process is straightforward and ultimately constitutes DMSMS risk management. As described in Section 1.1, the DMSMS management process has five tailorable steps. In the *Prepare* step (Section 3), a program office develops the foundations for DMSMS management and a DMP with resources to implement it. Through near-term monitoring and surveillance and longer-term technology forecasting of the *Identify* step (Section 4), a program office finds items with obsolescence risks. For the *Assess* step (Section 5), a program office determines whether it should open a case for the identified items, as well as when and at what level (i.e., the item or a higher level of assembly) to resolve the DMSMS issue. The program office determines the most cost-effective resolution during the *Analyze* step (Section 6) and then executes that resolution in the *Implement* step (Section 7).

Each of these steps applies throughout the system life cycle, from early technology development through sustainment. Although it is best to begin these activities early in the life cycle, they may be initiated at any point in the process. Robust DMSMS management is a dynamic process that continues over the system's life cycle. The five steps represent a sequence of discovery and resolution. Once a program office resolves one issue, it should move on to the next. When a program office has gone through its list of issues, it should start again; due to the dynamics of the market, something will have changed. A program office should repeat this cadence on an established rhythm until system retirement.

Figure 3 depicts the five DMSMS management steps. Within those steps, strategic processes increase the likelihood of implementing low-cost resolutions while delaying and preventing the occurrence of DMSMS issues in concert with system modification planning.[14] Strategic processes also include the use of evaluation results for the program office's DMSMS management processes to improve effectiveness and efficiency. Since the strategic processes relate to five DMSMS management process steps, Figure 3 depicts "strategize" as a circle encompassing them.

[13] DoD, *2022 National Defense Strategy of the United States of America Including the 2022 Nuclear Posture Review and the 2022 Missile Defense Review*, October 27, 2022, p. 19, available at https://apps.dtic.mil/sti/trecms/pdf/AD1183514.pdf.

[14] This document uses the term *system modification* to mean any change to the configuration of the system.

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Figure 3. Steps in the DMSMS Management Process



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Section 2. DMSMS Management Policy and Guidance

This section summarizes DoD DMSMS management policy and guidance. In addition to Department-level policy, the DoD Components also have their own DMSMS management policies and guidance. The Defense Standardization Program (DSP) DMSMS management website page contains policy and guidance documents.[15]

2.1 DMSMS Management Policy

2.1.1 High-Level DMSMS Management Policy

DMSMS management policy for the DoD has historically existed in DoD Instruction (DoDI) 4140.01, "DoD Supply Chain Materiel Management Policy," and volume 3 of DoD Manual (DoDM) 4140.01, "DoD Supply Chain Materiel Management Procedures: Materiel Sourcing," and DoDI 5000.91, "Product Support Management for the Adaptive Acquisition Framework." Recently, DMSMS management policy migrated from DoDI 5000.02, "Operation of the Adaptive Acquisition Framework," to DoDI 5000.85, "Major Capability Acquisition." Table 1 contains the DMSMS management–related policy language from these four DoD issuances.

Table 1. High-Level DMSMS Management–Related Policy

Document	Section	DMSMS Management Policy Statement
DoDI 5000.85	Section 3: Major Capability Acquisition Procedures	"3.1. GENERAL PROCEDURES. a. Program Planning. (1) A rapid, iterative approach to capability development reduces cost, avoids technological obsolescence, and reduces acquisition risk. Consistent with that intent, acquisitions will rely on mature, proven technologies and early testing. Planning will capitalize on commercial solutions and non-traditional suppliers, and expand the role of warfighters and security, counterintelligence, and intelligence analysis throughout the acquisition process."[16]
DoDI 5000.91	Section 4: General Product Support Procedures Over the Program's Life Cycle	"4.3. THE PSS [PRODUCT SUPPORT STRATEGY] AND THE LCSP [LIFE CYCLE SUSTAINMENT PLAN] … b. The LCSP is the primary program management reference governing operations and support planning and execution from program inception to disposal. An LCSP is required for all covered systems and is the principal document establishing the system's product support planning and sustainment, pursuant 10 U.S.C. § 4324 For covered systems, a detailed LCSP will include: (6) Sustainment risks, SCRM [supply chain risk management], and diminishing manufacturing sources and material shortage (DMSMS) risk management and proposed mitigation plans.

[15] The DSP DMSMS management website can be accessed at https://www.dsp.dla.mil/Programs/DMSMS/. That website also contains information and proceedings of past DMSMS-related conferences.

[16] DoDI 5000.85, "Major Capability Acquisition," November 4, 2021, available at https://www.esd.whs.mil/Portals/54/Documents/DD/issuances/dodi/500085p.pdf, p. 8.

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Document	Section	DMSMS Management Policy Statement
		(7) Engineering and design considerations, including DMSMS resilience, that support cost-effective sustainment for the system."[17] "4.11. ADDITIONAL PRODUCT SUPPORT ACTIVITIES: a. Materiel Management and Materiel Storage. The PM, through the PSM will evaluate and select materiel management solutions that balance support goals, total supply chain costs, and performance factors IAW [in accordance with] DoDI 4140.01, DoDI 4140.69, Volume 5 of DoD Manual 4140.01, DoDI 4245.15, and DoD Manual 4140.70. The PSM will use existing organic storage and warehousing facilities (DLA [Defense Logistics Agency] or other similar organic warehousing and storage) to the maximum extent practicable before establishing additional capacity at either organic or private warehouse or storage facilities. Additional requirements for DMSMS, SCRM, and materiel, equipment, and inventory management are detailed in DoDI 4140.01, DoDI 5000.64, and DoDI 5200.44. Serialized item management instructions are detailed in DoDI 8320.04."[18] "f. DMSMS The PM, through the PSM, will develop, ensure funding, and execute a DMSMS management plan and conduct proactive risk-based DMSMS management per that plan to identify current DMSMS issues, forecast future DMSMS issues, program and budget for resolving DMSMS issues, and implement those resolutions IAW DoDI 4245.15. Implementing DMSMS issue resolutions will take into account a parts management process that considers SCRM, supportability, loss of technological advantage, and obsolescence when selecting parts used in DMSMS resolutions. In addition, the PSM will use both current and forecasted DMSMS issues in developing product roadmaps for supportability."[19]
DoDI 4140.01	Section 2: Responsibilities	Assigns responsibilities to: "2.3. Director, Defense Pricing and Contracting. … c. Establishes procurement policies, procedures, and guidance to support timely solutions to diminishing manufacturing sources and material shortages (DMSMS), including obsolescence."[20] "2.5. USD(R&E) [Under Secretary of Defense for Research and Engineering]. … c. Provides technical advice and assistance, as necessary, to the ASD(S) [Assistant Secretary of Defense for Sustainment] on matters; … (2) Pertaining to the timely identification of technical solutions to DMSMS."[21] "2.7. DoD Component Heads… f. Establish programs for monitoring and mitigating the risk of: … (2) DMSMS, including obsolescence."[22]

[17] DoDI 5000.91, "Product Support Management for the Adaptive Acquisition Framework," November 4, 2021, available at https://www.esd.whs.mil/Portals/54/Documents/DD/issuances/dodi/500091p.pdf , p. 13.

[18] Ibid, p. 18.

[19] Ibid, p. 19.

[20] DoDI 4140.01, "DoD Supply Chain Materiel Management Policy," March 6, 2019, available at https://www.esd.whs.mil/Portals/54/Documents/DD/issuances/dodi/414001p.pdf, p. 5.

[21] Ibid, p. 6.

[22] Ibid, p. 7.

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Document	Section	DMSMS Management Policy Statement
DoDM 4140.01, Volume 3	Section 2. DMSMS Responsibilities	Assigns responsibilities to: "2.3. Secretaries of the Military Departments, Director, Missile Defense Agency (MDA), and Director, Defense Logistics Agency... f. Establish and implement, for their military department or agency, a standard strategy and program for diminishing manufacturing sources and material shortages (DMSMS) that: (1) Reduces or eliminates the cost and schedule impacts of all identified DMSMS problems. (2) Helps ensure that DMSMS problems do not adversely impact weapon system readiness."[23]
	Section 9: DMSMS	Identifies procedures for DoD Components, and more specifically for the Military Departments, DLA, and MDA, for: "9.1. DMSMS Requirements" "9.2. Actions for Minimizing the Impact of DMSMS" "9.3. Resolution of DMSMS Issues."[24]

2.1.2 Detailed DMSMS Management Policy

While the above high-level policy remains in effect; on November 5, 2020, the Department issued a more specific, stand-alone DMSMS management policy. DoDI 4245.15, "Diminishing Manufacturing Sources and Material Shortages (DMSMS) Management," promulgates DoD policy to:

a. *Establish and implement risk-based, proactive DMSMS management throughout the life cycle of all DoD items.*
b. *Evaluate all DoD system designs and redesigns for potential DMSMS issues that could arise during the life cycle of a DoD item.*
c. *Implement resolutions if necessary to minimize or eliminate risks and negative impacts (e.g., cost, schedule delays, readiness) from DMSMS issues throughout the life cycle of DoD items.*
d. *Implement improvements to DMSMS management processes throughout the life cycle of all DoD items across the DoD enterprise.*[25]

Furthermore, DoDI 4245.15 identifies responsibilities and procedures pertaining to DMSMS management. The instruction specifically identifies DMSMS-related responsibilities for:

- Under Secretary of Defense for Acquisition and Sustainment (USD(A&S));
- Assistant Secretary of Defense for Acquisition (ASD(A);
- Assistant Secretary of Defense for Sustainment (ASD(S));
- Deputy Assistant Secretary of Defense for Industrial Policy;
- Director, Defense Contract Management Agency;
- Under Secretary of Defense for Research and Engineering (USD(R&E));

[23] DoDM 4140.01, "DoD Supply Chain Materiel Management Procedures: Materiel Sourcing," Volume 3, Change 1, August 26, 2022, available at https://www.esd.whs.mil/Portals/54/Documents/DD/issuances/414001m/414001m_vol03.pdf?ver=at1glnYRAsxRWsnTFeiXAg%3D%3D, pp. 5–6.

[24] Ibid, pp. 33–37.

[25] DoDI 4245.15, "Diminishing Manufacturing Sources and Material Shortages (DMSMS) Management," November 5, 2020, available at https://www.esd.whs.mil/Portals/54/Documents/DD/issuances/dodi/424515p.pdf, p. 3.

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- DoD Component Heads; and
- Secretaries of the Military Departments; Director, DLA; Director, MDA; and Director, Defense Health Agency.[26]

The procedures section of DoDI 4245.15 identifies DMSMS management–related procedures for "acquisition and item management program offices and other DMSMS management performing organizations (DMPOs)."[27] These procedures address the following topics:

- Development and update of DMPs;
- The collection and maintenance of DMSMS case and operations data;
- The consideration of DMSMS during system design (i.e., new designs and redesigns);
- The identification, documentation, and mitigation of DMSMS risks;
- Programming and budgeting for DMSMS management operations and resolutions;
- The inclusion of DMSMS management–related provisions in contracts;
- The timely resolution of DMSMS issues to minimize cost, schedule, and readiness impacts;
- Communication with supply organizations and other external DoD Components to mitigate DMSMS issues and risk;
- Training the DMSMS management workforce and other stakeholders; and
- Evaluation of DMSMS management activities in engineering and logistics processes and practices.[28]

DoDM 4245.15 is a companion document to DoDI 4245.15 and expands on the responsibilities and procedures pertaining to DMSMS management identified in that Instruction. Specifically, DoDM 4245.15 adds DMSMS-related responsibilities for the DoD Chief Information Officer. The DoD CIO is responsible for providing information on how to address the impact of cyber supply chain risk management on DMSMS resolution implementation for digital capabilities.[29]The procedures section of DoDM 4245.15 expands on the management-related procedures for "acquisition and item management program offices and other DMPOs" addressed in DoDI 4245.15.[30]

2.2 DMSMS Management Guidance Throughout the System Life Cycle

As stated previously, the policy formerly captured in DoDI 5000.02, "Operation of the Adaptive Acquisition Framework" has migrated to a set of new Department instructions. The following new instructions contain policy pertaining to the acquisition pathways—

- DoDI 5000.81[31] governs the Urgent Capability Acquisition (UCA) pathway,
- DoDI 5000.80[32] governs the Middle Tier of Acquisition (MTA) pathway,
- DoDI 5000.85[33] governs the Major Capability Acquisition (MCA) pathway,

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[26] Ibid, pp. 4–7.

[27] Ibid, p. 7.

[28] Ibid, pp. 8–9.

[29] Ibid, p. 4–6.

[30] Ibid, pp. 7–19.

[31] DoDI 5000.81, "Urgent Capability Acquisition," December 31, 2019, available at https://www.esd.whs.mil/Portals/54/Documents/DD/issuances/dodi/500081p.pdf.

[32] DoDI 5000.80, "Operation of the Middle Tier of Acquisition (MTA)," November 25, 2024, available at https://www.esd.whs.mil/Portals/54/Documents/DD/issuances/dodi/500080p.pdf.

[33] DoDI 5000.85, "Major Capability Acquisition," November 4, 2021, available at https://www.esd.whs.mil/Portals/54/Documents/DD/issuances/dodi/500085p.pdf.

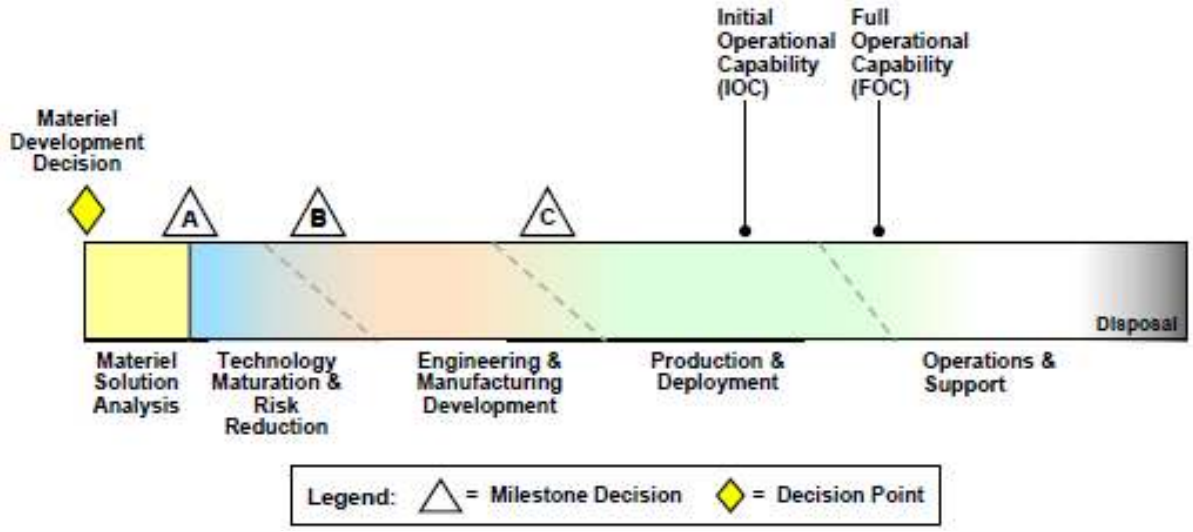
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- DoDI 5000.75[34] governs the Defense Business Systems (DBS) pathway,
- DoDI 5000.87[35] governs the Software Acquisition (SWA) pathway, and
- DoDI 5000.74[36] governs the Acquisition of Services pathway.

The DAU's "Adaptive Acquisition Framework" web page at: https://aaf.dau.edu/ provides more detailed policy and guidance pertaining to the acquisition pathways.

In the past, DMSMS management has mostly related to the MCA pathway (see Figure 4); however, a program office can tailor the best practices conveyed in this document to all acquisition pathways (except acquisition of services) as a function of risk. Section 2.2.3 describes this tailoring.

Figure 4. Phases of Major Capability Acquisition[37]



Given the long life cycles of defense systems, the exclusion of obsolete items from the designs of those systems and the continuous monitoring for obsolete items throughout the life of systems represent obvious best practices. Even in the early stages of the life of a system, program offices can examine a preliminary parts list to identify whether the system design contains any obsolete or near obsolete parts. Doing so offers an acquisition program office the opportunity to lower the cost of redesigns, identify items to design out before causing a detrimental impact, and provide guidance on what to preserve, in terms of technical data, tooling, and insurance spares to hedge against future obsolescence. Even when a program office cannot prevent the use of obsolete items, having properly vetted a system's design can enable the preparation of a mitigation strategy to address the known obsolescence at a lower cost. For all these reasons, DMSMS management, at the direction of the PM and PSM, should begin early and continue throughout the life cycle of the system. DMSMS problems are likely to be of increasing concern when the service life of a weapon system is extended and the product life cycle for high-technology

[34] DoDI 5000.75, "Business Systems Requirements and Acquisition," January 24, 2020, available at https://www.esd.whs.mil/Portals/54/Documents/DD/issuances/dodi/500075p.pdf.

[35] DoDI 5000.87, "Operation of the Software Acquisition Pathway," October 2, 2020, available at https://www.esd.whs.mil/Portals/54/Documents/DD/issuances/dodi/500087p.pdf.

[36] DoDI 5000.74, "Defense Acquisition of Services," June 24, 2021, available at https://www.esd.whs.mil/Portals/54/Documents/DD/issuances/dodi/500074p.pdf.

[37] This figure is adapted from DoDI 5000.85, "Major Capability Acquisition," November 4, 2021, available at https://www.esd.whs.mil/Portals/54/Documents/DD/issuances/dodi/500085p.pdf, p. 10.

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system elements decreases. Table 2 summarizes DMSMS management–related content outlined in guidance organized by life cycle phase.[38]

Table 2. DMSMS Management–Related Guidance Content

Life Cycle Phase	DMSMS Management–Related Guidance Content
Pre-Development	The DoD guidance, other than this document, does not specifically reference DMSMS management; however, program offices should obtain DMSMS management support[39] prior to the start of system development to: • Help draft mitigation strategies, possible technology refresh plans, and supportability risk analysis. • Research (in conjunction with contractors) planned parts lists, technology, materials, and the industrial base and determine potential obsolescence risks throughout the system's life cycle. • Include part selection criteria that consider DMSMS in the request for proposal (RFP).
Engineering & Manufacturing Development (EMD)	DoD guidance references DMSMS management responsibilities during EMD to reduce cost and prevent readiness delays by: • Incorporating technology management principles into design activities.[40] • Ensuring the intellectual property (IP) strategy captures accurate technical data including a comprehensive BOM.[41] • Ensuring the necessary funding and executing a DMP that encompasses proactive risk-based management for current and future DMSMS issues and taking necessary steps to resolve the issues.[42]
Production & Deployment	The DoD guidance, other than this document, does not specifically reference DMSMS management; however, program offices should obtain DMSMS management support during production and deployment to reduce cost and prevent readiness delays by: • Continuously monitoring to identify any changes in the system design materials and manufacturing processes that may result in DMSMS issues that impact production and operations and support activities. • Identify and develop recommendations to resolve DMSMS issues in a timely and cost-effective manner.
Operations and Support (O&S)	DoD guidance references DMSMS management responsibilities during O&S to reduce cost and prevent readiness delays by: • Addressing the requirement to obtain the technical data necessary for re-manufacturing, re-procurement, and sustainment engineering.[43] • Continuously monitor to identify any changes in the system design materials and manufacturing processes that may result in DMSMS issues that impact O&S activities.[44] • Assessing and revising the DMP to manage DMSMS issues as the system undergoes modernization.

[38] Although the vocabulary of MCA is used, the concepts of these phases apply to all pathways.

[39] Often this support is provided by a contractor.

[40] Office of the Under Secretary of Defense for Research and Engineering (OUSD)(R&E). *Systems Engineering Guidebook*, February 2022, available at https://www.dau.edu/sites/default/files/Migrated/CopDocuments/Systems-Eng-Guidebook_Feb2022-Cleared.pdf.

[41] Ibid.

[42] DoDI 5000.91, "Product Support Management for the Adaptive Acquisition Framework," November 4, 2021, available at https://www.esd.whs.mil/Portals/54/Documents/DD/issuances/dodi/500091p.pdf, p.19.

[43] Office of the Deputy Assistant Secretary of Defense for Product Support (ODASD[PS]), "PSM Guidebook," May 2022, available at https://www.dau.edu/tools/product-support-manager-psm-guidebook.

[44] Ibid.

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The following sections describe several more specific DMSMS management–related considerations for system development and system sustainment.

2.2.1 DMSMS Management Considerations for System Development

"Systems engineering comprises a methodical and disciplined approach for the specification, design, development, realization, technical management, operations, and retirement of a system."[45] DMSMS management encompasses both systems engineering and logistics to identify issues and implement resolutions in a way that avoids negative outcomes. The following two sections focus on two types of systems engineering activities—Systems Engineering Plans (SEPs) and SETRs[46]—that highlight additional DMSMS management–specific considerations.

2.2.1.1 Systems Engineering Plan

DoDI 5000.88, Engineering of Defense Systems, establishes policy that includes the requirement for Major Defense Acquisition Programs, acquisition category II and acquisition category III programs to have a SEP.[47] Lead systems engineers, "[u]nder the direction of the PM [Program Manager], develop a SEP to document and guide the program's specific systems engineering activities."[48] "The SEP documents technical risks, performance evolution strategy (including use of a modular open systems approach to the maximum extent practicable), processes, resources, metrics, SE [systems engineering] products, organizations, design considerations and completed and scheduled SE activities."[49] DMSMS issues represent risk to a system's schedule, performance, and life cycle cost. For this reason, program offices should incorporate DMSMS management into their SEPs, to ensure coordination with other planned technical processes and activities.

Section 3 of the *Systems Engineering Plan (SEP) Outline* describes the portions of a SEP to document various aspects of technical risks.[50] As one example, Section 3.2.1 of a SEP should incorporate the following content related to the technical risk, issue, and opportunity (RIO) management process:

- Risk management roles (roles and responsibilities associated with identifying risk),
- Risk issue management (risk tools, technical risk and mitigation planning, and risk reporting),
- Risk burndown, and
- Opportunity management.[51]

Unless waived by the approving authority, a SEP's software development approach should include software obsolescence as one of a number of activities for implementation,[52] Obsolescence represents one of several considerations for identifying severable components when "[t]he PM will use an appropriate open business model and system architecture that allows major system components to be

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[45] DoDI 5000.88, "Engineering of Defense Systems," November 18, 2020, available at https://www.esd.whs.mil/Portals/54/Documents/DD/issuances/dodi/500088p.pdf, p. 8.

[46] The concept of SEPs and SETRs are best practices that apply to all acquisition pathways.

[47] DoDI 5000.88, "Engineering of Defense Systems," November 18, 2020, available at https://www.esd.whs.mil/Portals/54/Documents/DD/issuances/dodi/500088p.pdf, p. 4.

[48] Ibid., p. 12.

[49] *Systems Engineering Guidebook*, available at https://ac.cto.mil/wp-content/uploads/2022/02/Systems-Eng-Guidebook_Feb2022-Cleared-slp.pdf, p. 7.

[50] Office of the Under Secretary of Defense for Research and Engineering, Department of Defense System Engineering Plan (SEP) Outline, Version 4.1, Washington, DC, May 2023, available at https://www.cto.mil/wp-content/uploads/2023/06/SEP-Outline-4.1.pdf, pp. 31–33.

[51] Ibid, p. pp. 31–32.

[52] DoDI 5000.88, "Engineering of Defense Systems," November 18, 2020, available at https://www.esd.whs.mil/Portals/54/Documents/DD/issuances/dodi/500088p.pdf, pp. 12–13 and p. 20.

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severable at the appropriate level for incremental addition, removal, or replacement over the system's life cycle."[53] Furthermore as a best practice, program offices should consider DMSMS management and the risk posed by DMSMS issues when assembling the relevant portions of a SEP.

2.2.1.2 Systems Engineering Technical Reviews

SETRs represent event-driven technical reviews and audits that occur throughout the system life cycle to provide leadership the opportunity to evaluate the program maturity and determine the status of technical risks associated with the cost, schedule, and performance goals of the system.[54] The *Systems Engineering Guidebook* specifically states that a program office should have a DMP in place and mitigate the risk of obsolescence during LRIP and FRP.[55] However, program offices would also have in place a number of additional DMSMS management–related issues of importance by the time of other SETR events. Table 3 summarizes specific issues of interest from a DMSMS management perspective at the time of each technical review.

Table 3. Summary of DMSMS Management–Related Issues of Interest by SETR

Review Type	Specific Issues of Interest from DMSMS Perspective
Alternative Systems Review (ASR)	DMSMS management planning has been initiated and is focused on the most likely preferred systems concepts. DMSMS impacts may be a consideration to ensure that the preferred system is cost-effective, affordable, operationally effective, and suitable and can be developed to provide a timely solution at an acceptable level of risk.
Systems Requirements Review (SRR)	The program office has begun to develop its DMSMS management strategy and plan, which begins to identify the roles and responsibilities of the government, prime/subcontractors, and third-party vendors. Some members of the DMT and contracting strategies have been identified. Technology development contracts require the delivery of data necessary for DMSMS management and define the contractor roles and responsibilities.
System Functional Review (SFR)	The DMP has been developed and a partial DMT has been formed. The development of DMSMS processes and metrics is underway.
Preliminary Design Review (PDR)	The DMP, including the documentation of all operational processes, has been formally approved by program office leadership. Monitoring and surveillance for DMSMS issues, using predictive tools and vendor surveys, is being conducted for notional or preliminary parts lists/BOMs. An assessment of whether and when obsolescence issues need to be addressed and if a resolution at a higher level of assembly should be considered, resolution determination, and resolution implementation have begun. Technology roadmaps and refreshment strategies are being factored into DMSMS management processes.
Critical Design Review (CDR)	Monitoring and surveillance for DMSMS issues, using predictive tools and vendor surveys, are being conducted based on the indentured BOM for the build baseline/final design. An assessment of whether and when obsolescence issues need to be addressed and if a resolution at a higher level of assembly should be considered, resolution determination, and resolution implementation are taking place. Technology roadmaps and refreshment strategies are being factored into DMSMS processes. Case management and the capture of metrics are taking place. EMD contracts require the delivery of data necessary for DMSMS management and define the contractor roles and responsibilities.
Production Readiness Review (PRR)	Monitoring and surveillance for DMSMS issues, using predictive tools and vendor surveys, are being conducted. An assessment of whether and when obsolescence issues need to be addressed and if a resolution at a higher level of assembly should be considered, resolution determination, and resolution implementation are taking place. Technology roadmaps and refreshment strategies are being factored into DMSMS processes. Case management and the capture of metrics are taking place.

[53] Ibid., pp. 2–26.

[54] *Systems Engineering Guidebook*, p. 1.

[55] Ibid, p. 86.

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Appendix B identifies specific DMSMS management–related questions a program office can use to support technical reviews. The DMSMS management–related questions offered in that appendix provide a basis for the DMSMS management community to inform DMSMS-related discussions prior to technical reviews and to highlight DMSMS issues to address during those reviews. Systems engineering checklists for technical reviews already incorporate DMSMS management–related questions. However, DMSMS management practitioners have undertaken an effort to expand upon them systematically.

The program office should review and oversee the DMSMS management efforts of its prime contractor and ensure the review of all obsolescence issues and their resolution before the decision to proceed to the next phase in the life cycle, particularly before moving into production. To enable this, program offices should incorporate the necessary language into contracts to ensure that they are able to review all designs and redesigns in relation to SETRs.

2.2.2 DMSMS Management Considerations for System Sustainment

> *The Life Cycle Sustainment Plan (LCSP) should be used for governing operations and support planning and execution from program inception to disposal. It serves as the "detailed product support plan, including sustainment metrics, risks, costs, and analyses used to deliver the performance-based best value strategy covering the Integrated Product Support (IPS) elements."*[56]

Through life cycle sustainment planning, readiness can be maximized, and product support can be provided at the lowest cost.[57] As a best practice, program offices incorporate DMSMS management into their life cycle sustainment planning. A program office's activities to monitor and assess the health of its system's supply chain should include DMSMS risk.[58] The following two sections focus on two types of sustainment activities—ILAs and SRs[59]—that highlight additional DMSMS management–specific considerations.

2.2.2.1 Life Cycle Sustainment Plan

The LCSP serves as a repository of the issues, analysis, and decisions associated with developing and executing a product support solution. PSMs are responsible (per 10 U.S.C. § 4324) for developing and maintaining an LCSP consistent with the product support strategy, beginning at Milestone A or an equivalent decision event for covered systems.[60]

Section 4.12.1 "Obsolescence Risk Management" of the *Life Cycle Sustainment Plan: Sample Outline* highlights obsolescence management, including DMSMS management as one type of consideration to address.[61] DMSMS issues can cause in-service problems and require funding to support the implementation of resolutions. As a best practice, a program office's planning for its system's sustainment should include planning for the inevitability of DMSMS issues and their mitigation. One element of this DMSMS-related consideration includes understanding and continuing to monitor a system's supply chain, particularly after production ends.

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[56] DoDI 5000.91, Product Support Management for the Adaptative Acquisition Framework, November 4, 2021, available at https://www.esd.whs.mil/Portals/54/Documents/DD/issuances/dodi/500088p.pdf, p. 12.

[57] Ibid.

[58] Ibid.

[59] The concepts of LCSPs and ILAs apply to all pathways.

[60] DoDI 5000.91, Product Support Management for the Adaptative Acquisition Framework, November 4, 2021, available at https://www.esd.whs.mil/Portals/54/Documents/DD/issuances/dodi/500088p.pdf, p. 12.

[61] ODASD(PS), *Life Cycle Sustainment Plan: Sample Outline*, Version 3.0, October 13, 2022, p. 40.

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2.2.2.2 DMSMS Considerations in Integrated Logistics Assessments and Sustainment Reviews

Although only a requirement for the MCA pathway, all acquisition pathways should use ILAs to ensure a focus on product support and sustainment planning.[62] An ILA represents an analysis of a program office's supportability planning, which serves as

> *[A]n effective and valid assessment of the program office's product support strategy, as well as an assessment of how well this strategy will lead to successfully fielding, operating and sustaining a system at an affordable cost….It is important to conduct ILAs early enough in the program life cycle, when the design can be influenced, and with the intent to achieve effective and efficient weapon system capability and life cycle management. Re-assessing the program sustainment planning at each MS, and periodically thereafter as the design matures, is critical to fielding a sustainable system.*[63]

Because DMSMS issues have a bearing on the sustainment of a system, DMSMS should be considered within ILAs and SRs. Table 4 summarizes the focus of the ILAs and SRs before fielding and during operations.

Table 4. Summary of ILAs and SRs

Assessment Type	Specific Issues of Interest from a DMSMS Perspective
Before-Fielding (ILA)	Monitoring and surveillance for DMSMS issues, using predictive tools and vendor surveys, are being conducted to identify immediate and near-term obsolescence issues associated with the system BOM. An assessment of whether and when obsolescence issues need to be addressed and if a resolution at a higher level of assembly should be considered, resolution determination, and resolution implementation are taking place. Technology roadmaps and refreshment strategies are being factored into DMSMS processes. Case management and the capture of metrics are taking place.
During-Operations (SR)	Monitoring and surveillance for DMSMS issues, using predictive tools and vendor surveys, are being conducted to identify immediate and near-term obsolescence issues associated with the system BOM. An assessment of whether and when obsolescence issues need to be addressed and if a resolution at a higher level of assembly should be considered, resolution determination, and resolution implementation are taking place. Technology roadmaps and refreshment strategies are being factored into DMSMS processes and reviewed for potential updates and adjustments. Case management and the capture of metrics are taking place.

Appendix C identifies specific DMSMS management–related questions a program office can use to support ILAs and SRs. As was the case for SETRs, the DMSMS management–related questions offered in the appendix provide a basis for the DMSMS management community to inform DMSMS-related discussions before ILAs and to highlight DMSMS issues to address during those ILAs. Checklists for assessment already incorporate DMSMS management–related questions, but DMSMS management practitioners have undertaken an effort to expand upon them systematically.

10 U.S.C. § 4323 requires the Secretary of each military department to conduct an SR of each covered system not later than five years after declaration of IOC of a major defense acquisition program and every five years thereafter throughout the life cycle of the covered system. One objective of the SR is to assess the effectiveness of the DMSMS and parts, materials, and processes (PM&P) management content and

[62] Guidance on this subject may be found in ODASD(PS) *Independent Logistics Assessment Guidebook*, April 2023, available at https://www.dau.edu/sites/default/files/Migrated/ToolAttachments/ILA%20Guidebook%20FINAL%2007%20April%202023.pdf.

[63] Ibid., p. 6.

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to support programmatic decisions and to make recommendations for improving that content.[64] For DMSMS management, effectiveness may be measured using data from mandatory record keeping requirements. For PM&P management, effectiveness will be more qualitative.

The SR assesses the product support strategy, performance, and O&S costs of the covered system. The results of the SR shall be documented in a memorandum by the relevant decision authority made available to the USD(A&S) within 30 days after the review is completed.[65] SR specifics are also called out in DoDI 5000.91, Product Support Management for the Adaptive Acquisition Framework (Section 4.10.b). The DoDI states that SRs:

- Are required for all active and in-service covered weapon systems;
- End five years before a covered system's planned end of service date;
- Will focus on statutory sustainment elements and track O&S cost growth; and
- Satisfy the requirement for ILAs after a program has achieved IOC.[66]

The ILA and SR are both important processes in the life cycle of a weapon system, but each focus on different phases of the system life cycle and are governed by different statutes, policy, and guidance.[67] While ILAs are not required by statute or policy after the FRP decision, if conducted at Component discretion after IOC, their results can inform and complement SRs.

2.2.2.3 Other System Sustainment Guidance for DMSMS Management

Additional guidance provides further information on system sustainment for the DoD enterprise. The following lists some of these guidance documents, their purpose, and how they relate to DMSMS management. DAU's tools catalog contains the complete interconnected suite of product support guidebooks.[68]

- ***Integrated Product Support Elements Guidebook***.[69] This guidebook provides additional guidance to PSMs, particularly regarding the 12 IPS elements.[70] Three sections devoted to the IPS elements—sustaining engineering, supply support, and information technology systems continuous support—mention DMSMS.
- ***Product Support Manager Guidebook***.[71] This guidebook provides PSMs with guidance on developing and implementing a program support strategy for their programs. A PSM, along with a PM, has the responsibility for planning for a risk-based approach to DMSMS

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[64] PM&P management was formerly called parts management. The name was changed to avoid confusion with the logistics discipline of materiel management. While all use of the term *parts management* in the text of this document have been changed, nonupdated reference document names and extracts continue to use the "parts management" terminology.

[65] 10 U.S.C. § 4323, available at https://www.govinfo.gov/link/uscode/10/4323.

[66] DoDI 5000.91, "Product Support Management for the Adaptive Acquisition Framework," November 4, 2021, available at https://www.esd.whs.mil/Portals/54/Documents/DD/issuances/dodi/500091p.pdf.

[67] DAU LOG Blogs, *ILAs and Post-IOC Sustainment Reviews: A Study in Contrasts*, November 13, 2019, available at https://www.dau.edu/blogs/ilas-and-post-ioc-sustainment-reviews-study-contrasts.

[68] DAU, "Integrated Product Support Guidebook Suite," available at https://www.dau.edu/IPS-Guidebook-Suite.

[69] DAU, *Integrated Product Support (IPS) Elements Guidebook*, March 2021, available at https://www.dau.edu/tools/Integrated-Product-Support-(IPS)-Element-Guidebook.pdf.

[70] The 12 integrated logistics support elements are: Product Support Management; Design Interface; Sustaining Engineering; Supply Support; Maintenance Planning & Management; Packaging, Handling, Storage, and Transportation Planning & Management; Technical Data; Support Equipment; Training & Training Support; Manpower & Personnel; Facilities & Infrastructure; and Computer Resources. DAU, *Integrated Product Support (IPS) Elements Guidebook*, March 2021, available at https://www.dau.edu/tools/Integrated-Product-Support-(IPS)-Element-Guidebook.pdf.

[71] ODASD(PS), PSM Guidebook, May 2022, available at https://www.dau.edu/tools/product-support-manager-psm-guidebook.

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management for the program office. An “Obsolescence/DMSMS Mitigation” section on key product support considerations refers to DMSMS.[72]

- ***Product Support Business Case Analysis Guidebook***.[73] This guidebook describes a methodology to assist in decision-making when considering product support alternatives. A table focused on analytic tools includes some DMSMS and obsolescence-related tools.[74]

2.2.3 Tailoring DMSMS Management Considerations for Different Acquisition Pathways

DMSMS management applies to five of the six acquisition pathways—MCA, UCA, MTA, DBS, and SWA. The first four pathways lead to the deployment of hardware systems and their associated software (with the DBS pathway being the most software intensive). The fifth pathway applies exclusively to software. The bulk of the remainder of this section deals with the first four pathways. The last part of this section briefly describes some unique DMSMS considerations associated with the SWA pathway.

There are differences among the four hardware-associated acquisition pathways:

- Terminology unique to each pathway,
- Different selection criteria for determining which pathway to follow, and
- Inclusion of phases differing in number and length.

Despite these differences, each of these four pathways encompasses similar generic life cycle activities—development, production and deployment, and O&S. These life cycle activities should incorporate DMSMS management operations.

The following describes the DMSMS management implications of those generic activities for each of those pathways. DMSMS management content for each pathway represents the tailoring of the best practices for DMSMS management operational processes described in Section 3 through Section 7 of this document. Tailoring of DMSMS management operational processes depends on:

- The specific hardware characteristics of the system of focus,
- The associated acquisition and sustainment strategies,
- The prevalence and complexity of software in the system, and
- The amount of risk that the program office is willing to tolerate.

The DMSMS management processes, summarized earlier in this section and detailed in Section 3 through Section 7 of this document support the MCA pathway. Consequently, very little tailoring is needed, and this section does not elaborate further on DMSMS management for the MCA pathway. The one exception is when it is useful to compare the application of DMSMS management with MCA to DMSMS management for the other three hardware-associated pathways.

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[72] Ibid, p. 140.

[73] DAU, *DoD Product Support Business Case Analysis Guidebook*, 2024, available at https://www.dau.edu/tools/product-support-business-case-analysis-bca-guidebook.

[74] Ibid, pp. 38–39.

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2.2.3.1 DMSMS Management Planning

Regardless of the chosen acquisition pathway, all program offices should have a DMT (although it may be small under certain circumstances) that oversees the execution of a DMP that answers some basic questions.

- What is the expected life cycle of the system? If a system will only be utilized for a short period, DMSMS management activities may be highly tailored because of the limited time horizon. For example, initial spares may have been purchased in quantities estimated to be sufficient to support the system through its planned life. Consequently, DMSMS management may be limited to the verification that those assumptions continue to hold.
- What items should be monitored proactively? From a risk-based perspective, the bulk of this list will reflect mission critical, high failure rate, and DMSMS-prone items without substitutes readily available in the commercial market.
- To what extent will the program office rely on its major suppliers to conduct DMSMS management? The answer to this question scopes the extent of DMSMS activities that the program office will conduct/oversee. This question applies only to the items selected for proactive monitoring.

  In the case of COTS assemblies, there may be a high reliance on suppliers for two reasons. First, BOMs are generally not available for COTS assemblies and therefore the complexity of DMSMS management operations that the government is able to have performed will be significantly limited. Second, commercial suppliers may automatically remove obsolescence from the items they sell because it is a good business practice to do so.

  For non-COTS items, the program office will typically want to have greater control over DMSMS management operations.

  A further consideration for both COTS and non-COTS items is the program office's assessment about the suppliers' capability to identify and resolve DMSMS issues in a timely and cost-effective manner.

The answers to these questions have an impact on what DMSMS management functions are and are not performed internally within the program office. Different pathways will have vastly different DMSMS management contract requirements. Regardless of the pathway selected, it is important to put the appropriate contract language in place as early as possible.

2.2.3.2 Urgent Capability Acquisition Pathway

The UCA pathway represents a set of highly tailored acquisition activities designed to support fielding of a quick reaction capability, usually within two years. There is a $525 million limit on the amount of research, development, test, and evaluation funding allowed for a single acquisition for this pathway.

Once an urgent requirement is approved, a UCA program is initiated. The *pre-development* phase determines the course(s) of action for fielding. This is similar in concept to a combination of the analyses performed prior to Milestone A and the *technology maturity and risk reduction* phase of the MCA pathway. Because the UCA development effort is so small and short, most if not all subsystems of the UCA will be mature during the *pre-development* phase. Thus, the supportability analysis conducted at that time should focus on DMSMS management planning. Considerations for answering the DMSMS management planning questions are as follows:

- The life of the system should generally be assumed to be short to coincide with the length of the contingency that generated the UCA. Although there can be a decision at a later point in time to extend system life, that decision will normally not be made at the time of pre-development activities.

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- Which critical items should be monitored is situation specific. DMSMS-prone items may not be of a high concern at this time because of assumptions about the system life. DMSMS may be considered unlikely over a system's short life cycle.
- Since most of the equipment will be commercial assemblies and the system life is expected to be short, it will be common to rely on commercial equipment suppliers to resolve any DMSMS issues that occur. However, a few subsystems will likely require more rigorous DMSMS management performed/overseen by the program office.

The *development* phase follows the *pre-development* phase of a UCA. The idea of design resilience applies to DMSMS management, however, given the minimal scope of the development effort, a significant portion of the design will have already been completed and assemblies will have been produced. Monitoring items in the system for obsolescence during *development* will be constrained by the lack of availability of parts lists associated with commercial assemblies and the reliance on commercial suppliers. Design changes to remove obsolete or near-obsolete items may not be feasible. Policy indicates that such deficiencies may be addressed later in the life cycle because of the urgency of the need to field the system. The UCA policy is also silent about the development of prototypes, but due to the compressed nature of the schedule, any prototypes are likely to be closer to engineering design models.

As a precaution, technology roadmaps should be obtained in the development phase (and it may even have started in *pre-development*). The roadmap should examine the technologies currently being used and forecast when they will become obsolete and what technology will replace them. The principal precautionary reasons for developing roadmaps are

- To verify that commercial suppliers will be able to maintain the viability of commercial assemblies over the expected service life and
- To prepare (both from a DMSMS-related technology refreshment perspective and a performance-improvement perspective) for service life extension if it were to occur.

The DMSMS management processes associated with the *production and deployment* and the *operations and support* phases continue the activities begun during *development*. Monitoring to identify issues may increase; any issues should be resolved before they impact the system as DMSMS issues can interfere with production. The constraints on data available for monitoring commercial assemblies may continue to apply and therefore monitoring processes will be tailored accordingly. Resolutions may involve refreshing the technologies as indicated by technology and product roadmaps.

For a UCA, a disposition analysis takes place after one year into the *operations and support* phase. At that point in time, a determination is made to either terminate the system, continue operations until the end of the current contingency, or transition to a program of record. The results of this decision inform technology and product roadmaps and other DMSMS management processes on the length of the planning horizon. When termination is near, DMSMS management involves far less forecasting and focuses on more short-term resolutions to the extent that they are necessary. If the system life is extended, then it is likely that there will be a transition to another acquisition pathway; DMSMS management activities should adjust as needed.

2.2.3.3 Middle Tier Acquisition Pathway

The MTA pathway applies to systems that have a level of maturity that allows them to be fielded within five years. MTA policy includes only two acquisition phases—*rapid prototyping* followed by *rapid fielding*. Depending on the maturity of the technology, some MTA programs may only have virtual prototyping, and some may skip the first phase entirely. When there is no *rapid prototyping* phase, production should begin within six months of MTA program initiation. DoDI 5000.80 indicates that DoD Components will develop a process for transitioning successful prototypes to new or existing acquisition programs for production,

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fielding, operations, and sustainment under the *rapid fielding* pathway or other acquisition pathway. This process will result in a transition plan, included in the acquisition strategy (AS), which provides a timeline for completion within two years of all necessary documentation required for transition, as determined by the milestone decision authority, after MTA program start.[75]

A tailored LCSP is a requirement for the approval of *rapid fielding* for major systems. Consequently, supportability, which includes DMSMS management, should be considered during *rapid prototyping* and in the time leading up to production. Even though no LCSP is required for non-major systems, supportability should not be ignored.

DMSMS management planning (including the development of a plan and the formation of a team) should occur immediately after MTA program initiation. The scoping questions listed for the UCA pathway apply.

- Since an MTA system will be in production for about five years and then in operation for some period beyond that, in general there is no reason to tailor DMSMS management activities based on the length of service. The life cycle is long enough to experience obsolescence issues that must be resolved.
- The items to be monitored will be selected from a risk-based perspective. Expectations are that monitoring will be needed for both COTS and non-COTS items.
- Because of the minimal amount of design and development time typically associated with an MTA, many items in the system are likely to be commercial or used on currently fielded systems. Therefore, many of the DMSMS management functions would be performed by suppliers early in the life cycle or are already being performed somewhere else in DoD. When this is not the case and later in the life cycle, the program office should plan for technology refreshment when commercial support is no longer available. There also may potentially be a need to prepare for greater in-house DMSMS management support and oversight.

Before a production decision, program office DMSMS management operations should focus on assessing DMSMS resilience in the designs that it controls and obsolescence in the build-to-print designs it controls. In an MCA acquisition, the program office has control over many more designs. Therefore, design reviews, smart parts selection, and obsolete part replacement will have been taking place, in conjunction with the prime contractor, as the designs mature. Since the final designs will be completed within a few months of program initiation, there are unlikely to be many opportunities for conducting these activities on preliminary versions of the design for the MTA pathway. There may also be limited options for eliminating obsolete or nearly obsolete items from designs. However, since production may last for more than four years, every effort should be made to do so.

Once in production and during O&S, the use of technology roadmaps should be a significant element of DMSMS management activities. For a UCA, using technology roadmaps was labeled precautionary. In the MTA pathway, it should be considered mandatory. Some COTS assemblies (especially electronics assemblies) are likely to become obsolete during more than four years of production. More will become obsolete during O&S. Consequently, there will be a need for technology refreshment. Furthermore, there may also be requirements for capability enhancement which will also be informed by technology roadmaps that reflect commercial and DoD research and development (R&D).

For other items, the program office should conduct/oversee activities to identify obsolescence issues as early as possible and resolve them before they impact the system.

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[75] DoDI 5000.80, "Operation of the Middle Tier of Acquisition (MTA)," December 30, 2019, available at https://www.esd.whs.mil/Portals/54/Documents/DD/issuances/dodi/500080p.pdf, p. 8.

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2.2.3.4 Defense Business Systems Pathway

The DBS pathway, as its name suggests, governs the acquisition of business capability by the DoD. Of the hardware-associated pathways described in this section, this pathway is the most heavily software oriented and most closely aligns with commercial best practices. Customization of commercial products is minimized by policy. No time limits on fielding are imposed and the size of the budget authority defines the decision authorities.

The first phase of the DBS pathway is *capability need identification*. Once a need is identified, a decision is made on whether to invest resources in the development of a capability to meet that need. There are no DMSMS management considerations in this phase which corresponds to pre-Materiel Development Decision (MDD) activities in the MCA pathway.

Solution analysis is the second phase of the DBS pathway. Activities in this phase generally align with pre-MDD analyses for the MCA pathway. The characteristics of the business system necessary to satisfy the needs is established. Industry analysis and market research play key roles in keeping requirements for new development to a minimum. Although supportability may be a consideration, the level of detail does not involve DMSMS management.

The *functional requirements and acquisition planning* phase follows *solution analysis*. Some parallels can be drawn to the *technology maturity and risk reduction* phase in the MCA pathway. In this phase, system requirements are determined, an acquisition approach is formulated, and an AS is developed. Prototypes may be used to demonstrate technological capabilities. DMSMS management planning activities begin at this point in the life cycle. Considerations for the DMSMS management planning questions are as follows:

- What is the anticipated life cycle of the system? Usually, DoD business systems are long-lived.
- What should be monitored? As was the case for the other pathways, the decision on what should be monitored is based on criticality and DMSMS risk.
- To what extent will the program office rely on industry to perform DMSMS management? Often this will be quite high for the DBS pathway, because of the reliance on commercial systems to meet DoD's needs. The availability of BOMs is typically quite low because of the commercial nature of the system and its subsystems.

Monitoring should also be initiated in this phase. Because of the prevalence of commercial software and commercial business equipment, support for many of the elements of the business system is often readily available. As a result, only a small number of items may need to be monitored. The program office will rely heavily on commercial suppliers to automatically resolve obsolescence issues for a time. Design resilience is generally not a large component of DMSMS management because most of the business system complexity is in the software, as opposed to hardware. One exception might be the discovery of a cybersecurity vulnerability. Such a situation should be addressed prior to deployment. Finally, technology roadmaps should also be obtained to identify when technology refreshments (as well as capability enhancements in product roadmaps) should be considered because of changes in commercial markets (or advancement from military R&D).

The fourth phase is *acquisition testing and deployment*. This phase aligns with the *engineering and manufacturing development* phase in the MCA pathway. Prototypes may be created as part of the development effort. Over the course of this phase, the business system is deployed, usually in two sub-phases—limited deployment and full deployment. Support activities are also baselined.

From a DMSMS management perspective, this is the point in the life cycle where proactive monitoring could begin as preliminary parts lists are delivered and refined for non-commercial equipment. Both long-term and short-term issues would be considered and resolved when deemed appropriate to do so.

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However, for the most part, due to the nature of DBS, typically (but there may be exceptions) the monitoring and resolving functions would be automatically done by the contractors providing software as part of their normal commercial business operations. With the current trend of using commercial cloud systems, DMSMS may not be a concern for hardware. If DoD decides to own and maintain business system hardware, it will typically be refreshed periodically. The DMSMS management community, through technology and product roadmaps, may contribute to the determination of an optimal refreshment cycle in combination with plans for capability enhancement through technology refreshment.

Capability support is the last phase, analogous to the *operating and support* phase for the MCA pathway. More items may have to be monitored when commercial suppliers are no longer willing to support the hardware and software in the DBS. Availability of data is a limitation on the program office's ability to do this. Consequently, there should be greater dependence on technology roadmaps. When the system is modified, DMSMS management activities should change accordingly.

2.2.3.5 Software Acquisition Pathway

The SWA pathway is designed for the acquisition of custom software for DoD. There are two phases to an SWA—*planning* and *execution*. The policy governing this pathway emphasizes the use of modern, iterative software development techniques and tools. Program offices acquiring software through the DBS pathway or independently via the MCA pathway may elect to follow this policy.

The SWA pathway includes two sub-pathways. The first applies to software installed on commercial hardware. The other pertains to situations where the software is embedded in a weapon system or other military unique hardware.

Because of the custom nature of SWAs using this pathway, DMSMS management considerations are often concerned with functional obsolescence of the acquired software because of several potential factors. For example:

- Changes to hardware or software in the system that interfaces with the acquired software.
- Changes in policy on the use or applicability of the acquired software.
- Discovery of new cybersecurity weaknesses or vulnerabilities in the acquired software.

The corollary to this situation applies as well. The installation of software acquired through this pathway may lead to the functional obsolescence of other hardware and software in the system.

DMSMS management planning and monitoring should therefore monitor and test for potential functional obsolescence issues during both the *planning* and *execution* phases. They may occur during software development, fielding, maintenance, and upgrade activities. When functional obsolescence is detected, appropriate resolutions should be immediately implemented.

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Section 3. Prepare: DMSMS Management Program Infrastructure

This chapter describes the *Prepare* step of the DMSMS management program. It encompasses establishing the foundations for robust DMSMS management, developing a DMP, forming a properly trained DMT to carry out all DMSMS activities, and establishing DMSMS operational processes—

- Secure resources for DMSMS management operations,
- Establish interfaces to advocate for DMSMS-resilient designs,
- Establish a DMSMS management evaluation process,
- Establish a Quality Management System (QMS),
- Establish a case monitoring and tracking process, and
- Establish supporting contracts.

As the first step, *Prepare* lays the groundwork for the other four DMSMS management process steps.

All the processes included in the *Prepare* step do not have to occur linearly. Aspects of these processes occur simultaneously or iteratively. For example, a DMT will usually begin to form while the DMP is being developed. Similarly, the establishment of DMSMS operational processes will often affect DMP content.

3.1 Establish the Foundations for DMSMS Management

The PM should establish the foundations for DMSMS management for the program office. This strategy sets the priorities for the DMSMS management approach to be pursued by the program office's DMT[76] and documented in its DMP.

Program office leadership may be tempted to delegate the establishment of the strategic direction to the DMT. However, doing so could result in a DMSMS program with a scope of effort documented in its DMP that does not match the funding obtained and available to support DMSMS management or program office management's expectations. Therefore, program office leadership should engage early (before the DMP is finalized) to define the objectives for DMSMS management, the roles and responsibilities of DMT members, and DMT operating guidelines as well as the procedures for and frequency of DMT meetings.

In addition, program office leadership should refine and elaborate upon the four overarching objectives of DMSMS management (Section 1.4). Finally, program office leadership should provide its expectations to the DMT and contractors with DMSMS responsibilities for the following:

- The relative relationships among DMT members;
- The relative relationships between government and contractor DMT members;
- Communication between DMT members and program office organizational entities that do not participate in DMT meetings;

[76] This document assumes that the program office's DMT will have the lead for DMSMS management. Having the lead however does not imply that the DMT (or even the program office itself) will be doing most of the work. In many instances, much of that effort is conducted by production or sustainment contractors. In such cases, the DMT is primarily responsible for oversight. Therefore, the priorities for the DMSMS management approach would also be pursued by contractors with DMSMS management responsibilities.

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- The artifacts, data, and deliverables to be produced; and
- The ground rules regarding when program office leadership and other key stakeholders should be briefed and the PM's role in decision-making.

More detailed information on DMTs can be found in Section 3.2.

Program office leadership should establish risk-based operating guidelines. For many program offices, operating guidelines will consist of similar, relatively generic statements such as the following:

- Maintain case management data on all DMSMS issues,
- Assess whether and when obsolescence issues need to be addressed and if a resolution at a higher level of assembly should be considered,
- Analyze ways to resolve the obsolescence issues at the level of assembly that is most cost effective over the life cycle,
- Monitor items and assemblies to identify obsolescence issues, and
- Oversee the implementation of the resolutions.

These generic operating guidelines should be refined by program office leadership to account for risk. To do so, program office leadership should work with DMT experts to address the following questions:

- ***Which subsystems should have priority for monitoring?*** Prioritization should be based on such considerations as mission criticality, safety, or known problem areas. This prioritization will assist in determining which BOMs and associated technical data (along with the necessary IP[77] rights to that data) the program office needs to obtain to support proactive monitoring for DMSMS issues. If a reactive approach to DMSMS management is sufficient for a particular subsystem, then it is not necessary for the program office to acquire or build that subsystem's BOM.
- ***What items within those subsystems should be monitored?*** The answer to this question is a function of the risks that program office management is willing to accept. These risks manifest when a determination is made of which items to proactively monitor (and by default, which items should be managed reactively) for a system or subsystem of interest as described in Section 4 as part of the *Identify* step of DMSMS management.

Program office leadership should also address the following questions to refine operating guidelines.

- ***When should DMSMS management begin?*** As a best practice, proactive DMSMS management should begin as early as the SRR, whenever preliminary parts lists become available. Pursuing proactive DMSMS management for electronic items early is important, because there have been numerous examples of obsolete electronic items being incorporated into designs, virtually assuring sustainment issues if they are not discovered. Because the market forces driving DMSMS issues for MaSME items operate much more slowly than the competitive commercial electronics market, one might question whether to begin DMSMS management for these items later in the life cycle. It remains a best practice, however, to begin proactive DMSMS management for MaSME items at the same time as electronic items, for several reasons. First, the earlier that monitoring begins, the larger the window of opportunity to address an issue, the larger the selection of less expensive resolutions, and the smaller the likelihood of experiencing schedule and readiness impacts. Second, since a program office will only be monitoring high-risk MaSME items, proactive monitoring of such items should begin early. Third, MaSME item monitoring can be integrated

[77] The DMSMS community does not typically have unique IP requirements. DMSMS needs (e.g., BOMs for monitoring and technical data about the item or its manufacturing process used in resolution determination) are usually included in the IP strategy by the logistics or engineering communities. While the DMSMS community works with BOMs on a continuous basis, other resolution-related technical data may only be needed occasionally. Therefore, from a DMSMS perspective, the IP rights for some data may be escrowed if that does not conflict with other communities' needs.

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with electronic item monitoring. Finally, early monitoring will enable designs containing high-risk MaSME items to be revised before it becomes much more costly to do so.

- ***How important is DMSMS management for software?*** While software does not become traditionally obsolete, the following issues may occur that affect the availability of the software and eventually impact the system:
  - The maintenance and support for the software is no longer available,
  - The hardware the software runs on becomes obsolete,
  - The environment and tools used to develop the software are no longer available,
  - The ability to build and test the software no longer exists, and
  - The skill set and knowledge necessary to support the software no longer exists.

  Just like hardware, many types of systems have little or no control over the supply chain for COTS software or the software development infrastructure they depend upon for development and support. In the COTS world, hardware and software have developed a symbiotic supply chain relationship where hardware improvements drive software developers to obsolete software, which in turn cause older hardware to become obsolete. By establishing a software obsolescence program, the following benefits are realized:
  - Identification of potential issues regarding software availability prior to system impact,
  - Increase in software personnel being cognizant of issues that were previously unknown,
  - Assistance with future planning of contracts,
  - Validation of software issue notification, and
  - Greater awareness of cybersecurity weaknesses and vulnerabilities.
- ***How should risk be taken into account?*** A risk-based perspective to monitoring is important because robust, proactive DMSMS management everywhere is not cost effective and will be unnecessarily time-consuming. A risk-based approach takes into consideration the differing sizes of (and associated number of items within) the systems and thus the workload required for monitoring for DMSMS issues across all types of items. At one end of the spectrum, everything could be proactively monitored to predict obsolescence before the item is no longer available. However, for many items, the impact of obsolescence is small because alternatives are readily available. A risk-based approach to DMSMS management strikes a balance among high-risk items that should always be proactively monitored, items that are broadly available over a long period of time, and everything in between. Additional information regarding levels of risk-based prioritization can be found in Section 4.1 and Section 4.3.2.

3.2 Form a DMT

The PM or PSM should charter the DMT and clearly identify and authorize its activities as well as the roles and responsibilities of its members inside and outside the context of the DMT. The DMT should represent both internal and external organizations that provide routine and recurring support to the DMSMS management program. In some cases, it may be appropriate for representatives of other system DMTs to participate if their DMPs and processes interact. In other cases, multiple layers of DMTs may exist (e.g., at the system level and for some subsystems); in these instances, subsystem level DMTs should participate in and leverage the program office DMT, as appropriate. The program offices near-term DMSMS management objectives will drive the composition of the DMT and the roles and responsibilities of the DMT members.

3.2.1 Roles and Responsibilities of DMT Members

The roles and responsibilities of the program office, the prime contractor, the OEMs, and independent subject matter experts (SMEs) are among those established by program office leadership in the DMSMS

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management strategy. The DMP further expands upon the roles and responsibilities of the DMT as well as its composition in overseeing and managing obsolescence, including software, to the extent that it is a strategic priority. For example, the DMP may include information on technology refreshment plans. Such plans may also include software upgrades. Also, they may include requirements for hardware and software vendor surveys or for tracking the extent to which the software satisfies information assurance requirements. Finally, appropriate contract language should be developed to carry out the prime contractor and/or OEM responsibilities.

The roles and responsibilities of the DMT are similar for every program office, but the level of effort required of the team will depend on the complexity of the system and the severity of its DMSMS issues. In general, the activities of the DMT are to gather the necessary data, develop and implement the DMSMS management processes that require those data, produce the management products that result, report metrics that measure the effectiveness of the DMSMS management program when compared to the defined objectives, and apply continuous improvement processes to DMT operations. The roles should be tailored to meet specific program office needs.

The composition of the team will depend on the complexity of the system as well as on other considerations. For example, some team members may have multiple duties. This may affect the amount of time team members can devote to DMT activities, or they may be assigned multiple roles on the DMT. As another example, if the responsibility for the system will be transferred to another agency or activity midway through the life cycle, the stakeholder who will ultimately bear responsibility for sustainment should participate in the DMT during all phases of the life cycle. The DMT composition may evolve over time. Early in the life cycle, before the CDR, a partial team may be sufficient.

If contractors and/or independent SMEs participate in the DMT, their roles and responsibilities should be identified. DMT oversight relationships with respect to contractors and/or independent SMEs can evolve and change depending on the phase of the life cycle, which may impact the DMT's composition. Ideally, a DMT should consist of the following roles:

- ***DMT lead***. The DMT lead is the spokesperson for the DMT who oversees and has the authority to control DMSMS management operations. The DMT lead is also the champion of DMSMS for the program office and, as a best practice, should be a full-time position. Both conditions being present signals the importance placed upon DMSMS management by program office leadership. The DMT lead is responsible for coordinating DMT meetings and managing corresponding action items, identifying potential sources of funding and funding availability, requesting funding and other resources to support the DMSMS management program, overseeing the DMSMS management support contracts and agreements, interfacing with the configuration control board (CCB), and reporting on DMSMS risks at technical, logistics, and programmatic reviews. This role should be filled from logistics, engineering, or program office management organizations by an individual who possesses sufficient knowledge of DMSMS issues to be able to communicate to decision makers on the importance of actions regarding DMSMS cases. This role may be filled by the DMSMS SME.
- ***Program office representative***. The program office representative represents the views of the PM on the DMT. Normally, this person would be the DMT lead and champion of DMSMS for the program office.
- ***DMSMS SME***. The SME coordinates the execution of DMT management processes and the development of DMT management products. This includes, for example, assessing obsolescence forecasts, processing business case analyses (BCAs), preparing budget forecasts, and presenting solution options to the DMT for discussion and concurrence. In addition, the SME assists with establishing the DMT and with developing and maintaining the program office's DMP as a living document. Furthermore, the SME monitors the effectiveness of the DMSMS management program and recommends ways to continually improve it by capturing and assessing metrics that accurately measure the success or failure of meeting the defined DMSMS management program objectives.

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- ***Engineering activity representative***. The DMT member representing the engineering function (including industrial, mechanical, electrical, and general engineering expertise, as well as a systems integrator perspective) is responsible for managing the incorporation of DMSMS-related technical data into government drawings, technical publications, and documentation. The engineering activity representative provides information to the DMT regarding resource requirements, systems integration engineering, and reliability and maintainability analyses on items selected for use on the system. The engineering activity representative also assesses the suitability and feasibility of proposed technical solutions. Early in the life cycle, the engineering activity representative may also include the prime contractor representative, if that is the source of greatest expertise.
- ***System security engineer (SSE)***. The DMT should include an SSE except in the highly unusual instance that system security is not a concern. The SSE is responsible for identifying security vulnerabilities and minimizing risk associated with those vulnerabilities. The SSE integrates contributions from other system engineering disciplines such as anti-tamper, defense exportability features (DEFs), hardware assurance, software assurance, SCRM, cybersecurity, security, intelligence, and counterintelligence specialties. The SSE uses the Program Protection Plan (PPP) to document a comprehensive approach to system security engineering analysis and security and engineering countermeasures to mitigate foreign adversary risk. Because of the risk of these disciplines and specialties imposing different security requirements on a system, they must interface with DMSMS management practitioners to contribute to the formulation and implementation of resolutions to DMSMS issues.
- ***SCRM representative***. Because of the elevated importance and likelihood of supply chain disruptions in both production and sustainment, a SCRM point of contact separate from the SSE may be warranted. SCRM is the systematic process of proactively identifying supply chain vulnerabilities, threats, and potential disruptions throughout the supply chain and implementing mitigation strategies to ensure the security, integrity, and uninterrupted flow of materials, products, and services as threats are identified or disruptions occur.[78]
- ***Logistics representative.*** Typically, this is the PSM or Lead Life Cycle Logistician from the program office. Within the DMT, the logistics representative should provide data and technical advice for developing the resolution in conjunction with system engineering. The representative would also ensure that resolution data is appropriately updated in logistics systems (e.g., provisioning systems), technical manuals, and associated training.
- ***Sustainment and maintenance activity representative***. The party or parties ultimately responsible for long-term sustainment and maintenance must participate in the DMT to ensure that the appropriate groundwork is laid to meet the long-term objectives of the DMP.
- ***Software SME***. The DMT should include a software SME, but all DMT members should possess a certain set of software obsolescence competencies. Even for those systems that do not place a high priority on software obsolescence, software expertise is valuable because hardware resolutions have the potential to affect software. Robust DMSMS management requires effective communication among all stakeholders. They must be made aware of hardware and software interdependencies and the potential impact of alternative resolutions, including the status quo. Having software expertise on the DMT facilitates the communications necessary to ensure that all resolutions are properly implemented.
- ***Supply support activity representative***. The supply support activity representative is an ad-hoc team member who provides his or her organization's viewpoint on DMSMS issues. The team may have several supply support activity representatives, for example, item managers from DLA. DLA involvement, in particular, can augment a DMT's efforts with research on DLA-managed items, additional but limited DMSMS management expertise and product

[78] ASD(S), Supply Chain Risk Management Taxonomy Version 2.0, January 22, 2025, available at https://www.dau.edu/sites/default/files/webform/documents/30116/DoD%20SCRM%20Taxonomy%20Version%202.0%20250122%20RELEASABLE.pdf.

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knowledge, and a cross-system perspective that can highlight impending, otherwise unforeseen problems and potential resolutions.

- ***Value engineering (VE) SME***. A VE representative is an ad-hoc team member who can offer advice regarding the best-value resolutions to pursue to address DMSMS issues, as well as how those resolutions are synchronized with roadmaps to integrate new technologies that add capability or improve supportability.
- ***Contracting office representative***. The DMT member representing the contracting office is an ad-hoc team member who provides guidance and administrative requirements for support contracts and agreements. This person also helps the DMT ensure that there is no ambiguity in the contractor's DMSMS management requirements.
- ***Prime/subcontractor representative***. Assuming this representation is in the company's contract, the prime/subcontractor representative ensures that OEMs fulfill their roles and responsibilities with respect to DMSMS management as established by the statement of work (SOW). The prime contractor representative may serve as the DMSMS management lead for subcontractors and present DMSMS issues and risks to the DMT. The prime/subcontractor representative's role may change depending on the phase of the life cycle.
- ***FMS representative***. The FMS representative is the conduit for the flow of information in both directions, including communication with the security cooperation/FMS organization within the program office. If a DoD program office is (or will be) responsible for any DMSMS management activities associated with FMS, a program office FMS representative should attend DMT meetings, as it is usually impractical to have FMS countries directly involved.[79] FMS partners need to have early awareness and understanding of current or potential obsolescence issues to make a timely decision among resolution options and have a channel to convey their obsolescence concerns to the program office. In many instances, resolutions will be cost-shared, and the development of resolutions (especially life-of-need buys) may be cheaper if they are based on both DoD and partner demand for obsolete items.[80] Absent such timely communication, information sharing, and coordinated resolution of partner DMSMS issues, suboptimal or inefficient outcomes are more likely.

Following are examples of other roles that the DMT could include, depending on the program office's DMSMS management infrastructure and objectives:

- Business financial management office representative,
- Intellectual property SME,
- Software license management group representative,[81]
- Environmental and materials engineering representative, and
- CCB representative.

Some roles may be combined, and some of the responsibilities may be deleted or new ones may be added over time. The most effective DMT organization allows for open communication among all team members, whether they are representing the government, the prime contractor, or subcontractors. Such open communication is critical for robust DMSMS management. For example, the following should be documented in the DMP and monitored:

- DMT meeting frequency,

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[79] In addition to export control or security limitations on what can be discussed, speculative ideas may be misinterpreted as planned and approved.

[80] FMS partner needs should be factored into the DoD purchase whether the buy is by the program office or by the supply system.

[81] Unlike hardware, software often requires a license or agreement. Although maintaining software licenses and maintenance agreements are not normally a DMT responsibility, the DMT may want to take responsibility if software presents a critical obsolescence issue for the program office. If a license management group is doing this work, the DMT should maintain an open line of communication with that group to remain cognizant of the status of licenses.

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- DMT action items (i.e., assigned responsibilities and suspense dates),
- DMT deliverables,
- DMT interfaces with other program office entities (e.g., IPTs, Business Financial Management, Contracting), and
- DMT plans for interfacing and sharing information with other program offices and Services.

The PM is responsible for DMSMS management for the system and should assign the relative roles of each DMT member using a RACI[82] chart (responsible, accountable, consulted, and informed). This commonly used practice helps ensure alignment with the program office leadership's established foundations for DMSMS management. These roles are defined as follows:

- ***Responsible (R)***. Identifies the individual or individuals who are responsible at each level of the organization to execute a specific assignment for an activity. The degree of responsibility is determined by the person accountable. There can be multiple "Rs" for one activity at each organizational level.
- ***Accountable (A)***. Identifies the individual who is ultimately accountable for the completion of the activity and who can say "Yes" or "No." There can be one and only one "A" for a decision or activity at each organizational level.
- ***Consulted (C)***. Identifies the individual who must be consulted before a decision or activity is finalized. This represents two-way communication. There can be multiple "Cs" for one activity at each organizational level.
- ***Informed (I)***. Identifies the individual who must be notified about the completion or output of the decision or activity. This represents one-way communication. There can be multiple "Is" for one activity at each organizational level.
- ***Not Informed (N)***. Identifies individuals who do not need to be notified about the completion or output of the decision or activity. There can be multiple "Ns" for one activity at each organizational level.

Table 5 is a notional example (rows and columns are not complete, and entries are hypothetical) of such a RACI chart, which shows the types of responsibility required relative to a set of roles and DMSMS management activities.

Table 5. Notional RACI Chart

Activity	Roles							
	DMT Lead	Program Office Rep.	DMSMS SME	Engineering Activity Rep.	Logistics Rep.	Supply Support Rep.	Contracting Office Rep.	Prime Contractor Rep.
Meeting coordination	A	I	C	I	I	I	I	I
Funding requirements	R	A	C	C	C	N	I	C
Future budget projections	R	A	C	C	C	N	C	C
DMSMS monitoring	A	I	R	I	I	I	N	C
DMSMS solution implementation	C	A	C	R, C	R	C, I	I	R, C
Contracting	C	R	I	I	N	I	A	C
Supply support	I	I	C	C	I	A, R	I	C, I

[82] Sometimes RACI is replaced with accountable, responsible, consulted, and informed (ARCI) to place more emphasis on who is accountable than on who is responsible.

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3.2.2 DMT Training Needs

All members of the DMT should be trained on their role supporting DMSMS management for the program office. Not all members of the DMT are expected to be DMSMS SMEs or reach a targeted competency level, but the DMT lead should identify minimum training requirements for DMT members based on the DMSMS management approach, available resources, and the roles and responsibilities of each DMT member.[83]

DAU Functional Area leaders have established a credentialing framework aimed at providing job specific skills.[84] The DMSMS management associated credential is "Parts and Material Life Cycle Management" applies to DMSMS management. It includes the following DAU courses[85]:

- LOG 0640 DMSMS: What the Program Manager Needs To Do and Why
- LOG 0650 DMSMS Fundamentals
- LOG 0660 DMSMS Executive Overview
- LOG 0630 Introduction to Parts Management
- LOG 0670 DMSMS Research Essentials
- LOG 0320 Preventing Counterfeit Parts in DoD Supply Chains
- LOG 0380 Provisioning and Cataloging
- LOG 0390 Additive Manufacturing Overview
- LOG 0400 Additive Manufacturing Case Studies
- LOG 0470 Sustaining Engineering
- LOG 0510 System Retirement, Disposition, Reclamation, Demilitarization, and Disposal
- CON 0040 Market Research
- CLE 019 Modular Open Systems Approach
- CLE 026 Trade Studies.

[83] If the DMT membership changes, the new members should receive training on DMSMS management and on their roles and responsibilities within the DMT.

[84] See https://www.dau.edu/training/credentials.

[85] DAU course numbers are in the process of changing. The future course numbers of the credential program courses are provided where known.

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Table 6 outlines training recommended for the different DMT roles. The DMT lead may use this as a guide but tailor it as necessary to meet the specific needs and constraints of the program office. The important thing is that DMT members have the appropriate knowledge and skill base to carry out their responsibilities effectively.

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Table 6. Recommended DMT Training

Role	LOG 0640	LOG 0650	LOG 0660	LOG 0630	LOG 0670	LOG 0320	LOG 0380	LOG 0390	LOG 0400	LOG 0470	LOG 0510	CON 0040	CLE 019	CLE 026
DMT lead	X	X	X	X	X	X	X	X	X	X	X	X	X	X
Program office representative	X	X	X	X	X	X	X	X	X	X	X	X	X	X
DMSMS SME	X	X	X	X	X	X	X	X	X	X	X	X	X	X
Engineering activity representative	X	X		X			X	X	X	X		X	X	X
SSE	X	X		X		X	X	X	X				X	X
Logistics representative	X	X			X	X		X	X	X	X	X	X	X
Sustainment and maintenance activity representative	X	X			X	X		X	X	X	X	X		
Software SME	X	X												X
Supply support activity representative	X	X			X	X	X	X	X	X	X	X		
VE SME	X	X	X											
Contracting officer representative	X	X		X							X			
Prime/subcontractor representative (if established by contract)	X	X	X	X	X					X	X	X	X	X
FMS representative	X	X									X			
Business financial management office representative	X	X								X	X			
Software license management group representative	X	X									X	X		
Environmental and materials engineering representative	X	X		X			X	X	X	X		X	X	X
CCB representative	X	X		X										

DMSMS SMEs should have the following knowledge, skills, and abilities:

- Knowledge of:
  - Logistics management and/or systems engineering, as well as an understanding of DoD policies and procedures as applied to DMSMS management, design interface, maintenance planning, and the acquisition and sustainment of a system;
  - Technical aspects of logistics elements and systems engineering principles and their impacts upon each other;
  - DMSMS management concepts and policies sufficient to provide guidance and direction to logistics and engineering personnel on issues related to or affected by DMSMS issues and concerns at an in-depth level;
  - Developing DMSMS management requirements and projecting funding requirements for an effective DMT at an in-depth level;

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- DMSMS case-tracking system and DMSMS metrics at an in-depth level;
- BCA processes in the DMSMS decision process;
- Military and contractor supply chains, especially for commodities of focus;
- Concepts, theories, and principles of system design, operations, and support;
- Technology roadmapping and its relationship to the program office modification planning (including technology refreshment and insertion) processes along with their interfaces with DMSMS resolution planning;
- Part selection criteria and their impact on DMSMS resilience;
- Relationships among design interface, maintenance planning, engineering design, and DMSMS considerations necessary to create and establish innovative and effective program office policies and procedures for systems as required by DoD activities and authorized FMS organizations at the functional level;
- DMSMS management for the development of agency policy, procedures, and processes for mitigating DMSMS issues at the functional level; and
- Critical items requiring system security requirements that affect design and DMSMS decisions.

- Skills in:
  - Interacting with senior government and industry executives, as well as with other logisticians, engineers, and PMs, both individually and in groups;
  - Resolving conflict and negotiating solutions to complex technical issues;
  - Developing and evolving collaboration among commands and agencies to maximize the attainment of efficiencies to determine best practices and leverage existing processes;
  - Communicating with others to interpret contractual requirements for performance-based logistics (PBL) for DMSMS management support packages;
  - Communicating with others about preventing obsolescence of critical items;
  - Communicating clearly and concisely, both orally and in writing; and
  - Perceiving relationships and effects between the subject under discussion and related areas of importance and bringing those relationships to the attention of all concerned.
- Ability to:
  - Provide recommendations to program offices and field support teams to assist with planning and developing the DMP, SOWs, contract language, and ILAs;
  - Provide focused management and coordination across multiple stakeholders in support of DMSMS management;
  - Chair and facilitate a DMT by developing annual goals and agendas and to direct the personnel to meet the established goals;
  - Identify, prioritize, and recommend solutions to the barriers that prevent a PM from establishing a robust DMSMS management program;
  - Apply advanced concepts and theories to DMSMS issues and tasks so they may be resolved effectively and efficiently; and
  - Develop and establish DMSMS management processes and guidelines for all personnel to follow.

Because the DMSMS competency does not exist in a vacuum, additional courses are needed to obtain informal foundational-level, advanced-level, and leadership-level competencies and experience associated with the roles and responsibilities of DMSMS management practitioners. Appendix G contains a comprehensive list of DAU courses associated with such DMSMS competency levels and can be used as a basis for improving DMSMS management workforce proficiency. The courses identified at the

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beginning of this section for the Parts and Material Life Cycle Management credential are recommended minimum requirements for DMT members and are also included in the appendix.

3.3 Develop a DMP

A DMP documents the foundations of a DMSMS management approach established by program office leadership and identifies the risks associated with deviations from the standard DMSMS management processes described in this document. As such, the DMP establishes a robust DMSMS management framework for a program office. Without an adequate plan, a program office cannot have effective DMSMS management. However, the DMP should be based on factors that are known or anticipated, not overly optimistic assumptions. As such, the DMP should be adjusted as actual conditions change.[86]

Developing a DMP requires detailed consideration of how DMSMS management principles should be integrated within the program office's mission. The DMP also describes the DMT and its duties (for detailed information on the DMT, see Section 3.2) within a set of tailored DMSMS management processes ideally designed to avoid miscommunication. The tailoring is a function of each program office's specific infrastructure, record keeping procedures, quality management plans, resources, priorities, and constraints (e.g., number of people, amount of funding, access to BOMs/parts lists and associated technical data, and ability to conduct vendor surveys on item availability). All program offices are required to have a DMP per the DoDI 4245.15.

As the responsible agent for developing the foundations of DMSMS management, the PM is ultimately responsible for final approval of the DMP. The plan should contain an approval page with the program manager's signature as well as the signatures of key reviewers (e.g., lead system engineer, product support manager, and DMSMS subject matter expert). These approvals demonstrate the senior program office leadership's agreement with and support for the actions prescribed in the plan. This is especially important when the competition for resources within a program office is high. While ultimately approved by the PM, DMT members develop the DMP and the DMT has management authority to put the approved DMP into action. As the DMP develops, members of the DMT ideally can be expected to more clearly articulate their future roles within the tailored DMSMS management activities of their specific program office. The DMP authorizes DMT members to carry out their DMSMS management-related duties; overriding what could otherwise be conflicting duties.

Formulation of the DMP should begin early in the life cycle—preferably immediately after program initiation for any of the acquisition pathways. DMPs are not static; they should be living plans that reflect the program office's changing circumstances. Changes within the program office, weapon system activities, and other events—such as modifying the roles and responsibilities of DMT members, revisions to DMSMS management contracts, changes to tools and processes, a life cycle extension, changes in procurement plans, or significant changes in operating tempo—will usually drive revisions to the DMP.

There is no prescribed length. The DMP should include only how the program office specifically intends to accomplish DMSMS management; it should not be a tutorial on DMSMS management. References to other documents should be made as much as possible to avoid duplication.

The DMP captures the DMSMS management experience and knowledge of the DMT by addressing a set of interrelated questions. The answers to these questions affect the near-term objectives of the DMP and the actions of the DMT in executing the DMSMS management process and its specific tasks. Process capability is an important consideration in answering these questions. Appendix D contains information on DMSMS management process capability, numbers the key processes associated with the five DMSMS

[86] This document places the DMP development process early in the *Prepare* step and indicates that modifications should be made when needed. Some program offices may determine that the DMP should be developed later in or at the end of the *Prepare* step.

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management steps, and identifies four levels of capability maturity for each process. A higher capability level indicates a more robust approach to DMSMS management. The DMP and the capability matrix should be aligned.

One method to document the alignment of the DMP and capability matrix is to create a table containing the program capability level process that links to the corresponding section in the DMP. The table should depict the program office's current capability level as well as the target capability level established by the program office. The table provides program office personnel with a way to quickly assess areas that require improvement and identify when the target capability level is achieved. Each program office decides on a target capability level that is based on its specific needs. The program office may have different target capability levels for different DMSMS operational processes, depending on its needs. The discussions of the capability levels associated with answering the questions posed in the DMP should include:

- The current associated process capability levels reflected in the DMP;
- The rationale for determining the desired capability levels;
- The desired capability levels, taking DMSMS risk into account;
- Barriers to reaching the desired capability levels (if not already achieved); and
- A plan to reach the desired capability levels (or rationale for the status quo or only partially achieving the desired levels).

Maintaining the status quo or partial achievement of the desired capability level may be acceptable based on the risk identified and the current programmatic posture; however, an increased risk, or lowered risk, warrants an updated review be conducted to determine if additional measures need to be employed to reduce potential DMSMS issues in the program office. The DMP should document the need to conduct periodic reviews, the frequency of the reviews, and a planned approach for evaluating all methods used to identify the desired process capability.

A second method to align the DMP with the capability matrix is to include the same information described for the first method as its own section of the DMP.

With the sole exception of a program office option to discuss capability levels throughout the DMP or to include capability information in a separate section, program offices are expected to use the following outline and format for the DMP unless they have a good reason to deviate from them. The subsections present a set of tailorable questions that should be considered and refer to the applicable rows of the capability matrix. The subsections have been mapped to the expected format where capability levels are presented in a single section.

Appendix E contains a DMP checklist and example plan that provide a starting point for creating and maintaining a DMP. The checklist helps those developing the plan and those tasked to review the plan avoid potential gaps that create additional DMSMS risk. In addition to the checklist and example plan in Appendix E, the Systems Planning and Requirements Software, a multi-service expert system, can guide a program office through the development of a DMP.

3.3.1 Purpose

- What are the system/program office's near-term and long-term DMSMS management objectives? (See Table 26: 3.1 in Appendix D.)

  At the most basic level, the DMSMS management objective is to ensure DMSMS-related cost, schedule, readiness, and availability impacts do not exceed an acceptable level. The specific definition of "acceptable" is a function of the size, complexity, and cost of the system, as well as the current life cycle stage, the AS, the LCSP, and the modification plans. Other

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examples of possible objectives include “no reactive DMSMS cases except where planned,” “avoid out-of-cycle redesigns,” and “keep annual resolution costs below a specific number.” The DMP should reference the key documents requiring the plan.

3.3.2 Scope and Applicability

- To what program office/systems does the plan apply? (See Table 26: 3.1 in Appendix D.)

  The program office to which the DMP applies should be described. Areas which should be addressed include acquisition designation and whether there is FMS. It should also be indicated whether this is an initial DMP or a revision.

  The system or subsystems covered by the DMP should be identified and described; the system descriptions should be brief and not a detailed description or concepts of operation, which are already covered in other system documentation (e.g., SEP, LCSP). The system description should include the quantity of systems planned and the timeframe over which they are intended to be supported.

  The determination of what is to be covered will ultimately require BOMs and other technical data for surveillance and analysis of those items which will be proactively managed. Technical data may also prove necessary to verify and validate DMSMS resolutions.

- For each applicable system, what is the AS, where is the system in its life cycle and what are the sustainment strategy, modification plans, and maintenance approach? (See Table 26: 3.1 in Appendix D.)

  The AS and life cycle phase of the applicable system can influence the types of DMSMS management operations activities taking place and the relative roles and responsibilities of DMSMS management stakeholders in performing those activities. The current AS and life cycle phase, as well as any major milestones, of the system should be indicated.

  The long-term sustainment strategy, modification plans, and maintenance approach affect the DMP objectives as well as the composition of the DMT and the availability of technical data. No simplifying assumptions (that count on uncertainties being treated in a specific way) should be made regarding long-term sustainment, modification, and maintenance responsibilities, since this will impact the types of activities that the program office needs to take now to ensure long-term sustainment and maintenance of the system. For example, if the DMP assumes that a contractor will provide sustainment support for the life of the system, the government might not have the appropriate DMSMS data to meet system sustainment requirements if in fact the long-term sustainment approach is for organic support.

3.3.3 DMSMS Management Approach

- For each applicable system, what will be the primary DMSMS management roles of contractors, program office personnel, and independent SMEs and how will the program office maintain a life cycle perspective for its DMSMS management approach?[87] (See Table 26: 3.1 and 3.2 in Appendix D.)

  Regardless of the relative roles of the government, the prime contractor, and independent SMEs in DMSMS management, the government is ultimately the responsible party. The government and its personnel should ensure that a life cycle perspective can be maintained. However, if a prime contractor or independent SMEs will be performing some aspects of DMSMS management, the program office should recognize that those entities’ responsibilities span the period of performance only. The DMP should include provisions that

[87] SMEs may be organizations or individuals outside of both the program office and the prime/OEMs that provide DMSMS management services such as obtaining or building BOMs, monitoring items for DMSMS issues, receiving and processing DMSMS notifications, creating and managing DMSMS cases, researching DMSMS issues and recommending resolutions, participating in or facilitating DMTs, assisting in the development of DMPs, and providing oversight of DMSMS-related activities.

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are in place for the program office to be able to have the information necessary to maintain a life cycle perspective.

As the system moves through the life cycle, the DMP should capture the program office's shifting focus from providing government oversight of contractor DMSMS management processes and delivery of management products to the government for acceptance to conducting organic DMSMS assessments and sustainment planning. For example, in the development function, the government should ensure that the contractor is minimizing obsolescence throughout the contract period of performance by selecting items that avoid or resolve hardware, material, and software obsolescence issues. During production and initial fielding, the government should ensure the contractor can meet the schedule and the government will be able to sustain the product over the long term. During O&S, the government may want contractor support for DMSMS management and to ensure that the system can be sustained until the next upgrade or replacement.

For many program offices, a prime contractor (and perhaps some OEMs) will have responsibility for performing day-to-day DMSMS management, particularly early in the system's life cycle. The program office should not try to duplicate prime contractor activities; the DMP should be aligned with what the contractor is doing based on its own internal DMP. The program office should not make assumptions about what the prime contractor is or is not doing. The facts can be obtained only from a careful examination of contract language, actual contractor processes, and data deliverables. When planned contractor responsibilities were not included on contract, the specific mitigations concerning the government assumption/ transfer of those responsibilities should be included in the plan.

If the prime contractor is effectively managing DMSMS risk and similar requirements are being flowed down the supply chain, the program office's role should be focused on oversight. The program office can also contract with independent SMEs to enable the government's oversight and other assigned roles.

- Where should the program office be reactive and where should the program office be proactive? (See Table 26: 3.1 in Appendix D.)

  While nearly everything will become obsolete or unavailable eventually, a robust DMSMS management approach does not imply being proactive everywhere. Program office leadership should set its expectations regarding the scope of proactive DMSMS management and the desired frequency for monitoring and surveillance activities.

  There are at least three types of information that characterize DMSMS risk related to proactivity and reactivity: 1) the overall percentage of the system and/or its subsystems that will be monitored; 2) criteria for those items viewed as high risk from a DMSMS perspective and require monitoring; and 3) the subsystems and commodity classes of items (e.g., software, material, MaSME, COTS, electronics, and support equipment) that the program office has determined it will not proactively monitor should be identified and an explanation of the associated risk provided. Section 4.1 and Section 4.3.1.2 provide additional detail on the prioritization of subsystems and items for proactive monitoring and surveillance.
- What mechanisms (i.e., product discontinuance notices [PDNs], predictive tools, vendor surveys, technology roadmaps, and DMSMS management information systems) will the program office use for monitoring, assessing, analyzing, and performing case management?[88] (See Table 26: 3.9 in Appendix D.)

  The sources from which the program office intends to receive PDNs (see Section 4.4.4) and the program office's planned process for managing those notices should be identified. The program should further note who is responsible for processing PDNs and its expectations

[88] A PDN is one type of a product change notice (PCN). PCNs also indicate issues that may affect the ability to use a part. For example, the manufacturing site could change, triggering the need for requalification.

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regarding how often and when PDNs are expected to be received (e.g., as soon as discontinuation intent is known, weekly, monthly, and so forth).

The DMP should identify all tools (see Section 3.4 and Section 4.2) stakeholders will use for DMSMS monitoring, assessment, analysis, and case management. For each tool, the organization to employ the tool should be identified, along with when the tool will be placed into service to perform the intended function and who will have access to the tool. It is particularly important that the government have at least visibility of the tools being used by a contractor providing DMSMS management services to the program office. Otherwise, the government will need to employ its own tools, which should also be documented.

Because not all items are conducive to monitoring via predictive tools, the DMP should document whether vendor surveys (see Section 4.4.2) or technology and product roadmaps will be used to monitor other types of items, such as software, MaSME items, and COTS assemblies.

- What contingency plans are in place for potential programmatic changes? (See Table 26: 3.2 in Appendix D.)

  A program office can and often does face any number of programmatic risks that can derail its plans. For example, a program office's funding by appropriation may vary. Its plan for contractor support for the life of the system may be determined to be less desirable and thus cause a shift to an organic method of support. A system may have its planned service life extended. Any of these (and other) programmatic events can impact DMSMS management. The DMP should identify any anticipated programmatic risks and describe whether plans have been developed to mitigate their DMSMS impact.

3.3.4 DMT

The DMT members conduct the core DMSMS operations for the program office. Members of the team typically represent core program teams and activities. The DMP should answer the following questions, discussed in Section 3.2, about the program office's DMT:

- Who are the stakeholders for the robust management of DMSMS issues for the program office (including other DMSMS management program offices that interact with the DMSMS management program office for the system in question)? (See Table 26: 3.3 in Appendix D.)
- What are the roles of the DMT members and who will fulfill those roles? (See Table 26: 3.3 in Appendix D.)
- What communication is required of the DMT internally and externally? (See Table 26: 3.3 in Appendix D.)

3.3.5 DMSMS Management Operations

- How will the program office's DMSMS management efforts be integrated with other strategies, planning, and reviews (e.g., product support strategies, the systems engineering design and review process, and modification planning)? (See Table 26: 3.4 and 3.6 in Appendix D.)

  There are numerous activities that take place throughout the life cycle of a system that could benefit from a DMSMS management perspective, while at times also providing inputs or opportunities for leverage that can improve DMSMS management. Example activities include—

  - Ensuring that the DMSMS community is establishing interfaces to advocate for DMSMS-resilient designs (see Section 3.4.2);
  - Assessing designs for DMSMS risk (see Section 4.5);

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- Providing the results of health assessments to inform and integrate DMSMS resolution funding with funding for program office modification plans (see Section 5.3 and Section 7.2);
- Participating in design reviews and the engineering change proposal (ECP) approval process; and
- Performing SCRM to identify, assess, and address potential obsolescence risks.

The DMP should identify in a few sentences these activities and how they will be integrated to the benefit of DMSMS management and the program office overall. The processes that will be used to ensure that the program office strategists and planners reach out to the DMT should also be included.

- How have DMSMS management processes been tailored for this plan? (See Table 26: 3.7 in Appendix D.)

  Although many general DMSMS management processes are transferrable from one program office to another at a generic level, the DMP should document processes that are tailored to meet the program office's specific needs at the working level. The DMP should consider the unique needs of the program office, the unique needs of each stakeholder, and the unique flow of communication required among the stakeholders to ensure that the process enables fulfillment of the DMP objectives. Figure 5 in Section 3.4 summarizes the processes. The DMP should only document areas where its processes differ from the standards.

  If a program office is unable to obtain the necessary data, the impact of not having this data should be documented and the DMP revised to reflect the absence of data, as appropriate. It should also be highlighted when a BOM will be constructed when it was not provided on contract (see Appendix J).

- How will the DMT provide oversight of contractor DMSMS management efforts? (See Table 26: 3.8 in Appendix D.)

  The DMT should establish the types of reporting and the level of involvement in the DMT by the contractor (IAW its contract) to ensure that the program office is informed of the contractor's performance in DMSMS management. These things are dependent on the degree of the contractor's responsibility for DMSMS management. If there is minimal responsibility, the reporting may only consist of DMSMS notifications and reports on actions accomplished. Alternatively, if there is significant responsibility, full case reporting, DMSMS health reports, budgetary information, and the like may be needed. Participation in the DMT should be similarly scaled. If the contractor is fully managing DMSMS, it should be very involved in the DMT.

- How will DMSMS issues be detected, tracked, and resolved? (See Table 26: 3.9 in Appendix D.)

  The DMP should briefly lay out how DMSMS operations will be conducted in the following areas. In many situations, some if not most of this work may be done by the contractor (IAW its contract). Even less detail would be needed in the government's plan when that is the case.

  - How BOMs and other programmatic data will be obtained. This should include the data sources (e.g., BOMs obtained on contract) and other inputs necessary to implement the DMSMS management processes, as well as how the necessary data will be obtained and used by DMSMS management processes to produce the DMSMS management products.[89]
  - The priority for loading subsystem BOMs into predictive tools. This describes the program office's risk-based methodology for prioritizing which subsystem BOMs and the items on

[89] It is not a good practice to aggressively pursue technical data and IP rights only when there is an issue. Such data should be obtained when it is available.

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those BOMs (see Sections 4.1 and 4.3.2.1) to load first into predictive tools. This should also include when the prioritized BOMs will be loaded and the frequency with which or events that could trigger the need to update.

- How item monitoring will be accomplished and at what frequency. Because not all items are conducive to monitoring via predictive tools, the DMP should document how vendor surveys (see Section 4.4.2) will be used to monitor other types of items, such as software, MaSME items, and COTS assemblies. Documentation should include what organizations will be performing vendor surveys and the frequency at which those surveys will be conducted.
- How DMSMS issue notifications will be received and processed. The item monitoring process will normally generate DMSMS notifications and there may be other paths for the program office to obtain these notifications. This area includes how the program office expects to receive notifications, how frequently it expects to receive them, and how it will determine their validity.
- How DMSMS cases will be initiated, managed, and maintained. This encompasses: the processes to assess the readiness, availability, cost, and schedule risks a DMSMS issue poses to the system (see Appendix I.3); the circumstances that require a case be established based on a DMSMS issue notification (see Section 3.4.5); how cases will be prioritized (see Section 5.2 and Section 5.3.2); how cases will be analyzed to determine the most cost-effective resolution, including actions at higher levels of assembly (see Sections 5.3.3 and Section 6); and how the DMT will oversee implementation of resolutions to ensure that all stakeholders carry out their assigned roles and responsibilities (see Section 7).

- How will DMSMS management processes, issues, and resolution options be communicated to foreign partners (when applicable) and DoD stakeholders external to the program office?

  The DMP should briefly summarize communication channels and frequency associated with DMSMS management for FMS equipment. Key points include:

  - What DMSMS activities will be performed by the DoD program office for FMS equipment (see Section 3.5.1.1);
  - When and how foreign partners will be notified (see Section 3.5.2);
  - How the involvement of other DoD program offices (e.g., for any GFP) will be managed (see Section 3.5.3);
  - How costs will be shared (see Section 3.5.1.2).

3.3.6 Funding

Resource needs (both DMSMS management operations and DMSMS resolution implementation) should be determined without fiscal constraints, taking into account the foundations for DMSMS management. If resource availability does not match the needs, the strategy should be modified and the associated risks should be documented in the DMP.

- What resources have been programmed and budgeted for DMSMS management operations? (See Table 26: 7.1 in Appendix D.)

  A program office should program and budget for the resources necessary to support DMSMS management operations as planned to be performed by a contractor, independent SME, and the program office itself (e.g., it's DMT). The costs associated with DMSMS management operations throughout the Program Objective Memorandum (POM) period should be documented in the DMP.

  Initial planning should not be resource constrained but rather should seek to address the foundations for DMSMS management established by the program office leadership. However, those initial DMP objectives, DMSMS management processes and products, and, to some extent, the DMT composition itself may need to be constrained if the program office's

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leadership cannot obtain sufficient funding (in-house and for contractors) to support the realization of that initial plan. In such an instance, program office leadership should explicitly acknowledge that it is accepting greater risk and appropriately revise the foundations for DMSMS management, before the final approval of the DMP.

- What resources have been programmed and budgeted for resolutions for known and anticipated DMSMS issues? (See Table 26: 7.1 in Appendix D.)

  To be able to pay for the implementation of resolutions to DMSMS issues, the program office must program and budget for the resources to fund those resolutions. The costs associated with the DMSMS resolution implementation throughout the POM period should be documented in the DMP as well as the availability of resources to meet this demand. Risks and mitigations associated with a mismatch between funding available and funding needs should be included in the DMP.

3.3.7 Contract Requirements

- What DMSMS management operations and/or resolution implementation requirements will be imposed on the contractor and/or independent SMEs? (See Table 26: 3.10 in Appendix D.)

  Specific DMSMS-related Contract Data Requirements Lists (CDRLs) and data item descriptions (DIDs) should be cited in this section of the DMP. In addition, specific SOW language can be cited to summarize the requirements imposed. These summaries should include performance assessment factors, incentives, and penalties. Examples of DMSMS management–related contract requirements can be found in Appendix F and in Standardization-related Document 26 (SD-26).[90]

  Since the prime contractor's work is based on contracts of limited time duration, the prime contractor's period of responsibility corresponds only to the period of performance of the contract. The program office should develop a provision through its DMP (supported by budgets and contracts, as necessary) that enables the government to manage DMSMS issues and their resolution across breaks in contracts (e.g., between production contracts, between a production contract and a sustainment contract, and between sustainment contracts) in a way that coordinates with expected contributions from the involved contractors. The ability to manage across contracts will require the ability to estimate the true LON (e.g., when the system retires and when the next technology insertion is scheduled, where technology insertion is a type of technology-related change that inserts or integrates technologies to improve the performance of the system).[91] The ability to procure and maintain the stocks of an item or assembly to satisfy through the LON is crucial. This could require the inclusion, in contract language, of a requirement for true LON (not life-of-contract) buys; provisions for handling LON stocks as government-furnished property; or clear, approved plans for alternate resolutions such as system modifications.

  Furthermore, the DMP must include exit strategies established by contract exit clauses to ensure both a smooth transition of responsibilities and the availability of technical data throughout the system's life cycle, not just until the end of the contract.

- Are there any desired DMSMS management requirements not incorporated into the final contract with the prime or OEMs? (See Table 26: 3.10 in Appendix D.)

  The final contract may not always include all the requirements that the DMSMS community sought; some requirements may be dropped during contract negotiations. The DMP should identify the requirements not included and the associated risk. In addition, the DMP should

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[90] Defense Standardization Program Office (DSPO), SD-26, *DMSMS and Parts Management Contracting Guide*, June 2023, available at https://quicksearch.dla.mil/qsdocdetails.aspx?ident_number=283456. Appendix F should be reviewed before reading the SD-26.

[91] Jay Mandelbaum and Christina Patterson, "Be Strategic!—Leverage Technology Insertion and Refreshment on DMSMS Issues," *Defense Acquisition Magazine*, DAU, July–August 2021, p. 38.

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address that risk to the extent possible. The foundations of DMSMS management should be changed accordingly.

3.3.8 *Metrics, Reporting, and Quality*

- What data elements will the program office collect? (See Table 26: 4.2, 4.3, and 5.1Table 25 in Appendix D.)

  There are two levels (see Section 3.4.3.1) of record keeping data elements, all program offices should collect the Level 1 data elements. The DMP should document those data elements it intends to collect. Such data elements focus on cost and operations efficiency. Consideration should also be given to the collection of compliance metrics which measure whether processes are being followed and whether they are working as intended.

  In addition, the DMP should document where the data will be maintained and how the data will be collected. This latter consideration is especially important because there are occasions where the DMT does not automatically receive feedback on whether a resolution was implemented or what resolution was implemented instead.

- How will the data elements be used? (See Table 26: 4.4 and 5.1Table 25 in Appendix D.)

  The DMP should document how the DMT will—

  - Use Level 1 data elements (or variations thereof) to develop metrics to improve DMSMS management operations, reduce DMSMS-related costs, prevent DMSMS issues from impacting readiness or schedule, and be in a strong position to explain and prove the benefits of the program office's DMSMS management efforts.
  - Use Level 2 data elements to (or variations thereof) to establish metrics that will enhance these benefits by enabling more complex analyses.
  - Use compliance metrics in the context of a QMS (see Section 3.4.4.2).

  The DMP should also indicate the periodicity at which the metrics will be reported, and to whom they will be reported. The DMP should also capture the sources that will be used to collect the data elements.

- What types of deliverables will the program office expect as outputs of its DMSMS management implementation approach?

  The DMP should identify the types of and schedules for deliverables needed to perform its oversight role and to be able to inform program office leadership decision-making. Such deliverables should include how sustainability analysis results will be provided. Other potential deliverables include notices of obsolescence, status of resolutions, and elements of supportability roadmaps that inform technology refreshments.

  Because DMSMS-related risk is not static and will fluctuate throughout the life of the system, the DMP should document a mechanism for reporting DMSMS issues and associated risk and as appropriate highlighting risk to program office leadership. For example, a top 10 risk ledger that is updated regularly or a risk management data base could be used. However, this reporting may be separate from the program office's formal risk management process.

3.3.9 *Process Capability*

- Where has the program office not achieved its desired process capability?

  The DMP should identify the desired process capability level required for the system and document any areas where the program office was unable to achieve the targeted level. The DMP should also record any specific DMSMS risks occurring as a result of not achieving the desired process capability, the specific area or areas of the system that are affected, and a plan to reach the desired capability levels. Finally, the DMP should include the rationale for not achieving the desired capability level in situations where it retained the status quo or only partially achieved the desired capability levels.

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- What are the process improvement plans?

  Knowing that the DMSMS-related risk is not static, the DMP should identify methods the DMT can employ to evaluate the process capability level and make improvements based on changes in the DMSMS-related risk to the system. Maintaining a status quo or partial achievement of the desired capability level may be acceptable based on the risk identified and the current programmatic posture; however, an increased risk, or lowered risk, warrants an updated review be conducted to determine if additional measures need to be employed to reduce potential DMSMS issues in the system. The DMP should document the need to conduct periodic reviews, the frequency of the reviews, and a planned approach for evaluating the methods used to identify the desired process capability.

3.4 Establish DMSMS Operational Processes

A process is any activity or set of activities that uses resources to transform inputs into outputs. Processes have objectives, inputs, outputs, activities, constraints, and resources. Three operational DMSMS management processes have already been discussed in this section: 1) establish the foundations for DMSMS management, 2) form the DMT, and 3) develop the DMP. As the DMT develops the DMSMS operational processes, the team must define the basic jobs needed to support the program office or other customers.[92] The team must then define and understand the inputs, outputs, activities, resources, constraints, and schedule. In fact, once the operational processes have been developed, key DMSMS management events should be included in not only the DMP, but all pertinent program office documentation including the integrated master plan (IMP), the integrated master schedule (IMS), the LCSP, Technology Development Strategy, Product Support Plan, PPP, and the AS.[93]

Tools are involved throughout all DMSMS operations to collect, aggregate, store, and report data, as needed, to produce DMSMS management products. It is not necessary (and not recommended) for the program office to develop or purchase its own tools. Program offices need to determine the right tool mix but can leverage existing DMSMS management systems and their associated service offerings to implement a new DMSMS management program.[94] These systems include management databases that trained DMSMS management practitioners use to integrate, analyze, forecast, and report on data collected from predictive tools, vendor surveys, critical materials analysis, and PDNs received from manufacturers (see Section 4.4.1 through Section 4.4.4, as well as logistics tools (see Section 5.1) and in-house case management tools (see Section 3.4.5).

DMSMS management processes can be categorized in many ways. Figure 5 shows the scheme used in this document. The DMT establishes and carries out all the operational processes shown in the figure, but each process is associated primarily with one of the major DMSMS management process steps. The tan shaded boxes indicate the strategic processes distributed across the five steps.

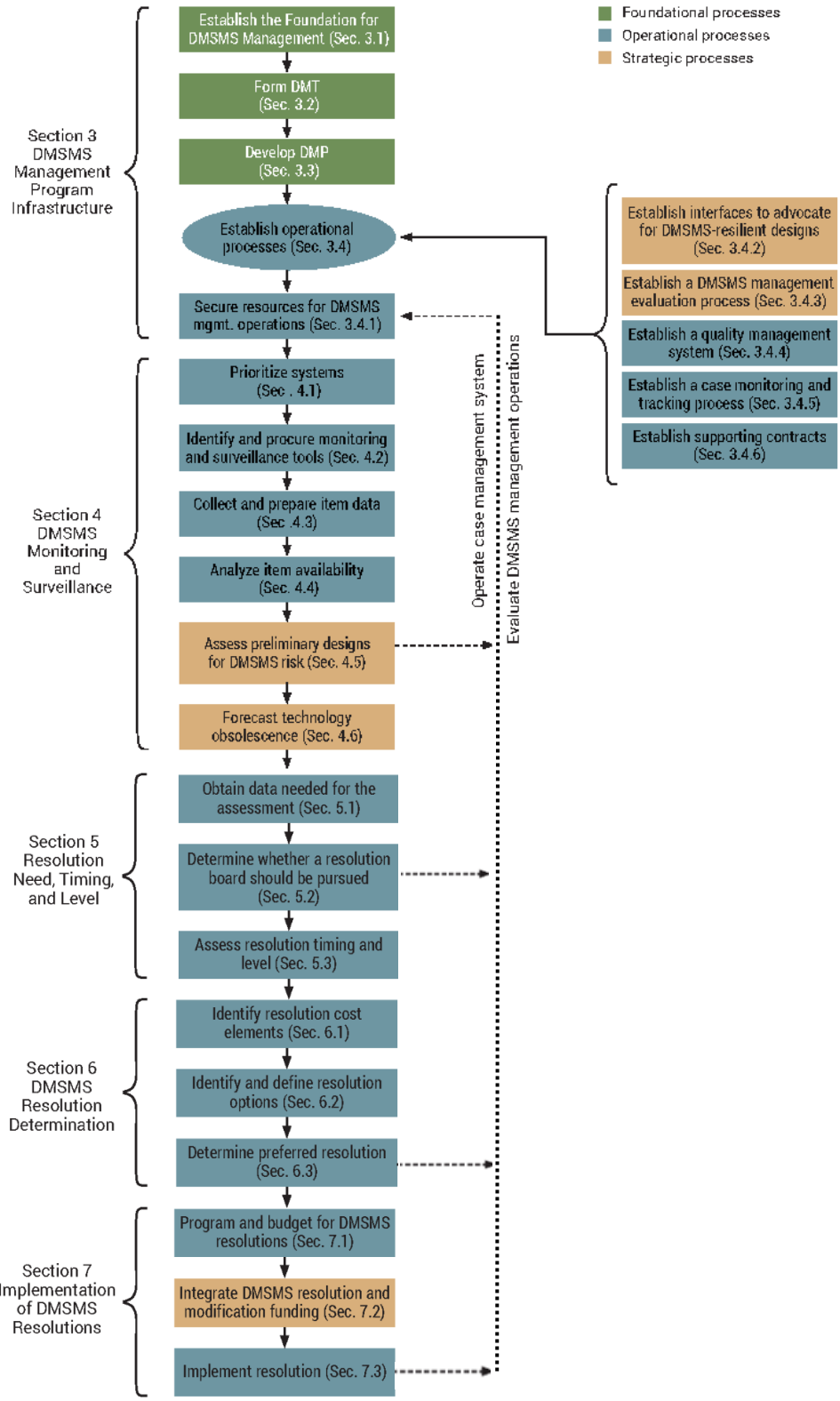
[92] As footnoted earlier, contractors may be the ones performing these processes and the DMT will be responsible for oversight. In these situations, contractor collaboration in the development of the processes is necessary.

[93] The PPP identifies critical program information (CPI) and critical components that may support DMSMS planning and prioritization.

[94] When determining the appropriate tool mix, the DMT should consider the tools already being used by the contractors.

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Figure 5. DMSMS Management Processes



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The following sections describe only those operational processes associated with the "*Prepare*: DMSMS management program infrastructure" step of the DMSMS management process, while the remaining operational processes are addressed in the remaining four sections of the main body of this document.

3.4.1 Secure Resources for DMSMS Management Operations

Section 3.2.1 discussed the roles and responsibilities of the DMT. It described how program offices use prime contractors, OEMs, and independent SMEs to perform the bulk of the work needed for DMSMS management operations. All other DMSMS management activities are performed as needed by program office personnel. Operations funding is required to support the day-to-day functioning of a program office's DMT, separate from the funding required for specific resolutions to identified DMSMS issues.[95]

This section discusses the determination of DMSMS management operations funding needs associated only with primes/OEMs and independent SMEs whose services are secured by contract. Funding requirements for program office staff are developed through non-DMSMS-related mechanisms.

Resources for DMSMS management operations contracts are normally obtained through the Component's programming and budgeting process. The specific procedures to follow are Component dependent and may not be the same across all program offices even within a single Component. Regardless of Component, the inclusion of DMSMS-related resource requirements for both operations and resolutions in the budgets of other activities, such as PM&P management, reliability and maintainability, or supportability is often a successful approach.

The next section lists DMSMS management operations funding drivers. Appendix H discusses best practices for programming and budgeting for DMSMS management operations.

3.4.1.1 DMSMS Management Operations Funding Drivers

The drivers of program office funding requirements for DMSMS management operations (regardless of who performs those tasks) can generally be classified into three categories. The first category primarily consists of one-time tasks focused around DMSMS management program initiation. The second includes both startup and recurring efforts associated with data collection and management, research, and forecasting where, for example, the startup tasks are repeated as more items are phased into active monitoring status. The third category mainly consists of recurring activities pertaining to data analysis and oversight.

- ***DMSMS management program initiation***. This category includes the following one-time activities or tasks that drive the need for resources:
  - Developing the DMP, including
    - Defining, documenting, and establishing DMSMS management processes;
    - Defining the required management products and articulating the required formats; and
    - Defining metrics for DMSMS management efficiency and effectiveness;
  - Establishing DMSMS quality management processes; and
  - Obtaining DMSMS analysis tools and case management databases and tracking systems.
- ***Data collection and management, research, and forecasting***. Resources needed for this category of startup and recurring activities are dependent on the following tasks:

[95] Section 7 discusses funding for implementing DMSMS resolutions.

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- Obtaining parts lists/BOMs or creating parts lists/BOMs (in instances where a parts list/BOM cannot be obtained);
- Formatting, cleaning up, and loading parts lists/BOMs;
- Analyzing BOMs to identify obsolete or near obsolete items;
- Researching obsolete or near obsolete items to verify their status and potentially identify alternatives;
- Surveying vendors to identify obsolete or near obsolete software, material, and mechanical and COTS items as a function of risk;
- Processing PDNs;
- Opening and documenting DMSMS cases; and
- Developing, analyzing, reviewing, and approving resolutions.

- ***Data analysis and oversight***. The following are illustrative recurring activities that consume resources:
  - Developing contract language;
  - Preparing DMSMS budgets;
  - Using technology roadmaps to forecast DMSMS issues and participating in discussions about them;
  - Facilitating and attending meetings, including travel and other logistics, as needed;
  - Conducting health assessments;
  - Preparing reports (both formal and informal) on the health of the system;
  - Reviewing metrics to identify where DMSMS management improvements are needed;
  - Reviewing designs to analyze DMSMS resilience;
  - Monitoring processes and quality management; and
  - Overseeing implementation of the resolutions.

Following are some additional drivers of the activities associated with the second and third categories.

- ***Monitoring and surveillance scope and level of effort***. Monitoring and surveillance are recurring tasks for the DMSMS management of any system. If the program office has adopted a risk-based approach to monitoring and surveillance, then decisions will have been made regarding which subsystems to monitor, which of their items and assemblies to monitor proactively, and where forecasts for technology obsolescence should be made. A program office might also consider the quality of the data available in the BOMs as a factor in determining the level of effort required to enable monitoring and surveillance, particularly in the beginning phases of a DMSMS management program. The scope and level of effort might fluctuate based upon the maturity of the DMSMS management program.

  Examples of startup tasks are obtaining or creating, formatting, and loading parts lists/BOMs and carrying out any additional research and analysis required to ensure the quality of the data for each subsystem being monitored. The scope of a DMSMS management startup effort is also broader, addressing the entire parts list/BOM, all vendors, and all parts, rather than only changes to or periodic updates to a subset of these. Because of those factors, the funding required for startup will be greater than that to maintain a steady-state DMSMS management effort. Table 7 compares the startup and steady-state efforts required for data cleanup, vendor surveys, and item research. Beyond the initial startup spike as new subsystems are phased in, the efforts associated with DMSMS monitoring and surveillance operations tend to be repetitive.

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Table 7. Comparison of Startup and Steady-State Effort for a Subsystem

DMSMS Monitoring and Surveillance Activity	Startup	Steady State
Data cleanup	Entire BOM	Only changes to the BOM
Vendor surveys	All vendors	Only on a time-phased, periodic schedule
Item research	All items	Only when certain conditions are met (e.g., item status changes, item has not been researched in a certain period, changes in the sources of an alternate item, packaging changes, or revisit of a previous "no action required" item)

More information on monitoring and surveillance activities is provided in Section 4.

- ***Assessment and analysis level of effort***. Assessment and analysis are recurring tasks for the DMSMS management of any system. The magnitude of the requirement is contingent upon the DMP and the number of items that the DMT chooses to assess. Also relevant is the level of detail required for periodic health assessments of the system from a DMSMS perspective (see Section 5.3), reporting requirements, meeting attendance, and so forth.
- ***Overseeing implementation***. The DMT's role does not end when a PM decides which resolution option to pursue. The final step of the DMSMS management process is implementation. In the *Implement* step, the DMT should be involved in two final processes: securing a source of funding for implementing the preferred resolution and ensuring that the actions required to implement the preferred resolution are taken. These are recurring tasks for all resolutions.

3.4.1.2 Supporting Best Practices

There are two best practices that support the performance of these responsibilities. The first best practice is to begin risk-based, proactive DMSMS management operations early in the life cycle and continue those activities throughout the entire life cycle.

Appendix B contains recommended DMSMS management questions for SETRs. According to that appendix, some aspects of DMSMS management operations should begin at the SRR. By the PDR, a forecasting/management tool and/or the results of manual research should be used to identify immediate and near-term obsolescence issues associated with the version of the BOM available at that point in the life cycle.

There are two implications of this from the perspective of programming and budgeting for DMSMS management operations. First, the programming and budgeting estimates should assume that DMSMS management operations begin early in the life cycle. Second is that these DMSMS management operations should be risk-based.[96] When identifying issues, priority should be given to the areas where DMSMS is likely to have the greatest impact. When low risk items are excluded, DMSMS management operations costs can be reduced, but there may be tradeoffs because being reactive can have a negative effect as discussed previously.

[96] See Sections 4.1 and 4.3.2.1. For many items, it is sufficient to be reactive because resolutions are readily available in a short period of time.

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Being proactive early implies that there is contract language for the prime (and its suppliers) to perform proactive, risk-based DMSMS management operations and as a result report potential DMSMS issues to the program office in a timely manner.[97]

The second best practice is to ensure that the personnel responsible for the program office's IP strategy programs and budgets for obtaining rights to the technical data needed to effectively perform DMSMS management operations and develop DMSMS resolutions.

While the costs of obtaining technical data are not part of DMSMS programming and budgeting, obtaining the data necessary to perform DMSMS management operations is a DMSMS programming and budgeting enabling best practice. The first key technical data requirement is indentured BOMs (with the original manufacturer's part numbers[98] at the component level) necessary for DMSMS monitoring. Before BOMs are finalized, parts lists should be required. The second important requirement is a technical data index along with an option to obtain (through a deferred ordering clause) specific data when needed to research DMSMS issues. This requirement could be used to ensure that all items are included on a BOM because it is a common problem for incomplete BOMs to be delivered. This type of index is like a drawing tree; however, a drawing tree is not always created.

A third related best practice is retaining technical data on older systems. Until a system is disposed of, there is always a chance that its service life will be extended, and the data will be needed again. Without the data, it will be necessary to rebuild from scratch.

3.4.2 Establish Interfaces to Advocate for DMSMS-Resilient Designs

Design is a complex task that must balance many performance, support, safety, environmental, security, and regulatory requirements and constraints. Because it is very difficult to balance all these things in a cost-effective way, the systems engineering process guides design tradeoffs, including the consideration of DMSMS management–related interests and concerns, to develop a design that takes all stakeholder considerations into account.

DMSMS management is one of numerous priorities and requirements competing for the attention of program office leadership. Part of the DMSMS management community's role (whether working inside the government or working in commercial industry in support of the development of a government system) is to educate program office leadership on the importance of identifying and addressing obsolescence issues as early in a system's life cycle as possible, often by addressing downstream implications. The DMSMS management community should not be thought of as a single-issue community on its own. Instead, DMSMS management should be approached as an integral part of reliability, maintainability, availability, and supportability. Indeed, the DMSMS management community might be able to build momentum and weight behind its recommendations by leveraging the interaction that other communities have with the PMs and chief engineers. Ultimately, the bottom line is that even if the DMT is unable to influence the design in a manner that eliminates "designing in" an item that is or has the potential to soon be obsolete, it will have at least identified DMSMS risks to continue to monitor for potential future mitigating actions.

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[97] The program office should be made aware of all DMSMS issues identified by a prime contractor that has been contracted to perform DMSMS monitoring, even when that same contractor may also be responsible for resolving certain types of classes of DMSMS issues. This visibility enables the program office to maintain oversight over the prime contractor's DMSMS management operations activities, as well as to be able to plan for any actions and/or resolutions that will be necessary to address a given DMSMS issue beyond the prime contractor's period of performance.

[98] This document uses the terms *part number* and *item number* interchangeably. For government record keeping and contractor reporting however, it is important to distinguish between an original component manufacturer (OCM) item number and an OEM item number. These terms are defined in Table 53. In this sentence, reference is made to the OCM. OEM item numbers are at a higher level of assembly.

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Members of the DMSMS community have a role in advocating for DMSMS considerations during design to generate a DMSMS-resilient design that reduces the likelihood of near-term DMSMS issues and increase the probability of a quick recovery when they occur. DMSMS resilience requirements should include some criteria to limit future DMSMS issues. The design requirements are achieved by selecting parts whose forecasted availability is five years or greater to the maximum practical extent for new designs, design changes, modifications, or configuration changes that propose new parts.[99] DMSMS design considerations are not automatically implemented. DMSMS-related design considerations require a business case. To facilitate this, there should be effective interfaces between the DMSMS management community and the design and systems engineering communities. These interactions ensure that design trades are made between design characteristics and that the required capability adequately accounts for obsolescence.

The remainder of this section provides guidelines regarding the DMSMS trade space for initial design and subsequent redesign phases resulting from DMSMS considerations. It discusses design considerations that should be promoted by the DMSMS community to delay the occurrence of DMSMS and increase the likelihood of low-cost resolutions.

DMSMS is one among many product support design considerations. Design decisions made early in the system development have a substantial impact on O&S costs later in the life cycle. As shown in Figure 6, a high percentage of the life cycle costs of a system are locked in based on early design decisions.

During the initial design process, performance, supportability, logistics, cost, and other considerations all have to be balanced and trades made to produce the optimal design. For a redesign effort, specifications and interfaces already exist, which may constrain the ability to determine an optimal design. DMSMS is one of the many considerations informing design and redesign decisions.

A DMSMS review of system or product designs provides opportunities to

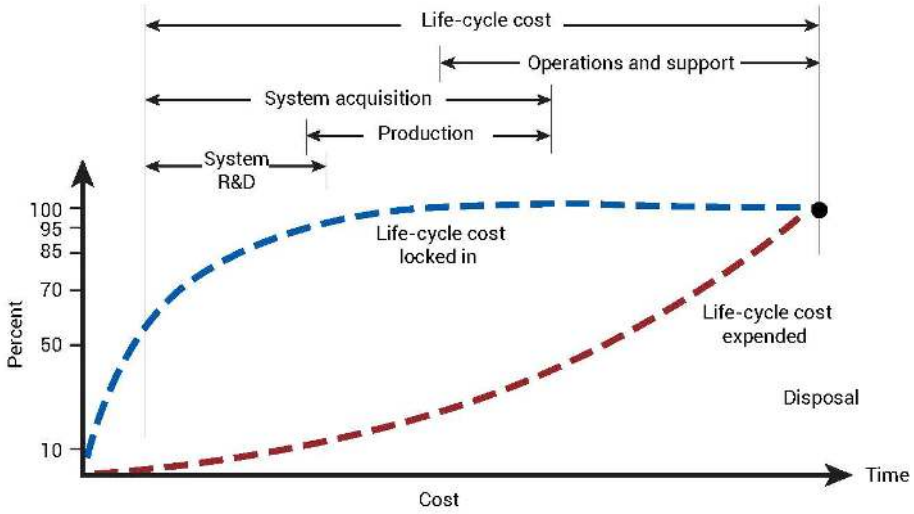
- Design out items that are high risk for various reasons and
- Plan in advance for technology refreshment.

For example, when the parts called for in a design are near their end of life (EOL), replacing them would be difficult or complex, and requalifying the system after replacing the items would be costly. As a best practice, program office leadership should actively set the tone for the importance of considering DMSMS risk even as early as the PDR by setting design goals in contracts that consider availability, supportability, producibility, reliability, maintainability, and technology refreshment over the fielded life of the product or system. These considerations should be built into a checklist for discussions with the chief engineer and PM leading up to each review. For example, the program office might challenge its design engineers to develop a design that will endure until year X, where “X” is the year in which either the next production block or technology insertion is scheduled. Considering DMSMS risk during design will help to avoid incorporating obsolete or soon-to-be obsolete items and enable a program office to more easily implement or integrate resolutions to and mitigate the impact of any DMSMS issues that do emerge during each phase of the life cycle. Failure to track and address these issues can have serious impact to cost, schedule, and even system viability.

[99] See illustrative requirement 17 in the SD-26, *DMSMS and Parts Management Contracting Guide*, June 2023, available at https://quicksearch.dla.mil/qsdocdetails.aspx?ident_number=283456.

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Figure 6. Relationship between Expended Life Cycle Cost and Locked-In Cost



Source: Wiley J. Larson and Linda K. Pranke, *Human Spaceflight: Mission Analysis and Design* (McGraw-Hill College, 1999).

DMSMS issues can be delayed or resolved in a simpler way if a system's design enables the substitution of readily available alternative items. Designers and systems engineers should consider the following design concepts to minimize DMSMS risk throughout the life cycle of a system or product:

- ***Future DMSMS issue resolution cost***. The previous discussion of DMSMS resilience emphasized the idea of selecting items whose EOL estimates are as far into the future as possible. While this is almost always a best practice to achieve DMSMS resilience in design, there are situations where designers should consider another factor in their item selection decision. The expected future cost of resolving DMSMS issues for an item may have an impact on the design. More specifically, if the DMSMS resolution portion of an item's life cycle cost is lower than that of an alternative item, then selecting that lower-cost item for the design may be the better DMSMS resilience decision, even when the alternative item has a longer anticipated time until it is obsolete. There are two key questions to answer when calculating the DMSMS portion of an item's life cycle cost.

  The first question is "W*hat is the timeframe over which to estimate DMSMS-related life cycle cost?*" The timeframe of interest for the item will not always be the planned life of the system in which the item will reside. Technology refreshments and technology insertions may affect how long the item will be used on the system. Therefore, the timeframe for estimating the item's life cycle cost should end when the item is planned to be removed from the system.

  The second question is "*What types of DMSMS resolutions are expected?*" Depending on the item selected, there could be significantly different costs associated with the resolution of a DMSMS issue. If the item is not technologically sophisticated, then it may be reasonable to predict that resolutions will involve a simple or complex substitute, especially when there is a robust industrial base able to manufacture the item. For more complex microelectronics items, a technology refreshment, redevelopment of the item, or even a redesign at a higher level of assembly may be more likely, especially when manufacturing processes have changed.
- ***Technology and item selection***. New technologies do not capture 100% of the market all at once; the new technology and the one it replaces will be in use for a time. However, the design should not include anything that is near the end of its functional life. A technology roadmap that anticipates the lifespans of technologies and synchronizes both technology refreshment and insertion is useful when designing systems, especially electronic systems. There are, however, tradeoffs associated with selecting new technologies. New technologies

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can be profoundly more effective at delivering an important defense performance parameter and thereby enable major changes in defense capability. So, it is desirable to be able to insert such new technologies. However, there is often a learning phase associated with a new technology in which issues are discovered.[100] Consequently, choosing the appropriate technology insertion timing where leading capability exists but where early phase problems have been corrected is essential. When feasible to do so, it is also a best practice to avoid the selection of sole-source items.

- ***Standardization and reuse***. An up-front assessment of the risk of obsolescence should influence parts selection during the design process.[101] Parts selection encompasses both the selection of new parts and the reuse of parts from previous designs. DMSMS issues can be delayed if part selection in design includes considerations to minimize DMSMS risks. Extending the interval between DMSMS issues and their projected impact on systems will decrease resolution costs over the system's life cycle, because there will be fewer issues to be resolved.

  The selection of new parts might seek to standardize the use of parts to the greatest extent possible and minimize the use of custom parts through the recommendation of alternatives. This is a tradeoff, however. Lower power and higher performance for an application-specific integrated circuit (ASIC) designed specifically for a task may be desirable where the volume or performance justifies the ASIC development. Using unique, highly specialized parts to meet performance requirements, will make DMSMS issues more prevalent during O&S. The risk assessment should also consider material selection, economic and regulatory trends, unique manufacturing processes, packaging schemes, and so on. Before including a part on a preferred parts list, the identified risks should be assessed and managed to make the BOM stable and sustainable. As the design stabilizes, it should minimize the number of OEM or OCM parts necessary for production. When non-preferred parts are used, their designs should be captured in the proper transportable computer-aided design models.[102] A somewhat higher level of standardization is standardized module development. For example, a common design can be reused over and over for many purposes, perhaps with minor variation in software or minor variation in connected sensors and actuators to enable use in many different applications. Similarly, this same platform can be repackaged in a different shape but otherwise the same design. The net result of standardized module design is, as above, higher volume of product, better factory support from suppliers, and more rapid increase of product reliability and producibility. In the context of DMSMS, the higher volume consumption of components enables a closer connectivity with suppliers to work DMSMS-related sourcing issues.
- ***Open systems architecture***—hardware. An open systems architecture employs technology-independent modular design tenets, uses widely supported and consensus-based standards for its key interfaces, and is subject to validation and verification, including test and evaluation, to ensure that key interfaces meet open standards. An open systems architecture thereby takes product roadmaps and technology insertion plans into account. Also, compared with design-specific approaches, it enables readily available alternative items to be used more easily in place of obsolete items if the substitutes have the same form/fit/function (F3) and interface as the ones they replace. Test interfaces must also be considered. An open systems architecture reduces DMSMS resolution costs, because it avoids expensive

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[100] For example: In electronic components these issues are often called errata. In specialized manufacturing, there can be issues with a manufacturing process that must be resolved before the product is sufficiently reliable for a specific application.

[101] DoDI 5000.88, "Engineering of Defense Systems," November 18, 2020, establishes a requirement for parts selection to consider DMSMS, available at https://www.esd.whs.mil/Portals/54/Documents/DD/issuances/dodi/500088p.pdf, p. 25.

[102] A very useful method of describing many firmware or logic hardware design is through a very high-speed integrated circuit (VHSIC) hardware description language (VHDL). A VHDL (assuming a satisfactory level of specificity) is easily ported from one generation to the next generation of technology. Life cycle costs may be reduced significantly through the proper use of VHDL design representation. When dealing with microcircuits, the most common hardware description language is VHDL.

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redesign by facilitating the insertion of advanced technologies.[103] Often, assuming no other proprietary technology is involved, it also enables multi-vendor competition, which will minimize the likelihood of future DMSMS issues.

- ***Open systems architecture***—software. An open and modular design enables the development of "plug-and-play" hardware and software that are interchangeable through industry-standard interface modules. To minimize DMSMS impacts, the software architecture of a system should be designed to take growth, evolving standards, and interfaces into consideration.[104] This provides for change while minimizing the impact on existing system functions. In addition, the design should allow for partitioning of the software into appropriate units that can be tested in isolation and should avoid making software dependent on the hardware through the appropriate isolation of drivers. Plug-and-play interfaces are desirable when appropriate. Low coupling (interdependent relationships) within a system allows a system to rely on information sharing to control, manage, and execute functions. When designing custom software and selecting COTS software, a program office will also want to carefully select interface standards and protocols that are the most stable and have the broadest support, as these will have greater staying power within industry. A program office should also seek to minimize the number of different interface standards and protocols applied across the weapons system, because this will simplify the design configuration and CM issues. The focus of software design can then be to meet the driver interfaces, rather than different, specific hardware items. Transportability of models that capture critical elements of the design is a consideration. The modules of an open system should be discrete, scalable, and reusable with low connectivity to the relationship between internal elements of different modules, simplifying and decreasing the number of interfaces required. Having high cohesion among module functions also enables multitasking and use of identical modules throughout the system. Significant complexities may be associated with using open systems architecture principles for a new software design being incorporated into an existing asynchronous system. One approach to software design is object-oriented design, which can increase portability and reusability of software. Finally, DMSMS issues will often require an update of standards-based protocols (such as Internet protocols). Because standards-based protocols are revised relatively often due to cyber defense issues, it is essential to recognize that the operating system and protocol stack are likely to be revised frequently and therefore any system should provide the required mechanisms for assured updates.
- ***Use of COTS assemblies***. COTS assemblies offer opportunities for reduced development time, faster insertion of new technology, lower procurement costs, and potentially, lower life cycle costs, due to a more robust industrial base. Consequently, DoD systems increasingly comprise COTS assemblies and software.

  Unfortunately, COTS items present a unique set of challenges for the management of DMSMS issues.[105] These challenges are due, at least in part, to the fact that these items are produced for the commercial market. For example, the rapid turnover of COTS items creates unique obsolescence-induced supportability issues for military systems, because OEMs are likely to replace or stop producing COTS items long before the life cycle of a system is complete. Furthermore, the DoD community has little influence over the far shorter life cycle of commercial products. Consequently, information on the future availability of COTS assemblies is hard to obtain or track. Changes during the system life cycle may not be documented, increasing the likelihood of CM and DMSMS issues. In addition, depending upon the system and program office management practices, requalification costs associated with replacing COTS assemblies may be significant. For that reason, the initial cost savings from the use of COTS assemblies may be offset by increased costs later in the life cycle

[103] The ability to use an open systems architecture design approach for a legacy system is limited if not anticipated in the initial design.

[104] Software is the primary focus of integration for the development of open, scalable, and adaptable systems.

[105] SAE International, EIA-933C, "Requirements for a COTS Assembly Management Plan," August 8, 2020, available at https://www.sae.org/standards/content/eia933c/, includes requirements for obsolescence management.

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when those assemblies have become obsolete or are replaced by a later-generation design. In short, it may or may not be appropriate to include COTS assemblies in critical paths or functions of a system.

Before including a COTS assembly in a design, the designer or PM should assess the risk and suitability (which should consider technology insertion and refreshment strategies).[106] A program office should avoid modification of COTS assemblies or software without careful consideration of the implications and alternatives. For example, modification could make the assembly or software nonstandard and incompatible with any standard updates to correct for deficiencies or errata, to add a feature, or to otherwise optimize performance. A PM would then be faced with choosing between the immediate costs of further revising the nonstandard assembly or software (to incorporate the update) or future high support costs and the inevitable obsolescence.

Managing DMSMS issues due to the introduction of COTS items in a system design calls for effective relationships among all relevant stakeholders: the COTS supplier, the system developer and integrator, the DMT, and the buyer (e.g., the item manager). The DMT must remember that all COTS items are subject to DMSMS issues, but some are prone to specific problems. For example, software, central processing units, memory chips, and disks change frequently. These specific COTS classes aside, a degree of obsolescence is always in place in the form of planned minor upgrades or refreshes, typically at the two- and four-year marks. Beyond that, a major upgrade—a next generation—should be expected at some time in the future.

- ***Use of alternative grade parts***. Alternative grade parts represent all grades of parts other than commercial grade and military grade, e.g., automotive grade. Alternative grade parts may be used instead of military grade parts in certain defense applications if the mission and application for these products can tolerate reduced short-term or long-term quality or reliability and a thorough understanding of how the alternate grade part will be used has been made. Many component engineers do not consider the application driven risk tolerance that alternative grade parts may have because they undergo less rigorous screening (quality, conformance, inspection, and testing) than that of a military grade part. Since alternative grade parts often receive significantly more screening than commercial grade parts, additional measures can be employed to reduce the risk of using alternative grade parts by ensuring reliability and performance in the intended application. One example may be conducting additional part qualification and screening and part derating to confirm the part can meet the specific intended application. Another example is the use of existing standards that can be used to predict performance and reliability (e.g., Automotive Electronics Council Q specifications). Effective measures can also be in form of process changes, such as planning for more frequent part refreshment, building in redundancy, and providing additional insulation against environmental extremes. Implementing such additional measures will likely add to overall part acquisition cost and time and that must be weighed against the benefits of using an alternative grade part. Benefits include improved technology or a reduction in size and weight. In addition, an alternative grade part may be more readily available from a larger supplier base than a military grade part, offering future reductions in lead times and costs and some protection against obsolescence.

[106] Ibid. This standard was originally issued by the Electronics Industries Alliance, which dissolved in February 2011 and no longer exists. The standard now belongs to SAE International, which is in the process of updating this document.

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The concept of PM&P management is closely related to these design considerations. At its simplest, PM&P management is concerned with choosing parts based on requirements specified in MIL-STD-11991B. The following quotations are excerpts from DoD's PM&P management guidance.[107]

> *Parts management is a systems engineering discipline for selecting parts and assemblies of parts, while accounting for the materials and processes used to manufacture them, throughout all phases of a system's (or equipment's) life cycle from initial design through disposal.*[108]
>
> *Part selection decisions should result in achieving application requirements throughout a system's life cycle. During design, part selection decisions are based on the thoughtful assessment and balancing of numerous, overlapping engineering design considerations (many of which are not the responsibility of the parts management practitioner, whether in government or industry).*
>
> *Part selection considerations include performance, cost, quality, qualification, reliability, maintainability, supportability, standardization, technology features and life cycle stage, manufacturing processes and producibility, Diminishing Manufacturing Sources and Material Shortages (DMSMS) risk, system security, cyber weaknesses and vulnerabilities, hardware and software assurance, supply chain risk, susceptibility to counterfeiting, unauthorized tampering, and use of hazardous materials.*

In addition to design concepts, the design tools themselves can impact future DMSMS issues. Modern defense systems are designed, built, and manufactured with extremely capable computer-aided design tools. These tools enable exhaustive checking of a design at each design phase. Whether electrical design, software design, firmware design, mechanical design, system design, system validation, production, or life cycle support, modern tools will be valuable in reducing the cycle time and eliminating many common types of errors. Some of these tools interface with DMSMS resources to provide alerts regarding the latest issues with components or subsystems or with current practices. Some are also able to check rules and checklists for best practices and identify alternate items that may be in an earlier stage of their life cycle. All can provide appropriate design documentation.

3.4.3 Establish a DMSMS Management Evaluation Process

The DMT should continually evaluate the effectiveness of its program office's DMSMS management operations measured against the program office's defined DMSMS objectives. This is accomplished by recording and periodically analyzing data. Although data do not provide an answer to program offices in and of themselves, data should be captured and analyzed to generate metrics for a DMSMS program. Metrics also indicate where a program office should investigate further.

The framework for DMSMS record keeping that will be introduced in Section 3.4.3.1 can be used strategically throughout the life cycle of defense systems. These records provide a basis for the identification, calculation, and analysis of DMSMS metrics that can be used. In general, benefits (summarized in Section 3.4.3.2) of using these records and the metrics derived from them include improving the efficiency and effectiveness of DMSMS management efforts, supporting DMSMS

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[107] For more information on PM&P management, see DSPO, SD-19, *Parts Management Guide: A Guidebook of Best Practices for Oversight of Parts Selection in Defense Systems,* December 2023, available at https://quicksearch.dla.mil/qsDocDetails.aspx?ident_number=119791, and MIL-STD-11991B, "General Standard for Parts, Materials, and Processes," October 3, 2023, available at https://www.dau.edu/sites/default/files/2024-01/MIL-STD%2011991B.pdf.

[108] The next update of the guidance will update the definition as follows: PM&P management is a systems engineering discipline for selecting and assuring performance of parts and assemblies of parts, while accounting for the materials and processes used to manufacture them, throughout all phases of a system's (or equipment's) life cycle from initial design through disposal. Note that all parts, materials, and processes used in the design, development, fabrication, procurement, and sustainment of DoD systems and equipment are included.

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management and resolution programming and budgeting decisions, reducing cost, and avoiding DMSMS impacts on schedule and readiness.

Some information collected may have only limited utility for a single program office, often because of a limited number of data points available. For instance, average cost and resolution time for a single program office is useful for self-assessment but cannot provide statistically significant data representing a wider set of program offices. It is important however to recognize that while program office management is the principal beneficiary of this record keeping framework and resultant evaluations, it is not the sole beneficiary from a strategic perspective. Higher-level organizations, such as Program Executive Officers (PEOs), Service Headquarters (HQ) organizations, and Office of the Secretary of Defense (OSD), profit as well.

For example, given evaluation and current record keeping practices, questions such as the following cannot be answered by today:

- How many program offices have DMPs?
- How many program offices program and budget for DMSMS management operations and resolutions?
- How many program offices are proactively monitoring for DMSMS issues? What portion of those program offices' system is proactively monitored for DMSMS issues?
- How much is DoD spending on resolutions for DMSMS issues?
- How much is DoD spending on DMSMS management operations?
- How would resolution costs change with greater DMSMS management proactivity across the board?

To answer such questions, it is therefore a best practice to establish and use a process to capture and evaluate a program office's DMSMS management operations data. By aggregating DMSMS case information across multiple program offices, higher level organizations can determine the need for overarching policy and guidance to both target the most important needs and measure whether the implementation of the policy and guidance is having the desired results.

3.4.3.1 Establish a DMSMS Record Keeping Framework

Data elements in the record keeping framework are kept, typically as part of a program office's case management system and to some extent from aggregate information about its DMSMS management program.[109] The data elements can be categorized into two broad information areas associated with DMSMS management of a given defense system as well as DMSMS management across an enterprise:

- ***Information associated with the cost of DMSMS resolutions and DMSMS management operations***. This includes characteristics of the obsolete item, the resolution and its cost, the amount paid for DMSMS management operations, and benefits associated with implementing the resolution. Such information is useful for cost estimating and programming and budgeting.
- ***Information associated with the efficiency of DMSMS management operations***. This includes case processing time, how DMSMS issues were discovered, and workload measures. Such information will help identify areas where process improvements are needed.

While the distinctions between these two information categories are relatively sharp, the data elements that fall within each are not entirely mutually exclusive. For example, some data elements associated with

[109] If any of the data will be provided by contractors, it should be asked for in an SOW so contractors can prepare for it.

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process efficiency might consider the magnitude of DMSMS management operations cost; similarly, how issues were resolved might impact case processing time. Furthermore, making inferences about an ROI requires both process efficiency and cost data.

All the identified data elements have direct, tangible benefits to program offices, PEOs, Service HQ, and OSD. No superfluous information is included. The data elements are divided into two hierarchical levels. Every program office should be collecting and analyzing the Level 1 data elements to improve DMSMS management operations, reduce DMSMS-related costs, prevent DMSMS issues from impacting readiness or schedule, and be in a strong position to explain and prove the benefits of its DMSMS management efforts. Level 2 data elements enhance these benefits by enabling more complex analyses, sometimes using data that may be more difficult to obtain. In addition, there are some benefits that provide greater visibility of DMSMS management throughout DoD and raise the gravity and significance of how it is regarded. Program offices with highly robust DMSMS management operations are encouraged to collect and use Level 2 data in addition to Level 1 data. Table 8 lists the Level 1 and Level 2 data elements. This table is shown here primarily to provide a formatted list of all the included data elements.[110] Appendix I.4 is a data dictionary for the Level 1 and Level 2 data elements. Section 3.4.3.2 summarizes specific ways in which the framework can be used strategically to benefit all stakeholders.

Table 8. Record Keeping Data Elements

Information Type	Level 1 Data Element Title	Level 2 Data Element Title
DMSMS cost–related data elements	• Type of Resolution Approved • Resolution Cost • Source of DMSMS Resolution Cost • Redesign Level • DMSMS Management Operations Cost Paid to Prime/OEM*# • DMSMS Management Operations Cost Paid to Independent SME Organizations*# • Management Operations Cost for Internal DMSMS-related Activities*#	• Commodity Type • Operating Environment of the Equipment* • Product Acquisition Cost • DMSMS Item Type • Value of Management Operations Activities Received at No Cost from a Centralized Service Source*# • Root Cause

[110] Although not specifically called out in Table 8, the name of the program office/system to which the data apply would also be needed.

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Information Type	Level 1 Data Element Title	Level 2 Data Element Title
DMSMS management operations efficiency–related data elements	• OEM Part Number • OCM Part Number • OEM/OCM Commercial and Government Entity (CAGE) Code • Nomenclature • Item Class • Case Proactivity Indicator • Date Alert Received • Date Case Opened • Date Resolution Submitted for Approval • Date Case Resolved • Date Case Closed • Date Implementation Needed • Resolution Avoided • Cost of Resolution Avoided • Number of Subsystems* • Number of Subsystems Monitored* • Components* • Components Monitored* • Reason Issue Was Discovered Reactively • Out-of-Cycle Redesign Avoided Indicator • Level of Out-of-Cycle Redesign Avoided	• LON Buy Preferred Indicator • Monitoring Techniques • Effect on Production Schedule • Effect on Logistics Response Time • Effect on Mission Capability • Resolution Avoided Implementation Time • Consumption Rate • Stock on Hand

Note: The system to which the records apply must also be identified. Most of the record keeping data elements are linked to individual DMSMS cases and therefore would be available from a suitable DMSMS case management system. An * depicts other readily available data elements needed to improve DMSMS management operations and understand their ROI that are not linked to individual cases because they represent the DMSMS management effort as a whole. Entries marked with a # indicate information a DMSMS contractor is not in a position to know, and must be collected by the program office; hence, these entries do not appear in DID DI-MGMT-82275A, “Diminishing Manufacturing Sources and Material Shortages Metrics Data.”

3.4.3.2 Benefits of Record Keeping

The beginning of Appendix I provides multiple examples of the benefits of this framework. The remainder of this section is a summary of those benefits.

3.4.3.2.1 Programming and Budgeting Benefits

DMSMS management costs at any level from program office to OSD should not be hidden or within some other function’s resource allocation. The inevitability of DMSMS issues and the extreme negative impacts associated with leaving them unresolved imply that the resources needed should be programmed and budgeted for explicitly.[111] Eight benefits of the record keeping framework that relate to improved

[111] The degree of explicitness would vary as a function of organizational level. This is not to suggest that there should be a budget line item or program element for DMSMS management activities. However, program office financial documents should identify DMSMS management resources precisely and this information should be visible at least at the PEO level.

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estimates of programming and budgeting requirements for DMSMS management—both resolutions and management operations follow:

- Greater fidelity cost estimates for DMSMS resolutions,
- Improved cost estimates that better take into account the timing and magnitude of inventory purchased for LON buys,
- Improved ability to develop and defend programming and budgeting requests for DMSMS resolutions,
- Improved evaluation of contractor cost estimates for resolving DMSMS issues,
- Improved understanding of the link between DMSMS management proactivity and risk,
- Improved cost and workload metrics for DMSMS management,
- Improved ability to develop and defend programming and budgeting requests for DMSMS management operations, and
- Improved evaluation of contractor cost estimates for conducting DMSMS operations.

3.4.3.2.2 Process Improvement Benefits

Increased DMSMS management effectiveness and efficiency resulting from process improvement generally implies lower DMSMS resolution costs. The following benefits apply:

- Improved efficiency for case processing time,
- Improving the effectiveness of item monitoring processes,
- LON buy process improvement, and
- Detection of anomalies in DMSMS resolution cost to identify areas for further analyses.

3.4.3.2.3 ROI Benefits

All the ROI-related benefits concern, in one way or another, assessing (either quantitatively or qualitatively) the value of risk-based, proactive DMSMS management. Some support provided to program offices from DMSMS management SMEs is not covered, such as improved contracts, better decision-making, the potential to obtain hard-to-find parts, the importance of finding the most accurate data, and so forth. The value of such support cannot be determined from the data elements in the framework. In addition, these types of functions represent additional reasons for securing comprehensive DMSMS management operations support and are often articulated by SME providers as rationale for determining which provider to select.

Figure 7 illustrates the difference between proactive and reactive DMSMS management. The earlier an issue is identified, the longer the window of opportunity to resolve it. A longer window of opportunity reduces the likelihood of readiness, schedule, and counterfeit-items or malicious-insertion impacts[112] and often the cost of the resolution. Cost is reduced because the early detection of the issue means more lower-cost options are available.

Benefits is this area are as follows:

- Improved estimates of the cost of being reactive,
- Improved estimates of the cost avoidance for being proactive,
- Improved understanding of DMSMS impacts on schedule,
- Improved understanding of DMSMS impacts on operational availability,
- Improved estimates of supply system impacts avoided by being proactive,

[112] Counterfeit/malicious insertion risks may be reduced because a program may still be able to work with primes and OEMs.

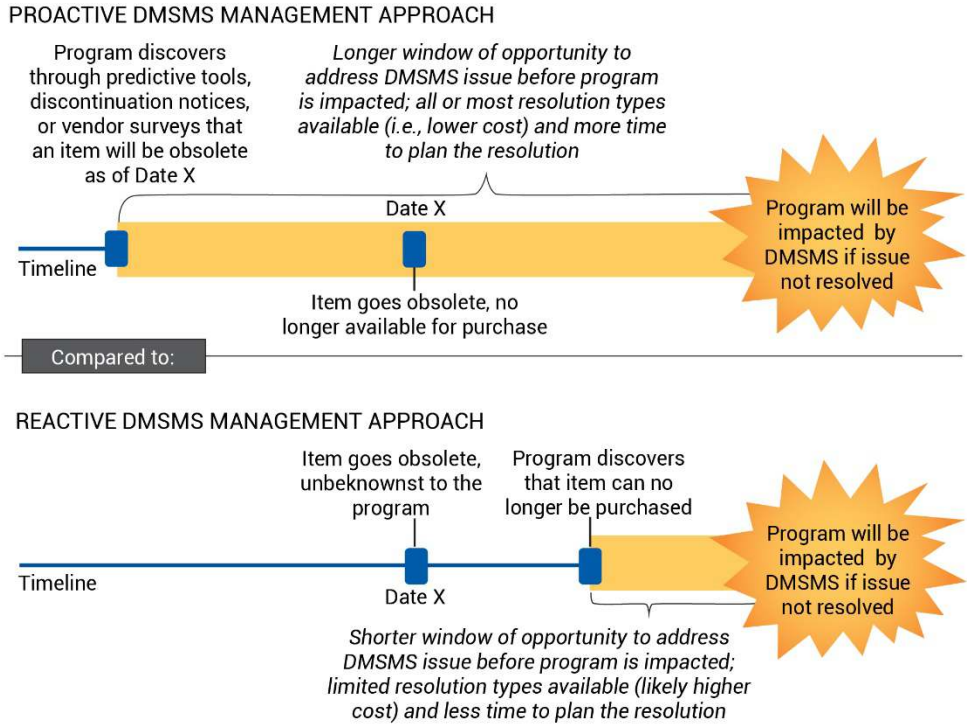
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- Improved estimates of improvements in resolution implementation time from being proactive, and
- Accurate determination of an ROI for DMSMS management.

Examining trends in such ROI-related areas provide insight on the cumulative impact of proactive, risk-based DMSMS management. This further enhances the value proposition for these associated benefits.

Each benefit has the same fundamental value proposition. For the program office, all the ROI-related benefits provide evidence of the value of fully funding DMSMS management operations. There are instances where the DMSMS management community has had difficulty convincing program office leadership of the importance of risk-based, proactive DMSMS management (relative to other needs), especially during sustainment where there may not be enough operations and maintenance (O&M) funds to meet requirements. The data elements for each of the following benefits should provide sufficient information to raise the priority of DMSMS management operations funding because it is the only way to mitigate the risks. The data should also be used for justification of funds requests.

Figure 7. How Proactive DMSMS Management
Increases the Window of Opportunity for Resolving a DMSMS Issue



The PEO perspective is twofold. A PEO organization should ensure that examples of the ROI calculations are disseminated among all the program offices in its portfolio to enforce full funding of DMSMS management operations. In addition, PEOs can use the data to persuade higher-level organizations to approve the requested amounts. A Service HQ would similarly use the data to justify funding requests to OSD as well as potentially establish policy. OSD could aggregate the data to demonstrate the need for high level attention and policy and guidance to all senior level decision makers.

All these ROI value propositions become more powerful when used in conjunction with the value propositions associated with developing programming and budgeting requirements for DMSMS resolution funding and improving DMSMS management process efficiency.

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3.4.4 Establish a Quality Management System

The DMSMS management should operate within a well-defined and functioning QMS. A QMS is an overarching framework that defines the organizational structure, responsibilities, methods, data management, processes, resources, customer satisfaction, and continuous improvement. DMSMS management processes must be controlled in a QMS to receive optimal benefits and ensure consistency over time regardless of who is performing them. A well-defined QMS can also ensure that the same high level of quality service and products can be produced regardless of personnel changes. This section focuses on control of DMSMS management processes. For contractors, it may be necessary to establish requirements to ensure that this is done.

3.4.4.1 Quality Plan

The QMS should call for a quality plan that defines the checks on the system or product necessary to ensure quality; that is, to ensure that processes and product are in control and meet defined requirements. An excellent means of controlling processes is to document the responsibilities and methods associated with the processes in a series of procedures or work instructions and to establish quality checks at optimal points within the process to ensure that the work product meets defined quality standards. Quality checks demonstrate verification of whether the process is operating as defined. The quality plan should include the identification, collection, and monitoring of compliance metrics to ensure that the process is successful. Compliance metrics provide information on whether the process is being followed or must be adjusted to meet the intended outcome.

Different entities may use specific nomenclature for the documented processes, such as standard operating procedures, or work instructions. The nomenclature chosen for this documentation does not matter.

A written procedure outlines how to perform a process. This level of documentation typically applies to the processes common across a function, such as DMSMS management. Because DMSMS management support can vary significantly from one platform to the next, a second tier of process definition should be developed. This second level, often known as work instructions, is used to define how to perform a task. Each platform team should develop its unique DMSMS management work instruction tailored to supporting its specific platform.

As an example of how this may function, consider the processes to collect and disseminate PDNs and obsolescence event data, often referred to as “Alerts.” PDNs are published by manufacturers to inform industry that an item is targeted for discontinuance. A procedure could be written for how to find, confirm, and document this obsolescence event data. The platform teams may take different actions in response to an alert; each team could write work instructions to describe its own specific process.

The general workflow in support of DMSMS management consists of data collection from many diverse sources, data compilation and analysis, risk assessment, and report or briefing development. Because data collection, manipulation, and analysis are at the heart of DMT activities, data standards should be clearly defined. Data standards define such aspects of data management as content accuracy, data content, and data entry standards.

To establish quality checks, the DMT should review all process inputs and outputs. For each data stream, whether input or output, the DMT must decide the acceptable characteristics. The DMT should document these characteristics and make them available to the DMT members who may create or process the data stream.

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The DMT determines the method to identify data that do not meet the defined standards. Quality checks can range from automated comparisons to defined standards to simply having an experienced team member review the work of a less-experienced DMT member.

The DMT should review the process flow and determine where to insert quality checks. These locations are the points at which errors can be identified and corrected before additional work is applied and before the customer is affected. The quality plan should include the inspection points, the inspection method, and the error correction mechanism.

To demonstrate a quality check for data content accuracy, consider the availability status of a highly complex electronic item. In general, the obsolescence of such an item, in contrast to an item that is of low complexity, has a greater impact on the mission of the platform, and resolution can be much more difficult. Therefore, the accuracy of the data concerning this type of item is critical. In such a situation, the DMT may decide that verifying the content accuracy of the availability status of this type of item may require manufacturer contact or no less than two predictive tool providers to report the availability status for the item. The quality check to ensure content accuracy could be to check that the availability was verified by contacting the manufacturer or using more than one predictive tool. For example, the DMT may decide that the item description must exactly match the approved list of item descriptions. The quality check would then determine whether, in fact, the entry for an item description matches the entry on the table of approved items.

This same principle can be applied in other data streams, such as the recording of mitigation efforts, often called case data. The DMT may decide that the implementation date for the resolution should be recorded. In this situation, a quality check would verify the presence of an implementation date.

3.4.4.2 Compliance

Compliance metrics measure the status of processes and activities within a quality plan.[113] Among the reasons that a process fails are budgeting too little time or too little money, inadequate planning, constantly changing goals, lack of process knowledge, and ineffective communication. Often when a process is in danger of failing, management is unaware of the problems.

One of the best tools for avoiding process failures is to track key indicators of process health. The data should be presented in a meaningful way to help process managers make the proper decisions, take corrective steps on processes, or both. It is also important to define the right measurable periods that can cover possible gaps in the control of the measuring indicators, as well as allow control of the situation upfront if a failure occurs within the measurable intervals.

Compliance metrics generally have a target value—such as an industry benchmark or regulatory guidelines—against which they can be compared. Five general criteria are typically used when defining compliance metrics for a process:

- Time,
- Cost,
- Resources (e.g., person-hours),
- Quality, and
- Actions.

[113] The compliance metrics discussed in this section are measuring the DMSMS management processes. This discussion does not include cost and operations efficiency metrics presented under case management.

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When compliance metrics are first applied to a process, it is often difficult to separate the categories of time, cost, and resources. Tracking compliance metrics that provide information on combinations of two or more of these concepts is a viable approach. As the QMS matures and the situation necessitates, compliance metrics can be redefined to provide more focused data. Below are two examples:

- Electronic item availability research is necessary for program offices that have item lists or BOMs. This research can be time-consuming. In general, it could be more cost effective and timely to obtain item status data from at least one predictive tool supplier. One measure of resource usage would be to track the percentage of items that the predictive tool companies recognize. By working with the predictive tool suppliers to increase the recognition rate, the team is effectively moving item availability research from an internal and more manual process to a subscription deliverable and, thus, is using resources more effectively.
- Data management in support of DMSMS management contains several distinct processes. Feedback on adherence to schedule could be obtained by tracking the time to perform the intermediary processes, such as the time from receipt of an item list or BOM to identification of the components to be monitored for availability. The time to perform the intermediary processes is then compared to a standard time established for this process. Values consistently over the standard indicate that a problem exists in the process. Results consistently under the standard indicate a need to adjust the standard because the process has been improved.

The quality criterion focuses on whether appropriate actions are taken in response to finding a process defect, not the existence of defects. The compliance metric chosen should provide insight on whether defects are tolerated or, even worse, ignored. In data management, a defect is a situation in which a defined standard is not met. Below are some examples:

- The DMT may require, in the quality plan, measurement of the conformance of configuration data to defined standards. This could be accomplished by tracking the number of defective configurations periodically and then showing trends. If the number of errors is higher than the acceptable quality level or increases over time, a problem exists with the quality of the CM process.
- The DMT may choose to open a case for each monitored item that has an obsolescence issue. The DMT could then track the number of items with obsolescence issues that do not have an associated case. In this situation, the quality of the program office support process is being measured.

The actions criterion focuses attention on identifying outstanding action items as a means of determining possible barriers to the process success. To use this criterion, the DMT should maintain an action item summary in support of the process steps. This action item summary should then be reviewed to develop compliance metrics:

- Any differences between the action completion date and the projected completion date may indicate that a problem existed for that task. The difference between the action completion date and the projected completion date should be compared to a calculated standard established to support that platform. This compliance metric is most meaningful for DMT members accustomed to setting reasonable projected completion dates.
- The number of open items measures multiple program office aspects. It may measure the skill of the PM in capturing the steps necessary to support the program office. It also may indicate that a project is experiencing difficulties in completing tasks.

Beyond these general compliance categories are some more intangible signs that a project may be in trouble. These signs include a general lack of interest in the project, poor communication among team members, a fear of talking about project problems, and a generalized lack of project advancement.

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To be successful, compliance metrics must be well thought out and consistently interpreted and applied. Finally, the DMT must act upon the conclusions on process health provided by the compliance metrics in a timely manner to correct or improve the process, metric, or both.

3.4.4.3 Process Analysis

A detailed analysis of the DMSMS management processes can be valuable both in eliminating defects in the process and in improving efficiency. One type of business process analysis that has been used successfully to improve a DMSMS management process is Lean Six Sigma (LSS), a methodology that employs a collaborative team effort to improve performance by systematically identifying and removing "waste." In the LSS context, "waste" means any nonproductive, obstructive, or error-causing part of a business process. These components of waste are called "defects." The purpose of an LSS analysis is to identify the defects in the process and to devise approaches to eliminating or mitigating them. The define, measure, analyze, improve, control (DMAIC) approach is commonly used for an LSS analysis and comprises the steps in the overarching process used by the analysis team. The final step is to put in place controls that ensure that defects are minimized and thus maintain the overall quality of the process.

Such an approach could be applied, for example, to the DMSMS management of COTS assemblies for which status is determined through vendor surveys. The functions that must be performed are to monitor items for potential obsolescence, identify items impacted, evaluate need for opening a DMSMS case, and implement a case when required. The process to accomplish these functions could be composed of four basic subprocesses:

- Use vendor surveys, vendor contacts, website analysis, and forecasting tools to discover potential obsolescence issues (market surveillance);
- Verify item identity, stock on hand, and demand trends for the item at issue;
- Determine if the obsolescence issue will impact the program office via analysis of demand and stockage levels; and
- Open and evaluate cases, determine preferred resolutions, and track resolutions.

COTS obsolescence management is characterized historically as a largely manual effort with extensive human reactions that are subject to error, not to mention inefficiencies. Variability and non-value-added steps result in additional labor, schedule delays, and increased costs. Such defects can then be effectively identified and addressed via LSS methodologies. Developing an automated database is a first step toward a faster, more efficient, and less error-prone process.

One DMT that conducted an LSS analysis of its DMSMS management process for COTS items identified 131 defects occurring in an approximate one-year period in market surveillance of COTS items. The LSS analysis team defined a "reactive defect" to be one wherein an identified obsolescence date occurred in the past and "updates within four-months defects" were instances in which the date occurred within four months of the current date (thus limiting the time available for case analysis and determination of resolution, if required). Process changes identified to address these defects resulted in substantial reductions in defects (25% or more) and attendant improvements in process efficiency on the order of 25%–30% or more (depending on the subprocess).

To implement an LSS project for DMSMS management, an LSS team is formed, and headed, if possible, by an LSS "black belt."[114] Team membership should comprise representatives knowledgeable in the various aspects of the process under scrutiny.

[114] If not available in house, a consultant should be engaged to advise the team.

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Assuming the DMAIC procedure will be followed, in the *Define* phase of the project, the team focuses on the overall process and identifies the overarching "problem" causing inefficiencies and errors. The team comprehensively examines the process and narrows down the areas of deficiency. An initial process map is created, and the data identified that can quantify the problem. It is particularly useful to document the value stream map (VSM) that describes the value-added workflow steps that produce required products. The VSM facilitates identifying non-value-added steps within the process for potential elimination or at least modification to create value. This phase ends with the drafting of a charter for the project containing a broad statement of the problem and the overall goals of the project. In addition, anticipated roadblocks are identified and a plan to overcome them formulated.

The *Measure* phase includes the development of a data collection method to capture pertinent aspects of the current processes and their outputs, the collection of data, and the establishment of a baseline for measuring improvements. For example, the time to accomplish a process segment might be a meaningful measure to examine. Another would be the trend in defect rate over time.

The *Analyze* phase determines the most critical root causes of defects from as many perspectives as possible. Team "brainstorming" sessions generate ideas to bring about process improvement. This process should be wide open, with the objective of identifying as many ideas as possible. The team then filters, sorts, combines, evaluates, and distills the improvement ideas into a feasible, most promising set.

In the *Improve* step, an implementation plan is formulated and initiated. If the list of desired improvements is lengthy, it will likely be necessary to time-phase implementation due to budgetary, operational, or other constraints, such as equipment or software availability.

Lastly, the *Control* step ensures that the improvements are being realized. A process of continuous improvement will seek to identify further improvements.

3.4.5 Establish a Case Monitoring and Tracking Process

The DMSMS management community deals with two types of cases. The first are cases based on known obsolescence (e.g., a PDN is received or predictive tool results). The second type of case could result from the review of a parts list for a preliminary design or forecasted obsolescence. The cases generated based on the review of a preliminary design should be used to inform the system designers so that, to the extent possible, obsolete items are not carried forward in the system design. If those obsolete items are designed out of the system, there is nothing further for the DMSMS community to do with respect to that case. In other instances, analyses of the underlying technology of items can forecast when those items are likely to become obsolete. Cases may be opened to assess and analyze these forecasted DMSMS issues.

Case management supports the DMSMS management–related activities in program offices in two interrelated ways as discussed in the following two sections:

- It maintains a source of information on the status of resolutions before a case is closed to enable a program office to track, monitor, and expedite the implementation of the resolution. A DMSMS case report contains the relevant information—CAGE,[115] part number, national stock number (NSN), item type, next higher assembly (NHA), and so forth—on DMSMS cases that are open (including all individual variants determined useful by the program office), closed, or closed but still being watched. Those reports include a synopsis of assigned priority, potential resolutions, selected resolutions, relevant points of contact, relevant metrics as defined by the DMT, and DMT action items relevant to each case.

[115] The CAGE codes are five-digit numbers used as a contractor identifier for firms doing business with DoD. These numbers are used as a method of identifying a given facility at a specific location.

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- It provides a repository of data on implemented DMSMS resolutions and the DMSMS management program itself that can be used 1) to formulate programming and budgeting needs and 2) to analyze and improve overarching operational effectiveness and efficiency (see Section 3.4.3). Associated reports can be used for publicizing DMSMS management successes and sharing data among other DoD platforms. Robust case management provides the basis for meaningful DMSMS program metrics. Effective outreach could help obtain funding both for DMT operations and for implementing resolutions to DMSMS issues.

One purpose of case management is to track and manage DMSMS issues from initial identification to implementation of a resolution. A program office will need to document its approach (i.e., any criteria to be applied) regarding when to open an obsolescence case. Should it be for every item for which an EOL notice has been issued? Just for those items for which an EOL notice has been issued and current stocks will not support the system through end of need? Only for those items that will require a DMSMS resolution? Ideally, a program office might want to open a case on every item for which there is an EOL notice, even if a resolution is not necessary, because it will provide more complete documentation and inform future DMSMS management efforts for the program office.

To facilitate DMSMS case management, the DMT should consider the use of a tracking tool or database built upon a case sheet, consisting of basic and status data, for each DMSMS issue. Depending on how software intensive a system is, a program office's case management database may include some additional data elements associated with software. The development of a case sheet would therefore mark the beginning of the case management process. One prime contractor undertook the development of a case management database to standardize and facilitate what was previously a time-consuming manual process that required months to complete a basic status update for a case. Having a standard case management database enabled the program office to expedite the resolution of issues and ensure configuration control in the presentation of DMSMS issue case information across the program office.

The tracking tool or database should support functions such as the following:

- Tracking cases by case number for future reference (including an ability to link to a previous, related case).
- Tracking all appropriate part information and nomenclature (configuration and vendor parts) for manufacturer data, including last sale date and demand.
- Documenting information on an item's higher assemblies and the criticality of impact.
- Documenting the results of research and other engineering notes.
- Selecting or identifying a particular DMSMS resolution or set of resolutions for each case.
- Checking whether another platform is affected (potentially across the entire Department).
- Assigning action items to individuals or organizations related to a case. For example, who is responsible for working what aspects of the case.
- Tracking the length of time to identify and resolve DMSMS issues (e.g., the timing for the next case management action and the alignment of such actions with key dates, such as the dates by which a bridge buy will need to be made).
- Determining the status of a DMSMS case, such as the following:
    - *Open*. Cases that are actively being worked. Below are potential variants to "open" cases that a program office might want to consider tracking further:
        - *Open: Under Investigation*. Cases that are open and one or more DMT members are actively investigating the identified DMSMS issue. Such an investigation will first seek to validate the existence of a DMSMS issue that has the potential to impact the system and then to identify a recommended resolution.
        - *Open: Decision Pending*. Cases that are open and for which the recommended resolution has been determined and is awaiting final program office decision to

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proceed and/or funding. Information on opened cases requiring no action should be captured and fed back into the *Identify* phase of the DMSMS management process, so that the case will not have to be investigated again, unless some new information comes to light.

        - *Open: Implementation Pending*. Cases that are open and for which the decision to proceed and funding have been obtained, but implementation of the approved resolution has yet to begin.
        - *Open: Under Implementation*. Cases that are open and for which the approved resolution is being actively implemented.
    - *Closed*. Cases for which the approved resolution has been fully implemented and fielded.
    - *Watch List.* Cases that are closed, but where the program office has chosen to place further scrutiny to monitor if assumptions regarding obsolescence and resolutions remain consistent with new realities in the future. For example, it is important to know if demand for an item is greater than the assumptions used to calculate the size of an LON buy until the next technology refreshment/insertion. This would also apply to cases whose implementation is taking longer than expected.
- Facilitation of communication with the DMT regarding the status of a DMSMS case.

Some program offices may track a resolution until it is completely implemented and fielded. Other program offices may stop tracking a resolution once it has been funded, rather than tracking it through fielding, due to the length of time for implementation. This is a bad practice. It is important for the DMT to know whether a resolution was implemented to be able to determine optimal resolutions for future DMSMS issues.

In addition, depending on the level of detail needed, program offices may combine open and pending resolutions. The decision about the level of detail to be tracked should be made when the program office establishes its case management process. Consideration should also be given to record keeping needs as discussed in the next section.

3.4.6 Establish Supporting Contracts

Contracts and logically similar agreements among government organizations are the mechanisms for enabling the DMSMS management operations discussed in this document. Internal agreements among government organizations may or may not require funding.

3.4.6.1 Obtaining a Provider for DMSMS Management Operations Support

Responsibilities may change significantly, depending on how the prime contractor is supporting DMSMS management activities (see Appendix F). In many instances, the prime contractor is responsible for most of these activities. The program office should understand that the contractor's "standard" DMSMS management process is based on its own internal guidance and best practices. Thus, a program office should not assume that a contractor will have operational procedures that mimic what is in this document.

As a best practice, program office leadership should strive to use its personnel as efficiently as possible to implement DMSMS management. For example, in-house engineering personnel should not be diverted to perform routine, day-to-day DMSMS management activities. Leadership should instead rely on individuals with more specialized expertise and experience who are located within the prime contractor or the OEM and on independent SMEs. These existing resources and resident expertise should be leveraged by program offices to the greatest extent possible. The prime contractor has the most in-depth knowledge of the system and, therefore, should also be involved throughout the system life cycle.

It is likewise a best practice for a program office to employ independent SMEs, even if the prime contractor is already involved in DMSMS management. Independent SMEs can 1) help the government

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oversee the prime contractor, ensuring consideration of a life cycle perspective; 2) provide an independent verification of issues and a neutral third-party validation of resolution recommendations; 3) provide access to specialized tools, processes, data, and unique supplier relationships that may not be available to the prime contractor; 4) advise a program office on formulating contract language, securing BOMs and other technical data, developing an SOW for DMSMS management activities for a prime contractor, and other responsive, tailored support to specific needs; 5) serve as a central linkage to DMSMS management activities and best practices in other program offices; and 6) provide a conduit to improve access to supplier data in a competitive situation. Independent SMEs may also prove helpful during sustainment, if the government is entirely responsible for sustainment support and the prime contractor has little or no role.

Although each DMT should have a DMSMS SME, it is not always necessary to find that expertise within the program office. Centralized DMSMS SME teams reside within various organizations across DoD. These teams have in-house DMSMS management expertise and well-established processes that any program office can easily leverage to implement a DMSMS management program. In addition to having an established knowledge base and documented processes that enable robust DMSMS management, some of these teams own DMSMS management systems that experienced DMSMS practitioners use to integrate, analyze, and report on DMSMS-related data collected using predictive tools, vendor surveys, and PDNs received directly from manufacturers.

3.4.6.2 Obtaining DMSMS Management Operations Data

Contractual or similar arrangements must also be made to obtain DMSMS management data (see Appendix F).[116] Effective DMSMS management requires data for the following DMSMS management functions:

- Selecting subsystems to monitor (see Section 3.1 and Section 4.1),
- Selecting the items to be monitored from a risk-based perspective (see Section 4.3.2.1),
- Monitoring the at-risk items and forecasting when obsolescence will occur (see Section 4.4),
- Assessing the impact of known and predicted obsolescence (see Section 5.3),
- Developing recommendations to resolve DMSMS issues (see Section 6.2),
- Tracking whether the assumptions behind LON buys are holding (see Section 7.3 and Section 3.4.5), and
- Using technology roadmaps and health assessments as input into the timing for system modifications.

Data beneficial in conducting these functions include:

- Subsystem characteristics, for example:
  - Safety-related,
  - Mission criticality, and
  - Overall cost.
- Indentured BOMs or parts lists that provide the OEM CAGE code and the manufacturer and OEM parts numbers.
- Supportability data (mostly item-related, some associated with the systems using the item):
  - Operating tempo,
  - Failure rate,

[116] Contractors may consider some data to be proprietary. Therefore, obtaining such data may need to be negotiated in the contract SOW. See Appendix F for information.

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- Consumption rate/reliability data,
- Washout rate,
- Inventory on hand,
- Asset visibility (on fielded systems, salvageable),
- Number used on system,
- Number of systems manufactured and fielded over time,
- Maintenance and repair strategies,
- Provisioning data (e.g., mission essentiality, criticality) (Appendix O defines and addresses the provisioning process),
- Potential DMSMS resolution cost, and
- Supply chain information (e.g., backorders, sources, cost).

- Manuals and drawings:
  - Maintenance and repair manuals and
  - Hierarchical drawings and data sheets.
- The item itself is also a data source.
- DMSMS prediction data, EOL estimate, prior experience with DMSMS issues on the system (qualitative), technology roadmaps, modification plans.

Table 9 maps the different types of data into the DMSMS management functions to show data needs by function. Although there are numerous data elements portrayed, the BOM is far and away the most important. It answers the question what items are on the system. If that information is unknown, DMSMS management (and even reactive DMSMS management) is extremely limited. In such a situation, even if DMSMS information were obtained, without a BOM, there is no clear way to determine if the obsolete item is on the system.

Table 9. Data Needed to Perform DMSMS Management Functions

Functions Where Data Are Needed	Indentured BOMs or Parts Lists	Supportability Data	Manuals and Drawings	The Item Itself	DMSMS Prediction Data
Select most important subsystems and items within them to monitor	The existence of a BOM or parts list is a necessary factor to facilitate selection Need to know the universe of items to select from	Provisioning data to assess the risk of subsystems and items and to construct BOMs or fill in gaps as needed Cost, demand, sole source and foreign suppliers are measures of risk The repair strategy impacts what is included in provisioning data which in turn impacts the items to be monitored If the above data is unavailable, use of an algorithm is the fallback approach	To construct BOMs or fill in gaps as needed		Historical data pertaining to DMSMS risk of subsystems and items or similar subsystems and items

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Functions Where Data Are Needed	Indentured BOMs or Parts Lists	Supportability Data	Manuals and Drawings	The Item Itself	DMSMS Prediction Data
Monitor	BOMs are the basis for monitoring				What is obsolete (or is about to become so) in predictive tools databases, surveys, and EOL estimates
Assess	Indenture structure is used in an impact analysis at the item and higher-level assembly levels	Operating tempo, inventory, consumption/ reliability, and number fielded to examine demand to determine when a DMSMS issue will impact the system			What is (or is about to become) obsolete in predictive tools databases, surveys, and EOL estimates
Recommend	Indenture structure to determine the level of the resolution (i.e., at the item level or higher-level assembly level)	Operating tempo, inventory, consumption/ reliability, and number fielded to examine demand to size an LON buy Suppliers, maintenance and repair strategies, asset visibility, and resolution cost are considerations in an analysis of alternatives (AoA)	To suggest a resolution	To pursue reverse engineering and redesign	Technology roadmaps and modification plans
Track		Operating tempo, inventory, consumption/ reliability, and number fielded to test if LON buy assumptions are holding			
Plan		Operating tempo, inventory, and consumption/reliability to estimate future demand			Technology roadmaps and modification plans to develop a strategy to deal with inevitability of DMSMS issues and improve long-term DMSMS resilience

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3.5 Establish and Encourage Communication Processes for FMS Equipment

The five-step DMSMS management processes described in this document applies to both DoD and FMS equipment. However, FMS-related barriers may inhibit or even prevent these processes from being proactively carried out for FMS equipment.

The window of opportunity for resolving a DMSMS issue on military equipment is the time between issue identification and the date that the issue will have an impact on the equipment. During this time, three steps of the DMSMS management process should be completed.

- Assessing whether a resolution is needed, when it is needed, and the level of assembly at which the resolution should be applied;
- Analyzing the situation to determine the preferred resolution option; and
- Implementing the preferred resolution (which includes securing the necessary resources from funds for the FMS case).

Proactive DMSMS management identifies issues earlier and therefore increases the likelihood of completing these steps within the window of opportunity. Because DoD program offices are very concerned with the operational availability of DoD equipment and funding for resolving DMSMS issues on FMS equipment may not be immediately available, resolution implementation on FMS equipment sometimes lags implementation on DoD equipment. That lag decreases the proactivity of DMSMS management on FMS equipment and FMS-related barriers may slow the implementation even more, potentially affecting the availability of partner equipment. Therefore, this document section suggests FMS-specific considerations that should be incorporated into DoD's five-step processes to avoid these barriers as much as possible.

FMS-specific processes often must be repeated for each FMS case. There will generally be differences among FMS cases for different international partners and sometimes differences in the terms and conditions of similar cases with the same partner. Cost sharing arrangements among multiple partners may magnify the extent of these differences.

The FMS process begins with discussing foreign partner military needs and potential DoD sales to meet those needs. These discussions are formalized by the foreign partner in a Letter of Request (LOR) identifying what it would like to purchase—equipment and/or support—and estimated costs. In turn, a formal Letter of Offer and Acceptance (LOA) defines the terms and conditions of an FMS case between the U.S. and a foreign partner. Ideally, where applicable, an initial LOA will be for both the purchase of equipment and some number of years of support/sustainment, including DMSMS management. Subsequent LOAs may cover continuing support of that equipment if the partner country wants it. DMSMS management support could be continued proactively or could be limited to technical assistance or excluded altogether if, for instance, the foreign partner wants to sustain some items organically or there is a separate contract with the prime contractor/OEM to provide support.

The following three subsections identify areas where communication should be established in the prepare step to encourage DMSMS management of FMS equipment to be as proactive as possible, through a combination of efforts by DoD program office leadership, DMSMS management practitioners, and security cooperation points of contact in the program office.

3.5.1 Communication with LOA Drafters

The DMT should make efforts to learn of potential foreign sales as early as possible during the LOR development process through periodic contact with the people responsible for FMS in the program office. The DMT should then establish contact with LOA drafters to provide DMSMS management information

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discussed in the following subsections. In many instances, the PSM may be the one to establish contact. Having this information in the LOA is key to the successful performance of DMSMS management on FMS equipment, since it is the official mechanism through which the DoD program office is authorized to conduct DMSMS management on that equipment. *The information should be provided to the LOA drafters regardless of whether it was specifically requested*.

3.5.1.1 Including all Necessary DMSMS Management Requirements in the LOA

Unless other arrangements have been made, the DMT and the PSM should try to convince LOA drafters that DMSMS management should be explicit in the LOA. Rationale include the following:

- DMSMS management is a normal part of development, production, and sustainment of a system as evidenced by actual spending on it in the last few fiscal years; and
- Partners rely on the FMS equipment for their defense and therefore every effort should be made to maintain its supportability and availability.

Specifically, the DMT and the PSM should encourage that the following be included in the LOA[117]:

- A recognition of the need for DMSMS management because of the inevitable impacts of obsolescence on the supportability and availability of the FMS equipment. If DMSMS management is not mentioned in the LOA, the program office may not be able to provide proactive support, and consequently, the LOA may have to be modified at a future date. Also, the funding for DMSMS management could be misunderstood as being included in other sustainment support or similar functions. DMSMS risks should be clearly understood by the foreign partner.
- An explicit statement of the DMSMS management activities that the program office will perform. When there are particular actions (e.g., obsolescence monitoring, verifying whether a supply issue is caused by obsolescence, the extent of technical assistance to help a foreign partner resolve an issue) that the program office must perform, the proper language should be explicit in the LOA to enable the program office to undertake that work. Furthermore, being explicit about the DoD program office responsibilities helps avoid differing expectations about who is responsible for providing certain services or funding certain aspects of DMSMS management. Such differing expectations may arise from communication breakdowns or misunderstandings. Explicit and detailed language in the LOA reduces the likelihood of misunderstandings.
- An identification of data to be provided by the foreign partner that is needed by the DoD program office to perform its responsibilities in the LOA. Depending on the circumstances, the program office may need foreign partner-specific technical data to perform the identify, assess, and analyze steps of the DMSMS management process.[118] For example, the foreign partner's equipment may have a different design configuration than systems in the current DoD inventory, or if the foreign partner equipment interface specifications may be different than those used by DoD equipment:
  - Configuration management data may be needed to determine whether a resolution applies to a particular partner (see Section 4.7.2).
  - Technical specifications or BOMs often are needed to verify whether a DMSMS issue exists (see Section 4.7.1) or to develop resolution options (see Section 6.4.2).[119]

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[117] This is consistent with The Defense Security Cooperation Agency's Total Package Approach (TPA) as documented in its Security Assistance Management Manual. TPA is designed to ensure that FMS equipment can be operated and maintained in the future.

[118] Sometimes the OEM has technical data not shared with DoD or the foreign partner.

[119] BOMs are critical because they enable 1) proactive monitoring of items that are "FMS peculiar" or no longer used by DoD and 2) resolution negotiations with the current source.

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- Logistics and supply system data may be needed for proper analysis of impact and short- and long-term resolution options.

3.5.1.2 Estimating DMSMS Management Costs for the LOA

The LOR, as an FMS partner responsibility, should be the first place where cost estimates appear. DMSMS cost estimates in the LOR are more likely to be included in the subsequent LOA. When the LOR is being prepared, it is extremely important for the program office DMSMS management practitioners to provide their best estimate of total DMSMS management costs for the entire LOA period of performance. The PSM should insist that these estimates be made and strongly indicate the need to include them in both the LOR and LOA. If some costs are omitted from the LOA:

- Elements of DMSMS management with no cost estimate may be excluded from the LOA and not performed;
- The foreign partner's expectation of the DMSMS management work to be performed may differ from the DoD perception;
- There may be differences of opinion regarding who pays for certain DMSMS management activities; and
- Proactive DMSMS management may not be performed adequately.

There are two types of cost to consider depending on what the program office is responsible for per the LOA. The smaller component is the cost of DMSMS management operations strictly associated with the FMS equipment. For example, this might be the cost for the program office to provide technical assistance and necessary information to the foreign partner to verify or resolve a DMSMS issue itself. The more significant component when applicable is the partner's share of the cost to implement DMSMS resolutions for their FMS equipment throughout the LOA period of performance.

If the FMS case covers resolutions during the LOA period of performance, the cost for FMS equipment should be estimated in a manner similar to the way that the program office estimates the resolution cost for DoD equipment for its own programming and budgeting purposes, taking potential cost sharing considerations into account. Guidance on that is in Sections 3.4, 4.6, and 7.1 of this document. The organizations that will perform the work, e.g., organic depots or prime contractors, should be involved in developing the estimate.

Estimating resolution costs over the LOA period of performance may be difficult due to factors common to DoD and FMS equipment such as the following:

- Unanticipated obsolescence;
- Uncertainty on the year that resolutions will be funded (which may differ by partner);
- The foreign partner's operating timeframe for the FMS equipment may differ from DoD's operating timeframe;
- Deferral of resolution implementation; and
- Changes to the timing and content of future technology refreshments and insertions.

There can be additional complications for estimating resolution costs for FMS equipment such as the following:

- When an FMS case involves supporting old equipment no longer in DoD's inventory, the amount of obsolescence may be unclear, and the extent of redesign requirements may not be fully understood.
- There are many valid ways of determining a foreign partner's fair share of DMSMS resolution costs. Any approach is acceptable if there is a consensus that the arrangement is

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equitable.[120] But determining fair share requirements in advance is often difficult because it is hard to predict which foreign partners will be involved.

- More than one DoD program office may be involved. LOA cost estimates should include both the primary platform being purchased as well as government furnished property (GFP) included in the sale that is managed by a different DoD program office. Examples of GFP are test equipment, ground support equipment, and payloads.

Consequently, the program office's DMSMS management cost estimate for the LOA should be complete and conservative and may have to be adjusted if the LOA process is prolonged. LOA funding should include everything necessary to buy and sustain military equipment during the LOA's period of performance. Underestimating foreign partners sustainment costs leads to insufficient funds to maintain key defense systems and/or capabilities. Relying on management reserve funding or funds transfers from one element of the LOA to another are not best practices.[121]

One particular cost estimating scenario, when applicable, should not only be priced in the LOA, but also explained in detail to the foreign partner. There may be FMS platforms being manufactured with obsolescence issues that are not applicable to DoD assets because of a *prior* life-of-need buy that did not consider newly produced equipment strictly for foreign sales. This situation may occur when:

- A current LOA for the production of FMS equipment was initiated after a life-of-need buy was made for some combination of *prior* FMS cases and DoD needs; and
- Prime contractor/OEM has sufficient inventory of the obsolete items to fulfill the FMS production deliveries, but not the sustainment of that equipment.

To avoid delivering equipment that is unsupportable because of obsolescence, the partner should be informed of the situation and the cost of the LOA adjusted to include whatever is needed to make the platform sustainable during the period of performance of the LOA.

3.5.2 Communication with Foreign Partners

Communication with foreign partners begins when an LOR is being discussed and lasts throughout the period of performance of the FMS case. The DMT (with PSM help if needed) should encourage the establishment and the use of bi-directional communication—both foreign partner requests to learn about DoD DMSMS management processes and program office DMSMS management practitioners' efforts to inform foreign partners of current and projected DMSMS issues and resolution options.

Using the (direct and/or indirect) communication channels that apply to the specific FMS case, the DMT should urge foreign partners to request information on DMSMS management processes so that the partners have confidence in the following:

- DoD considers DMSMS management to be very important during both system development and sustainment.
- The DoD DMSMS management process is rigorous and thorough, and partners' needs are fully considered.
- DoD's need for and use of technical data on foreign partner equipment is *integral* to its DMSMS management process.

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[120] Typically, the number of systems owned by the foreign partner is an important determinant of the cost share.

[121] Because of the unpredictable aspects of DMSMS, cost estimates may be too high or too low. The DoD program office DMSMS practitioners should remind FMS partners of this through the appropriate communications channel and suggest that partners make arrangements to enable transferring funding to or from DMSMS activities in the LOA based on actual costs.

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- Nothing will be overlooked when performing DMSMS management on FMS equipment consistent with the LOA.
- Proactive processes are in place to anticipate future issues and early resolution planning is underway.

These confidence building activities apply between the program office's DMSMS management lead and both partner officers stationed in the U.S., as well as in-country leadership, *and should be repeated as personnel change*.

To enable early and knowledge-based decision making by foreign partners, the DMT should also promote establishing frequent and thorough processes to inform partners about:

- The status of ongoing DMSMS monitoring and resolution implementation for their equipment;
- Current and future DMSMS issues along with any changes to future obsolescence projections;
- Possible courses of action to resolve or defer DMSMS issues along with the pros and cons of each; and
- Planned capability improvements and technology refreshments to delay future obsolescence.

3.5.3 Communication with Other Involved DoD Program Offices

As indicated previously, multiple DoD program offices may be involved in an FMS sale. Such sales may be managed as a single FMS case that includes related government furnished property (GFP) or multiple FMS cases. When other GFP is included in a single FMS case, the DoD program office for the overall platform being purchased is responsible for ensuring that DMSMS management is performed on the GFP. Usually, a matrix team, established by the DoD platform program office, can facilitate sufficient communication if the team meets on a regular basis. The matrix team meeting agenda may go beyond DMSMS management. The situation can become more complicated when the GFP is from another service.

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Section 4. Identify: DMSMS Monitoring and Surveillance

This chapter describes monitoring and surveillance, which constitute the *Identify* step of the DMSMS management process. This second step requires a program office to monitor and survey its items and materials for EOL notices or other indicators of potential discontinuance. DMSMS monitoring and surveillance should begin as early as possible during the design phase and continue throughout the entire life cycle of the system. This section describes the monitoring and surveillance processes (whether accomplished within the government, by a contractor, or preferably as a combined team):

- ***System prioritization***. This process entails determining the scope and focus (e.g., which subsystems of the system are of most interest, due to criticality, operational safety, or associated DMSMS-related costs) for the DMSMS management effort.
- ***Identification and procurement of monitoring and surveillance tools***. This process entails identifying and procuring (or procuring access to) the DMSMS predictive forecasting and associated data collection and management tools to support the DMSMS management program.
- ***Collection and preparation of item data***. This process encompasses collecting (or if necessary, creating) BOMs and item data and the prioritization of items to eliminate those that can be easily replaced (such as fasteners) through items availability analysis. It also includes preparing the BOM/parts list and loading into a predictive tool to analyze item availability.
- ***Analysis of item availability***. This process includes combining market research and the use of predictive tools to determine initial, and subsequent, item availability baselines for immediate and near-term obsolescence issues for the program office.
- ***Assess preliminary design for DMSMS risk***. This process reviews new and revised designs for DMSMS risk. It focuses on initial parts selection and recommends mitigating actions when high risk parts have been identified.
- ***Forecast technology obsolescence***. This process uses technology management to forecast obsolescence risk for the system and tentatively identify when issues may potentially occur. It sets the stage for mitigation planning through technology refreshment or insertion.

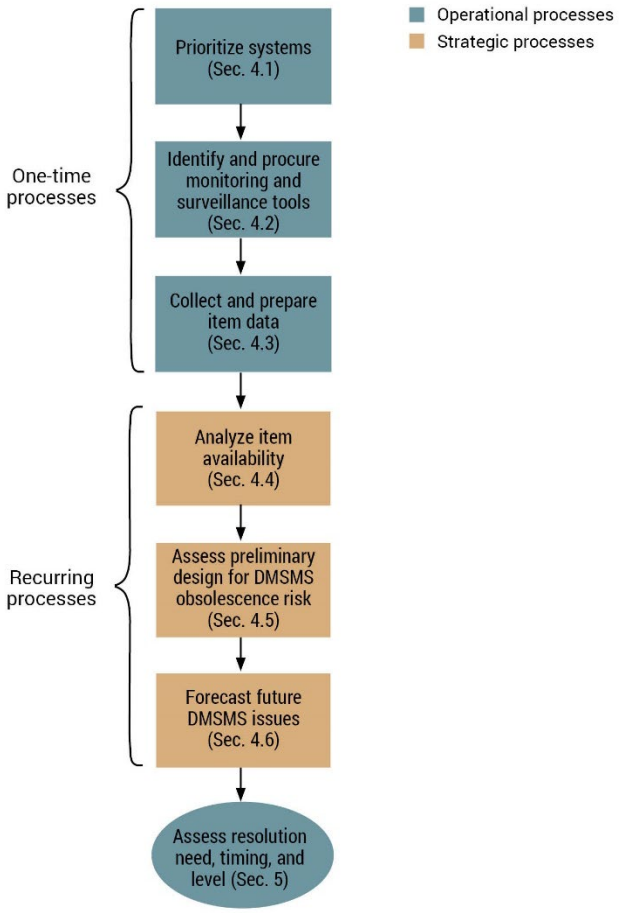
At the end of the *Identify* step, a preliminary health assessment report (see Section 5.3.1) could be generated to inform program leadership of the immediate results.

Figure 8 identifies the one-time processes and the recurring processes associated with DMSMS monitoring and surveillance. For the most part, system prioritization, identification, and procurement of monitoring and surveillance tools and the collection and preparation of item data are one-time processes. However, depending on when the prioritization was done, new data on DMSMS issues may lead to additional systems being given a high priority.[122] The major data cleanup effort during DMSMS management start-up for the program office will also require upkeep to maintain consistency with configuration changes once DMSMS management has reached a steady state.

[122] Since these processes are generally performed once, they could be placed in the *Prepare* step. This document includes them in *Identify* step because of their close association with the identify function. Overall process flow is unaffected if they are inserted at the end of the *Prepare* step or at the beginning of the *Identify* step.

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Figure 8. DMSMS Monitoring and Surveillance Processes



The other three processes—analyze item availability, assess preliminary design for DMSMS risk, and forecast technology obsolescence—recur when warranted throughout the life of the system.

4.1 Prioritize Systems

Robust DMSMS management may require monitoring and surveilling thousands of items simultaneously. It could take months or even years, depending on the size of the system, the availability and format of data, and the program office's manpower to load all BOMs into a predictive tool. Prioritizing the scope and focus for the DMSMS management program, using a risk-based approach, is crucial for a complex platform, which typically has many subsystems, each with multiple units with multiple assemblies, which in turn include many items and software. Prioritization in this process is not ranking subsystems to monitor; rather, it is making a "yes" or "no" DMT determination of what portions of the system to actively monitor, when, and at what frequency.

Because it is prohibitively expensive to monitor everything, a risk-based perspective should be taken to determine the priority of what to monitor. Some program offices may not even be able to obtain the resources needed to monitor at the desired level of risk. When that occurs, available funding will determine how much risk to accept. However, accepting too much DMSMS risk has led to large,

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unexpected resolution costs. Initial prioritization of the portions of the system on which to focus the DMSMS management effort can be based upon the following:

- ***Safety***. A top priority for the scope and focus of a DMSMS management program is any subsystem containing a critical characteristic whose failure, malfunction, or absence could cause a catastrophic failure, loss, or serious damage resulting in an unsafe condition. Special attention should be paid to aircraft, missiles, rockets, and airborne systems, as well as to other systems that involve command, steerage, and propulsion of ships or land vehicles. Similar safety concerns on other systems should be identified by the program offices.
- ***Mission criticality***. Another top priority for the scope and focus of a DMSMS management program is any system—whether a primary mission system or an auxiliary or supporting system—whose operational effectiveness and operational suitability are essential to successful completion of the mission or to aggregate residual combat capability. Such systems are critical, because if the system fails, the mission likely will not be completed, especially if there is a known single point of failure or a significant impact on NHAs.
- ***DMSMS-related costs***. Any subsystem experiencing or expected to experience frequent or expensive DMSMS-related issues should be monitored. Considerations for identifying subsystems under this criterion, before actual data are available, include unique fit or materials, closed architecture, modified COTS assemblies, high electronics content, high redesign costs, single source, low reliability, or hard-to-support software.
- ***Existing problems/historically troublesome***. If DMSMS management is already underway, program office management may already have a sense of those subsystems and software that cause the most headaches. If a DMSMS management program is just starting, program office management might still be able to look to predecessor platforms to identify areas that have consistently proven to be trouble spots. Data to review in these instances include the reliability (e.g., low mean-time-between-failure rates or high mean-time-to-repair rates) of the assemblies, software, and other items. Another area of interest are those items that are common across platforms and, therefore, could have a potentially large impact if they fail.
- ***Life Cycle phase***. The system prioritization process may vary as a function of life cycle phase:
    - For systems in design and production, actual data may not be available to understand where high costs or frequent DMSMS issues are occurring. There may be only some near-term indications of such areas based upon ongoing monitoring and surveillance. These areas are expanded as the system matures. It is especially important to identify DMSMS issues during design or production, because decisions made during those phases can significantly affect the system's life cycle. Furthermore, when obsolete items are not eliminated from product designs, higher risk distributors are more likely to be used to obtain items that are no longer in production. This adds to the risk of finding counterfeit parts in the DoD supply chain and, more important, in DoD weapon systems used by the warfighter.
    - Over time, the sustainment strategy may evolve; consequently, the mix of organic and contractor roles may change.
    - Once a subsystem has been fielded, there is a greater potential that an obsolescence impact on that subsystem could be felt directly by the warfighter in terms of readiness. This may be less of an issue if a program office knows that it has sufficient spares availability.
- ***Sustainment strategy***. A system's sustainment strategy reflects the maintenance or support concept of operations for that system. Such strategies consider impacts on system capability requirements, responsiveness of the integrated supply chains across government and industry, maintenance of long-term competitive pressures on government and industry providers, and effective integration of system support that is transparent to the warfighter and provides total combat sustainment capability. The DMT should be particularly concerned with

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these issues if the government is providing sustainment support. If a contractor is required to resolve DMSMS issues, then the DMT's primary role is to oversee the contractor's efforts.

- ***Availability of technical data***. Although a program office may prefer to implement robust DMSMS management over all priority subsystems, the reality is that not all the BOM/parts list data may be available (or can be constructed) to do so, particularly at the start-up of DMSMS management for a system. In such instances, program offices will not be able to scrutinize priority subsystems until sufficient data become available. However, a program office should not postpone or avoid pursuing the necessary data for known, more complex and troublesome subsystems.
- ***Vulnerability to supply chain exploitations***. The potential for supply chain compromise is similar to safety and mission criticality. When an item is in short supply, less trustworthy sources may be used to meet demand. Using these sources increases the chance of exploitation of a supply chain vulnerability, leading to compromised items on DoD systems and in the supply system, potentially affecting safety and mission effectiveness. Consequently, SCRM should prioritize items where the supply chain exhibits weaknesses in asset-based security controls, hardening against digital threats, or life cycle security management.

Each program office will need to determine the factors of most importance to prioritize its subsystems for DMSMS monitoring and surveillance. Among the factors of interest to a program office, not all will be of equal importance. To address this, a program office might wish to develop a weighting scheme to help sort the system priorities for its monitoring and surveillance effort. A DMSMS management program that already is well underway, and for which historical data are available, could consider establishing a method that considers the likelihood and consequence of a particular subsystem being degraded due to obsolescence.

This prioritization will help determine which BOMs the program office needs to obtain to support proactive monitoring for DMSMS issues. If a reactive approach to DMSMS management is sufficient for a particular subsystem, then it is not necessary for the program office to acquire or build that subsystem's BOM.

4.2 Identify and Procure Monitoring and Surveillance Tools

The program office should identify and procure predictive obsolescence tools and associated data management tools needed to support DMSMS monitoring and surveillance. Predictive tools may be particularly useful for analyzing certain types of items, such as electronics; however, these tools have limited capability for other types of items, such as mechanical hardware or COTS assemblies. Most DMSMS predictive tools perform the same core functions of monitoring the availability of electronic items in the BOM and forecasting their obsolescence. Each tool has a set of loading criteria and formats, output report formats, and other information that can be ascertained from the loaded BOM.

Beyond predictive obsolescence tools, BOM data management tools, configuration tools, logistics data collection tools, data storage and retrieval tools, and report generation tools are all needed for monitoring and surveillance. Selection criteria include reliability, user friendliness, cost, and usability by multiple systems. As discussed in Section 3, DMSMS management can include both proactive functions and data collection and management functions.

4.3 Collect and Prepare Item Data

Once the focus and scope of the DMSMS management program have been determined by the prioritization of subsystems based upon mission criticality, operational safety, and so on, the data necessary to support item availability analysis and, ultimately, whether and when an obsolescence issue should be addressed should be identified and collected. Indeed, program office leadership should ensure that the data to support DMSMS management are obtained. Item data, including parts and software lists/

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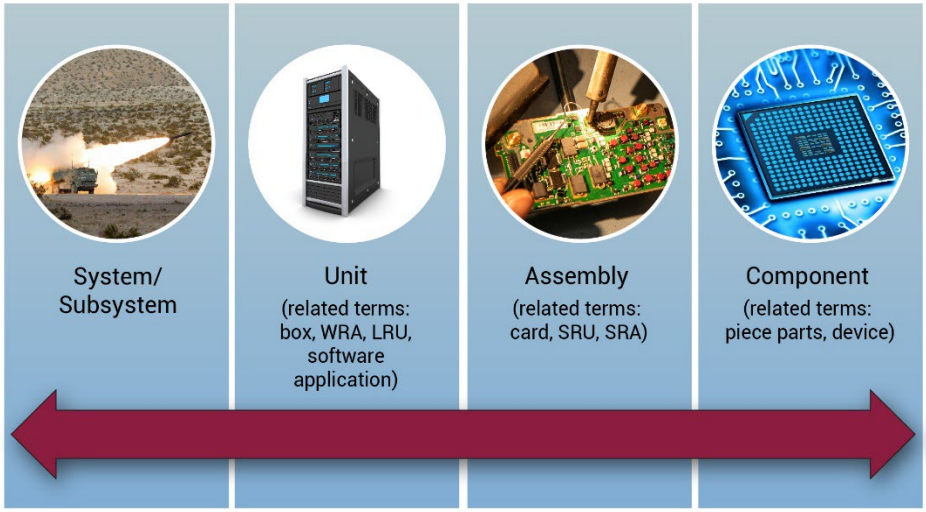
BOMs and additional information obtained from market surveys, are used to analyze item availability, resulting in a list of system items that have immediate, or anticipated, near-term obsolescence issues.

4.3.1 Item Data Collection

4.3.1.1 Hierarchy of System Items

To adequately and cost effectively address obsolescence for a program office, the DMT may have to monitor, assess, and resolve DMSMS issues at different and multiple levels within a system. Figure 9 illustrates the hierarchy of system items. As one moves from left to right across the figure, the system is decomposed into increasingly smaller items, from unit to assembly to component. For each of the items of the system, additional related terms are also provided. So, for example, when referencing the component level, a program office might use other terms like *piece parts* and *device*.

Figure 9. Hierarchy of System Items



Note: LRU = line replaceable unit, SRA = shop replaceable assembly, SRU = shop replaceable unit, WRA = weapon replaceable assembly.

4.3.1.2 Different Types of Item Data

Different types of items are likely to be incorporated into the design of any system. Therefore, the DMT needs to be aware of how different types of data may need to be collected or even suggest different means of collecting or developing and managing the data. The following sections contain such information for both COTS and hardware—electronic and MaSME items and for software.

4.3.1.2.1 Hardware—Electronic and MaSME Items

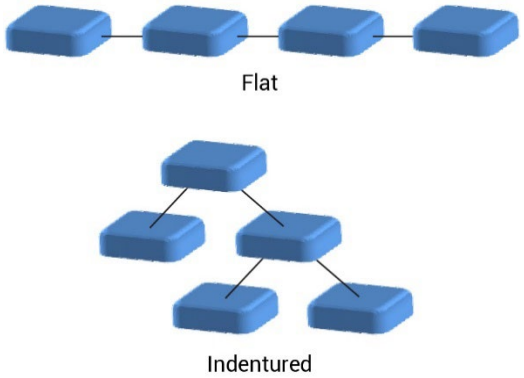
For hardware—electronic and MaSME items data, a parts list or BOM is an indispensable data resource for robust DMSMS management. BOMs are the basis for monitoring for DMSMS issues, and as such, they are far and away the most important data type necessary for proactive DMSMS management (and even reactive DMSMS management). Without a parts list or BOM, item availability analysis, an assessment of whether and when to address an obsolescence issue, and the continuous prediction of discontinuance by a DMSMS management program are impossible.[123]

[123] This does not imply that the government must have a BOM. DMSMS can be managed by the prime contractor. See Appendix F.

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A BOM identifies the materials, components, and assemblies used in making a unit. It answers the question: what items are on the system? The list may be in a flat format or an indentured format. A flat BOM is a simple list of items, while an indentured BOM shows the relationships (generally in a top-down breakout format) of components to assemblies to units to the system. Figure 10 depicts the two formats.

Figure 10. Comparison of Flat and Indentured BOMs



Because it provides a bigger picture for identifying and weighing resolution options for an identified DMSMS issue, an indentured BOM format is preferred over a flat format (although there could be IP issues obtaining an indentured BOM from some suppliers). For example, when analyzing item availability, a flat BOM enables the identification of only the number of obsolete items within the unit; it would not provide any indication of whether some of the items are on the same assembly. Not knowing the effect of the identified, immediate, and predicted obsolescence issues on the system's item hierarchy limits resolution options. In some cases, it may be more cost effective to perform a minor redesign of an assembly, rather than undertaking LON buys of multiple components within that assembly. An indentured BOM enables the program office to more readily visualize the relationships of identified obsolescence issues within the system and to use this information to inform the identification and determination of potential resolution options.

In addition to the configuration (indenture) information conveyed through an indentured parts list or BOM, useful item data pertaining to the components, assemblies, and units of the system include the following (sometimes difficult to obtain) elements:

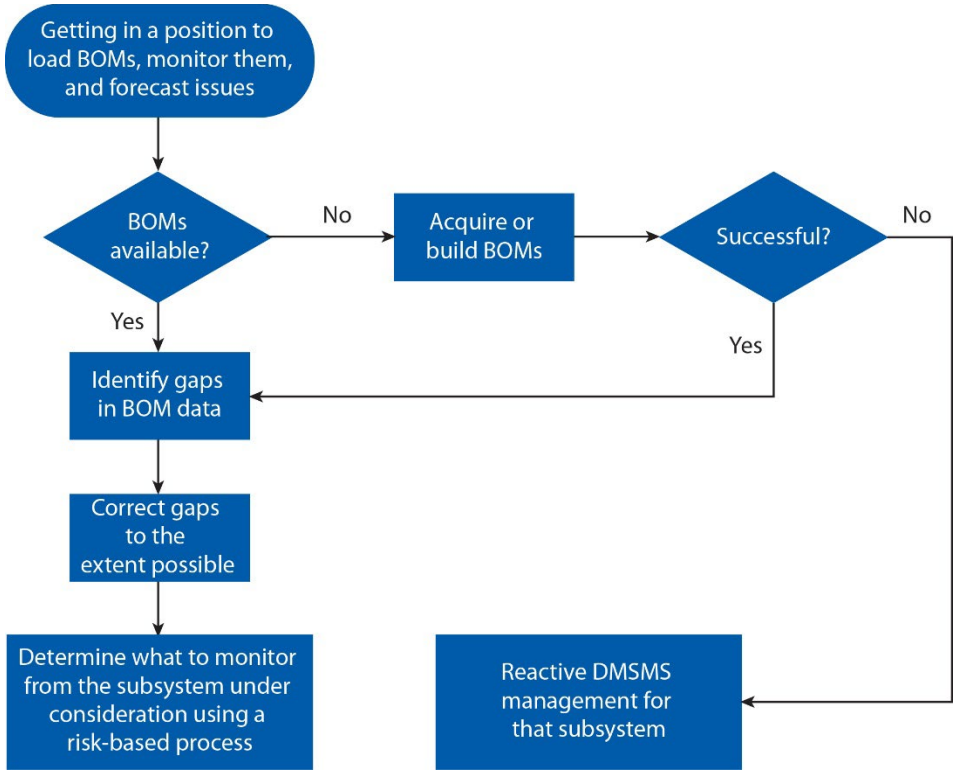
- OEM-approved alternatives,
- OEM technical manuals,
- OEM DMSMS mitigation efforts underway,
- OCM part number,
- Sources of active manufacturing,
- Actual or projected EOL,
- Function (active vs. passive, complexity),
- Type (custom, hybrid, proprietary),
- RoHS/Pb-free information, and
- F3 details.

For MaSME items listed on a BOM, the items of interest may be reduced after production is completed by using a Provisioning Master Record (PMR), which includes only those items that are purchased by the DoD supply system. PMRs also provide additional information that can be used to further reduce the number of items to monitor, as described under Section 4.3.2.1.

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One of the first tasks is to obtain the BOMs for the system. Figure 11 portrays a high-level process flow of the activities of this task. It focuses on obtaining BOMs and how access to BOMs and the quality of those BOMs will impact the program office's DMSMS management approach. The objective of the process is to position the DMSMS management providers to load BOMs into predictive tools, monitor the items in the BOMs, and forecast DMSMS issues.

Figure 11. Preparing for Proactive, Risk-Based DMSMS Management



The DMT's work regarding this task begins with identifying whether BOMs are available.

- *If the program office already has the BOMs or a mechanism in place for accessing them*, then information gaps within them are identified, the gaps are closed to the extent possible, and finally the specific items to be monitored are identified using a risk-based approach (see Section 4.3.2.1).
- *If the program office does not have the BOMs or a mechanism in place to access them,* the BOMs will need to be acquired or developed, where feasible to do so. After such BOMs are acquired or developed, then the same process steps apply as for when BOMs are available. When BOMs cannot be acquired or developed or gaps cannot be closed, items are monitored reactively, which means DMSMS issues will not be identified until there is a failed attempt to purchase an obsolete item.

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The best situation is one in which the government has an established contractual requirement for the BOMs (and for notional BOMs or parts lists during design).[124] Contractual language, including data rights issues, is important to establish up front between the program office and prime contractor. Prime contractors often have to negotiate with OEMs for access to their BOMs, so it is not valid to assume that a prime contractor will automatically be able to make these available to the program office. When a contractual requirement is not in place early in the system's life cycle, all is not lost. Appendix J contains information on considerations (including IP concerns) for program offices that find themselves in the position of needing to acquire (when not contractually required up front) or develop BOMs.

BOMs for COTS assemblies are not usually readily available and may not be cost effective to obtain if available.[125] However, for a COTS-intensive system, a program office may want to investigate whether BOMs can be obtained and, if so, develop the cooperative arrangements necessary to ensure delivery of those COTS BOMs, if cost effective.

4.3.1.2.2 Software

A complete list of software items should be required by contract with associated CDRLs and DIDs. If that was not the case, there are other existing artifacts that could have CDRLs that may provide some or all the necessary information. The cybersecurity (information assurance) engineering DID requests product, version, and manufacturer data for software.[126] CM documents, drawings, and technical data packages (TDPs) are other potential sources for identifying the software in a system. For a third source, software is often a part of the software version description.[127] A fourth source is a data rights disclosure letter if it is a requirement on the contract. This letter lists all areas for which the government does not have full data rights, including commercial software applications and contractor proprietary software. A final source is a software licensing management group, if one exists.

Just as an indentured BOM shows hierarchical interrelationships among items, software interdependencies should also be captured. For each listed software element, the software and hardware that depend on it and the software and hardware upon which it is dependent should both be identified. Software interdependencies may not be hierarchical; there can be cross-system relationships. An understanding of these relationships will not be found in a BOM; it is best achieved through discussions with systems engineers and/or software developers or may be identified in interface control documents. Consideration should be given to modifying the DMSMS DID or creating a new one for software.

Given the possibility that multiple sources may be queried to develop lists of software items and their interrelationships with hardware and other software items, it is important to be cognizant of the following challenges that could inhibit DMSMS management for software:

- Inaccurate or incomplete part numbers, nomenclatures, and software descriptions and

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[124] To understand the data rights, see the original procurement contract and any follow-on contracts. The contracts usually contain specific detail on the data rights for items delivered as contained in DD250 forms. Using product data for government purposes, such as monitoring integrated BOM part numbers for EOL warnings and using product data for competitive reprocurement are significantly different. DoD should obtain technical data rights commensurate with the sustainment strategies of the systems used in its global defense missions so that it can ensure they remain affordable and sustainable. For more information about data rights, refer to the *Army Data & Data Rights Guide*, August 2015, available at https://www.dau.edu/sites/default/files/Migrated/CopDocuments/Army%20Data%20and%20Data%20Rights%20Guide%201st%20Edition%204%20Aug%202015.pdf.

[125] In contrast, BOMs may be available for non-developmental items (NDIs) designed for the government.

[126] DI-MISC-80508B, "Technical Report - Study/Services," revised November 14, 2006, available at https://quicksearch.dla.mil/qsDocDetails.aspx?ident_number=204915.

[127] DI-IPSC-81442A, "Software Version Description," revised January 11, 2000, available at https://quicksearch.dla.mil/qsDocDetails.aspx?ident_number=205921.

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- Inconsistent versions captured (e.g., decimal points captured not matching OEM development and product releases).

4.3.2 Item Data Preparation

By this point, the program office has identified and collected all relevant item data as well as selected and procured the appropriate DMSMS predictive tools. Before loading the parts lists/BOMs into the tools, the program office should take several final steps to prepare the item data for recurring analysis of item availability. First, the parts lists/BOMs should be reviewed to identify the items on which the program office's DMSMS monitoring and surveillance activity will focus. This is a second prioritization filter—the first being the prioritization of subsystems on which to focus the DMSMS management effort (see Section 4.1)—that considers the criticality and safety as well as the vulnerability of particular items within the system design.

Because of rapid technology changes, robust DMSMS management and proactive monitoring for electronic items has generally been the primary focus of DoD's DMSMS management guidance and effort. Being the center of attention, however, does not necessarily equate to adequate funding. Program offices often struggle to obtain and maintain budgets for proactive DMSMS management in this area.

Narrowly targeting the items to monitor can have a large impact on the cost of a program office's DMSMS management effort, thereby focusing attention on the higher risk items. For a program office dealing with a larger number of items, applying this second prioritization filter could eliminate large portions of the BOM from the monitoring requirement. Culling out lower-vulnerability items from a DMSMS monitoring effort may be less critical if the program office is not dealing with many items.

4.3.2.1 Hardware—Electronic and MaSME items

Parts lists and BOMs may contain any number of items (e.g., fasteners) that do not need to be analyzed with a predictive tool because of the availability of so many alternatives. As described in Section 3.1, to make more effective use of limited resources, program offices should adopt a risk-based approach to proactive monitoring. For items that are listed on parts lists and BOMs, this section contains more details on the application of the first two strategic determinations—applying algorithms based on life cycle estimates and further analysis of uncategorized items—associated with determining which items to monitor.

The first determination involves applying algorithms based on life cycle estimates. In some instances, a supply system health analysis is performed to offer up a broad-based supportability picture and identify low-, medium-, and high-risk candidates from a support perspective. The results of applying the first determination to hardware—electronic and MaSME items listed on parts lists/BOMs are the following three categories of items:

- ***Items to definitely monitor***. These items include certain item classes known to have a high propensity for obsolescence issues. These item types include electronic COTS assemblies (e.g., networking gear, computers), active components, radiofrequency components, programmable devices, memory, microprocessors, ASICs, hybrids, and custom electronic assemblies. Assemblies that contain sole-source items that are in low demand also should be proactively monitored.[128] Custom passive items are also prime candidates for monitoring. In addition, if a design contains materials with chemical properties that are a function of the design, are sole source, and/or are otherwise potentially threatening to the environment, these materials should be monitored. All electromechanical items should also be included in

[128] Obsolescence of complex assemblies is often caused by obsolescence of their critical items.

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a program office's monitoring efforts. This subset of item types generally introduces high risk to a system if the program office chooses not to monitor them.

- ***Items not to monitor***. There are two types of items not to monitor:
  - *Items where no further action is required*. This subset of item types includes standard/ common industrial items, such as mechanical components, connectors, cabling, and certain consumables, that typically do not present a significant risk, because most of these items are easily and quickly replaced when they become obsolete. Generally, these items can be eliminated from monitoring. Some circumstances, however, warrant a DMT's monitoring of these types of items. For example, some items may have something unique about their operating environment, may be identified by the DMT as important, or may require extensive requalification if replaced. The DMSMS SME and engineering activity representative should understand the associated risk before choosing not to monitor such items and should revalidate that decision periodically. Information from suppliers may be needed to fully understand such risks.
  - *Items where preparations should be taken*. Custom-fabricated items (e.g., fenders or castings) that will no longer be produced after final delivery also should not be monitored for DMSMS issues; however, logistics managers and PMs should ensure that enough of these items are acquired for system sustainment through system disposition. As a safeguard, the program office should obtain sufficient documentation to enable the reacquisition of custom-fabricated items in case of future need, through new acquisition contracts.[129]
- ***Items for which not enough is known to determine the need for monitoring***. The final category of items is uncategorized items, because not enough information is known to determine whether the program office should monitor these items.

For this final group of uncategorized electronic and MaSME items, a program office has three options from which to choose:

- ***Monitor all these items***. This is a low risk for being caught off guard with an obsolescence issue, high-monitoring cost approach.
- ***Do not monitor any of these items***. This is a high risk for being caught off guard with an obsolescence issue, low-monitoring cost approach.
- ***Conduct further analyses to determine which items to monitor***. This approach optimizes the risk associated with being caught off guard with an obsolescence issue and monitoring cost.

From a risk-based and resource-constrained perspective, the latter option should only lead to monitoring those uncategorized items where the negative effects of a reactive approach are both most likely and most severe. The decision to pursue a reactive approach to DMSMS monitoring implies that the system will experience no severe ill effects from waiting until an item cannot be obtained before seeking a resolution. A reactive approach should be sufficient unless significant risks are present.

In applying the second determination, a program office performs additional analysis to determine which of the items from the "uncategorized" list to proactively monitor. Three risk categories should be used to determine where a proactive approach should be taken for a particular material or item. These risk categories are as follows:

- ***Item criticality***. This risk category addresses the degree to which an item (whether it is an assembly or a component used to repair an assembly) is critical to the functionality of the

[129] DAU's *Systems Engineering Brainbook* provides guidance on the technical data and the manufacturing processes documentation necessary to reacquire custom fabricated items, https://www.dau.edu/tools/se-brainbook.

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system and ultimately the operational readiness of the unit employing that system. Quantitative and qualitative considerations for this risk category include the following:

- Critical safety item,
- Mission criticality,[130]
- Item essentiality code,
- High demand (perhaps in top 10%), and
- High cost.

The first three considerations are direct indicators of criticality. Obviously, items that are mission-critical or critical safety items meet the high criticality criterion. While critical safety items may be clearly identified, mission-criticality items are sometimes more ambiguous. The item essentiality code is an attempt to help classify these items, but it is not always accurate. There also should be a high correlation between high-cost and/or high-demand items and criticality. These considerations, depending on the system, can therefore serve as factors of interest in their own right or in combination as a reasonable proxy for mission criticality when no other data are available.

Provisioning and other DLA databases can also be used to identify potentially critical items that should be monitored. One indicator of criticality is the Weapon System Essentiality Code field. Values of 1, 5, 6, and 7 could be considered critical. Those values are defined as follows:

- 1: Failure of the item will render the end item inoperable.
- 5: The item is needed for personal safety.
- 6: The item is needed for a legal, climatic, or other requirement peculiar to the planned environment.
- 7: The item is needed to prevent impairment of or temporary reduction of operational effectiveness of the end item.

Another indicator of criticality is whether the item is technically critical by reason of tolerance, fit restrictions, application, nuclear hardness properties, safety, or other characteristics that affect identification of the item. The Criticality Code field provides some data on these factors. Values of E, F, H, M, and S could be considered critical as follows:

- E: The item is an aviation critical safety item and is nuclear hard.
- F: The item is an aviation critical safety item.
- H: The item is nuclear hard.
- M: The item is nuclear hard, and it has other critical features such as tolerance, fit restrictions, or application.
- S: A failure of the item will result in serious damage to equipment, or serious injury or death to personnel.

The Special Procedures Code (SPC) field also provides data on some of these factors. Pertinent SPC values are 1, 3, 4, and 5. They are defined as follows:

- 1: Aircraft launch and recovery equipment.
- 3: Navy critical safety items.
- 4: Intercontinental ballistic missile item.

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[130] For the Army, the Rand Corporation has developed a readiness indicator that could contribute to this factor. The Rand Readiness Indicator uses an Army deadline report that shows the number of times a part has appeared in that report and the total number of days that the part was in the report. See Eric Peltz, Patricia Boren, Marc Robbins, and Melvin Wolff, *Diagnosing the Army's Equipment Readiness: The Equipment Downtime Analyzer*. (Santa Monica, CA: RAND, 2002), available at https://www.rand.org/content/dam/rand/pubs/monograph_reports/2010/MR1481.sum.pdf, pp. 46–48.

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- 5: Army critical safety item.

In addition, requisition data may provide insight on criticality because a requisition identifies high priority orders, orders that imply a non-mission capable supply (NMCS) or equivalent situation, and the tactical status of the unit placing the order as indicated by the Joint Chiefs of Staff project code (e.g., wartime conditions). Requisition data also provide some insight on the impact of DMSMS risk on systems. Highest priority orders are those with priority designator code (PDC) values of 01, 02, or 03. When one of these PDC values has been assigned, the required delivery date (RDD) will indicate a mission capable (MICAP) incident in three circumstances:

- When the RDD is 999, implying expedited delivery;
- When the first character of the RDD is N, implying NMCS status; and
- When the first character of the RDD is E, implying anticipated NMCS status.

When a Navy requisition is associated with a casualty report, there may be a “W” in the fourth position from the right of the requisition number.

- ***Supply chain vulnerability***. This risk category represents a key difference between electronic items and MaSME items. In the former case, the item often becomes obsolete because of technology changes. For the latter, obsolescence is usually related to a source going out of business or changing its product line. Quantitative and qualitative considerations for this risk category include the following:
  - Source related (e.g., no identified source, sole source, or only foreign sources),
  - Financial health of the supplier (e.g., as measured by Dunn and Bradstreet),
  - Persistent backorders (perhaps as indicated by an increasing number of backorders for at least eight consecutive months),
  - Long customer wait times (perhaps top 10%),
  - Recent substantial price increase,
  - Time since last order (perhaps if more than three years),
  - Low demand, and
  - Life cycle of the items.

  The first three considerations examine the supply chain directly. A supply chain is potentially vulnerable if there is no source, just one source, or only foreign sources identified. Even if there are multiple sources, all unhealthy suppliers are a situation of concern. In some high-risk situations, it may be useful to conduct a specific financial analysis on a particular supplier, since sources such as Dunn and Bradstreet are not always current. Similarly, long customer wait times, especially when there are persistent backorders are indicators of a potential problem.

  The fourth and fifth considerations not only exacerbate the risk further but also indicate other less obvious supply chain vulnerabilities and scarcities. Long wait times may indicate more serious problems. Similarly, a sudden price increase may imply something is changing in the supply chain that will affect availability. Either of these factors may represent an early warning of future problems.

  If a long time has passed since the last order, there may be substantial uncertainty about supply chain vulnerability. A source may make a financial decision that it is not profitable to keep producing an item, if the demand signal for that item is low. Finally, if the item has a short technological life cycle, its supply chain is more vulnerable.

  As was the case with critical items, provisioning and other DLA databases have fields that indicate potential supply chain vulnerabilities and consequently should be monitored. Risks and fields indicating those risks are as follows.

  - *Items with explicit procurement risks*:

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Suspended, cancelled, obsolete, or discontinued items fall into this category. The Reference Number Variation Code (RNVC) field is one indicator of this. A value of "9" implies that the item of supply is inactive.

When DLA receives a discontinuation notice from an item's last viable source, the DLA Columbus DMSMS Office sets a DMSMS flag. The flag may also be set when a Military Department transfers an item to DLA that was indicated to be obsolete by the transferring Military Department. Presumably, the RNVC would be set to 9 if this happened.

The Acquisition Advice Code (AAC) field may indicate that an item is not procurable. When the AAC is set to "V," future procurement is not authorized, but there is inventory in stock. When there is no inventory on hand and future procurement is not authorized, the AAC is set to "Y." In the case of NSNs where the AAC is "X," the services have not provided the necessary information to DLA in a timely manner to enable procurement. This code may be used on a temporary basis.

- *Items with limited sources*:

  There may be a difference between what DLA data indicates as a sole source and an actual commercial industry sole source since there could be sources unknown to DLA. Sole source items can be identified in DLA data when three conditions are met.

  - First, the Acquisition Method Code field must be 3, 4, or 5. A value of "3" means that the item must be acquired directly from the actual manufacturer, whether the prime contractor is the actual manufacturer. When the Acquisition Method Code is "4," DLA must acquire, for the first time, directly from the actual manufacturer rather than the prime contractor who is not the actual manufacturer. Finally, if DLA, can only acquire through the prime contractor although the engineering data identifies another source, the Acquisition Method Code is "5."
  - Second, there is only one item associated with that NSN.
  - Third, the Reference Number Category Code–RNVC combination is either 3-2, indicating a primary source, or 5-2, indicating a secondary source.

  Data rights issues may also prevent DLA from procuring an item from another source. The Acquisition Method Suffix Code (AMSC) field identifies this situation. When the government's rights to use data in its possession is questionable, the AMSC is set to "A." An AMSC of "D" indicates that the data needed to procure the item is not available. If the rights to use the data needed to make a purchase are not owned by the government and cannot be purchased, an AMSC value of "P" is assigned. Lastly, when the AMSC is "R," the data or rights to use the data needed for purchase from additional sources are not owned by the government and it has been determined that it is not economical to purchase them.

  A CAGE code beginning with an "alpha" character indicates a foreign source. There is a risk if an NSN is only associated with a foreign source item.

- *Low inventory items*:

  NSNs where the inventory is zero and some yet to be determined long lead time and/or high demand present a DMSMS risk. A subcategory for zero inventory items that represents even greater risks are backordered items. The risks increase when the backorders are persistent.

- ***Time to implement a resolution***. The risk category addresses how long it will take to implement a resolution to a DMSMS issue for an item or material in comparison to the stocks on hand. When there is more than enough stock on hand and the time to implement is short, the risk would be viewed as low; however, if there is a long lead time to implement a resolution and the stocks on hand are not sufficient, the risk would be viewed as high. Quantitative and qualitative considerations for this risk category include the following:

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- TDP availability for structural, mechanical, and electrical items or electronic items (e.g., not available or limited data) or availability of the material specification for an engineered material[131] (knowledge of material composition will shorten cycle time);
- Source controlled;
- Manufacturing difficulty;
- Long lead time to requalify;
- Manufacturing cycle time;
- Availability of tooling and test equipment;
- Cost to implement a resolution; and
- Defense unique.

If technical data are not available, reverse engineering will be required. This takes a long time and adds significantly to the risk. Reverse engineering a MaSME item or electronic item is almost always feasible. Many items can be reverse engineered in three to six months, but some take much longer. Also, source-controlled items (e.g., items where the allowable or qualified sources are listed on the drawing) typically involve a longer time to implement a resolution.

Measuring manufacturing difficulty is subjective. Factors such as the need for specialized skill and high capital equipment costs are indicators of potential manufacturing difficulties for all items. Sometimes this is a reason for a source-controlled designation. For structural, mechanical, and electrical items and electronic items, manufacturing difficulty may be associated with unique manufacturing processes and/or demanding requirements, such as extreme tolerances. For materials manufacturing, difficulties may be associated with the presence of hazardous materials or processes, which require special handling; the use of other exotic materials that are not commercially available and with little demand outside this application; demanding requirements (e.g., long shelf life, compatibility with other materials, replacement materials must be the same material, performance outside of normal operating environment); and material that cannot be reliably recycled for use in another form.

Other measures of the time to implement axis include a long lead time for requalification, manufacturing cycle time, and/or the lack of tooling or test equipment. For example, shipboard testing may be proposed, which requires scheduling ship availability. The cost to implement a resolution can be an indicator of the time required. Finally, resolutions for defense unique items are likely to require more time to implement.

Figure 12 depicts a risk cube illustrating where proactive monitoring of uncategorized items is important. Using this risk cube, if an item is not mission or safety critical, then there will be little operational impact from a DMSMS issue. Resolutions for such items can be developed and implemented without time sensitivity. If the item is critical, but the supply chain is robust, the likelihood of a DMSMS issue is also small, because other suppliers should be able to satisfy demands. Even if the supply chain is not healthy for a critical item, proactive monitoring is only needed when the time to implement a resolution exceeds the length of time covered by the on-hand inventory for that item. A best practice is that a program office should only expend resources to proactively monitor those previously uncategorized items that are high risk across all three risk categories (in other words, red/red/red).

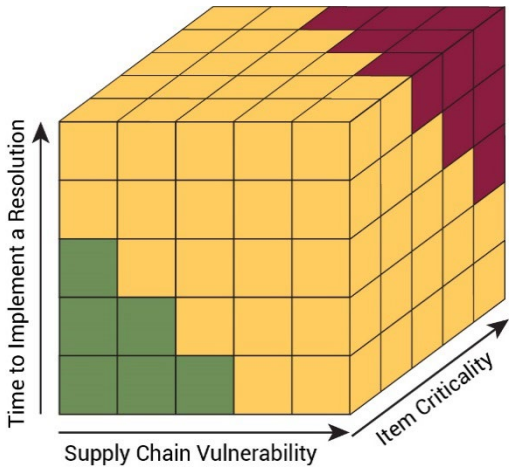
Once a program office has determined which of the above considerations within the three risk categories to use, a best practice is to categorize the items that were previously uncategorized based on the use of applicable algorithms. These items should be evaluated based upon those considerations suitable for evaluation through automated databases. Each consideration should also be assigned a weight. For all items being evaluated, the weights of all considerations should be summed and the items above a

[131] An engineered material is designed to perform a specific function and is composed of multiple raw materials.

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predetermined cut-off point should be monitored, while those below that cut-off point should not. Both the weights assigned to the considerations and the cut-off point should be determined by the program office, based on its unique needs.

Figure 12. Risk Cube for Determining Where Proactive Monitoring of Previously "Uncategorized" Items Is Important



Regardless of what level of risk analysis is conducted, an SME could adjust the classifications, including the "do not monitor" items. This evaluation will take into account risk cube considerations that could not be adequately measured by automated databases, if applicable. There may also be some known vulnerabilities. For example, the technical members of the DMT may be aware of items that are known to be a problem and items with pending environmental or safety regulations that may limit their availability and use in any area of the world where the system operates. As a bottom line, the DMSMS SME and engineering activity representative should understand the associated risk before choosing not to monitor any such items and should revalidate that decision periodically.

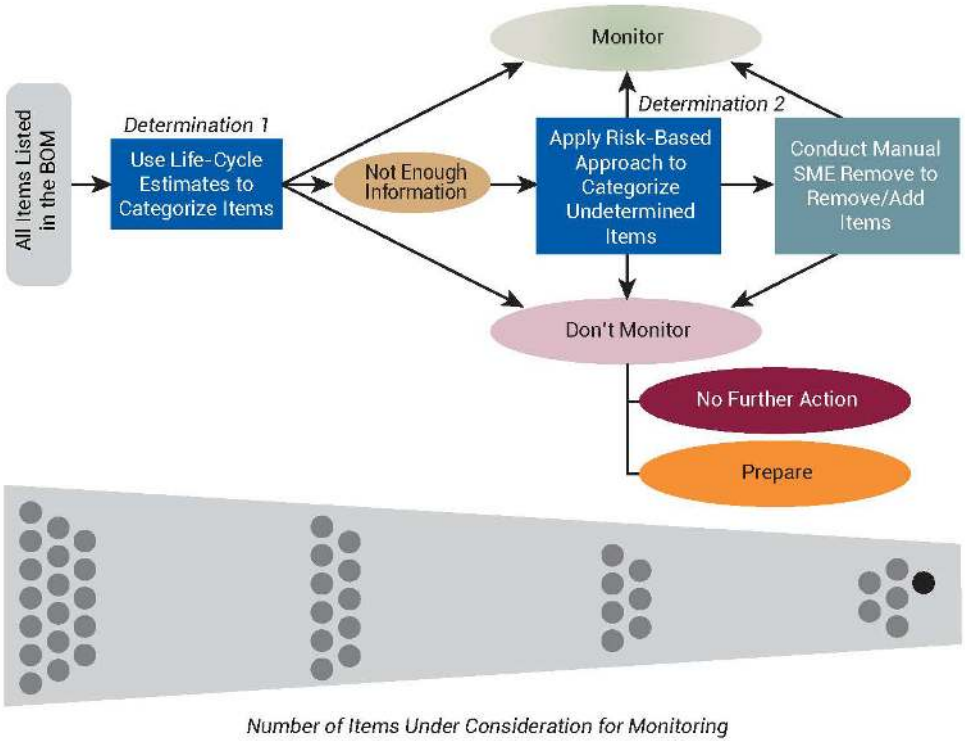
Because not all parts and components have the same obsolescence risk, the DMT should have an obsolescence management strategy for every item. The program office should carry forward the "definitely monitor" and "uncategorized" items that have been determined to be high risk across all three risk categories for availability analysis. The strategy for the "not to monitor" items is to find an alternative when they become obsolete, because ample replacements are available commercially, or to make the necessary preparations should a resolution be required down the road.

Once a program office has categorized and prioritized the items on its BOMs, the data for these items can be loaded into predictive tools where appropriate.

Figure 13 summarizes the entire process described in this section.

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Figure 13. Summary Illustration of Risk-Based Approach for Determining Which Items and Materials on the BOM to Monitor



4.3.2.2 Software

As part of this second layer of prioritization, a program office should also use a risk-based approach to determine where proactive monitoring for software obsolescence makes sense. Knowledgeable software experts should be consulted as needed. Taking a risk-based approach on what to monitor implies higher priority should be given to software that has a critical impact on the ability of the system to perform its missions. Other risk-based monitoring considerations for software include effect on safety, past problems with software, information assurance and cybersecurity, the complexity of interdependence with hardware and other software, and the frequency of updates.

A primary driver of obsolescence is the customization of defense system software for specific COTS operating system, middleware, and application software. Particularly regarding software, once a decision is made on what to monitor, there should be a risk-based determination of which versions or revisions to track, because further refinements (lower-level revisions) may fix only minor errors and not affect functionality or major vulnerabilities.

4.4 Analyze Item Availability

When all the items are analyzed for obsolescence (as determined via either predictive tool usage or vendor surveys), the magnitude of the program office's immediate and near-term DMSMS issues will begin to surface. Regardless of the method used, engineering analysis and judgment remain key factors in identifying DMSMS issues.

These items' availability status results represent a snapshot in time and, therefore, must be repeated throughout the life of the system, in response to the identification of new obsolescence notices, a vendor survey, or a regularly scheduled update to the predictive tools. If possible, a program office should receive daily DMSMS notifications that pertain to the electronics in its systems. Quarterly or annual alerts

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or vendor surveys may suffice for COTS items, but may be too late for electronic items, especially if a program office has only 30 days to make an LON buy. Update frequency may also differ for different commodity types, based on the availability of information, the rapidity of the technology's evolution, and the risk that the item or material poses to the system and mission.

Analyzing item availability should focus on identifying the items in each of the following three categories for further assessment:

- Items that are no longer available and for which no alternatives are available,[132]
- Items for which discontinuation notices have been issued, but some are still available, and
- Items projected to be out of production in the near future (where the time horizon is specified by the program office).

Sections 4.4.1–4.4.4 describe the four methods—predictive tools, vendor surveys, critical materials analysis, and PDNs—that prompt a refresh of a system's item availability status for hardware—electronic and MaSME items. Section 4.4.5 after that discusses some special considerations for software.

4.4.1 Predictive Tools

Some items (especially electronic items) are more readily analyzed using predictive tools. In contrast, most predictive tools do not cover MaSME items and COTS assemblies. Most program offices will not be in a situation in which they independently evaluate and purchase a license for using a predictive tool. More typically, either a parent organization will have bought a centralized subscription to one or more tools, or the program office will choose a DMSMS management provider (e.g., its prime contractor or independent SME)[133] that has its own set of tools. Tools, however, may be a selection factor for independent SME organizations and the program office may need to obtain a license for its prime contractor or OEMs.

In determining which predictive tools to use, a program office should consider the following desirable attributes:

- Manage accurate configurations,
- Enable real-time assessments of availability for items qualified for the system,
- Identify obsolescence issues and specific quantities per affected assembly,
- Identify after-market sources of supply,
- Create or generate timely alerts on production change notifications and PDNs,
- Enable real-time views of current item availability analysis,
- Provide highly accurate data,
- Provide flexible data input and query options,
- Allow comprehensive reporting options,

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[132] The fact that a predictive tool indicates the existence of an alternative item does not guarantee that the item will work successfully in legacy systems. The conversion of original hard-copy drawings to digital drawings for legacy systems may make it difficult to know why a particular source's item was chosen over another source's item that appears to be similar or the same. The hard-copy drawings may have indicated a difference that was not captured digitally. Therefore, the DMT should check with the engineering authority before concluding that an assessment of whether and when to address obsolescence and at what level is not needed.

[133] If a program office decides to have its prime contractor or OEMs perform monitoring and surveillance using DMSMS tools, the government must have access to the outputs of those tools for two key reasons: 1) to allow the government sufficient visibility for effective oversight and 2) to enable it to readily assume DMSMS management responsibilities if DMSMS management roles change. There may be IP issues if the outputs contain data that is proprietary to the prime contractor or a supplier to the prime contractor.

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- Contain many key items for the prioritized subsystems (or be able to add them),
- Contain information about counterfeits and information assurance and cybersecurity requirements,
- Rapidly develop obsolescence case sheets, providing streamlined and complete status of obsolete item issues when integrated with a DMSMS management system,
- Provide engineers with the data needed to evaluate and implement resolutions,
- Share notes and resolutions across all managed platforms and systems,
- Enhance productivity by minimizing the impact on engineering staffs while rapidly providing critical data needed for decision-making,
- Provide excellent customer service, and
- Provide good value.

A specific tool, alone, will not recognize all items in a BOM. An informal study of two predictive tools found that one of them successfully recognized only 71% of the items being researched by the team,[134] and the other recognized only 72% of the items being researched. When comparing the availability reported by the two predictive tools, the study found that the tools disagreed regarding the obsolescence status of 4% of the items being researched. There are legitimate reasons for these statistics. Different tools use different algorithms and philosophies to identify and report obsolescence. Also, the electronics industry changes rapidly, and new items are added daily. Furthermore, update schedules for the predictive tools vary, sometimes resulting in discrepancies in item availability status between tools. Therefore, if funding allows, and if practicable, the program office should use more than one predictive tool or a tool containing two or more predictive sources.

If the tools disagree regarding the obsolescence status of an item, then additional manual research is needed to confirm whether the item has an immediate or near-term obsolescence issue. If the item does not present an immediate or near-term obsolescence issue, it does not need to be assessed for DMSMS impacts. Even if the predictive tools agree that an obsolescence issue exists with respect to a particular item, a manual check should be done to confirm that finding.

Predictive tools may not provide an obsolescence status for some items. This may be due to an incorrect item number, a lack of identifying information, or the way the tool provider collects data. Also, the item type or the item's manufacturer may not be monitored by the tool. Some items may not be included in a tool's database. Items with unknown availability must not lead to a false sense of security. Additional work is needed to determine their availability; the risk will determine if this additional work is value added. It may be that data errors can simply be corrected to enable the predictive tools to forecast item availability. In other cases, manual research may be necessary. For example, the OCM, if known, should be contacted. Otherwise, inquiries should be made down the supply chain until the OCM can be identified and source control drawings can be accessed. If the item number is correct, another predictive tool may be used. Tool providers allow users to submit requests for items to be added to their library of monitored parts. Certain restrictions apply, but providers usually will add catalog item numbers at a subscriber's request. Subscribers of these tools should take full advantage of this to reduce the amount of manual research required for future BOM monitoring and receipt of EOL notifications.

During start-up, a program office may face a substantial manual research effort to perform an initial cleanup of the data and to confirm the obsolescence-related findings generated by predictive tools. Scoping the types of items to be monitored based on the application of a second prioritization filter (as

[134] The study was done approximately ten years ago. It reviewed all the systems (ranging from missiles to aviation and in all phases) monitored by the U.S. Army Aviation and Missile Research Development and Engineering Center. Based on that review, the study calculated recognition rates for the two predictive tools used by the center.

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described in Section 4.3.2.1) can help a program office reduce the level of effort it will need to employ to clean up and manually research such unknowns and verify statuses. Later, once a program office's DMSMS management has entered a steady state, research will need to be conducted only when certain predetermined conditions occur (e.g., item status changes, an item has not been researched in a certain amount of time, changes in sources of alternate items, packaging changes, and periodic revisit of previous "no action required" items).

Predictive tools should be used throughout the life cycle. Early in design, they should be used on notional BOMs or preferred parts lists; both are good sources of items that are likely to be used in production. Early design for new systems is usually based on existing designs being developed by the OEM. The starting point is rarely a predominately blank technical drawing.

4.4.2 Vendor Surveys

Predictive tools may not be able to forecast the availability of some items (such as COTS assemblies and MaSME items). This will also be the case for materials such as alloys, epoxies, glues, tapes, cooling fluids, and adhesives. In these instances, the best way to analyze availability is through vendor surveys, phone calls, emails, and vendor websites. To help make the research process repeatable, it is a best practice to document the steps taken to research items and decisions and agreements made.

It is helpful to develop a vendor survey questionnaire to manually interact with COTS and hardware—electronic and MaSME items manufacturers and knowledgeable material specialists, establish a database to capture and track the survey information, and determine the frequency to make contact for updates (again, prioritized based on criticality). Contacts for these surveys can be made through phone calls or email communication. A program office should be mindful to conduct these surveys in the least burdensome manner possible to increase the likelihood of responses. Developing relationships with vendors often improves response time and willingness to participate. Some data can also be collected by reviewing manufacturers' websites and other web research. This activity is particularly relevant to COTS items; manufacturers will often provide the status for the current item's planned life cycle, especially when a next-generation version is intended.

The following list provides some examples of vendor survey information and questions that a program office can include in its vendor survey questionnaires for electronic items.[135] The types of vendor survey questions suggested here for electronic items are generally applicable to MaSME items, as well; however, additional questions can be added to determine if someone other than the OEM is a likely candidate for providing future support for those items.

- What is the product name?
- What is the company name?
- What are the CAGE codes?
- What is the part number?
- What is the contact information?
- Is this item currently in production? If no, when did production end? If this product is no longer in production, can the government still purchase it? If yes, how many? When is the last

[135] The answers to these questions and any follow-up questions should be provided to the appropriate technology road-mapping community in the program office.

Although more questions can be asked, the response rate is likely to be higher if the vendor survey is brief. Additional questions specific to different types of software are provided in Section 4.4.5.

ANSI/VITA 53.0-2010, *Commercial Technology Market Surveillance*, is a reference to consider for additional questions to pose to a manufacturer (production start/end dates, end-of-support dates, failure, warranty, distributers, design changes, and so forth).

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date that the product can be purchased? If currently in production, when do you anticipate end of production?

- If you are not currently planning an end of production date for this product, please provide an estimate, based on similar products, past history, technology/item obsolescence, and so forth. (Keep in mind that this date is used for supply planning purposes only.)
- How long after the end of production will the government be able to have this product repaired? What's the typical cost to repair this item?
- Once production has been discontinued on the product, how much stock (in time) is typically available for sale? Are there considerations regarding shelf life (may be of particular interest for materials) of which the government should be aware?
- When this product is discontinued by your company, will you enter into an agreement with an after-market vendor so that customers can still buy the product? If yes (for this product or for other similar ones), please indicate the name of the vendor and give a point of contact.
- Is there a replacement or a planned upgrade to the product? Is the new item equivalent in terms of form, fit, and function? If so, what is the new product's part number and cost?
- What are your technology update plans? What additional capability will the new technologies provide?
- What warranty does the product have? What is the warranty length and can the length or start time be adjusted to allow for integration and deployment? What extended warrantees are available, and at what cost?
- What is the list price of the product and its lead time?

A key step in developing an obsolescence management strategy for MaSME- and COTS-based systems is to compile a list of equipment and items in the system and group them by the original manufacturer. With such a list, the DMT can make one phone call to each OCM and OEM to obtain obsolescence information about numerous items. Another helpful hint for a contractor that has been tasked by the government to survey vendors is to obtain a letter of permission to seek this information from the government and share it with the vendor. With this letter, vendors will likely be more cooperative in sharing information. The program office (regardless of whether in-house or contracted out) should decide how often to contact the OCMs and OEMs; the appropriate frequency will depend on the criticality of each system, general life cycle expectations, and other DMT-determined factors.

It is a best practice to conduct vendor surveys twice per year for electronic items, due to the rapid pace of change characterized by the market. For MaSME items, market changes occur much slower. While this might support the argument that vendor surveys could be conducted less frequently for MaSME items, there are several arguments against such an approach. First, conducting different vendor surveys (potentially for the same vendor) at different frequencies could lead to extra work and unnecessary complications for a program office's DMSMS management effort. Second, the marginal cost of conducting more frequent vendor surveys would likely be small, assuming that there are not a very large number of contacts to be made. Finally, material obsolescence caused by changes to environmental regulations or geo-political disruptions in supply can happen very quickly. For this reason, it could be beneficial to be monitoring MaSME items on a more frequent basis. Ultimately, a program office should make its own determination on the frequency of vendor surveys for non-electronic items, based on obsolescence risk, resources, and the criticality/safety associated with those items.

Performing research on the managed software items is the most time-consuming effort within a software obsolescence program. The frequency of researching/refreshing each item must be determined by taking into account the 1) associated risk, 2) personnel, 3) number of items being tracked, and 4) funding. The

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following are the recommended frequencies for research and updates depending on the known risk of the software:

- High Risk: quarterly.
- Moderate Risk: semi-annually.
- Low Risk: annually.

4.4.3 Critical Materials Analysis

It is important to be aware that under some circumstances, the vendor survey and research approach may not be sufficient. This could be the case if the suppliers of the items listed on the BOM are unaware of issues associated with materials within their items' supply chains. It is because of these instances that the importance of the program office's third strategic decision from the foundations of DMSMS management (see Section 3.1) is highlighted. This third determination reflects the program office's determination of whether to investigate critical materials in the supply chain (those not identified in a BOM) or in a manufacturing process.

A critical material could be hazardous, exotic, or otherwise be supply constrained. Such a material can either be embedded within an item that is listed on a BOM or required as part of the manufacturing of an item on a BOM. The incorporation of critical materials into a system will usually be done at a low level in the supply chain—a level that is below an item being surveyed or whose status is being determined by a predictive tool and from a company that may not even know that its product is destined for a DoD system. Potential changes and disruptions at the lower-tier material level may not be immediately apparent and understood when an item's status is being determined (especially for low-demand items) and, consequently, potential DMSMS concerns may not be identified early enough to resolve in a timely or cost-effective manner. For example, a buyout of a key critical material producer may lead to a major impact if there are plans to consolidate production lines. Given examples such as this, knowing whether there are critical materials present in or used in the manufacturing process of a MaSME item in the BOM will improve the analysis of availability of that item or material. The same is true for electronic items analyzed through predictive tools or other vendor surveys.

Program offices often do not know which critical materials are in their supply chains. Only minimal information can be learned about imbedded critical materials from TDPs, material safety data sheets, and an item's technical characteristics. Even if the critical materials in an item's supply chain were known, in nearly all cases, issues with these critical materials will affect multiple program offices and systems. It would therefore be inefficient if every program office independently conducted research to identify obsolescence concerns for critical materials in its own supply chain and then determine how to mitigate any issues that may be discovered from its own perspective. Such efforts should be centralized in coordination with all other stakeholders.

Program offices should devote resources to identify material issues in lower-tier suppliers based on their perceived risk. First, a program office can identify the set of lower-tier critical materials with which it is concerned; these would be the critical materials that are anticipated to have the greatest potential impact, if there should be an issue obtaining or being able to use a given material. Second, a program office can strive to better understand the extent to which issues associated with the lower-tier critical materials in the supply chain may impact monitored item availability.

To identify the lower-tier critical materials with which to be concerned, a program office has two options: 1) create a master list of all critical materials or 2) create a targeted list of critical materials for which the availability of that material can be anticipated to be uncertain because of pending regulatory change or other potential supply disruption. In selecting an approach, a program office should keep in mind that

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hazardous materials (which often represent a large fraction of all critical materials) may fall into one of three categories: 1) their use is prohibited, 2) their use is restricted, or 3) their use is otherwise tracked.

Regardless of which hazardous material category applies, a predictive tool or a vendor survey will accurately capture the obsolescence risk status of an item on a BOM that uses a hazardous material, as long as there have been no recent changes in the categorization of that hazardous material within that item. For example, if the material is prohibited and has been for a period of time, the impact of that material on the items that it is in will already likely be known and those items will most likely be obsolete. In other words, the status of the item will already reflect the known categorization of its hazardous material. If the material is also known to be restricted or tracked, supply chain availability will be understood; there may or may not be DMSMS concerns. Only when the material's category has recently changed or there is a strong likelihood of a change in the near future would there be a need to conduct further research to ensure that the end item status accurately accounts for these risks.

Consequently, it is sufficient for most program offices to create a list of hazardous materials where the availability of that material can be anticipated to be uncertain; however, uncertainty does not imply that the material should not be used. A targeted list of hazardous materials should be based on an understanding of and remaining up to date on the latest regarding the following:

- ***Uncertainties associated with environmental restrictions pertaining to materials***. The Chemical and Material Risk Management Program (CMRMP) scans the industrial landscape for chemicals and materials that may be regulated or restricted and identifies potential impacts to DoD mission readiness, the defense industrial base supply chain, the Foreign Military Sales program, or North Atlantic Treaty Organization interoperability. The CMRMP focuses on emerging chemicals (ECs)—chemicals relevant to the DoD that have been determined to be a perceived or real threat to human health or the environment with new or changing toxicity or regulatory standards because of new science, detection capabilities, or exposure pathways. DoDI 4715.18 establishes the requirement and process for identifying, assessing, and managing risks associated with ECs. ECs that are associated with DoD activities but do not pose a high risk to the DoD enterprise are included on an EC Watch List and monitored for possible changes. ECs that pose a high risk to DoD are included in the EC Action List.[136]
- ***Material vulnerability uncertainties associated with the Department's National Defense Strategy and national emergency planning***. DLA's Strategic Materials Office maintains a strategic and critical materials (SCM) "watch list," which focuses on materials of concern.[137] The list is compiled from nominations by DoD Components, other parts of the Executive Branch, Congress, SMEs, and other interested parties. Criteria for nomination include evidence of "weak links" in important material supply chains for defense and/or critical civilian applications. The list is used as the basis for assessments to recommend possible materials for purchase by the National Defense Stockpile. The Strategic Materials Office produces such assessments on an ongoing basis. Assessment results show that not every entry on the "watch list" is a potential shortfall. A shortfall does not imply a peacetime shortage—it only implies a possible shortage associated with DoD reconstituting its losses of that material (or end-items that use that material) during a national emergency. Therefore, from a DMSMS perspective, materials of interest could be limited to new additions to DLA's identified shortfalls, or significant changes in the quantity and value of a previous shortfall material. Context for material shortfalls is provided in the assessments. A careful reading of

[136] The EC Watch List and Action List are routinely updated and the latest version can be requested using the CMRMP mailbox: osd.pentagon.ousd-atl.mbx.cmrmp@mail.mil. Further information about the CMRMP can be found at the main webpage (https://www.denix.osd.mil/cmrmp/).

[137] See https://www.dla.mil/HQ/Acquisition/StrategicMaterials/Materials/.

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assessment documents can provide insight into the details associated with each assessment to determine a subset of materials on which DMSMS may choose to focus.

While it will likely be sufficient for most program offices to create a targeted list, based on the sources described above, some program offices may still judge that all critical materials are of concern. In this case, there are several additional sources of data that can be used to support this effort:

- ***The 2016 National Aerospace Standard (NAS) 411-1, Hazardous Material Target List (HMTL)***. The HMTL has been developed to support and enhance the management of risks associated with the use of hazardous materials in products and services consistent with NAS 411, Hazardous Materials Management Program, and Military Standard (MIL-STD) 882E, Department of Defense Standard Practice: System Safety, in which a Hazardous Materials Management Plan is described as a management task. The HMTL includes prohibited and restricted materials. A consolidated, tracked materials list is under development. NAS 411-1 may be purchased online from the Aerospace Industry Association.
- ***Aerospace and Defense Declarable Substance List (AD-DSL)***. The International Aerospace Environmental Group (IAEG) develops an AD-DSL to identify chemical substances for which aerospace and defense supply chain supplier reporting is necessary. The AD-DSL addresses substances that are:
  - "Restricted in articles,"
  - "Restricted in substances and mixtures,"
  - "Declarable in articles," and
  - "Of interest."[138]

  This list can be found at the IAEG website's "Materials and Substances Declaration for Aerospace and Defense" web page.[139] This web page includes further links to the 2015, 2017, and 2019 AD-DSL in multiple download formats.
- ***Registration, Evaluation, Authorisation, and Restriction of Chemicals (REACH)***. The European Union's REACH regulation controls the use of and imposes reporting requirements for certain hazardous substances in Europe. The following European Commission website contains information on substances for which additional restrictions have been adopted and specific chemicals that require additional legislation: https://ec.europa.eu/growth/sectors/chemicals/reach/restrictions_en and https://ec.europa.eu/growth/sectors/chemicals/specific-chemicals.
- ***Restriction of Hazardous Substances***. European and various other countries have issued RoHS regulations that limit the material content of electronic and electrical equipment.[140]
- ***Regulation of Chemicals***. The Toxic Substances Control Act of 1976 provides the Environmental Protection Agency with authority to require reporting, record keeping, and testing requirements, and restrictions relating to chemical substances and/or mixtures.[141]
- ***Critical Supply Chains***. Executive Order 14017, February 24, 2021, Executive Order on America's Supply Chains, contains policy objectives focused on strengthening the resilience of America's supply chains.[142]

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[138] IAEG, "Aerospace and Defense Declarable Substance List," Version 5.0, March 19, 2021, p. 1, available at http://www.iaeg.com/chemicalrpt/addsl/.

[139] See http://www.iaeg.com/chemicalrpt/addsl/.

[140] Regulated substances are available at http://www.rohsguide.com/.

[141] The Environmental Protection Agency website contains information on the production, importation, use, and disposal of specific chemicals available at https://www.epa.gov/laws-regulations/summary-toxic-substances-control-act.

[142] The Department of Commerce drafted a list of products in the supply chains of critical sectors and subsectors is available at https://www.trade.gov/data-visualization/draft-list-critical-supply-chains.

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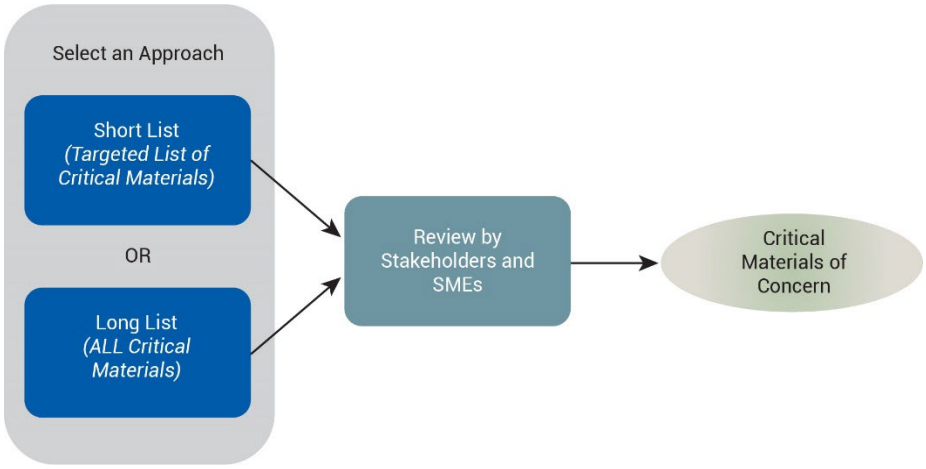
Regardless of which approach is taken to compile a list of lower-tier critical materials of concern, DMSMS management stakeholders should have an opportunity to contribute. Anyone on the DMT, including material and environmental engineers in the program office, may become aware of a potential material-related supply chain issue. Other potential sources include component organizations with material SMEs and/or organizations that conduct industrial base analyses.

A system's list of critical materials of concern may be shortened over time as a function of what the DMT learns about whether these critical materials are being used on the system or not. Consequently, there will typically be three classes of materials on a system's list of critical materials of concern:

- Those materials that are known to be on the system,
- Those suspected to be on the system, and
- Those where their presence on the system remains unknown (this is likely to be the most prevalent class, particularly at the beginning of a program office's DMSMS management effort).

Figure 14 illustrates the first step of a proactive approach for issue identification for critical materials in the lower tier of the supply chain. This first step focuses on selecting those materials that will serve as the critical materials of concern for the program office.

Figure 14. Select the Critical Materials of Concern for the Program Office



For items in the BOM, robust DMSMS management seeks to be proactive in problem identification to discover potential issues well before they materialize. This maximizes the time available to implement a resolution, thereby increasing the likelihood of finding an inexpensive resolution and minimizing any ill effects on schedule and readiness. From this perspective, once a program office has identified the critical materials of interest, the next set of activities should follow a risk-based, proactive problem identification path to highlight "the unknown unknowns" of critical materials' impact on item availability.

A typical proactive approach using predictive tools and vendor surveys and research would not normally be used to better understand the extent to which critical materials issues in the supply chain may impact monitored item availability. Such an approach to problem identification is very labor intensive and the data do not indicate that the DMSMS risks and potential impacts are severe enough to justify the investment needed to be proactive in that way.[143] Even if the government DMT were able to identify and contact lower-tier vendors, these contacts may not generate a complete response as the government will be

[143] This may change in the future if a greater number of substances become subject to environmental restrictions.

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seeking proprietary information that the company is unlikely to share unless it is in the company's best interest to do so (e.g., the company wants the government help to solve the problem).

In the absence of any exacerbating circumstances indicating high risk, the most cost-effective DMT approach for risk-based, proactive problem identification for lower-tier critical materials (and materials used in manufacturing processes) is to encourage and engage in communications among key critical material stakeholders to share tacit knowledge on the subject.[144] Considerations regarding how these types of communications can be fostered include the following:

- ***Establishing an agenda item for critical material supply chain issues for every DMT meeting***. This will engage the prime contractor and any OEMs on the DMT. These DMT members may be aware of some critical materials in the supply chain, as well as potential issues with these materials because of any chemical profiling they are performing on their products. Such chemical profiling may be due to regulations or because it has been determined important for them to do this from a business perspective (e.g., sales may otherwise be affected). Furthermore, primes and OEMs have the most detailed engineering understanding of the items on the system. Therefore, they will play an important role in determining a resolution, should that be necessary. A DMT agenda item will also engage the organic depots involved in sustainment. Although the depots may not normally have a representative on the DMT (Section 3.2.1 lists them as ad-hoc members), someone on the DMT will be in the position to communicate with them. This is an important avenue of communication because there is a supply chain associated with depot activities and, consequently, the depots may be aware of the presence of critical materials in the system and critical materials issues analogous to those known by the primes and OEMs. Depots may even be in a position to query their supply chains about issues.
- ***Engaging with other stakeholders to prepare for a DMT discussion***. Other stakeholders to engage include the following:
  - DLA's Strategic Materials Office.[145]
  - The Office of the Assistant Secretary of Defense (Industrial Base Policy) (OASD[IBP]) has several functions that could be pertinent to these discussions. These functions pertain to the Defense Production Act and Title III,[146] Executive Order 14017, Securing America's Defense-Critical Supply Chains,[147] Industrial Assessments and Industrial Base Analysis and Sustainment,[148] and Critical Energetic Materials Working Group.[149] Program offices focused on energetic materials can coordinate with OASD(IBP) Kinetic Capabilities Policy, Analysis, & Transition Office. The Kinetic Capabilities Policy, Analysis, & Transition Office conducts analysis and maintains internal data on critical energetic materials.[150]
  - The Manufacturing Technology (ManTech) program anticipates and closes gaps in manufacturing capabilities, and as such represents another potential stakeholder of interest.[151]

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[144] A more proactive approach is to include a provision in contracts to collect data on any material of interest in the supply chain. This is a more costly approach and could be integrated with potential programmatic environment, safety, and occupation health evaluation–related efforts. See Appendix F.

[145] This office can be contacted using the "Contact Us" link at the bottom right of its website available at https://www.dla.mil/HQ/Acquisition/StrategicMaterials.aspx).

[146] See https://www.businessdefense.gov/ibr/mceip/dpai/dpat3/index.html.

[147] See https://www.businessdefense.gov/eo14017.html.

[148] See https://www.businessdefense.gov/ibr/mceip/icam/index.html.

[149] See https://dsiac.org/tag/critical-energetic-materials-working-group/.

[150] See https://www.businessdefense.gov/ibr/pat/index.html.

[151] See https://www.dodmantech.mil/.

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- CMRMP. This program produces environmental risk alerts, which are maintained by the DoD Environment, Safety, and Occupational Health Network and Information Exchange.[152]
- USD(R&E), Assistant Secretary of Defense (Science and Technology), Science and Technology Program Protection (STPP) has functional responsibility for Science and Technology (S&T) and acquisition program protection that could be pertinent to DMSMS discussions.[153]
- Major OEMs in the supply chain. This would apply particularly in the case of those OEMs who are not already represented on the DMT but who may be profiling chemicals on their own products.
- Material engineers in the program office.
- Environmental engineers in the program office. As part of the program office's programmatic environment, safety, and occupation health evaluation requirements, hazardous materials, waste, and pollutants on the system must be identified.[154] Environmental engineers may also be briefed periodically on regulatory changes that might affect the system.
- Organizations that conduct materials research or perform industrial base sector analysis, and/or cross-cutting materials SMEs for the components.
- The Strategic Environmental Research and Development Program and the Environmental Security Technology Certification Program conduct assessments and develop and demonstrate innovative, cost-effective, and sustainable solutions to meet DoD's environmental and installation energy and water challenges, including management of polyfluoroalkyl substances.[155]

Figure 15 illustrates the second step of the proactive approach for issue identification for critical materials in the lower tier of the supply chain, as described above. This step focuses on the DMT assembling materials-related stakeholder knowledge and experience (both internal and external to the program office) to inform DMT meetings. In that way, materials-related red flags can be raised, as necessary, to help mitigate against being caught off guard.

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[152] This program can be accessed using the "Menu" link at the top right of the CMRMP webpage at https://www.denix.osd.mil/cmrmp/home/.

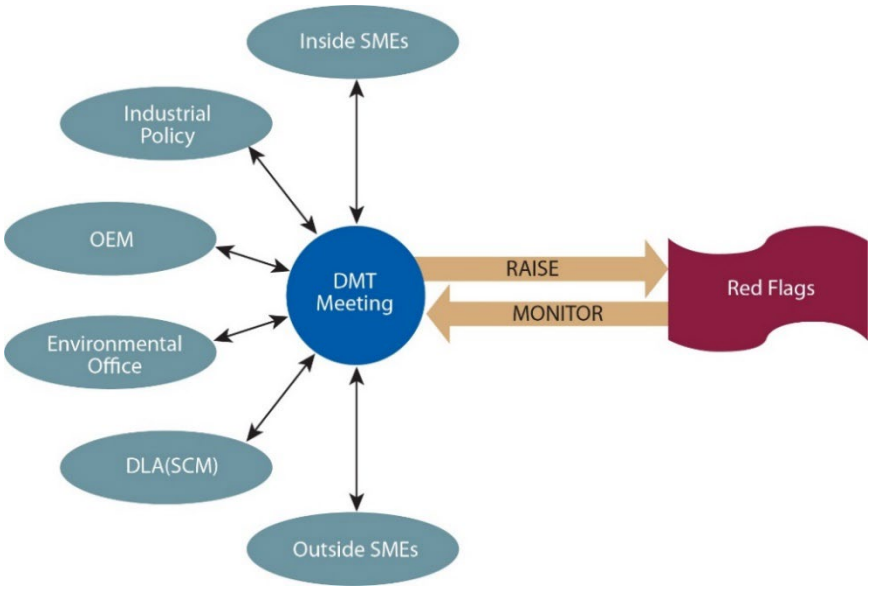
[153] The functional responsibilities pertaining to the identification and prioritization of critical technology elements in S&T projects and acquisition programs is located at https://rt.cto.mil/stpp/.

[154] DAU, *Systems Engineering Brainbook*, "System Safe (Including Environment and Occupational Health (ESOH)" located at https://content1.dau.edu/DAUMIG_se-brainbook_189/content/Design%20Considerations/System-Safety.html.

[155] See https://serdp-estcp.org.

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Figure 15. Identification of Potential DMSMS Issues Associated with Critical Materials



Stakeholder engagements provide opportunities to identify issues related to critical materials in the following ways:

- Encourage all stakeholders to keep their ears open—primes and OEMs may be encouraged to more proactively monitor their supply chains,
- Learn the potential issues others are aware of and what they are concerned about,
- Learn what is being done,
- Learn what conversations are taking place,
- Actively share all information among the stakeholders, and
- Ultimately put the stakeholders in a position to anticipate changes in regulations and other market-driven disruptions to the critical materials supply chain.

Once a potential problem is discovered, these engagements will force a conversation on what to do about the problem. A resolution that develops a drop-in replacement that is compliant and meets performance specifications is highly desirable. One of the stakeholders may be willing to take the lead in researching the issue and recommending a course of action. There may be circumstances where the prime and an OEM agree to resolve the problem because there is a clear business case for that to happen—this is most likely to occur when the material is included via a company-owned specification or drawing.

By engaging high-level organizations, a DoD-wide initiative may be established. Some DMT research may be warranted but typically would not occur until other options have been pursued. Regardless of who does the research, some potential data sources are as follows:

- Industry associations.
- Organizations that track both recent and pending domestic and international regulation changes. As one example, the National Aeronautics and Space Administration (NASA) has established a Principal Center for Regulatory Risk Analysis and Communication (RRAC) that conducts regulatory reviews and performs risk analyses that can be accessed via email subscription or through the RRAC website.[156]

[156] NASA, "EMD Principal Centers," last updated June 29, 2020, available at https://www.nasa.gov/emd/emd-principal-centers.

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- REACH, RoHS, and conflict minerals data associated with items.
- Other technical data, for example, annotations on drawings may lead to educated guesses about tracking or identifying critical material suppliers.
- Full material disclosures published on company websites.
- OEM contractual deliverables indicating potential DMSMS issues and supply issues.
- Just as proactive DMSMS monitoring should begin by the time of the program office's PDR, so should these engagements and chemical profiling activities. They may lead to a design change if substances are included where there is a concern about anticipated regulation. Also, they help establish a baseline of understanding critical material content and issues during sustainment.

4.4.4 Product Discontinuance Notices

The DMSMS management program should receive automated industry obsolescence notices and DMSMS alerts from the selected predictive tools, Government Industry Data Exchange Program (GIDEP), and DLA. Although overlaps will occur, all three sources should be used to maximize completeness and timeliness. Any PDNs received should be validated before any action is taken. In addition, the DMT should query manufacturers' websites, build relationships with OCMs (similar to the vendor survey relationships), and access other federal supply sources such as GIDEP and DLA to identify data and notifications on item availability. The remainder of this section focuses on alerts and external triggers for item availability analysis updates from GIDEP and DLA.

4.4.4.1 Government Industry Data Exchange Program

A DMSMS management program should leverage GIDEP membership early in its life cycle and a member of the DMT should become trained in its usage. GIDEP is a cooperative activity between government and industry participants seeking to reduce or eliminate expenditures of resources by sharing essential technical information during the research, design, development, production, and operational phases of the life cycle of systems, facilities, and equipment. For complete requirements, and to become a member, see the GIDEP website.[157]

GIDEP is a useful tool to support monitoring and surveillance, because it has developed a part batch search routine that permits GIDEP participants to send and compare parts lists to the part identifiers in the GIDEP database. Parts lists are protected so that only GIDEP operations center personnel have access. Batch processing is available only to registered GIDEP participants. Additionally, GIDEP members can search the GIDEP site for key words such as DMSMS, Stop Shipment or Suspect Counterfeit to identify associated alerts and parts lists to compare to a parts list outside GIDEP.

Also, as a GIDEP member, a program office can get "push mail," which is generated, as a convenience, to provide GIDEP participants with an overview of information without having to access the database. If a part or title in the list is of interest, the corresponding document can be retrieved through direct database access. All GIDEP representatives are automatically eligible to receive push mail. Users may also be granted access with their representative's approval. Representatives can either access the push mail registration online to update their profile or to assign distribution to their users. Once users have been granted access to push mail, they can update and change their own distribution or email online. As part of push mail, members can receive weekly summaries that list documents committed to the database during the week cited. The list includes the document number, date, designator, title, and abstract.

[157] See https://www.gidep.org.

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Members can also request parts lists that represent all part identifiers (manufacturer, government, specification, drawing, model, base, and NSNs) either contained within or cross-referenced to all documents entered into the GIDEP database during the week cited. This allows a program office to check its parts against the GIDEP-generated weekly parts list without having to create reports itself. A program office may then enter the database to retrieve only those documents of interest to the program office.

GIDEP can also provide support when developing resolution options for a DMSMS issue. The GIDEP Urgent Data Request is a service available to any authorized GIDEP user as well as the public. The service enables the user to enter two types of queries—a source of supply and a request for information:

- A source of supply request is a mechanism for locating hard to find or obsolete items that are no longer available through traditional sources. The item may be an entire assembly or a material used in its manufacture. If multiple sources of supply are found, the user can select the most cost-effective supplier. Significant time savings to resolve an issue are achievable if any source of supply is found. If no source of supply can be found, the query may lead to potential new production sources.
- A request for information may help an activity with resolutions to an obsolescence problem by finding technical or experiential data or other information that apply to the issue. For example, the request for information may ask for test, calibration, design, maintenance, or failure data. Having such data may help determine the viability of a substitute part.

In summary, GIDEP does not have the ability to predict which parts will become obsolete, but it can provide a program office with a no-cost means to find out which NSN parts[158] of interest already have discontinuation notices against them. Program offices can also use GIDEP's batch processing to ensure that the program office will receive discontinuance notices that match system parts and may provide the ability to assist with identifying unmatched parts.

4.4.4.2 Defense Logistics Agency

DLA also provides PDN alerts to subscribers—including military services, government agencies, FMS customers, and industry (with .mil email accounts and common access card capability)—through its shared data warehouse.[159,160] These DLA-generated alerts contain information not available through GIDEP, such as DLA usage and weapon system coding. For DLA-managed items, additional analyses are done to determine resolution options ranging from requesting users to determine quantities for LON buys to examining options to emulate microcircuits using its Generalized Emulation of Microcircuits (GEM) and Advanced Microcircuit Emulation (AME) programs.[161]

Access to DLA's websites allows a program office to search the following:

- ***Qualified Manufacturers Lists (QMLs)/Qualified Products Lists (QPLs)***. The data provided in this search are updated as changes occur and may contain information not reflected in the hard-copy version. A program office's search will always return the latest information available at that time. QMLs/QPLs are also available in the Acquisition Streamlining and Standardization Information System (ASSIST) Qualified Products Database. DLA updates the lists as necessary and is charged with requalifying vendors every two years.
- ***Standard microcircuit cross-reference***. This search provides a cross-reference of microcircuits covered by standard microcircuit drawings, MIL-M-38510 specifications, and

[158] Many designs contain parts without NSNs.

[159] The email address to become a subscriber is dmsms@dla.mil.

[160] DLA's Obsolescence Data Repository is a centralized repository for resolution data and information.

[161] SRI International, "AME GEM," available at http://www.gemes.com.

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vendor item drawings. If a program office prefers to use the cross-reference data on a local computer, a standard microcircuit lookup table can be downloaded.

- ***Military specifications (MilSpecs) and drawings***. This website provides courtesy copies of documents managed at DLA. If a program office cannot find a document here, it may not be managed at DLA. For a complete list of all DoD MilSpecs, refer to DLA's document automation and production service.
- ***Standard microcircuit drawings***. A list of standard microcircuit drawings is available to download.

4.4.5 Special Considerations for Software

The following software management best practices should be taken into consideration to establish and maintain a robust software obsolescence program:

- A software background/understanding is preferred for personnel who will be researching the software since a lot of the research relies heavily on either contacting the vendors or visiting vendor websites.
- Periodically re-evaluate survey questions to determine which questions are hardest to convey to the vendors. Re-wording/rephrasing/reordering questions can assist in getting a better response from the vendor.
- Ensure research is value added by meeting with the people who use the research to determine if additional questions would be helpful, if answers to current questions are not beneficial and should be removed or if the report format/content requires modification to assist with decision-making.
- Document research process used. If using a website to research software, document the steps performed to re-create the research (for new personnel/refreshing current research).
- Document research agreements/decisions made on what items are to be researched for each system and what items will not be managed.

The following sections provide best practices for conducting research as a function of the type of software item. Similar to hardware, provider/developer websites may be a valuable source of the information needed. Also, it is a best practice to document the steps taken to research items and decisions and agreements made to help make the research process repeatable.

4.4.5.1 COTS Operating System, Middleware, and Application System

The use of COTS software is increasing. It is often less expensive and quicker to integrate a COTS software product than to develop a custom product. COTS software may be monitored primarily by keeping track of licenses and support agreements,[162] analyzing technology and product roadmaps and projected new release information, participating in user groups,[163] tracking new interface standards, and conducting vendor surveys of the rapidly changing market to evaluate competitive products as a future replacement option.[164] Qualified sources of support for each element of software should be identified using software vendor survey questions like the following:

- What is the basis for changing version/revision levels? Are they updated regularly, and when?

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[162] To achieve economies of scale, organizations should consider having a higher-level organization obtain licenses and support. This is not a DMSMS management activity.

[163] User groups are also a source of information on error and vulnerabilities.

[164] There is a question of whether this function should be done at the program level or enterprise level for COTS software, because the same software may be used by multiple programs. In addition, the need for software vendor surveys may not be as great compared to hardware because much more software update information is available on the web.

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- How are patches and updates announced? How are they distributed?
- What is the current version/revision of the software?
- When was this product first available for sale?
- Has {your version/revision} been discontinued?
- How long will license agreements for {your version/revision} be obtainable?
- Will license downgrades be available? For how long?
- How long will you support {your version/revision}?
- Will third-party support be available after that? For how long?
- What is the planned product EOL?
- Is there a planned replacement product?
- Is the planned replacement backward compatible?
- What are the different technical characteristics between the old and new version?
- What operating systems are compatible with the software?
- What are the minimum hardware requirements (if any), for example, processor speed, communications interfaces, or memory?

Another aspect of proactive obsolescence management for COTS software is information assurance. DoD security bulletins may also be monitored.

4.4.5.2 Custom Operating System, Middleware, and Application Software

Because licenses do not usually apply to custom applications, the key information that can be tracked is viable continuation of support when there are both contractual and in-house elements. Surveys may not be the best mechanisms to obtain information. Program office sustainment personnel may be in a good position to identify potential software obsolescence risks. Key questions for consideration include the following:

- What is the current version/revision of the software?
- Do you still have the ability to modify the software?
- Is the source code repository maintained?
- Are the development tools maintained?
- Are there any third-party items?
- Are you able to compile those third-party items?
- What is the planned product EOL?
- What is the planned product end of support?
- Will third-party support be available after that? For how long?
- Is there a replacement product?
- Is the planned replacement backward compatible?
- What development and testing hardware and software infrastructure are needed to maintain the software?

4.4.5.3 Open-Source Operating System, Middleware, and Application Software

If open-source software is used, the government and/or OEM should assume configuration control for the source code. An analogy can be made to custom software, but there are differences. The expertise for making changes is likely to be found in the open-source community, not with the OEM. Consequently, proactive software obsolescence management may consider monitoring changes made to the open-source version (often found on the website), because using the newer version of the software may be

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necessary to support changes to the older code being used by the government. Licensing may not be an issue but the terms and conditions for using the open-source software should be reviewed by a legal team because, for example, there may be a requirement to provide any modifications to the entire open-source community.

4.4.5.4 GOTS Operating System, Middleware, and Application Software

Government off-the-shelf (GOTS) software is a subset of COTS software; therefore, the same considerations may apply. Licensing is unlikely to be an issue. The vendor survey would be conducted with the appropriate government entity.

4.4.5.5 COTS Firmware

Product changes that only upgrade COTS firmware may impact the system. When buying items, the program office may not realize that firmware changes have been made and that those changes may not be fully compatible with the rest of the system even though there is no issue with the hardware in which it is embedded. In this situation, the item is a functional group—a combination of hardware and software. The item becomes obsolete when either the hardware or the firmware becomes obsolete in a way that affects the system.

COTS firmware changes may be tracked by monitoring the item itself as a functional group. If the hardware item is tracked through vendor surveys, a question about the firmware version or revision should be included. If the hardware item is monitored with a predictive tool, depending on the risk to the system, it may be important to include that hardware item in a vendor survey. Potential questions include the following:

- What is the current version of the firmware?
- Is it still in production?
- When was the firmware last updated? What was the reason for the update?
- When is the next scheduled update? What is the reason for the update?
- What is the estimated remaining market life?
- How are updates announced and how are they distributed?
- How many years of maintenance will be offered for the older version?

The need to use specific firmware to avoid a critical impact to the system should be documented; for example, the requirement may be identified in an engineering drawing.

4.4.5.6 Custom Firmware

There are no obvious considerations for proactive software obsolescence management because the program office should be aware of changes to the firmware it controls. Therefore, there is no need to monitor such firmware.

4.5 Assess Preliminary Designs for DMSMS Risk

Selecting obsolete parts and/or hazardous or exotic materials into a system design should be avoided. Assessing preliminary designs, whether for a new design or redesign, for obsolescence risk is a best practice to minimize this occurrence. Assessments evaluate the DMSMS resilience of a design early in a system's life cycle or early in the course of a redesign. Program offices should generate such design assessments based upon the review of the design's preliminary parts list. This can identify the wider potential for obsolescence issues in the overall system design. This type of assessment might be of particular use during early phases of the life cycle and/or during the early stages of DMSMS management for a program office, when the system design may still be in flux and/or a complete parts list or BOM data

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are not yet available. A more comprehensive system health assessment, as described in Section 5.3.1, may be more suitable once a system design is mature and stable. Regardless, an assessment of a preliminary design might also be useful at other times during the system life cycle to provide a summary assessment of the system's obsolescence risk.

Ideally, an assessment of a preliminary system design should be aligned to coincide with major technical design reviews or design changes. The assessments should be conducted and provided to program office leadership before such design reviews to identify DMSMS issues ahead of time and, ideally, to help prevent the incorporation of an obsolete or near-obsolete item into a design. This could be either an obsolete item itself (e.g., a die) or its packaging. Assessments of preliminary designs should not, however, be restricted to supporting the design reviews. They should be conducted whenever new parts lists are received. Table 10 is a sample template for organizing information for an assessment of a system's preliminary design. Each program office should tailor the information and format of such an assessment to best suit its needs.

Table 10. Sample Template for an Assessment of Preliminary Designs for Obsolescence Risk

Risk Area	Value	Impact (High, Medium, Low)			Red/ Yellow/ Green
		Cost	Schedule	Performance	
Early preparation					
DMP adequate and appropriately funded (as judged by logistics and reliability) throughout the life cycle	Not applicable				
Technology roadmap available and integrated into DMSMS management process, and vice versa	Not applicable				
Ability to monitor					
Percentage of critical BOMs/parts lists available (i.e., a risk-informed decision has been made to monitor)					
Percentage of available critical BOMs/parts lists being monitored					
Item obsolescence					
Percentage/number of items obsolete (or with EOL notice)					
Percentage/number of items at high risk (custom items, ASICs, hybrids, sole source, and so forth)					
Number of obsolescence cases open, closed, and pending (i.e., no assessment has occurred to determine whether to open a case)					

The assessment should highlight risk in terms of its impact on cost, schedule, and performance. The first two rows of Table 10 are especially important early in the life cycle. They indicate the status of two key enablers of robust DMSMS management. The next two rows measure the scope of the DMSMS management efforts underway relative to where it should be from a risk-based perspective. Finally, the last three rows portray a high-level view of the extent of obsolescence in the system.

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When high risk items are identified, mitigating actions should be taken. There should be interactions with the parts selection, system engineering, and program office management communities to either replace high risk parts, or establish a risk mitigation plan (e.g., make an LON buy). As a result, there will be:

- Maximum use of parts with most of their life cycle remaining,
- Trades conducted between part life and technology refreshment plans,
- Consideration of trades between standard parts and innovations, and
- System designs more resistant to DMSMS impacts.

Programs should also consider the effects and risk of incorrect provisioning on a program. There is a risk that errors or inconsistencies in provisioning data may unnecessarily lead to a temporary material shortage or potentially even a DMSMS issue. Risk may be based on stocking levels and lead time to buy an item, such as when an item is not required to be on the shelf and the lead time to purchase the item is long. Risk may also be based on something like an incorrect supplier being identified. DMSMS practitioners should analyze provisioning data for such instances and understand that provisioning data alone is not sufficient for DMSMS management.

4.6 Forecast Future DMSMS Issues

Forecasting DMSMS issues is an important element of the *Identify* step of DMSMS management. The only way to delay DMSMS issues is to address forecasted obsolete items strategically before they become a problem. Strategic plans can be implemented on an item-by-item basis or by more comprehensive modification plans.[165] *Modification* is one term to describe configuration changes to a system. Other terms include (block) upgrades, (preplanned) product improvements, service life extension, modernization, and technology refreshment and insertion (defined later in this section). Plans that address anticipated DMSMS issues prevent operational or production issues so long as implementation occurs before those items can no longer be purchased or stocks of those items run out.

The processes for the *Identify* step contribute to forecasting future DMSMS issues. The predictive tools for DMSMS management track the obsolescence status of tens of thousands of items. The infrastructure behind those tools links to suppliers to obtain information on whether items have been discontinued, whether a discontinuation notice has been issued, and the official discontinuation date, if applicable. Furthermore, some predictive tools use the part introduction date, average life of the technology in the item, and other factors to estimate an EOL date. Vendor surveys asking for EOL dates and other parts research information help monitor items not covered by predictive tools. Finally, monitoring financial data for certain manufacturers may provide insight on the viability of the company and its products in the future. All this information is further supplemented with product roadmaps, market research, and technology roadmaps to develop a picture of when items will no longer be produced.

Developing and maintaining product roadmaps support modification planning (on any scale) and DMSMS issue forecasting. A roadmap is an information organizing framework that brings together diverse issues in a common view.[166] This SD uses the term *product roadmap* to represent the evolution of the product (or system) resulting from an acquisition program, including all funded, as well as planned but not yet funded, modifications. An acquisition program is “a directed, funded effort that provides a new, improved, or continuing materiel, weapon, information system, or service capability in response to an approved

[165] For items with a robust commercial capability, plans can entail finding another source in lieu of a redesign.

[166] Petrick, Irene J., *Developing and Implementing Roadmaps: A Reference Guide*, Pennsylvania State University, nd.

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need. …"[167] Often, a program office has a portfolio of product roadmaps that can be applied to an individual item (hardware or software), assembly, subsystem, or the entire system.

A product roadmap for the entire system aggregates the lower-level roadmaps that it encompasses. In a DoD program office, such an aggregation occurs in an IMS that captures the tasks, events, and accomplishments necessary to implement the changes in the product roadmap. DAU defines an IMS as follows:

> *An integrated and networked multi-layered schedule of program tasks required to complete the work effort captured in a related Integrated Master Plan (IMP). The IMS should include all IMP events and accomplishments and support each accomplishment closure criteria.*[168]

Whereas the IMP is

> *An event-driven plan that documents the significant accomplishments necessary to complete the work and ties each accomplishment to a key program event.*[169]

Other types of roadmaps contribute to product roadmaps (see Appendix K for a more detailed description). Product improvement roadmaps apply to changes in capability to address a new threat, safety deficiencies, information assurance, or the introduction of new statutory or regulatory requirements. Supportability roadmaps encompass improvements in the ability to maintain or sustain the product, increase its reliability, or extend service life. Product roadmaps, therefore, contain all planned product and supportability improvements for the system.

Finally, technology roadmaps capture technology changes that enable product improvement and, supportability roadmaps. In program offices, the technologies in technology roadmaps are usually selected solely to drive desired product and supportability improvements. The DMSMS community uses technology roadmaps and the technology management efforts underlying them as inputs to forecast and verify issues during the *Identify* step in the DMSMS management process (the output).

- The *Identify* step finds current and near-term DMSMS issues. While some near-term issues are discovered through predictive tool forecasts and vendor surveys, technology roadmaps developed from market research and technology development activities may expand coverage and extend the forecast timeline.
- One aspect of technology management, as an important element of supply chain management throughout the life cycle, creates a strategic understanding of the supplier base, enabling an assessment of planned system technology developments and their effects on the competitiveness and viability of essential industrial and technological capabilities. Such an understanding furnishes a picture of the forecasted health of the industrial base and its ability to develop, produce, maintain, and support the system. That picture of forecasted industrial base health serves as an early warning of potential DMSMS issues not only for the system as it is configured today but also as it may be modified in the future.

Figure 16 summarizes the relationships among roadmaps and forecasts of DMSMS issues.

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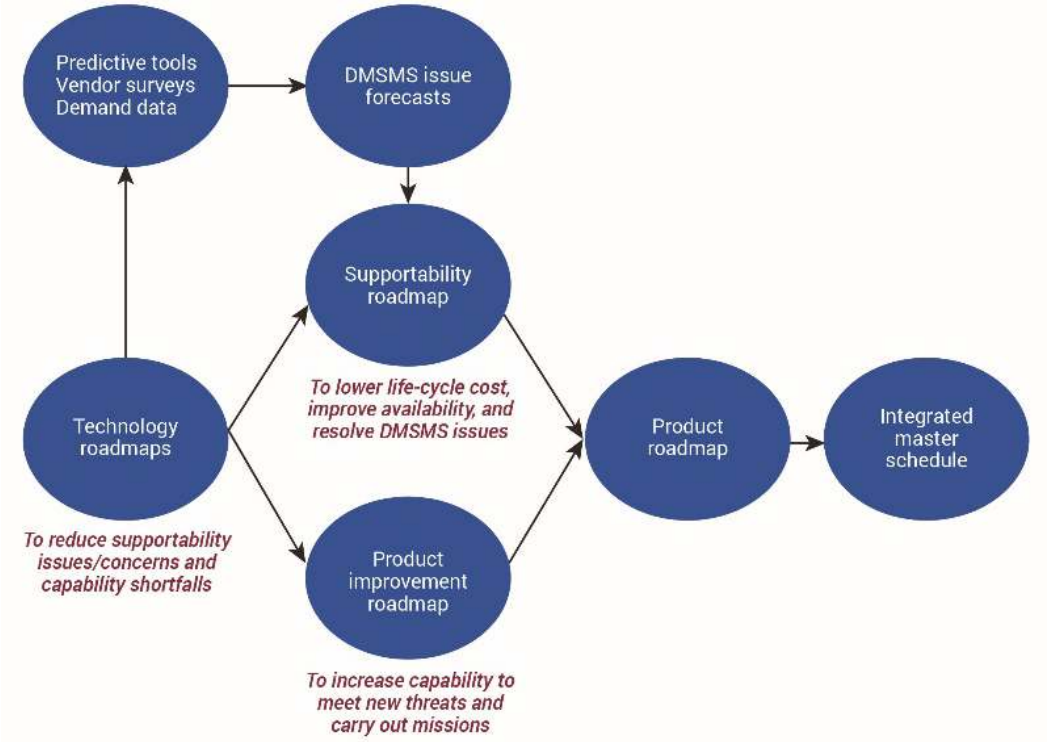
[167] DAU, *Glossary*, July 21, 2020, available at https://www.dau.edu/glossary/Pages/Glossary.aspx#!both|A|26790.

[168] Ibid.

[169] Ibid.

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Figure 16. Relationships among Roadmaps and DMSMS Forecasts



Technology roadmaps enable the following:

- Incorporating acquisition and life cycle sustainment strategies into the product roadmap,
- Minimizing the cost of resolving future obsolescence issues,
- Utilizing state-of-the-art technologies to increase reliability,
- Lowering sustainment costs, and
- Increasing warfighting capability to meet evolving requirements throughout an indefinite service life.

Determining when to upgrade or make modifications requires considering cost tradeoffs. From a single item perspective, the cost of the upgrade at a specified time can be compared to the cost of the upgrade at a later time plus the cost of mitigating the DMSMS issues until that later upgrade date. Such cost comparisons become increasingly complicated when multiple items are involved. Changes in capability also affect the decision regarding when to upgrade.

4.7 Additional Considerations for FMS Equipment in the Identify Step

4.7.1 Verifying DMSMS Issues

In some instances, FMS equipment buyers raise potential DMSMS issues, often based on their inability to purchase an item. The DoD program office may be asked to provide technical assistance to determine whether the situation is the result of a temporary disruption in the supply system (meaning the country

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should reorder the item) or a permanent unavailability due to obsolescence that could potentially affect DoD as well.[170]

Program office DMSMS management practitioners may need country-specific technical data to verify the existence of a DMSMS issue. To prepare for such situations, it is a good practice to explain the need to the foreign buyer while the LOA is being prepared and to include provisions for obtaining the data in the LOA itself. (See Section 3.5.1.1.) In some situations, foreign buyers may be reluctant to provide such data. In these cases, the program office should clearly describe the potential repercussions of not providing the technical data in that situation. The final decision on releasability will be made by the foreign buyer.

4.7.2 Determining Whether a Foreign Partner is Affected by a DoD DMSMS Issue

Unknown or imprecise information about foreign configurations could impede determinations about the applicability of a DoD DMSMS issue to a foreign partner. Without knowing the exact system design configurations of foreign partner equipment, there may be uncertainty on which foreign partner to inform about a DMSMS issue identified for DoD equipment (as well as a resolution).

This situation is similar to the one described in Section 4.7.1 on verifying issues. The principal difference is that system design configuration data applies to and may be different for each foreign partner. Establishing the need for the foreign partner's configuration management data in the LOA enables configuration management processes to be put in place at the start of an FMS case. If this has not been done, consider providing issue notification to all partners and have each partner determine its applicability. This practice may not be practical in some limited circumstances (e.g., there are only two partners, and only one is affected) because there may be restrictions on what can be disclosed to one partner about something that applies to another partner.

4.7.3 Informing Foreign Partners in a Timely Manner

There may be instances where the foreign partner is given late notification of a DMSMS issue. This situation could delay resolution decisions and implementation. This is especially significant when a life-of-need buy is being implemented because the delay may preclude FMS partner requirements from being included.[171] The foreign partner could then be forced to fund a more expensive resolution to make the platform sustainable (e.g., a redesign involving NRE and new parts).

Delays may result from inefficient communication processes associated with verifying DMSMS issues and determining whether those issues affect foreign partners. There may be:

- Delays in partner notification of the DoD program office of potential obsolescence issues;
- Delays with the DoD program office verification efforts;
- Delays with the DoD program office determining which partners are affected by a DMSMS issue; and
- Delays with back-and-forth communication associated with the above situations.

Communication processes with the foreign partners contributing to such delays should be made as efficient as possible.

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[170] Sometimes an assembly cannot be purchased but it can be repaired. This is not a DMSMS situation.

[171] Precluding partner requirements may also increase the unit price of the item for all participating partners.

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Delays in informing the foreign partner may also result from communication inefficiencies internal to the program office between DMSMS practitioners and the security assistance/FMS points of contact. Both groups should meet on a regular schedule (e.g., semi-annually or more frequently).

- Security assistance/FMS points of contact in the program office should be aware of all obsolescence issues.
- DMSMS issues that impact FMS cases should be discussed with all DMSMS practitioners.

While Section 3.2.1 summarized the responsibilities and indicated the importance of an FMS representative on the DMT, there may be no formal process for informing security assistance/FMS points of contact of DMSMS issues. The security assistance/FMS points of contact may rely more on *ad hoc* ways such as informal networks, contacting program management such as the product support manager, contacting the DMSMS management lead, and attending monthly meetings with the prime contractor. Relying on these methods could result in missed opportunities. A single individual such as the DMSMS management lead should be made responsible for informing security assistance/FMS points of contact of DMSMS issues. Identifying obsolescence issues only in FMS meetings is not recommended.

One possible reason for this *ad hoc* situation is that equipment sales tend to begin towards the end of production after DMSMS management has been operating for years. As a result, FMS may sometimes be treated as an afterthought by engineering and others. Although invited to DMSMS management meetings, security assistance/FMS points of contact in the program office do not always attend because their networks are not formalized. Security assistance/FMS points of contact do not necessarily know which meetings are important and which are not. Formalizing and enforcing FMS participation in DMT meetings should help avoid this situation. It is a good practice to always include FMS on the DMT agenda.

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Section 5. Assess: Resolution Need, Timing, and Level

The *Assess* step of the DMSMS management program examines the potential effects that a DMSMS issue, at any level of a system, may have on cost, schedule, availability, and readiness. Most DMSMS issues result in a combination of these effects and, ultimately, all if left unaddressed:

- Cost impacts may be experienced in any stage of the life cycle. The impact is measured as 1) the additional cost that must be paid to resolve the issue, 2) the change in support costs (it will cost the program office less if reliability is improved), and 3) the difference in the cost of items before and after resolution. This third element of cost may be positive or negative, depending on the resolution pursued. If a more expensive alternative item is used, then the cost will be higher.
- Schedule impacts are usually associated with the design or production phases of the life cycle or with modifications during the sustainment phase, because obsolescence may delay design or manufacturing activities.
- Availability and readiness impacts normally occur during sustainment. DMSMS issues may affect the mission capability of a system, or they may prevent the system from being used altogether.

The purpose of the assessment is to answer three questions:

- Should a resolution to the problem be pursued? Or should a case be opened?
- Which problem should be addressed first? Or when should the resolution be started?
- At what level of assembly should a resolution be considered?

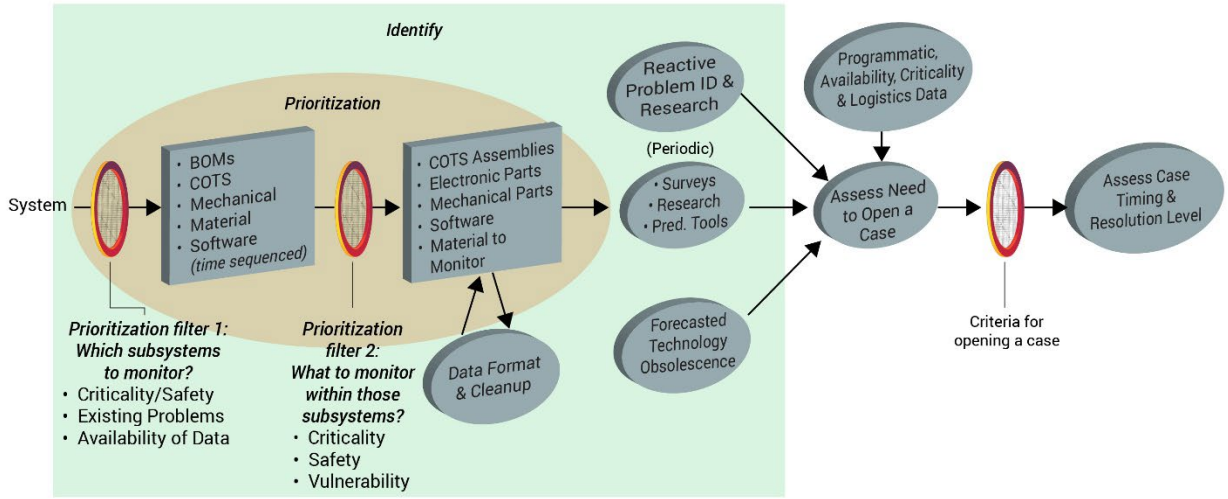
The DMSMS community does not answer these questions on its own. Data collection and research on the potential resolution level are carried out in conjunction with other DMT members. In particular, the assessment should be done in partnership with program office and prime/OEM logisticians because inventory data and demand data are essential elements for determining when an impact will occur. If the demand data are not based on field experience, then the program office and prime/OEM engineers must be involved to assess reliability. Determining the level at which the resolution should be considered involves several technical engineering considerations about resolution feasibility and complexity as well as DMSMS information about other items in the higher-level assemblies.

Figure 17 summarizes the DMSMS management activities leading up to the *Assess* step. As the figure shows, an assessment may be initiated for the following reasons:

- The results of predictive tool analyses or vendor surveys indicate a problem.
- The program office receives a PDN.
- A change occurs in a health assessment because of an increase in demand for obsolete items in inventory.

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Figure 17. Initiation of the Assessment Step



The remainder of this section describes the data and assessments needed to help provide answers to the above three questions. As a best practice, as much data as possible should be gathered to increase the rigor of the analysis. However, in many cases, some of the data may not be available. The DMT should do the best job possible with the data it has. When the DMT uses assumptions to compensate for missing data, the results of the analysis will be subject to greater uncertainty.

5.1 Obtain Data Needed for the Assessment

Programmatic, availability, criticality, and logistics data, along with the results of an item's availability analysis, support the DMSMS assessment process and, ultimately, resolution determination. The data should be refreshed regularly (as changes are made to the systems being monitored) to ensure that the most up-to-date data are used for DMSMS assessments and program office decision-making. In some cases, the data may be updated with the receipt of EOL notices for an item or set of system items or with the update of predictive tools or vendor surveys.

The data collection process differs slightly as a function of the acquisition phase. Early in the design phase, item data may be notional and based on a preferred parts list. Programmatic data may have less certainty early in a system's life cycle. Predicted reliability data should be used until better data can be derived from operational use. Actual logistics data will be available only during sustainment.

Logistics and programmatic data may be acquired from the program office, logistics databases, item managers, OEMs, and depots (contractor and organic). FMS unique and common equipment should also be taken into account. Of note, the services and DLA may obtain this type of information from their own logistics tools and databases if DoD equipment is common with FMS equipment. Those data enable the DMSMS management program to assess when an item with a DMSMS issue may no longer be available or supported, and which mitigation resolution is most feasible and cost effective.

5.1.1 Programmatic Data

Below are the different types of programmatic data needed for an assessment. It would be useful to have similar data on other systems using the obsolete item.

- ***Life Cycle phase***. If the system is in the design or production phase, the overall life cycle risk is significant, and emphasis on obsolescence issues at this point will have a significant impact on the total ownership cost of the system. However, an obsolescence issue

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discovered in the sustainment phase may not be as significant if the system is scheduled for disposal or if the replacement system is ready to be fielded. Industry tends to be interested in collaborating with DoD to solve an obsolescence issue during the design and production phases of acquisition; however, such collaboration can be difficult once the production line has gone cold.

- ***Planned technology insertions or refreshments for the subject item/assembly/software element and the next higher unit***. This information is important for an obsolescence issue at any point in the life cycle, because it presents an opportunity to eliminate the requirement for the problem item. It is important to understand whether the planned technology insertion or refreshment is funded. If no resources are programmed, then technology insertion or refreshment is unlikely to occur. It is also important to understand that DMSMS management is not the main driver of technology insertion or refreshment. DMSMS management simply leverages this information to determine risk and, where appropriate, to recommend a resolution option. However, DMSMS issues can affect the technology refreshment's scope and schedule, both positively and negatively, after they are initially established (see Section 4.6).
- ***Planned end of system life***. EOL data are used for inventory-related calculations. If the system is in design or production, the system EOL may not be known. Even during sustainment, the EOL may be uncertain, because of unplanned service life extensions, which in turn affect inventory requirements and may have potential DMSMS impacts. If the service life is extended, DMSMS situations with no operational impact before the extension may have a significant operational impact because of the extension. Nevertheless, the only approach is basing DMSMS assessments on official plans.
- ***Number of systems in use over time through the end of system life***. This number is used for inventory-related calculations. If the system is in design or production, only near-term numbers may be available.
- ***Planned average operating hours per system***. This number is used to help calculate demand for the item. If the system is in design or production, future average operating hours may not be available. In that case, it may be possible to make estimates based upon historical data for similar systems or on the planned operating tempo for the system.

5.1.2 Availability Data

Availability data are needed at the item, assembly, and unit levels. For software elements, the program office should track licenses, end-of-support dates, and frequency of updates. Availability should be identified at the lowest level possible, with an assessment of the impact at the next higher levels to better understand the risk and to help identify the most efficient cost resolution option. The DMT should differentiate between items that are currently unavailable and items forecasted to be obsolete in the near term (within two or three years). If authorized substitutes are available, there is no immediate obsolescence risk.

5.1.3 Criticality Data

Like availability data, criticality data are needed at the item, assembly, and unit levels. The first process in the *Identify* step (see Section 4) is to prioritize systems according to their mission criticality and safety-related features. Those same criticality factors apply in assessments. Furthermore, item (hardware and software) criticality is often determined by the criticality of its function. Examples of items with critical functions are microprocessors, microcontrollers, memory, ASICs, and field-programmable gate arrays. Finally, the cost of the item is a criticality factor.

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5.1.4 Logistics Data

Systems managed and repaired organically may have access to logistics data, assuming the data are captured and archived. Each military service has a logistics management system and item managers who have access to and understand logistics data. The contractor will have the data for program offices that employ contractor logistics support (CLS); the government should arrange to have access to this information via contract requirements and deliverables. The following are examples of logistics data:

- ***Item cost***. This represents what the program office should expect to pay for the purchase of the item.
- ***Demand for the items, assemblies, and units with DMSMS issues***. This applies primarily to items in sustainment, unless the same items are used in the same way on other systems in the inventory. It is important to note that demand for many items is often only recorded at the assembly or higher level, which can mask the demand for individual components within the assembly. This occurs, for example, when an assembly depot level reparable (DLR) enters a depot for repair, but the particular components which were replaced to repair the item were not recorded. This is a bad practice that inhibits data-driven DMSMS management and is discussed further in Appendix L.2.
- ***Reliability of the items, assemblies, and units with DMSMS issues***. This should be the same as the demand data for items in sustainment. When in design or production, when no demand data have been collected, the manufacturer's stated reliability may be used, but it introduces more uncertainty into the assessment.
- ***Inventory for the items, assemblies, and units with DMSMS issues***. Inventory may be found in the service depot (either contractor or organic), production facility, and DLA facilities. The portion of the inventories should be identified for the system in question versus that for other platforms. Data on inventory due in, backlogged orders, and the length of time on back-order are also relevant. If the system is in design or production, inventory is most likely available from contractors.
- ***Maintenance philosophy for the items, assemblies, and units with DMSMS issues***. Some items may be repairable, while other items may be disposed of when they fail. The availability of or the development of a source of repair can reduce the risk that all these elements must be investigated.
- ***Repair history***. This indicates how often the item is repaired based on the number of hours in service. Engaging with vendors on an increasing repair turn-around-time can help identify an impending DMSMS issue before it is fully realized.
- ***Survival rates***. This represents the fraction of time repair is economically feasible.
- ***Wear out rates***. This represents the fraction of time repair should not be made because the item would be expected to fail very quickly after being returned to service.

5.2 Determine Whether a Resolution Should Be Pursued

Just because a predictive tool indicates that a particular item is obsolete, or anticipated to be obsolete by a particular date, does not automatically translate into a DMSMS issue for which a program office should pursue a resolution. A program office will want to first validate the risk by examining when the item will no longer be available, the stocks on hand for the item, and the expected time to implement a resolution. The factoring in of the time to implement a resolution may be of particular importance when addressing a MaSME item. For example, there may be instances where an item, perhaps particularly a MaSME item, is not yet obsolete, but the time to realize a resolution would be very long. Depending on the exact facts of the situation, it may be worthwhile for the program office to either increase its on-hand inventory or take an action that would reduce the time to resolve the issue should it arise.

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One way to answer this question is to identify when a resolution should *not* be pursued. Clearly, no resolution is needed if enough items are on hand to meet all future demands. However, because the level of "all future demands" is never certain, the level of risk should be considered, as illustrated in these situations:

- ***Situation 1***. If the system is in sustainment and there have never been demands for the item facing a DMSMS issue, then there is usually a low risk for not pursuing a resolution although the consequences of a DMSMS issue should also be considered.
- ***Situation 2***. If the system is in sustainment and calculations show that enough inventory of the item is on hand to last until the system is retired or until a technology refreshment replaces the item, then the risk of not pursuing a resolution is low, but it should be evaluated further. For instance, an individual program office should also keep in mind that the item inventory available may not be held for only its own specific needs; therefore, inventory levels should be monitored periodically. Reliability data are an extremely useful input in the assessment of risk at any level in the configuration: the higher the reliability, the greater the availability of the item and, therefore, the lower the risk of an obsolescence impact. For example, if a circuit card assembly (CCA) seldom has to be replaced or repaired (highly reliable), obsolescence issues at the item level will not be as high risk as an obsolescence issue on a card that is continually being repaired or replaced. This information, if available, should be used in the overall assessment. If the EOL date for the item is uncertain and the item is in high demand, a best practice is to keep it on the list of problems to be addressed, but with low priority. Conversely, for items with a known EOL date and low-demand items, the risk of not pursuing a resolution is relatively low.
- ***Situation 3***. The risk of not pursuing a resolution could be considered low if there are reclamation opportunities to recover a sufficient quantity of the item to satisfy the projected demand for that item.
- ***Situation 4***. While a system is in the design or production phase, a constant supply of items is usually required. The rare exception is when there is a high degree of confidence that all items needed for production and sustainment have already been procured. The uncertainty of such an analysis would be enormous.

As illustrated through the situations above, when hardware—electronic and MaSME items become obsolete, there may be stockpiles that last for a while. A resolution may not be needed at all, depending on the days of supply on hand. Loss of a software license will usually have a more immediate impact. Assuming the software is mission or safety critical, a resolution should be pursued. Similarly, an information assurance issue with the software has an immediate impact as the software can no longer be used without a waiver.

Loss of software support is more complex. If obsolete software has never been changed and no errors have been uncovered or no changes are anticipated, then it also may be safe not to pursue a resolution for some period of time. The software may continue to operate correctly until the end of system life if the underlying layers can be sustained. Consequently, the cost of changing the software becomes a consideration. Requalifying systems after a software change can be more extensive than after a hardware change due in part to the complex nature of the required testing.

In the case of firmware changes, there is a question of whether there will be an effect if a new functional group is introduced into the system. A resolution should be pursued based on the risk in making changes in the functional group application.

When an issue of environmental compliance is identified regarding a material, the program office will need to evaluate how long the item in which that material is resident will remain available.

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5.3 Assess Resolution Timing and Level

5.3.1 Conduct a Health Assessment

Health assessments provide insights that support the assess step. After the design becomes mature, health assessments should be conducted regularly throughout the remainder of system development, production, and sustainment.[172] They offer a program office a comprehensive accounting of its current and projected hardware and software obsolescence issues.

5.3.1.1 Hardware Health Assessments

In its most basic form, a health assessment breaks out individual items, documenting, by year, the starting quantity balance, predicted/actual usage, and ending quantity balance of that item over a certain timeframe, say ten years. This information enables a program office to make a preliminary estimate of when a component will no longer be available to satisfy a demand and therefore when to begin implementing a DMSMS resolution. Table 11 is a simplified example of the type of format that could be used for reporting the results of a health assessment. Program offices should tailor the content and format to best meet their specific needs.

Table 11. Basic Template for a Health Assessment Report

Item No.	Item Type	Sub-system	Status Characteristics	FYx	FYx +1	FYx +2	FYx +3	FYx +4	FYx +5	FYx +6	FYx +7	FYx +8	FYx +9
123	Micro-processor	1	Starting balance	4	3	2	0	−1	−2	−3	−5	−6	−7
			Predicted/actual usage	1	1	2	1	1	1	2	1	1	1
			Ending balance	3	2	0	−1	−2	−3	−5	−6	−7	−8
456	Amplifier	1	Starting balance	135	122	108	92	75	55	33	8	−18	−44
			Predicted/actual usage	13	14	16	17	20	22	25	26	26	26
			Ending balance	122	108	92	75	55	33	8	−18	−44	−70
789	Touch screen	2	Starting balance	16	15	14	13	11	10	9	8	7	5
			Predicted/actual usage	1	1	1	2	1	1	1	1	2	1
			Ending balance	15	14	13	11	10	9	8	7	5	4
211	Motherboard	2	Starting balance	12	10	7	4	2	−1	−4	−7	−9	−12
			Predicted/actual usage	2	3	3	2	3	3	3	2	3	3
			Ending balance	10	7	4	2	−1	−4	−7	−9	−12	−15
222	Graphics CCA	2	Starting balance	11	11	11	11	11	10	10	10	10	10
			Predicted/actual usage	0	0	0	0	1	0	0	0	0	0
			Ending balance	11	11	11	11	10	10	10	10	10	10
233	Ethernet interface	2	Starting balance	18	14	11	7	3	−1	−5	−9	−13	−17
			Predicted/actual usage	4	3	4	4	4	4	4	4	4	4
			Ending balance	14	11	7	3	−1	−5	−9	−13	−17	−21
244	Serial input/output CCA	2	Starting balance	2	−38	−83	−128	−173	−218	−263	−308	−353	−398
			Predicted/actual usage	40	45	45	45	45	45	45	45	45	45
			Ending balance	−38	−83	−128	−173	−218	−263	−308	−353	−398	−443
255	Notebook computer	2	Starting balance	11	10	9	7	6	5	4	2	1	0
			Predicted/actual usage	1	1	1	1	1	1	2	1	1	1
			Ending balance	10	9	7	6	5	4	2	1	0	−1

Legend:

- Sufficient assets to support more than 5 years
- Zero quantity reached within 4 years
- Insufficient assets (0 or negative)
- Sufficient assets to support next 5 years
- Zero quantity reached within 3 years

Appendix L provides a comprehensive description of a health assessment. It is a best practice to have a contractual requirement for the prime contractors/OEMs/sustainment providers to perform the calculations described in the appendix, because DoD may not have visibility of contractor held stocks. For a mature

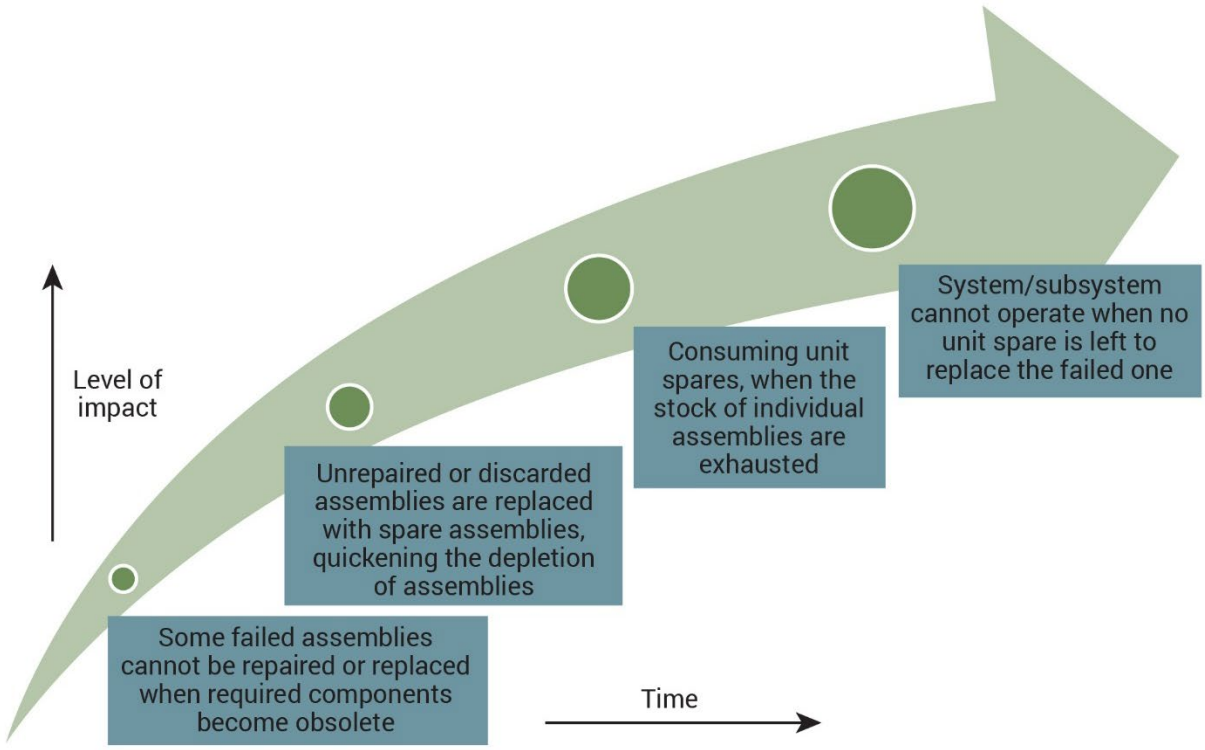
[172] The DMSMS management community has not established standard terminology for the efforts described as health assessments. Other terms include obsolescence impact assessment, DMSMS impact assessment, tombstone charts, sustainability assessments, supportability analysis, and operational impact assessments.

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system in sustainment that relies heavily on organic support, the DMT may be the most appropriate source for these calculations.

Employing the Appendix L methodology enables an improved estimate of when a DMSMS issue will affect the system rather than merely the component. This is accomplished by aligning the steps in the health assessment with the supportability impact levels displayed in Figure 18.

Figure 18. Supportability Progression after a Component Can No Longer Be Procured



The sustainment system is designed to keep weapon systems operating. Failed items at any level in a system's hierarchy are replaced by operational ones stocked in the supply system. The failed items are then either discarded or repaired. In the former case, new items are bought for the supply system; in the latter case repaired items are returned to the supply system for issuance when needed. When obsolescence occurs, fewer repairs are feasible, and replacements cannot be obtained. Health assessments portray the progression of how failures at the component level affect the availability of assemblies, units, and ultimately systems/subsystems for components that can no longer be procured.[173]

A health assessment assumes that spares are stocked at all levels (i.e., component, assembly). If that is not the case, then a step in the progression disappears and the impact occurs sooner. When a component fails, it will cause the assembly that houses that component to fail. Under normal circumstances, the failed component is replaced, and the assembly is ready to be reissued where needed.[174] The DoD supply system will order another component to replace the one used to fix the assembly.[175] If the component is obsolete, then it cannot be replaced in the supply system. Therefore, the

[173] This hierarchical terminology is used to explain the different steps in the health assessment. The term *item* is not used because that term is employed throughout this document in the more general sense to apply at any place in the hierarchy.

[174] The use of wear out and survival rates discussed in Section 5.1 is a consideration in the health assessment process.

[175] While inventory models are somewhat more sophisticated than this, the same principle applies. Inventory models usually specify a reorder point and lot size (the number of components to procure) for every item.

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next time that component fails, the only way to maintain operations is to replace the entire assembly with a spare that is in inventory.[176] Once spare assemblies are depleted, spare units must be used to maintain operations. When the spare units are exhausted, the MICAP rate for the system/subsystem begins to decline.

Understanding the overall risk of an obsolescence issue depends on understanding when the issue will affect the system. This can be accomplished only through an understanding of the supply availability of the items within the system. An analysis of the logistics and programmatic data provides a snapshot in time of current inventory levels and usage rates to identify the timeframe available (sometimes referred to as days of supply) to identify and implement a resolution. The ability to develop a resolution—within the timeframe of availability levels and while a replacement item is available—is directly related to the risk of experiencing some negative impact because of a DMSMS issue.

To determine the potential for a future shortage of a particular item, the DMT must estimate the future demand for the item and determine whether the existing stocks (including items due in and items on back-order) will meet that demand. Mathematical methods, applied at the component or NHA level and accepted by the logistics community, are available for calculating future demand.[177] Estimating the future demand for items is risky, due primarily to two key assumptions that must be made: projected future operational hours and reliability. The relative risk varies from decision to decision, but it is real and should be expressed to higher level decision makers when proposing resolutions. The risk introduced by assumptions is always higher before a system is deployed, because no reliability data based on actual operations are available. If the subject item will be removed from the system by a planned technology insertion, then the period of requirement ends when the item is replaced. Once the demand is established for the period required, the DMT can simply subtract the demand from the available stock (including due ins from prior orders and planned procurements of items that are not yet obsolete) to determine if and when the shortage will affect the system.

Applying the full Appendix L methodology also provides insight on the level of assembly where the resolution should be applied by including other obsolete and obsolescent[178] items and technology roadmaps and forecasting considerations within the highest level of assembly at which the resolution could be considered. This in turn could have an impact on priority and timing.

The health assessment results should be an input to the modification planning process because those results may have an impact on both the timing and scope of modification plans. For example, if there is a planned modification for an item in year X, but the health assessment indicates an earlier impact in year X-1, consideration should be given to changing the modification timing or implementing a short-term resolution such as an LON buy. Another example is a situation where the marginal cost of incorporating the resolution of an obsolete item into a modification is much smaller than the cost to resolve the issue independent of the modification (see Section 7.2).

5.3.1.2 Software Health Assessments

There has not been sufficient experience with software health assessment to develop best practices on how to quantify the level of risk to a system due to software obsolescence. Therefore, it is up to the program office to define the levels of risk that should be represented in a health assessment.

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[176] In theory, replacement assemblies could be procured, but this is unlikely when at least one component needed for its manufacture is not available.

[177] Many articles are related to this topic; see the material available at https://www.dau.edu/acquipedia-article/supply-chain-demand-forecasting.

[178] About to become obsolete.

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The concepts of a starting balance, usage, and an ending balance do not apply to software. However, in developing a software analog to the health assessment portrayed in Table 11, those concepts can be replaced by a measure of the risk that obsolescence will have a negative impact on the system. The rows of such a table could correspond to every tracked software item or a set of tables could be developed for tracked software items by subsystem.

Some possible software obsolescence situations are listed below. The health assessment could use all the elements in this list to characterize risk by year for each software element. Alternatively, the situations described could be combined in a way that is useful to the program office to characterize the risk as low, moderate, and high or some other similar scale. Finally, the health assessment could also be presented at a more aggregate level showing, by fiscal year, the number of tracked software items at the different levels of risk or in the different risk situations. Some common risk situations are as follows in order of increasing risk for the specified version of the software item:

- Most current version; supported with updates for security, new capability, and error correction; available for purchase.
- Not the most current version; supported with updates for security and error correction; not available for purchase.
- Not the most current version; supported with error correction only; not available for purchase.
- Not the most current version; supported with error correction only through the period of performance of an existing support agreement; not available for purchase.
- Not the most current version; no support; not available for purchase; no history of needing support for error correction.
- Not the most current version; no support; not available for purchase; with a history of needing support for error correction.

5.3.2 Which Problem Should Be Addressed First?

A step-by-step process can be used to prioritize problems based on their impact on the system. Several ways exist to develop such a prioritized list. The example here is based on knowledge of piece-part electronics, CCAs, and the black boxes the circuit cards populate. A similar approach for assessing risk of mechanical assemblies, materials, and COTS assemblies can be derived from these steps. There is one situation, however, that may make prioritization of the problem unnecessary—an external organization will resolve the issue. This may be the case for a common item that is shared among many platforms. The DMT may become aware of such a situation through its interfaces with DMSMS management activities in other program offices. This may also be the case for sub-tier materials where a higher-level organization is pursuing a DoD-wide resolution. In either situation, the DMT should evaluate whether that external resolution will meet both its schedule and technical requirements.

The steps below are based on assessing the impact of circuit card obsolescence issues given knowledge of the devices (piece parts) on the card. These steps include general statements, such as rank by “x” or adjust the rankings as a function of “y.” There is no set formula for these rankings and adjustments. They are based on the experience of the person making the assessment:

1. The first step considers both an analysis of piece-part availability and the results of the calculation of the timeframe until impact using the logistics data and the programmatic data. The order in which availability and logistics analyses are evaluated is determined by the risk assessor. Initially, the cards could be ranked by some combination of the total number of obsolete parts per card, the number of obsolete mission- or safety-critical parts per card, and the number and distribution of unique parts. The rankings could then be adjusted by the days of supply and the average monthly demand for both the parts and the card if the system is in

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sustainment. If the system is in design or production, reliability data, if available, could be used for the average monthly demand. Inventory levels would be the contractor's stock level.

2. The second step is to adjust the rankings based on the near-term obsolescence risk for the parts and the card and on the number of sources or alternatives available for the parts on the card. These data could be generated using predictive tools.
3. A third step is to adjust the rankings based on the maintenance philosophy for the parts and the card. If the circuit card is repairable and the obsolete parts are highly reliable, then the risk of the part causing the card to be unavailable is not as great as if the card is a throw-away and no more cards can be produced, due to an obsolete part (even if it is reliable). For example, a repairable circuit card with critical obsolete items may not rank as high risk if the inventory levels of the circuit card are high and the usage rate is low, whereas a card not considered highly complex may be a greater risk based on low inventory levels and high demand rates. The bottom line is that no matter how simple the card, if a spare is not available, then the unit is out of commission. Based on this additional logistics information, the risk priority of the cards in a given unit may change. In design or production, this step might not affect the rankings very much, but the factors addressed are a consideration.
4. A final step is to examine programmatic data concerning product improvement plans or other mitigation efforts already underway. If a modernization plan calls for the replacement of the unit or if a refreshment of any of the circuit cards is planned, the risk priority may change yet again. For example, if a CCA is identified as high risk with low inventory levels, but a replacement unit is scheduled to be fielded, the risk may not be as great. The timeframe for fielding and the ability to support the card through other means until that time may reduce the risk and therefore the priority of the card. These factors would probably not have much effect for a system in design or production.

If information is available about the electronic piece parts that populate a unit, but no structure is available to understand the breakdown from the unit to the card level, then the previous four steps are completely analogous to Situation 1. Only the risk at the part level and this translation to the unit level can be evaluated. The number of obsolete items, the complexity of the function of any obsolete items, and the near-term obsolescence risks, along with logistics information at the unit level and the programmatic data, will identify the risk. For example, assume the list of parts contains 100 unique items. The availability analysis identifies no current obsolescence issues; however, several critical part functions are predicted to be obsolete in less than two years. The unit was just fielded, with production to continue for two more years and no near-term plan to replace it. In this situation, the near-term obsolescence elevates the risk of the unit, but the DMT has some time to plan the resolution options. Evaluating the availability of alternates for the high-risk items and working with the program office and the prime contractor to develop a path ahead will reduce the DMSMS risk of this unit.

The same four steps could be used to assess a COTS or a mechanical assembly for which part data are unlikely to be available. However, the analysis would be much less granular. Instead of using predictive tools, the DMT would need to derive availability data from vendor surveys. For step 1, the availability data would simply be that the assembly is obsolete, and the logistics data would be similar to the data above. Step 2 would consider just the near-term obsolescence risk for the assembly, and steps 3 and 4 would be analogous to the above. For example, assume the supplier survey indicates that the box will be available for another year and that a replacement is planned for when the box is discontinued. However, this replacement is not backward compatible; therefore, some non-recurring engineering is required and possibly some testing to evaluate the use of the new unit in the system. From the logistics input, the DMT determines that the demand rate is low, with enough inventory to support the item for another 18 months if the demand rate does not increase. The risk may be assessed as low, given the availability of a replacement and the current inventory levels.

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In another case, only one manufacturer supplies ball bearings for aviation platforms. The Industrial Capability Assessment (ICA) of the manufacturer indicates both financial and workforce well-being. However, the ICA also indicates that the manufacturer has limited capacity to surge and that all aviation platforms across the military services use this one source of ball bearings. If current inventory levels indicate a six-month supply at the current operating tempo, and if DoD plans to increase the operating tempo, the ball bearings could be a high-risk item for availability (material shortage) even though not obsolete. The supplier might have problems meeting delivery schedules if multiple systems also experience an increase in operating tempo. If an additional source for these bearings is being developed, but qualification of the new supplier is still two years out, this manufacturer and this item would be considered medium risk and monitored. Evaluation of the increase in flying hours, along with reliability of the bearings, can identify possible future shortages.

Even though software can function for a long period of time with no support and without any adverse impact if underlying layers are stable, the loss of a software license should be addressed immediately. The same holds true for software no longer meeting information assurance requirements or a firmware change that affects system operation. When determining the priority under a loss-of-support situation, consideration should be given to the number and frequency of updates, the number of different versions currently being used on the system, and the age of the versions in use.[179]

5.3.3 At What Level Should a Resolution Be Applied?

If a system will be affected by a DMSMS issue, the DMT should determine where in the item's hierarchy the resolution should be applied. For example, the subject item may be one of many items within its hierarchy that have DMSMS issues. In such cases, it may be expeditious to replace or redesign the assembly rather than resolving the problems with each individual item. The same factors should be considered in this analysis:

- Number and difficulty of DMSMS problems in the hierarchy,
- Reliability of items in the hierarchy,
- Expense of repair within the hierarchy compared with redesign or replacement,
- Life cycle of other items in the hierarchy, and
- Potential for enhancing mission capabilities by redesigning or replacing items.

Most of these factors can be analyzed by the DMT if sufficient data exist. The importance of integrating item availability data with logistics and life cycle data cannot be overemphasized when analyzing the impact of a DMSMS issue. The first factor is largely a numbers game. If the cost to implement numerous DMSMS resolutions exceeds the cost to redesign the NHA, then the redesign should be considered.[180] The second and third factors are similar, in that one must compare the cost of continued operation of the existing item to that of implementing and maintaining a new item. This calculation is more involved, but in the end, it is a simple evaluation of which resolution provides the most bang for the buck.

The fourth factor requires a more subjective judgment, because item life cycles are not a strictly objective measure and because educated guesses are required to predict DMSMS problems. One must look at various sources of information and determine if the risk of future obsolescence and its accompanying costs exceed the benefits of resolving known problems now. This analysis is often the basis for planning

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[179] Software can degrade through configuration incompatibilities. While all the individual software elements may be fine, over time, the combination of these elements can be incompatible and lead to system failure.

[180] A single item may be a constituent component in multiple higher-level assemblies. This may change the cost calculation, because multiple higher-level assemblies may need to be redesigned. The most cost-effective option could be a combination of resolutions at the item level and at higher levels of assembly.

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technology refreshes and may result in a decision to resolve the DMSMS problem for a limited time (LON buy).

The last factor will require the input of other program offices, and potentially higher level, personnel. If future mission demands require new equipment or different capabilities, it may be expeditious to implement those features now rather than to wait.

For software, the assessment is more complicated. A hardware health assessment is relatively linear in that item obsolescence will affect its NHAs. Non-linear, secondary and tertiary effects of software obsolescence are common. Consequently, software dependencies—those elements of the system potentially affected by changes to software—are usually more complex and far reaching than those of hardware. Understanding software (and hardware, for that matter) dependencies is crucial for determining the most cost-effective level of resolution.[181] Because of this, the answer to this question should be determined on a case-by-case basis.

[181] That is why identification of the interdependencies was included in the *Identify* phase of robust DMSMS management.

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Section 6. Analyze: DMSMS Resolution Determination

This chapter discusses the *Analyze* step; specifically, it explains how to find the best resolution option. The resolution determination process is iterative; the analysis is updated as new issues are identified and prioritized. Typically, new issues to be resolved are added at every DMT meeting. The following sections address identifying the cost elements associated with estimating implementation costs, identifying and defining resolution options, and determining the preferred resolution option. The resolution determination process is the same whether the DMSMS issue is related to hardware—electronic and MaSME items—or software.

6.1 Identify Resolution Cost Elements

To determine the best resolution, a program office must first understand that resolution's total implementation cost. That cost is the sum of all applicable cost elements associated with that resolution. For example, a resolution may require anything from simple drawing and technical manual updates to full development and testing of new designs to be implemented in a system. If the actual costs for cost elements are known, those costs should be used to develop a more accurate account of the costs required to implement a resolution or series of resolutions. Actual costs give a program office the most accurate account of the funding required to mitigate obsolescence and represent an important metric.[182] Although using actual costs for resolutions is preferred, actual costs may not be readily available, and obtaining actual costs may be cost-prohibitive. Therefore, each program office should develop average costs for each applicable cost element.

The following cost elements should be considered when determining the total cost for resolving a DMSMS issue:

- ***Non-recurring engineering***. The cost of the design and development and the changes from the old to the new configuration.
- ***Engineering and engineering data revision***. Cost of modifying drawings and other data to reflect the new configuration.
- ***Purchase of engineering, design, or technical data***. Cost of purchasing technical data required for support.
- ***Qualification of new items***. Testing and evaluation cost to choose a new item.
- ***Revision of test procedures***. Cost of updating test procedures to accommodate any new testing requirements of the selected solution.
- ***Software changes***. Cost of updating software because of the selected solution and including software updates to test equipment.
- ***Start-up costs***. Non-recurring engineering costs to develop production or repair capabilities.
- ***Testing***. Cost of testing requirements for the selected resolution to ensure system compatibility.
- ***Tooling, equipment, test equipment, or software.*** Cost of repairing and maintaining equipment.

[182] Over time, this metric can be referenced for projecting budget requirements for implementing resolutions (see Section 7).

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- ***Computer programs and documentation***. Costs of new software and documentation to support the new item.
- ***Interim support.*** Contractor cost to maintain a product until a permanent resolution can be implemented.
- ***Spares.*** Cost to procure spares for sustainment.
- ***Supply and provisioning data***. Cost to update logistics data to ensure support of selected resolution.
- ***Support and test equipment***. Cost to provide the repair center with any required support or test equipment.
- ***Technical manuals***. Cost to provide any manuals and documentation to repair centers.
- ***Training and trainers***. Cost to develop and maintain training for the new equipment.
- ***Other***. Any other costs as required.

The prevalence of counterfeit parts and the use of Pb-free solder in the electronics industry also affect the costs and risks to resolve a DMSMS issue. When a DMSMS resolution option involves purchasing an electronic item from sources other than authorized suppliers (i.e., OCM, OEM, authorized or franchised distributer, or authorized or approved after-market manufacturers), additional testing must be done to ensure that counterfeit parts do not enter DoD's supply chain. Therefore, the average testing cost must be included. See the following commercial standards and a DoDI for more information on counterfeit prevention:

- AS5553, "Counterfeit Electrical, Electronic, and Electromechanical (EEE) Parts; Avoidance, Detection, Mitigation, and Disposition"
- AS6081, "Fraudulent/Counterfeit Electronic Parts: Avoidance, Detection, Mitigation, and Disposition—Distributors"
- AS6171A, "Test Methods Standard; General Requirements, Suspect/Counterfeit, Electrical, Electronic, and Electromechanical Parts"
- AS6174A, "Counterfeit Materiel; Assuring Acquisition of Authentic and Conforming Materiel"
- AS6462C, "AS5553C, Counterfeit EEE Parts; Avoidance, Detection, Mitigation, and Disposition Verification Criteria"
- AS6496, "Fraudulent/Counterfeit Electronic Parts: Avoidance, Detection, Mitigation, and Disposition—Authorized/Franchised Distribution"
- DoDI 4140.67, "DoD Counterfeit Prevention Policy."

6.2 Identify and Define DMSMS Resolution Options

Resolution options fall into three broad categories: existing material (logistics), substitutes (engineering), and redesign (engineering). These broad categories indicate the level and amount of research required to implement a resolution. As a program office progresses through the various resolution categories, the amount of research and number of cost elements required to implement a resolution increase. Resolutions under the existing material (logistics) category require actions to secure availability of existing supply. Substitute (engineering) resolutions require engineering involvement to qualify or implement. Redesign resolutions usually require all aspects of engineering and qualification to implement new or highly modified equipment. Table 12 contains the standard definitions and examples of each type of resolution, in order of complexity.

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Table 12. DMSMS Resolution Options, Definitions, and Examples

Resolution	Definition	Examples
No solution required	No solution is required, because existing stock contained in government- or contractor-maintained inventories will satisfy future demands for the product or because the existing software may be used indefinitely without any anticipated repercussions. This is often the result of planned technology refreshment, redesign, or system retirement.	It is determined that sufficient stock of an item exists in current government- or contractor-maintained inventories to support the system until its next technology refreshment. It is determined that firmware embedded in obsolete hardware will remain functional until the hardware is replaced and existing hardware stocks are sufficient to meet system requirements through the end of service date. An Urgent Data Request query identified a vendor that had new items with a government acceptance stamp on them that had previously been sold as excess.
Approved item	The obsolescence issue is resolved by using items already approved on the drawing and still in production but typically not currently used on the system.	Research indicates that the drawing includes a reference to another approved item that is still available. Supply is directed to procure the other approved item. The media used to store the software is no longer readable (e.g., floppy disks). The software is digitally ported to a compact disc.
LON buy[183]	A sufficient quantity of the discontinued item is purchased to sustain the product until its next technology refreshment or the discontinuance of the host assembly. The quantity purchased should consider demands from all users. Because this resolution uses an approved item, no testing or drawing changes are required. The source of supply can be residual stock from the original manufacturer, shelf stock from distributers, sponsor-owned material, and so forth. Costs for packaging, storage, and transportation should be considered in the analysis for selecting resolutions. This is sometimes referred to as a life-of-type buy, bridge buy, or lifetime buy. For software, sufficient licensing and/or support is obtained for the LON, assuming the LON is short enough to ensure that the vendor will remain in business.	Based on historical usage rates, it is determined that 165 diodes are required to sustain the system until it is decommissioned. Sufficient inventory of the discontinued item is then purchased from an approved distributor and stored for use as needed. An LON buy can also be made during design or production. Production material and associated spares can be procured when an obsolescence issue occurs early in the life cycle. A license downgrade is negotiated with the software vendor, which enables the users to expand or extend authorized use of an older product by purchasing additional licenses of the latest version and applying those licenses to the older product until it is retired. A particular adhesive used in production of circuit cards went obsolete. Enough adhesive was purchased to meet demand until a new adhesive could be qualified.

[183] When planning an LON buy, a consideration is that the LON quantity may be hard to predict if there are no funded modification plans. When an OEM makes an LON buy, the program office should understand how DLA inventory was or was not considered.

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Resolution	Definition	Examples
Repair, refurbishment, or reclamation	The obsolescence issue is resolved by doing one of the following: Instituting a repair or refurbishment program for the existing item or assembly, whether through a depot repair, a repair contract with the original manufacturer, or support from a third party. Instituting a reclamation program to reclaim items from marginal, out-of-service, or surplus materiel. Costs for restoring reclaimed materiel from electrostatic discharge damage, handling damage, and heat damage from unsoldering should be considered.[184] Obtaining access to the software source code, development tools, and the human resource skills necessary to change it to ensure continued support.	A program office has sufficient items or assemblies to support the system, if they are refurbished. A private company is identified that has this capability, and a contract is awarded to repair these assets for the system's remaining service life. Hybrids are salvaged from an earlier configuration of the NHA, repaired, and used for future repairs on higher assemblies. Because of scrap steel shortage, it was difficult to maintain a source for high explosive munitions bodies. A process was developed to decontaminate and mill surplus munitions projectiles. The original vendor allows the customer to purchase the source code and the development tools to maintain it and will provide software engineering support for a fee.
Extension of production or support	The supplier is incentivized to continue providing the obsolete items. This may involve long-term agreements to procure specific quantities of items. One-time costs may be associated with setting up this resolution. Those costs should be included in any cost and cost avoidance by being proactive calculations. For software, long-term licensing and/or support agreements are obtained.	The DMT works with the manufacturer to resolve any obsolescence problems with a COTS assembly's piece-parts or raw materials, so the original COTS assembly can still be manufactured. The government obtains the COTS assembly BOM from the OEM, resolves piece-part obsolescence, and then provides the needed parts to the OEM as government-furnished property to facilitate continued manufacture and repair. The DMT works with the manufacturer or software vendor to extend the warranty or support period, thus extending the useful life of the product. A third party is contracted to continue support on a software application. A vendor creates a custom item number that freezes hardware and firmware at a specific version/revision level to ensure that future supply meets the original requirements.

[184] The salvage or reclamation process for used authentic items may impact the item's internal integrity. In addition, salvaged or reclaimed items may be less reliable because of age and because of their storage and shipping. Consequently, significant testing should be conducted before using a reclaimed item. Also see DoDM 4160.21, *Defense Materiel Disposition: Property Disposal and Reclamation*, Volume 2, Change 3, September 30, 2019, available at https://www.esd.whs.mil/Portals/54/Documents/DD/issuances/dodm/416021m_vol2.PDF?ver=2019-09-30-130020-577 for reclamation procedures.

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Resolution	Definition	Examples
Simple substitute	The item is replaced with an existing item that meets all requirements without modification to either the item or its NHA and requires only minimal qualification. Typically, this implies use of a commercial item or NDI that is an F3 substitute. Associated costs are largely administrative. This is sometimes referred to as an alternate.	The original oil specified in the drawings is no longer available. Another company makes oil with equivalent characteristics and was approved as a substitute with minimal evaluation. An F3 equivalent item is available for substitution but has not yet been approved or added to the control drawing. The TDP of an intrinsically suitable, but different, item (e.g., a more reliable version or an existing item) is evaluated. A rebadged COTS product is discontinued by its vendor, but the source item is still available from its OEM under a different part number. The deployed version of an operating system is no longer supported. The supported version is installed as an upgrade and meets all the current requirements. A previously emulated device from an aftermarket developer is substituted for the original item. Software is rehosted to operate correctly with new application hardware or software.
Complex substitute	A replacement item that has different specifications but requires no modification of the source product or the NHA, is researched and validated. The substitute may be the result of a redefined military requirement.	An optical coupler approved in the source control drawing is no longer made. An engineering search finds four couplers with similar characteristics. After qualification, two are approved for the application. The suggested sources table in the source control drawing is changed to authorize the new items. The current operating system is obsolete. The replacement operating system does not meet all the specifications of the current version and must be thoroughly tested. A military requirement was restated or revised to allow for the use of a substitute item (e.g., an automotive grade item) from a commercial source. Another software product is used to replace the obsolete software. A magnetic tape with an obsolete fire-resistant coating was replaced with a tape with a similar fire-resistant coating that had to be fully tested and qualified before use.
Development of a new source	A new manufacturing or production source for the item is established using technical data without affecting the NHA. If the government has not already obtained access to the technical data, the necessary technical data must be purchased or obtained by the government and provided to the new source. First-article testing is required along with any necessary testing to ensure that the new item functionally meets all requirements when installed in the system.	A virtual machine environment card is discontinued by its original manufacturer. Another manufacturer is contracted to purchase drawing packages, manufacturing equipment, and production rights to continue production of the card. A manufacturer is approached to purchase specifications and production rights to resume production of a mechanical item (e.g., a diesel engine) discontinued by the original manufacturer.

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Resolution	Definition	Examples
Design refreshment	The original item is replaced with a new item developed using existing technical data and without significantly affecting the NHA. The government already possesses the bulk of the technical data for the unrefreshed item, or the data is generally available at no cost or can be purchased. First-article testing is required along with any necessary testing to ensure that the new item functionally meets all requirements when installed in the system. The manufacturing source for the new item may be the original manufacturer or a new source.	A special fabrication project in an organic facility is initiated to develop and produce an item. The firmware for a circuit card is no longer available and must be rewritten using different tools. The function of an obsolete ASIC (using original design data) is retargeted into a field-programmable gate array. Minor board re-layout may be required to accommodate the new solution and accompanying support components. A lower-voltage static random-access memory (SRAM) is found to replace an obsolete SRAM. An interposer and voltage regulator are added to maintain the initial design without requalification.
Redesign–NHA	The affected item's NHA must be significantly modified. Only the NHA is affected, and the new design will not affect anything at a higher level in the system.	An obsolete component for which a viable F3 replacement cannot be found requires a redesign of the circuit card on which the component is found. The operating system of a single board computer is obsolete and no longer supported by the manufacturer. Policy dictates that it can no longer be used on DoD systems. The new version of the software will not run on the existing hardware. A replacement board that runs the new version of the operating system is available and will not require changes to other equipment. Some of the associated software must be modified to accommodate the new operating system. A refrigeration system that used a banned Freon refrigerant had to be redesigned to use an approved refrigerant.
Redevelop the item	The original item is replaced with a new item developed without the benefit of existing technical data and without affecting the NHA. The new item may be developed by emulating, reverse engineering, designing a replacement based on the original manufacturing designs and processes, or designing a different item based on the original or new requirements. The manufacturing source for the new item may be the original manufacturer or a new source.	An aftermarket developer creates a device that emulates the original device or a new way to manufacture the item is developed. The software application is redeveloped because of an obsolete compiler or obsolete modeling tools. For example, using digital thread data, obsolete weapon system parts can be printed as replacements at a reduced cost. This can also prevent system downtime that impacts Service readiness levels. A newer technology replacement is designed for an existing obsolete CCA board.
Redesign–complex/ system replacement	A major assembly redesign affects assemblies beyond the obsolete item's NHA and may require that higher level assemblies, software, and interfaces be changed.	Aircraft radar was replaced to use a different operating frequency. Many obsolescence issues were eliminated in the new design. The operating system of a server must be replaced due to policy changes. The new operating system will require hardware changes to multiple hardware and software configurations. A vehicle's diesel engine became obsolete, requiring the replacement of the entire drive train for the vehicle, because the old transmission was not compatible with the new engine.

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Table 13 identifies the cost elements that apply to each resolution. Each "X" indicates a cost element that is likely to be part of the listed resolution and may need to be considered when evaluating costs. There may be some differences when software is an issue when applying the DMSMS cost elements to the resolution options. Because the cost element terminology is very broad, the differences are small. However, no data exist to support whether an average cost estimate for a software resolution will be the same or different than the cost of a hardware resolution.

Table 13. Cost Elements as Applied to DMSMS Resolution Options

Cost Element	Existing Material (Logistics)				Substitute (Engineering)			Redesign (Engineering)			
	Approved Item	LON Buy	Repair, Refurbishment, Reclamation	Extension of Production or Support	Simple Substitute	Complex Substitute	Development of a New Source	Design Refreshment	Redevelop the Item	Redesign—NHA	Redesign–Complex/System Replacement
Non-recurring engineering			X			X	X	X	X	X	X
Engineering, engineering data revision					X	X	X	X	X	X	X
Purchase of engineering, design, or technical data			X			X	X	X	X	X	X
Qualification of new items					X	X	X	X	X	X	X
Revision of test procedures			X			X	X	X	X	X	X
Software changes						X	X	X	X	X	X
Start-up costs (after-market, and so forth)			X	X			X	X	X	X	X
Testing			X		X	X	X	X	X	X	X
Tooling, equipment, test equipment, or software			X			X	X	X	X	X	X
Computer programs/ documentation			X		X	X	X	X	X	X	X
Interim support								X	X	X	X
Supply/provisioning data					X	X		X	X	X	X
Support/test equipment			X			X	X	X	X	X	X
Technical manuals			X		X	X		X	X	X	X
Training/trainers			X			X		X	X	X	X
Spares (optional)			X				X	X	X	X	X
Other (as required)	X	X[185]	X[186]	X	X	X	X	X	X	X	X

6.3 Determine the Preferred DMSMS Resolution

6.3.1 Overall Process

The resolution determination process uses various outputs from monitoring and surveillance and health assessments that determine if and when an issue will affect the operational availability of the system.

[185] Optionally includes the one-time cost of the items themselves since it is a programming and budgeting consideration.

[186] Optionally includes the cost of the item's repair, refurbishment, or reclamation itself.

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Figure 19 illustrates the major activities and tasks of the resolution determination process. One important, initial activity in this process is to determine whether there is an external organization that is pursuing a resolution to a DMSMS issue in a manner that meets the program office's schedule and technical requirements. If the external resolution process meets all the system's requirements, then the program office only needs to monitor that the resolution process is on track. Otherwise, the program office should pursue the other major activities of the resolution process.

The resolution process should also consider the requirements to transition from one system life cycle phase or contract to another. For example, resolutions in the design phase may include short-term actions until a longer-term option can be implemented in the production phase. This complicates the process because the analysis may need to be conducted over multiple time increments (e.g., between now and the end of production and between the end of production and a planned technology insertion).

If the health assessment indicates that a resolution is needed, one or more of the resolutions listed in Table 12 can be applied regardless of whether the DMSMS problem is mechanical, material, software, or electronic in nature.[187] While all these resolution types apply to different item types, the frequency of use of a particular DMSMS resolution may differ depending on commodity type. Table 14 illustrates the observed distribution of resolution types by commodity—electrical (28.4%), electronics (63.2%), and mechanical (8.4%)—using data collected through a 2014 Department of Commerce DMSMS Cost Survey.[188] Although implementation may vary drastically for different types of issues, the overall resolution determination process is the same.[189] Not all resolutions can be applied to a given DMSMS problem. Only those that can be applied are considered viable.[190] For instance, most of the resolution options are not viable for a forged impeller body that has become obsolete; the only viable resolutions may be the identification of a new source or redesign. However, the process to determine the viable resolutions (often the most cost effective) is the same, whether the problem is a CCA or a specific chemical used in the manufacturing process that has become obsolete because of environmental restrictions.

[187] The National Defense Stockpile should be a consideration for a DMSMS issue relating to raw materials. The raw materials may already be stockpiled, or they may be added to the stockpile in the future. For more information, see 50 U.S.C. § 98 (available at https://www.govinfo.gov/link/uscode/50/98) and refer to the DLA Strategic Materials office.

[188] Appendix M contains more detailed survey data.

[189] Technology and product roadmaps can provide information that influences options for implementing different resolutions.

[190] All requirements (performance, safety, security, and so forth) must be met for a resolution to be viable. A viable resolution must also address all second-order derivative effects.

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Table 14. Distribution of DMSMS Resolutions by Part Commodity

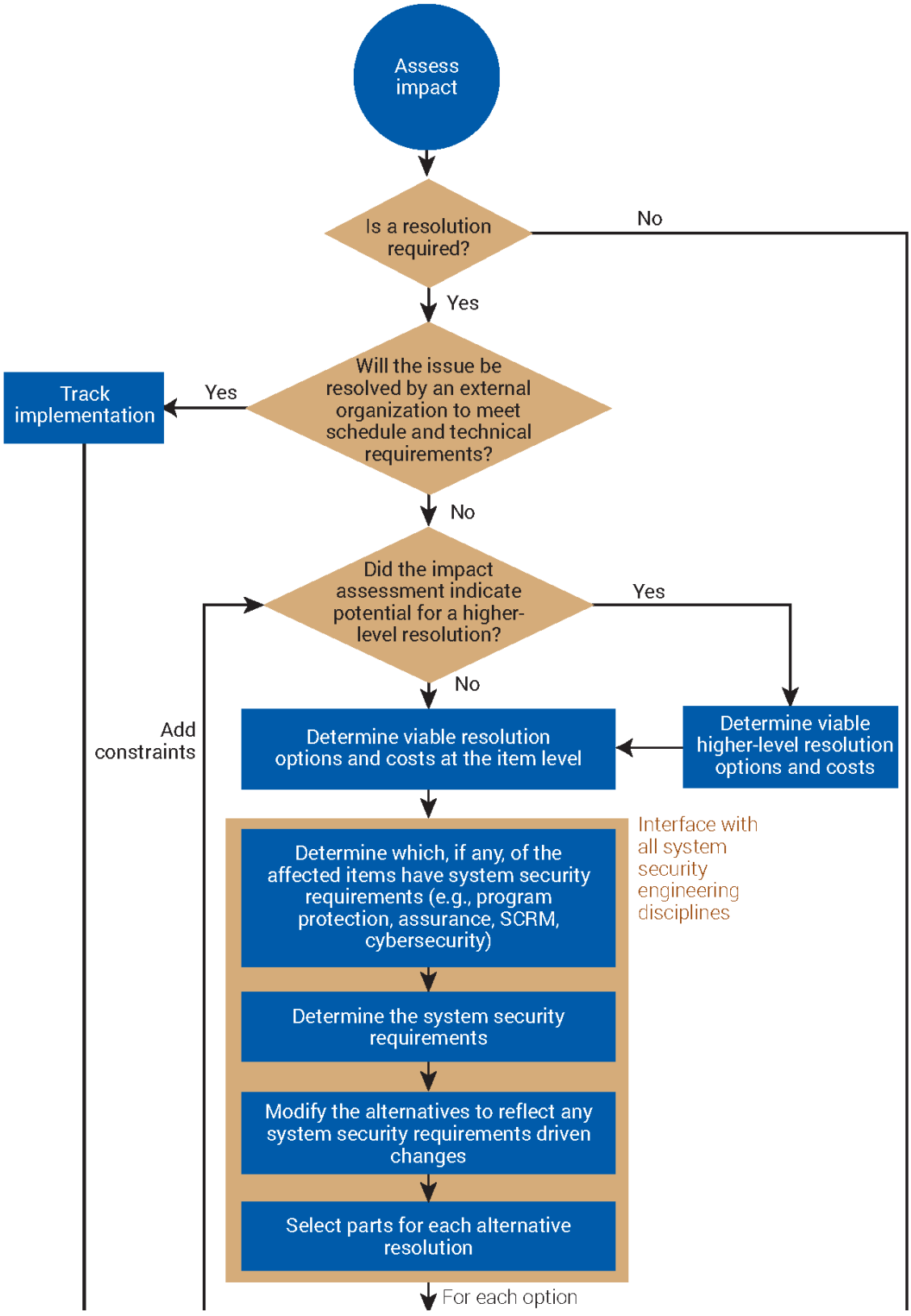
DMSMS Resolution	Electrical	Electronics	Mechanical	Total
Approved items	987	316	236	1,539
LON buy	27	633	6	666
Simple substitute	190	1,233	77	1,500
Complex substitute	34	331	45	410
Extension of production or support	27	68	3	98
Repair, refurbishment, or reclamation	1	38	1	40
Development of a new source, design refreshment, and redevelop the item[191]	14	103	10	127
Redesign—NHA	2	132	4	138
Redesign—complex/system replacement	12	31	1	44
Total	1,294	2,885	383	4,562

Source: DSPO, *Diminishing Manufacturing Sources and Material Shortages: Cost Metrics* (Fort Belvoir, VA: DSPO, February 2015), p. 12.

[191] The Department of Commerce survey asked for data in a single resolution type: development of a new item or source.

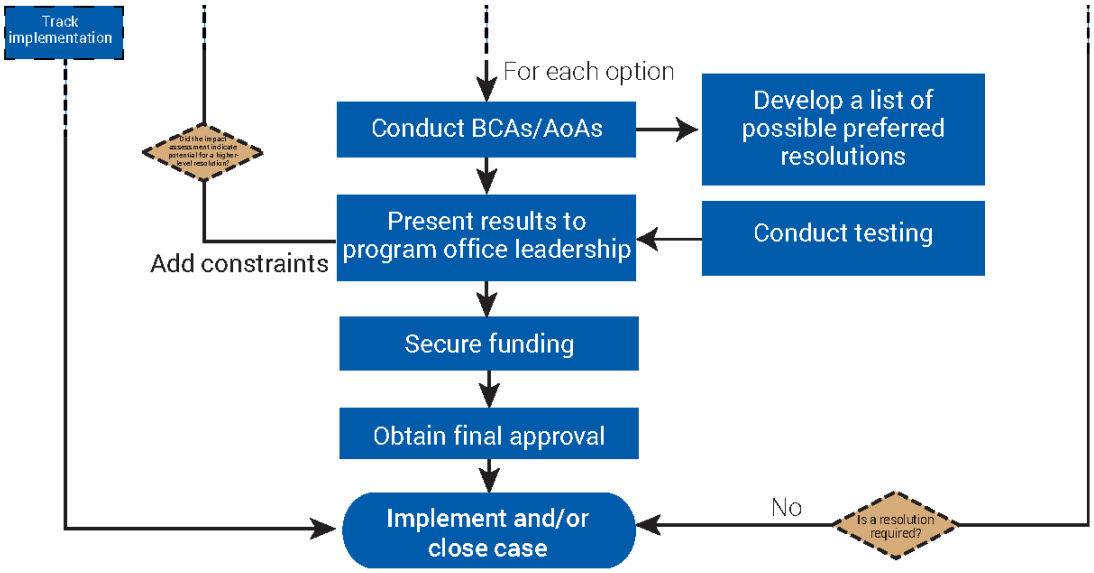
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Figure 19. DMSMS Resolution Determination Process



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Figure 19. DMSMS Resolution Determination Process *continued*



The remainder of this section discusses the lower portion of Figure 19 in greater detail.

- Section 6.3.2 expands upon the determination of viable resolution options
- Section 6.3.3 discusses system security engineering aspects of the resolutions
- Section 6.3.4 provides information on conducting BCAs/AoAs
- Section 6.3.5 expands upon interactions with program leadership to obtain initial resolution approval
- Section 6.3.6 deals briefly with preparations for implementation (implementation is the subject of Section 7).

6.3.2 Determining Viable Resolutions

There is no single best way to incorporate the guidance of this document into a repeatable process for determining viable resolution options. The remainder of this section lays out factors that should be considered. When the DMSMS issue also involves FMS equipment, the other countries that will have to implement the resolution should be kept informed.

All approaches begin with an analysis of the situation. Much of the logistics and programmatic data will have already been collected in the *Assess* step discussed in Section 5. Examples of other information program offices should gather include:

- What is the LON for the item, accounting for both DoD and FMS equipment if applicable?
- How many of the items are still available? Consideration should be given to stock at organic and commercial depots, stock expected to be produced in the future (perhaps with a known end-time for further production), and stock that may be cannibalized from out-of-service assets (such as would be found at “boneyards”).
- Are substitute items listed in drawings or other technical data?
- In the case of software, how long is product support expected to be available? For hardware, for how long are repair services expected to be available?

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6.3.2.1 Role of Design Considerations

During the initial design of a system, systems engineers select parts for the system's design, considering the design, production, operation, support, and disposal of the system throughout its life cycle. This part selection process entails the balancing of these design considerations (e.g., the application, standardization, technology, qualification, producibility, performance, cost, DMSMS risk, reliability, maintainability, supportability, cyber weaknesses and vulnerabilities, supply chain risk, susceptibility to counterfeit and unauthorized tampering, use of hazardous materials, and other factors) to identify the parts that best satisfy the requirements for the system's design. These design considerations should also be considered when evaluating resolution options for any DMSMS issue that arises after the initial design of the system.

Design considerations impact the feasibility of resolutions. For example:

- ***Real-time software***. Many times, defense products use software that performs a real-time function. Often this is software for signal processing or signal analysis that has a specific time available to accomplish a specific task. This means that the failure to complete the signal processing in the available time means that information is lost, or worse, the system will crash. DMSMS issues can often have a profound impact on the portability of real-time software. Even when the real-time properties of various software modules are well documented, and latencies of interrupted service requirements are documented, the need to use a new and different processor or components with different memory speed due to obsolescence of an earlier or different processor component can result in considerable impact to real-time performance. Furthermore, real-time performance validation can be extremely difficult, particularly when real-time failures only occur under rare combinations of interrupted conditions. Whether a new processor performs slower or faster, the interaction with external interfaces is certain to be different, and it is occasionally difficult for drivers to fully compensate for speed changes. It is essential that the design, development, and maintenance of software systems that have real-time components maintain very accurate analysis and models of system timing so that as processors and technology evolve, the real-time performance can be easily validated, and software can be ported. There are also important implications with respect to maintaining an understanding of execution time statistics of each software module, and the corresponding understanding of selected compiler optimizations and coding style, to maintain real-time performance when DMSMS issues cause design updates.
- ***Validation and production testing***. Product testing advances today include many sophisticated capabilities to assess analog and digital subsystem performance. When a DMSMS issue occurs, the resolution may impact the means for testing. Today's digital subsystems are tested with Joint Test Action Group and other interface validation. If a design or redesign causes interface changes, or timing changes, there are likely to be impacts to both validation tests and production line tests. It is essential to assure that testability of validation and production line tests remain resilient to these changes. This is especially important so that counterfeit components and marginal designs are quickly detected, identified, and corrected as part of a DMSMS redesign. To assure this, the testing process must retain documentation of traceability between what functions are being tested and how the tests relate to requirements, so that system behavior changes after a DMSMS redesign can be properly understood and validated. When the test equipment itself is affected by a DMSMS issue, validation considerations go beyond testing the function of the equipment. To deal with the potential for malware being introduced into the test equipment, validation should ensure that the equipment is performing all the tests in the way that they should be performed. This is beyond the scope of typical validation testing.

6.3.2.2 The Role of Roadmaps

Roadmaps provide important information on the need for resolutions in the future as well as technical details on resolutions associated with immediate issues.

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6.3.2.2.1 Direct Effect on Formulating Resolutions

Product roadmaps offer information on the following:

- How long an item will be in the system (i.e., the length of need) and
- When planned modifications to the system will be made and the extent of those changes.

While some of this information is in product improvement and product supportability roadmaps, all the information should be integrated in the overall product roadmap and the IMS. If this is the case, the DMSMS community does not need to use product improvement or product supportability roadmaps; the product roadmap suffices.

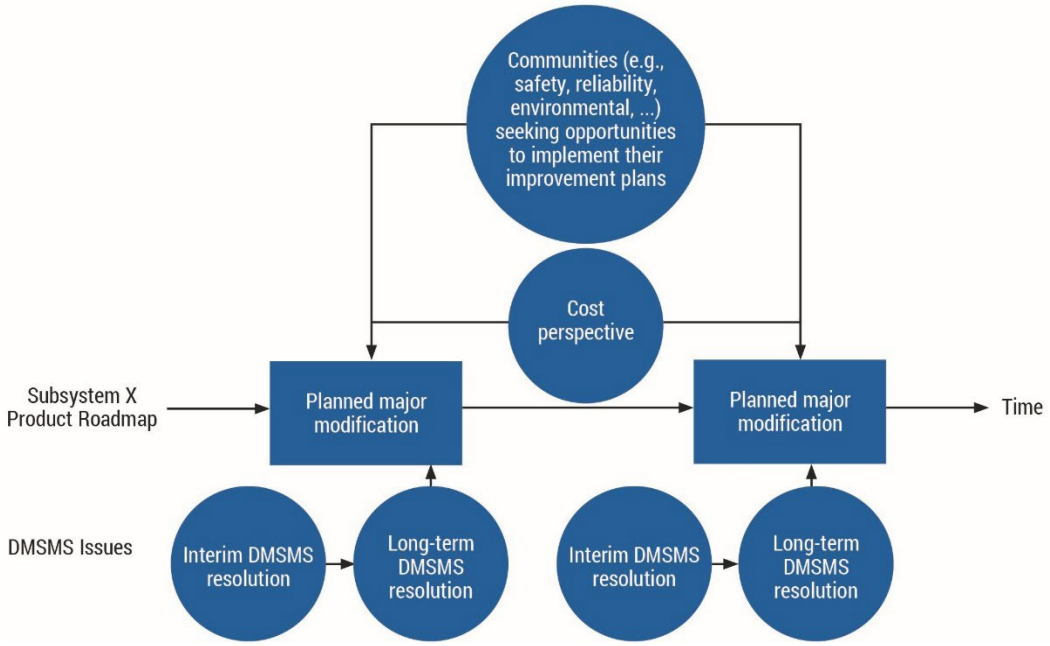
The DMSMS community uses product roadmaps for the following purposes:

- ***Sizing LON buys***. In some situations, the selected resolution is a LON buy. To calculate the quantity to purchase, information is required on how long the item must be supported. If the item will be removed from the system because of a planned future modification, then the LON would decrease to the point at which the modification is completed (or nearly completed). If the purchase were sized without the product roadmap, too many would be procured, and money would be wasted. In addition, item demand from FMS equipment should be factored in the quantity to purchase if there is an LOA in place that includes and funds such purchases. It is important to recognize that DoD and each foreign partner often have different timeframes for operating the equipment involved in the FMS case. There may be confusion over the meaning of a phrase like “sustain a system until a specific date.” The phrase may be interpreted as meaning that no system will be operating after that specific date, or it may be interpreted to mean that equipment retirement will begin at that specific date. When calculating the size of a LON buy, DoD should ensure that each partner bases its calculation on the anticipated demand over its own timeframe for operating the equipment. (See Section 7.3 for more information on sizing LON buys.)
- ***Evaluating competing resolutions***. Product roadmaps assist evaluating competing resolutions. For example, if an LON buy and a substitute part were both viable, the product roadmap could be instrumental in making a final decision.
- ***Planning a multi-phase resolution that delays or postpones a redesign***. This planning would focus on a DMSMS issue that is associated with an item that is or could be affected by the planned modification in the product roadmap (e.g., the resolution chosen without knowledge of the product roadmap might be a redesign of the item). If such an item were changed or eliminated by the modification, then performing a redesign earlier may be avoidable. A substitute (or an LON buy or reclamation) may be feasible and sufficient to offer an interim resolution before the modification occurs. Even if the item were not affected in the planned modification, waiting until the modification to begin the redesign can reduce costs by performing the redesign in conjunction with the other work. Furthermore, when the modification and redesign are combined, a redesign at a different level of assembly may be more cost-effective. The same set of interim resolutions may apply (see Section 6.3.4 for more information on formulating a cost-effective resolution).

Figure 20 illustrates how short- and long-term DMSMS resolutions can be the most cost-effective approach to supportability because they enable synchronization with planned roadmap events. To enable this synchronization, individuals evaluating DMSMS resolutions must first know what the planned roadmap events are, which may focus on design changes to resolve other support function concerns (e.g., reliability, safety, environmental), encompass, as well as their timing and magnitude. Strategically, those functions attempt to meld their perspectives into the larger product improvement standpoint. Cost influences the ability to include all desired changes.

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Figure 20. Interactions with Supportability and Product Improvement



The DMSMS community uses product roadmaps to evaluate the type, scope, and timing of resolutions. Therefore, these roadmaps affect the programming and budgeting of the resources necessary to implement resolutions—not only the number of resources necessary, but also the responsibility for securing them. If the DMSMS resolution is implemented by itself, then the DMSMS community is responsible. However, in situations where the DMSMS resolution is implemented in conjunction with a larger change to the product, then the IPT responsible for the modifications may be responsible for programming and budgeting for the modification and the DMSMS resolution.[192] Just as DMSMS may affect programming and budgeting by the responsible IPT (see Section 7.2 for more information on integrating DMSMS resolution and modification funding), it may affect the scope and timing of the change as described in the next section.

6.3.2.2.2 Interfaces with Modification Planning in Formulating Resolutions

There are two main drivers for modification plans (i.e., plans to change the configuration of the system design) are numerous. Two of the main drivers are as follows:

- ***Modification plans to improve supportability (including the resolution of DMSMS issues)***. This type of modification plan is often characterized as addressing technology refreshment needs. Technology refreshment is described as "the periodic replacement of COTS items … to assure continued supportability of that system through an indefinite service life. Technology refreshments can be strategically applied to prevent the occurrence of DMSMS issues preemptively or to minimize them significantly."[193] In DoD, the limitation to apply technology refreshment only to COTS items is not applicable. It can apply to custom electronics as well.
- ***Modification planning to improve capability***. This type of modification plan is often characterized as addressing technology insertion needs. Resolutions to DMSMS issues are, by definition, not intended to improve capability, except perhaps when a minor capability

[192] It is unusual for two organizations to be responsible for obtaining the funding for a single body of work.

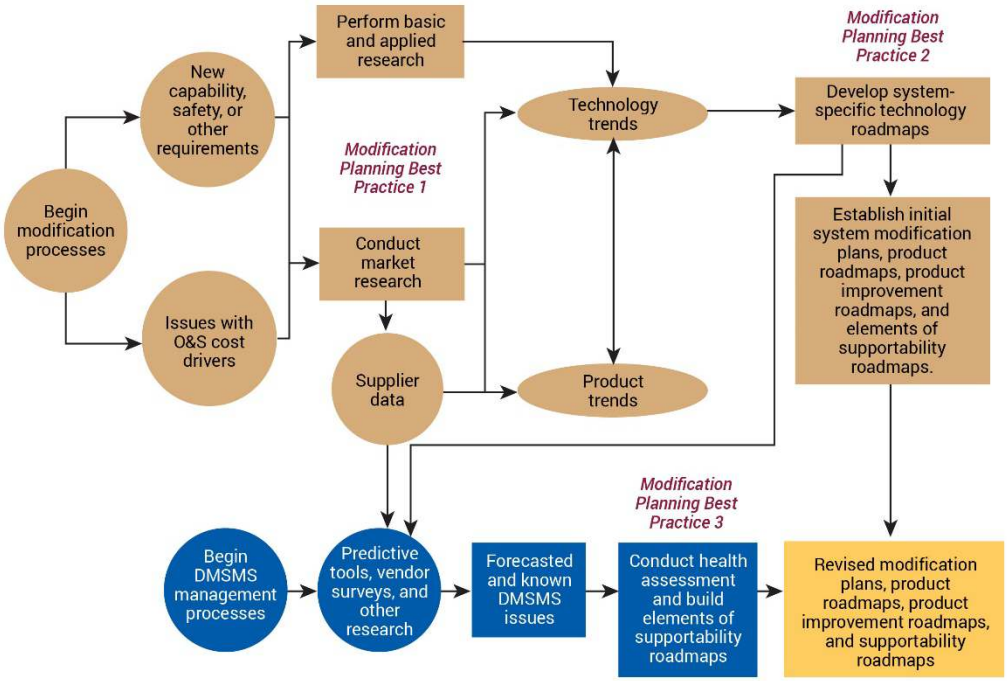
[193] Pete Pizzutillo, "Technology Refreshment—A Management/Acquisition Perspective," July 2001.

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increase results incidentally. A program office may, however, leverage a planned technology insertion to simultaneously implement needed DMSMS resolutions.

Figure 21 Illustrates the interactions between modification planning, roadmapping, and DMSMS management activities. The tan portions are associated with modification planning processes while the blue portions pertain to DMSMS management–related activities. The yellow portion represents general processes taking both into account.

Figure 21. Interactions among DMSMS Management, Roadmapping, and Modification Planning



The left side of the figure (where the reader should start) deals with technology management and O&S cost drivers. The upper right side depicts the formulation of technology roadmaps and modification plans. The bottom shows interactions with DMSMS management that refine modification plans. The following bullets discuss various elements of this figure through a set of enabling best practices (shown on the figure) for DMSMS management and modification planning. The first two of these enabling best practices indicate ways a program office's DMSMS management community should leverage its modification planning. The third best practice, which includes significantly more explanatory material, discusses how DMSMS management may influence modification planning.

- ***Modification Planning Best Practice 1***. Utilize market research to minimize near-term DMSMS issues. Market research identifies areas where new technologies and products will be introduced. Modification planning should strongly consider incorporating these new technologies and products in the system. Doing so enables an AS and a life cycle sustainment strategy that minimizes the cost of resolving future obsolescence issues while incorporating state-of-the-art technologies to increase reliability, lower sustainment requirement costs, and increase warfighting capability to meet evolving requirements throughout an indefinite service life.
- ***Modification Planning Best Practice 2***. Utilize technology roadmaps to guide modification planning. Technology roadmaps (often developed external to the program office) and system modification are closely linked. Technology roadmaps show when technologies are mature enough to be applied to a system, implying the potential for future DMSMS issues and the approximate timing for technology insertion or refreshment. Thorough technology roadmaps,

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therefore, should form an initial basis for modification plans to avoid some obsolescence issues before they materialize and resolve other issues while achieving other product improvement and supportability benefits.

- ***Modification Planning Best Practice 3***. Use DMSMS health assessments to refine modification plans. Health assessments are discussed in Section 5.3.1 and Appendix L. DMSMS health assessments should be made for LRUs, boxes, COTS, or other levels of assembly that contain items vulnerable to obsolescence. The health assessments portray when individual items are expected to become obsolete and when that obsolescence may affect the assembly based on analysis of PDNs, OEM surveys, the results generated by predictive tool algorithms, knowledge of typical commercial technology life cycles, and technology and product roadmaps. Given this forecast of when and what in the system will be affected by obsolescence, the DMT can recommend DMSMS resolutions to eliminate the problem or defer impact. The modifications (developed jointly by the PSM and the DMSMS community) necessary to prevent the ill effects forecasted in these heath assessments are ultimately translated into the supportability roadmaps once funded and scheduled.

Figure 22 represents the notional results of a health assessment of a subsystem with five primary parts. The red, yellow, and green coloring scheme in the figure connotes the size of the inventory of the parts in the supply system. Green implies adequate, yellow notes a concern, and red means unfulfilled demands. The black vertical line denotes the optimal time for a technology refresh because all the parts inventories are adequate or of concern and some are soon to be unfilled. As a function of risk, the optimal time could be earlier because the effect of the shortfalls would not be acceptable or later if the effect of the shortfalls would be minimal.

Figure 22. Notional Results of a Health Assessment

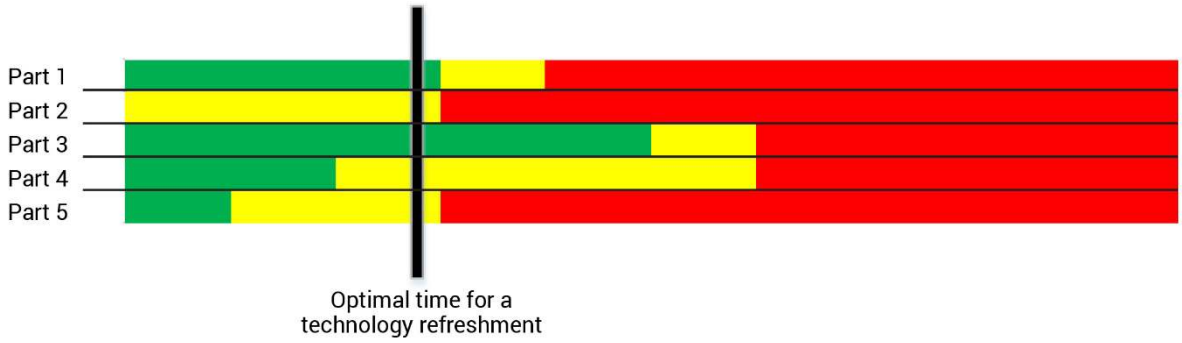


Figure 33 in Appendix K and Figure 22 are linked.[194] Assume that configuration 5 in Figure 33 is a technology refreshment. In theory, the optimal time would coincide with the start of configuration 5 in Figure 33. But, in practice, this may not necessarily be the case. The DMSMS management community is not responsible for managing the content and schedule for a program office's technology refreshments. The DMT may recommend interim DMSMS resolutions to delay the technology refreshment initiation for a limited time. The optimal technology refreshment date is that point when the sum of the cost of individual resolutions is greater than the cost to redesign.

[194] The relationship between the date of impact estimated by the DMSMS management community and comparable points from a technology roadmap is not predictable in advance. The underlying data used by the DMSMS management community is different. Furthermore, the impact date may be a multiple-year range with additional uncertainty about whether obsolescence will occur in the timeframe.

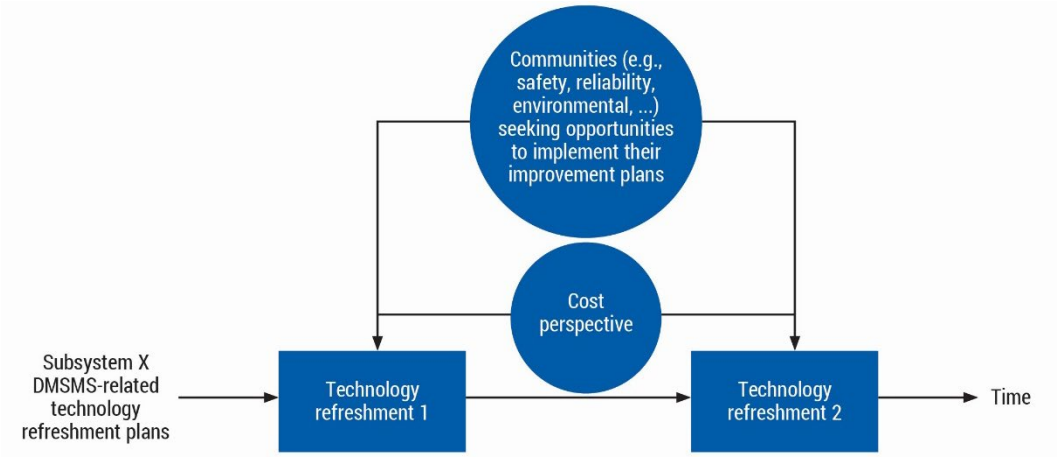
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From a general availability and cost perspective, combining technology refreshment with technology insertion is often convenient. The DMSMS community can suggest changing the schedule for planned technology insertion if supportability to the planned insertion start date cannot be extended.

It is helpful if the engineering and product support members of the DMT, who contribute to the health assessment and interface with their counterparts, also contribute to modifications plans. A program office can avoid significant DMSMS-related resolution costs in its POMs and budgets by selecting optimum system modification dates. For example, a modification plan to upgrade a product should simultaneously seek to eliminate obsolete or near obsolete items (as identified via a health assessment) because resolving a DMSMS issue simultaneously with other changes to the system design is more cost effective than a standalone, out-of-cycle redesign only removing obsolescence.

Figure 23, based on Figure 22, illustrates this point. These interactions are more likely during sustainment when capability enhancements are less likely and instances of in-supportability are more frequent.

Figure 23. Necessary Technology Refreshment May Enable Capability Improvements



Robust DMSMS management will lower the costs associated with obsolescence issues. However, even in the best program offices, individual DMSMS resolutions are often suboptimal. LON procurements are problematic because of limited contractual horizons and uncertainties in estimating the total requirement over the remainder of the system life cycle. Finding or qualifying alternative items may work for a time, but such approaches rarely use new technologies and capabilities. Unplanned redesigns are costly. Therefore, refining the scope and timing of modification plans based on DMSMS health assessments is a best practice to further reduce DMSMS-specific programming, budgeting, and readiness effects throughout the life cycle.

6.3.2.2.3 Repeating the Process

The activities in this section recur, although some may not do so as frequently as the activities for analyzing item availability. Circumstances for *using* roadmaps include integrating programming and budgeting for technology refreshment and insertion, formulating phased resolutions, forecasting DMSMS issues, and sizing LON buys.[195] The DMSMS community should always use the latest version of roadmaps whenever the circumstances create the need to do so. However, in some instances, a roadmap can change in a way that affects a prior DMSMS management decision.

[195] See Sections 7.2 and 7.3.

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Once executed, changing an LON buy decision may not be possible. If a roadmap change indicates that an LON was too large, then eventually excess inventory will occur. If a roadmap change indicates that the LON buy was too small, then inventories should be watched closely, and a supplemental resolution should be considered. Whenever any roadmap changes, the DMSMS community should review those changes and how they may impact future resolutions and their associated funding and update them as required. If budget updates are not feasible, reallocating resources in existing budgets should begin immediately. Forecasts of future DMSMS issues should be reevaluated to account for supportability roadmap changes.

Product improvement roadmaps are updated as needed, typically when the funding profile changes or unanticipated capability needs materialize. Product roadmaps may also be revised if the roadmap for the requisite technology changes. For example, a technological breakthrough could enable an unplanned, but desirable, capability. Similarly, the timing of technology and manufacturing maturity could accelerate or extend the product improvement schedule.

Supportability considerations (and, therefore, supportability roadmaps) could change at a frequency different from product improvement roadmaps. If supportability roadmaps are combined with the product improvement roadmaps, the pace of change would be a combination of the two.

Supportability roadmaps can often change because of funding availability. ROI is usually a primary consideration in funding any such effort. However, an upfront investment (paid back over time) is typically necessary. Obtaining sufficient funding for these upfront investments is difficult and even when upfront funding is attained, it is sometimes diverted to other higher priority items.

Supportability roadmaps may also be affected whenever the analysis of item availability changes or technology roadmaps change (if utilized). For example, supportability issues can occur when demand for the item changes. Similarly, an unanticipated discontinuation notice may be released because of the introduction of an updated version of the item or insufficient sales to make a viable business case for maintaining production.

Technological breakthroughs or setbacks affect technology roadmaps. Changes in market conditions influence the timing of new market offerings. Cognizant IPTs should have their technology roadmaps updated periodically to reevaluate their product improvement and supportability roadmaps. These updates require an external relook at DoD technology development progress and updated market research efforts.

6.3.2.3 Initial Formulation of Viable Resolution Options

Having identified the basic facts and other considerations, viable resolution options can be formulated. Table 12 provides categories of resolution options and it is recommended that the DMT go through each row in this table from top to bottom to identify applicable and feasible options. As resolutions become more complex, their implementation becomes more costly.[196] Many of the factors (e.g., near-term cost, total life cycle cost, level in the system, mission factors, planned technology refreshes or upgrades, results of health assessments, and terms and conditions of contract) used to analyze the operational impact should be used to help determine which resolutions are viable. The overall resolution determination process should consider all viable resolutions at the lowest level of indenture possible.

[196] An LON buy may appear to be the least costly and simplest option to implement. However, before this resolution may be used, many obstacles must be addressed. Limitations may be imposed on the size of an LON buy (see Section 7.3). Also, contractors cannot typically be obligated to procure stock beyond the life of their contract, so the government would need to procure and maintain a stock of the needed item. Also, because reliability and end of system life are estimates, accurately determining the quantity of an item to buy is nearly impossible. These obstacles may result in the determination that an LON buy is not an option unless it would be used as an interim resolution until another alternative is implemented.

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However, if the health assessment indicates that a resolution at a higher level of assembly may be preferable, resolution at that higher level of assembly should also be considered. In some cases, a resolution at a higher level will have a higher ROI, because it may resolve numerous issues at once or provide other improvements (e.g., improve reliability).

At this point in the process, implementation and costs do not need to be elaborated in detail. The only requirement is that each resolution option identified meets the technical requirements of the systems in which items will be used. For instance, if using a substitute as a proposed resolution, costs and availability need not be determined, but the item must have engineers' approval for use. In other cases, such as a redesign, it is assumed the end-product will meet all technical specifications (because these will be written into the redesign contract).

The first row of Table 12 defines the "No solution required" DMSMS resolution option. Sometimes a program office seems to face a serious DMSMS issue but finds that further information contradicts that initial conclusion. Regardless, a resolution option of doing nothing now and assessing the possibility that external factors resolve the DMSMS issue should always be considered. While this option involves uncertainty, DMTs should enact due diligence and assess this option. It may be attractive, for instance, if the risk to mission of a materiel shortage is low (considering all risk categories identified in Figure 12) and the costs of all other resolution options are high.

The other resolution categories in Table 12 may also be informed by the analysis of the situation. For instance:

- An LON buy is an option if enough stock is available for purchase to meet the lifetime need of the item. The *cost* of an LON buy, to include storage costs, will be determined later. At this point, only the feasibility of an LON is being evaluated.
- There are multiple types of redesigns and technology refreshments in Table 12. While there may not be a refresh or redesign planned that will resolve the DMSMS issue, it may be possible to schedule one that takes advantage of planned modification events for other items to achieve economies of scale.
- Other entries in Table 12, such as the development of a new source of supply, will always be an option. The use of multi-phase options, such as a partial LON buy followed by a redesign, *can* also be considered, but there are many possible combinations.

6.3.3 Considering System Security Engineering

System security engineering should also inform the determination of viable resolution options.

6.3.3.1 Terminology

System security engineering integrates contributions from system security engineering disciplines[197] such as anti-tamper, DEFs,[198] hardware assurance, software assurance, SCRM, program protection, and cybersecurity and contributions from security specialties[199] such as personnel security, industrial security, physical security, and information security. These contributions include intelligence and

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[197] The Department of Defense Technology and Program Protection Guidebook of July 2022, available at https://aaf.dau.edu/storage/2024/08/TPP_Guidebook.pdf, uses the term *specialties* instead of *disciplines*. To better distinguish them from the traditional security specialties, this document uses the term *disciplines*.

[198] DEF is an element of technology security and foreign disclosure. It encompasses anti-tamper and the removal of technologies or capabilities prohibited for export. Since the former is already included among the system security engineering disciplines and the latter is only pertinent to legal restrictions on DMSMS resolutions associated with FMS, DEF is not discussed further in this document.

[199] Security specialties are applicable to classifying and protecting parts or information about parts after they are selected and used. As such, security specialties are typically a security manager's responsibility and are not discussed further in this document.

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counterintelligence threat assessments, vulnerability assessments, and the identification of protection measures or technical mitigations to address those vulnerabilities. The outcome is a comprehensive program and system protection within the constraints of cost, schedule, and performance while maintaining an acceptable level of risk of exploitation.

System security engineering is defined by DoDI 5200.44 as "an element of system engineering that applies scientific and engineering principles to identify security vulnerabilities and minimize or contain risks associated with these vulnerabilities."[200] As such, system security engineering also translates protection measures into system security requirements taking broader systems engineering constraints and decisions into account.

In practice, however, there may be overlap among the functions performed by the different system security engineering disciplines and the scope of their responsibilities may vary. Because of the risk of these disciplines imposing different security requirements on a system, they must interface both among themselves and with DMSMS management practitioners to contribute to the formulation and implementation of resolutions to DMSMS issues.

- Anti-tamper encompasses systems engineering activities intended to prevent or delay exploitation of CPI in U.S. defense systems in domestic and export configurations to impede countermeasure development, unintended technology transfer, or alteration of a system due to reverse engineering.[201] CPI are "U.S. capability elements that contribute to the warfighters' technical advantage, which if compromised, undermines U.S. military preeminence. U.S. capability elements may include, but are not limited to, software algorithms and specific hardware residing on the system, its training equipment, or maintenance support equipment."[202]
- All DoD systems require some degree of hardware and software assurance. Hardware assurance is "an activity to ensure a level of confidence that microelectronics (also known as microcircuits, semiconductors, and integrated circuits, including its embedded software and/or IP) function as intended and are free of known vulnerabilities, either intentionally or unintentionally designed or inserted as part of the system's hardware and/or its embedded software and/or IP, throughout the life cycle."[203] Software assurance is "the level of confidence that software functions as intended and is free of vulnerabilities, either intentionally or unintentionally designed or inserted as part of the software throughout the lifecycle."[204] When vulnerabilities in the items in DoD systems are exploited, the ill effects include the following:
    - Memory can be read or changed,
    - Malicious circuitry can be injected,
    - Data can be revealed to attackers,
    - Encryption protections can be disabled, and
    - Counterfeits can enter the DoD supply system and be installed on DoD systems.

SCRM is a key enabler for hardware and software assurance. SCRM is "the process for managing risk by identifying, assessing, and mitigating threats, vulnerabilities, and disruptions to the DoD supply chain from

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[200] DoDI 5200.44, *Protection of Mission Critical Functions to Achieve Trusted Systems and Networks*, February 16, 2024, available at https://www.esd.whs.mil/Portals/54/Documents/DD/issuances/dodi/520044p.pdf.

[201] DAU, *Glossary*, July 21, 2020, available at https://www.dau.edu/glossary/anti-tamper.

[202] DoDI 5200.39, *Critical Program Information (CPI) Identification and Protection Within Research, Development, Test, and Evaluation (RDT&E)*, May 28, 2015, change 3 effective October 1, 2020, available at https://www.esd.whs.mil/Portals/54/Documents/DD/issuances/dodi/520039p.pdf.

[203] DAU, *Glossary*, July 21, 2020, available at https://www.dau.edu/glossary/hardware-assurance.

[204] Ibid.

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beginning to end [of a system's life] to ensure mission effectiveness. Successful SCRM maintains the integrity of products, services, people, and technologies, and ensures the undisrupted flow of product, materiel, information, and finances across the life cycle of a weapon or support system. DoD SCRM encompasses all sub-sets of SCRM, such as cybersecurity, software assurance, obsolescence, counterfeit parts, foreign ownership of sub-tier vendors, and other categories of risk that affect the supply chain."[205] Supply chain illumination is an important SCRM element.

- Program protection is another important assurance enabler. Program protection is "the integrating process for managing risks to advanced technology and mission-critical system functionality from foreign collection, design vulnerability or supply chain exploit/insertion, and battlefield loss throughout the acquisition lifecycle."[206] Program protection planning identifies CPI, threats to compromise CPI, and the associated risk. It is DoD policy that "CPI will be identified early and reassessed throughout the Research, Development, Test, and Evaluation program so that CPI protections requirements and countermeasures may be identified and applied as the CPI is developed and modified throughout the lifecycle as needed."[207] Furthermore, "CPI protection measures will be integrated and synchronized, then documented within the PPP in accordance with [DoDI 5200.39]."[208]
- Cybersecurity is "the prevention of damage to, protection of, and restoration of computers, electronic communications systems, electronic communications services, wire communication, and electronic communication, including information contained therein, to ensure its availability, integrity, authentication, confidentiality, and nonrepudiation."[209]

6.3.3.2 Identifying Items with System Security Engineering Requirements

Critical items needing system security requirements to reduce the risk of exploitation must be identified during design. Those items also impact initial part selection, configuration, BOMs, and supplier selection prior to production. The PPP should be the primary source for identifying critical items.[210] However in practice, there may be *some* additional critical items, not explicit in the PPP, that are similarly vulnerable to exploitation in a way that degrades the level of assurance. For example, not every logic-bearing microelectronics device is typically explicit in a PPP. Reasons for such omissions include:

- Critical components may be hard to identify,
- Identifying components associated with CPI is partially subjective,
- Items needing protection may be identified at too high of a level of assembly,
- The PPP may not be current,[211]
- Program offices often attempt to minimize CPI because of constraints on funding and people, and

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[205] DoDI 4140.01, *DoD Supply Chain Materiel Management Policy*, March 6, 2019, available at https://www.esd.whs.mil/Portals/54/Documents/DD/issuances/dodi/414001p.pdf, p. 21.

[206] DAU, *Program Protection Plan Outline and Guidance*, Version 1.0, July 2011. A similar definition is provided in the DAU, *Glossary*, available at https://www.dau.edu/glossary/program-protection, as "the integrating process for managing risks to DoD warfighting capability from foreign intelligence collection, from hardware, software, and cyber vulnerability or supply chain exploitation, and from battlefield loss throughout the system life cycle."

[207] DoDI 5200.39, *Critical Program Information (CPI) Identification and Protection Within Research, Development, Test, and Evaluation (RDT&E)*, May 28, 2015, change 3, effective October 1, 2020, available at, available at https://www.esd.whs.mil/Portals/54/Documents/DD/issuances/dodi/520039p.pdf.

[208] Ibid.

[209] DAU, *Glossary*, available at https://www.dau.edu/glossary/cybersecurity.

[210] DoDI 5000.83, available at https://www.esd.whs.mil/Portals/54/Documents/DD/issuances/dodi/500083p.pdf, introduced a S&T Protection Plan that also addresses the protection of critical technology elements. The S&T Protection and Program Protection Plan are two phases of protection that now covers S&T projects and developmental programs. The intent is for the S&T plan to naturally transition to a program protection plan when a program of record is established.

[211] A good practice to help mitigate this situation is to link the PPP update schedule with key LCSP review milestones.

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- The scope of program protection activities could be limited because other system security engineering disciplines have primary responsibility in a particular organization.[212]

Therefore, to address all system security risks to critical items, a first step is to supplement PPP-derived items with items identified by SMEs from all the system security engineering disciplines.[213] Then, the larger risk management function in a program office should help determine ways to mitigate the risk of compromise to *all* the items with system security requirements. The following policy statement illustrates that point.

> *Use assured suppliers or appropriate SCRM countermeasures for system elements that perform mission-critical functions, in accordance with DoDI 5200.44. … Technical mitigations for mission-critical functions and critical components must, at a minimum, include:*
>
> *(a) Software assurance.*
>
> *(b) Hardware assurance.*
>
> *(c) Procurement strategies.*
>
> *(d) Anti-counterfeit practices.*[214]

6.3.3.3 Relationship to DMSMS Resolutions

Resolving a DMSMS issue usually involves a change to the system's configuration or design, replacing one item with another, a change to the BOM, or a change to suppliers. Those changes may

- Replace or otherwise affect critical items with security requirements,
- Incorporate new items with their own unique security requirements, and/or
- Incorporate new suppliers or foreign investors that may introduce new supply chain vulnerabilities.[215]

The following illustrates these points for each DMSMS resolution:

- Approved item resolution: If the approved item was not used in the initial manufacturing of the system, but is only shown on a drawing, there could be a new supplier or a vulnerability may have been discovered in the new item after the initial drawing was made.
- LON buy or extension of production or support resolutions: If a production line must be restored, new manufacturing equipment may introduce vulnerabilities, there may be new opportunities for tampering, or the configuration of an assembly may change.[216]
- Repair, refurbishment, or reclamation resolution: These processes may introduce new suppliers and new components in an assembly, which could also introduce vulnerabilities and new supply chain risks.
- Simple substitute, complex substitute, or development of new source resolutions: All imply there will be a new supplier and probably a new item, which could also introduce vulnerabilities and new supply chain risks.

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[212] This situation could also be mitigated in the SEP and/or LCSP by scoping its update process to encompass all system security engineering disciplines.

[213] While it would be good if this were a standard system security engineering process, as discussed below for DMSMS resolutions, it is possible to seek information about the need for protection on a case-by-case basis.

[214] DoDI 5000.83, *Technology and Program Protection to Maintain Technological Advantage*, July 20, 2020, change 1 effective May 21, 2022, , available at https://www.esd.whs.mil/Portals/54/Documents/DD/issuances/dodi/500083p.pdf.

[215] This can occur outside of a DMSMS situation. In certain critical applications, such an event may trigger a reassessment of risk.

[216] This is less likely for an LON buy.

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- Design refreshment, redesign—NHA, redevelop the item, and redesign complex resolutions: All imply a redesign at some level and there will consequently be new parts and usually new suppliers, which could also introduce vulnerabilities and new supply chain risks.

Because all the changes inherent to a DMSMS resolution can affect the risk of compromise, the selection and implementation of a resolution should trigger a relook at risk to determine the need for different or additional protection, assurance, cybersecurity, and/or SCRM measures.

The DMSMS management community (which is a combination of program office personnel, independent SMEs, and prime and subcontractor personnel) uses a multi-disciplinary DMT for the following:

- Formulating potential resolutions,
- Ensuring they are feasible,
- Conducting AoAs or BCAs on them,
- Taking all risks into account, and
- Making recommendations on a preferred resolution to the resolution approval authority.

Therefore, all protection, assurance, cybersecurity, and SCRM measures resulting from the need to resolve a DMSMS issue must be considered by the DMT because those measures may impact implementation in a way that changes the system configuration. And consequently, the following disciplines should be included in the DMT and the associated decision-making process. The SSE should lead the system security engineering team.

- ***All the system security engineering disciplines as previously described, taking into account their scope of responsibilities as established by the program office***.
- ***Developmental and operational testing***. Developmental testing plans and conducts vulnerability testing and verifies and validates system security requirements. Developmental testing also ensures that adequate counterfeit detection testing is performed on the parts selected and purchased for the system.[217] Operational testing would be necessary for (re)qualification.
- ***PM&P management***.[218] As discussed in Section 3.4.2, PM&P management is a systems engineering discipline for selecting and assuring performance of parts and assemblies of parts, while accounting for the materials and processes used to manufacture them, throughout all phases of a system's (or equipment's) life cycle from initial design through disposal. From a security engineering perspective, some of the questions to be considered during parts selection when deciding whether to use a part or not will vary as a function of criticality, the application of the part within the design, program duration, risk that the program office is willing to accept, and other factors.

The material in Figure 19 between the boxes for "determining a viable resolution" and "conducting BCAs/AoAs" represents notional, high-level system security engineering activities within the DMSMS resolution determination process. When a DMSMS case is opened, in the top part of Figure 19, the DMT considers many possibilities. After evaluating their feasibility from various viewpoints (e.g., technical, cost, schedule, meeting performance requirements), a smaller set of alternative resolutions are identified for further analyses.

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[217] Ibid.

[218] DMSMS impacts often stem from part selection decisions, which can be affected by a disciplined PM&P management process to help limit DMSMS effects. DoDI 5000.88, , available at https://www.esd.whs.mil/Portals/54/Documents/DD/issuances/dodi/500088p.pdf, requires program managers to implement a PM&P management process to address these and other concerns and suggests including requirements in contractor SOWs for most program phases, including technology maturation, development, and production. This requirement can be effectively met by requiring a PM&P Management Plan (approved by the government) in accordance with MIL-STD-11991.

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These initial efforts are generally performed without detailed knowledge of system security engineering considerations. Figure 19 does not suggest that the DMSMS practitioner should examine the PPP or other documents to determine whether or where or how system security requirements apply. That is not a best practice because:

- The DMSMS practitioner may not be in a position to interpret the PPP,
- The PPP has not been updated, or
- Key components may fall outside the scope of the PPP.

Therefore, Figure 19 next indicates that the DMSMS practitioner should coordinate with the appropriate SMEs among the system security engineering disciplines.

The most efficient way to coordinate will vary by organization. In some instances, all the system security engineering disciplines should be included. In other situations, the program protection community may have already considered the requirements of all the other disciplines. The coordination process will also identify the following:

- Needed program protection, cybersecurity, and/or SCRM countermeasures;
- Instances where existing protections and assurance levels are sufficient; and finally
- Additional protections that should be put in place to lower risk to an acceptable level.

The outcomes of using this information include the following:

- Identification of the affected items that should have system security requirements;
- Determination of those requirements;
- Modification of the alternatives to reflect those requirements; and
- Selection of items that meet their allocated requirements within the alternatives balancing the need for protection, cybersecurity, assurance, and SCRM with other selection criteria as appropriate.

In some instances, detailed risk assessments may be needed to achieve these outcomes.

6.3.4 Conducting BCAs/AoAs

The section discusses the blocks in Figure 19 beginning with conducting BCAs/AoAs and ending at the point just before a list of possible preferred resolutions is presented to the program manager.

At this point in the Figure 19 flowchart, each alternative should have been modified as necessary on the basis of assurance, supply chain, CPI protection, and other security risks associated with the items that will be affected (e.g., items to be replaced, items where suppliers will change, or existing items that will interface with items that change) by implementing the resolution. After all viable resolutions are examined in this way, the DMT should ensure that those resolutions satisfy all technical requirements.

All viable options (including the status quo) are then analyzed further using either an AoA or a BCA to determine which resolution or set of resolutions gains the best ROI. For the DMSMS management community, a BCA is a formal, structured approach for examining the costs, benefits, and risks of different alternatives. It requires background research and data collection and management. It also requires a thorough understanding of the quality and completeness of the data and of any assumptions made.

The standard criterion for comparing alternatives on an economic basis is net present value (NPV), the discounted monetized value of expected net benefits (benefits minus costs). NPV is computed by assigning monetary values to benefits and costs, discounting future benefits and costs using an appropriate discount rate, and subtracting the sum total of discounted costs from the sum total of

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discounted benefits. For DMSMS resolution alternatives, the one with the highest NPV (e.g., lowest life cycle cost) is preferred, because the benefit of mitigating the DMSMS condition—that is, avoiding negative impacts on system operational readiness—is the same for each alternative.

DMSMS management practitioners treat an AoA is a simplified version of a BCA. An AoA does not require as much detailed analysis as a BCA to determine the most viable resolution. Typically, an AoA is used in place of a BCA when a low cost and risk of the resolution can be estimated accurately up front without an in-depth analysis.

6.3.4.1 Budget/Resources

A program office must be able to calculate resolution costs at an acceptable level of fidelity to perform an AoA or BCA. To decide on which resolution to pursue, the cost calculations do not necessarily need to be exact; they just have to be consistent enough to establish an ordinal ranking. To ensure that the funding is sufficient to support the implementation of the selected resolution, a program office will need better fidelity. Table 15 shows average costs in FY24 dollars associated with implementing each of the DMSMS resolution options. These data were primarily compiled from submissions to a 2014 Department of Commerce survey of government and commercial DMSMS management programs.[219] While it is preferable for a program office to estimate specific costs for resolutions, the costs cited in the table can be used to make preliminary cost estimates when a program office is gathering more detailed information. Other uses of Table 15 include an initial judgement of the validity of resolution cost estimates and cost avoidance calculations when no better information is available.

Table 15. Average Cost Associated with Implementing Each DMSMS Resolution Option

Resolution Option	Average (FY24 Dollars)[220]
Approved item	$1,258
LON buy	$6,399
Simple substitute	$15,379
Complex substitute	$31,068
Extension of production or support	$31,143
Repair, refurbishment, or reclamation	$79,492
Development of a new source	$322,098
Design refreshment	$938,476
Redesign—NHA	$1,336,220
Redevelop the item	$2,043,388
Redesign—complex/system replacement	$12,578,942

Calculating averages from the original Department of Commerce data for redesigning an NHA is problematic. Depending on the level of assembly of the obsolete item, data used to develop the average

[219] DSPO, *Diminishing Manufacturing Sources and Material Shortages: Cost Metrics* (Fort Belvoir, VA: DSPO, February 2015). The data to develop the average costs for development of a new source, design refreshment, and redevelop the item includes information collected from companies, examples collected from the government, data from the original Department of Commerce survey, and professional engineering judgement. The numbers derived from these sources are consistent with other data in the table, sensible from an engineering perspective, and sufficient for the uses for which the table is intended.

[220] Resolution costs adjusted for inflation using a factor of 1.8% per "National Defense Budget Estimates for FY 2016—Office of the Under Secretary for Defense (Comptroller)," Table 5-2: Pay and Inflation Rate Assumptions—Outlays, 54; then adjusted to FY22 dollars using the 2022 DoD Greenbook, Column: Total DoD Excluding Defense Health Program, Table 5-4: Department of Defense Deflators—Total Obligation Authority by Public Law Title (Base Year = 2022); and finally adjusted to FY24 dollars using the 2024 DoD Greenbook, Column: Total DoD Excluding Defense Health Program, Table 5-4: Department of Defense Deflators—Total Obligation Authority by Public Law Title (Base Year = 2024).

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cost for redesigning an NHA could represent simpler and, therefore, less expensive redesigns than those in the data for calculating the average cost for development of a new item or source.

Appendix M can be used, with caution, to modify these averages based on more specific circumstances. The appendix is a three-part table that contains the complete results from the Department of Commerce survey. The rows of the table show the resolution options subdivided by environment (aviation, ground, shipboard, space, and undersea). The columns show the commodity type (electrical, mechanical, and electronics) subdivided by item type (assembly, component, raw material, and software). Entries in the table are average cost in FY24 dollars and sample size. Little confidence should be placed in entries with a low sample size.

Both an AoA and a BCA should account for life cycle costs for each applicable time increment.[221] When possible, multiple resolutions sequenced over time for implementation through the end of need should be considered. This allows a program office time to implement the resolution with the best ROI if barriers exist at the time of notification. For example, if a CCA with an ASIC is obsolete, and the impact on operational availability is projected to occur within six months (based on stock, demand, and reliability information), the DMT may determine that the resolution with the best ROI is to develop a new source of supply. However, if developing a new source will take at least one year after qualification, the DMT will need another resolution to cover the development time; for example, if stock is still available, then an LON buy could be implemented.

Section 3.4.5 introduced the concept of a watch list wherein inventory levels are monitored to determine, in part, how well demand assumptions made during resolution determination are holding. This concept is especially important in determining the preferred resolution to a DMSMS issue, especially when the resolution may be an LON buy or reverse engineering or redesign of the item followed by an LON buy.

The fixed non-recurring engineering cost required to establish a production line for an obsolete item can be more than two orders of magnitude greater than the variable cost to manufacture the item. If the LON buy quantity purchased before a last order date is too small, it may be necessary to make such a large non-recurring investment to support the system in the future. This compares to the relatively small cost of making a larger LON buy before the item is discontinued and possibly having excess inventory when the need for the item disappears. AoAs and BCAs should consider such long-term comparisons; they should not be limited to examining only the difference between different types of resolutions.

Similarly, when the resolution is to re-develop the item or design refreshment, some quantity of the new item must also be purchased. The fixed non-recurring engineering costs will be amortized over whatever that quantity is. As part of the resolution, actions should be taken to reduce the risk of incurring similar fixed costs in the future. It may be necessary to purchase a larger quantity of the item, similar to an LON buy. The unit cost will decrease substantially because the fixed costs are being amortized over a larger number of items. Also, by obtaining the TDP for the item, it should be possible to lower re-order costs in the future. Again, such long-term approach considerations should be part of the AoA or BCA when determining the resolution.

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[221] For an LON buy, the DMT should consider whether it must purchase a minimum quantity. If that quantity is greater than the expected need, the program should try to identify other potential users as partners. As a resolution option, an LON buy could be implemented at the component level or for a higher level of assembly. In most instances, component level purchases would be significantly less costly. There could however be situations where the government might consider purchasing the higher-level assembly, e.g., when a component is obsolete, there is no repair process for the assembly, and the government does not have the tech data for the assembly, the government would have to buy the assembly. Therefore, when an LON buy is a viable resolution option, that option should be expanded to include all LON buy possibilities along with their costs and risk.

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6.3.4.2 Other Considerations

The cost avoidance from being proactive is also a factor in both an AoA and a BCA. The validity of the previous methodology used for calculating DMSMS cost avoidance has been questioned and consequently this document establishes a new best practice for calculating cost avoidance.[222] The cost avoidance by being proactive as it relates to DMSMS resolutions should be the difference between the cost of the reactive resolution avoided and the actual proactive resolution cost. For example, if the proactive resolution was a simple substitute and the reactive resolution would have been a redesign, the cost avoidance from being proactive would be the difference in the cost between a redesign and a simple substitute. There would however be no cost avoidance if the resolution is a redesign because the reactive resolution would be the same. Similarly, there is no cost avoidance if the issue were identified reactively. (See Appendix I.3.2 for a more detailed explanation.)

Once the program office has identified the implementation cost for the viable resolutions, the program office can calculate the breakeven points and ROI to determine which resolution is the most cost effective. That resolution, however, may not be the best option when risk is taken into account. All identified risks associated with the resolutions should be captured and a proper weighting factor should be associated with each risk. Some risks will require a higher weighting factor than others. The following are among the risks to consider:

- ***Technical***. Risk associated with the ability to develop or implement a resolution while still maintaining performance within the specification.
- ***Supply chain***. Risk associated with the financial viability of the resolution provider that will be maintaining the capability.
- ***Financial***. Risk associated with the availability of funding required to implement a resolution within a specified time period.
- ***Schedule***. Risk associated with implementing a resolution before operational availability is affected.

Application of these risks in the decision-making process is subjective. In some instances, a high-cost resolution with low risk is preferable to a low-cost resolution with high risk. For example, testing and qualifying an alternate item that uses technology like that in the obsolete item may not be the best choice, because there may soon be a shortage of that alternate. Instead, it may be better to develop a substitute item using more current technology.

After the DMT has developed a list of possible preferred resolutions, all appropriate testing should be conducted to ensure the technical viability of the referred resolution. Next, program office leadership should decide which are acceptable.

6.3.5 Obtaining Initial Program Office Leadership Approval

This section discusses the feedback loop in Figure 19 that may result from presenting all the preferred resolution possibilities to program office leadership. Program office leadership may introduce additional constraints or considerations that refine the list of preferred resolutions before a final decision on the preferred resolution is made and funding is secured. To do this, leadership should understand the timeline for implementation events and their relationship to the system's capability to perform missions.

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[222] Problems with the previous methodology are threefold. First, there was no prohibition for including resolutions for problems that were found reactively. Second, there was no reason to believe that the next viable resolution would have happened or for that matter would have been feasible had the issue been discovered reactively. Finally, the process for determining the cost of the avoided resolution was not clearly defined, which could result in very large cost avoidance figures and discrepancies in the cost avoidance for the same or similar resolutions between different systems.

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All the required implementation events, and when they need to occur, should be identified to evaluate the "time to implement a resolution" risk category depicted in Figure 12. Examples of such implementation events include:

- When to place an LON buy order, and the expected receipt date of the items.
- When to begin work on a redesign, and the expected date the redesigned item will be ready for use in deployed systems.
- In the case of establishing a new source of supply, events to be scheduled include the procurement of technical specifications for the part, solicitation of offers from potential new suppliers, selecting a new supplier, and the beginning of new production.
- When to begin qualification and testing procedures for new items, as applicable. Resolution options requiring qualification and testing can be identified in Table 14.

Program office leadership should use the implementation timeline to assess risks that will be born during implementation that might impact mission readiness. For example, the planned completion time of a redesign, if scheduled too late, risks the depletion of current stocks of critical items before the new item is ready for deployment. If not having the part will make a critical weapon system NMCS, then the DMT should consider changing the redesign timeline to minimize this risk. In the context of Figure 12, this latter scenario involves reducing the time to implement a resolution to mitigate a high degree of risk for critical items. However, requiring work to be completed earlier will likely come with an associated increase in cost. It also may lead to a change in the level of assembly at which the resolution is implemented.

Trading off costs and risks is an inexact practice that can involve both quantitative techniques and subjective judgement calls. Nevertheless, a few basic principles should be considered:

- While costs are always quantitative, cost estimates are almost always inexact. Program offices should strive to quantify the degree of uncertainty in their estimates. All else equal, a resolution option with less uncertainty in its cost estimate is better than one with more uncertainty.
- Risk is difficult to quantify, but at minimum program offices should rank-order the resolution options in terms of risk. This enables narrowing the scope of options under consideration, as options that are more risky and more costly than another can be discarded (such options can be said to be "dominated" by another).
- A more quantitative treatment of risk would assign a numerical risk value to each option. These values could be set by, e.g., specifying criteria that contribute to risk, assigning a weight to each criterion, and evaluating options based on which criteria they meet.
- With numerical values for costs and risks, program offices can specify acceptable additional costs to reduce risk by a set amount and calculate which option provides the best cost-to-risk tradeoff.
- Alternatively, cost-to-risk can be assessed via a judgement call after eliminating dominated options.

These additional constraints and considerations from program office leadership may lead to another iteration of the process for determining viable resolution options and conducting BCAs/AoAs for each of the remaining viable resolution options.

The resulting list of viable resolution options may include combinations of those listed in Table 13. For example, the DMT may have assessed an option to redesign an item or assembly and determined that the redesign cannot be completed before current stocks are depleted. The use of a complex substitute that might not have been previously considered a viable resolution option could be combined with a redesign as a more attractive option to consider.

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6.3.6 *Preparing for Implementation*

The final three blocks of the Figure 19 flowchart are discussed primarily in Section 7. Appendix N and Section 7.1 address estimating resolution costs to inform programming and budgeting. After program office leadership has come to an initial decision of the preferred resolution, funding is secured and leadership gives final approval. If funding is unavailable or must be diverted from a high priority activity, it is possible that some elements of the preferred resolution determination process will need to be repeated to develop an alternative approach to mitigating the issue.

6.4 Additional Considerations for FMS Equipment in the Analyze Step

Although DMSMS management processes for DoD and FMS equipment are similar, there are additional considerations when working with foreign partners (e.g., technology security, information assurance, International Traffic in Arms Regulations). The DMSMS management practitioners in the program office should interface with the foreign partners before finalizing any resolution recommendations.

6.4.1 *Resolving DMSMS Issues Related to Equipment*

A particularly challenging (but common) situation is an FMS case calling for a DoD program office to resolve or provide technical assistance that enables the foreign partner to resolve DMSMS issues on older (or modified) equipment that is not in DoD's inventory. This situation may arise when:

- The original FMS case was for older or FMS unique configuration equipment;
- Foreign partners may not have implemented DoD-adopted modifications because of high cost, lack of requirements for the enhancements, or timing (e.g., partner plans may call for upgrades in the future);
- Partner resolutions to DMSMS issues may differ from DoD recommendations because of cost; or
- Partners may wish to maintain technical capabilities developed by their domestic companies over the years and rely on those capabilities for support in the future.

DoD engineers will normally take extraordinary measures to develop resolution options significantly less expensive than a major redesign, especially when the LOA only contains funding for technical assistance. However, as more and more of these circumstances occur, the technology and configuration of partner equipment moves further away from the DoD equipment configuration and ultimately the DoD may no longer be able to provide support. Other factors that could come into play include:

- DoD has no support infrastructure for that equipment;
- The production lines (or test equipment) to make replacement parts or conduct repairs are themselves obsolete;
- The maintenance providers no longer find it economical to provide support services; or
- DoD does not have the technical data or does not have sufficient rights to the data to find an alternative support provider and the OEM may be unwilling to provide data rights or may charge too high a price for the data or reverse engineering may be too expensive or too time consuming.

At some point, resolutions of DMSMS issues short of an expensive redesign can become impractical because of equipment age and the length of time the equipment has been in service by the foreign partner. Furthermore, neither the DoD nor the foreign partner would pay for a high-cost redesign, because it would be better and less expensive to modernize to achieve greater capability.

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While this situation may sometimes be unavoidable, using the communication processes discussed in Section 3.5.2 may help. DMSMS management practitioners in the program office should periodically provide the following to FMS partners:

- Forecasts of the future state of parts and equipment availability using a tombstone chart or a similar mechanism;
- Courses of action (e.g., follow DoD recommendations, take a different resolution approach, do nothing) to deal with projected availability issues and costs; and
- Pros and cons of the different courses of action that encompass resolution and future support costs, change in capability to meet partner defense needs, impacts on availability, impacts on interoperability with DoD equipment, etc.

This is especially important close to the point in time when the DoD plans to completely retire an equipment type or system. The program office should determine the DMSMS-related actions (e.g., LON buys, use of ECPs, ...) that partners not retiring that system should take to enable future support and then give partners the opportunity to take such actions as part of the current FMS case.

6.4.2 Access to Technical Data

As discussed in Sections 4.7.1 and 4.7.2, DoD may need technical data for a foreign partner's design configuration to verify DMSMS issues and determine issue applicability to partners. Access to technical data is even more important when dealing with the DMSMS issues related to equipment or any circumstances where there are differences between DoD equipment and partner equipment. Technical data may include BOMs, information about the partners' repair capability, available support through non-DoD mechanisms (e.g., prime contractors), or spares on-hand. Unavailability of technical data may reduce the efficiency of providing DMSMS management support leading to sub-optimal results or perhaps an increase to the time and cost needed to resolve an issue. For example, the feasibility of resolutions such as identification of another source, recommending an upgrade to fix the problem, or recommending a smaller scale redesign (like at the part level) to fix the problem may change.

Access to logistics data may also be important in determining the preferred resolution. For example, the stock on hand and demand rates indicate when the DMSMS issue will impact the FMS equipment. The impact date may affect the need for an interim resolution.

As previously indicated, an understanding by the foreign partner of the reasons for providing DoD program offices access to technical data, its design configuration, and inclusion of provisions in the LOA to obtain that data are key. The right contract language between the partners and their equipment/support providers might be necessary to solve the technical data issue. Illustrative DoD contract language for program offices in the SD-26, may also help foreign partners obtain the technical data needed by DoD program offices.

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Section 7. Implement: Implementation of DMSMS Resolutions

The DMT's role does not end when a PM decides which resolution option to pursue. The final step of the DMSMS management process is implementation. In the *Implement* step, the DMT should be involved in three final processes: programming and budgeting for implementing the preferred resolution, integrating DMSMS resolution and modification funding, and implementing the preferred resolution.

In some cases, contracts with the prime contractor (during design and production) or a logistics provider (during sustainment) may include a requirement for the contractor to fund DMSMS resolutions. (Appendix F contains more information on contracting.) This situation is complex:

- The definition of "end-of-life" can differ depending on one's perspective. If the contract requires the contractor to buy additional items to resolve a DMSMS issue, the contractor will normally be concerned only with demands up to the end of the contract period of performance, whereas the government will likely be interested in addressing an issue through the "end-of-need." Unless the government specifically defines this, it can remain open to interpretation. The government should not expect the contractor to buy enough items to last until the end of need without additional funding.
- If the current contract requires the contractor to resolve DMSMS issues, the contractor may not fund the most cost-effective resolution from the program office's perspective. The contractor will determine the resolution based on its own business case calculations. However, depending on its relationship with the government, the contractor may factor the government's long-term needs into the calculation if the government informs the contractor of those needs. If the program office included a requirement for the contractor to fund the most cost-effective resolution from a program office perspective in its RFP for the current contract, the contractor may bid a much higher price to compensate for uncertainty, unless the parties are able to create a contractual incentive to minimize the contractor's financial risk. Consequently, the program office should be prepared to negotiate with the contractor on which resolution option to implement and should be prepared to provide additional funding if it is not included in the contract. These negotiations are enhanced by a strong government-industry relationship and full government awareness of the DMSMS services provided by the contractor.

7.1 Program and Budget for DMSMS Resolutions

The focus of the programming and budgeting information that follows is for the implementation of resolutions for known and forecasted DMSMS issues for every item in the system whether it is monitored or not.[223] When resolutions are implemented in phases (e.g., an LON buy in conjunction with a longer-term resolution), programming and budgeting should take both into account. The DMT can assist program office leadership in developing its programming and budgeting submissions to be able to fund DMSMS resolutions where necessary as well as to ensure that obsolescence is incorporated in the program office's modification plans.

Although a program office may not know the specific DMSMS problems that it will encounter in the future, experience has proven that DMSMS issues are inevitable and thus a program office will face multiple DMSMS problems annually. Therefore, obtaining funding in the execution year is not a best practice.

[223] Separate resource requirements for performing DMSMS management operations were discussed in Section 3.4.1 for the "*Prepare*" step.

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Without advance planning for these funds, the necessary resources may not be available and the resulting failure to mitigate DMSMS issues can lead to unacceptable performance, degradation of system reliability and availability, schedule slippage, and increased costs.

Appendix N contains detailed information pertaining to the following programming and budgeting for DMSMS resolutions topics:

- Best practices,
- Considerations for funding DMSMS resolutions in the year of execution,
- Leveraging working capital funds (WCFs) to fund DMSMS resolutions, and
- Other resources (including FMS cases) that may be available to finance DMSMS resolutions.

Programming and budgeting for DMSMS resolution funding is only necessary when a resolution is implemented on its own or DMSMS is the principal purpose of a larger work effort that also includes non-DMSMS–related activities. According to interviews with the Comptroller Offices, DMSMS resolutions are often implemented in conjunction with other efforts (e.g., capability enhancement, technology insertion, technology refreshment, planned maintenance, life extension, planned modifications, and so forth). This occurs because it is convenient to combine efforts from the perspective of system availability. Also, it is often less costly to combine work efforts on the same subsystem. When the work efforts are combined, and DMSMS is not the driving purpose for the work efforts, then some of the programming and budgeting best practices suggested in this document may not be applicable.

7.2 Integrate DMSMS Resolution and Modification Funding

Modifications create opportunities to resolve DMSMS issues as notionally shown in Figure 24. The figure represents before and after modification tombstone charts. Normally, entries in the chart would depict the number of items remaining with green representing a safe number, yellow implying a near-term concern, and red indicating not enough items available (i.e., a negative number). To illustrate notionally how a modification could reduce DMSMS issues, only color scheme changes are portrayed in the figure.

When considering resolution funding needs in conjunction with modification efforts, typically programming and budgeting is the responsibility of the modification planners. This section describes best practices for DMSMS resolution programming and budgeting in connection with modification planning.

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Figure 24. Notional Change in Obsolescence Status after Modification

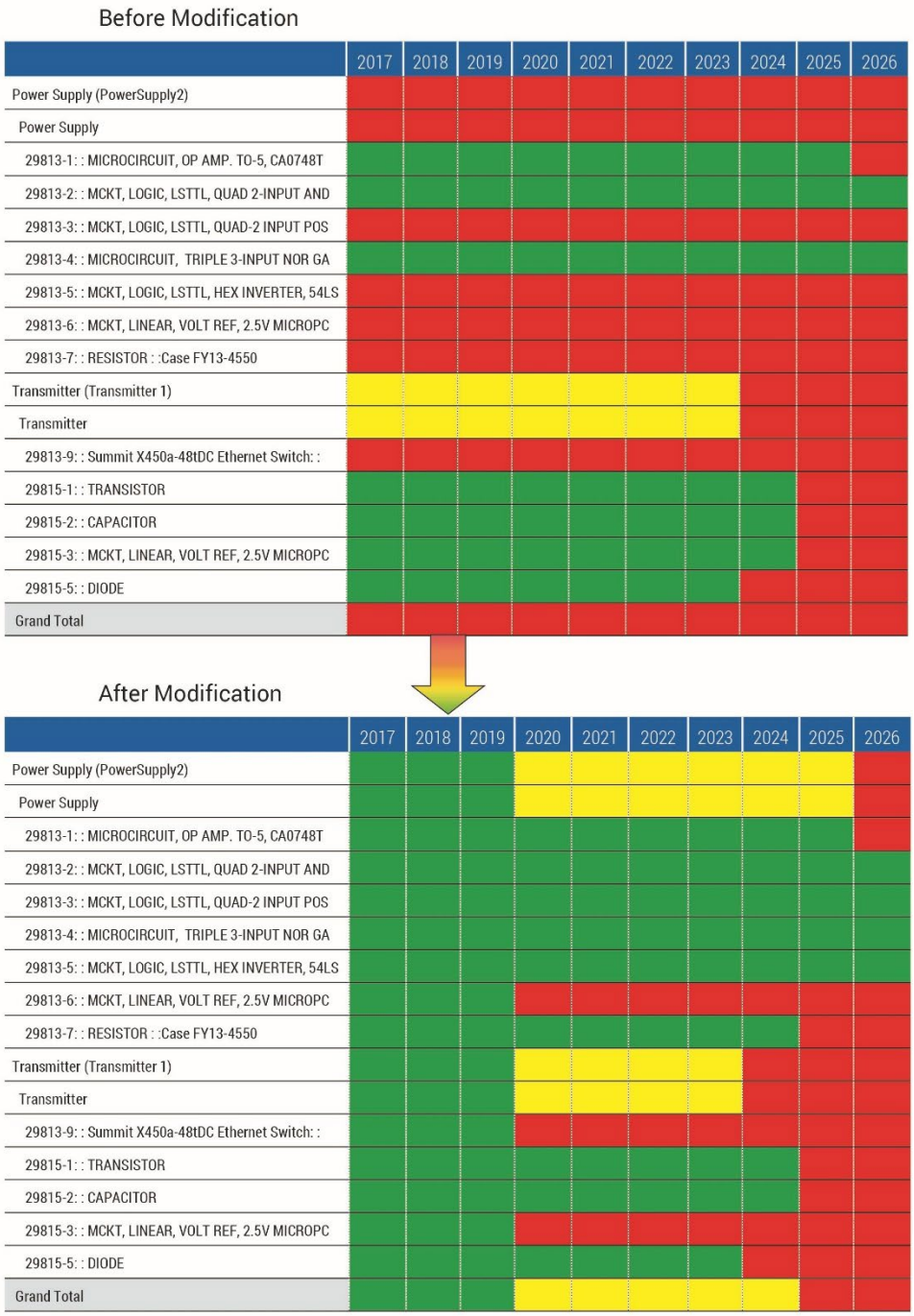
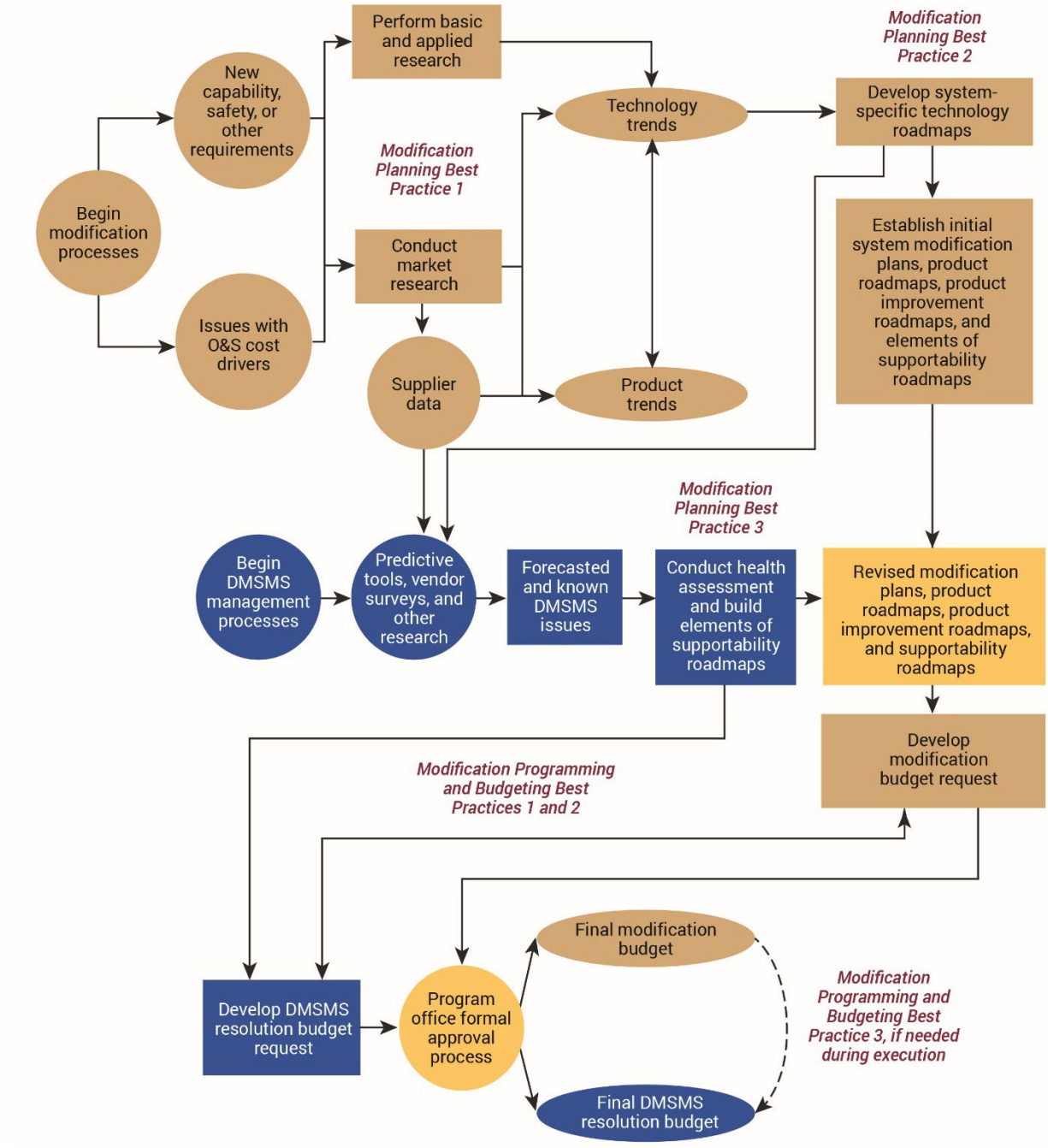


Figure 25 is an extension of Figure 21 in Section 6.3.2.2. The top of Figure 25 concerns the interactions among DMSMS management, roadmaps, and *modification* planning. The bottom section adds *programming and budgeting* considerations and illustrates the interactions between modification planning and DMSMS management activities that yield modification plans and DMSMS resolution programs and budgets. The gray-shaded portions of this figure are associated with modification while the blue shaded portions pertain to DMSMS management. The tan portions represent general programming and budgeting processes that impact modification and DMSMS management.

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The upper left of the figure (where the reader should start) deals with technology and supply chain management. Then upper right depicts the formulation of modification plans. The formulation of DMSMS resolution POMs and budgets is represented in the lower left, and their consideration or integration into modification POMs and budgets is illustrated on the lower right. Modification planning (and the funding of those plans) has a significant influence on programming and budgeting for DMSMS resolutions.

Figure 25. Interactions of DMSMS-Related and Modification-Related Programming and Budgeting



The following discusses new elements of this figure through the introduction of a set of additional modification programming and budgeting best practices. These best practices are shown on the figure to

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indicate ways in which a program office's DMSMS management community should leverage modification POMs and budgets for DMSMS resolutions.

- ***Modification Programming and Budgeting Best Practice 1***. Consider approved modification POMs and budgets when calculating the size of LON buys.

  Funded modification plans may identify a timeframe for phased DMSMS resolutions for obsolete items no longer needed once the modification has been implemented. If any such items impact the system before the modification, then an interim resolution must be put in place. If that interim resolution is anything other than an LON buy, there normally are no programming and budgeting implications related to the modification. For an LON buy, the period of need is defined by when the DMSMS issue will impact the system and when the modification effort will be implemented. The period bounded by these two dates has implications on the quantity of the item to be purchased.

  Except for costs associated with holding inventory, the resolution cost over time for an LON buy is very small, assuming the items will eventually be used. However, LON buys create programming and budgeting implications for the year the buy is executed, especially if the item has a high cost (see Appendix N.2). Consequently, the best practice from a programming and budgeting perspective is to take funded modification plans into account when sizing LON quantities. However, there may be uncertainty with modification plans. Funding may not be in place because of the uncertainty of start dates or lack of agreement about the scope of the effort. Some situations have a high probability (but not certainty) of funding. These factors should be considered when programming and budgeting for a safety level in the size of an LON buy as discussed in Section 7.3.

- ***Modification Programming and Budgeting Best Practice 2***. Consider funded modification plans when formulating DMSMS resolution-specific POMs and budgets to avoid duplication.

  The DMSMS management community should be able to articulate how much is being spent on resolutions for DMSMS issues. Questions can be raised about the extent to which resolutions are funded through modification. Modification plans may resolve a DMSMS issue if the obsolete item is removed solely because of the desired system improvement. Modification plans can also be augmented to resolve a DMSMS issue at a cost substantially lower than the cost to resolve the same issue in the absence of the modification effort.

  It is a best practice for the DMSMS management community to know about the former to avoid unnecessary effort in determining resolution options. It is also a best practice for the DMSMS management community to propose the latter as part of normal operating processes in the program office. Following these best practices should avoid duplication among DMSMS management and modification programming and budgeting activities.

- ***Modification Programming and Budgeting Best Practice 3***. Consider adjusting modification efforts to initiate DMSMS resolutions to avoid an immediate impact on the system when emergency and unpredicted, execution year obsolescence occurs.

  While many DMSMS issues are discovered well in advance, DMSMS issues also routinely occur unexpectedly during budget execution. Some of these issues may require significant near-term resources to resolve before there is an impact to the system. As discussed in Appendix N.2 one potential source of resources is a funded modification effort. In many cases, there would have been formal programming and budgeting for the modification effort; however, other (sponsors) sources of funds may be applicable if that avenue for programming and budgeting is unsuccessful.

  The DMSMS management community should be knowledgeable of both the technical (engineering) authority for the modification effort and the key financial decision makers in the program office. The technical authority may be able to identify how the scope of the modification can be adjusted to resolve the new issue or how the modification effort can be slowed to free some funding that can be temporarily applied to a pressing DMSMS issue. The financial points of contact would have to approve funding for either of these technical alternatives.

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7.3 Implement DMSMS Resolutions

Upon acceptance and funding, the case enters the implementation phase. This phase should follow the program office's standard process for updating configurations and engineering modifications. Some changes may be largely clerical and not require a specified process for updates, while other changes will require a formal change process. For instance, most updates that affect major configuration changes or engineering modifications flow through the appropriate level of the ECP process. The standard ECP process ensures that all changes and qualifications satisfy the system's requirements.

It is usually a mistake for the DMT to assume that the program office's standard processes will be problem free. As a best practice, the DMT should be involved in the following ways during implementation:

- Ensure that all stakeholders understand their roles and responsibilities for implementation. These roles and responsibilities should have been established when the DMT was formed.
- Ensure that the implementation steps are defined.
- Verify that appropriate technical actions (e.g., qualification of the new item or procurement of the item) were successfully carried out.
- Monitor the process.
- Obtain feedback on the project status to ensure maintenance of full operational availability during implementation. If the project is behind schedule, the DMT may be required to determine supplemental mitigation actions.
- Update BOMs being monitored to reflect the configuration changes once the project is completed.

In some cases, the DMT may have difficulty performing these functions. A champion in the program office is critical to implementation success. The champion should be at a high enough level to assure the appropriate level of attention and be knowledgeable about the importance of an obsolescence program to take ownership of it and justify it to program office leaders. The champion is often the catalyst who brings all the functional disciplines together toward the common goal of managing the availability of the system and resolves the difficulties the DMT faces when carrying out its DMSMS management responsibilities.

In some cases, the DMT is asked for advice on procedures to deal with issues that arise during implementation. Below are examples of some issues, along with some considerations about ways to resolve them:

- ***Improving the priority of DMSMS management with the contracting officer***. The DMT should invite the contracting officer to its meetings and explain his or her roles and responsibilities. The DMT should ensure that the contracting officer understands what is needed and the associated urgency.
- ***Buying in advance of need***. 31 U.S.C. § 1502(a) establishes a limitation on the funds that can be expended on an LON buy and consequently may limit the quantity that can be procured. The statute's wording is as follows:

  (a) The balance of an appropriation or fund limited for obligation to a definite period is available only for payment of expenses properly incurred during the period of availability or to complete contracts properly made within that period of availability and obligated consistent with section 1501 of this title. However, the appropriation or fund is not available for expenditure for a period beyond the period otherwise authorized by law.

  To obtain an exception to this limitation, generally, a "bona fide need" statement must be documented for the General Counsel's office. That statement should explain the DMSMS issue and its impact if not resolved and describe how and why the resolution option was determined. In the absence of any specific organizational guidance, a best practice is to include the following in the statement:

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- Statement of the problem,
- Analyses showing the inability to find an alternate part or a resolution that does not require a significant CM change,
- Analyses that show why an LON buy is the most cost-effective resolution,
- Description of the expected impacts if an LON buy is not executed,
- Forecast of the quantities of the item needed to support production and sustainment of both DoD and foreign partner assets along with an explanation of any assumptions made,
- Documentation of the discontinuation date and the date that impact begins, and
- Identification of the funds to be used for the procurement action.

Procuring more than two years of supply in a stock fund.[224] 10 U.S.C. § 3070 may also be an issue in sizing a sufficiently large LON buy for an obsolete item that is managed by a stock fund. The statute reads as follows:

(a) Two-Year Supply.—The Secretary of Defense may not incur any obligation against a stock fund of the Department of Defense for the acquisition of any item of supply if that acquisition is likely to result in an on-hand inventory (excluding war reserves) of that item of supply in excess of two years of operating stocks.

(b) Exceptions.—The head of a procuring activity may authorize the acquisition of an item of supply in excess of the limitation contained in subsection (a) if that activity head determines in writing—

(1) that the acquisition is necessary to achieve an economical order quantity and will not result in an on-hand inventory (excluding war reserves) in excess of three years of operating stocks and that the need for the item is unlikely to decline during the period for which the acquisition is made; or

(2) that the acquisition is necessary for purposes of maintaining the industrial base or for other reasons of national security.

For DLA managed items, the exceptions section in the statute is automatically initiated as part of the process described in Appendix N.3 if the following information is available:

- Justification for future demand projections,
- Limitation on the ability to purchase (e.g., the discontinuation notice), and
- Requisite approvals (or statements of need) from buyer leadership.

For Service-managed items in the DoD supply system, LON buys are much more unusual since the Services manage typically expensive DLRs.

- ***Calculating sufficient stock to end of need***. If all demands for the item are well understood, determining the amount of stock required through the end of need is a straightforward calculation involving the use of a statistical distribution with the appropriate confidence limits, operating tempo, number of units in service, and failure rates (either actual or predicted).[225] The end-of-need date for the item must also be understood. It could be the retirement date of the system, the point at which a funded modification will remove the obsolete item from the system, or the implementation of a longer-term resolution.

  The greatest ambiguity is whether that end-of-need date will later change because plans for the system's life or modifications change, leading to an overestimation or an underestimation of the quantity needed. If there is underestimation, there could be a need to implement a

[224] 31 U.S.C. § 1502(a), available at https://www.govinfo.gov/link/uscode/31/1502, does not apply since the stock fund corpus is not subject to the obligation limitations expressed in the statute.

[225] Normally, the Poisson distribution is used.

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second interim resolution which may be expensive because options are likely limited. With overestimation, there will be excess inventory.

Another issue with the calculation is the uncertainty in the input. Calculating the LON quantity may not account for all relevant data, leading to underestimates in quantity. For example:

- If an organization that manages a WCF is making the purchase, it just considers its own customer demand experience in sizing an LON buy. Some programs may have bought the item in the past outside of WCF processes (e.g., through logistics support contractors or direct commercial purchases), so there may not be a record of the demand. Being very concerned about avoiding excess inventory, an organization that manages a WCF may not adequately consider past purchases made outside its WCF. Similarly, there may be instances where a provisioning error does not record a system as a user of an item being considered for an LON buy. Demand from such a system may not be fully reflected in sizing an LON buy. Furthermore, there may also be demands from production lines.
- The LON buy quantity may be reduced if there is a shelf life limitation for the item.
- FMS demand, where applicable, may not be easily determined (see Section 4.7.3).

Program offices should take as many of these considerations as possible into account when calculating an LON requirement. Generally, it is better to make conservative assumptions and consider the inclusion of a safety level of items since the cost of underestimating the quantity needed is almost always significantly higher than the cost of buying some excess inventory.

- ***Determining the appropriate contract vehicle***. A contract must be in place with the organization that will implement the resolution (e.g., the organization performing the non-recurring engineering or the organization that will sell the items). Restrictions exist on the use of all appropriations. In some cases, additional procurement funds are necessary; in others, research, development, test, and evaluation funding is required for redesign, material substitution, or qualification of a new source. The contracting office can provide advice on this subject.
- ***Managing inventory***. Some issues may be associated with receiving, inspecting, and storing items. Programs should consider these concerns early in the process; if these problems cannot be solved, an LON buy resolution option may not be viable. Options for storage locations include suppliers, prime contractors, and DoD component storage facilities.[226] There have been situations where program offices have used their own funding to contract commercially for storage.
- Another related concern is that if the item has low demand, a WCF-funded storage location could dispose of the item if it concludes that there is excess inventory. A program office may have to take action to prevent that. In addition, for items that are controlled by an organization that manages a WCF, but used in various systems, a program office may not be able to protect assets it needs from being procured by another program office.

7.4 Additional Considerations for FMS Equipment in the Implement Step

When all the following apply, foreign partner funding may not be available to resolve DMSMS issues for FMS equipment:

- LOA is not signed or even if signed, money is not yet available to the executing program office;
- DoD production has ended or is about to end;
- Resolving the DMSMS issue is not necessary for DoD requirements (production or sustainment); and

[226] For example, White Sands Missile Range in New Mexico has established a facility and processes for storage.

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- Resolving the DMSMS issue is not necessary for delivering FMS equipment.

If the contractor/OEM is not willing to resolve risk to FMS equipment, efforts to resolve the issue will be delayed until the FMS case is in place and funded. Even when the foreign partner is willing to pay, there is no simple mechanism to establish a contract to get the work done. This contracting issue becomes more complex when there are several major subcontractors for a given system.

In addition, there may be difficulty integrating foreign partner funding with DoD funding. This can be especially difficult for LON buys. There may be delays in obtaining partner funding, multiple partners may be involved, and cost sharing will be affected when there are changes to which partners are involved. The difficulty in integrating foreign partner and DoD funding will be exacerbated when there is a limited time window to make the buy.

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Appendix A. Obsolescence and Its Relationship to DMSMS

Despite no difference between DMSMS management and obsolescence management from a *process* perspective, obsolescence is not synonymous with DMSMS from a *dictionary* perspective. Obsolescence is less definitive and is situationally dependent in that an item can be obsolete from one perspective, but not from another. For example, the Intel 8086 was introduced in 1978 and was one of the first 16-bit microprocessors. Although it is no longer used in modern computers, it is still used in some industrial and embedded systems. Similarly, vacuum tubes were crucial to the development of such products as radio, television, and radar. Today they are obsolete for these applications; however, one type of vacuum tube is still used in high-end audio amplifiers for its ability to generate desirable sound effects.

In the context of DMSMS management, an item is obsolete if it is no longer current and its function is superseded by something new. Below are key underlying causes of an item being out-of-date:

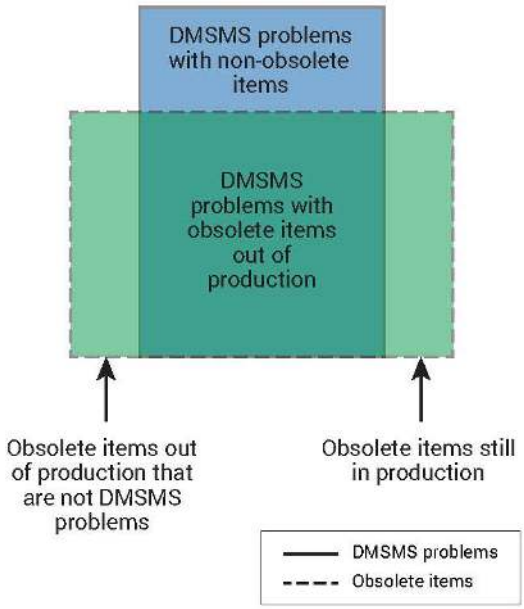
- ***Technology***. A technology is obsolete when the use of a newer technology becomes broadly preferred over the old, even if the old technology still functions and can be produced and purchased for certain unique situations. For example, videocassette recorders and players became technologically obsolete, when DVD recorders and players superseded them.
- ***Function***. An item is obsolete functionally if it no longer functions as intended because of hardware, software, or requirements changes to the item. Such an item may still be available commercially. For example, a videocassette tape (especially one in beta format) could be obsolete functionally, because players are no longer available for purchase to extract the tape's content.
- ***Regulation***. Regulations that ban the use of items or substances lead to their obsolescence. For example, Freon use has been banned because of its ozone-depleting characteristics. Similarly, the purchase of rare-earth elements such as neodymium or ytterbium from China has been banned.
- ***Supportability***. An item may be obsolete if it is no longer supportable. An example is commercial software, which continually needs product support to correct errors, to defend against vulnerabilities, and, in some cases, to maintain licenses. Unsupportable software is obsolete. Beyond software, if the necessary item test capability is no longer available, then the item may no longer function properly and, therefore, could be considered obsolete.
- ***Market demand***. A product becomes obsolete when there is no longer a demand for it, because, for example, it is no longer desirable even though it may still be available for purchase. Leisure suits are an example. Another example is a product that is no longer profitable to produce because of low demand.

Obsolescence may be planned or unplanned. An example of planned obsolescence is a relatively new home computer printer that is technologically equivalent to the latest ones on the market but requires proprietary ink cartridges that are different from the ones used in the most recent model of the printer. The manufacturer may deliberately stop manufacturing those cartridges (thereby making the printer functionally obsolete) to force people to purchase the most recent printer model.

A high degree of overlap exists between obsolete items and DMSMS problems. Figure 26 notionally depicts their relationship.

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Figure 26. Notional Relationship between DMSMS and Obsolescence



An item may be obsolete, but if it is still in production, there is no DMSMS issue if the production capability or capacity can meet the demand. For example, hand pumps for water are still made for places where power is not available for an electric pump. Another example is a situation of an obsolete computer that is out of production, but it is not a DMSMS issue because there is sufficient stock in inventory to meet all future demands. In the first example, no DMSMS case would be opened because the item is still in production. In the second example, a DMSMS case would be opened, but no resolution would be needed. Changing the circumstances of the second example can create a situation in which a DMSMS problem will evolve over time. If there were not sufficient inventory to meet future demands, DoD might be able to repair the computer instead of replacing it. If that were the case, there would be no DMSMS problem. However, a DMSMS issue would occur if the repair parts were also obsolete, if the know-how (e.g., skills) to make the repair was lacking, or if the ability to test the system (e.g., the test equipment) after the repair was unavailable.

Finally, a non-obsolete item may have a DMSMS issue. For example, market factors may drive a supplier out of a particular line of business, a supplier may declare bankruptcy, a natural disaster may affect production, or a buyout of a sole-source provider may lead to temporary or permanent termination of production of a particular product. Some temporary DMSMS issues may be due to allocation of a scarce item. In this situation, some systems may be faced with a DMSMS issue, while others may not.

Not all obsolescence results in DMSMS issues, and not all DMSMS issues result from obsolescence. However, most DMSMS issues result from some form of obsolescence.

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Appendix B. DMSMS Management Questions for SETRs

This appendix contains DMSMS management questions intended for use by DMSMS management practitioners to prepare for six of the SETRs of primary importance:

- Alternative Systems Review (ASR),
- Systems Requirements Review (SRR),
- System Functional Review (SFR),
- Preliminary Design Review (PDR),
- Critical Design Review (CDR), and
- Production Readiness Review (PRR).

The questions are designed for the program office, but many also apply to prime contractors and subcontractors. The questions are presented in five tables. Table 16–Table 20 contain questions pertinent to the five DMSMS management steps: *Prepare*, *Identify*, *Assess*, *Analyze*, and *Implement*. They are further broken down by DMSMS management operations processes.

These questions are focused on engineering and logistics processes associated with performing DMSMS management on DoD equipment and FMS equipment, if applicable. Technically, these processes are identical for DoD and FMS equipment. Therefore, no FMS unique material is proposed.

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Table 16. DMSMS Management Questions for SETRs: *Prepare*

Process	ASR	SRR	SFR	PDR	CDR	PRR
Establish the foundations for DMSMS management		Has program office leadership identified expectations for DMSMS management operations and deliverables?	Has program office leadership updated expectations for DMSMS management operations and deliverables?	Has program office leadership updated expectations for DMSMS management operations and deliverables?	Has program office leadership updated expectations for DMSMS management operations and deliverables?	Has program office leadership updated expectations for DMSMS management operations and deliverables?
		Has program office leadership defined the roles and relative relationships among DMT members?	Has program office leadership updated the roles and relative relationships among DMT members?	Has program office leadership updated the roles and relative relationships among DMT members?	Has program office leadership updated the roles and relative relationships among DMT members?	Has program office leadership updated the roles and relative relationships among DMT members?
		Has program office leadership determined a risk-based perspective to DMSMS management by identifying criteria for which systems to monitor and which items to monitor within those systems?	Has program office leadership updated the risk-based perspective to DMSMS management by updating criteria for which systems to monitor and which items to monitor within those systems?	Has program office leadership updated the risk-based perspective to DMSMS management by updating criteria for which systems to monitor and which items to monitor within those systems?	Has program office leadership updated the risk-based perspective to DMSMS management by updating criteria for which systems to monitor and which items to monitor within those systems?	Has program office leadership updated the risk-based perspective to DMSMS management by updating criteria for which systems to monitor and which items to monitor within those systems?
Form a DMT			Has a partial DMT been formed?	Has a partial DMT been formed?	Has the full DMT been formed?	Has the full DMT been formed?
			Do all identified DMT members understand their roles and responsibilities and have adequate training to fulfill their roles and responsibilities?	Do all identified DMT members understand their roles and responsibilities and have adequate training to fulfill their roles and responsibilities?	Do all identified DMT members understand their roles and responsibilities and have adequate training to fulfill their roles and responsibilities?	Do all identified DMT members understand their roles and responsibilities and have adequate training to fulfill their roles and responsibilities?

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Process	ASR	SRR	SFR	PDR	CDR	PRR
Develop a DMP		Has the program office started to develop its plan for addressing and managing the impact of DMSMS issues?	Has the program office established a robust DMSMS management program that identifies obsolescence due to DMSMS issues before critical items are unavailable?	Has a government DMP been formally approved by program office leadership?	Is the government DMP being implemented and updated, as necessary?	Is the government DMP being implemented and updated, as necessary?
		Does the draft DMP identify the roles and responsibilities of the prime/subcontractor and third-party vendors?	Does the draft government DMP identify the roles and responsibilities of the prime/subcontractor and third-party vendors?	Does the approved government DMP identify the roles and responsibilities of the prime/subcontractor and third-party vendors?		
		Have the roles and responsibilities of the government, prime/ subcontractor, and third-party vendors been established?	Have the roles and responsibilities of the government, prime/ subcontractor, and third-party vendors been established?	Have the roles and responsibilities of the government, prime/ subcontractor, and third-party vendors been established?	Are the roles and responsibilities of the government, prime/ subcontractor, and third-party vendors being executed?	Are the roles and responsibilities of the government, prime/ subcontractor, and third-party vendors being executed?
		Is the government conducting sufficient oversight when contractors are responsible for executing DMSMS operational processes?	Is the government conducting sufficient oversight when contractors are responsible for executing DMSMS operational processes?	Is the government conducting sufficient oversight when contractors are responsible for executing DMSMS operational processes?	Is the government conducting sufficient oversight when contractors are responsible for executing DMSMS operational processes?	Is the government conducting sufficient oversight when contractors are responsible for executing DMSMS operational processes?
Establish DMSMS operational processes			Is the process of defining and documenting all DMSMS operational processes in the government DMP underway?	Have all DMSMS operational processes been defined and documented in the government DMP?		

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Process	ASR	SRR	SFR	PDR	CDR	PRR
			Are DMSMS considerations incorporated into pertinent program office documentation?	Are DMSMS considerations incorporated into pertinent program office documentation?	Are DMSMS considerations incorporated into pertinent program office documentation?	Are DMSMS considerations incorporated into pertinent program office documentation?
Secure resources for DMSMS management operations			Have current and out-year DMSMS management operations budgets been estimated and identified?	Have current and out-year DMSMS management operations budgets been established, approved, and funded?	Have current and out-year DMSMS management operations budgets been established, approved, and funded?	Have current and out-year DMSMS management operations budgets been established, approved, and funded?
Establish interfaces to advocate for DMSMS-resilient designs	Are DMSMS impacts a consideration when analyzing alternative systems to help ensure that the preferred system is cost effective, affordable, operationally effective, and suitable and can be developed to provide a timely solution to a need at an acceptable level of risk?	Have interfaces been established with design engineering for the consideration of DMSMS resilience characteristics?	Have interfaces been established with design engineering for the consideration of DMSMS resilience characteristics?	Have the design engineering interfaces been successful in employing DMSMS resilience principles in the designs?	Have the design engineering interfaces been successful in employing DMSMS resilience principles in the designs?	Have the design engineering interfaces been successful in employing DMSMS resilience principles in the designs?
		Have interfaces been established with the PM&P management community for parts selection to enhance DMSMS resilience?	Have interfaces been established with the PM&P management community for parts selection to enhance DMSMS resilience?	Have the PM&P management interfaces been successful in selecting items for the designs that will enhance DMSMS resilience?	Have the PM&P management interfaces been successful in selecting items for the designs that will enhance DMSMS resilience?	Have the PM&P management interfaces been successful in selecting items for the designs that will enhance DMSMS resilience?

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Process	ASR	SRR	SFR	PDR	CDR	PRR
Establish interfaces to advocate for DMSMS-resilient designs *(continued)*				Is DMSMS management a consideration when the system design approach is being determined to minimize the impact on supportability and sustainability?	Is DMSMS management a consideration when the system design approach is being determined to minimize the impact on supportability and sustainability?	Is DMSMS management a consideration when the system design approach is being determined to minimize the impact on supportability and sustainability?
			Are the following addressed: Order of precedence for parts selection (e.g., use of QML parts, particularly for applications requiring extended temperature ranges) Selection of parts relatively early in their life cycle Minimized use of custom parts Requirement for a preferred parts list and parts control before detailed design to minimize obsolescence issues?	Are the following addressed: Order of precedence for parts selection (e.g., use of QML parts, particularly for applications requiring extended temperature ranges) Selection of parts relatively early in their life cycle Minimized use of custom parts Requirement for a preferred parts list and parts control before detailed design to minimize obsolescence issues?	Are the following addressed: Order of precedence for parts selection (e.g., use of QML parts, particularly for applications requiring extended temperature ranges) Selection of parts relatively early in their life cycle Minimized use of custom parts Requirement for a preferred parts list and parts control before detailed design to minimize obsolescence issues?	
Establish a DMSMS management evaluation process			Has planning begun for reporting DMSMS metrics?	Has the plan been established and approved?	Are metrics being reported to program office leadership and other higher-level commands?	Are metrics being reported to program office leadership and other higher-level commands?

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Process	ASR	SRR	SFR	PDR	CDR	PRR
Establish a DMSMS management evaluation process *(continued)*				Are DMSMS metrics being used for programming and budgeting, improving process efficiency, and determining ROI and other benefits?	Are DMSMS metrics being used for programming and budgeting, improving process efficiency, and determining ROI and other benefits?	Are DMSMS metrics being used for programming and budgeting, improving process efficiency, and determining ROI and other benefits?
Establish a QMS			Has a quality plan been established and approved?	Is the quality plan being executed to drive continuous improvement?	Is the quality plan being executed to drive continuous improvement?	Is the quality plan being executed to drive continuous improvement?
Establish a case monitoring and tracking process			Has the program office defined the record keeping framework to 1) track information about the resolution implementation process and status and 2) provide information about DMSMS cost and management operations efficiency?	Has the program office identified how it will capture and track information about the resolution implementation process and status?	Is the program office management tracking and monitoring resolution implementation?	Is the program office management tracking and monitoring resolution implementation?
				Has the program office developed or obtained a DMSMS case tracking and program record keeping database?		

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Process	ASR	SRR	SFR	PDR	CDR	PRR
Establish supporting contracts		Are prime contractors flowing down DMSMS management requirements to their subcontractors and are those subcontractors flowing down requirements to their supply chains in a similar way?	Are prime contractors flowing down DMSMS management requirements to their subcontractors and are those subcontractors flowing down requirements to their supply chains in a similar way?	Are prime contractors flowing down DMSMS management requirements to their subcontractors and are those subcontractors flowing down requirements to their supply chains in a similar way? Is a CDRL included for the delivery of the prime contractor's DMP?	Are prime contractors flowing down DMSMS management requirements to their subcontractors and are those subcontractors flowing down requirements to their supply chains in a similar way?	Are prime contractors flowing down DMSMS management requirements to their subcontractors and are those subcontractors flowing down requirements to their supply chains in a similar way?
		Are there exit strategies in the contracts that require all sustainment providers to ensure no item EOL issues are unresolved at the completion of the period of performance?	Are there exit strategies in the contracts that require all sustainment providers to ensure no item EOL issues are unresolved at the completion of the period of performance?	Are there exit strategies in the contracts that require all sustainment providers to ensure no item EOL issues are unresolved at the completion of the period of performance?	Are there exit strategies in the contracts that require all sustainment providers to ensure no item EOL issues are unresolved at the completion of the period of performance?	Are there exit strategies in the contracts that require all sustainment providers to ensure no item EOL issues are unresolved at the completion of the period of performance?
		Have the roles and responsibilities of the government, prime/ subcontractor, and third-party vendors been established as contractual requirements?	Have the roles and responsibilities of the government, prime/ subcontractor, and third-party vendors been established as contractual requirements?	Have the roles and responsibilities of the government, prime/ subcontractor, and third-party vendors been established as contractual requirements?		

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Process	ASR	SRR	SFR	PDR	CDR	PRR
Establish supporting contracts *(continued)*				Have DMSMS management operations contracts been put in place with the prime contractor, independent SMEs, and appropriate OEMs and OCMs?	Have DMSMS management operations contracts been put in place with the prime contractor, independent SMEs, and appropriate OEMs and OCMs?	Have DMSMS management operations contracts been put in place with the prime contractor, independent SMEs, and appropriate OEMs and OCMs?
				Have contractual provisions been put in place to obtain the data and IP rights necessary for effective DMSMS management?	Have contractual provisions been put in place to obtain the data and IP rights necessary for effective DMSMS management?	Have contractual provisions been put in place to obtain the data and IP rights necessary for effective DMSMS management?

Table 17. DMSMS Management Questions for SETRs: *Identify*

Process	ASR	SRR	SFR	PDR	CDR	PRR
Prioritize systems			Are mission criticality, operational safety, and DMSMS-related costs being considered to identify and prioritize the systems and subsystems to be monitored?	Are mission criticality, operational safety, and DMSMS-related costs being used to identify and prioritize the systems and subsystems to be monitored?	Are mission criticality, operational safety, and DMSMS-related costs being used to identify and prioritize the systems and subsystems to be monitored?	Are mission criticality, operational safety, and DMSMS-related costs being used to identify and prioritize the systems and subsystems to be monitored?
Identify and procure monitoring and surveillance tools			Have DMSMS forecasting and associated data collection and management tools or service providers been researched?	Have DMSMS forecasting and associated data collection and management tools or service providers been researched and selected?	Have DMSMS forecasting and associated data collection and management tools been reviewed to determine their continued suitability for sustainment? Have tool selections been made to supplement, as necessary?	Have DMSMS forecasting and associated data collection and management tools been reviewed to determine their continued suitability for sustainment? Have tool selections been made to supplement, as necessary?

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Process	ASR	SRR	SFR	PDR	CDR	PRR
Collect and prepare item data				Have the items associated with critical system functions been identified? Is a CDRL included for the delivery of the system BOM?	Have the items associated with critical system functions been updated?	Have the items associated with critical system functions been updated?
				Have notional BOMs for the system been acquired IAW DI-MGMT-82274?	Have indentured BOMs for the system been acquired IAW DI-MGMT-82274?	Have indentured BOMs for the system been acquired IAW DI-MGMT-82274?
				Are critical materials of concern within the supply chain being considered?	Are critical materials of concern within the supply chain being considered?	Are critical materials of concern within the supply chain being considered?
				Does the program office have a strategy for obtaining the following: Design disclosed items, including subtier hardware indenture levels F3/proprietary design items, including subtier hardware indenture levels Items that are single source and those for which the government cannot obtain data rights and the associated corrective action plans are identified?	Has the program office obtained the following: Design disclosed items, including subtier hardware indenture levels F3/proprietary design items, including subtier hardware indenture levels Items that are single source and those for which the government cannot obtain data rights and the associated corrective action plans are identified?	Has the program office obtained the following: Design disclosed items, including subtier hardware indenture levels F3/proprietary design items, including subtier hardware indenture levels Items that are single source and those for which the government cannot obtain data rights and the associated corrective action plans are identified?
				Has the notional BOM been loaded into the selected forecasting/ management tool and/or service to perform an initial DMSMS items availability assessment?	Has the build baseline/final design BOM been loaded into the selected forecasting/ management tool and/or service to perform a DMSMS items availability assessment?	Has the BOM been regularly updated and reloaded into a DMSMS forecasting/management tool and/or service to perform periodic DMSMS items availability assessments?

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Process	ASR	SRR	SFR	PDR	CDR	PRR
Collect and prepare item data *(continued)*				Have preliminary lists of items, software, and materials to monitor been prepared?	Have lists of items, software, and materials to monitor been updated?	Have lists of items, software, and materials to monitor been updated?
Analyze item availability				Are the results of selected forecasting/ management tool or manual research being used to identify immediate and near-term obsolescence issues associated with the notional BOM? For any DMSMS issues identified, are they addressed and mitigated before establishment of the build baseline/final design BOM?	Have the results of selected forecasting/ management tool or manual research been used to identify immediate and near-term obsolescence issues associated with the build baseline/final design BOM? For any DMSMS issues identified, are they addressed and mitigated before acceptance and approval of the build baseline/final design BOM?	Are the results of selected forecasting/ management tool or manual research being used to identify immediate and near-term obsolescence issues associated with the BOM?
					Is the program office receiving obsolescence forecasts on a scheduled basis?	Is the program office receiving obsolescence forecasts on a scheduled basis?
					Are PDNs being received regularly?	Are PDNs being received regularly?
				Are vendor surveys being conducted on a regular basis?	Are vendor surveys being conducted on a regular basis?	Are vendor surveys being conducted on a regular basis?
Assess preliminary designs for DMSMS risk				Are “quick look” health assessments being conducted to evaluate the DMSMS resilience of designs early in a system’s life cycle?	Are “quick look” health assessments being conducted to evaluate the DMSMS resilience of designs early in a system’s life cycle?	

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Process	ASR	SRR	SFR	PDR	CDR	PRR
Assess preliminary designs for DMSMS risk (*continued*)				Are design changes being made to mitigate the risks identified in the "quick look" health assessments?	Are design changes being made to mitigate the risks identified in the "quick look" health assessments?	
Forecast technology obsolescence			Is a formal technology roadmap and insertion/ refreshment strategy being developed for all, or portions of, the system?	Has a formal technology roadmap and approved insertion/refreshment strategy been developed?	Has a formal technology roadmap and approved insertion/refreshment strategy been developed and funded?	Has a formal technology roadmap and approved insertion/refreshment strategy been reviewed for potential updates and adjustments?
			Does the technology roadmap and insertion/ refreshment strategy focus on and address the identification of critical items, materials, and technologies, as well as emerging technologies?	Does the roadmap and insertion/refreshment strategy identify critical items, materials, and technologies, as well as emerging technologies?	Does the technology roadmap and insertion/ refreshment strategy identify critical items, materials, and technologies, as well as emerging technologies?	Does the technology insertion/refreshment plan identify critical items, materials, and technologies, as well as emerging technologies?
			Is the technology insertion/refreshment plan being used to determine the timeframe for potential DMSMS operational impacts?	Is the technology insertion/refreshment plan being used to determine the timeframe for potential DMSMS operational impacts?	Is the technology insertion/refreshment plan being used to determine the timeframe for potential DMSMS operational impacts?	Is the technology insertion/refreshment plan being used to determine the timeframe for potential DMSMS operational impacts?

Table 18. DMSMS Management Questions for SETRs: *Assess*

Process	ASR	SRR	SFR	PDR	CDR	PRR
Obtain data needed for the assessment			Have programmatic, logistics, availability, and criticality data needs for health assessments been identified?	Have programmatic, logistics, availability, and criticality data needs for health assessments been updated?	Have programmatic, logistics, availability, and criticality data needs for health assessments been updated and collected?	Have programmatic, logistics, availability, and criticality data needs for health assessments been updated and collected?

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Process	ASR	SRR	SFR	PDR	CDR	PRR
Determine whether a resolution should be pursued			Are formal analyses being conducted to determine whether a resolution should be pursued?	Are formal analyses being conducted to determine whether a resolution should be pursued?	Are formal analyses being conducted to determine whether a resolution should be pursued?	Are formal analyses being conducted to determine whether a resolution should be pursued?
Assess resolution timing and level			Are formal analyses being conducted to determine when a resolution should be pursued?	Are formal analyses being conducted to determine when a resolution should be pursued?	Are formal analyses being conducted to determine when a resolution should be pursued?	Are formal analyses being conducted to determine when a resolution should be pursued?
			Are DMSMS operational risks being identified and prioritized at various levels of assembly?	Are DMSMS operational risks being identified and resolved at various levels of assembly?	Are DMSMS operational risks being identified and resolved at various levels of assembly?	Are DMSMS operational risks being identified and resolved at various levels of assembly?
			Is the monitoring of usage and anticipated demand for items and materials being considered in health assessments?	Is the monitoring of usage and anticipated demand for items and materials being considered in health assessments?	Is the monitoring of usage and anticipated demand for items and materials being considered in health assessments?	Is the monitoring of usage and anticipated demand for items and materials being considered in health assessments?

Table 19. DMSMS Management Questions for SETRs: *Analyze*

Process	ASR	SRR	SFR	PDR	CDR	PRR
Determine the preferred DMSMS resolution				Are DMSMS issues being identified and addressed during the initial item availability analysis before acceptance and approval of the notional BOM?	Are resolutions to DMSMS issues being identified and addressed during the item availability analysis before acceptance and approval of the build baseline/final design BOM?	Are resolutions to DMSMS issues being identified and addressed before low-rate production?
				Is a BCA or AoA being performed (including ROI calculations) as part of resolution determination?	Is a BCA or AoA being performed (including ROI calculations) as part of resolution determination?	Is a BCA or AoA being performed (including ROI calculations) as part of resolution determination?
				Have all costs associated with a resolution been considered?	Have all costs associated with a resolution been considered?	Have all costs associated with a resolution been considered?
				Do mitigation strategies clearly address the entire system life cycle (not just the contract period)?	Do mitigation strategies clearly address the entire system life cycle (not just the contract period)?	Do mitigation strategies clearly address the entire system life cycle (not just the contract period)?

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Process	ASR	SRR	SFR	PDR	CDR	PRR
				Has resolution determination taken into account that the most cost-effective resolution may be found at a higher level of assembly?	Has resolution determination taken into account that the most cost-effective resolution may be found at a higher level of assembly?	Has resolution determination taken into account that the most cost-effective resolution may be found at a higher level of assembly?

Table 20. DMSMS Management Questions for SETRs: *Implement*

Process	ASR	SRR	SFR	PDR	CDR	PRR
Program and budget for DMSMS resolutions			Are DMSMS record keeping data elements being used to generate metrics in support of funding requests?	Are DMSMS record keeping data elements being used to generate metrics in support of funding requests?	Are DMSMS record keeping data elements being used to generate metrics in support of funding requests?	Are DMSMS record keeping data elements being used to generate metrics in support of funding requests?
			Are projected current and out-year DMSMS resolution budgets being developed using a sound analytical basis that is persuasive enough to obtain necessary funding?	Have projected current and out-year DMSMS resolution budgets been established using a sound analytical basis that is persuasive enough to obtain necessary funding?	Have projected current and out-year DMSMS resolution budgets been established using a sound analytical basis that is persuasive enough to obtain necessary funding?	Have projected current and out-year DMSMS resolution budgets been established using a sound analytical basis that is persuasive enough to obtain necessary funding?
			Is funding being sought based on projected resolution budgets?	Have the resolution budgets been approved and funded?	Have the resolution budgets been approved and funded?	Have the resolution budgets been approved and funded?
Integrate DMSMS resolution and modification funding			Is DMSMS resolution programming and budgeting done in coordination with modification planning (technology insertion/ refreshment)?	Is DMSMS resolution programming and budgeting done in coordination with modification planning (technology insertion/ refreshment)?	Is DMSMS resolution programming and budgeting done in coordination with modification planning (technology insertion/ refreshment)?	Is DMSMS resolution programming and budgeting done in coordination with modification planning (technology insertion/ refreshment)?
			Are DMSMS health assessments being considered as a basis for adjusting the scope or schedule of modification planning (technology insertion/refreshment)?	Are DMSMS health assessments being considered as a basis for adjusting the scope or schedule of modification planning (technology insertion/refreshment)?	Are DMSMS health assessments being considered as a basis for adjusting the scope or schedule of modification planning (technology insertion/refreshment)?	Are DMSMS health assessments being considered as a basis for adjusting the scope or schedule of modification planning (technology insertion/refreshment)?

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Process	ASR	SRR	SFR	PDR	CDR	PRR
Implement DMSMS resolutions				Are the DMSMS impacts on the notional BOM, identified during the item availability analysis resolved?	Are the DMSMS impacts on the build-baseline/final design BOM, identified during the item availability analysis resolved?	Are funded DMSMS resolutions being implemented on a timely basis?

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Appendix C. DMSMS-Related Questions for ILAs and SRs

This appendix contains DMSMS-related questions intended for use by DMSMS management practitioners to prepare for ILAs before-fielding and during-operations. The questions are presented in five tables. Table 21–Table 25 contain questions pertinent to the five DMSMS management steps: *Prepare*, *Identify*, *Assess*, *Analyze*, and *Implement*.

These questions are focused on engineering and logistics processes associated with performing DMSMS management on DoD equipment and FMS equipment, if applicable. Technically, these processes are identical for DoD and FMS equipment. Therefore, no FMS unique material is proposed.

Table 21. DMSMS Management Questions for ILAs and SRs: *Prepare*

Process	Before-Fielding (ILA)	During-Operations (SR)
Establish the foundations for DMSMS management	Has program office leadership identified required operations and deliverables, defined DMT member roles and responsibilities, and developed a risk-based approach to DMSMS management?	Has program office leadership updated required operations and deliverables, defined DMT member roles and responsibilities, and developed a risk-based approach to DMSMS management?
Develop a DMP	Has the program office established a robust DMSMS management program that identifies obsolescence due to DMSMS before items are unavailable?	Is the DMSMS management program being executed per the formal approved DMP?
	Has a formal DMP been approved and signed by leadership?	Is the government DMP being updated, as necessary?
	Does the government DMP identify the roles and responsibilities of the prime/ subcontractor and third-party vendors?	Has the government DMP been updated to identify the roles and responsibilities of the prime/subcontractors and third-party vendors as necessary?
	Have these roles and responsibilities for the prime/subcontractor and third-party vendors been established?	Have these roles and responsibilities for the prime/subcontractor and third-party vendors been established?
	Is the government conducting sufficient oversight when contractors are performing DMSMS operational processes?	Is the government conducting sufficient oversight when contractors are performing DMSMS operational processes?
Form a DMT	Has the DMT been formed?	Has the DMT been formed?
Establish DMSMS operational processes	Have all DMSMS operational processes been defined and documented in the government DMP?	Have all DMSMS operational processes been defined and documented in the government DMP?
	Are DMSMS management considerations incorporated into pertinent program office documentation?	Are DMSMS management considerations incorporated into updates of pertinent program office documentation?
Secure resources for DMSMS management operations	Have current and out-year DMSMS management operating budgets been established, approved, and funded?	Have current and out-year DMSMS management operating budgets been established, approved, and funded?
Establish interfaces to advocate for DMSMS-resilient designs	Have the design engineering interfaces been successful in employing DMSMS resilience principles in the designs?	

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Process	Before-Fielding (ILA)	During-Operations (SR)
Establish interfaces to advocate for DMSMS-resilient designs (*continued*)	Have the PM&P management interfaces been successful in selecting items for the designs that will enhance DMSMS resilience?	
	Is DMSMS management a consideration when the system design approach is being determined to minimize the impact on supportability and sustainability?	Is DMSMS management a consideration when the system modification approach is being determined to minimize the impact on supportability and sustainability?
	Are the following addressed: Open system architecture Order of precedence for parts selection (use of QML parts, particularly for applications requiring extended temperature ranges) Selection of parts relatively new in their life cycle Minimized use of custom parts Requirement for a preferred parts list and parts control before detailed design to minimize obsolescence issues Identification of shelf and operating life requirements Identification of technology life expectancies?	Are the following addressed: Open system architecture Order of precedence for parts selection (use of QML parts, particularly for applications requiring extended temperature ranges) Selection of parts relatively new in their life cycle Minimized use of custom parts Requirement for a preferred parts list and parts control before detailed design to minimize obsolescence issues Identification of shelf and operating life requirements Identification of technology life expectancies?
Establish a DMSMS management evaluation process	Are metrics being reported to program office leadership and other higher-level commands?	Are metrics being reported to program office leadership and other higher-level commands?
	Are DMSMS metrics being used for programming and budgeting, improving process efficiency, and determining ROI and other benefits?	Are DMSMS metrics being used for programming and budgeting, improving process efficiency, and determining ROI and other benefits?
Establish a QMS	Is the quality plan being executed to drive continuous improvement?	Is the quality plan being executed to drive continuous improvement?
Establish a case monitoring and tracking process	Has the program office defined the record keeping framework to 1) track information about the resolution implementation process and status and 2) provide information about DMSMS cost and management operations efficiency?	Has the program office defined the record keeping framework to 1) track information about the resolution implementation process and status and 2) provide information about DMSMS cost and management operations efficiency?
	Has the program office developed or identified a DMSMS case tracking database?	Has the program office developed or identified a DMSMS case tracking database?
	Is the program office using DMSMS metrics to track resolution implementation?	Is the program office using DMSMS metrics to track resolution implementation?
Establish supporting contracts	Have the roles and responsibilities for the prime/subcontractor and third-party vendors been established as contractual requirements where applicable?	Have the roles and responsibilities for the prime/subcontractor and third-party vendors been established as contractual requirements where applicable?
	Where applicable, are there exit strategies in the contracts that require all sustainment providers to ensure no item EOL issues are unresolved at the completion of the period of performance?	Where applicable, are there exit strategies in the contracts that require all sustainment providers to ensure no item EOL issues are unresolved at the completion of the period of performance?

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Process	Before-Fielding (ILA)	During-Operations (SR)
Establish supporting contracts *(continued)*	Is a CDRL included for the delivery of the prime contractor's DMP?	

Table 22. DMSMS Management Questions for ILAs and SRs: *Identify*

Process	Before-Fielding (ILA)	During-Operations (SR)
Prioritize systems	Are mission criticality, operational safety, and DMSMS-related costs being used to identify and prioritize the systems and subsystems to be monitored?	Are mission criticality, operational safety, and DMSMS-related costs being used to identify and prioritize the systems and subsystems to be monitored?
Identify and procure monitoring and surveillance tools	Have DMSMS forecasting and associated data collection and management tools or service providers been researched and selected?	Have DMSMS forecasting and associated data collection and management tools or service been reviewed to determine their continued suitability for sustainment? Have tool selections been made to supplement, as necessary?
Collect and prepare item data	Have the items associated with critical functions been identified?	Have the items associated with critical functions been updated?
	Is a CDRL included for the delivery of the system BOM?	
	Have indentured BOMs for the systems been acquired IAW DI-MGMT-82274?	
	Has the program office obtained the following? Design disclosed items, including subtier hardware indenture levels and F3/proprietary design items, including subtier hardware indenture levels Items that are single source and those for which the government cannot obtain data rights and the associated corrective action plans are identified?	
	Has each indentured BOM been loaded into the DMSMS forecasting/management tool?	Has the BOM been regularly updated and reloaded into a DMSMS forecasting/ management tool or service?
	Have items, materials, and software been identified for monitoring?	Have items, materials, and software been identified for monitoring?
Analyze item availability	Are the results of the selected forecasting/ management tool or manual research being used to identify immediate and near-term obsolescence issues associated with the BOM?	Are the results of the selected forecasting/ management tool or manual research being used to identify immediate and near-term obsolescence issues associated with the BOM?
	Is the program office receiving obsolescence forecasts on a scheduled basis?	Is the program office receiving obsolescence forecasts on a scheduled basis?
	Are vendor surveys being conducted?	Are vendor surveys being conducted?
	Are PDNs being received regularly?	Are PDNs being received regularly?
Assess preliminary designs for DMSMS risk	Are "quick look" health assessments being conducted to evaluate the DMSMS resilience of a design early in a system's life cycle? Is the design changed to accommodate the results?	Are "quick look" health assessments being conducted to evaluate the DMSMS resilience of a design early in a system's life cycle? Is the design changed to accommodate the results?

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Process	Before-Fielding (ILA)	During-Operations (SR)
Forecast technology obsolescence	Has a formal technology roadmap and approved insertion/refreshment plan been developed and funded?	Has a formal technology roadmap and approved insertion/refreshment plan been reviewed for potential updates and adjustments?
	Does the technology roadmap and insertion/ refreshment strategy focus on and address the identification of critical items, materials, and technologies, as well as emerging technologies?	Does the technology roadmap and insertion/ refreshment strategy identify critical items, materials, and technologies, as well as emerging technologies?
	Is the technology insertion/refreshment plan being used to determine the timeframe for potential DMSMS operational impacts?	Is the technology insertion/refreshment plan being used to determine the timeframe for potential DMSMS operational impacts?

Table 23. DMSMS Management Questions for ILAs and SRs: *Assess*

Process	Before-Fielding (ILA)	During-Operations (SR)
Obtain data needed for the assessment	Have programmatic, logistics, availability, and criticality data needs for impact assessment been identified or updated and collected?	Have programmatic, logistics, availability, and criticality data for impact assessment been updated and collected?
Determine whether a resolution should be pursued	Are formal analyses being conducted to determine whether a resolution should be pursued?	Are formal analyses being conducted to determine whether a resolution should be pursued?
Assess resolution timing and level	Are formal analyses being conducted to determine when a resolution should be pursued?	Are formal analyses being conducted to determine when a resolution should be pursued?
	Are DMSMS operational risks being identified and resolved at various levels of assembly?	Are DMSMS operational risks being identified and resolved at various levels of assembly?
	Is the monitoring of usage and anticipated demand for items and materials being considered in a health assessment?	Is the monitoring of usage and anticipated demand for items and materials being considered in a health assessment?

Table 24. DMSMS Management Questions for ILAs and SRs: *Analyze*

Process	Before-Fielding (ILA)	During-Operations (SR)
Determine the preferred DMSMS resolution	Are resolutions to DMSMS impacts being identified?	Are resolutions to DMSMS impacts being identified?
	Is a BCA or AoA being performed (including ROI calculations) as part of the resolution determination?	Is a BCA or AoA being performed (including ROI calculations) as part of the resolution determination?
	Have all costs associated with a resolution been considered?	Have all costs associated with a resolution been considered?
	Do mitigation strategies clearly address the entire system life cycle (not just the contract period)?	Do mitigation strategies clearly address the entire system life cycle (not just the contract period)?
	Has resolution determination considered that the most cost-effective resolution may be found at a higher level of assembly?	Has resolution determination considered that the most cost-effective resolution may be found at a higher level of assembly?
	Are system security engineering functions contributing to the formulation and resolution of DMSMS issues?	Are system security engineering functions contributing to the formulation and resolution of DMSMS issues?

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Table 25. DMSMS Management Questions for ILAs and SRs: *Implement*

Process	Before-Fielding (ILA)	During-Operations (SR)
Program and budget for DMSMS resolutions	Is funding to mitigate DMSMS risk being identified and obtained?	Is funding to mitigate DMSMS risk being identified and obtained?
	Have projected current and out-year DMSMS resolution budgets been established using a sound analytical basis that is persuasive enough to obtain necessary funding?	Have projected current and out-year DMSMS resolution budgets been established using a sound analytical basis that is persuasive enough to obtain necessary funding?
	Have these projected resolution budgets been approved?	Have these projected resolution budgets been approved?
	Have these resolution budgets been approved and funded?	Have these resolution budgets been approved and funded?
Integrate DMSMS resolution and modification funding	Is DMSMS resolution programming and budgeting done in coordination with modification planning (technology insertion/refreshment)?	Is DMSMS resolution programming and budgeting done in coordination with modification planning (technology insertion/refreshment)?
	Are DMSMS health assessments being considered as a basis for adjusting the scope or schedule of modification planning (technology insertion/refreshment)?	Are DMSMS health assessments being considered as a basis for adjusting the scope or schedule of modification planning (technology insertion/refreshment)?
Implement DMSMS resolutions	Are funded DMSMS resolutions being implemented on a timely basis?	Are funded DMSMS resolutions being implemented on a timely basis?

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Appendix D. DMSMS Program Capability Levels

Program capability levels can be a powerful tool for improving DMSMS management effectiveness and therefore their use in the DMP is a strongly recommended best practice. The DMP should include plans for achieving target DMSMS capability levels.

This appendix contains information to help guide a decision on the appropriate level for a program office. Table 26 identifies the program capability levels for each DMSMS management step and management operational processes. The levels are defined as follows:

- Level 1 represents minimal DMSMS management capability. Practices are largely reactive.
- Level 2 represents a DMSMS management capability greater than Level 1. Practices are somewhat proactive in situations where proactive practices are needed.
- Level 3 represents a DMSMS management capability greater than Level 2. Proactive and some strategic practices are used when needed.
- Level 4 represents a robust DMSMS management capability. Strategic efforts are being applied throughout all steps of the process.

A program office should use the table as the basis for determining the current state of its DMSMS management practices. This is done by examining each row of the table and identifying what is being done. If the program office does not have a DMP, then it is effectively below capability Level 1. The DMP should provide a basis for systematically progressing through the capability levels to achieve the program office's target. Several factors should be considered when determining the appropriate target capability level for a program office:

- A lower capability level could be sufficient near the end of a system's life cycle.
- A higher capability level might be needed for more complex systems, because such program offices are likely to encounter additional DMSMS issues. However, smaller program offices may be seriously affected, depending on the technologies used.
- A higher capability level cannot be achieved without significant DMSMS management subject matter expertise and DMSMS management training for the entire DMT.
- Not every DMSMS management process must be at the same capability level.
- A program office cannot immediately move from a low capability level to a high capability level; the transition should be gradual.
- Resource constraints may exist, either for a single program office or for a group of program offices.

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Table 26. DMSMS Program Capability Levels

Step	Process	Level 1	Level 2	Level 3	Level 4
Prepare (Section 3)	3.1 Establish the foundations for DMSMS management	Foundations for DMSMS management not established	Individual DMT members set foundations for DMSMS management in an ad hoc way	DMT sets foundations for DMSMS management, leadership partially aware	Program office leadership sets foundations for DMSMS management
	3.2 Develop a DMP	No or partial DMP	DMP developed	DMP approved and signed by program office leadership	DMP approved and signed by program office leadership and updated periodically
		No government oversight of contractor activities	Minimal government oversight of contractor activities	DMP calls for limited government oversight of contractor activities	DMP calls for extensive government oversight of contractor activities
	3.3 Form a DMT	No DMSMS point of contact established	DMSMS point of contact established (but retains other duties); meets with other stakeholders on ad hoc basis	Full-time DMSMS point of contact established; meets with other stakeholders on ad hoc basis	Full DMT formed including all stakeholders; stakeholders understand their roles and responsibilities
		No independent DMSMS management SME	Limited funding for the use of an independent DMSMS management SME	Independent DMSMS management SME funded to assist the government with overseeing the prime contractor, give an independent perspective on issues and resolutions, and provide general DMSMS management advice	Independent DMSMS management SME team funded to assist the government with overseeing the prime contractor, give an independent perspective on issues and resolutions, and provide general DMSMS management advice
		DMSMS point of contact has limited training	DMSMS point of contact trained	DMT trained	DMT members have advanced DMSMS training
	3.4 Establish DMSMS operational processes	DMSMS management operational processes entirely ad hoc and reactive; DMSMS management considerations occasionally incorporated in program office documents	DMSMS management operational processes defined, but not documented; DMSMS management considerations sometimes incorporated in program office documents	DMSMS management operational processes defined and documented, and processes are proactive when needed; DMSMS management considerations sometimes incorporated in program office documents	DMSMS management operational processes defined and documented, and processes are proactive when needed; DMSMS management considerations incorporated in program office documents

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Step	Process	Level 1	Level 2	Level 3	Level 4
Prepare (cont.)	3.5 Secure resources for DMSMS management operations	No DMSMS management operations-earmarked funding	Funded to operate at Level 2	Funded to operate at Level 3; funding shortfall and impact identified and reported to decision makers	Funded to operate at Level 4; funding shortfall and impact identified and reported to decision makers
	3.6 Establish interfaces to advocate for DMSMS-resilient designs	DMSMS design characteristics and parts selection criteria not being raised to design engineering	DMSMS design characteristics and parts selection criteria rarely considered by design engineering	DMSMS design trades sometimes being made; parts selection criteria being consistently applied	DMSMS design trades being made; parts selection criteria being consistently applied
		DMSMS resilience not a design consideration	DMSMS resilience rarely made a design consideration	DMSMS resilience sometimes considered in design trades	DMSMS resilience fully considered in design trades
		DMSMS resilience not considered in parts selection	DMSMS resilience rarely considered in parts selection	DMSMS resilience sometimes considered in parts selection	DMSMS resilience always considered in parts selection
	3.7 Establish a DMSMS management evaluation process	No record keeping	Limited record keeping, limited attempt to evaluate program office	Detailed records kept. Metrics aggregated and analyzed to improve program office performance, to calculate ROI, and to support programming and budgeting	Metrics widely accepted and used by program office management and higher-level organizations
	3.8 Establish a QMS	No quality plan	Quality plan established, limited attempt to improve process	Robust quality plan established; corrective actions and continuous process improvement occurring	Quality assurance metrics established; corrective actions and continuous process improvement occurring
	3.9 Establish a case monitoring and tracking process	No record keeping or resolutions not being tracked	Ad-hoc record keeping and some resolutions being tracked	Record keeping formalized and some resolutions being tracked	Record keeping formalized and resolutions being tracked
	3.10 Establish supporting contracts	Marginal contract requirements covering some aspects of DMSMS management	Contract requirements established for all significant and applicable aspects of DMSMS management	Contract requirements, to include exit clauses, established for all significant and applicable aspects of DMSMS management and flowed down the supply chain	Contract requirements, to include exit clauses and incentives, established for all significant and applicable aspects of DMSMS management and flowed down the supply chain

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Step	Process	Level 1	Level 2	Level 3	Level 4
Prepare (cont.)	3.11 Establish communication processes with FMS partners	LOA does not explicitly mention DMSMS management; communications channels are ad hoc	LOA explicitly includes DMSMS management; nature of the effort is not included in the LOA but has been discussed with the foreign partner; regular meetings planned.	LOA includes details of DMSMS management responsibilities; nothing explicit about any technical data needs, but the subject has been discussed with the foreign partner; regular meetings planned.	LOA includes details of DMSMS management responsibilities and technical data needs; clear distinction made on who does what; both subjects discussed with the foreign partner on a regular basis; frequent meetings planned; avenues established for out-of-cycle communications.
Identify (Section 4)	4.1 Prioritize systems	No prioritization of subsystems	Ad hoc prioritization of subsystems for DMSMS management execution	Subsystems prioritized for DMSMS management execution	Materials and software explicitly considered
		No prioritization of items	Ad hoc prioritization of items for DMSMS management execution	Items prioritized for DMSMS management execution	Materials and software explicitly considered
	4.2 Identify and procure monitoring and surveillance tools	No predictive or data management tools in use	Predictive tools and data management tools only partially in place	Predictive tools and data management tools in place	Comprehensive DMSMS management tools in place
	4.3 Collect and prepare item data	Only miscellaneous item data collected; everything driven by PDNs or the inability to procure the item	BOM data collected, but may not be indentured	Indentured BOM data collected (including software and materials); vendors surveyed for assemblies, mechanical parts, software, and materials	Indentured BOM data collected (including software interface specifications and materials); vendors surveyed for assemblies, mechanical parts, software, and materials
		BOM data errors not fully corrected	BOM data errors corrected	Some items and materials prioritized, and ad hoc determination made regarding what to exclude from proactive monitoring	All items and materials prioritized, and formal determination made regarding what to exclude from proactive monitoring
	4.4 Analyze item availability	No predictive tool usage	Predictive tools used occasionally	Results of predictive analyses for electronic items examined continually	Results of at least two predictive tools examined continually for electronic items; vendors surveyed periodically for MaSME items and software

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Step	Process	Level 1	Level 2	Level 3	Level 4
Identify (cont.)	4.5 Assess preliminary designs for DMSMS/ obsolescence risk	No assessments of preliminary designs	Some assessments of preliminary designs	Assessments of most preliminary designs	Assessments of preliminary designs always conducted
	4.6 Forecast technology obsolescence	No technology roadmaps developed	Technology roadmaps created	Technology insertion and refreshment plans being developed	Technology roadmaps used to develop technology insertion and refreshment plans
Assess (Section 5)	5.1 Obtain data needed for the assessment	No programmatic, logistics, availability, and criticality data collected	Little programmatic, logistics, availability, and criticality data collected	Some programmatic, logistics, availability, and criticality data collected for impact assessment	Comprehensive programmatic, logistics, availability, and criticality data collected for impact assessment
	5.2 Determine whether a resolution should be pursued	Ad hoc determination made with no analysis	Resolution pursued in all cases	Some logistics and programmatic data and vendor surveys being used to determine when an operational impact will occur	Extensive logistics and programmatic data and vendor surveys (including software and materials) being used to determine when an operational impact will occur
	5.3 Assess resolution timing and level	Ad hoc; only when PDN received	Only parts availability considered	Some logistics and programmatic data and vendor surveys being used to determine when an operational impact will occur	Extensive logistics and programmatic data and vendor surveys (including software and materials) being used to determine when an operational impact will occur
		No priorities	Rough priorities being assigned on an ad hoc basis	Priorities being assigned, but little consideration given to next higher levels of assembly	Specific priorities being assigned; next higher levels of assembly being examined for operational impact
Analyze (Section 6)	6.1 Determine the preferred DMSMS resolution	Ad hoc; limited cost data used	AoA conducted using unrefined cost factors	BCA conducted using refined cost factors, tailored to the specific problem	Resolution options determined at item level and for higher levels of assembly BCA used where necessary
Implement (Section 7)	7.1 Program and budget for DMSMS resolutions	No resolution budget; funding sought on case-by-case basis	Fixed amount of funding for resolutions not based on any kind of projection; when funds expended, resolution costs sought on case-by-case basis	Resolutions budget funded based on projections of issues; out-year budgets unfunded	Active engagement in obtaining other sources of funding; out-year budgets programmed

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Step	Process	Level 1	Level 2	Level 3	Level 4
	7.2 Integrate DMSMS resolution and modification funding	No health assessments conducted	DMSMS resolution programming and budgeting uses health assessments to establish resolution timing	DMSMS resolution programming and budgeting done in conjunction with modification planning	DMSMS resolution programming and budgeting done in conjunction with modification planning; health assessments and LON buys used to adjust modification plans
	7.3 Implement DMSMS resolutions	No follow-up	Ad hoc oversight of execution	Limited oversight of execution	Comprehensive oversight of execution

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Appendix E. DMSMS Management Plan Checklist and Example

As introduced in Section 3.3, this appendix includes a checklist and an example plan that provides a starting point for creating and maintaining a DMP. According to DoDI 4145.15, every DoD DMSMS management organization must have a DMP. The DMP is essential for ensuring long-term sustainability and readiness of weapon systems, minimizing or avoiding out-of-cycle redesigns, and ensuring material readiness.

E.1 DMP Checklist

The DMP checklist serves as a tool to assist in developing DMPs. It addresses all DMSMS management sections found in Section 3.3. The checklist also assists program office personnel tasked with reviewing and approving the DMP with the information they need to understand if all areas of DMSMS management areas are addressed and to provide a strong foundation for managing DMSMS risk that may impact the program.

DMP Sections			References
Title Page	Y	N	**General Best Practice**
Does the title page include the name of the program/system, the version of the DMP, the date of the DMP, and distribution and handling requirements?			
Approval Page	Y	N	**SD-22 Sec. 3.3**
Does the approval page have the signature blocks for the approver?			
Record of Changes Page	Y	N	**General Best Practice**
Is there a Record of Changes page and does it document any changes including the date of the change, the specific change made, and the identification and initials of the person who made the change?			
Table of Contents	Y	N	**General Best Practice**
Is there a Table of Contents that identifies each section of the DMP?			
Does the Table of Contents identify each appendix or annex (e.g., Abbreviations)?			
Is there a List of Tables that identifies each type of table used in the DMP?			
Is there a List of Figures that identifies each figure used in the DMP?			
DMP Body			
Note: The bulleted information below the questions provides greater explanation on additional information that needs to be included or should be considered for inclusion in the DMP. A more in-depth discussion is provided in the corresponding SD-22 section identified in the Reference column for each section of the DMP.			

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DMP Sections			References
1. Purpose	Y	N	**SD-22 Sec. 3.3.1**
Does the DMP clearly define near- and long-term DMSMS objectives aligned with program life cycle phases and sustainment goals? • The primary objective is to ensure DMSMS-related cost, schedule, readiness, and availability impacts do not exceed an acceptable level. • Other possible objectives include "no reactive DMSMS cases except where planned," "avoid out-of-cycle redesigns," and "keep annual resolution costs below a specific number."			DoDM 4245.15, 3.2.c.(2)
2. Scope	Y	N	**SD-22 Sec. 3.3.2**
Is there a brief description of the program office/system and subsystem(s) the DMP pertains to? • Name of the systems and subsystems and a short description. • The quantity of systems planned and the timeframe they will be supported. • The program's acquisition designation. • Whether there is FMS and the countries involved. • Whether this is an initial DMP or a revision.			DoDM 4245.15, 3.2.c.(3)(a)
Does the DMP contain pertinent information from the acquisition strategy and where the system is in its life cycle, the Acquisition Category (ACAT) level, milestone dates, how the system and subsystem (if applicable) will be used, the identification of stakeholders, and a description of the long-term sustainment strategy, modification plans, and maintenance approach?			DoDM 4245.15, 3.2.c.(3)(c)
3. DMSMS Management Approach	Y	N	**SD-22 Sec. 3.3.3**
Is there a description of the primary DMSMS management roles of contractors, program office personnel, and independent SMEs and how the program office will maintain a life cycle perspective for its DMSMS management approach? • Identify if the prime contractor or other independent DMSMS SME organizations will perform DMSMS management, the period for which they will be responsible, and how the program office will provide oversight.			DoDM 4245.15, 3.2.a.(3)
Does the DMP specify proactive DMSMS activities (e.g., predictive tools, market surveys. forecasts) and clearly define conditions where reactive measures are acceptable? • Identify the desired frequency for proactive monitoring and surveillance activities. If reactive, does the DMP outline procedures that will be used to evaluate and prioritize DMSMS items associated by the level of risk?			DoDM 4245.15, 3.2.c.(3)(a)
Is there a description of what mechanisms (i.e., PDNs, predictive tools, vendor surveys, technology roadmaps and insertions, and DMSMS management information systems) the program office will use for monitoring, assessing, analyzing, and resolving DMSMS issues in a timely manner? • Identify the sources where PDNs will be obtained. • Identify who will be responsible for processing PDNs and how often they will be expected to be received and reviewed (e.g., as soon as known, weekly, monthly, or other defined timeframe).			DoDM 4245.15, 3.2.c.(6)(d)

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DMP Sections			References
• Document all tools that all stakeholders will use for DMSMS monitoring, assessments, analysis and case management. • Document all tools the government program office will use. • Document whether vendor surveys or technology and product roadmaps will be used to monitor other types of items such as software, MaSME items, and COTS assemblies.			
Does the DMP identify the contingency plans for potential program changes (e.g., variations in funding, change in acquisition strategy, and planned service life extension)? • Identify and document how often the DMP will be reviewed to evaluate the status and effectiveness of DMSMS management activities. • Note: The frequency of reviews will depend on program manager guidance, Service specific guidance, acquisition pathway, and type and level of risk. Recommend a review be conducted at least annually.			DoDM 4245.15, 3.2.c.(4)(g)
4. DMSMS Management Team (DMT)	Y	N	**SD-22 Sec. 3.2.1, 3.2.2, and 3.3.4**
Are all internal and external stakeholders (including the OEM/Prime) involved in DMSMS management identified, including coordination across other programs DMTs? • Include the organization the stakeholder is assigned to. • Include other DMSMS management program offices that interact with the DMSMS management program office for the system/sub-system covered by the DMP.			DoDM 4245.15, 3.2.c.(3)(f)
Are the roles and responsibilities of the DMT members and the person(s) filling those roles identified? • Consider using a RACI chart or risk cube.			DoDM 4245.15, Sec. 3.2
Does the DMP identify the training requirements for each DMT member? • Include both the required and optional training for the DMT members.			
Are the methods the DMT will use to communicate both internal to the team and external to the program office identified? • Identify the specific method that will be used to communicate. • Include the frequency of meetings and other required communication. • Include an agenda of primary topics that will be addressed during meetings.			DoDI 4245.15, Sec. 3. h.
5. DMSMS Management Operations	Y	N	**SD-22 Sec. 3.2.1, 3.3.5, 3.3.6, and 3.5**
Is there a description of how the program office's DMSMS management efforts will be integrated with other strategies, planning, and reviews (e.g., product support strategies, systems engineering design and review process, and modification planning)? • Explain the interfaces established to advocate for DMSMS resilient designs.			DoDM 4245.15, 3.2.c.(1),(2),(2)(a),(3)(d), and Glossary

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DMP Sections			References
• Include how the results of health assessments are used to inform and integrate DMSMS resolution funding with funding for program office modification plans. • Document whether the DMT is participating in design reviews and the ECP approval process. • Include an explanation of how the DMT will perform supply chain risk management activities (e.g., cybersecurity, software assurance, counterfeit parts, foreign ownership of sub-tier vendors, and other categories of supply chain risk) to identify, assess, and address potential obsolescence risks.			
Is there a method or process identified to address how other program activities will coordinate with the DMT?			DoDM 4245.15, 3.2.(C)(1), (5)(c)
Does the DMP explain how the DMT will provide oversight of contractor DMSMS management efforts? • Include the types of reporting and the level of involvement in the DMT by the contractor. • Note: The level of involvement will be based on what the contract or statement of work requires.			DoDM 4245.15, Sec. 3.2 (C)(6)(c)
Is there an explanation of how DMSMS issues will be detected, tracked, and resolved? • Explain how BOMs and other programmatic data will be obtained. • Describe the risk-based methodology used to prioritize the loading of subsystem BOMs into predictive tools. • Describe when the prioritized BOMs will be loaded and the update frequency. • Describe the process used to monitor items and the frequency of monitoring. • Provide a description of the impact if the program office is unable to obtain the necessary BOM data. • If applicable, outline steps required to identify how a program can construct a BOM from other available data if it is not provided under contract. • Document how vendor surveys will be used to monitor other types of items, such as software, MaSME items, and COTS assemblies. • Address how DMSMS cases will be initiated, managed, and maintained.			DoDM 4245.15, Sec. 3.; 3.2.(C)(6)(d)–(g), (c)(7)(c)
Does the DMP explain how DMSMS management processes, issues, and resolution options will be communicated to foreign partners (when applicable) and DoD stakeholders external to the program office? • Briefly summarize communication channels and frequency associated with DMSMS management for FMS equipment. This should include: ○ The DMSMS activities the DoD program office will perform. ○ When and how foreign partners will be notified. ○ How the involvement of other DoD program offices will be managed. ○ How costs will be shared.			

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DMP Sections			References
6. Funding			
Does the DMP identify what resources have been programmed and budgeted for DMSMS management operations? • Document the costs associated with DMSMS management operations throughout the POM period.			
Are the costs associated with the DMSMS resolution implementation throughout the POM period, as well as the availability of those resources, documented in the DMP? • Include the costs associated with the DMSMS resolution implementation throughout the POM period as well as the availability of those resources. • Document the risks and mitigations associated with a mismatch between funding available and funding needs.			DoDM 4245.15, Sec. 3.6.(a)(1)
7. DMSMS Contract Requirements	Y	N	**SD-22 Sec. 3.3.7**
Does the DMP identify the DMSMS management operations and/or resolution implementation requirements that will be imposed on the contractor and/or independent SMEs? • Summarize the statement of work requirements and list the DMSMS-related CDRLs and DIDs. • Include performance assessment factors, incentives, and penalties. • Identify (supported by budgets and contracts, as necessary) a method that enables the government to manage DMSMS issues and their resolution across breaks in contracts (e.g., between production contracts, between a production contract and a sustainment contract, and between sustainment contracts) in a way that coordinates with expected contributions from the involved contractors. • Identify a contract exit strategy. ○ An Exit Strategy is useful for planning and managing the obsolescence of parts and materials in a system or subsystem, particularly when those parts are no longer manufactured or readily available. ○ This strategy is also crucial for ensuring continued functionality, minimizing disruptions, especially in cases where a system or program is going from CLS managed to organic.			DoDI 4245.15, Sec. 3.f.(4) DoDM 4245.15, Sec. 3.2.(a)(8)(b)(c)
8. DMSMS Metrics, Reporting, and Quality	Y	N	**SD-22 Sec. 3.3.8**
Are the data elements the program office will collect identified (e.g., Level 1 and Level 2 record keeping data elements)?			DoDM 4245.15, Sec. 3.2.(d)(4); Table 1 & Table 2
Is there a description of how the Level 1 and Level 2 data elements will be used? • Use Level 1 to develop metrics to improve DMSMS management operations. • Use Level 2 to establish metrics that will enhance these benefits by enabling more complex analyses. • Use contractor compliance metrics to measure DMSMS processes and activities.			DoDM 4245.15, 3.2.d.(4); Table 1 & Table 2

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DMP Sections			References
• Include the periodicity at which metrics will be reported, and to whom they will be reported. • Document the sources that will be used to collect the data elements.			
Is there a description of the types of deliverables (e.g., how sustainability analysis results will be provided, notices of obsolescence, status of resolutions, and elements of supportability roadmaps that inform technology refreshments) the program office will expect as outputs of its DMSMS management implementation approach? • Identify the schedules for deliverables needed for the DMT to perform its oversight role and to be able to inform program office leadership decision-making. • Document a mechanism for reporting DMSMS issues and associated risk and as appropriate highlighting risk to program office leadership.			DoDM 4245.15, (c)(7)(c)
9. Process Capability	Y	N	**SD-22 Sec. 3.3.9**
Are gaps in DMSMS process maturity identified? • Record any specific DMSMS risks that result from not achieving the desired process capability, the specific area or areas of the program that are affected, and a plan to reach the desired capability levels. • Include the rationale for not achieving the desired capability level in situations where it retained the status quo or only partially achieved the desired capability levels.			
Is there a roadmap for process improvement? • Identify methods the DMT can employ to evaluate the process capability level and make improvements based on changes in the DMSMS-related risk to the system. • Include the requirement to conduct an updated review to determine if additional measures need to be employed to reduce potential DMSMS issues in the program office if the risk should change. • Document the need to conduct periodic reviews, the frequency of the reviews, and a planned approach for evaluating the method(s) used to identify the desired process capability.			

E.2 DMP Example

Preparing a DMP is covered in depth in Section 3.3, which should be read prior to creating a plan to get an understanding of the many facets of DMSMS management and to ensure that the final plan satisfies all of the needs of the organization.

The example provides a structure that follows the DMSMS management steps found in Section 3.3 and contains minimum content that may be tailored to meet individual program needs. Your parent component organizations may have additional policy, guidance, templates, or tools to assist in developing a DMP. Ensure that these resources, if used, are consistent with DoD policy and guidance.

In situations where a single DMSMS management organization manages multiple systems with variations in life cycles, acqusition strategies, etc., the differences can be documented in annexes or appendices to

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the plan. An example of the content that can be documented in an annex or appendix is provided below. Only the differences need to be addressed in the annex or appendix. The body of the DMP, organized as follows, should be modified to reference the annexes or appendices appropriately and contain the core approaches the DMSMS management organization is pursuing.

1. Details about the scope and applicability of the annex
   a. A brief description of the system including how and where it will be used
   b. The ACAT
   c. The current life cycle phase
   d. Salient information from the acquisition strategy including the contracting approach, the sustainment strategy, and the maintenance approach
   e. Any planned milestone dates
   f. Major stakeholders such as FMS or other services. The remainder of the annex need only address differences from the plan in the subject areas below.
2. DMSMS management approach
3. DMT
4. DMSMS operations
5. Funding details
6. Contract details
7. DMSMS Metrics, Reporting, and Quality Management Systems

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Diminishing Manufacturing Sources and Material Shortages (DMSMS) Management Plan (DMP) for:

Undersea Wildlife Detection System BSS-001

Version 2.0 June XX, 2025

Prepared for Undersea Wildlife Detection Program SEA 0AB

Prepared By:

John Okuma, SEA 0AB

Add the appropriate distribution and destruction statement here.

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Approval Page

Submitted By:

_____ _____

Bill Dawia Date
DMSMS Team Lead

Reviewed By:

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Hank Garfield Date
Product Support Manager

Approved By:

_____ _____

Tom Fieldmaster Date
Program Manager

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This document is to be reviewed and updated annually or as needed.

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Appendix A – Acronyms

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1. Purpose

The purpose of this plan is to document and implement an effective DMSMS management program that the Undersea Wildlife Detection Program (UWDP) will use to minimize the impact of DMSMS issues on the Undersea Wildlife Detection System, BSS-001. This plan defines procedures for DMSMS risk identification; government, contractor, and DMSMS roles; and data integration and resolution strategies across the program life cycle. Additionally, this plan ensures the BSS-001 system will reduce or avoid out-of-cycle redesigns, mitigate obsolescence risk, and sustain operational availability for U.S. and FMS customers. During development and production, the plan focuses on designing DMSMS resilient systems and eliminating or minimizing production delays due to parts unavailability. As the program moves to sustainment, emphasis shifts to focusing on preventing DMSMS impacts to the operational mission.

2. Scope and Applicability

2.1 System Description

The BSS-001 system consists of a physical unit and integrated software that displays real-time information on existing shipboard tactical systems. The surface unit provides a communication link between the undersea units. The surface unit can control the communication of up to six undersea units. The BSS-001 system detects and identifies undersea wildlife and provides the tactical systems the location, type, and quantity of the wildlife found. The BSS-001 is based on a commercial product deployed in both US and foreign fishing fleets, however, there are significant differences in the military version. The BSS-001 system will be deployed on all surface combatants to provide tactical information on the location of undersea wildlife. The UWDP is developing additional units under LOAs in support of two FMS customers, Liechtenstein and Transnistria. There is currently no expectation of any cross-service applicability.

2.2 Acquisition Strategy and Milestone Dates

The BSS-001 system follows the MCA Pathway and is designated as an ACAT IV Monitor (M) program. BSS-001 is post Milestone A and is scheduled for a Milestone B review on 25 September 2026. Approximately 300 BSS-001 systems will be built with a planned initial deployment date of October 2027 and final delivery of all units by January 2030. The system is currently planned to be in service until 2050.

The BSS-001 system is being developed by FishFinder, Inc. using a firm fixed price contract and is planned to be produced with the same type of contract. Incentives for DMSMS management that exceed the current statement of work requirements will be included in each contract. Sustainment of the BSS-001 will be a mixture of contract and organic, with the organic sustainment focused on the non-COTS aspects of the system. The Defense Logistics Agency will manage logistics for consumables and the Naval Supply System Weapon Systems Support (NAVSUP WSS) will manage reparable items.

3. DMSMS Management Approach

3.1 Primary DMSMS Management Roles

The UWDP will serve as the primary DMSMS manager; however, because the prime contractor will be responsible for designing the system and maintaining the commercial parts of it, FishFinder, Inc. has been tasked to also manage DMSMS. FishFinder, Inc. is tasked to provide the government access to its internal DMSMS management system, DMX. Given the complexity of the system, Naval Surface Warfare Center (NSWC) Biloxi is also tasked as a DMSMS SME organization. NSWC Biloxi's DMSMS management tool, OMNICRON, will be used for loading and monitoring BOMs, handling DMSMS notifications, and managing DMSMS cases through deployment and potentially in sustainment. The prime contractor is tasked to participate in DMT meetings and to provide DMSMS notifications and case statuses on a regular basis. NSWC Biloxi personnel will review the deliverables and monitor the performance of the prime contractor. Any issues regarding contractor performance will be communicated to UWDP's program leadership for action if needed. The roles and responsibilities of each stakeholder

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are defined to ensure continuity, ownership, and accountability for proactive DMSMS identification, resolution planning, and communication across acquisition phases.

3.2 Prioritization of Subsystems and Items for Proactive Monitoring and Surveillance

The DMT will primarily focus on proactive surveillance of electronic components and assemblies, working closely with FishFinder, Inc., through production. Mechanical items that pose a significant risk to the mission will also be managed proactively. NSWC Biloxi will monitor the supply system for all reactively managed items for indications that a DMSMS event has occurred and work with the DMT to take actions necessary to prevent impacts. Once the system transitions to sustainment, the government will assume full responsibility for all DMSMS management activities. Early, proactive identification of DMSMS risks expands available resolution options, creates opportunities for an increased number of lower-cost alternatives, and enhances operational readiness.

3.3 Monitoring and Case Management

The UWDP's engineering team collaborates with FishFinder, Inc. in the development of technology roadmaps and plans for updating and refreshing the system that supports long-term system viability. To proactively monitor risk, the DMT will work closely with the program's engineering team to identify technology trends that may signal early obsolescence or other DMSMS issues that will impact system availability and readiness once it is fielded. The DMT will actively contribute to and remain knowledgeable of changes to the product roadmaps to ensure the program plans for and adopts new technologies, to avoid costly mistakes, and to ensure the DMP addresses technology obsolescence. The engineering team will ensure that all program technology plans are stored on the UWDS information portal. Access to the portal can be granted to DMT members with the correct Common Access Card (CAC) or External Certification Authorities (ECA) certificates. This information will be used for ongoing monitoring, planning, and case management.

3.4 Contingency Planning

There are several areas of concern for contingency planning: funding shortfalls, workload surges, or changes to delivery quantities and system end-of-service dates that could delay the system's production or transition to sustainment. If the funding provided falls below the amount needed to sustain operations as planned or if the DMSMS workload exceeds projections, the DMT will coordinate with leadership to evaluate options—including securing additional funding, re-prioritizing tasks, or reducing the level of service provided by the team. The DMT will not duplicate activity that FishFinder, Inc. is already funded to perform under contract ensuring efficient use of government resources. To minimize potential impacts, the DMT will maintain close communication with program leadership regarding end of service or deliverable changes. All impacts and mitigations will be documented and communicated to leadership to support timely decision-making and risk mitigation.

4. DMSMS Management Team (DMT)

4.1 Team Composition and Responsibilities

The DMT is the central coordinating body for all DMSMS activities. It includes representatives from the program office, the prime contractor, and external SMEs. Together they serve as the interface between engineering, logistics, supply chain, production, sustainment, and leadership stakeholders. The DMT's responsibilities include maintaining awareness of DMSMS risks, executing resolution actions, reviewing contractor performance, and informing acquisition decision points. Members are responsible for liaising with their respective organizations to drive coordinated responses. The DMT structure ensures that all technical, contractual, and life cycle stakeholders are represented. Training requirements, knowledge sharing practices, and team continuity plans are defined to support consistent, life cycle–aligned DMSMS execution.

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Table 1–DMT Roles and Responsibilities provides details of the team's membership and their roles and responsibilities. Each team member monitors the progress of tasks and actions assigned by the DMT to their respective teams and report them to the DMT.

Table 27–DMT Roles and Responsibilities

Organization	Role	Responsibility
UWDP	DMT Lead	Defines activities of the team and ensures DMSMS management is conducted IAW the DMP.
NSWC Biloxi	DMSMS SME	Interface between NSWC Biloxi personnel and UWDP. Maintains the BOM and case data in OMNICRON and communicates issues to the DMT. Collects and compiles metrics and provides reports to the DMT.
UWDP (Engineering)	Engineering Activity representative	Interface between the DMT and the engineering team. Ensures that DMT actions are carried out.
UDP (Logistics)	Logistics representative	Interface between the DMT and the logistics team. Ensures that DMT actions are carried out.
FishFinder, Inc.	Prime Contractor representative	Interface between the DMT and the contractor. Insures DMSMS issue and case data are communicated. Works with the DMT to resolve DMSMS cases.
NAVSUP WSS	Supply support activity representative	As the system moves to sustainment, NAVSUP WSS will work with the DMT to ensure supportability concerns are properly addressed.
UWDP (Contracting)	Contracting Officer's Representative	An ad hoc team member. Ensures contracting issues are addressed. Responsible for formal communication with the contractor.
UWDP (Business Office)	Business Office	An ad hoc team member. Responsible to communicate financial needs between the Business Office and the DMT.
FMS Office	FMS	Interfaces between the DMT and the FMS customers.

4.2 DMT Training Requirements

All DMT members are required to take the courses listed in *Table 2–Required DMSMS Training*. The training in *Table 3–Optional DMSMS Training* is recommended for full-time team members.

Table 28–Required DMSMS Training

LOG 0640–DMSMS: What Program Management Needs To Do And Why
LOG 0650–DMSMS Fundamentals
LOG 0660–DMSMS Executive Overview
LOG 0670–DMSMS Basic Component Research

Table 29–Optional DMSMS Training

LOG 0630–Introduction to Parts Management
LOG 0320–Preventing Counterfeit Parts in DoD Supply Chains
LOG 0380–Provisioning & Cataloging

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LOG 0390–Additive Manufacturing
LOG 0400–Additive Manufacturing Case Studies
LOG 0470–Sustaining Engineering
LOG 0510–System Retirement, Disposition, Reclamation, Demilitarization, Disposal
CON 0040–Market Research
CLE 019–Modular Open Systems Approach
CLE 026–Trade Studies

4.3 DMT Communication

Effective communication among DMT members and the offices they represent is crucial to efficient and effective operation of the team. Consistent engagement with program office leadership ensures that contracting and funding impacts are recognized and resolved throughout the program's life cycle.

DMT meetings are held monthly via teleconference. The meetings will include the following topics:

- Review of action items
- The status of existing cases
- Emergent DMSMS issues

The DMT also participates in the annual program review to present their status and report on important DMSMS issues.

5. DMSMS Management Operations

5.1 Integration with Other Strategies, Planning, and Reviews

The UWDP's SEP defines a process by which BOMs are routed through the DMT for evaluation early in the design phase. The DMT will work with systems engineers to ensure designs take into consideration a number of factors as discussed in SD-22. These factors can include the maturity of the items, the anticipated end of life of the items, the types of interfaces used, and commonality.

5.2 DMSMS Processes

The DMSMS management processes used by the DMT are consistent with those in SD-22 and address the unique needs of the program office and each stakeholder to ensure the communication is sufficient to meet all DMP objectives.

5.3 Contract Oversight

The UWDP receives reports from the prime contractor as described in Section 7 of this DMP. NSWC Biloxi reviews the reports and assesses the performance of the contractor concerning completeness and accuracy of data provided, timeliness of DMSMS notifications, case resolution proposals and cost, and efficiency. The results of the assessments are provided to the DMT lead who takes action when needed. When the prime contractor proposes a solution to a DMSMS issue that is not likely to prevent impacts throughout the life of the system, the DMT reviews the rationale for the solution and determines what course of action to take. In some cases, the contractor may be required to select an alternate, longer life solution. If that is not possible, the DMT will resolve the issue.

5.4 Detecting, Tracking, and Resolving DMSMS Issues

5.4.1 Obtaining BOMs

The UWDP has instituted contractual requirements with FishFinder, Inc. to deliver BOMs for all their deliverables. The BOMs will be delivered once the designs are considered mature and will be updated every six months. BOMs received are loaded into OMNICRON and evaluated for current and forecasted

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DMSMS issues as well as other DMSMS factors. Any issues found are reported to the design engineers for remedy. ECPs are routed to the DMT once they are finalized.

5.4.2 Prioritizing BOM Loads

Given the finite resources of the DMT, it is possible that the order of loading BOMs into OMNICRON will need to be prioritized. The DMT will utilize a methodology similar to that described in SD-22 using the following list, in order of precedence, to set the priority of each BOM:

- BOMs associated with safe operations
- Mission essentiality
- Complexity of the BOM.

5.4.3 Prioritizing Item Monitoring

An assessment of the risks associated with the types of items used in the BSS-001 system was conducted in May 2024. The types of items were categorized and each category of items was assigned a risk weight based on the likelihood of obsolescence within the anticipated life of the system and the difficulty and cost of resolving the issue. The assessment resulted in a list of item types that will be monitored (proactive) and others that will not (reactive). This list is stored on UWDP's information portal and is used by NSWC Biloxi to determine which items to load and monitor in OMNICRON.

5.4.4 BOM Monitoring and DMSMS Issue Notification

BOMs loaded in OMNICRON are monitored in two fashions: electronically using commercial sources of information and manually using direct email or telephone contact. The former monitoring is nearly real time in frequency; the latter is conducted in six-month cycles. DMSMS notifications resulting from this monitoring are validated and then relayed to the DMT for initial impact evaluation. The prime contractor is contractually obligated to inform the DMT of DMSMS notifications within 10 days of their discovery.

5.5 Case Initiation and Management

Upon determination that a DMSMS issue will have a negative impact on the BSS-001 system, a case is created in OMNICRON. Case management follows the basic process described in the Assess section of SD-22. Cases are stored and managed in OMNICRON. Cases are reviewed during DMT meetings and acted on by members of the DMT and their respective teams and organizations. Cases can have three statuses: Open, Resolved, and Closed. Resolved cases are open cases that are awaiting implementation of the resolution. Closed cases have been fully implemented including documentation. When there is the possibility that the resolution of a case may need to be reevaluated (e.g., a Life of Need buy may come up short), the case is placed on a watch list in OMNICRON. Watch lists have automatic triggers based on things like stock levels and elapsed time. Case action items are tracked in OMNICRON.

Cases are prioritized based on criteria to ensure the DMT is focused on the most important cases. Prioritization attempts to quantify the effects a DMSMS issue may have on cost, schedule, availability, and readiness.

Factors considered include the following:

- The number of manufacturers
- Manufacturing difficulty
- Lead time
- Cost
- Potential impact to production or sustainment
- Impacts on software.

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5.6 Foreign Military Sales

UWDS is developing additional units under LOAs in support of two FMS customers, Liechtenstein and Transnistria. The DMT previously negotiated the following language in LOAs with both Liechtenstein and Transnistria:

- The need for DMSMS management because of the inevitability of obsolescence;
- An explicit statement of the DMSMS activities that the program office must perform; and
- An identification of the data that will be provided by the foreign partner that is needed by the program office to perform the work.

The DMSMS lead and, as necessary, DMT SMEs, will communicate directly with FMS customers concerning DMSMS issues that may affect system development, production, and sustainment. This communication will occur quarterly through the FMS liaison officer or designated program office FMS representative. DMSMS issues that are identified as having significant impacts will be relayed immediately to the lead systems engineer and program office manager with a recommendation to notify the FMS customer in a timely manner.

6. DMSMS Funding

6.1 DMSMS Management Operations Programming and Budgeting

DMSMS operational funding is comprised of several components: funds to maintain the UWDP DMT, funds for the prime contractor, and funds for the DMSMS SME Organization, NSWC Biloxi. Funding levels for operational funding are expected to be higher in FY28–30 and then diminish as the program moves into sustainment. Details on funding levels for the next five years can be found on line 13 of the Sustaining Engineering tab of the Logistics Requirements and Funding Summary (LRFS). The LRFS is located in a restricted section of the UWDP's information portal. Budget proposals are presented and reviewed at the annual program review meeting.

6.2 DMSMS Resolution Funding

The DMT uses case and DMSMS forecasting information from OMNICRON and from health assessment reports delivered by FishFinder, Inc. to develop budget forecasts for resolving DMSMS issues. This data contains projected resolution costs based on estimates to resolve both known and forecasted issues and estimates for when resolutions funding will be needed to avoid impacts on scheduled deliveries and on sustainment. DMSMS resolution funding will largely be consumed by the prime contractor to implement resolutions. A small amount will go to UWDP's logistics and engineering teams. Funding levels for resolutions are anticipated to increase over the five years as the equipment ages. Details on funding levels for the next five years can be found on line 11 of the Sustaining Engineering tab of the LRFS.

7. DMSMS Contractual Requirements

7.1 Contractual Requirements

The following DMSMS management requirements are found in the DMSMS section of the SOW:

- DMSMS Management Plan
- Participation in DMT
- Item monitoring activities
- Research and analysis of resolutions
- Resolution Funding
- Flowing down DMSMS management requirements to Subcontractors
- Monitoring, managing, and reporting subcontractor DMSMS capability
- Delivery of an as-built configuration list
- Delivery of BOMs for DMSMS monitoring

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- Delivery of a DMSMS issue mitigation plan
- Delivery of a DMSMS health analysis report
- Exit plan.

Details of these requirements can be found in the contract or by reviewing SD-26, which will also be used in developing future contracts. These requirements will be tailored to suit the current acquisition phase and the UWDP's acquisition strategy.

7.2 Contract DMSMS Management Approach

The UWDP's approach to DMSMS management includes both a government and a contractor component. FishFinder, Inc. will be conducting DMSMS management during production of the BSS-001. Information will be exchanged between the UWDP's DMT and the contractor to ensure all DMSMS issues are resolved to the satisfaction of both parties.

The following CDRLs are found in the current contract:

- 0052 – DMSMS Management Plan
- 0053 – DMSMS Subcontractor Health Report
- 0055 – As Built Configuration List
- 0061 – DMSMS Case Data
- 0062 – DMSMS Bill of Materials (BOM)
- 0063 – DMSMS Case Mitigation, Cost, and Budgeting Data
- 0067 – DMSMS Health Assessment Report
- 0072 – DMSMS Operations Transfer Plan

8. DMSMS Metrics, Reporting, and Quality

8.1 Data Collection

The DMT will collect data as described in DoDM 4245.15. This data will be used for reporting both internally and to higher-level organizations such as PEO Undersea Sensors and the Navy's DMSMS Lead in the Deputy Assistant Secretary of the Navy for Logistics (DASN(L)) office. Most of the records will be obtained from OMNICRON. A compilation of the records and the reports generated are stored in the DMSMS section of the UWDP's information portal.

8.2 Obtain, Store, and Access the Data for Required Level 1 DMSMS Management Record Keeping Data Elements and Optional Level 2 Data Elements

The DMT will collect data applicable to the scope of the DMSMS management effort. The data will be used for reporting both internally and to higher-level organizations as needed. The DMT will analyze the Level 1 and consider use of Level 2 data elements as DMSMS management operations mature and become more robust and coordinate this with the PM. Detailed information, to include the data dictionary for the Level 1 and 2 data elements, are found in DoDM 4245.15, Tables 1 and 2, and Appendix H.4 of SD-22. Most of the data items will be obtained from OMNICRON and through CDRLs.

The DMT will use results of the analysis conducted for the Level 1 data elements and develop metrics that are designed to account for areas in DMSMS management operations, areas responsible for excessive DMSMS-related costs, and any DMSMS issues impacting schedule and system readiness.

8.3 DMSMS Deliverables

Table 4–DMSMS Reports details reports regularly generated by the DMT, who receives the reports, and their frequency. The content of these reports is based on Appendix I of SD-22.

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Table 30–DMSMS Reports

Report Name	Recipient	Frequency
Opened, Resolved, and Closed Cases	DMT	Monthly
Cumulative Cost Avoidance	DMT, UWDP Leadership	Monthly
Navy DMSMS Cost Metrics	DMT, UWDP Leadership, DASN (L)	Annually
DMSMS Resolution Cost	PEO Undersea Sensors, DASN (L)	Annually
DMSMS Mgt. Operations Cost	PEO Undersea Sensors, DASN (L)	Annually
DMSMS Case Processing Time	DMT	Monthly
Monitoring Effectiveness	DMT	Monthly
LON Buy Efficiency	DMT	Monthly

9. Process Capability

9.1 Desired Process Capability

The UWDP and DMT have conducted an assessment and established the target program capability levels for each DMSMS management step and operational process and has used the results of this assessment to inform the content of this DMP. The assessment identified the current program capability level, the target program capability level, the steps required to achieve the target level and the risk of not achieving the target level. The UWDP and the DMT have documented the results of their assessment as an appendix to the DMP. The DMT will provide UWDP a monthly progress status update and immediately notify UWDP should there be a significant change that warrants immediate action.

9.2 DMT Process Improvement

The UWDP is dedicated to process improvement in all areas. *Table 5–Process Improvement Reports and Usage* details reports the DMT uses for process improvement.

Table 31–Process Improvement Reports and Usage

Report Name	Purpose
DMSMS Resolution Cost	Used by PEO and Navy to compare the cost of similar resolutions across programs
Monitoring Effectiveness	Used by the DMT to identify areas for improvement in the monitoring process
LON Buy Efficiency	Used by the DMT to determine if the LON buy process is working effectively

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Appendix A – Acronyms

Define all the acronyms used within the DMP in this appendix.

Abbreviation	Definition

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Appendix F. Contracting

DMSMS management can be performed 1) by a program office using government personnel (and contractor technical support), 2) by the prime contractor and its subcontractors,[227] or 3) by independent DMSMS management SME organizations.[228] The entity managing DMSMS issues for a program office has no bearing on the robustness of that effort.

Regardless of who performs DMSMS management, the government remains responsible for ensuring that DMSMS risk is managed effectively. The government must remain in a position to carefully oversee how the prime contractor and its subcontractors do for DMSMS management, including identifying and resolving current and near-term obsolescence issues. Simply receiving an item obsolescence report at a design review is not sufficient oversight; the government should have a thorough understanding of and maintain visibility into the DMSMS management processes being used by the prime contractor and its subcontractors.

Effective management is established by the DMP, which emphasizes how a robust DMSMS management effort will reduce future obsolescence-related costs and minimize detrimental impacts on materiel readiness, operational mission capability, and safety of personnel. The DMP identifies all the program office's DMSMS management planning objectives, the approach to be pursued, and the entities that will perform the functions necessary to pursue the approach (see Section 3.2). If contractors perform the work, the DMP should describe the nature and extent of government oversight.

The DMSMS management responsibilities of contractors can vary greatly. Under certain circumstances, nearly all DMSMS management activities (other than program office oversight) will be performed by contractors. This is often the case when dealing with complex systems prior to sustainment. In other situations, the government may perform most DMSMS management functions. This generally applies to commercial items or to systems maintained organically. Contract language must be tailored to these two extremes and everything in between.

When a program office decides to use a prime contractor (and its subcontractors) or an independent SME organization for DMSMS management, contracts containing DMSMS management–related requirements (including government access to sufficient DMSMS management–related information), combined with government oversight, provide the basis for ensuring that DMSMS management is effective. This appendix provides a broad overview of the factors a program office should consider incorporating into contracts with a prime contractor (and its subcontractors) or an independent contractor provider of DMSMS management. It also discusses considerations for deciding whether to contract for DMSMS management.

The SD-26 provides much more detailed guidance. Its purpose is to assist DoD program offices in preparing DMSMS management provisions in contracts by providing representative contract language for various aspects of DMSMS management. The tables in the SD-26 contain illustrative contract language, and related information keyed to specific DMSMS management tasks, for use in preparing a consolidated performance work statement or SOW for contract DMSMS management activities. The following is a list

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[227] The OEM should be responsible for flowing down DMSMS management–related contractual requirements to its suppliers and to require those suppliers to flow down DMSMS management–related requirements through their supply chains in a similar way. This appendix assumes that is the case whenever the OEM is discussed as a potential provider of DMSMS management functions.

[228] An independent SME organization could be part of a government DMSMS management SME team, as discussed in Section 3.2.1.

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of the tables included in Appendix A of the SD-26 which suggests ways to fill out the CDRLs for different scenarios:

- Table 1. Illustrative Contract Language for Areas of Potential DMSMS Requirements;
- Table 2. Clause Applicability during Each Acquisition Phase;
- Table 3. Clause Applicability Based on Government Involvement in DMSMS Management;
- Table 4. Notional Decision Drivers for Table 3;
- Table 5. DMSMS Topics for Non-DMSMS Contract Sections;
- Table 6. DMSMS CDRLs and DIDs; and
- Table 7. Non-DMSMS CDRLs and DIDs that could or should exist in the contract.

The SD-26 can be used when relevant or tailored for the specific approach to DMSMS management the program office has taken. The wording should be tailored to the specific program office to blend with its AS, product support concepts, competition strategy, technical data strategy, and IP strategy (all likely contained in the program office's AS).

It is useful to read this appendix before reading the SD-26.

F.1 Determining Whether to Contract for DMSMS Management

F.1.1 General Factors to Consider

Early in a system's life cycle, a decision should be made regarding who will have primary responsibility to perform DMSMS management in support of the program office. This section includes factors a program office should consider in determining whether to keep DMSMS management solely within the government or to contract with a prime contractor (and its subcontractors) or an independent SME organization.

F.1.1.1 Availability of Resources

Using a contractor (whether a prime contractor and its subcontractors or an independent SME organization) for DMSMS management requires funding. In some instances, the program office may already have funding available associated with a design, production, or sustainment contract. In those cases, such contracts and funding could be used to support a prime contractor and its subcontractors as the DMSMS management provider for the program office. Using an independent SME organization for DMSMS management normally requires a separate contract. If funding available for such contracts is limited, internal program office staff should be used to provide DMSMS management functions.

F.1.1.2 DMSMS Management Capability

The capability of the organizations being considered to perform DMSMS management should be a key consideration. Two basic capability questions apply to all organizations:

- What kind of capability and access does the organization have regarding the receipt of PDNs or the ability to manually research items?
- Have other program offices used this organization? How have these other program offices evaluated the organization's performance?

From a prime contractor's perspective, if there is a robust DMSMS management capability in place, then the program office might want to leverage that existing capability, rather than establishing a duplicate government-run DMSMS management program. In this case, the government might want the prime contractor to manage and mitigate DMSMS issues for the program office, while the program office establishes a surveillance strategy for the government to oversee the prime contractor's DMSMS

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management efforts. However, if a prime contractor's internal DMSMS management program does not appear to be sufficiently effective, then the government may need to establish its other more robust DMSMS management program before a contract requirement is created.

If a program office is considering the use of an independent SME organization, it should take into account the additional capabilities such an independent SME organization might offer, as compared to other DMSMS management providers. Below are some questions that a program office should consider:

- Is the independent SME organization offering DMSMS management capability that is not available through the prime contractor or the program office itself? If so, what additional capability is being offered?
- Is an independent SME organization needed if the prime contractor and its subcontractors have an existing, robust DMSMS management capability?

F.1.1.3 Life Cycle Phase

During the design and production phases of a system's life cycle, the program office already has a prime contractor under contract to develop and manufacture the system. DMSMS management and mitigation should inherently be a part of those efforts; therefore, the prime contractor and its subcontractors are in a natural position to perform these activities. In such a case, the government should ensure that a contractual requirement exists for the prime contractor to develop and deliver a DMP establishing its DMSMS management objectives and approach for the program office in alignment with the DMP for the program office. There is, however, the potential for conflicting forces to be at work, if DMSMS management responsibility is assigned to the prime contractor. The business objectives of any for-profit company include lowering costs and increasing revenue, whereas implementing DMSMS management practices requires expending time and resources. The contractor's leadership must understand that DMSMS management is a good business practice, because it makes products more attractive to buyers through the reduction of total ownership cost. Consequently, the contractor's products will have a competitive advantage.

Effective DMSMS management in the design phase is critical. If obsolescence is "designed in," then the program office will face large costs to mitigate these problems later in the life cycle. The prime contractor and its subcontractors, operating under comprehensive government oversight, are in a good position to manage DMSMS issues during design, because of their familiarity with the current configuration of the design (including the potential for rapidly changing items).

Although the system's parts lists and BOMs should be stable by the time the system has entered the production phase, the government will still have limited experience with the system. Consequently, the prime contractor may continue to play a key role in DMSMS management. One of the areas that the government should include in its oversight is laying the appropriate groundwork to transition DMSMS management to the entity most appropriate to perform that role during sustainment.

Typically, the government uses a combination of three different sustainment strategies:

- In-house support through a service depot or supply system or through DLA,
- Non-PBL contractor support services, and
- PBL contracts.

In the first case, the government will typically not use the prime contractor and its subcontractors for DMSMS management. The government will either use its own in-house capability or combine its in-house capability with an independent SME organization. The latter two cases do not preclude the use of organic service providers through public-private partnerships, especially for PBL contracts where such partnerships are commonly formed.

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The use of non-PBL contract support depends on the scope of work. For example, a contractor operating a repair depot on either a cost-reimbursable or fixed-price basis could be asked to perform DMSMS management. If the work is more limited (e.g., interim or emergency support), then DMSMS management could be out of scope for that contract.

PBL is a sustainment strategy that places primary emphasis on optimizing system support to meet the needs of the warfighter. PBL specifies outcome performance goals of systems, ensures that responsibilities are assigned, provides incentives for attaining those goals, and facilitates the overall life cycle management of system reliability, supportability, and total ownership costs. It is an integrated acquisition and logistics process for buying system capability. Generally, PBL contracts are long term (five–ten years) and require that the provider manage many aspects of product support throughout the life cycle. A properly structured PBL contract contains DMSMS management requirements.

In a theoretical sense, PBL incentivizes the provider to maintain a proactive DMSMS management program to achieve the required performance outcomes. This is not always true in practice. The contractor will take the most cost-effective approach to meeting its performance requirements within the terms and conditions of its contract. This approach may not be the most cost effective for the government when the contract is completed, which is why DMSMS management should be specifically called out in the PBL arrangement. The government should ensure that its DMSMS management interests are adequately covered in the contract, especially at and near the end of the period of performance.

F.1.1.4 Conflicts of Interest

Whenever the government contracts for services, it must determine whether any potential conflicts of interest exist and then manage those situations effectively. For example, there could be a situation in which a non-government DMSMS management provider has a business interest (e.g., potential additional revenue) regarding a specific resolution option, as compared to other options. This situation does not necessarily preclude the use of that DMSMS management provider; however, it does place an additional burden on the government to appropriately oversee and understand the potential repercussions of decisions and to factor them into the program office's decision-making process.

F.2 An Overview of DMSMS Activities in Contracts

When contracting for DMSMS management, a program office should develop a contract that clearly conveys DMSMS management requirements. (The program office also should state, in its RFP and other communications, that DMSMS management–related criteria will be used in source selection). This ensures that the contractor knows its specific responsibilities for DMSMS management and that the government has access to the information it needs for adequate oversight.

F.2.1 Representative Management Activities by Acquisition Phase

The following is a list of representative DMSMS management activities—by acquisition phase—that a program office should consider when developing contracts to cover DMSMS management:

- ***Design and development***
  - The prime contractor and its subcontractors should minimize obsolescence throughout the contract period of performance by selecting products that will avoid or resolve hardware, software, and firmware obsolescence issues. This may be pursued through various DMSMS resilience design considerations, such as selecting technologies or items that are not near their EOL, PM&P management, open systems design, and so on, as described in Section 3.4.2.
  - The prime contractor should select parts whose forecasted availability (predicted years to EOL) is five years or greater to the maximum practical extent for new designs, design

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changes, modifications, or configuration changes that propose new parts. The contractor shall provide an exception report containing a list of all parts where the five-year threshold is not met and the reasons why the part was selected. The list of parts not meeting the five-year threshold must be analyzed to understand why it was selected, including any mitigation actions or anticipated upgrades. The prime contractor and its subcontractors should determine the most cost-effective resolution to obsolescence issues. For the purposes of the contract, hardware, software, and firmware should be considered obsolete when the item can no longer be procured from the OCM as identified in the current TDP.

- The prime contractor and its subcontractors should flow down DMSMS management requirements to their suppliers, who should flow down requirements in a similar manner.
- The prime contractor should deliver a parts list and indentured BOM (or notional versions, if that is all that is available at the time), IAW DI-MGMT-82274, to the program office at agreed-upon points in the technical schedule.
- The prime contractor and its subcontractors (and possibly an independent third-party contractor, if one is to be used) should monitor the availability of items (with agreed-upon frequency of update) and provide the results to the program office. The government should be notified of pending and emergent obsolescence issues, supplier recall notices, and emergent vendor-implemented changes.
- The prime contractor and its subcontractors should resolve and document any DMSMS issues before the delivery of the design. (Supplemental funding for the contractor may be necessary.)
- The prime contractor should deliver a production plan for LRIP.
- The prime contractor should deliver a supportability roadmap (with agreed-upon frequency of updates).
- The prime contractor should deliver a description of how the system is envisioned to be supported after fielding, including the process for assigning the source of repair.
- The prime contractor and its subcontractors (and possibly an independent third-party contractor, if one is to be used) should participate in the government-contractor DMT (with frequency of face-to-face and telephone communications specified).

- ***Production***
  - The prime contractor and its subcontractors should minimize obsolescence throughout the contract period of performance by selecting suppliers that will avoid or resolve hardware, software, and firmware obsolescence issues.
  - The prime contractor and its subcontractors should determine the most cost-effective resolution to obsolescence issues. For the purposes of the contract, hardware, software, and firmware should be considered obsolete when the item can no longer be procured from the OCM as identified in the current TDP.
  - The prime contractor and its subcontractors should flow down DMSMS management requirements to their suppliers, who should flow down requirements in a similar manner.
  - The prime contractor should deliver a parts list and indentured BOM, in accordance with DI-MGMT-82274, to the program office if it has not already been delivered.
  - The prime contractor and its subcontractors (and possibly an independent third-party contractor, if one is to be used) should monitor the availability of items (with agreed-upon frequency of update) and provide the results to the program office. The government should be notified of pending and emergent obsolescence issues, supplier recall notices, and emergent vendor-implemented changes.
  - The prime contractor and its subcontractors should solve and document any DMSMS issues before delivery of the system for fielding.

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- The prime contractor and its subcontractors (and possibly an independent third-party contractor, if one is to be used) should participate in the government-contractor DMT (with frequency of face-to-face and telephone communications specified).
- The prime contractor and its subcontractors should develop and execute a plan to transition DMSMS management to the sustainment provider.

- ***End of production***
  - For systems that are required for many decades of continuous service, an in-depth DMSMS study should be performed before the contractual production line schedule ends because when production ends, obsolescence and item unavailability often increase dramatically. The study should determine potential future "end item" readiness impacts with and without the suppliers' production line support.
  - A BCA for extending production lines at a reduced output rate at some level of assembly should be made to evaluate options for reducing future resolution costs.
- ***Sustainment***
  - The sustainment provider should minimize obsolescence throughout the contract period of performance by selecting suppliers that will avoid or resolve hardware, software, and firmware DMSMS issues.
  - The sustainment provider, especially in the PBL case, should determine the most cost-effective resolution to DMSMS issues. For the purposes of the contract, hardware, software, and firmware should be considered obsolete when the item can no longer be procured from the OCM as identified in the current TDP.
  - The sustainment provider, especially in the PBL case, should flow down DMSMS management requirements to suppliers, who should flow down requirements in a similar fashion.
  - The sustainment provider (and possibly an independent third-party contractor if one is to be used) should monitor the availability of items (with agreed-upon frequency of update) and provide the results to the program office. The government should be notified of pending and emergent DMSMS issues, supplier recall notices, and emergent vendor-implemented changes.
  - The sustainment provider (and possibly an independent third-party contractor if one is to be used) should participate in the government-contractor DMT (with frequency of face-to-face and telephone communication specified).
  - The sustainment provider should recommend DMSMS resolutions.
  - The sustainment provider should work with the program office in areas such as asset management, preservation of tooling and support equipment, and reclamation or use of retired assets as requested.

F.2.2 Exit Clauses

Exit clauses do not fit neatly into DMSMS management in contracts by acquisition phase or by function. Nevertheless, exit clauses for DMSMS management are a critical element in any contract, including PBL contracts. The primary purpose of this type of clause is to mitigate the risk of DMSMS issues at the end of the contract period of performance. Exit clauses require the contractor to ensure all known and forecasted DMSMS issues have been identified and have mitigation plans, so that the program office is not left with a system that is not supportable or sustainable due to DMSMS issues. By requiring the contractor to deliver all accumulated (non-proprietary) DMSMS to the program office in a specified format, exit clauses ensure that the information needed to manage DMSMS issues is provided. They establish procedures and timeframes to ensure the orderly and efficient transfer of performance responsibility upon completion or termination of the contract. The exit clauses should require delivery of those items, identified in the SOW or statement of objectives, within the negotiated contract price.

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Exit clauses are necessary but not sufficient to guarantee the transition of DMSMS management responsibilities from one provider to another. As part of its contractor oversight, the program office should develop an understanding of all DMSMS management activities being performed by the contractor. In that way, the program office will be in the best position to ensure that effective DMSMS management continues throughout a transition.

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Appendix G. Developing DMSMS Management Workforce Competencies

This appendix outlines the recommended training required to achieve informal foundational-level, advanced-level, and leadership-level competencies and experience associated with the roles and responsibilities of DMSMS management practitioners.

Foundational training provides an individual with basic knowledge of the processes and procedures required to establish and maintain a robust DMSMS management program. An individual with foundational competency is not expected to be proficient in analyzing DMSMS issues or in leading a DMSMS management program. An individual with foundational competency should perform DMSMS analysis only in conjunction with an individual possessing advanced-level or leadership-level competency. An individual with foundational competency should review all data before they are submitted for approval. An individual with foundational competency should assist with DMSMS management functions under the supervision of an individual with leadership-level competency.

An individual with advanced-level competency is capable of leading, conducting, explaining, and defending the results of any analyses that he or she has led. DMSMS analysts with advanced-level competency should submit analyses to a person with a DMSMS management leadership-level competency for approval. A DMSMS analyst with an advanced-level competency should be capable of establishing and maintaining a robust DMSMS management program with minimal oversight.

An individual with leadership-level competency is well versed, trained, and experienced in DMSMS analyses, applications, and management practices. This is the desired level for DMT leaders. A leadership-level analyst will have developed and led numerous DMSMS efforts and must be conversant in all aspects of DMSMS management processes and policy. The leadership-level analyst is ultimately responsible for planning the overall DMSMS management effort for a program office.

DMSMS management competency is not developed in a vacuum. It must be obtained in conjunction with DAU Defense Acquisition Workforce Improvement Act (DAWIA) certifications for government employees and a company analog for industry.[229]

To achieve DMSMS management foundational-level competency, an individual should have the following DAU training beyond DAWIA lower tier certification in any career field:

- **DMSMS: What The PM Needs to Do and Why (LOG 0640**)
- **DMSMS Fundamentals (LOG 0650)**
- **DMSMS Executive Overview (LOG 0660)**
- **Introduction to Parts Management (LOG 0630)**
- **DMSMS Basic Component Research (LOG 0670)**
- Defense Logistics Agency Support to the PM (LOG 0020)
- **Market Research (CON 0040)**
- Market Research for Engineering and Technical Personnel (CLE 028)
- COTS Acquisition for Program Managers (CLM 025)

[229] The DAWIA was initially enacted as Public Law 101-510 in November 1990. Most of the act is codified in 10 U.S.C. § 1701–1765, available at https://www.govinfo.gov/link/uscode/10/1701.

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- **Preventing Counterfeit Electronic Parts from Entering DoD Supply System (LOG 0320)**
- Counterfeit Prevention Awareness (LOG 0620)
- Fundamentals of Systems Acquisition Management (ACQ 1010)
- Fundamentals of Test and Evaluation (TST 102)
- Intro to Systems Engineering for Testers (TST 1100).

To achieve DMSMS management advanced-level competency, an individual should have the following DAU training beyond DAWIA upper tier certification in any career field:

- All lower tier certification requirements
- Product Support Business Case Analysis (LOG 0150)
- Planning, Programming, Budgeting, and Execution (BFM 0050)
- Independent Logistics Assessments (LOG 0160)
- Reliability and Maintainability (CLE 301)
- Contracting for the Rest of Us (CLC 011)
- **Provisioning and Cataloging (LOG 0380)**
- Supportability Analysis (LOG 0120 or LOG 211V)
- Introduction to Data Management (CLM 071)
- Improved Statement of Work (CLM 031)
- Technical Reviews (CLE 003)
- **Modular Open Systems Approach (CLE 019)**
- Risk Management (CLM 093, PMT 0170, ISA 220, BCE 0420)
- **Sustaining Engineering (LOG 0470)**
- **System Retirement, Disposition, Reclamation, Demilitarization, & Disposal (LOG 0510)**
- Applied Systems Engineering for Testers (TST 2100)
- Introduction to DoD Software Lifecycle Management (LOG 0270)
- **Additive Manufacturing Overview (LOG 0390)**
- **Additive Manufacturing Case Studies (LOG 0400)**
- Performance-Based Logistics (LOG 2350)
- Lead Free Electronics Impact on DoD Programs (LOG 0070).

To achieve DMSMS management leadership-level competency, an individual should have the following DAU training beyond DAWIA Life Cycle Logistics upper tier certification:

- All upper tier certification requirements
- Intellectual Property and Data Rights (CLE 068)
- Program Protection Planning Awareness (ACQ 160)
- Technical Data Management (LOG 2150)
- **Trade Studies (CLE 026)**
- Forecasting Techniques (CLB 026)
- Configuration Management (LOG 2040)
- Reliability Centered Maintenance (RCM) (LOG 0300).

The DAU courses bolded above are included in requirements to obtain the Parts and Material Life Cycle Management credential.

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Appendix H. Programming and Budgeting for DMSMS Management Operations

As a best practice, program office leadership should program and budget for all DMSMS management operations for data collection and management, research, and forecasting using a risk-based approach and in consultation with the DMT and contractors supporting DMSMS management. That risk-based approach determines the subset of items to be proactively monitored and thereby drives the level of effort required. Four enabling best practices are discussed below.

H.1 Convince Program Office Leadership of the ROI

While this enabling best practice is not about programming and budgeting per se, it is a key enabler. In some instances, program office management does not recognize the importance of DMSMS management operations. Reasons for this include competing program office priorities, unfamiliarity, and a belief in DMSMS-related myths. The DAU Course LOG 0640 entitled "What Program Management Needs to Do and Why," debunks the following myths. All the following statements are *false*:

- ***Myth #1***. My system is not in sustainment yet, so obsolescence issues do not exist.
- ***Myth #2***. A design's use of COTS assemblies provides built-in obsolescence immunity.
- ***Myth #3***. DMSMS is just another drain on a program office's budget.
- ***Myth #4***. My system has a performance based AS, so the prime handles obsolescence.
- ***Myth #5***. PBL forces the vendor to address obsolescence efficiently.
- ***Myth #6***. My system has hired independent DMSMS SMEs, so they handle the resolutions.

Despite these myths, many PMs and PSMs recognize proactive DMSMS management operations as a priority and understand that it will cost a great deal more to deal with problems reactively. However, program office management activities are extremely demanding and time consuming. During design, development, and production, their focus is on meeting cost, schedule, and performance goals. In sustainment, program office management is heavily concerned with operational activities and readiness. Unfortunately, even when the importance is recognized, insufficient priority may be given to DMSMS management operations funding and the program office is forced to be reactive.

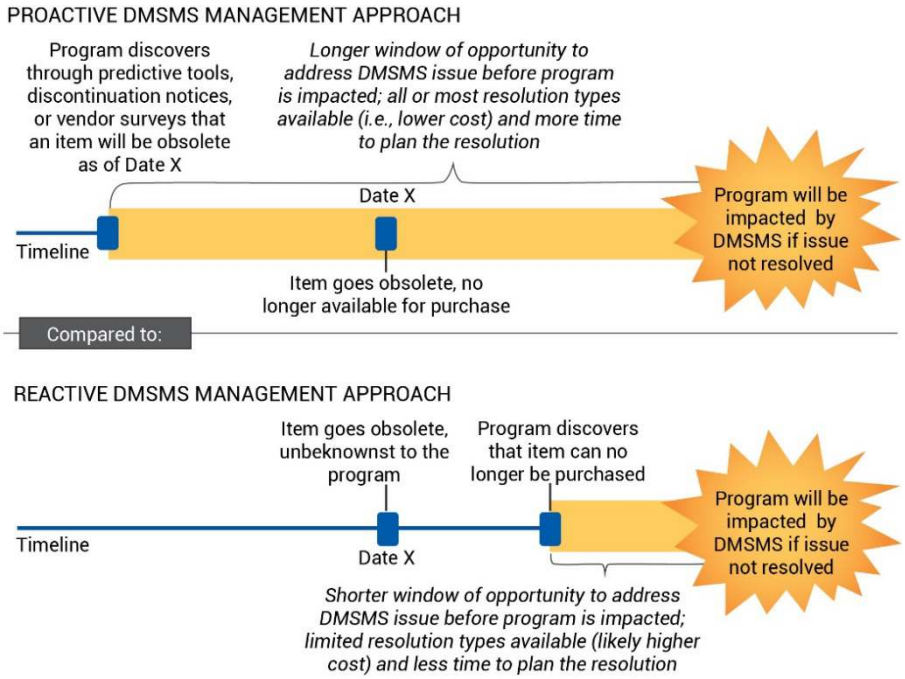
The ultimate reason for giving priority to DMSMS management is that it extends the window of opportunity for resolving DMSMS issues. Figure 27, reproduced here for convenience from Section 3.4.3.2.3, depicts the way in which proactive DMSMS management increases the window of opportunity for resolving a DMSMS issue before it impacts a DoD system. A longer window of opportunity improves key aspects of ROI—increasing the likelihood of implementing a lower cost resolution and decreasing the likelihood of the affected system suffering schedule delays or lower operational availability.

The window of opportunity, highlighted in yellow in Figure 27, can be used as an argument to convince program office management of the ROI for fully funding DMSMS management operations. Tangible benefits of DMSMS management activities can be articulated quantitatively. A program office may be able to demonstrate conclusively how a DMSMS issue identified because of proactive DMSMS management (before a failed attempt to procure and before a manufacturer's discontinuation notice would have been received) avoided a more costly resolution or prevented a schedule slip. Such a demonstration would require proof that the resolution selected proactively would not have been feasible at a later point in time as well as an estimate of what would have been done had action been delayed. For example, the

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following may be used to convince program office management of the need to assign sufficient priority to DMSMS management operations funding.

Figure 27. How Proactive DMSMS Management
Increases the Window of Opportunity for Resolving a DMSMS Issue



- ***Show specific examples of significant cost impacts by being reactive***.

  It is best to use actual experience from the specific program office. If none are available, a DMT should seek examples from larger service working groups. LOG 0640 contains the following example:

  - The OEM for a program office identified component obsolescence issues pertaining to three boards within a box and a module of a system. The parts with obsolescence issues were critical to the system. The obsolescence had not been addressed previously because more immediate program office interests and priorities, focused on operational test and evaluation, took priority. When the system was in the operational test and evaluation phase and ready to begin LRIP, the previously identified obsolescence issues were projected to impact production within three years. The OEM proposed a resolution with a price of $11.6 million. Working with the OEM, the program office was able to isolate the immediate obsolescence issues to five components. For four of the components a complex substitute was possible, but the F3 components still required testing and approval. The fifth component required a more extensive redesign. This resulted in a revised OEM proposal of $6.6 million to resolve the obsolescence associated with the five components. If the program office had acted to resolve obsolescence when it was initially identified, LON buys for the five components covering the quantities necessary for six future lots, could have been purchased for $518,000, avoiding more than $6 million in costs.

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- ***Show specific examples of readiness or schedule impacts associated with reactivity***.

  As stated under cost impacts, where possible, use examples of significant schedule and readiness impacts associated with the specific program office or another program office in the service. LOG 0640 contains the following examples:

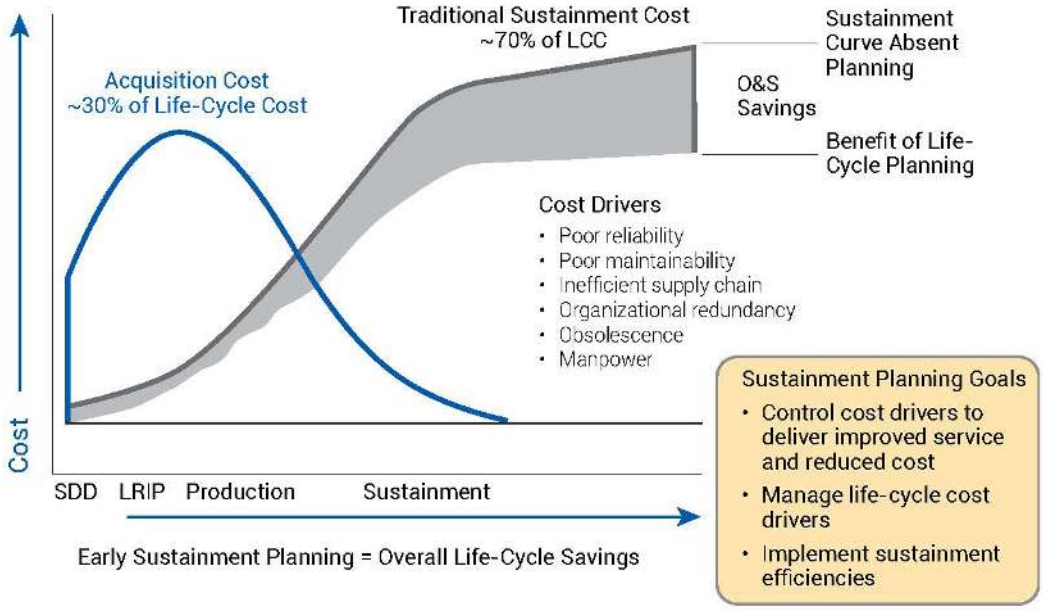
  - *Schedule impact*. A system was about to enter another manufacturing cycle to support the next phase of a multi-year production contract. When the OEM went to contract with a subcontractor for the next lot of media convertors for that system, it was determined that the media converters couldn't be manufactured due to obsolescence. Because there was no F3 alternate, the program office was forced to redesign and transition to a new configuration. The new configuration required requalification and was replaced by attrition. The implementation plan required the program office to support two design configurations during the replacement and sustainment period. This redesign effort stalled production for three to four months, which delayed the overall schedule for the program office a corresponding amount. The entire obsolescence resolution effort cost $800,000 to $1 million.
  - *Readiness impact*. During the production contract for one air platform, an obsolescence issue was identified pertaining to a display LRU. The program office decided to not allow this obsolete LRU to shut down the entire production line. Instead, production of the air platform continued and these platforms were delivered to the field with the plan to install the LRU once available. Unfortunately, the absence of this LRU made the air platform mission incapable and the result, at least for a time, was the non-availability of MICAP systems to support training and unit activations to theater.

- ***Explain how proactive DMSMS management operations can lower the life cycle cost***.

  Figure 28 notionally illustrates that the bulk of life cycle cost is incurred during sustainment. Robust, proactive DMSMS management operations may provide a basis for lowering that life cycle cost curve, also as depicted in the figure as the "benefit of life cycle planning." There is, of course, the complication that the current PM may believe that any effect on life cycle cost will be far in the future and therefore choose to spend current resources on current problems and not invest in DMSMS management operations. Such a belief is erroneous as illustrated by the first two examples; the impact could be very short term.

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Figure 28. Notional Depiction of How Including DMSMS Management Operations in Life Cycle Planning May Lower Sustainment Costs



Source: Hot Topics in DoD Logistics, Briefing by Kristin K. French, Acting Assistant Secretary of Defense for Logistics and Materiel Readiness to the Product Support Managers Workshop, June 6, 2017.

Note: LCC = life cycle cost; SDD = system development and demonstration.

- Explain how proactive DMSMS management operations are key to accurate programming and budgeting not only for DMSMS resolutions but also for modifications.

  Robust DMSMS management operations benefit program offices beyond providing advance notice of obsolescence. One such benefit is the development of DMSMS health assessments. These health assessments estimate when some assembly or unit will no longer be supportable because of either current or anticipated obsolescence, taking both stock on hand and expected demand into account.[230] These estimates inform programming and budgeting not only for DMSMS resolutions but also for scheduling technology refreshments, capability improvements, and/or other modifications. Without this information, programmed and budgeted amounts may be insufficient and out-of-cycle modifications (redesigns) may become necessary to resolve obsolescence issues for the system to remain operational between funded modifications (especially during sustainment).

All these examples (along with ones from the program office itself) should at least be sufficient for the PM to allocate some start-up funding to DMSMS management operations. The result of doing this should almost always clearly demonstrate its value and establish a desire to continue (and increase where needed) funding.

H.2 Evaluate the Funding Needed

Another enabling best practice is to develop a basis for evaluating the level of funding needed for proactive DMSMS management operations. There are situations where, as a result of a service-wide centralized contract, program offices may obtain independent SME support at no cost. When this occurs, it of course is not necessary to POM/budget for that support.

[230] See Section 5.3.

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The cost of DMSMS management services provided by primes/OEMs/logistics support providers can vary widely. Factors contributing to this variation include contractor experience and in-house capabilities, type and level of services provided, and the ability to separate DMSMS management functions from other similar functions. For example, a single person may be performing PM&P management, DMSMS management, and engineering functions. It's a best practice to use independent SMEs to technically evaluate the DMSMS management operations functions and associated costs in proposals. In addition, these independent SMEs should also provide technical support to the actual contract negotiations.

Section 3.4.6.1 describes why external SME services should be performed by both the prime contractor/ OEM and independent SMEs. DMSMS management operations costs should be separately estimated for each of these entities because they are funded through different mechanisms. The prime contractor/OEM efforts are normally funded as part of the larger design/development/production/PBL/sustainment contracts.[231] Independent DMSMS management SMEs are normally funded via separate contract vehicles.

The DMT can assist program office leadership in developing its program and budget submissions for funding DMSMS management operations. Two approaches for making these estimates are as follows:

- ***Estimate based on specific activities***

  Section 3.4.1.1 listed the specific activities encompassed by DMSMS management operations. The cost of independent SMEs depends on the services required/provided, the number of items to be monitored, and whether the effort is in a start-up or steady-state mode among other factors. Funding needs could be developed through a determination of the person hours required to perform the work and then multiplying those hours by the appropriate rates. Program office management and engineering can help in making such estimates. They can provide insight on what will be monitored proactively, risk reduction activity, and resource constraints. Nevertheless, this is a complex task and program office personnel may not have sufficient experience to approximate necessary expenditures.

  If a program office is currently receiving independent SME support, then it should be in a good position to estimate future needs. If a program office is not receiving independent SME support, it is a best practice to begin with a person-year of effort (unless there is a reason to do otherwise).[232] After the first year, the program office will be in a better position to estimate future funding requirements based on the value of the support being provided, the functions to be performed, and the expected level of effort to perform them.

- ***Estimate based on rules of thumb***

  The following planning factors will help determine the cost of using external SMEs (i.e., primes, OEMs, and/or independent SMEs) to conduct DMSMS management operations. It is highly likely that initial person-hour estimates will not be accurate; consequently, all are subject to revision.

  - Number of parts lists/BOMs to be addressed by the DMSMS management effort (dividing these into critical, mission essential, and non-mission critical may provide a program office with the opportunity to phase in the assessment and analysis, and corresponding costs, of less critical ones over time, rather than all at once).
  - Whether the items to be addressed are primarily electronic or primarily COTS or MaSME.

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[231] Separating DMSMS management costs from DMSMS resolution costs may be difficult because these vehicles may also include the implementation of resolutions depending on the terms and conditions of the contract. Appendix F contains more information on contracting.

[232] For example, a mechanical system with a low risk of obsolescence may be able to start with less than a person-year. Highly complex systems may want to begin higher, but an evaluation must be made of whether it is feasible to ramp-up support quickly enough to usefully spend more than one person-year.

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- Number of person-hours required to perform the non-recurring startup tasks for each BOM.
- Number of person-hours required to perform recurring steady-state tasks to monitor each parts list/BOM.
- Number of person-hours required to perform recurring steady-state tasks to assess and analyze each parts list/BOM.
- Number of person-hours required to perform strategic processes such as technology roadmapping, design resilience reviews, and long-term health assessments.
- Number of person-hours required to generate reports.
- Person-hour costs for participating in meetings.

When estimating external SME person-hours, the level of service is a very important consideration. The DMSMS management program initiation, data collection and management, research and forecasting, and data analysis and oversight activities listed above represent full service. While it is not a best practice, some program offices may not have funding for all aspects of full service, especially those associated with data analysis and oversight.

With this information, program offices can estimate the funding required to support DMSMS management operations to provide input to formulating the program office's programming and budgeting submissions. A spreadsheet and model originally developed by the Naval Air Systems Command (NAVAIR) can be used to help estimate certain DMSMS management costs for electronic units (e.g., LRUs or WRAs). The planning factors were developed over five years and were based on averages so they are not tailored to the level of service or the degree of automation. Consequently, for example, recurring costs (e.g., vendor surveys) not specifically identified are embedded into the planning factors. Table 32, excerpted from the model itself, shows the planning factors incorporated. The top half of the table shows startup and other non-recurring activities, while the bottom half shows only recurring tasks.

Table 32. Planning Factors for Estimating DMSMS Operations Cost for Electronic Boxes

Startup and Other Non-Recurring Activities	One Time Hrs.	Total Hrs./BOM	Tech	Engineer	Database Manager	PM
Obtain BOM		0				
Format BOM for Loading into DMSMS		22	22			
Quality Assurance BOM		4	2	2		
Load BOM into the DMSMS Predictive Tool		1	1			
Clean Gray Components		40	32	8		
Additional Research and AoA, as required		24	12	10		2
Established Database and Tracking System	40				40	
Total Hours	40	91	69	20	40	2
Recurring Activities		**Total Hrs./BOM**	**Tech**	**Engineer**	**Database Manager**	**PM**
Monitor BOM		2	2			
Evaluate effect of alerts on BOMs		3	2	1		
Additional Research and AoA, as required		3	1	2		
Cost Analysis		3	1	1	1	
Update BOM Files		3	2		1	
Assemble Health Data		9	6		3	
Assemble Trend and Reliability Data		12	9	3		
Perform Health Evaluation		8	2	6		

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Startup and Other Non-Recurring Activities	One Time Hrs.	Total Hrs./BOM	Tech	Engineer	Database Manager	PM
Perform Trend and Reliability Evaluation		15	3	9		3
Total Hours		58	28	22	5	3

The NAVAIR spreadsheet is designed to estimate person-hours, not costs; the user is required to input an average cost per person-hour. Assumptions on how BOMs are phased in over time are built in. Person-hours for status reporting, meetings, training, technology insertion planning, and travel expenses are also required inputs.

The spreadsheet shown in Table 32 shows a 40.5% ratio between start-up and steady state associated with a BOM for an electronics box.[233] The level of service represented is somewhere between minimum and full.

Table 33 portrays some other possible person-hour ratios associated with level of service, start-up versus steady state, and COTS versus electronics BOMs.[234] The data used to build Table 33 were based on approximately 1,000 items being monitored. The person-hour requirement for a heavily electronics–oriented BOM for a minimum service level in the steady state has been normalized to 1. Program offices may find this information useful in developing and evaluating estimates of DMSMS operations costs.

Table 33. Person-Hour Ratios for Developing and Evaluating Estimates of DMSMS Operations Cost

	Start Year	Steady State
Electronics		
Minimum Service	1.3	1.0
Full Service	2.1	1.5
COTS		
Minimum Service	2.7	2.0
Full Service	3.7	3.0

H.3 Establish a Dedicated POM/Budget Line

All the enabling best practices in this section emphasize the importance of funding DMSMS management operations because of their contributions to lowering cost, maintaining schedule, and preventing readiness degradation due to DMSMS issues. Burying funding for these functions in a POM/budget line for another purpose, without clearly documenting the resources allocated to DMSMS management activities and the rationale for doing so, is *not* a best practice. Attempts to hide this funding should not be made. In fact, the opposite is true; highlighting the funding and relying on the DMT to oversee the activities informs program office leadership that attention is being paid to a serious management concern. This is especially important during sustainment when O&M funding may be difficult to obtain.

Dedicated DMSMS management operations POM/budget lines are preferable to an allocation of some larger lines. Dedicated lines elevate the visibility and importance of DMSMS management operations to both program office leadership and financial management communities (both internal and external to the program office). In addition, in the case of funding cuts, the DMSMS management activities are less likely to be reduced if they are separately identified.

[233] This is a ratio of 184 to 131 because the start-up period includes 131 non-recurring hours plus 53 recurring hours for a single BOM.

[234] The average ratio of start-up to steady state for electronics BOM in Table 33 is 135% which is not a substantial difference from the Table 32 result of 40.5%.

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As discussed previously, primes/OEMs/logistics support providers as well as independent SMEs provide DMSMS management operations services. From a programming and budgeting perspective, all the material associated with this enabling best practice applies to independent SMEs. Therefore, there should be an associated spend plan. When DMSMS management operations are also being provided by others, these activities will likely be a small part of a much larger contract (e.g., a production contract or a PBL contract). In that case, programming and budgeting will normally be for the total contract (and probably the spend plan as well). While it is still preferable to have a separate contract line item for DMSMS management operations, the most important best practice is to ensure that the contracts contain the right requirements and that the DMSMS management operations-related requirements are not dropped during contract negotiations.

H.4 Communicate the Value of Funding DMSMS Management Operations

This enabling best practice involves communication of DMSMS programming and budgeting considerations to all stakeholders. DMSMS management practitioners do not solely control the effectiveness and efficiency of DMSMS management–related activities in program offices. Everyone in a program office contributes. Programming and budgeting can only be successful if that is the case. Outreach to key stakeholders is the first essential step. While the need for outreach is mentioned elsewhere in this document, that material is organized by best practice. This section is organized by the stakeholder community interacting both inside and outside the program office. It summarizes the information that should be conveyed and how the stakeholder community can help with DMSMS management–related programming and budgeting.

- ***Program office management***. Although this section uses the term program office management, three specific elements of the program office are principally involved. The PM is the ultimate decision maker. The technical aspects of the programming and budgeting request should largely be coordinated and supported by the chief engineer and the PSM. First and foremost, the importance of DMSMS management operations (its inclusion in contracts and its associated funding) should be explained to program office management. Program office management must recognize the potential impact of pursuing DMSMS management reactively on cost, schedule, and readiness.

  Program office management also needs to understand the value of a stand-alone programming and budgeting line item for DMSMS management operations in terms of their contributions to proactivity and efficient program office operations. Because technical data is necessary to perform proactive DMSMS management operations, program office management must be able to support DMSMS-based contract clauses for this information and the processes that use the data.
- ***Contracting***. Similar to program office management, the contracting community needs to understand the importance of DMSMS management operations and resolving DMSMS issues. Consequently, it becomes imperative to prevent the elimination of associated clauses during contract negotiation processes. Also associated with DMSMS management–related contract clauses (including clauses for technical data), the contracting officer should be made aware of the value of independent DMSMS management SMEs to evaluate contractor proposed technical activities and their associated cost. Finally, the contracting community should explain what it needs from the DMSMS management community to satisfy the DMSMS management community's requests.
- ***Financial management***. Along with program office management stakeholders, the financial management community also must appreciate the importance of having stand-alone line items for DMSMS management and resolution-related programming and budgeting. In addition, the financial management community is also a source of key programming and budgeting information of interest to the DMSMS management community. They can provide information about the process, the types of appropriations that can be used for DMSMS

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management activities and help make the justifications for the funding as persuasive as possible. As was the case for contracting, the financial management community should explain what it needs from the DMSMS management community.

- ***WCF organizations***. In some cases, organizations that manage WCFs are the first to learn about an obsolescence issue because of an inability to issue a contract to repair an assembly on a system. This is most likely to occur with low demand items. DMSMS management practitioners in program offices should ensure timely communication channels exist with WCF organizations so that the program office learns about such issues as soon as possible. Some WCF organizations also perform DMSMS management operations associated with identifying issues in advance (e.g., equipment specialists). This is another area where there should be communications with the program office to ensure that DMSMS management operations are adequately resourced and that all stakeholders are aware of DMSMS issues, regardless of who discovers them.
- ***IPTs***. This is a generic heading because every program office may have a different IPT structure. IPT stakeholders include those organizations that are concerned with both readiness and system development and modification. DMT interfaces with various IPTs should be done on those IPTs' own terms. For example, it is generally more effective for DMSMS management SMEs to attend the meetings of other IPTs than to invite various IPTs to attend the program office's DMT meetings. The DMSMS management community should convey how it can help the various IPTs and then what the IPTs need to do to enable that support. This could eventually lead to a situation where various IPTs contact the DMT to provide key information.

  DMSMS health assessments should be communicated to the appropriate IPTs as an input to their modification planning. Since the various IPTs may drive technical data requirements, the DMT should inform those communities of its needs to be proactive. While the engineering and product support organizations should be represented on the DMT, it is important to emphasize the communication of DMSMS needs to these organizations. All technical information flowing from the DMT to the various IPTs should be coordinated with engineering and product support.

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Appendix I. Benefits from the Record Keeping Framework

The record keeping framework was designed with program offices in mind. By using this framework, program offices will have an improved capability to do the following:

- Prepare and justify DMSMS resolution budget requests;
- Prepare and justify budgets for DMSMS management operations;
- Conduct AoAs DMSMS resolution approaches;
- Identify, evaluate, and potentially reduce case implementation time;
- Determine how far in advance DMSMS resolutions should be budgeted;
- Evaluate both the cost and schedule for alternative resolutions and contract amendments to implement the preferred resolutions;
- Evaluate contract costs for DMSMS management activities;
- Quantify the cost of being reactive;
- Assess how DMSMS management affects readiness and schedule;
- Assess the differences between reactive and proactive DMSMS management on the supply system;
- Assess the ROI from being proactive;
- Enhance DMSMS management process effectiveness;
- Evaluate effectiveness of the DMSMS resolution process in implementing lower cost resolutions; and
- Enable higher level metrics reporting.

Commensurate benefits exist for other organizations. Based on program offices' use of the framework, PEOs and higher-level service organizations will have an improved capability to do the following:

- Evaluate DMSMS resolution budget requests,
- Evaluate DMSMS management operations budget requests,
- Enhance the LON buy process,
- Assess how DMSMS management affects readiness and schedule,
- Assess the ROI from being proactive,
- Monitor the performance of DMSMS efforts and find areas to target for improvement,
- Enhance DMSMS management process effectiveness,
- Evaluate effectiveness of the resolution process in implementing lower cost resolutions,
- Enhance resolution cost sharing,
- Monitor the performance of DMSMS efforts and find areas to target for improvement, and
- Enable higher-level metrics reporting.

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In addition, higher-level service organizations and OSD will have an improved capability to do the following:

- Benchmark and promulgate best practices for DMSMS management operations and resolution implementation processes;
- Benchmark, develop, and promulgate DMSMS management cost/workload factors;
- Assess how DMSMS management affects readiness and schedule;
- Enhance DMSMS management process effectiveness;
- Evaluate effectiveness of the DMSMS resolution process in implementing lower cost resolutions;
- Develop cost factors or cost estimating relationships (CERs) to improve DMSMS resolution programming and budgeting techniques after the design has been completed and before the design has been completed for integration into system engineering design tools;
- Facilitate the sharing of DMSMS resolutions for common items throughout the enterprise;
- Assess the ROI from being proactive;
- Assess the magnitude and trends of DMSMS resolution costs and take appropriate actions to reduce them; and
- Improve pertinent aspects of DMSMS management policy and guidance.

Appendix I.1–Appendix I.3 provide a substantially more detailed explanation of the value proposition along with examples. The appendixes are associated with the following benefit areas, respectively:

- Programming and budgeting,
- Process improvement, and
- An ROI for DMSMS management.

For each individual benefit, a subset of all the record keeping data elements provided in Table 8 of Section 3.4.3.1 is shown. The subset only contains the data elements related to that benefit. Appendix I.4 provides a data dictionary of record keeping terms.

I.1 Programming and Budgeting Benefits

I.1.1 Greater Fidelity Cost Estimates for DMSMS Resolutions

I.1.1.1 Value Proposition

Table 34 shows the Level 1 and Level 2 data elements necessary to obtain this benefit.

Table 34. Data Elements Needed to Obtain Greater Fidelity Cost Estimates for DMSMS Resolutions

Information Type	Level 1 Data Element Title	Level 2 Data Element Title
DMSMS resolution cost-related data elements	• Type of Resolution Approved • Resolution Cost • Source of DMSMS Resolution Cost • Redesign Level	• Commodity Type • Domain where Equipment Operates • DMSMS Item Type • Root Cause
DMSMS management operations efficiency-related data elements	• Item Class	

While it is always preferable to use actual resolution costs or costs based on estimates derived from an analysis of a specific DMSMS problem, there are many instances where another approach is taken. For

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example, cost estimates can be based on factors published in Table 15.[235] The value proposition for collecting, compiling, and analyzing the Level 1 and Level 2 data elements indicated in Table 34 is associated with the latter: the development of improved cost factors for DMSMS resolutions. These cost factors could be used for programming and budgeting purposes as illustrated in Appendix I.1.3–Appendix I.1.5.

The value proposition for program offices and PEO organizations is that they can benefit from access to improved DoD cost factors generated by OSD or Service HQ, based on the analysis of the identified data elements collected over many systems. Under some circumstances, program offices and PEO organizations could improve cost estimates by using their specific data to independently refine the DoD cost factors for their own use.

The value proposition for OSD and Service HQ is based on access to data from all program offices. Analysis of this data would enable OSD to provide improved DMSMS management guidance through periodic updates and refinements of the DoD cost factors. A multi-dimensional table to break down average resolution cost by characteristics of the platform (e.g., Domain where Equipment Operates) and the obsolete item (e.g., DMSMS Item Type or Commodity Type) could be generated. Costs could be further broken down by the root cause of a DMSMS case, which is useful for cost planning if the likelihood of different root causes can be predicted (e.g., items using significant amounts of hazardous material are more likely to have environmental or safety concerns as a root cause). This could be done on a routine basis without reliance on (and funding for) an occasional survey. This more systematic data collection would enable OSD to publish more statistically significant resolution cost factors. In addition, it may be feasible for OSD to develop CERs to refine cost estimates even further, thus enabling an improvement over what the DoD cost factors provide today. Service HQ would then be able to update Service-specific guidance on the use of the cost tables/CERs. Lastly, an analysis of the distribution of costs for a given resolution may identify a need to further refine resolution types and consequently add greater fidelity to the resolution cost factors.

I.1.1.2 Examples

The following illustrates how the data elements could be used to benefit PEO organizations and program offices:

- Use of Level 1 data elements would lead to more accurate cost estimates by providing more up-to-date information on resolution costs. For example, the February 2015 version of the SD-22 indicated that the average cost of a redesign at the NHA level was $109,200.[236] Guidance from a previous version of the SD-22 stated that the average cost of a major redesign was $467,000.[237]
- Use of Level 2 data elements would lead to more accurate cost estimates by providing data to refine or tailor the overall averages. Specifically, information on the commodity type, domain where equipment operates, type of item, redesign level, and whether the item was COTS could be used. For example, Appendix L indicates that the average cost for a redesign at the next higher level of assembly for aviation equipment is $1.61 million and $0.28 million for shipboard equipment. Analysis of actual resolution costs pertaining to a specific system or sets of systems could provide a basis for even further refinement of the Table 15 cost factors.

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[235] The reason for including the source of the DMSMS resolution costs is to have a better understanding of what the resolution cost data element represents. That understanding can impact how any analysis of record keeping data elements should be interpreted and used. This comment applies to the remaining descriptions of record keeping benefits.

[236] DSPO, SD-22, Diminishing Manufacturing Sources and Material Shortages—A Guidebook of Best Practices for Implementing a Robust DMSMS Management Program, February 2015, p. 99.

[237] Ibid, September 2009, p. 27.

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Examples of benefits to OSD and Service HQ are as follows:

- Periodically updating and refining resolution cost guidance enables the above benefits to PEO organizations and program offices. The data element on the source of the resolution cost is important to consider. Costs that are generated from Table 15 itself should obviously not be considered in the update process.
- Both Level 1 and Level 2 data elements could be used to further refine the DMSMS resolution types. For example, in previous versions of this document, the average cost to develop a new item or source was based on 127 data points from the 2014 Department of Commerce survey. An examination of a histogram of Level 1 data for the 127 data points shows that approximately 20% of the resolutions cost more than $750,000 and some resolutions exceeded $5 million. This implied that the "development of a new item or source" resolution category was too broad, so it was replaced by three resolution types based on limited data. Data collected against the Table 34 data elements would ultimately refine these estimates and should facilitate more statistically sound analyses of resolution types for all high-cost resolution categories.[238]
- It may also be possible to use Level 2 data to develop CERs as a function of the extent of variability that can be explained by commodity type, domain where equipment operates, type of item, redesign level, and whether the item was COTS.

I.1.2 Improved Cost Estimates for Making LON Buys

I.1.2.1 Value Proposition

Table 35 identifies the three data elements that would be used to take LON buy quantities into account in programming and budgeting.

Table 35. Data Elements for Estimating Cost of Purchasing Inventory for an LON Resolution

Information Type	Level 1 Data Element Title	Level 2 Data Element Title
DMSMS resolution cost-related data elements		• Domain where Equipment Operates • Product Acquisition Cost
DMSMS management operations efficiency-related data elements	• Number of Subsystems	

The value proposition for collecting and analyzing these LON buy data elements is derived from the absence of guidance on how to program and budget for such purchases. The DoD LON buy resolution cost factor does not consider the item purchase itself. The same is true for an LON buy conducted in combination with another resolution—e.g., the reverse engineering cost factor does not take into account any items purchased in conjunction with a reverse engineering resolution.

All the published DMSMS resolution cost factors attempt to measure the average non-recurring costs of a resolution over the life cycle of a DoD system. In the case of an LON buy (either alone or in conjunction with another resolution), the upfront one-time cost of purchasing the items is excluded because that cost would have also been incurred if the item were not obsolete (i.e., the life cycle cost is zero).[239] There is, however, a possibility that a substantial amount of unplanned funding could be needed in the year that the entire purchase is executed. The data elements shown in Table 35 provide some insight into this.

[238] Analyses of low-cost resolution categories may not be worthwhile if cumulatively they only account for a small percentage of total resolution costs.

[239] Inventory holding cost would be part of the non-recurring cost.

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The value proposition for program offices is that there would be data to help estimate and justify a modification to programming and budgeting requests for DMSMS resolutions including cost increases associated with LON buys in the year of execution.[240] If such adjustments were made and the requests approved, program offices would have a diminished need to take extraordinary measures to identify sources of funding on a time critical basis (since the opportunity for LON buys is constrained by the last date of sale as published in an EOL notice).

The program office value proposition should however take into account several additional factors:

- Programming and budgeting for spikes is only applicable in situations where both the quantity of items needed *and* the price of the item is high.
- Since spikes are hard to forecast, provisions must be in place to reallocate the funding to other priorities if the need does not materialize.
- LON buys are only selected as the most cost-effective resolution when justified by a BCA or an AoA.

For PEOs, the value proposition is based on oversight of the program offices in their portfolios. The oversight would encompass ensuring that anticipated LON buy spikes are included in programming and budgeting requests and that those requests are well documented to enable the PEO organization to justify the request to higher level decision makers. The value proposition for Service HQ and OSD would also be based partly on justifying budget requests. And there is one additional benefit: data would be available to develop and promulgate appropriate policy and guidance on this subject.

I.1.2.2 Examples

OSD guidance would not be based on averages across all program offices. Averages would not account for spikes in the execution year of a budget. As indicated earlier, the guidance would only apply to high-volume and high-cost situations. For example, LON buy costs for an inexpensive component on a high-volume ground-based system might not impact budget execution very severely. Similarly, the need to buy a highly reliable, high-cost electronic item on a nuclear propulsion system might not be disruptive either due to the low volume.

The data element for the total number of subsystems in the system would have to be linked to the number of systems fielded (which is not one of the data elements) to determine volume. The environment in which the subsystems operate could potentially be used to identify high-cost situations (or the likelihood thereof). After some analysis of the data, a table of factors could be generated.

Programs offices would see how their experience compares to the entries in such a table to determine whether they need to program and budget for spikes in LON buys, and if so, estimate how much to request for such contingencies. Data extracted from a guidance table should be adjusted by the specific circumstances of the system itself. If the program office has never experienced a spike, then it probably should not program and budget for one. If a program office has substantial LON buys every year, it should give more weight to individual experience than the factors in a guidance table.

[240] If budget requests were based on historic averages, average LON buy costs would be included if the purchase amounts were included. However, averages may not account for spikes.

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I.1.3 Improved Programming and Budgeting for Resolutions

I.1.3.1 Value Proposition

Table 36 shows the Level 1 and Level 2 data elements necessary to better develop and defend programming and budgeting requests for DMSMS resolutions. The only difference between Table 34 and Table 36 is that a Level 2 data element for the "Cost of LON Items" is included in Table 36.

The value proposition to program offices for collecting, compiling, and analyzing the Level 1 and Level 2 data elements indicated in Table 36 is that they can use the improved cost estimates (as discussed in Appendix I.1.1) to develop more accurate and more defendable programming and budgeting submissions for DMSMS resolutions including LON buys. Likewise, PEO organizations and Service HQ would be in a better position to evaluate those submissions and defend them to higher level approving authorities. OSD would be in a better position to approve the programming and budgeting figures.

Table 36. Data Elements Needed to Obtain Greater Fidelity Cost Estimates for DMSMS Resolutions

Information Type	Level 1 Data Element Title	Level 2 Data Element Title
DMSMS resolution cost-related data elements	• Type of Resolution Approved • Resolution Cost • Source of DMSMS Resolution Cost • Redesign Level	• Commodity Type • Domain where Equipment Operates • Production Acquisition Cost • DMSMS Item Type • Root Cause
DMSMS management operations efficiency-related data elements	• Item Class	

I.1.3.2 Examples

Program offices could use the data elements to apply one of two common approaches (or a combination thereof) to build programming and budgeting submissions for DMSMS resolutions. The first approach is to analyze past trends in resolution cost for their particular system and then extrapolate those trends into the future. The Table 36 "Resolution Cost" data element collected over time supports such an approach. The second approach is based on predicting the types of resolutions that will occur by year and using the resolution cost factors to estimate the total cost in any given year. One way of predicting the types of resolutions in a given year is to analyze the "Type of Resolution Approved" data element over time to develop averages or trends. More sophisticated methods using predictive tools and vendor surveys can also be used to predict resolutions over time.

A variation on the second approach is to predict only redesigns and use the cost factors, further refined by the "Redesign Level" data element to develop programming and budgeting submissions. The viability of this variation is also something that can be determined by program offices. The 2014 Department of Commerce survey data indicated that 95% of total resolution costs were related to redesign. If individual program offices can use the "Resolution Cost" and "Type of Resolution Approved" data elements over time to verify that redesigns are the principal driver of resolution cost for their systems, then it would only be necessary to predict redesigns in the future and then include a factor for everything else. For example, if redesigns normally accounted for 90% of the resolution costs, the predicted resolution cost in a given year based only on redesigns could be increased by 11% to generate an estimate for the total DMSMS resolution cost.

Program offices could enhance the above calculations by analyzing and considering the Production Acquisition Cost data element when they believe it is necessary to do so. From a programming and budgeting perspective, the impact in a given year may be significant if appropriated funds are needed to

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execute an LON buy.[241] For example, if a system is in production with large quantities of an expensive item yet to be purchased, and the item is expected to be highly reliable and unlikely to be redesigned, the LON costs in the fiscal year of purchase are likely to be significant. Similarly, a small dollar value system that does not incur many DMSMS problems could experience high LON buy requirements if it involves an expensive component.

Examples of the benefits to stakeholders beyond the program office are straightforward. The other Level 2 data elements (i.e., commodity type, domain where equipment operates, item type, and root cause) can also be used to further enhance resolution prediction. These data elements could indicate trends affecting obsolescence. They involve using the data elements to verify program office calculations and strengthening the justification for funding. There is one additional benefit for OSD. This document is currently silent on the consideration of LON buy costs in programming and budgeting. The Table 36 data elements could be used to improve guidance in the future.

I.1.4 Improved Evaluation of Contractor Costs for Resolutions

The Table 34 data elements also apply here. A further enhancement of the value proposition associated with greater fidelity cost estimates for DMSMS resolutions (see Appendix I.1.1) is that program offices would have additional information useful for evaluating cost proposals to resolve DMSMS issues. This could strengthen the program office's contract negotiating capability and potentially result in lower contract cost.

For example, if a cost estimate were very close to the associated cost factor, a program office might have a certain degree of confidence in its acceptability. In situations where the cost proposal is considerably less than the cost factor indicates, the program office may want to take steps to ensure that the contractor understands the complexity of the resolution. Finally, when the cost estimate is well above the cost factor, the program office should attempt to understand why that is the case. The program office should be prepared to either negotiate the price downward or seek other bidders.

I.1.5 Improved Understanding of Proactivity and Risk Linkages

I.1.5.1 Value Proposition

Linking DMSMS management proactivity with risk provides an indication of the extent to which the level of DMSMS management operations is sufficient to mitigate DMSMS risk. Table 37 shows the Level 1 and Level 2 data elements necessary to obtain this benefit.

Table 37. Data Elements Needed to Obtain an Improved Understanding of the Link between DMSMS Management Proactivity and Risk

Information Type	Level 1 Data Element Title	Level 2 Data Element Title
DMSMS resolution cost-related data elements	• Type of Resolution Approved • Resolution Cost • Source of DMSMS Resolution Cost	• Commodity type • Root Cause
DMSMS management operations efficiency-related data elements	• Nomenclature • Item Class • Case Proactivity Indicator • Reason Issue Was Discovered Reactively	

[241] DoD cost factors do not include these costs because 1) there could be large variation and 2) the life cycle cost is zero because the items would eventually be purchased over time based on failure rate.

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These data elements provide information on the impact of the risk being taken with respect to the extent of item monitoring that a program office is funding.[242] The value proposition for program offices is that they will have a quantitative basis for either increasing or decreasing the level of monitoring activity and associated funding.

The data elements show the cases that were opened reactively because of a conscious decision by the program office not to monitor the affected items. This is a direct measure of the risk taken as part of the risk-based approach to DMSMS management. How those issues were resolved along with the associated resolution costs enables an assessment of whether too much risk is being taken. This assessment should lead to a decision of whether the level of DMSMS monitoring is sufficient.

The data elements also provide an indication of whether too many resources are being applied to DMSMS management. The goal of proactive DMSMS management is not to have zero cases opened reactively. There are commodity products (e.g., resistors, capacitors) that become obsolete frequently. The resolution is normally to identify another off-the-shelf item that has the same specifications. This can be accomplished very quickly; and consequently, there is generally no need to try to identify such issues far in advance. Resources spent on proactive monitoring of such items may be inefficient.

The value proposition for PEOs is based on an evaluation of the amount of risk being taken by program offices within the PEO organization's portfolio to indicate whether any of the program offices is taking too much or too little risk. Adjustments can be made based on quantitative data. A more detailed examination of the impact across program offices within the portfolio could help make an informed judgement regarding an appropriate level of risk.

Service HQ and OSD could examine such data to obtain a broad understanding of the extent to which risk-based, proactive DMSMS management activities occur on a broad scale. Such information could inform new policy and guidance.

I.1.5.2 Examples

The basis of the difference between the value propositions for Level 1 and Level 2 data is some additional refinement in terms of determining whether an obsolete item should have been monitored. Level 2 data allows for a slightly more refined examination. Level 1 data determines whether an item is commercial or not and it also provides the item nomenclature. Level 2 data add the commodity type (e.g., electronics, mechanical) and root cause of the DMSMS case being opened.

The first two program office examples below use only Level 1 data. The third and fourth program office examples also use Level 2 data.

- Assume a program office opened 20 DMSMS cases in a given year, and that five of those cases were opened reactively for commercial electrical items that were deliberately not monitored. Further assume that in all five cases, the resolution was a simple substitute. There is no indication that the program office is accepting too much risk.
- On the other hand, if two of the resolutions were costly redesigns for commercial electronics items, the same conclusions should not be immediately drawn. There would be a strong indication that too much risk is being taken in terms of what should be monitored.
- A third circumstance might be that none of the 20 cases were opened reactively and all cases were resolved with a simple substitute. If for example, the items were all mechanical, it would

[242] Monitoring is usually the basis for developing an initial cost estimate for DMSMS management operations. In the first year, estimates may be based on experience from similar systems. After the first year, more refined estimates can be made based on experience from the system itself.

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be a good idea to investigate in more detail whether some items are being monitored unnecessarily.

- Finally, assume a case was opened reactively for a part and proved costly, but the root cause of the case was an unexpected change in relations with a sole foreign producer of the part. There were no other indications a resolution would be expensive should a DMSMS case arise. Because the change in relations was unexpected (such as with a previously friendly nation), the PEO may have been correct, from a risk perspective, in not monitoring this part.

Examples for PEO organizations would be similar. PEOs would be looking for circumstances where the level of DMSMS management activities should be adjusted for program offices within their portfolio. This would be accomplished by determining whether too much or too little risk was being taken by different program offices. If two similar program offices were having dissimilar results, then that would be a further indication that the DMSMS management activities of at least one of the program offices should be changed.

Service HQ, PEO organizations, or OSD might be interested in using the data in aggregate. For example, a histogram of the types of resolutions used for reactively opened cases could be generated. The level of high-cost resolutions implemented could be used as an indicator of how much cost could be avoided with greater investment in proactive DMSMS management operations.

I.1.6 Improved DMSMS Management Cost and Workload Metrics

I.1.6.1 Value Proposition

Table 38 shows the Level 1 and Level 2 data elements necessary to obtain improved costs and workload metrics.

Table 38. Data Elements Needed to Obtain Improved Cost and Workload Metrics for DMSMS Management

Information Type	Level 1 Data Element Title	Level 2 Data Element Title
DMSMS management cost-related data elements	• DMSMS Management Operations Cost Paid to Prime/OEM • DMSMS Management Operations Cost Paid to Independent SME Organizations • Management Operations Cost for Internal DMSMS-Related Activities	• Value of Management Operations Activities Received at No Cost from a Centralized Service Source
DMSMS management operations efficiency-related data elements	• Number of Subsystems • Number of Subsystems Monitored • Components • Components Monitored	

The value proposition for these data elements is that program offices will have a much better basis for estimating what DMSMS management operations might cost, although there might be a large variance among program offices. The only guidance available today is based on a limited sample size and only provides relative cost estimates.

Having much more refined DMSMS management operations cost factors published in guidance will be far more useful to program offices. The data elements portrayed in Table 38 would provide specific cost estimates by breaking down cost by provider and obtaining a much more refined measure of the workload involved. Finally, the new DMSMS management operations cost factors would be more accurate since they would be based on data from hundreds of program offices.

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The value proposition for PEO organizations is very similar to that for program offices. PEOs would be able to compare DMSMS management operations costs among the program offices in their portfolios to identify opportunities to reduce cost based on the cost factors. Furthermore, by comparing similar program offices, other efficiencies may manifest themselves.

For OSD, the value proposition will be enabling the issuance of comprehensive guidance. For Service HQ, there is an opportunity to tailor the results potentially based on the availability of a centralized DMSMS management service provider.

I.1.6.2 Examples

The difference between the value proposition for the Level 1 and Level 2 data elements is that the Level 2 data element also provides the program office with information about support resources provided at no cost to the program office (when that situation applies). Therefore, the Level 2 data element could provide additional insight on the total cost of DMSMS management operations at the program office although it does not impact program office programming and budgeting.

Program office and PEO examples are relatively straightforward. They would be able to develop cost estimates based on their specific situation.

Given the data elements shown in Table 38, OSD and Service HQ would be able to publish DMSMS management operations cost factors. However, using the current unpublished guidance, additional considerations (e.g., start-up versus steady state, level of DMSMS management operations services) could be integrated into new cost factors. Furthermore, using some other data elements not included currently in Table 38, other refinements could be developed. For example, there may be a difference between management operations costs as a function of system complexity which may be visible from the domain where equipment operates data element. Such a hypothesis could be tested before the guidance is published.

I.1.7 Improved Programming and Budgeting for Operations

Table 39 shows the Level 1 and Level 2 data elements necessary to better develop and defend programming and budgeting requests for DMSMS management operations.

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Table 39. Data Elements Needed to Obtain an Improved Ability to Develop and Defend Programming and Budgeting Requests for DMSMS Management Operations

Information Type	Level 1 Data Element Title	Level 2 Data Element Title
DMSMS management and resolution cost-related data elements	• Type of Resolution Approved • Resolution Cost • Source of DMSMS Resolution Cost • DMSMS Management Operations Cost Paid to Prime/OEM • DMSMS Management Operations Cost Paid to Independent SME Organizations • Management Operations Cost for Internal DMSMS-Related Activities	• Commodity Type • Root Cause
DMSMS management operations efficiency-related data elements	• Nomenclature • Item Class • Case Proactivity Indicator • Number of Subsystems • Number of Subsystems Monitored • Components • Components Monitored • Reason Issue Was Discovered Reactively	

The value proposition is straightforward. Program offices will be in a better position to obtain the DMSMS management operations resources they need through strong justification for what is needed. PEOs will be in a better position to oversee the programs in their portfolios and defend programming and budget requests to Service HQ. In turn, Service HQ would be in a better position to defend funds requests to OSD and OSD would have greater confidence in the accuracy of the requests.

Examples are also straightforward. Initial estimates would be data driven. That justification would be seen by higher level organizations. The difference between Level 1 and Level 2 data is greater refinement to the estimates themselves, as discussed in Appendix I.1.6. Certain data elements in Table 40 will not be observed during programming and budgeting, such as the type of resolution, resolution cost, and root cause. However, by collecting these data elements once realized, planners can infer relationships between them and aspects of the parts used in the system, which in turn improves future programming and budgeting; if certain aspects of a system are learned to be correlated with particular root causes, resolution types, and resolution costs, planners can use that information to inform future expected DMSMS costs.

I.1.8 Improved Evaluation of Contractor Costs for Operations

The Table 39 data elements also apply here. Not all companies have the same skills and capabilities for DMSMS management operations. Some companies are far more experienced than others. There have been instances where DMSMS management operations cost proposals have been vastly different. Therefore, having guidance available on DMSMS management operations cost factors (see Appendix I.1.6) would provide program offices material that could be used to evaluate cost proposals for DMSMS management operations, taking into account the extent of DMSMS management operations conducted inside the program office. This could strengthen the program office's contract negotiating capability and potentially result in lower contract cost.

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As discussed in Appendix I.1.3, if a cost estimate were very close to the cost derived from guidance, a program office might have a certain degree of confidence in its acceptability. In situations where the cost proposal is considerably less than the cost factor indicates, the program office may want to take steps to ensure that the contractor understands the program office's DMSMS management operations requirements. Finally, when the cost estimate is well above the cost factor, the program office should attempt to understand why that is the case. The program office should be prepared to either negotiate the price downward or seek other bidders.

I.1.9 Identifying DMSMS Trends to Improve OSD Strategy

Table 40 shows the Level 1 and Level 2 data elements necessary to identify important trends in DMSMS cases that may inform strategic decision-making regarding DMSMS management at the OSD-level.

Table 40. Data Elements Needed to Identify DMSMS Trends to Improve OSD Strategy

Information Type	Level 1 Data Element Title	Level 2 Data Element Title
DMSMS management and resolution cost-related data elements	• Type of Resolution Approved	• Root Cause
DMSMS management operations efficiency-related data elements	• OEM CAGE Code • OCM CAGE Code • Item Class • Case Proactivity Indicator	

These data elements, collected across multiple program offices and cases, will allow OSD to identify factors that commonly underly DMSMS cases. Identifying key factors may in turn inform funding and other policy decisions. Examples of how each data element may lead to a valuable insight follow:

- ***Type of Resolution Approved and Case Proactivity Indicator***. If it's found that many DMSMS cases are being resolved retroactively via technology refreshes, this may encourage OSD to fund more research into optimal technology refresh planning to reduce costs and operational impacts of reactive tech refreshments.
- ***OCM/OEM CAGE Code***. If a certain CAGE code, or multiple CAGE codes owned by a common supplier, have many DMSMS cases, this may be a sign to engage in executive-level dialogue with the supplier to improve DMSMS management.
- ***Item Class***. If certain item classes are seen to be more prone to DMSMS cases, OSD may consider policies or guidance to minimize their use or at least devote more proactive DMSMS management for such items.
- ***Root Cause***. An upward trend in DMSMS cases caused by foreign trade policy may lead OSD to consider advocating for increased manufacturing in the United States and/or friendly nations. As another example, a trend in DMSMS cases caused by technology could, as in a previous example, lead the OSD to fund more research in optimal technology refresh planning.

I.2 Process Improvement Benefits

I.2.1 Improved Efficiency for Case Processing Time

I.2.1.1 Value Proposition

Table 41 shows the Level 1 and Level 2 data elements necessary to obtain this benefit. The efficiency related data elements are all associated with case processing time. The remainder of the data elements are concerned with characteristics of the situation that could have an impact on case processing time.

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Table 41. Data Elements for Case Processing Time Improvement

Information Type	Level 1 Data Element Title	Level 2 Data Element Title
DMSMS resolution cost-related data elements	• Type of Resolution Approved	• Commodity Type • Domain where Equipment Operates • DMSMS Item Type • Root Cause
DMSMS management operations efficiency-related data elements	• Item Class • Date Alert Received • Date Case Opened • Date Resolution Submitted for Approval • Date Case Resolved • Date Case Closed	

The value proposition for case processing time is associated with comparing the times required to open, determine the optimal resolution, submit the resolution for approval, receive approval for the resolution, and implement the resolution. The comparisons would be made as a function of the type of resolution with Level 1 data only and as a function of the commodity type, the domain where equipment operates, the DMSMS item type, and root cause for Level 2 data. When comparisons indicate potential issues, further analyses would be initiated.[243]

From a program office perspective, actual times could be compared to expected times to determine whether any part of the process is taking too long. If there are no expectations, then the program office could compare itself to the benchmarks developed from other program offices' data. If this comparison were to indicate that the program office is taking longer, investigations into the process itself could be initiated to identify potential problems and initiate process improvement. This is the basis of the value proposition to the program office.

PEO organizations could compare case processing times reported by the program offices within its portfolio. This could indicate those program offices within the portfolio that are being less efficient than desirable in their DMSMS management approach; therefore, identifying program offices that could benefit from additional attention. Like program offices, a PEO organization could benefit from having access to a larger data repository to provide a more broadly populated set of case processing time benchmarks.

OSD and Service HQ would be able to assess whether anything can be gleaned from an analysis of case processing times, aggregated across organizations and program offices, as a function of the data elements in Table 41 to highlight DMSMS management efficiencies and inefficiencies. This would provide evidence to support desirable changes/initiatives to improve DMSMS management policy and guidance. For OSD and Service HQ, there could be a project to identify a further breakdown of the timing to identify contract lead time and programming and budgeting lead time. Once these are factored out, the differences between case processing time might not be meaningful. Alternatively, this could imply that OSD should concentrate policy and guidance on the non-DMSMS management–related aspects of processing time (e.g., decision-making, contracting, and programming and budgeting).

[243] Some might think that if the case is closed before the implementation is needed, then there is no concern about processing time. If this were the only criterion for process improvement, inefficient processes could continue and they may eventually negatively impact the program when a rapid implementation is needed. For that reason, the implementation need date is not included in Table 41.

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I.2.1.2 Examples

Several years ago, an unpublished analysis was conducted pertaining to this case processing time topic. The analysis included only the time required to implement resolution types based on data for several hundred cases. Observations made from that data raised questions which serve as examples for the following types of questions that a larger, more complete set of data might answer:

- The average time to close an LON buy was 497 days. Since most LON buy cases are opened when a discontinuation notice is received and the typical amount of time given to make a final purchase is less than one year, why would the *average* be one year and four months? There were also examples where it took up to four years to close some of the LON cases. Given that LON buys are generally time sensitive, why was that the case?
- The average time to implement a simple substitute resolution was 319 days, the average for a complex substitute was 387 days, and the average time for development of a new source was 198 days. Why is the time difference between simple and complex substitutes so small? Why does the seemingly much more complex development of a new source take so much less time than a simple substitute?
- The Level 1 and Level 2 data elements suggested in Table 41 could explain some of the observed time differences.

Table 56 contains the results of a more recent calculation of the average time to close a case. Why are these values substantially different from the numbers in the above two bullets?

In addition, when comparing the data from one program office to the aggregate data, performance trends could be identified and, if needed, corrected.

I.2.2 Improving the Effectiveness of Item Monitoring Processes

I.2.2.1 Value Proposition

The Level 1 and Level 2 data elements associated with this benefit are listed in Table 42. As was the case for Table 41, the operations efficiency data elements provide insights into performance and the cost-related data elements represent potential consequences of poor performance.

Table 42. Data Elements for Improving Item Monitoring Process Effectiveness

Information Type	Level 1 Data Element Title	Level 2 Data Element Title
DMSMS resolution cost-related data elements	• Type of Resolution Approved • Resolution Cost • Source of DMSMS Resolution Cost • Redesign Level	
DMSMS management operations efficiency-related data elements	• Reason Issue Was Discovered Reactively	• Monitoring Techniques

The two DMSMS management operations efficiency-related data elements shown in Table 42 are intended to capture information about the effectiveness of a program office's DMSMS monitoring processes. If monitoring is inefficient, cases may be opened reactively where they could have been opened proactively. For proactive cases, there is also the possibility that the source of the information used to open a case was not the most efficient (i.e., did not indicate an item discontinuation in a timely manner). If so, then there would be a possibility that the case could have been opened earlier.

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The value proposition for the program office has three primary components:

- First, Level 1 data could be used to conduct a risk analysis to determine whether the items managed reactively led to resolutions with a pronounced adverse impact on cost or readiness (see Appendix I.3). This may lead to an adjustment to the risk-based approach to increase the number of items monitored proactively.
- Second, using Level 1 data, analyses of the monitoring process should be conducted for specific instances where proactive processes did not identify a DMSMS issue before a failed attempt to buy the item to measure the effectiveness of the tools and techniques used in monitoring (quality). These data can help answer the question "Why did the tools miss this particular instance?"
- Third, through an examination of resolution costs as a function of how a DMSMS issue was discovered proactively, insights may be obtained about the efficiency of one type of monitoring versus another. This type of examination would require Level 2 data. For example, if one program office has a lower rate of reactive issues for monitored parts than another program office, then further investigation could be done to determine what tools are being used.

The two latter components of the value proposition may require the analysis of multiple program offices because most cases in a given program office will be opened because of something identified by the most efficient monitoring technique being used by that program office. That program office would therefore need to analyze data from other program offices to determine the efficiency of other monitoring processes not being used. These analyses could identify actions to enable the earlier detection of DMSMS issues.

The latter two components of the value proposition also apply to PEOs. PEO organizations would also be better able to compare process efficiency because of proactivity. This could provide an indication of those program offices within the portfolio that are being less efficient than desirable in their DMSMS management approach, therefore identifying program offices that could benefit from additional attention.

The value proposition for OSD and Service HQ would be realized through an ability to use factual evidence to a greater degree when making determinations of the extent and effectiveness of proactivity associated with the monitoring techniques across all program offices. "Extent" indicates the degree to which program offices were being proactive. "Effectiveness" indicates the extent to which cases that should have been opened proactively were in fact opened reactively. This data could be used to identify causes of reactivity and thereby improve policy, guidance, and best practices, especially when cost or readiness impacts can be associated with that reactivity.

Furthermore, since one of the monitoring areas deals with surveillance of LON stocks, to determine whether the quantity procured remains sufficient, an additional value to OSD and Service HQ is associated with the methods and processes to calculate and procure those items. If a pattern emerges, new policy and guidance may result.

I.2.2.2 Examples

One possible analysis could correlate actual program office proactivity data based on DMSMS monitoring approaches (e.g., Are they using predictive tools? Do they rely solely on the GIDEP?) with resolution costs. It could result in the discovery that some less proactive monitoring techniques lead to a greater number of expensive resolutions such as redesign. As a result, efforts could be undertaken by program offices to become more proactive or improvements could be made to the efficiency of the less effective monitoring techniques. In addition, the analysis might reveal that some monitoring methods may not be worth the added cost for the results they provide.

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Another analysis could be conducted to study the pattern of reactive cases. If, for example, it was found that items were not being monitored, then an increased level of DMSMS management operations funding might be needed. If there were a correlation between reactive cases and the scope or technique of monitoring, then something could be done to increase the effectiveness of the monitoring by altering the scope or technique.

Associated with the monitoring of LON buy quantities, another study could focus on how well LON buy quantities are determined. Some organizations who conduct LON buys may have a history of miscalculating demand, resulting in too few or excess stock of the items. That might prompt a more detailed examination of that organization or better guidance on computing the size of LON buys.

I.2.3 LON Buy Process Improvement

Table 43 identifies the single Level 2 data element needed to achieve this benefit in conjunction with Level 1 cost data.

Table 43. Data Elements Improving LON Buy Processes

Information Type	Level 1 Data Element Title	Level 2 Data Element Title
DMSMS resolution cost-related data elements	• Type of Resolution Approved • Resolution Cost • Source of DMSMS Resolution Cost • Redesign Level	
DMSMS management operations efficiency-related data elements		• LON Buy Preferred Indicator

The value proposition here requires Level 2 data; it is associated with the LON buy process within an organization. This process may involve many stakeholders internal and external to a program office. The LON preferred indicator identifies cases where an LON buy was the most cost-effective resolution but could not be executed. Principal reasons why an LON buy was not feasible include sufficient funds were not available, there was not enough time to execute the process, the process was broken, or a limitation was imposed on the quantity to be procured.

The resolution cost data may be used to estimate how much more is being spent over the life cycle of a system because of such an inability to execute. If these amounts were significant, program offices, PEO organizations, and Service HQ would use this information to improve their LON processes. If the primary obstacles to LON buys were legislative or regulatory, OSD could initiate modifications.

I.2.4 Detection of Anomalies in DMSMS Resolution Cost

It is beneficial to detect anomalies because they are an indication that further analyses are needed. For this benefit, the Level 1 and Level 2 data elements are shown in Table 44. There are no operations efficiency data elements included.

Table 44. Data Elements for Detecting Anomalies in DMSMS Resolution Cost

Information Type	Level 1 Data Element Title	Level 2 Data Element Title
DMSMS resolution cost-related data elements	• Type of Resolution Approved • Resolution Cost • Source of DMSMS Resolution Cost • Redesign Level	• Commodity Type • Domain where Equipment Operates • DMSMS Item Type • Root Cause

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Appendix I.1 discussed benefits associated with developing resolution cost factors and using resolution cost data to support programming and budgeting. In this case, an examination of the differences in resolution cost under somewhat similar circumstances across program offices and more aggregate organizations could be made. Level 1 data elements enable such a comparison. Level 2 data elements portray some of the reasons why there may be differences and thereby enable a much more detailed analysis.

The value proposition of program offices would be in situations where Service HQ or OSD published guidance about the range of costs to expect for DMSMS resolutions. A program office might find that its costs were high. This could indicate that there is something inefficient (or overly expensive) about the way resolutions are implemented. At minimum, it would generate questions that should be answered. The PEO organization could make similar comparisons, probably just among the program offices in its portfolio.

OSD and Service HQs would use the entire data repository to assess how DMSMS resolution costs are broken down among organizational entities. There are valid reasons why one Service may be spending different amounts relative to another Service or one set of platforms is spending more or less than another set of similar platforms. OSD and Service HQs would identify potential differences, verify that the differences are significant, and attempt to determine the root causes. This could lead to changes in policy, guidance, or best practices. Service HQs may choose to analyze just their own portions of the data repository to do this.

I.3 ROI-Related Benefits

The value proposition associated with all the ROI-related benefits is similar. It is based on the ROI concept itself, there will be some benefit or return resulting from the investment of resources in risk-based, proactive DMSMS management.

The benefit may be monetary, schedule-related, or availability-related. In the latter case, some benefits may apply directly to avoiding impacts to supply system operations such as:

- Fewer backorders or
- On-hand stock representing something below the optimal level of days of supply.

Another indirect benefit associated with operational availability is reducing the number of days needed to implement a resolution. The shorter the implementation time, the greater the likelihood of avoiding any ill-effects.

Inherent in this value proposition is the improved capability to demonstrate the benefits of risk-based, proactive DMSMS management. From a program office perspective, articulating such benefits increases the likelihood of successfully programming and budgeting for the resources needed to conduct the right amount of DMSMS management operations. The same holds true at the PEO level, however PEOs also would obtain sufficient evidence to be convinced that they are supporting important and necessary functions. Service HQ and OSD would aggregate such data. In doing so, they would increase their understanding of the value of DMSMS management activities as well as issue policy and guidance to institutionalize the necessary level of support.

I.3.1 Estimating the Cost of Being Reactive

Table 45 lists the data elements needed for estimating the cost of being reactive. This benefit is achievable with Level 1 data.

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Table 45. Data Elements for Estimating the Cost of Being Reactive

Information Type	Level 1 Data Element Title	Level 2 Data Element Title
DMSMS resolution cost-related data elements	• Type of Resolution Approved • Resolution Cost • Source of DMSMS Resolution Cost	
DMSMS management operations efficiency-related data elements	• Case Proactivity Indicator • Reason Issue Was Discovered Reactively	

Appendix I.2.2 focused on the adequacy of risk-based DMSMS management and monitoring and made suggestions for improvement. This section adds the refinement of estimating the cost of being reactive. Appendix I.1.3 discussed the fact that the bulk of DMSMS resolution costs are associated with redesigns because redesigns are costlier than other DMSMS resolutions. Therefore, insight on the cost of being reactive may be obtained from just redesigns rather than all cases. The following three steps may be taken:

- Extract all cases where the resolution was a redesign at the item, NHA, or major subsystem level;
- Select the subset of those redesigns where cases were opened reactively based on the Case Proactivity Indicator; and
- Calculate the total cost of that subset of the redesigns.

The cost of redesigns because of cases that were opened reactively represents an estimate of the cost of being reactive. The estimate may be high because some redesigns might have occurred even if the case had been opened proactively. Since the cases were not opened proactively, it is not possible to know when a redesign would still have been the resolution. In addition, this representation assumes that when a lower cost resolution was feasible reactively, a similar cost resolution would have been feasible proactively. The data elements are not able to provide any additional insight.

The data element for the reason the issue was discovered reactively is included because it can be used to make inferences about the reason for incurring a significant cost of being reactive.

I.3.2 Estimating the Cost Avoidance for Being Proactive

A new best practice is to define cost avoidance as the difference between the cost of the resolution implemented when a DMSMS issue is found proactively and the cost of the resolution had the DMSMS issue been found reactively. This requires that those responsible for determining cost avoidance perform an analysis of which solution would have been chosen had the issue been discovered reactively.

The process for determining the cost avoidance requires that the resolution of a proactive DMSMS issue has been implemented and that the costs are known or can be estimated. The estimated cost of the reactive resolution is based on two factors:

- The date that the DMSMS issue was likely to be identified reactively. This will usually be the date when the existing supply of the item reaches its reorder point based on the demand rate.
- An analytical determination of the cost of the resolution had the DMSMS issue been found reactively.

The cost avoidance is then calculated by subtracting the cost of the proactive resolution from the estimated cost of the reactive resolution. In situations where a single DMSMS issue affecting multiple assemblies or multiple DMSMS issues are solved with a single solution, the cost of the reactive solution would be determined by summing the costs for each unique assembly or solution, taking into consideration that common costs should only be applied once.

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There may be situations where no cost avoidance will be realized.

- The reactive and proactive resolutions were the same. For example, there are common situations where microelectronics parts numbers are changed because of new external packaging or the elimination of tin-lead finishes. Resolutions for these situations range from just changing the part number to a board layout change. In a proactive situation, the part change would be identified early; in a reactive situation, the part change would be identified after a failed attempt to purchase the former part number. Nevertheless, the resolutions are the same and there is no cost avoidance from being proactive.
- When the proactive resolution was a redesign. It is assumed that a redesign is always an option whether the issue was found proactively or reactively.
- When the DMSMS issue was identified reactively, since cost avoidance only applies to issues found proactively.
- In cases where the proactive resolution used was not the least expensive for some programmatic reason, there is the potential for there to be a negative cost avoidance. If this occurs, the cost avoidance could be discarded or the same assumptions used in the proactive solution could be applied to the reactive solution and the costs determined using those assumptions.

Table 46 lists the Level 1 data elements needed to estimate the cost avoidance for being proactive. There is a close relationship between the cost of being reactive as discussed in Appendix I.3.1 and the cost avoidance for being proactive. For Appendix I.3.1, the perspective is from a reactive program office considering what would have been the case (i.e., how much money it could have saved) had it been proactive. In this section, the perspective is from a proactive program office estimating what would have been the case (i.e., how much spending it avoided) had it been reactive.[244]

Table 46. Data Elements for Estimating Cost Avoidance for Being Proactive

Information Type	Level 1 Data Element Title	Level 2 Data Element Title
DMSMS resolution cost-related data elements	• Type of Resolution Approved • Resolution Cost • Source of DMSMS Resolution Cost	
DMSMS management operations efficiency-related data elements	• Case Proactivity Indicator • Resolution Avoided • Cost of Resolution Avoided	

One difference from Appendix I.3.1 is that the data elements extended the focus to all resolutions, not just redesigns. In this situation, the resolution avoided data element identifies situations where the case was opened proactively, and potentially a different resolution would have been required if it had been opened reactively. Determining what that resolution would have been involves: identifying when in the future there would have been a failed attempt to purchase; assessing which of the current feasible resolutions would still be feasible at that future date; and selecting which of those feasible resolutions would have been the cost effective choice.[245] The cost avoidance for being proactive would be the sum over all cases of the

[244] The cost avoidance of being proactive cannot be calculated accurately for reasons like those stated in Appendix I.3.1, for why the cost of being reactive cannot be calculated precisely.

[245] In Appendix I.3.6, the resolution avoided data element also refers to the proactive resolution that could have occurred had the DMSMS issue been identified proactively. Determining what that resolution would have been involves: identifying when discontinuation information could have been known in the past; assessing what additional resolutions would have been feasible at that time; and selecting the resolution that would have been the most cost effective at that time.

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differences between the cost of the resolution avoided and the actual resolution cost. Such cost estimates should be made using the following in descending order of precedence:

- Estimates provided in the process of determining the best resolution to use or from other sources (e.g., the prime contractor);
- Actual costs used for similar resolutions or similar items within the program office (the term *similar items* means same type of item, same domain where equipment operates, and same complexity);
- Average resolution costs for similar items that were generated within the program office; and
- Average resolution costs for similar items generated by higher level organizations.

I.3.3 Improved Understanding of Schedule Impacts

The Level 1 and Level 2 data elements associated with identifying DMSMS impacts on schedule are shown in Table 47.

Table 47. Data Elements for Understanding DMSMS Impacts on Schedule

Information Type	Level 1 Data Element Title	Level 2 Data Element Title
DMSMS resolution cost-related data elements	• Type of Resolution Approved	• Root Cause
DMSMS management operations efficiency-related data elements	• Case Proactivity Indicator • Reason Issue Was Discovered Reactively • Date Case Closed • Date Implementation Needed	• Effect on Production Schedule

The Level 2 data element's effect on schedule indicates the number of months that a production schedule for fielding an augmenting capability was delayed because of DMSMS issues. While it is difficult to monetize such delays, such data should have an impact at all levels of aggregation in terms of the value proposition stated earlier. The resolution type, root cause, or both for a DMSMS issue may partially explain the number of months production is delayed.

Two Level 1 data elements could provide supplemental information. A comparison of the date closed and the implementation date needed indicates whether there was an impact to the program office. While the impact could be production schedule, which is captured in the Level 2 data element, the impact may be elsewhere because the implementation need date is defined as the date that there will be an impact on the program office. For example, the impact could be related to operational availability or the impact could drive a cannibalization action that ultimately affects cost.

Three other data elements included in Table 47 may provide a better understanding of both Level 1 and Level 2 impacts. For example, analysis of the correlation between the type of resolution and whether the case was opened proactively or reactively (and the associated reason) could help explain why the delay occurred. Depending on the results of that analysis, potential process improvements (as well as policy and guidance changes) could be initiated. The analysis could also simply indicate that delays may be a result of the inevitability of DMSMS issues even when managed proactively.

I.3.4 Improved Understanding of Operational Availability Impacts

Table 48 presents the Level 1 and Level 2 data elements relating to this benefit. The data are analogous to those in Appendix I.3.3, where schedule delays are now replaced with mission capability impacts.

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Table 48. Data Elements for Understanding DMSMS Impacts on Operational Availability

Information Type	Level 1 Data Element Title	Level 2 Data Element Title
DMSMS resolution cost-related data elements	• Type of Resolution Approved	
DMSMS management operations efficiency-related data elements	• Case Proactivity Indicator • Reason Issue Was Discovered Reactively • Date Case Closed • Date Implementation Needed	• Effect on Logistics Response Time • Effect on Mission Capability

The inclusion of the logistics response time data element provides further insight. While mission capability is affected by logistics response time, a logistics response time delay may not be sufficient. For example, an analysis of differences in the occurrence of logistics response time and mission capability impacts should lead to finding out about risk of mission capability impacts sooner.

I.3.5 Estimating Supply System Impacts Avoided by Being Proactive

The Level 1 and Level 2 data elements listed in Table 49 help demonstrate the benefits of DMSMS management proactivity on the supply system.

Table 49. Data Elements for Estimating Supply System Impacts Avoided by Being Proactive

Information Type	Level 1 Data Element Title	Level 2 Data Element Title
DMSMS management operations efficiency-related data elements	• Case Proactivity Indicator • Resolution Avoided	• Resolution Avoided Implementation Time • Consumption Rate • Stock on Hand

There is a relationship between backorders in the supply system and operational availability. Depending on the level of safety stock and availability of workarounds such as cannibalization, backorders will eventually lead to decreased operational availability. Therefore, backorders avoided can be a more sensitive indicator of the benefits of risk-based, proactive DMSMS management.

Reactive DMSMS management will lead to a greater number of backorders of the obsolete item than proactive DMSMS management because there is a shorter implementation window from issue discovery to impact. The implementation activities only begin when the supply system is at its reorder point and additional quantities are not available for purchase. Backorders occur if the demand during the implementation period exceeds the stock available at the reorder point.

A backorders avoided (BOA) metric, which would only apply to situations where the case proactivity indicator specifies proactive, should be calculated as follows:

- Determine the estimated quantity of the item that would be required during the time it takes to implement the resolution that would have been chosen had the issue been discovered reactively (resolution avoided data element). This is simply the resolution avoided implementation time data element multiplied by the consumption rate date element.
- Subtract the stock on hand data element to estimate BOA.

A closely related metric is days of supply impact avoided (DSIA). DSIA is BOA divided by the consumption rate data element. For both metrics, when working with cases that involve multiple solutions, calculate the quantities for each item and sum them before performing the final proactive or reactive calculation. Finally, use the same items for both the proactive and reactive calculations.

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Some caveats on the use of both metrics are as follows:

- The metrics are only applicable during sustainment since they are related to availability and the supply system.
- The metrics assume there are no backorders for the proactive resolution. This is the goal of proactive DMSMS management, but sometimes may not necessarily be the case. Coordination between the program office and the inventory manager helps minimize supply availability impacts for the proactive resolution.
- The metrics may portray inaccurate results if the reactive resolution time is less than the proactive resolution time. This could happen in a situation where different assumptions (often in terms of risk tolerance) are employed to resolve a resolution reactively. For example, a proactive resolution might be the use of a complex substitute where there may be some development work and substantial testing. If the reactive resolution involved a waiver of a qualification test (which is also a complex substitute), the resolution time could be substantially less.

I.3.6 Estimating Improvements in Resolution Implementation Time from Being Proactive

Another factor that could have an impact on availability is the time necessary to implement a resolution—the longer the time, the higher the likelihood of an impact. Table 50 show the Level 1 and Level 2 data elements needed to examine the differences between reactive and proactive DMSMS management.

Table 50. Data Elements for Estimating Improvements in Resolution Implementation Time from Being Proactive

Information Type	Level 1 Data Element Title	Level 2 Data Element Title
DMSMS management operations efficiency-related data elements	• Date Case Opened • Date Case Closed • Case Proactivity Indicator • Resolution Avoided	• Resolution Avoided Implementation Time

There are two ways to calculate resolution time decrease.

- When the case is opened proactively (i.e., the case proactivity indicator data element signifies proactive), the resolution time decrease from being proactive is the reactive resolution avoided implementation time minus the actual proactive resolution implementation time. The actual proactive resolution time would be the time elapsed between the data elements for date case opened and date case closed.
- When the case is opened reactively (i.e., the case proactivity indicator data element signifies reactive), the potential resolution time decrease that could be achieved by proactive DMSMS management is the reactive resolution time minus the proactive resolution avoided implementation time. The actual reactive resolution time would be the time elapsed between the data elements for date case opened and date case closed.

For both metrics, the formula varies slightly when working with cases that involve multiple solutions. Determine the implementation times for each case and sum them before performing the final calculation. Also, use the same items for both the proactive and reactive calculations. When aggregating at the program office level, sum the factors for all cases being aggregated and calculate the metrics using those summed factors.

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Some caveats on the use of both metrics are as follows:

- As indicated in the previous section, there may be instances where the reactive resolution implementation time is less than the proactive resolution implementation time.
- The fact that there is a difference between the two implementation times may not have any impact on availability. The key is whether the resolution was implemented before the supply system exhausted its inventory.

I.3.7 Determining an ROI for DMSMS Management

A calculation of an ROI for DMSMS management can be made with the Level 1 and Level 2 data elements itemized in Table 51.

Table 51. Data Elements for Determining an ROI for DMSMS Management

Information Type	Level 1 Data Element Title	Level 2 Data Element Title
DMSMS management and resolution cost-related data elements	• Type of Resolution Approved • Resolution Cost • Source of DMSMS Resolution Cost • DMSMS Management Operations Cost Paid to Prime/OEM • DMSMS Management Operations Cost Paid to Independent SME Organizations • Management Operations Cost for Internal DMSMS-Related Activities	• Value of Management Operations Activities Received at No Cost from a Centralized Service Source
DMSMS management operations efficiency-related data elements	• Case Proactivity Indicator • Resolution Avoided (Cases where issue was found proactively only) • Cost of Resolution Avoided	

It is important to understand that the concept of ROI does not apply to a single case; it pertains to a DMSMS management program. ROI is generally defined as the money earned (return) because of an investment. The formula for ROI is as follows:

$$ROI = \frac{Return - Investment}{Investment} \; x \; 100$$

The data elements in Table 51 include most of the data elements from Appendix I.3.2, which pertains to the cost avoidance of being proactive. It is that cost avoidance of being proactive that equates to the return part of the ROI calculation. Data elements for the total cost of DMSMS management operations are also included in Table 51. They represent the investment part of the equation. Applying the basic ROI formula to DMSMS management should then be the ratio of the two as shown:

$$ROI = \frac{DMSMS\ Cost\ Avoidance - Total\ Cost\ of\ DMSMS\ Management\ Operations}{Total\ Cost\ of\ DMSMS\ Management\ Operations} \; x \; 100$$

I.4 Record Keeping Data Element Dictionary

Table 52 shows the Level 1 DMSMS management and resolution cost-related data elements.

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Table 52. Level 1 DMSMS Management and Resolution Cost-Related Data Elements

Name of Field	Data Collected	Description of Data Collected
Type of Resolution Approved	No solution required	See Table 12 for definitions.
	Approved item	
	LON buy	
	Simple substitute	
	Complex substitute	
	Extension of production or support	
	Repair/refurbishment/reclamation	
	Development of a new source	
	Design refreshment	
	Redesign of the NHA	
	Redevelop the item	
	Redesign complex/system replacement	
Resolution Cost	Cost to develop and implement (exclude LON item costs and costs to procure non-developmental and test items)	Costs related to engineering and design of the solution, include Engineering/Engineering Data Revision, Qualification of New Items, Software Development or Modification, Startup Costs, Tooling/Equipment/Software and one-time costs related to the implementation of the solution: Computer Programs/Documentation, Interim Support, Supply/Provisioning Data, Support/Test Equipment, Technical Manuals, Training/Training Equipment, Installation Costs. The cost of items for LON purchases and of the actual items beyond testing items should not be included. Note: For a multi-phase resolution, capture this information for each phase.
Source of DMSMS Resolution Cost	Actual costs	This item is selected if the development and implementation cost are actual costs.
	Estimated costs	This item is selected if the development and implementation cost are estimated costs.
	Costs from DoD cost tables	This item is selected if the development and implementation cost are derived from Table 15.
Redesign Level (if redesign involved in the resolution)	Component (piece parts, device, commercial item)	A smaller, self-contained part of a larger entity. These are the lowest level items used in an assembly.
	Assembly (card, SRA, SRU), may be a commercial or NDI item)	Assemblies are items built from components.
	Subsystem (boxes, WRA, LRU)	Subsystems are complete functional items built from assemblies, and items.
DMSMS Management Operations Cost Paid to Prime/ OEM	The amount paid to the prime contractor/OEM for DMSMS management operations	DMSMS management costs only. Does not include the cost to resolve DMSMS issues.

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Name of Field	Data Collected	Description of Data Collected
DMSMS Management Operations Cost Paid to Independent SME Organizations	The amount paid to independent SME organizations for DMSMS management operations	DMSMS management costs only. Does not include the cost to resolve DMSMS issues.
Management Operations Cost for Internal DMSMS Activities	Amount to fund internal DMSMS operations	DMSMS management costs only. Does not include the cost to resolve DMSMS issues.

Table 53 shows the Level 1 DMSMS management operations efficiency–related data elements.

Table 53. Level 1 DMSMS Management Operations Efficiency–Related Data Elements

Name of Field	Data Collected	Description of Data Collected
OEM Item Number	OEM item number	The item number assigned by the OEM of the parent assembly of the item. If the referenced item is software or firmware, then the OEM item number refers to the item that the software or firmware is used on.
OCM Item Number	OCM item number	The item number assigned by the original manufacturer of the item being reported.
OEM CAGE Code	OEM CAGE code	The CAGE code of the entity providing the item referenced in the OEM data element.
OCM CAGE Code	OCM CAGE code	The CAGE code is a unique identifier assigned to suppliers to various government or defense agencies, to government agencies, and various organizations. CAGE codes provide a standardized method of identifying a given facility at a specific location.
Nomenclature	Nomenclature	Name of item.
Item Class	Commercial item	Commercial items include any item of a type customarily used by the general public or by non-governmental entities for purposes other than governmental purposes that has been sold, leased, or licensed, or offered for sale, lease, or license to the general public (see the Federal Acquisition Regulation [FAR] 2.101).
	NDI	See FAR 2.101 and definitions section of 48 C.F.R. § 2.101.
	Developmental item	A developmental item (DI) is any item that is not commercial and not NDI.
Case Proactivity Indicator	Reactive	Items found to be obsolete after a failed attempt to purchase them or no bid on repair work because of obsolescence.
	Proactive	Items determined to be obsolete before an attempt to purchase them. For this document, proactively implies the item was found to be obsolete because of a discontinuation notice from any source, such as a predictive tool, a vendor survey, research on the item, or an attempt of an LON buy.

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Name of Field	Data Collected	Description of Data Collected
Reason Issue Was Discovered Reactively (Only When Case Proactivity Indicator Was Reactive)	Not monitored by choice	Items which are not monitored at the program office's choice.
	Vendor survey failed to identify	Items monitored by a vendor survey for obsolescence, which are determined to be obsolete after a failed attempt to purchase them.
	Predictive tool failed to identify	Items monitored by a predictive tool for obsolescence, which are determined to be obsolete after a failed attempt to purchase them.
	Discontinuation notice not received	Items dependent on a notification from the manufacturer or other source, but which are determined to be obsolete after a failed attempt to purchase them.
	Data error	Items proactively monitored, but the monitoring failed because of data errors.
	Other (provide details)	All other circumstances where products are monitored.
Date Alert Received	Date alert received	The date the DMSMS alert was received.
Date Case Opened	Date case opened	The date the case was opened to determine resolution.
Date Resolution Submitted for Approval	Date resolution submitted for approval	The date the resolution for the case was submitted for approval.
Date Case Resolved	Date case resolved	The date a resolution for the case was approved.
Date Case Closed	Date case closed	Date the case was fully implemented. In situations where the implementation requires installation/retrofit over long periods, this can be the date when the development work was complete and the first installation/retrofit is successfully completed.
Date Implementation Needed	Date implementation needed	The date by which a case must be implemented to prevent an impact on the system.
Resolution Avoided	No resolution required	See Table 12 for definitions. *Note:* For a proactive resolution, the resolution avoided is the resolution that would have been implemented had the DMSMS issue been opened reactively at the point in the future where there would have been a failed attempt to purchase the item. For a reactive resolution, the resolution avoided is the resolution that could have been implemented had the DMSMS issue been identified and opened proactively at the point in the past where information such as PDN could have been discovered. When the type of resolution approved is no resolution required, then the resolution avoided is also no resolution required.
	Approved item	
	LON buy	
	Simple substitute	
	Complex substitute	
	Extension of production or support	
	Repair/refurbishment/ reclamation	
	Development of a new source	
	Design refreshment	
	Redesign of the NHA	
	Redevelop the item	
	Redesign complex/ system replacement	

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Name of Field	Data Collected	Description of Data Collected
Cost of Resolution Avoided	Cost to develop and implement (exclude LON costs) the resolution avoided	Costs related to engineering and design of the resolution, include non-recurring Engineering/Engineering Data Revision, Qualification of New Items, Software Development or Modification, Startup Costs, Tooling/Equipment/Software and one-time costs related to the implementation of the solution: Computer Programs/Documentation, Interim Support, Supply/ Provisioning Data, Support/Test Equipment, Technical Manuals, Training/Training Equipment, Installation Costs. The cost of items for LON purchases and of the actual item beyond testing items should not be included.
Subsystems	Total number of subsystems in the system	Total number of subsystems whether monitored or not. Subsystems are items that are built from assemblies and in some cases components. Components are the lowest level items used in assemblies.
Subsystems Monitored	Total number of subsystems in the system being monitored	Total number of subsystems that are monitored.
Components	Total number of unique components	Unique components within the system/subsystem.
Components Monitored	Total number of monitored components	Components that are being monitored within the system/ subsystem.

Table 54 shows the Level 2 DMSMS cost-related data elements.

Table 54. Level 2 DMSMS Cost-Related Data Elements

Name of Field	Data Collected	Description of Data Collected
Commodity Type	Software	Software is a sequence of instructions written in a programming language that computers, microprocessors, microcontrollers, or other processing devices can interpret and execute. Types of software include operating systems, applications, drivers, firmware, and utilities. Market categories of software include commercial, government, open-source, and freeware. Programming languages range from machine code (device language) to 4th and 5th generation languages. Each generation tends to further abstract the language from the device.
	Electronics	Electronic devices control electrical energy by electrically manipulating the flow of electrons. Electronics encompasses electrical circuits that involve active electrical components such as vacuum tubes, transistors, diodes, integrated circuits, associated passive electrical components, and interconnection technologies. Commonly, electronic devices contain circuitry consisting primarily or exclusively of active semiconductors supplemented with passive elements; such a circuit is described as an electronic circuit.
	Electrical/electro-mechanical	Electro-mechanical devices carry out electrical operations using moving parts. Electro-mechanical devices deal with the generation, distribution, switching, storage, and conversion of electrical energy to and from other energy forms using wires, motors, generators, batteries, switches, relays, transformers, resistors, and other passive components.
	Mechanical	Mechanical devices are machines or parts of machines which are primarily related to or controlled by physical forces. Examples include bearings, machined devices, castings, valves, screws, bolts, panels, and so forth.

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Name of Field	Data Collected	Description of Data Collected
	Materials	Materials are the substance or substances of which a thing is made or composed. Materials can include raw, refined, or manufactured items that are used in the manufacture of other items. Examples include glues, metal, fabric, minerals, gases, liquids, chemicals, paints, and so forth.
Domain Where Equipment Operates	Space	Items that are used on equipment that operates in the space domain.
	Air	Items that are used on equipment that operates in the air domain.
	Land	Items that are used on equipment that operates in the land domain.
	Maritime or Sea	Items that are used on equipment that operates in the maritime or sea domain.
	Cyberspace	Items that are used on equipment that operates in the cyberspace domain.
Product Acquisition Cost	Cost of items purchased as part of a resolution	The total cost of actual items purchased for an LON buy resolution or for other resolutions where some quantity of the new items was purchased in conjunction with any non-recurring engineering and testing to develop that new item (e.g., the funded resolution may have been reverse engineering but a quantity of the new reversed engineered item was purchased at the same time).
DMSMS Item Type	Component (piece parts, device, commercial item)	A component is a smaller, self-contained part of a larger entity.
	Assembly (card, SRA, SRU, may be a commercial item) or NDI	Assemblies are items built from items.
	Subsystem (boxes, WRA, LRU)	Subsystems are complete functional items built from assemblies and items.
Value of Management Operations Activities Received at No Cost from a Centralized Service Source	Value of resources provided from a centralized command source	DMSMS management costs only. Does not include the cost to resolve DMSMS issues.
Root Cause	Technological Obsolescence	Technological progress causing diminishing demand, and hence supply, of parts based on older technology.
	Loss of Supplier	A supplier stops producing a part/material for business reasons such as the bankruptcy of the supplier or a change in business strategies.
	Functional Obsolescence	An item, although still available commercially, no longer functions as intended because of hardware, software, and/or requirements changes to the system.
	Offshoring	Parts or materials have been moved offshore making the product unavailable for use.
	Policy	Parts or materials are no longer available due to foreign or domestic policy.
	Materials	Materials used directly in a part's manufacturing process are no longer being produced, affecting the ability to procure the item. Note this does not cover material shortages due to policy, environmental, or safety concerns.

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Name of Field	Data Collected	Description of Data Collected
	Environmental/ Safety	Materials used in a part's manufacturing process are no longer being produced due to a deliberate choice related to environmental or safety concerns. Note this does not cover material shortages due to policy or trade concerns.
	Process Change	A part/material is still being produced, but the process to produce it has changed, requiring requalification before it can continue to be used.
	Other	All other root causes.

Table 55 shows the Level 2 DMSMS management operations efficiency–related data elements.

Table 55. Level 2 DMSMS Management Operations Efficiency–Related Data Elements

Name of Field	Data Collected	Description of Data Collected
LON Buy Preferred Indicator	Yes	This should be limited to situations where an LON would have been the best approach but was unavailable (e.g., we were notified too late, there was no budget, procurement time took too long, and so forth).
	No	
Monitoring Techniques (Only When Case Proactivity Indicator Was Proactive)	Vendor survey	The need for a DMSMS case was established by a monitoring process where the manufacturer of the item is contacted directly to determine the production status of a part.
	Predictive tool	The need for a DMSMS case was established by a monitoring process where the production status of a part is determined using a predictive tool.
	DLA notice	The need for a DMSMS case was established by obtaining a notification from DLA.
	GIDEP notice	The need for a DMSMS case was established by obtaining a notification from GIDEP.
	OEM notice	The need for a DMSMS case was established by obtaining a notification from the prime contractor or component manufacturer.
	Prior LON buy inadequate quantity	The need for a DMSMS case was established when the items purchased for a previous LON buy solution did not last as long as planned.
Effect on Production Schedule	Number of months of schedule slip	Months of production delay for fielding a capability because of the issue that initiated the DMSMS case.
Effect on Logistics Response Time	Number of days of extended logistics response time	Days of logistic response time increase because of the issue that initiated the DMSMS case.
Effect on Mission Capable Rate	Percentage change in MICAP rate	Percentage points that MICAP rate decreased because of the issue that initiated the DMSMS case.

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Name of Field	Data Collected	Description of Data Collected
Resolution Avoided Implementation Time	Estimated time to implement the resolution avoided	Days required to implement the resolution avoided (either a proactive one or a reactive one), i.e., the total time required to close a DMSMS case. Most DMTs consider a DMSMS case closed when all activities are complete, including changes to drawings and technical manuals. When a resolution will take a long time to complete, for instance, when it is being back fitted by attrition, the DMSMS case can be considered closed when the initial installations are complete and planning for future installations are in place. This data element should be determined based on engineering judgement. In the absence of other data, Table 56 may be used with caution to estimate the implementation time based on the resolution avoided. The information in Table 56 was derived from an analysis of 39,535 different cases. While this table reflects actual case resolution times, the impact and the impact timeline also influence the results. For example, if an issue were predicted to occur in five years, the personnel working this DMSMS case may elect to defer resolution actions, especially if a preliminary assessment indicates a simpler solution or no sense of urgency. Furthermore, the criteria for case closure were not defined. For example, some DMTs may consider closure at the approval point, others at contract implementation and others at complete documentation update (e.g., technical manuals, engineering drawings, etc.).
Consumption Rate	Demand per quarter	The expected quarterly demand for the obsolete item based on all systems using the item in question, not just the program office maintaining records.
Stock on Hand	Stock on hand in the supply system	Estimated stock that would be on-hand in the supply system at the time that the DMSMS issue was reactively identified. Typically, this is the reorder level including all safety stock.

Table 56. Mean Days to Close a DMSMS Case

Solution Type	Mean
Approved Part	179
Life of Need Buy	187
Simple Substitute	136
Complex Substitute	225
Extension of Production or Support	440
Repair, Refurbishment, or Reclamation	265
Development of a New Source[246]	190
Design Refreshment194	190
Redesign—NHA	301
Redevelop the Item194	190
Redesign—Complex/System Replacement	644

[246] As described in Section 6, the resolution “Development of a New Item or Source” has been replaced by three resolution types. The mean time data shown in this table were based on that previous resolution type.

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Appendix J. Considerations for Acquiring or Building a BOM

J.1 Introduction

This appendix focuses on the portion of the *Identify* step (Section 4, more specifically Section 4.3.1.2.1) that introduces the topic of obtaining BOMs. The objective of this appendix is to highlight considerations for a program office that is faced with acquiring (when not contractually required up front) or developing BOMs to support proactive DMSMS management.

The appendix contains five sections. First, there is a section that addresses precursor considerations that apply to BOMs. More specifically, these considerations include IP and the importance of ensuring that there is a business case that supports the acquisition or development of a BOM as described in the second section. The third section then documents best practices for acquiring a BOM if it was not delivered on contract.[247] The fourth section documents best practices for creating or building a BOM. Finally, the fifth section describes processes for identifying and correcting gaps in BOM data.

The Annex to Appendix J shows potential data elements for a BOM in priority order.

J.2 Precursor Considerations

J.2.1 Intellectual Property

The PM is responsible for creating an IP strategy at the program office's inception as part of the acquisition strategy and for including it at milestone B (or a similar decision point) as part of the LCSP. The IP strategy and the DMP should be in alignment respective to BOMs (or other technical data) and contract language to support the appropriate use rights to ensure DoD is able to use the data in the manner needed.[248] Obtaining a BOM, either on the original development contract or after the fact, does not mean that DoD can automatically use the BOM for DMSMS management purposes. There must also be an agreed upon determination of DoD's use rights, which could entail restrictions on how DoD can use the information. To act upon its use rights, DoD must obtain the data via a contract deliverable.

J.2.1.1 Establishing a Contractual Requirement for Data Access

Establishing a contract requirement to acquire the data is therefore, the first step in obtaining a BOM.[249] All DoD contracts must specify the technical data to be delivered under a contract and delivery schedules for the data.[250] There are several ways in which the requirement can be established. Typically, the contract requirement is explicitly expressed as a CDRL and an associated DID. Without such an explicit

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[247] This document is focused on BOMs for non-commercial hardware. Commercial item monitoring is usually focused on the item itself and is carried out via vendor surveys.

[248] DoD Instruction 5010.44, Intellectual Property (IP) Acquisition and Licensing, October 16, 2019, p.10

[249] If a company objects to a technical data requirement in a contract, the program office and its contracting officer must resolve the situation during the contract negotiation process. Once the contract is signed, there have been instances where companies refused to deliver the technical data called for via the requirements on a contract because of IP concerns. It is up to the individual program office to determine whether to dispute this. If there was a contract requirement to deliver the data, the contractor does not have much leverage. More than likely, the contractor would have negotiated the requirement out of the contract if the issue were important enough.

[250] 48 C.F.R. § 252.227-7013, available at https://www.ecfr.gov/current/title-48/chapter-2/subchapter-H/part-252/subpart-252.2/section-252.227-7013.

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requirement, the government may compel the contractor to deliver data up to three years after the acceptance of all items called for in the contract under the FAR "Additional Data Requirements"[251] or the Defense Federal Acquisition Regulation Supplement (DFARS) "Deferred Ordering"[252] clauses. Lastly, the government may include a specifically written contract clause for access to data.[253]

Any of these requirements can be established in the original contract or in a contract modification. Absent one of these contractual mechanisms to obtain a BOM, the contractor is not required to provide it to the government, regardless of use rights, because data delivery requirements may have a cost associated with the packaging and delivery of the data.

J.2.1.2 Ensuring the Government's Rights to Use Data Matches the Program Office's Needs

The second step is to ensure that government rights to use the data meet a program office's needs. Three types of technical data rights are most pertinent to contract requirements to deliver data needed for DMSMS management[254] as described in 10 U.S.C. § 3771(b):

- ***Unlimited Rights***. "…in the case of an item or process that is developed by a contractor or subcontractor exclusively with Federal funds (other than an item or process developed under a contract or subcontract to which regulations under section 9(j)(2) of the Small Business Act (15 U.S.C. § 638(j)(2)) apply), the United States shall have the **unlimited right** (*emphasis added*) to- (A) use technical data pertaining to the item or process; or (B) release or disclose the technical data to persons outside the government or permit the use of the technical data by such persons."[255]
- ***Limited Rights***. "…in the case of an item or process that is developed by a contractor or subcontractor exclusively at private expense," the government's data rights may be restricted to **limited rights** (*emphasis added*). In that case, the government cannot "release or disclose the technical data outside the Government, use the technical data for manufacture, or authorize the technical data to be used by another party"[256] without the written permission of the contractor.
- ***Government Purpose Rights***. "…in the case of an item or process that is developed in part with Federal funds and in part at private expense, the respective rights of the United States and of the contractor or subcontractor in technical data pertaining to such item or process shall be established as early in the acquisition process as practicable…the United States shall have **government purpose rights** (*emphasis added*) in technical data pertaining to an interface between an item or process and other items or processes" except as provided in other paragraphs in this section…"the United States shall have **government purpose rights**

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[251] 48 C.F.R. § 52.227-16, available at https://www.ecfr.gov/current/title-48/chapter-1/subchapter-H/part-52/subpart-52.2/section-52.227-16.

[252] 48 C.F.R. § 252.227-7027, available at https://www.ecfr.gov/current/title-48/chapter-2/subchapter-H/part-252/subpart-252.2/section-252.227-7027.

[253] DFARS 227.7108 (48 C.F.R. § 227.7108, available at https://www.ecfr.gov/current/title-48/chapter-2/subchapter-E/part-227/subpart-227.71/section-227.7108) provides some guidance on contractor data repositories that relate to access. Even beyond the question of government use rights, access to DMSMS management data is problematic. Only accessing BOM data without taking delivery of the BOM itself can be problematic if the data is in a form that cannot readily be input into a DMSMS monitoring tool. Consequently, DoD would have to possess the data in a usable format at some point in the process. Furthermore, access would normally be limited to the duration of a contract. DoD uses BOMs throughout the life cycle of the subsystem, not the period of performance of the contract. There is no standard contract clause governing rights to data that is accessed but not formally delivered.

[254] Specifically negotiated license rights allows government purpose rights to be tailored. For commercial items where the government funded none of the development, standard commercial license rights apply. Restricted rights associated with computer software documentation are not discussed.

[255] 10 U.S.C. § 3771(b), available at https://www.govinfo.gov/link/uscode/10/3771.

[256] Ibid.

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(*emphasis added*) in technical data pertaining to a modular system interface developed exclusively at private expense or in part with Federal funds and in part at private expense and used in a modular open system approach pursuant to section 4401 of this title" except as provided in other paragraphs in this section.[257] Government purpose rights implies that the government has rights to "use, modify, reproduce, release, perform, display, or disclose technical data within the Government without restriction; and release or disclose technical data outside the Government and authorize persons to whom release or disclosure has been made to use, modify, reproduce, release, perform, display, or disclose that data for United States Government purposes."[258]

It might seem that it is easy to determine whether an item or process is developed in whole, in part, or not at all with federal funds. That is not the case. There can be disagreements over what it means to develop an item.[259] For example, there have been situations where the contractor developed a breadboard design with private funds and the government funded the update to a design that can be manufactured, but did not obtain unlimited rights. Also, developed exclusively at private expense encompasses "costs charged to indirect cost pools, costs not allocated to a government contract, or any combination thereof."[260] That means that both independent R&D and bid and proposal funds are included under private expense even though some of those costs are recoverable.

Consequently, government contracts normally include an assertions table which identifies:

- The technical data to be furnished with restrictions,
- The basis by which the restrictions have been asserted, and
- The asserted rights category.[261]

The assertions table may be limited to the data required by CDRL. However, the contractor often includes everything known at the time an offer is made to the government. The assertions table may be revised when new data is generated in the performance of the contract. If the government chooses to exercise the deferred ordering clause, the assertions table may be updated to reflect this request as part of a contract modification.[262]

Assertion tables are normally prepared at the lowest practical level (doctrine of segregability). That means that the data rights for an assembly are defined by the sum of the data rights associated with the subassemblies. The government therefore may only have limited rights to certain subassemblies that are essential to the performance or maintenance of the assembly.

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[257] Ibid.

[258] 48 C.F.R. § 252.227-7013, available at https://www.ecfr.gov/current/title-48/chapter-2/subchapter-H/part-252/subpart-252.2/section-252.227-7013.

[259] Developed is defined in 48 C.F.R. § 252.227-7013(a)(7), available at https://www.ecfr.gov/current/title-48/chapter-2/subchapter-H/part-252/subpart-252.2/section-252.227-7013.

[260] 48 C.F.R. § 252.227-7013(a)(8), available at https://www.ecfr.gov/current/title-48/chapter-2/subchapter-H/part-252/subpart-252.2/section-252.227-7013.

[261] 48 C.F.R. § 252.227-7017, available at https://www.ecfr.gov/current/title-48/chapter-2/subchapter-H/part-252/subpart-252.2/section-252.227-7017.

[262] Deferred ordering applies only to data generated in the performance of the contract. The contractor is compensated for converting the data into a prescribed form, reproduction, and delivery.

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A program office may challenge[263] the rights asserted in the assertions table.[264] The challenge can literally occur at any point in time.[265] The steps in the process include the following:

- A pre-challenge request for information by the government contracting officer.
- A formal challenge by the government contracting officer.
- The contractor response (considered a claim under the Contracts Disputes Act). Contractors are required to maintain records (time sheets, project records, engineering records, and so forth) to justify the validity of restricted data rights.[266]
- The contracting officer's final decision.
- Contractor appeal or suit.

When there is a strong disagreement regarding asserted rights, litigation may be the only way to resolve the situation. Program offices often choose not to recommend litigation over data rights and inform the contracting officer that the government should accept limited use rights. Reasons for this decision include the time and money involved to litigate, the difficulty in convincing multiple levels of management to support the litigation[267] (sometimes in the presence of contractor lobbying efforts to avoid litigation), maintaining a good working relationship with the contractor, and the absence of immediate consequences to the current PM.

To avoid disagreements, the government should understand what the use rights on technical data will be before the data is delivered with a marking on it. It is therefore a best practice for the government and the contractor to come to a common understanding about use rights to avoid misunderstandings after the data is delivered. In fact, where possible, the results of that common understanding should be specified in the contract CDRL that calls for the data.

Use rights are identified by the contractor's marking on the data.268 When the clause at 48 C.F.R. § 252.227-7013 is placed on a contract and the contractor "desires to restrict the government's rights in technical data [the contractor must] place restrictive markings on the data."[269]

Unlimited rights or government purpose rights are more than sufficient for DMSMS monitoring (and resolution) purposes. Limited rights are problematical because many DoD program offices use commercial predictive tools to determine the obsolescence status of items being monitored. At issue is whether the use of a commercial predictive tool implies that the data is being released "outside the

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[263] IP challenges virtually never are concerned with who owns the data. The contractor always retains title to the data according to FAR and DFARS clauses. Only the terms of the government's license to use the data are at issue.

[264] 48 C.F.R. § 252.227-7037, available at https://www.ecfr.gov/current/title-48/chapter-2/subchapter-H/part-252/subpart-252.2/section-252.227-7037.

[265] Data rights for products or processes do not automatically change over time or expire. There are instances where the original manufacturer has gone out of business or no longer has any interest in selling the product or process. Under these or similar circumstances, DoD can only change data rights with the permission of the company (or its traceable successor company or escrow agent) that originally established the data rights.

[266] 48 C.F.R. § 252.227-7013, available at https://www.ecfr.gov/current/title-48/chapter-2/subchapter-H/part-252/subpart-252.2/section-252.227-7013.

[267] Ultimately, the Department of Justice would have to agree to litigate.

[268] Per 48 C.F.R. § 227.7103-10(c)(1) (available at https://www.ecfr.gov/current/title-48/chapter-2/subchapter-E/part-227/subpart-227.71/section-227.7103-10), "Technical data delivered or otherwise provided under a contract without restrictive markings shall be presumed to have been delivered with unlimited rights." There also are instances where companies provide technical data with improper markings (e.g., proprietary). In that situation, the government should challenge the markings, but in the meantime the government must act as if there are limited use rights.

[269] 48 C.F.R. § 227.7103-10(b)(1), available at https://www.ecfr.gov/current/title-48/chapter-2/subchapter-E/part-227/subpart-227.71/section-227.7103-10.

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government." Several arguments can be made for why the use of commercial predictive tools is allowed under limited data rights.

- Such tools take a parts list with the manufacturer's part number as input. If this is the only information being input to a predictive tool, and the parts list is not associated with anything, then an argument can be made that the BOM itself is only being used inside the government.
- A covered government support contractor is exempted from the limited rights restriction "outside of government." A covered government support contractor is defined as a contractor under a contract where "the primary purpose of which is to furnish independent and impartial advice or technical assistance directly to the Government in support of the Government's management and oversight of a program or effort (rather than to directly furnish an end item or service to accomplish a program or effort)."[270] It can be argued that the use of a commercial predictive tool is technical assistance directly to the government.
- Technically, when using predictive tools, the government is not releasing the data to anyone. The predictive tool provider would assure the government that its data is secure. Compromising the data would therefore require an overt act by the predictive tool provider.
- There are two exceptions that allow the government to provide technical data for use by an outside contractor: 1) when the data include only F3 information (along with qualification requirements) and (2) when the data is necessary for operations, maintenance, installation, or training (other than detailed manufacturing or process data). An argument can be made that the data is necessary for O&M.[271]

None of these arguments has been tested in court. Consequently, if the government only has limited rights to a BOM, the government may want to consider obtaining contractor permission to use the data in a predictive tool.

J.3 Ensuring There Is a Business Case to Acquire or Build a BOM

The "Acquire or Build the BOM" blocks in Figure 29 implicitly include a BCA to determine whether it is worthwhile in a situation where it has been determined that it is desirable to proactively monitor a subsystem per the above criteria, but the subsystem BOM has not previously been obtained via contract. Such a BCA should consider the following:

- How much would it cost to acquire or construct the BOM? Cost encompasses both obtaining the data and converting the data into a form that is usable by predictive tools.
- How severe are the risks if the subsystem were monitored reactively? Proactive DMSMS management increases the window of opportunity to implement a resolution to a DMSMS issue. The longer the window of opportunity, the greater the likelihood that 1) the issue will not impact a production schedule or readiness and 2) the cost to implement the resolution will be lower. Reactive DMSMS management minimizes the window of opportunity. The second factor is an assessment of the likelihood of a severe negative repercussion for the particular types of items found in the subsystem. Considerations in making this assessment include the following:

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[270] 48 C.F.R. § 252.227.7013(a)(5) , available at https://www.ecfr.gov/current/title-48/chapter-2/subchapter-H/part-252/subpart-252.2/section-252.227-7013.

[271] Although not the subject of this document, this footnote provides a brief discussion of the implications of limited data rights on implementing DMSMS resolutions. (There are no implications for unlimited or government purpose rights.) Limited rights data cannot be used to support resolutions unless the resolution will be implemented using only government or government support contractors. This is very unusual. The exception that allows the government to provide technical data for use by an outside contractor to implement a resolution when the data is necessary for operations, maintenance, installation, or training (other than detailed manufacturing or process data) may be sufficient. Unfortunately, the terms in the exemption are not defined in law. Furthermore, a contractor can legally challenge government release of limited use data under either of these exceptions because there can be disagreements on whether the conditions were met. Such a challenge may result in a lawsuit.

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- What are the items in the subsystem where a longer window of opportunity will significantly reduce the likelihood of a severe negative outcome (i.e., unrepairable and/or expensive items where the time to implement a resolution is lengthy)?
- What is the likelihood that there will be a DMSMS issue associated with any of these items within their LON? Some factors affecting this likelihood include the life cycle of the underlying technology, reliability, and supply chain vulnerability. The likelihood should be considered over the entire period that the subsystem will be operational. An event that is unlikely to occur in the next year may be much more likely to occur within five years. Furthermore, if more than one item is under consideration (from the first bullet), then the likelihood of multiple events should be taken into account. If all these likelihoods are vanishingly small, then it may not be worthwhile to attempt to acquire or construct the BOM. On the other hand, if the likelihoods are large or even small but not insignificant, then acquiring the BOM data should be considered further.
- How large is the negative outcome likely to be, measured in dollars, relative to the cost to acquire or construct the BOM, if a DMSMS issue is identified reactively, rather than proactively? If it's relatively inexpensive to acquire or construct the BOM data, then it may be worthwhile to pursue it. If the possible cost of a severe negative repercussion is very high as compared to the cost of acquiring the BOM then a decision maker may try to acquire or construct the data even when the likelihood of a DMSMS issue is low.

- To what extent would the acquired or constructed BOM reduce the risks? The third factor affecting the decision is whether acquiring the BOM data will sufficiently reduce the risk of a severe negative repercussion. Different ways of acquiring the data have different degrees of risk reduction. Risk reduction is affected by the level of detail (completeness) that will be available in the BOM, the currency of the data, and the accuracy of the data. For example, if it is unlikely that the acquired or constructed BOM will contain enough information to proactively monitor the items of concern, it may not be worthwhile to pursue the BOM.

J.4 Acquiring a BOM

If a BOM was not delivered on contract, it is a best practice to attempt to acquire the BOM before trying to build it. Figure 29 shows a process for doing that.

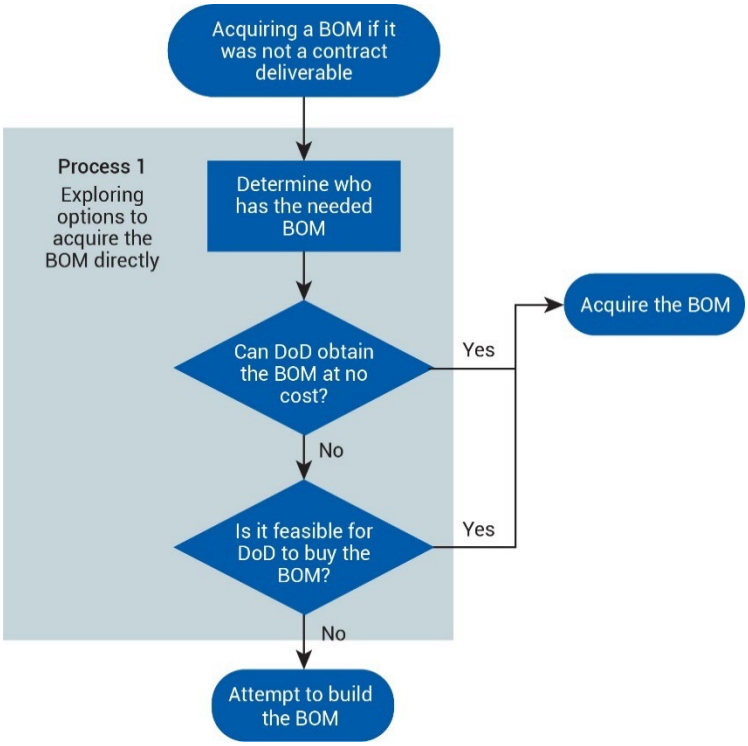
- ***Acquiring a BOM if it was not a contract deliverable***. In this context, the term *BOM* refers to an engineering BOM or EBOM. The EBOM shows the items and materials used along with their OEM CAGE code/manufacturer's part number. The SD-26 references the DID that should be used when purchasing a BOM on a development contract. To some extent when acquiring a BOM using the Figure 29 process (but much more so when building a BOM using the process in the next section of this document) it is not necessary to pursue all the data elements with equal priority. Section 4.3.2.1 identifies potential data elements to pursue, indicates which of those data elements are essential, and suggests levels of desirability for the non-essential ones.

  Two outcomes are possible from the Figure 29 process: the data are obtained or there is an attempt to build a BOM using the process in the next section.
- ***Determine who has the needed BOM data***. Unless the subsystem is part of a legacy system where there is no industry support, it is likely that some commercial entity will have a BOM. The prime may have the data (if the prime is the OEM or has obtained the data from the OEM). Of course, the OEM (or its traceable successor company or escrow agent) should have it. If the item is government furnished, then the data may reside within another program office or if not, with the OEM that produced the item for that program office. Lastly, it is also possible that another DoD program office already has a BOM.
- ***Can DoD obtain the BOM at no cost?*** There are several aspects to this question.
  - First is the case where another program office has already obtained the same (or even a similar) BOM and the program seeking to obtain a BOM can possibly acquire it from that

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other program office. There are no data rights issues when the government program office with the BOM has unlimited or government purpose rights to the data. Limited data rights do not preclude one government program office from giving a BOM that it purchased to another government program office that uses the same or a similar subsystem. Under limited data rights, it is a best practice for there to be a formal agreement between the two program offices that includes a statement of the data rights under which the receiving program office must operate.

Figure 29. A Process for Acquiring a BOM If It Was Not Delivered on Contract



- Second is the situation where the DMSMS management service provider works directly with the owner of the data. Sometimes the data may be obtained by simply asking for it. Explaining that what is being requested is not the delivery of the entire TDP, but rather specific data fields as outlined in DI-MGMT-82274, DMSMS Life Cycle Management Data, or by arguing that it is important for the government to have the data for national security purposes. DoD may also agree to provide its DMSMS monitoring findings to the OEM to help it manage its product in return for the BOM. There is no legal basis for these types of agreements, but they occur in practice. If such an agreement to obtain the data is reached, the people involved should honor the terms of the agreement which would normally be restricted to DMSMS monitoring purposes.[272] The program office can also solicit support from the Defense Contract Management Agency, who may be able to obtain the BOM from the owner of the data. In these situations, providing the BOM is solely at the discretion of the contractor. The contractor may terminate the agreement at any time.
- A third possible scenario (which is similar to the second) occurs when the contractor wants to implement an ECP or obtain a waiver or deviation on the DMSMS item. The government may need a BOM to approve the change. The rights conveyed on the use of the BOM would be determined by the markings on the BOM. As indicated earlier, the

[272] In some instances, companies have asked for a non-disclosure agreement that specifies how the data can be used.

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contractor should not be able to mark the BOM in any way that it wants. There should have been a discussion and agreement between the contractor and the government on what the markings will be.

- A fourth scenario is the situation where there was no disagreement that the government owns the data (i.e., it was developed entirely at federal expense). If the data exist, the government can ask to see it and copy it.

- ***Is it feasible for DoD to buy the BOM?*** There are multiple dimensions to the word "feasible."
  - Was there an option to buy in the contract, and can that option be exercised?
  - Is the data owner willing to sell it? If the data owner learns that DoD will be conducting a physical product audit (PPA) to construct a BOM or considering a reverse engineering project, then the data owner may become more willing to provide the data at a lower cost.
  - If DoD can buy the data, does it have the money and a contracting vehicle to do so? One approach for a contracting vehicle is a contract action to perform an ECP. The associated (modification of a) contract could be an opportunity to add a CDRL to purchase the data. If it happens that the ECP is on the assembly where the data are needed, then adding the CDRL in the contract is even easier. As previously indicated, the usage rights must also be stipulated.

If a program is not able to convince an OEM to share or offer for purchase, the required BOM data and it is not available through another source (e.g., other government program office), then a program office may need to consider building a BOM or BOMs to enable proactive DMSMS management.[273] In addition, a DMT should plan to leverage future redesigns and modifications to upgrade system capability as opportunities to include the requirement for BOMs as a contract deliverable for those new efforts.

J.5 Building a BOM

Figure 30 addresses a process for building a BOM.

The BOM building process first attempts to use drawings, then provisioning data, then manuals, and ultimately may rely on a PPA. This ordering of the sources of technical data goes from greatest to least detail. The drawings purchased may not be complete and may not be at a low enough level of detail. Provisioning data are limited to those items to be purchased through the supply system and often may not have a sufficient level of detail about subcomponents. Manuals usually do not provide the appropriate level of detail either. PPAs can be expensive and depending on how they are conducted, may have even less detail because part numbers may not be visible.[274] If the PPA is not destructive, the resulting BOM will miss many of the items. If the PPA is destructive, it will be even more expensive. Every one of these technical data sources may also be out of date.

Usage rights on a constructed BOM are defined by the usage rights of the technical data sources. If limited rights data were used, the government's rights to use the resulting BOM would be similarly limited. Usage rights apply to drawings and manuals. The government would decide on the use or disclosure of provisioning data because it belongs to the government.

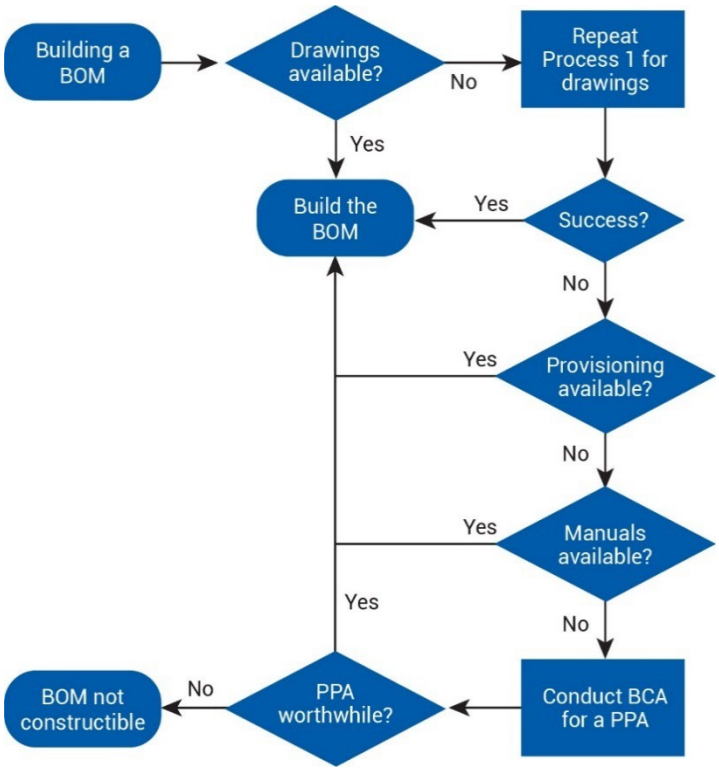
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[273] Another possibility is that the supplier who owns the BOM is willing to perform the proactive DMSMS management functions desired by the program office in an acceptable way and agrees to share the results with the government. In this situation, it may not be necessary to build a BOM.

[274] If a PPA is performed, the program office should ensure that it uses the current version of the item. Older versions may contain obsolete components that have more recently been eliminated.

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Figure 30. A Process for Building a BOM



In general, there are no usage rights limitations on BOMs obtained from a PPA (or TDPs obtained through reverse engineering).[275] A key exception is for patented articles where financial penalties may be imposed.[276] Since markings are required on any patented article, any patent identified in a PPA or a reverse engineering effort should be analyzed to ensure that there is no patent infringement.

- ***Drawings available?***[277] Reaching this block of Figure 30 implies that the initial application of Process 1 (who has the data needed?) did not result in obtaining a BOM. It assumes that the next step is to try to build a BOM from drawings, assuming that the drawings were delivered to DoD and are at the appropriate level of detail. The difference between access and possession could also be important here because predictive tools require input to be formatted properly. In addition, there may be some ambiguity whether access is sufficient because the drawings that DoD accesses may be marked proprietary (and consequently, building a BOM may not be an allowed use). In addition, part of this process step includes checking whether the assembly where a drawing is needed has been provisioned elsewhere. Another program may have drawings and be in a position to share under the same conditions as discussed above for sharing of BOMs.
    - Drawings provide a hierarchical representation of items in a system. The top-level drawing shows the major subsystems and lower-level drawings give details about the items in that subsystem ultimately down to the level of assembly (or component) applicable to the specific situation for which a CAGE code and part number are provided. That lowest level is dependent on what was purchased and the repair or maintenance

[275] Some procurement contracts (more so with commercial items) prohibit reverse engineering of the items being bought.

[276] 28 U.S.C. § 1498, available at https://www.govinfo.gov/link/uscode/28/1498, establishes that financial penalties may be imposed on U.S. government infringements of patents issued by the U.S. Patent and Trademark Office.

[277] This question is asked from the perspective of building a BOM using drawings, not from the perspective of how drawings can support recommending a resolution.

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strategies for the subsystem or assembly. For example, if the assembly were to be replaced and never repaired, then having the drawings stop at the assembly level would be sufficient. There may be situations where a repair could be feasible if obsolescence on one part were resolved. Typically, more detailed information would be sought only after the assembly is obsolete.[278] Various types of drawings may be found at that lowest level. It could be a source control drawing which are mainly used to provide CAGE code and part number for sole source items. Vendor item control drawings are typically found for items that can be purchased from multiple sources. These drawings identify potential suppliers and include quality and performance requirements as well as interchangeability data. Specification control drawings give performance characteristics and physical dimensions of the item.

- ***Repeat Process 1 for drawings***. This step is included in Figure 30 for completeness. If Process 1 did not lead to obtaining a BOM, then it is very unlikely that it will be successful in obtaining drawings (less likely to be given, since drawings contain even more detailed data and the cost of drawings will almost certainly be more than the cost of a BOM). The one mitigating factor is that drawings have many users beyond the DMSMS management community in a program office. Therefore, there may be more leverage for Process 1 to be successful.
- ***Provisioning data available?*** If a BOM could not be constructed using drawings, the next option is to attempt to use provisioning data. Provisioning data encompasses the range and quantity of support items necessary to operate and maintain an end item of materiel for an initial period of service.[279] All DoD systems have provisioning data to the extent that they are supported by the DoD supply system. This will be limited for situations where sustainment is being implemented through CLS or PBL. Also, if systems are provisioned at high levels of assembly, then generally there will not be a sufficient level of detail to build a BOM, although, another system may be provisioned at a lower level of detail and therefore may be a source for the data necessary to build a BOM assuming DoD has sufficient rights to enable the sharing of that data.

Given the potentially substantial benefit to the DMSMS management community from the provisioning process, there should be closer relationships between these two communities. It may be useful to include provisioners and item managers in a program office's DMT. The DMSMS management community should also encourage provisioners to include as much lower-level detail as possible.

- ***Manuals available?*** While DoD normally has maintenance and repair manuals consistent with its support strategy, these manuals may not be kept up-to-date and may not reflect the most current support strategy. In addition, these manuals often do not have the level of detail necessary to construct a BOM for all subassemblies, especially the ones that are provisioned at a high level of assembly. Lastly, the questions about sufficiency of access versus possession discussed for drawings also apply here. Process 1 is not applicable; manuals are generally not available after the fact. Manuals are normally created because the customer buys them.

  There can be situations where a contractor is providing sustainment and consequently there may not be much government provisioning data, but repair and maintenance manuals exist and provide some data down to the level of repair. It is also possible for the provisioner to include data below the level of repair. In general, every potential source of technical data should be considered, if a BCA indicates that it is worthwhile to expend the effort.
- ***Conduct BCA for a PPA***. The purpose of this BCA is to determine whether a PPA is worthwhile. Appendix J.2 described BCA considerations. Those considerations also apply to

[278] It is uncertain whether a manufacturer would provide data years after the original contract work is completed.

[279] DoDM 4140.01, DoD Supply Chain Materiel Management Procedures: Demand and Supply Planning, Volume 2, November 9, 2018.

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drawings, provisioning data, manuals, as well as attempts to buy the data. The BCA is mentioned explicitly for a PPA for two reasons: 1) a PPA may be expensive and 2) it may not provide a sufficient level of detail. PPAs are often nondestructive, because there are additional expenses associated with requirements to restore the assembly back to a serviceable condition. If the assembly was obsolete, then it may be a very precious asset since there may be very few in the supply system. Even if it is not obsolete, the assembly could be very expensive to replace. A nondestructive PPA is limited to items that can be readily seen and identified. Items on a card may be visible, but the identification markings may be hidden. Typically, only 50%–60% of the items can be determined from a non-destructive PPA. A destructive PPA will increase accuracy and nearly all the items should be able to be determined.[280]

J.6 Identifying and Correcting Gaps in BOMs

J.6.1 Identifying Gaps in BOMs

Figure 31 expands upon the "Identify gaps in BOM data" block in Figure 11 to determine the adequacy of either a delivered BOM or a constructed one.

The principal blocks in Figure 31 are as follows:

- ***Conduct high-level risk analyses***. Whenever a BOM (or any other DMSMS management–related contract deliverable for that matter) is delivered on a contract, or acquired through the processes in Figure 29, or a BOM is built using the processes depicted in Figure 30, a high level analysis is conducted to determine its acceptability.[281] Some key acceptability criteria with respect to its adequacy are as follows:
  - Whether it is complete (i.e., are there any major omissions).
  - Whether it is structurally and hierarchically sound. For an indentured BOM, structural errors imply that an item is both the parent and child of another item.
  - Whether it is up to date.
  - Whether key item information is missing or in error (e.g., CAGE code, manufacturer's part number, nomenclature).[282]

  A determination of acceptability usually happens quickly. Unfortunately, a program office's ability to assess these issues is limited at the point of time that the deliverable is received. While missing data elements are relatively easy to spot and structural soundness can be determined from an automated analysis, it may not be clear if there are major omissions, it is up-to-date, or there are data errors. Sometimes these latter three concerns can be identified by inconsistencies among documents (e.g., BOMs, tombstone charts, drawings). However, finding such inconsistencies requires a more extensive examination of data that does not usually occur before a potential obsolescence situation is detected.

  If no gaps are detected, the Figure 29 process is complete. In that case, the next steps would be to determine what items to monitor using a risk-based process and then load the selected items into a predictive tool and/or commence vendor surveys as appropriate. When gaps are identified, then the Figure 29 process continues.

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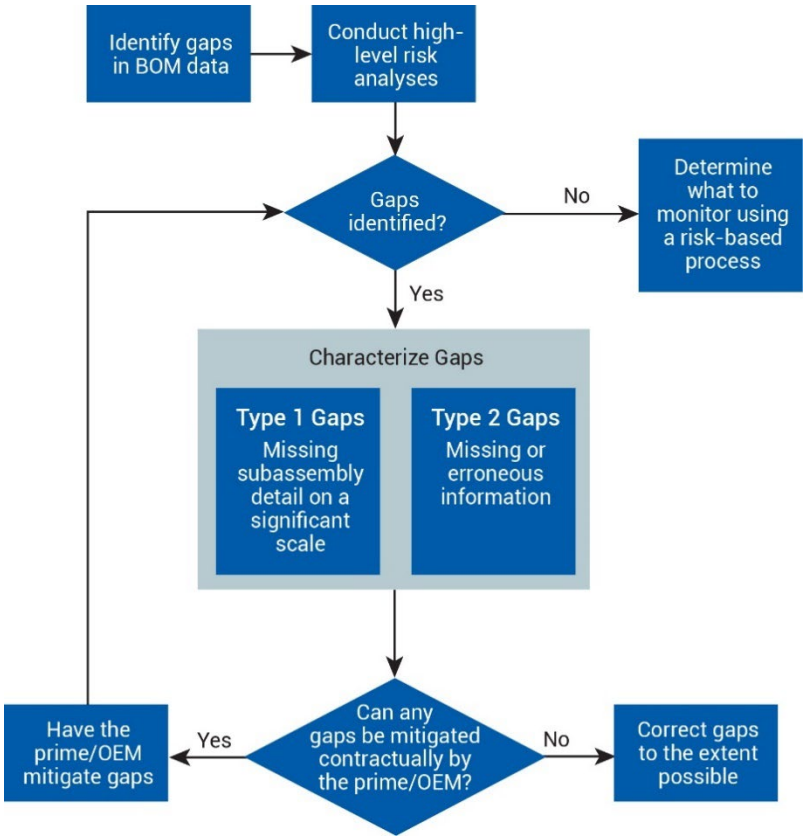
[280] A destructive PPA could be the first step in reverse engineering. Reverse engineering must also identify the function of all the items. That additional information is not necessary to build a BOM.

[281] Although not portrayed in Figure 29, it is a best practice to apply the Figure 29 process to drawings, provisioning data, and manuals to identify acceptability of contract deliverables, identify gaps, and mitigate those gaps where possible.

[282] Three of the acceptability criteria for BOMs are relevant across the board: completeness, whether it is up-to-date, and whether key information is missing or in error. The criterion for indenture issues is not relevant.

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Figure 31. Process for Identifying Gaps in BOMs Where Additional Data Are Needed



- ***Characterize the gaps***. The gaps correspond to the acceptability criteria. They are categorized into two different types because the mitigation approach is different.
  - *Type 1 Gaps*. These gaps are large. In this instance, BOMs would be lacking subassembly detail on a significant scale. The mitigation for such gaps is an attempt to construct the missing sections of the BOMs using the processes in Figure 11.[283]
  - *Type 2 Gaps*. These gaps are characterized by missing or erroneous information.[284] They include the following:
    - Lower-level assemblies need to be identified,
    - Missing manufacturer's part numbers or other data omissions,
    - Indenture issues,
    - Out-of-date data, and
    - Data errors (e.g., an incorrect part number).

  The first and the last two of these gaps may be hard to identify. In the former case, it is difficult to recognize missing parts or assemblies. Asking for a drawing index on contract helps identify such gaps. For the latter, maintaining data in its most current version is difficult. Even a PPA may be out-of-date if updated items have not been fully fielded.

[283] Type 1 gaps for drawings, provisioning data, and manuals would be the same ones as above for BOMs. These gaps are large. In this instance, drawings/provisioning data/manuals would be lacking subassembly detail on a significant scale.

[284] The Type 2 gaps for drawings, configuration data, and manuals would be like those for BOMs. Lower-level COTS assemblies need to be identified, data omissions, and out-of-date data all apply. Only indenture issues are not applicable.

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Processes and techniques for closing these gaps are further illustrated in the following section.[285]

- ***Can any gaps be mitigated contractually by prime/OEM?*** Obviously, if the BOM was constructed per Figure 11, the answer is no. If the BOM were a contract deliverable, then DoD should not accept the deliverable and the contractor should mitigate the gaps. Even if the deliverable was accepted by DoD, the prime/OEM should be asked to mitigate the gaps, commonly as a deliverable on a subsequent contract. It has also been the case that DoD did not specify BOM requirements adequately and the delivered product was not usable. In this case, the only remedies are to rely on the good graces of the contractor or modify the contract.

J.6.2 Correcting Gaps to the Extent Possible

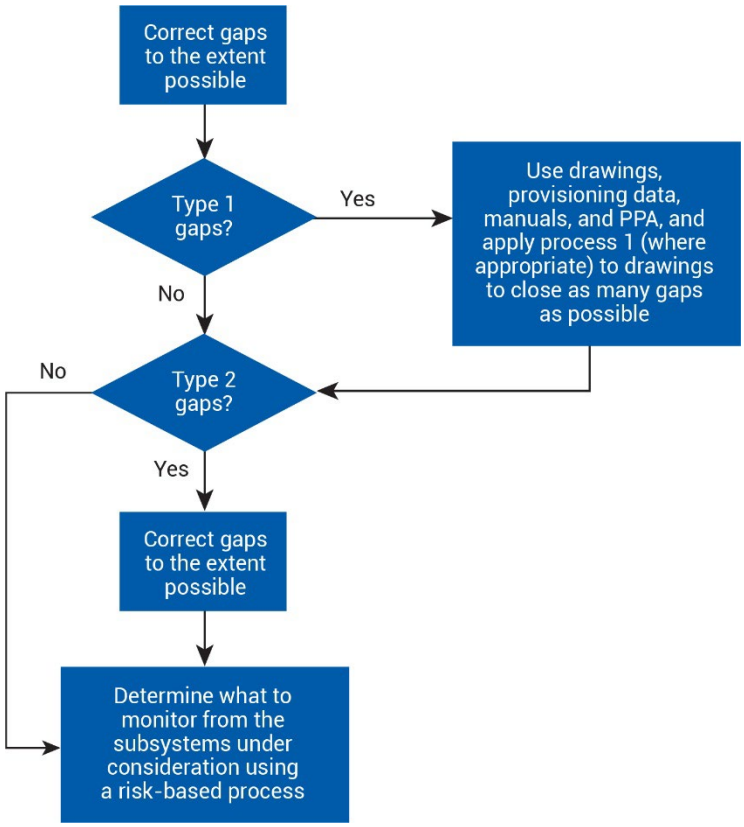
Figure 32 is a continuation of Figure 31. Once gaps have been identified, Figure 32 depicts a process for correcting those gaps. This process ends at either (a) the point at which a risk-based process is used to determine what to monitor or (b) the point at which sufficient data cannot be obtained or constructed to allow proactive monitoring and therefore a reactive DMSMS management approach for that subsystem is the only possibility.

Drawings, provisioning data, and manuals often do not contain sufficient useful information to build a BOM or to fill in gaps. Although depicted linearly in Figure 32, the process of closing gaps in BOM data is labor intensive and highly iterative. All sources of data are used. Other manuals and drawings may be available elsewhere in the DoD supply or maintenance systems. This is especially true for provisioning data; another program may have provisioned the same assembly at a lower level of detail. It may be necessary to identify actual part numbers associated with NSNs and then research those part numbers in greater detail.

[285] Mitigation approaches for drawings, configuration data, and manuals include asking the prime or OEM to mitigate the gaps before acceptance, modify the contract to buy the needed information, and the Figure 31 process.

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Figure 32. Closing Gaps in BOM Data



A brief discussion of the principal blocks in Figure 32 follows.

- ***Type 1 gaps?*** Type 1 gaps are closed by using a process similar to the one for constructing a BOM. Drawings, configuration data, manuals, and potentially a PPA are examined/ considered to try to obtain the missing data. If that is unsuccessful, Process 1 within Figure 29, may be used to try to obtain the necessary drawings. Ultimately as many Type 1 gaps as possible are closed. Any remaining portions of the system for which Type 1 gaps still exist will be dealt with reactively. Although not explicit in Figure 32, as discussed previously, before any significant effort to close Type 1 gaps is undertaken, a BCA should be conducted.
- ***Type 2 gaps?*** Type 2 gaps are normally addressed by research. As stated earlier, these gaps are characterized by lack of COTS detail, missing data, or structural issues. Often, Type 2 gaps are associated with a lack of detail or questions associated with individual items. Several research avenues are possible, the order through which they are pursued is often an individual preference, sometimes constrained by access to the source. Similar to Type 1 gaps, before any significant effort to close Type 2 gaps is undertaken, a BCA should be conducted. Potential research avenues are as follows:
  - *Internet research*. This is a common first course of action. Many researchers collect as much data as possible before having discussions with an SME.
  - *Supply system analysis*. Previous discussions explained how provisioning data may allow BOMs to be built or augmented. However, for items that are provisioned, a significant amount of additional data is available and this individual data may be sufficient to close the gap or if not, point the researcher in a direction to close the gap. Even though the item may not have been provisioned by the system in question, the item may have already been provisioned by another system and consequently the data may provide a point of contact associated with that other system who can provide the data to resolve the Type 2 gap.

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- *Contact an SME*. The vendor is one source of expertise. Equipment specialists in the supply system or program office are another possibility. There may be others in the program office who can also provide the needed information. It may not always be necessary to obtain more data. SMEs may also be in a position of assessing the risk of not obtaining the data, and, if that risk is small, making further effort unnecessary.

- ***Correct gaps to the extent possible***. At this point in the process, all existing Type 1 and Type 2 gaps will be closed to the extent possible using the approaches described above. The result will be a BOM with as much detail that can be obtained.

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Appendix J–Annex. Essential and Desirable Data Elements to be Obtained If Constructing a BOM

Table 57. BOM Fields, Definitions, and Essentiality

Field Name	Definition	Essentiality/ Desirability
Nomenclature	The name or designation of the referenced item.	Essential
Manufacturer's Item Number	The complete orderable number assigned by the actual manufacturer of the referenced item. This number will be used to monitor the item for obsolescence issues.	Essential
Manufacturer's Name	The full legal name of the entity that actually manufactures the referenced item.	Essential
Manufacturer's CAGE	The CAGE code of the entity that actually manufactures the referenced item.	Essential
OEM Item Number	The item number assigned by the OEM. This may be the same as the manufacturer's number but more often than not it is different. Sometimes referred to the Configuration Part Number or the Drawing Number.	Most desirable
Indenture Level	A number that represents the specific level of the top-down structure specified on the BOM.	Most desirable
NHA Item Number	The OEM item number of the referenced item's parent NHA. If the referenced item is software or firmware, then the NHA is the item the software is used on.	Most desirable
NSN	NSN of the referenced item.	Most desirable
Software/ Firmware Version	The designator assigned to the version of software/firmware used on the referenced item.	Second highest desirability
OEM Name	The full legal name of the entity that is providing the referenced item to the government. Sometimes referred to as the Prime Integrator or Prime.	Third highest desirability
OEM CAGE	CAGE code of the OEM.	Third highest desirability
Revision Designator	The designator of the revision of the referenced item.	Third highest desirability
Reference Designator	A reference designator identifies the referenced item. ASME Y14.44-2008 and IEEE-315-1975 provide details for the proper configuration and use of reference designators.	Third highest desirability
Find Item or Index Number (FIIN)	FIIN refers to the TDP drawing ordinal number that gives an ID tag to the referenced item in an item list (list of materials, BOM, parts list).	Third highest desirability
NHA CAGE	CAGE code of the OEM of the NHA.	Third highest desirability
Quantity	The count of the referenced item per NHA.	Third highest desirability
Provisioning Status	The referenced item number has been provisioned in a services supply system (Yes/No).	Third highest desirability
APL, AEC, or RIC	Allowance Parts List (APL), Allowance Equipage Code (AEC), or Repairable Identification Code (RIC) for the referenced item.	Third highest desirability
COG	Cognizance (COG) code: A two-item alphanumeric designator used by the Navy to provide supply management information.	Third highest desirability
Manufacturer's Address	The physical address of the item's manufacturer, including country.	Third highest desirability

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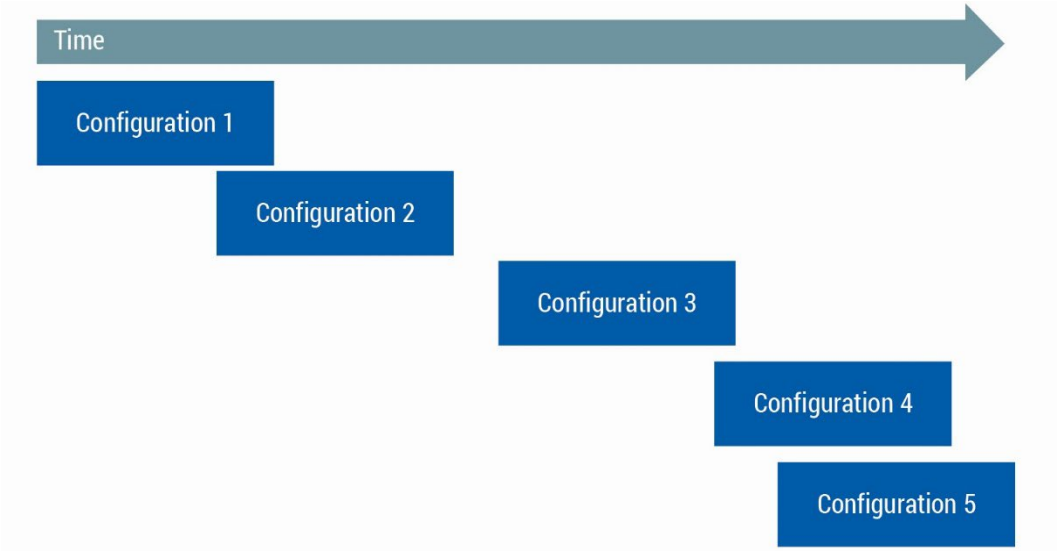
Field Name	Definition	Essentiality/ Desirability
Manufacturer's Website	The web address of the item's manufacturer.	Third highest desirability
Alternate For	The OEM item number of the item the referenced item is the alternate for.	Third highest desirability
Monitoring Status	Identifies whether the referenced item is being proactively monitored for DMSMS issues.	Third highest desirability
Item Class	Identifies whether the item is COTS, NDI, or DI.	Third highest desirability
Item Type	Enter the type of the referenced item number based on the following: An assembly is comprised of components and potentially other assemblies (e.g., a circuit card or a transmission). A component is a single item of manufacture used to make an assembly (e.g., a microcircuit or a screw). Software is an item of a computer system that consists of data or computer instructions, in contrast to the physical hardware from which the system is built. Material is the products used to manufacture the assembly (e.g., wire, fiberglass, steel, glue).	Third highest desirability

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Appendix K. Building Roadmaps

While additional supporting information may be included, Figure 33 depicts a notional, minimalistic product roadmap. It portrays when the product's (system's) configuration changes over time.[286] The length of the boxes in the time dimension could denote the start and end date for implementing the configuration changes.[287] Other supporting information may include a description of the change, contract information for making the changes (indicating funding), the number of systems with each different configuration, and the life cycle of a system configuration.

Figure 33. Notional Product Roadmap



Source: Adapted from Irene J. Petrick, *Developing and Implementing Roadmaps: A Reference Guide*, Pennsylvania State University, nd.

Several factors can drive configuration changes reflected in a product roadmap and IMS:

- Inability to support the product, potentially coupled with the need to extend the service life of the system,
- Changes to the threats the product faces or its mission,
- Safety deficiencies,
- Additional information assurance or software security requirements,
- Introduction of new statutory or regulatory requirements (e.g., environmental or security requirements),
- Insufficient reliability or maintainability (especially for drivers of operations and sustainment costs as part of a deliberate cost reduction initiative), and
- Changes to technology enabling additional capability or enhanced supportability.

[286] The large overlap between configurations 4 and 5 indicates that the configuration changes are associated with different subsystems or different technologies associated with the two configurations.

[287] The configuration remains operative after the end time indicated on the figure.

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Roadmaps vary in level of detail, format, and coverage. A program office might not maintain a single, multipurpose roadmap that encompasses all the factors driving change at some level of aggregation (e.g., for a subsystem). Therefore, the notional depiction in Figure 33 could represent the configuration changes for the entire system or any subset of the entire system. When the product roadmap represents the entire system, or major elements thereof, all the configuration changes for an assembly or subsystem can be collapsed into a single line.

In addition, how program offices label various types of roadmaps, as a function of their purpose, varies. This appendix describes the following types of roadmaps:

- Product improvement roadmaps apply to changes in capability to address a new threat, safety deficiencies, information assurance, or the introduction of new statutory or regulatory requirements.
- Supportability roadmaps encompass improvements in maintainability or sustainability of the product, product reliability, or service life.
- Technology roadmaps capture technology changes that enable the two previous types of roadmaps and, consequently, the technologies are driven by product and supportability improvements.

In DoD, product improvements (and by implication, product improvement roadmaps) are the principal driver of product roadmaps. Product improvement roadmaps should reflect the lowest-cost path to the desired capability levels. Supportability roadmaps deal with aspects of the O&S cost.

The generic form of Figure 33 also applies to a notional product improvement roadmap or a notional supportability roadmap. However, a notional technology roadmap looks different. The following sections describe product improvement, supportability, and technology roadmaps in more detail.

K.1 Product Improvement Roadmaps

A product improvement roadmap for capability improvements, safety enhancement, or other new requirements considers the following:

- When new capabilities need to be in place,
- When the technology needed to achieve the new capability requirements will be demonstrated successfully in relevant or operational environments (i.e., sufficient maturity, according to the technology readiness level scale[288]),
- When the associated hardware and software changes will be producible, and
- Planned funding.

As is the case for product roadmaps, product improvement roadmaps can exist for different levels of a system's design hierarchy. These varied product improvement roadmaps fall under the purview of different stakeholders. For example, the PM is responsible for the roadmap for the entire system. Lower-level improvement roadmaps are the responsibility of a cognizant IPT. Depending on the program office, the IPTs may be associated with a subsystem (e.g., fire control) or a functionality or condition (e.g., readiness). Not every subsystem has an IPT, especially during sustainment, and not every IPT has a product improvement roadmap.

The concept of product improvement aligns closely with technology insertion. Technology insertion integrates mature technologies with capability requirements and logistics planning to expand system

[288], Office of the Executive Director for Systems Engineering and Architecture "Technology Readiness Assessment (TRA) Guidebook," June 2023, p. 59 , available at https://www.cto.mil/wp-content/uploads/2023/07/TRA-Guide-Jun2023.pdf.

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capability as well as increase readiness, lower life cycle costs, and reduce the logistics footprint. A program office can avoid significant costs by determining optimum technology insertion timeframes. For example, a redesign to upgrade a product should simultaneously seek to eliminate obsolete or near-obsolete parts (as identified via a health assessment [see Section 5.3.1]) because it is usually more cost effective to resolve DMSMS issues in conjunction with other design changes rather than as standalone, out-of-cycle redesigns.

Depending on how a program office is organized and the stage in the system's life cycle, product improvement roadmaps contain none, all, or some supportability roadmap data. In some instances, no product improvements are planned and, therefore, no product improvement roadmap is created. This normally occurs toward the end of the system's sustainment phase. When product improvements are planned, the product improvement roadmap may be a convenient place to maintain all product changes, including supportability changes, thereby eliminating the need for a separate supportability roadmap. In some program offices, the product improvement roadmap only includes the supportability changes associated with the product improvement work package, requiring a separate supportability roadmap for other changes generated by the cognizant IPT or the PSM.[289]

K.2 Supportability Roadmaps

The first and most important element of the supportability roadmap focuses on resolving DMSMS issues before they affect the availability of the system. System non-availability occurs because items can no longer be purchased and there are not enough in the supply system to sustain the system through its expected life. As such, the DMSMS community supplies the associated roadmap data (see Section 4.6). Availability often causes major cost implications for the supportability roadmap.

The second element of a supportability roadmap encompasses any planned modifications to reduce O&S costs.[290] Changes would be designed to affect the major cost drivers. Modifications may also be planned to reduce downtime. Readiness suffers when poor reliability affects system up-time. The general process for developing the supportability roadmap includes the following:

- Identify the principal cost drivers and reliability inhibitors (that are not cost drivers),
- Determine and price alternative approaches for addressing the issues associated with those drivers,
- Perform a cost-benefit analyses to estimate potential life cycle savings and break-even points,
- Present the alternatives to decision-makers,
- Secure funding to implement the changes approved by the decision makers, and
- Add the information to the overall product roadmap, IMS, and product improvement roadmaps.

The concept of multiple levels of roadmaps applies to supportability. The PSM is responsible for the overall system. At lower levels, the responsible IPTs share the responsibility with the PSM. When no associated IPT exists, responsibility falls entirely on the PSM. The maintenance and DMSMS communities supporting the PSM develop options to meet the desired objectives.

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[289] Since supportability is a PSM responsibility, as a best practice, the PSM should have the lead role in supportability roadmaps. However, the cognizant IPT has significant influence over its content.

[290] Many program offices find implementing this type of change difficult because it requires an upfront investment to reduce costs over the life cycle.

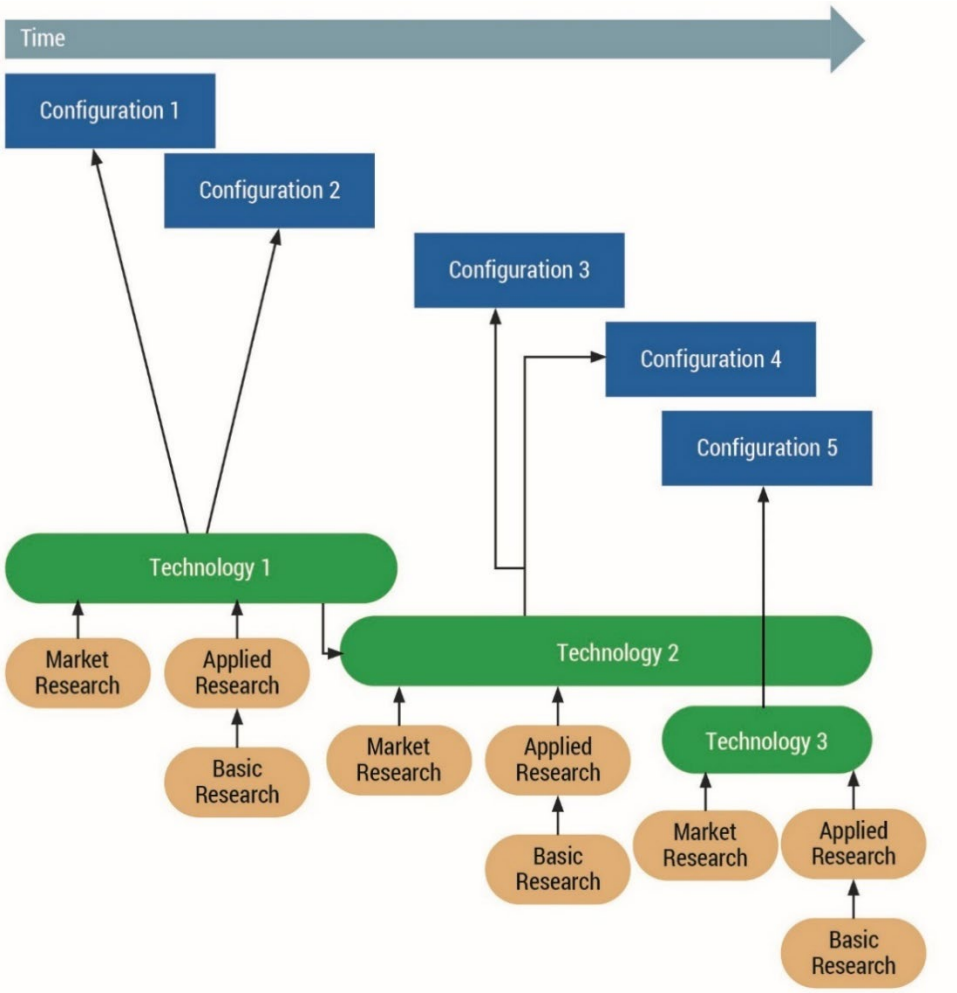
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K.3 Technology Roadmaps

Both product improvement and supportability (including technology refreshment) considerations are based on specific technology advancements. Therefore, product improvement and supportability roadmaps determine which technology roadmaps are needed. As these technology roadmaps are planned and pursued, they enable capability and supportability improvements. Technology roadmaps uncover alternative technology paths for meeting performance targets.[291]

Figure 34 superimposes a notional (and minimalistic) technology roadmap onto Figure 33. It indicates how basic, applied, and market research (technology management) contribute to technology maturation (for three notional technologies) and then when the mature technologies are incorporated into the configuration changes. A more complete roadmap would contain the specifics, such as what the technologies are, what they provide, and relevant specifications.

Figure 34. Notional Technology Roadmap and Interactions with a Product Roadmap



Source: Adapted from Irene J. Petrick, *Developing and Implementing Roadmaps: A Reference Guide*, Pennsylvania State University, nd.

[291] Marie L. Garcia and Olin H. Bry, *Fundamentals of Technology Roadmapping*, SAND97-0665 (Albuquerque, NM: Sandia National Laboratories, April 1997).

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An actual technology roadmap can portray more than three technologies, depending on the level of aggregation of the product roadmap supported. In this notional example, technology 2 is a replacement for technology 1 and technology 3 is independent of the other two. The length of the technology ovals along the time dimension could represent the period of sale for those technologies. In the notional picture, technology 1 is available for implementation into configurations 1 and 2. (Other technologies and factors could contribute to those configuration changes.) Toward the end of the technology 1 life cycle, technology 2 becomes available as a replacement and remains available through the end of the configurations depicted. Technology 3 becomes available for configuration 5 implementation.

A technology roadmap codifies the results of technology management activities. Unlike product improvement or supportability roadmaps, technology roadmaps (and their underlying technology management activities) are typically developed external to the program office but can be funded in part by the program office. Effective technology management begins with a strategic understanding of the market and its trends. Market research for a program office, particularly before a system begins the sustainment phase of its life cycle, entails collecting information about existing and emerging technologies, products, manufacturers, and suppliers. It has two components:

- ***Market surveillance***. A continuous canvassing of the commercial market to identify existing and future technologies, vendors' products, and market trends with the potential to meet existing and emergent requirements from a strategic perspective. Market surveillance methods include searching the internet, attending trade shows, reading technology publications, hiring consultants, issuing requests for information from prospective manufacturers and suppliers, visiting manufacturer and supplier facilities, and viewing product demonstrations.
- ***Market investigation***. A focused process of finding and evaluating whether specific technology products meet functional requirements. Market investigation includes system obsolescence profiling to plan for continued support or replacement of soon-to-be obsolete products. This product-level information and the associated budget requirements form the basis for sustaining the operation or functionality of a system. Market investigation methods include beta testing; prototyping; testing for compliance, conformance, and compatibility; and querying of manufacturers and suppliers about product obsolescence status.

Market research occurs in all system life cycle phases, enabling the acquiring activity to do the following:

- Anticipate obsolescence situations due to rapid and asynchronous product changes;
- Plan and budget using a broader range of product obsolescence management options;
- Maintain insight into technology trends, as well as internal product changes by the manufacturer, and test the effects of those changes on the system;
- Assess the quality of a manufacturer and the effect on a system of a product's change, including its suitability for the user, information security characteristics, and supportability; and
- Determine the manufacturer's support period and inventories for a particular product.

Ignoring market research increases the likelihood of poor product and technology selections as well as an inability to predict and mitigate obsolescence impacts, leading to out-of-cycle redesigns. This can negatively affect system performance, schedule, and cost.

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Technology management also encompasses basic and applied research activities to develop new technologies to close military capability gaps driven by product improvement needs. The process is summarized in a few basic steps.[292]

- Assess technology performance requirements.
- Define the scope of the technology development in terms of the initial technology gaps to be closed.
- Establish a technology development strategy.
- Initiate research to mature the technologies.
- Evaluate the effect of the technologies on the gaps.
- Continue research and evaluation until the technology is demonstrated to close the initial gaps in a relevant environment.
- Repeat the process as further capability enhancements are needed or low-cost technology discriminators that improve mission performance become available.

The combination of the results of market research, an understanding of current and future mission needs, and DoD-planned basic and applied research activities enables the development of a technology roadmap. The lead for obtaining a technology roadmap is the cognizant IPT and the PSM for supportability elements. Some program offices only minimally use technology roadmaps, especially late in the system's life cycle.

[292] Adapted from Robert Phaal, Clare J.P. Farrukh, and David R. Probert, *Technology Roadmapping—A Planning Framework for Evolution and Revolution*, May 26, 2003.

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Appendix L. Health Assessment Methodology

Health assessments may be conducted for several purposes.

- ***Proactive DMSMS monitoring***. Hardware and software health assessments may be used to identify items where more intensive monitoring should be applied because of both an increased likelihood of near-term obsolescence and the potential ill effects of obsolescence on readiness (or operational availability) if no mitigating action is taken within the window of opportunity for that action.
- ***Monitoring inventory levels***. As determined by the hardware health assessments, the dates when obsolescence issues will affect the system are based in part on assumptions about demand and inventories of obsolete items (at inventory control points, elsewhere in DoD's possession, and/or available from commercial sellers as appropriate). These dates may change because of uncertainties in those assumptions and consequently the timing of mitigating actions will be affected. The health assessment should identify items of high interest for the watch list (as discussed in Section 3.4.5).
- ***Resolution prioritization***. Hardware and software health assessment results are an important consideration in the determination of when a DMSMS issue will have an impact. The impact date is a key determinant of when the resolution should be initiated.
- ***Programming and budgeting***. Hardware and software health assessment supports the determination of when funding for a DMSMS resolution should be programmed/budgeted by estimating when that DMSMS issue will impact readiness. The impact analysis itself provides substantive material for use in program/budget justification as well the development or update of technology refreshment/insertion plans and associated funding.

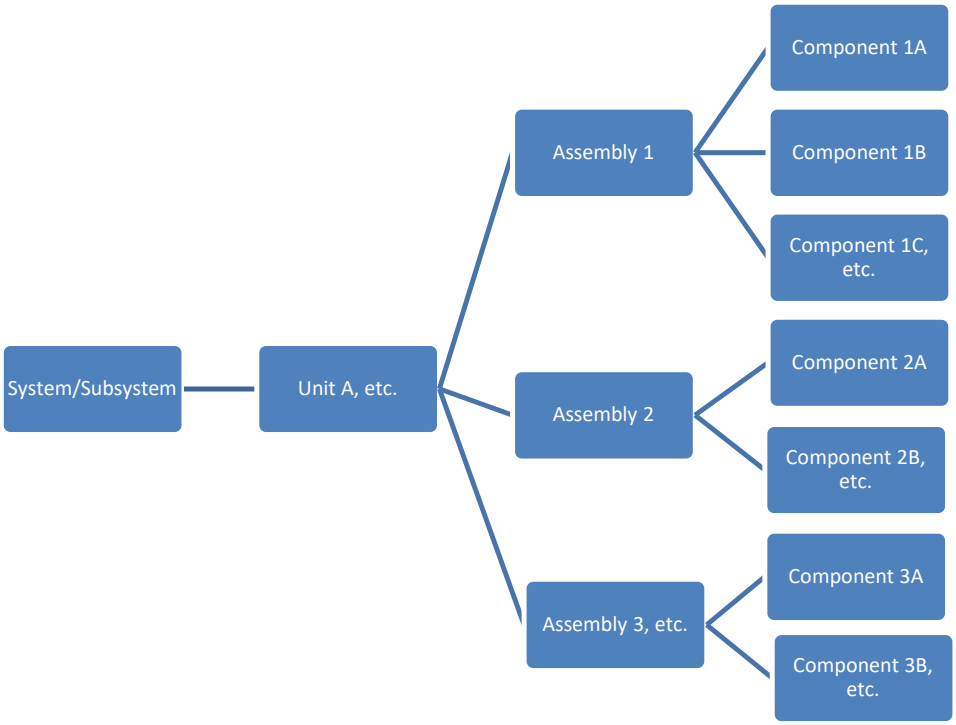
The following sections portray the steps involved in making a health assessment. Obtaining the data needed for steps 2, 3, and 4 can be difficult. The information necessary may not be centralized and the quality of the data can vary as a function of where it is found. In some cases, the required data may not have been recorded at all. Appendix L.2 briefly discusses a common cause of the absence of required data, and the implications for DMSMS managers and policy makers.

L.1 Step 1: Establish a System/Subsystem Hierarchy

The first step of the health assessment lays out the pertinent system/subsystem hierarchy or items (as shown generically in Figure 35) that will be monitored for DMSMS issues because of a risk-based assessment. For example, the system/subsystem under consideration might be a radar. Below that, a unit could be a power supply. A regulator assembly is one of the assemblies within the power supply and components within that may be a voltage regulator or a radio frequency amplifier. Although not depicted graphically, there can be more units, assemblies, and components within the system/subsystem.

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Figure 35. Generic Health Assessment Hierarchy



A health assessment can be conducted for various portions of the system hierarchy as a function of the objectives of the assessments.

- Health assessments associated with cases begin with lower-level components that can no longer be procured or have a PDN associated with them. The objectives of this health assessment are principally to determine the priority of implementing a resolution (as a function of the impact date) and to develop the data needed to determine the appropriate level of assembly for the resolution. To accomplish these objectives, the hierarchy should stop at the highest level of assembly where a resolution may occur. This is most often at the unit level. The components considered in the hierarchy should not only include those that cannot be purchased or those with a PDN, but also encompass those projected to become obsolete as described in Section 4.6.
- Health assessments may also be conducted at higher levels of assembly that cut across subsystems and units. Such health assessments may be thought of as the sum of the health assessments done at the individual component level and for projected item obsolescence that is too far in the future to open a case (e.g., resource expenditures are not warranted at the current time). These health assessments are used to better integrate DMSMS resolutions with modifications plans and funding.

L.2 Step 2: Calculate the Depletion Year for Each Obsolete Component

The second step of the health assessment calculates a depletion[293] year for all components that can no longer be purchased. Table 58 portrays a notional way of making that calculation. It shows the initial on-hand inventory for each component. The formula just subtracts the annual demand from the inventory

[293] The calculations shown in the remainder of this appendix implicitly use averages for failure rates, wear out rates, and so forth. These rates will be governed by a Poisson probability distribution. When making these calculations, a program office may want to increase its confidence level that depletion will not occur earlier than estimated by adding a standard deviation to the averages.

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levels at the end of the prior year. By subtracting the average annual demand for each component, the year in which the component is depleted (i.e., its on-hand inventory is negative and shaded in the table) can be determined.

The calculations are made over the expected operational life of the system/subsystem. For this example, the EOL of the system/subsystem is FY24. If a component is projected to have inventory beyond that point, its depletion year is shown as EOL + 1 or FY25. The depletion years that occur before the EOL are highlighted.

There are two aspects of demand data: production and sustainment. For any given component, both aspects of demand may apply. Early in the life cycle, the demand may be solely production based. During production, the demand is likely to be a combination of the two because some units will already be fielded while others are still being produced. Late in the life cycle, the demand may be entirely driven by sustainment. The Table 58 example makes a simplifying assumption that does not apply in all situations. Future demand does not have to be constant. Neither the production schedule nor failure rates which drive demand during sustainment are constant over time.

Table 58. Notional Calculation of the Depletion Year of DMSMS Components

Components	Initial Total Inventory	Annual Demand	Depletion Fiscal Year
Comp 1A	92	15	FY21
Comp 2A	1	0	FY25
Comp 2B	85	22	FY18

Comps	FY15	FY16	FY17	FY18	FY19	FY20	FY21	FY22	FY23	FY24
1A	77	62	47	32	17	2	(13)	(28)	(43)	(58)
2A	1	1	1	1	1	1	1	1	1	1
2B	63	41	19	(3)	(25)	(47)	(69)	(91)	(113)	(135)

The production aspect of demand should be determined from the production schedule. Early in the sustainment phase of the life cycle, demand will be based on predicted reliability, but as field-level usage becomes available, empirical measures of demand should be used. Demand for sustainment must take into account all users of the component in question that draw upon the sources used for the initial inventory determination, not just the specific program conducting the health assessment. Therefore, the usage aspect of demand data should be obtained from the organization that manages the component (either DLA or a Military Department supply organization).[294]

Obtaining component-level demand data can be challenging for multiple reasons, such as, data may not be centralized and may be of questionable quality. One reason for data being decentralized is that programs have been known to purchase some components directly from commercial sources. For a program that uses a common component that's purchased commercially by other programs, demand estimates based on DLA or Military Department organizations will be too low. It's unlikely a program office will have any basis for adjusting for this situation.

Another common challenge for gathering component demand history occurs when an assembly which is a DLR is sent in for repairs, and the assembly is returned from the depot without specifying which components needed to be replaced or repaired. In such a situation, all that is known is that at least one

[294] The program office demand could also be considered. If the program office is the only user, then the program office demand should be used. (The program office demand should equal the supply system demand unless the program office makes purchases in the commercial market. However, if there are other users and inventory cannot be reserved, then using the program office demand will lead to an overly optimistic calculation of the item quantity required.

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component within the assembly experienced a demand. It cannot be determined how frequently individual components are demanded, which leads to uncertainty as to whether an obsolete component has sufficient inventory to last until the system/subsystem's EOL. To rectify these situations, as a best practice, inventory managers (whether DLA, Military Components, CLS, or others submitting items for repair) should require DLR contracts to provide data on the specific components that were replaced or repaired.

In addition to demand, the initial total component inventory is needed to determine depletion years and should include as many sources as possible. Beyond DoD supply systems, contractors and potentially DoD depots may hold inventory (especially when they are performing maintenance on the assembly or there is new production). Individual programs may hold inventory as well. To the extent feasible, such inventory data should be included.

L.3 Step 3: Calculate the Depletion Year for Each Obsolete Assembly

The fact that a component's stock level is depleted does not imply that the supportability impact will occur immediately. There may be spare assemblies that can be used to defer the impact. Therefore, the third step of the health assessment determines the year that the assemblies containing obsolete components are expected to be depleted. The notional computations for this step are shown in Table 59 and Table 60 which are built on the results determined in Table 58. The underlying idea behind these calculations is that when an assembly fails, it is returned to the supply system for repair and a new/refurbished assembly is provided to the user.[295]

Table 59. Notional Calculation of the Depletion Year of DMSMS Assemblies (Part 1)

Assembly	Qty in Stock	Qty in Repair	Other Qty	Wear Out Rate	Survival Rate	Annual Demand	Obs Degrader	EOL FY	Obs Impact FY
Assembly 1 (contains components 1A, 1B, etc.)	6	12	1	0.02	1.00	36.00	0.50	24	21
Assembly 2 (contains components 2A, 2B, etc.)	5	15	3	0.14	0.88	18.00	0.50	24	18

Table 60. Notional Calculation of the Depletion Year of DMSMS Assemblies (Part 2)

	FY15	FY16	FY17	FY18	FY19	FY20	FY21	FY22
Calculated Wear Out Rate	0.02	0.02	0.02	0.02	0.02	0.02	0.52	0.52
Demand/Year	36	36	36	36	36	36	36	36
Add-on Assemblies/Year	4	3	2	—	—	—	—	—
Reusable Units	35	35	35	35	35	35	17	17
Assembly 1 Year End Inventory	22	24	25	24	23	22	3	(15)

Calculated Wear Out Rate	0.14	0.14	0.14	0.64	0.64	0.64	0.64	0.64
Demand/Year	18	18	18	18	18	18	18	18
Add-on Assemblies/Year	—	—	—	—	—	—	—	—
Reusable Units	14	14	14	6	6	6	6	6
Assembly 2 Year End Inventory	14	9	4	(8)	(20)	(32)	(44)	(56)

[295] If the assembly is not repaired (i.e., it is thrown away) then its depletion year should have been determined in step 2 as there would be no need to look at components in the assembly.

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Table 60 shows most of the inputs to the calculations. Beyond the year that an obsolete component is depleted (calculated in Table 60 and highlighted yellow in Table 59), there are three types of input as follows:

- ***Inventory sources for assemblies***. Three sources are shown. The quantity in stock are the assets ready for issue. The quantity in repair are the ones being refurbished. The other stock would include any other potentially available assets (e.g., in the pipeline, test units). Table 60 also allows for an increase in the number of assemblies due to decommissioned units and cannibalization to the extent it can be determined.
- ***Assembly demand***. As discussed in step 2 at the component level, there could be both production and sustainment aspects of the demand and the comments made about determining the component demand (and its potential changes over time) apply to the assembly level as well.
- ***Ability to repair failed assemblies***. Not all assemblies are repairable. Table 60 indicates three factors that should be used to determine whether a repair can be made.
    - A wear out rate indicates the fraction of assemblies that cannot be repaired because it would be expected to fail very quickly after being returned to service. For the example from Table 58 that is extended into Table 59, wear out rates of 0.02 and 0.14 were used for assembly 1 and assembly 2, respectively.
    - Some assemblies cannot be economically repaired.[296] The survival rate indicates the fraction of assemblies that can be economically repaired. Table 59 uses 1.0 and 0.88 for assemblies 1 and 2, respectively.
    - An obsolescence degrader is defined as the fraction of the assembly repairs that require an obsolete component. Table 59 uses 0.5 for both assemblies 1 and 2. In the calculation of an assembly depletion date, the obsolescence degrader only has an impact on an assembly when one of its obsolete components has been depleted. According to Table 58, those dates are FY22 and FY18, respectively. For the notional calculation, the obsolescence degrader is added to the wear out rate in Table 60 since both have the same effect on available inventory. For assembly 1, this adjustment begins in FY22 and for assembly 2, it is FY18.

Assembly depletion year calculations are shown in Table 60. The basic formula being used is that the end of year assembly inventory is equal to the starting inventory plus any add-ons from decommissioning and/or cannibalization minus losses associated with failed assemblies that cannot be repaired. The losses are determined as the difference between the total number of failed assemblies and the total failures reduced by the survival rate and by 1 minus the calculated wear out rate. The formula for the losses per assembly is as follows:

(failures/year) × (1 – calculated wear out rate) × (survival rate).

For the first year of the calculation, the starting inventory is the sum of the quantity on-hand, other assemblies available, and the number of assemblies in repair reduced by those that cannot be repaired. The formula for the starting inventory is as follows:

(quantity in stock + other quantity + (quantity in repair – quantity that cannot be repaired)) × (1 – calculated wear out rate) × (survival rate).

As can be seen (tan shaded area) in Table 60, the depletion year for assembly 1 is FY22 and the depletion year for assembly 2 is FY18. These are the years that the inventory becomes negative.

[296] For an obsolete item, it may be 1.0 to avoid impacting operational availability. Even though it would cost more to repair the item than the item is worth, it may be worthwhile to repair it because a replacement cannot be purchased.

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L.4 Step 4: Calculate when the Obsolete Unit Affects Readiness

The third step explained why a depleted component does not always translate into its higher-level assembly being depleted at the same time. There may be spare assemblies. The same holds true for a depleted assembly. There may not be a supportability concern because there may be spare units. Consequently, the fourth step of the health assessment calculates the year that the number of units containing the obsolete assemblies falls below the number required by fielded assets. That event implies that the unit will be insupportable because the MICAP rate will be affected. The notional computations for this step are shown in Table 61, built from the results given in Table 60.

Step 4 computations and assumptions at the unit level are nearly the same as those for step 3 at the assembly level. In some cases, when a unit fails, the organization level repair can determine which assembly has failed, but in other cases, the failed assembly cannot be determined. The former case would be reflected in the demand for assemblies and included in Table 60. Table 61 applies to the latter case where failed units are replaced with new/refurbished units from the supply system.

Table 61. Notional Calculation of the Depletion of DMSMS Units

Unit Spares Qty	Assemblies in Unit	Assembly Depletion FY		FY15	FY16	FY17	FY18	FY19	FY20	FY21	FY22	FY23
4			Annual Unit Demand	2.5	2.5	2.5	2.5	2.5	2.5	2.5	2.5	2.5
	Assembly 1	22	Year End Inventory	22	24	25	24	23	22	3	(15)	(33)
	Assembly 2	18	Year End Inventory	14	9	4	(8)	(20)	(32)	(44)	(56)	(68)
			Calculated Wear Out Rate	0.10	0.10	0.10	0.40	0.40	0.40	0.40	0.65	0.65
			Reusable Units	2.03	2.03	2.03	1.35	1.35	1.35	1.35	0.79	0.79
			Year End Units	13.0	12.0	11.0	9.0	7.0	5.0	3.0	1.0	—
			Systems in Service	10								
			Unit Wear Out Rate	0.1	10% of failed units are beyond repair							
			Assembly 1 Obs Degrader	0.25	Chance unit fails because of assembly 1							
			Assembly 2 Obs Degrader	0.3	Chance unit fails because of assembly 2							
			Unit Survival Rate	0.9	90% of the reparables are economically worthwhile to repair							

The inputs to the unit depletion calculation parallel the inputs to the assembly depletion calculation.

- ***Inventory sources for assemblies***. Table 61 differs from Table 60 in that it combines all sources into a single input. The number of units in the supply system is assumed to be four. No add-on assets are included in Table 61, but they could be because of decommissioned systems/subsystems. To measure the supportability, one additional input is needed: the number of units in place on fielded systems (assumed to be ten for this notional example).

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- ***Unit demand***. The annual demand for units is shown as 2.5 annually.[297] This figure would include both production and sustainment analogous to the way it was done for assemblies in Table 59 and Table 60 and for components in Table 58.
- ***Ability to repair failed units***. Analogous to Table 60, this ability is reflected by a unit wear out rate (0.1) and a unit survival rate (0.9). In addition, obsolescence degraders for the obsolete assemblies are used. The figure 0.25 is the probability that the unit failure is due to assembly 1, and the figure 0.3 is the probability that the unit failure is due to assembly 2. These figures are added to the wear out rate when the associated assembly is depleted, FY22 and FY18 respectively, as calculated in Table 60 and highlighted in Table 61.

The Table 61 calculation for the depletion of units is also identical to the assembly calculation in Table 60. The basic formula being used is that the end of year unit inventory is equal to the starting inventory minus losses associated with failed units that cannot be repaired. The losses are determined as the difference between the total number of failed units and the total failures reduced by the survival rate and by 1 minus the calculated wear out rate. The formula for the losses per assembly is as follows:

(annual unit demand) × (1 – calculated wear out rate) × (unit survival rate).

For the first year of the calculation, the starting inventory is the sum of the quantity on-hand and the number of units in the fielded systems/subsystems. The formula for the starting inventory is:

(unit spares quantity + systems in service +
(annual unit demand)) × (1 – calculated wear out rate) × (unit survival rate).

When unit spares are depleted, further failures attributable to one of the obsolete assemblies leads to readiness degradations. As indicated by the tan shading, if no mitigating action is taken, in FY18, only nine of the ten fielded assets will have an operational unit. Two years later, only half of the units will be operational.

[297] For example, demand may be calculated as a function of operating tempo and mean time between failures.

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Appendix M. Complete Department of Commerce Cost Survey Results

This appendix can be used, with caution, to modify the averages shown in Table 15, based on more specific circumstances. The appendix is a three-part table (see Table 62, Table 63, and Table 64) that contains the complete results from the 2014 Department of Commerce survey. The rows of the table show the above resolution options subdivided by environment (aviation, ground, shipboard, space, and undersea). The columns show the commodity type (electrical, mechanical, and electronics) subdivided by item type (assembly, component, raw material, and software). Entries in the table are average cost (adjusted to FY24 dollars) and sample size. Little confidence should be placed in entries with a low sample size.

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Table 62. Number of DMSMS Resolutions Reported and Average Cost (FY24 $s) by Type, Commodity, and Environment (Part 1)

	Electrical					
	Assembly		Component		Total	
	No.	Average	No.	Average	No.	Average
Approved Parts	86	$3,584	901	$258	987	$547
Aviation	10	$1,945	191	$1,215	201	$1,251
Ground	76	$3,799	0	—	76	$3,799
Shipboard	0	—	0	—	0	—
Space	0	—	710	—	710	—
LON Buy	3	$7,484	24	$16,166	27	$15,201
Aviation	0	—	1	$73,911	1	$73,911
Ground	3	$7,484	3	$18,239	6	$12,862
Shipboard	0	—	0	—	0	—
Space	0	—	20	$12,967	20	$12,967
Simple Substitute	0	—	190	$4,298	190	$4,298
Aviation	0	—	89	$6,432	89	$6,432
Ground	0	—	4	$27,392	4	$27,392
Shipboard	0	—	0	—	0	—
Space	0	—	97	$1,390	97	$1,390
Undersea	0	—	0	—	0	—
Complex Substitute	1	$167,272	33	$20,183	34	$24,510
Aviation	0	—	0	—	0	—
Ground	1	$167,272	29	$18,725	30	$23,677
Shipboard	0	—	4	$30,748	4	$30,748
Undersea	0	—	0	—	0	—
Extension of Production or Support	27	$35,010	0	—	27	$35,010
Aviation	27	$35,010	0	—	27	$35,010
Ground	0	—	0	—	0	—
Shipboard	0	—	0	—	0	—
Repair, Refurbishment or Reclamation	1	$9,725	0	—	1	$9,725
Aviation	0	—	0	—	0	—
Ground	1	$9,725	0	—	1	$9,725
Shipboard	0	—	0	—	0	—
Undersea	0	—	0	—	0	—
Development of a New Item or Source is no longer used						
Redesign–NHA	1	$417,034	1	$25,934	2	$221,485
Aviation	1	$417,034	1	$25,934	2	$221,485
Ground	0	—	0	—	0	—
Shipboard	0	—	0	—	0	—
Redesign Complex/System Replacement	12	$27,379,841	0	—	12	$27,379,841
Aviation	12	$27,379,841	0	—	12	$27,379,841
Ground	0	—	0	—	0	—
Shipboard	0	—	0	—	0	—
All for Part Type	142	$2,349,295	1152	$5,830	1294	$262,995

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Table 63. Number of DMSMS Resolutions Reported and Average Cost (FY24 $s) by Type, Commodity, and Environment (Part 2)

	Electronics									
	Assembly		Component		Raw Material		Software		Total	
	No.	Average	No.	Average	No.	Average	No.	Average	No.	Average
Approved Parts	47	$3,028	269	$1,023	0	—	0	—	316	$1,321
Aviation	31	$80	207	$386	0	—	0	—	238	$346
Ground	1	$5,891	45	$971	0	—	0	—	46	$1,079
Shipboard	15	$8,929	17	$8,670	0	—	0	—	32	$8,929
Space	0	—	0	—	0	—	0	—	0	—
LON Buy	86	$4,650	547	$6,695	0	—	0	—	633	$6,417
Aviation	44	$5,685	392	$6,862	0	—	0	—	436	$6,743
Ground	10	—	75	$2,305	0	—	0	—	85	$2,034
Shipboard	32	$4,681	68	$2,341	0	—	0	—	100	$3,090
Space	0	—	12	$53,336	0	—	0	—	12	$53,336
Simple Substitute	91	$4,330	1141	$19,751	1	$25,934	0	—	1233	$18,618
Aviation	80	$3,128	989	$21,149	1	$25,934	0	—	1070	$19,805
Ground	0	—	105	$9,568	0	—	0	—	105	$9,568
Shipboard	11	$13,071	47	$13,071	0	—	0	—	58	$13,071
Space	0	—	0	—	0	—	0	—	0	—
Undersea	0	—	0	—	0	—	0	—	0	—
Complex Substitute	63	$26,072	268	$36,454	0	—	0	—	331	$34,479
Aviation	40	$5,303	265	$36,756	0	—	0	—	305	$32,631
Ground	4	$326,819	3	$9,855	0	—	0	—	7	$190,978
Shipboard	19	$6,483	0	—	0	—	0	—	19	$6,483
Undersea	0	—	0	—	0	—	0	—	0	—
Extension of Production or Support	9	$17,829	54	$14,457	5	$253,631	0	—	68	$32,488
Aviation	9	$17,829	51	$15,179	5	$253,631	0	—	65	$33,889
Ground	0	—	3	$2,162	0	—	0	—	3	$2,162
Shipboard	0	—	0	—	0	—	0	—	0	—
Repair, Refurbishment, or Reclamation	12	$115,734	26	$75,781	0	—	0	—	38	$88,397
Aviation	0	—	4	$227,790	0	—	0	—	4	$227,790
Ground	1	$49,793	1	$960,840	0	—	0	—	2	$505,316
Shipboard	5	$164,225	21	$4,681	0	—	0	—	26	$35,363
Undersea	6	$86,316	0	—	0	—	0	—	6	$86,316
Development of a New Item or Source data is no longer used										
Redesign–NHA	56	$2,155,035	72	$848,791	0	—	4	$2,918,159	132	$1,465,664
Aviation	16	$2,091,709	25	$1,866,744	0	—	1	$1,755,965	42	$1,949,807
Ground	33	$2,566,945	5	$213,382	0	—	3	$3,305,556	41	$2,333,970
Shipboard	7	$357,914	42	$318,513	0	—	0	—	49	$324,142
Redesign Complex/ System Replacement	22	$7,453,358	9	$10,327,307	0	—	0	—	31	$8,287,731
Aviation	11	$6,950,787	7	$12,810,926	0	—	0	—	18	$9,229,730
Ground	3	$1,886,421	2	$1,634,640	0	—	0	—	5	$1,785,708
Shipboard	8	$10,231,996	0	—	0	—	0	—	8	$10,231,996
All for Part Type	410	$815,143	2465	$97,016	6	$215,681	4	$2,918,159	2885	$203,231

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Table 64. Number of DMSMS Resolutions Reported and Average Cost (FY24 $s) by Type, Commodity, and Environment (Part 3)

	Mechanical								All Res. Types/ Environment	
	Assembly		Component		Raw Material		Total			
	No.	Average	No.	Average	No.	Average	No.	Average	No.	Average
Approved Parts	0	—	228	$4,376	8	$12,053	236	$4,637	1539	$1,333
Aviation	0	—	0	—	3	$8,645	3	$8,645	442	$814
Ground	0	—	226	$4,336	1	$5,814	227	$4,343	349	$3,794
Shipboard	0	—	2	$8,980	4	$16,167	6	$13,771	38	$9,694
Space	0	—	0	—	0	—	0	—	710	—
LON Buy	1	$14,912	5	$6,577	0	—	6	$7,966	666	$6,787
Aviation	1	$14,912	0	—	0	—	1	$14,912	438	$6,915
Ground	0	—	5	$6,577	0	—	5	$6,577	96	$2,947
Shipboard	0	—	0	—	0	—	0	—	100	$3,090
Space	0	—	0	—	0	—	0	—	32	$28,106
Simple Substitute	3	$8,103	72	$7,921	2	$49,922	77	$9,020	1500	$16,311
Aviation	0	—	0	—	1	$2,593	1	$2,593	1160	$18,766
Ground	3	$8,103	72	$7,921	0	—	75	$7,929	184	$9,287
Shipboard	0	—	0	—	0	—	0	—	58	$13,071
Space	0	—	0	—	0	—	0	—	97	$1,390
Undersea	0	—	0	—	1	$97,251	1	$97,251	1	$97,251
Complex Substitute	5	$34,796	19	$26,037	21	$28,313	45	$28,072	410	$32,949
Aviation	0	—	1	$33,714	4	$38,576	5	$37,604	310	$32,711
Ground	0	—	0	—	1	$42,277	1	$42,277	38	$54,986
Shipboard	5	$34,796	18	$25,611	15	$17,888	38	$23,771	61	$18,843
Undersea	0	—	0	—	1	$129,668	1	$129,668	1	$129,668
Extension of Production or Support	0	—	2	$28,190	1	$25,934	3	$27,438	98	$33,029
Aviation	0	—	0	—	1	$25,934	1	$25,934	93	$34,129
Ground	0	—	0	—	0	—	0	—	3	$2,162
Shipboard	0	—	2	$28,190	0	—	2	$28,190	2	$28,190
Repair, Refurbishment, or Reclamation	1	$3,286	0	—	0	—	1	$3,286	40	$84,304
Aviation	0	—	0	—	0	—	0	—	4	$227,790
Ground	1	$3,286	0	—	0	—	1	$3,286	4	$255,912
Shipboard	0	—	0	—	0	—	0	—	26	$35,363
Undersea	0	—	0	—	0	—	0	—	6	$86,316
Development of a New Item or Source data is no longer used										
Redesign–NHA	2	$172,459	2	$651,167	0	—	4	$411,813	138	$1,417,085
Aviation	1	$181,535	1	$1,118,387	0	—	2	$649,961	46	$1,818,148
Ground	0	—	0	—	0	—	0	—	41	$2,333,970
Shipboard	1	$163,382	1	$183,947	0	—	2	$173,664	51	$318,240
Redesign Complex/System Replacement	1	$1,491,183	0	—	0	—	1	$1,491,183	44	$13,340,202
Aviation	0	—	0	—	0	—	0	—	30	$16,489,774
Ground	1	$1,491,183	0	—	0	—	1	$1,491,183	6	$1,736,621
Shipboard	0	—	0	—	0	—	0	—	8	$10,231,996
All for Part Type	16	$602,730	334	$13,430	33	$25,733	383	$39,108	4562	$206,404

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Appendix N. Programming and Budgeting for DMSMS Resolutions

As introduced in Section 7.1, this appendix includes best practices and additional information related to programming and budgeting for DMSMS resolutions. Four topics are organized as follows:

- N.1 Best practices,
- N.2 Considerations for funding DMSMS resolutions in the year of execution,
- N.3 Leveraging WCFs to fund DMSMS resolutions, and
- N.4 Other resources that may be available to finance DMSMS resolutions.

N.1 Best Practices

These six enabling best practices encompass processes and procedures for determining funding requirements, taking other sources of funding into account, the type of appropriations that may be used, justification, communications, and the very important need to utilize POM/budget line items dedicated to this subject. Section 7.2 provides additional information on the relationships between DMSMS resolution funding and modification planning.

N.1.1 Estimate the Funding for DMSMS Resolutions

The first enabling best practice is for program offices to establish processes and procedures to estimate the funding required to resolve its known and anticipated DMSMS issues within the programming and budgeting horizon. Known DMSMS issues are the items that are obsolete at the time that programming and budgeting takes place. However, given the inevitability of obsolescence, additional DMSMS issues will materialize during the programming time horizon. Since there is a lack of certainty about what items will become obsolete and how those issues will be resolved, the term *anticipated obsolescence* is used in this document. This best practice discusses resolution cost estimating techniques for programming and budgeting for both known and anticipated issues.[298]

Regardless of uncertainty, program offices should resist falling victim to the myth that predicting DMSMS resolution costs is not possible or too hard. Successful programming and budgeting submissions must be convincing to decision makers; therefore, submissions should be based on a reasonable estimation approach. Although additional criteria may be applicable as a function of the specific situation, an approach should have the following characteristics:

- Realistic and defensible assumptions are used to calculate cost estimates in a repeatable way.
- The methodology is easy to understand and explain.
- Estimates should be methodologically sound and based on as much actual data as possible.
- More than one estimating method is considered.

The timeframe for routine programming/budgeting is from two to six years in the future. Typically, budget estimates are developed two years in advance of budget execution. Often, at the point in time when a DMSMS issue is identified, it is too late to impact the current (or execution) year or even the following year without reprogramming funds because the budgets have been finalized (or if not finalized, out of the

[298] Section 6.3 describes the AoA used to determine the preferred resolution.

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Department's control).[299] When cost estimates for the second year are approved, they become the basis for the next budget. Cost estimates beyond that year are part of the POM.

Estimates should be developed by the DMT. Several approaches can be used to help estimate resolution costs and, therefore, inform the program office on what is required to support resolution funding. Despite diligent efforts to apply an appropriate cost estimation method for the system, stakeholders understand that funding requirements for DMSMS resolutions and corresponding budget requests, which are based on the best available data, will almost always be either too high or too low. Because DMSMS management is not standalone from a financial management perspective, there will always be opportunities to use extra funding for other reliability, maintainability, or supportability issues. If program office resources are not sufficient to implement a preferred resolution, other alternatives are possible.

Table 65 summarizes different approaches that could be pursued to estimate resolution funding. These approaches are listed by the extent of analytical rigor in deriving the estimate (with the most rigorous at the bottom). Each program office will have to make its own assessment regarding the level of effort and degree of difficulty it is willing to pursue to estimate DMSMS resolution costs to inform programming and budgeting. A key consideration is the amount of rigor necessary to successfully defend the funds requests.

Table 65. Description of DMSMS Resolution Cost Estimation Approaches

Approach	Description
Estimate by prior trends	Calculate and use the historical average costs and trends for each resolution type
Estimate by technology segment	Analyze technology segments and what is known of their lifespans to determine when and where obsolescence is predicted to impact a design
Estimate by cases	Analyze specific items to estimate discontinuation time

Building programs/budgets using any one of these approaches implies assumptions about the timing of DMSMS impacts so that appropriate resolution funding is in place to avoid those impacts. These three approaches are discussed in the next three sections. In all cases, the final estimate should be adjusted by SME opinion. The DMT should develop recommendations in conjunction with the prime/OEM/logistics support provider, where applicable. When a commercial company is involved, generally it will make initial recommendations on the resolutions. The DMT considers these recommendations but goes through the analysis process as described in Section 6.3 to determine its recommended resolutions.

N.1.1.1 Estimate by Prior Trends

These estimating methods apply both for a well-established program office and a new program office. The new program office would use data from a similar system.

The simplest way of making these estimates uses an annualized average of historical cost data as the estimated requirements for each year in the programming timeframe. A slightly more complex approach takes trends into account. Generally, obsolescence events increase with age because the useful life of items is finite. Estimates are based on an extrapolation of historical data based on trend lines or moving averages and SME opinion to adjust for factors not captured in the trends. For example, when a product roadmap indicates a near-term change, SME adjustments will be necessary. The extrapolations should also take into account any changes in the extent of proactive monitoring being conducted, in which case, the extrapolations should be limited to the portions of the historical data reflecting the current approach to DMSMS management. These methods work best for a system in sustainment. To develop a scenario for DMSMS issues expected during design or production, the DMT should consider the contractor's plans for

[299] Reprogramming is always possible, however, that should not be a consideration in program/budget formulation.

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modifications, information from manufacturers, the age of the technology in active electronic items, and relevant experience with similar systems.

A third approach divides historical data into categories. For example, historical data may be categorized by type of resolution and separate extrapolations may be made for each resolution type and then combined into a single value for each year. The potential advantage to this more detailed calculation is that it may provide SMEs with more information for making adjustments. If no adjustments are made, extrapolations by category may not improve the accuracy of the total annual estimate.

The NAVAIR developed model discussed in Appendix H.2, also has the capability to estimate resolution costs for electronic boxes based on historical averages. The problem with basing program/budget estimates solely on historical data, even after allowing for expert adjustments, is that program/budget decision makers may not find this approach convincing.

N.1.1.2 Estimate by Technology Segment

Another technique in use today estimates total annual DMSMS resolution costs and DMSMS management costs (as discussed in Section 3.4.1) for the entire program/budget timeframe. This technique involves technology segments. This approach benefits from contributions from the systems engineering team, who has knowledge of system life and the difficulty of resolving issues. Linkages to the product support community also help. The first step in the technology segment approach is to segregate the equipment into technology segments with predictable life cycles. Examples of technology segments include analog to digital conversion, amplifiers, backplanes and chassis, and so forth. For each of these segments, the expected life cycle (or technology refreshment period) and the breakdown of DMSMS resolutions by type is estimated.[300] By allocating assemblies to each technology segment a projection of total annual DMSMS cost can be developed. The following bullets summarize the computations:

- Estimate the number of years in the life cycle for *each* technology segment.
- Based on historical data, use SMEs to estimate the *average fractions* of resolutions by type for the technology segment. These data should be estimated by the DMT based on an extrapolation of historical data for the entire system.
- Identify assemblies with a high probability of obsolescence impact and determine their technology segment.
- Derive CERs for monitoring and surveillance costs, engineering integration and kit development, kit procurement and installation, support product data updates, and management, taking the distribution of resolutions into account.[301]
- Calculate the results of the CERs for each of the items with a high probability of obsolescence and sum over all items to obtain an estimate of total annual cost.

One drawback to this approach is that it can be time consuming. A second drawback is that programs may have little existing data available to use to derive the CERs. There may be potential simplifying assumptions, but none are in general use.

N.1.1.3 Estimate by Cases

There are two building blocks for estimating program/budget resolution requirements based on cases:

- ***Costs and timing associated with known DMSMS issues***. This building block pertains to already obsolete items that can no longer be procured and to items where a PDN or equivalent has been received.

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[300] Just three aggregate resolution types may be used—low cost, medium cost, and high cost.

[301] In theory, different CERs may be derived.

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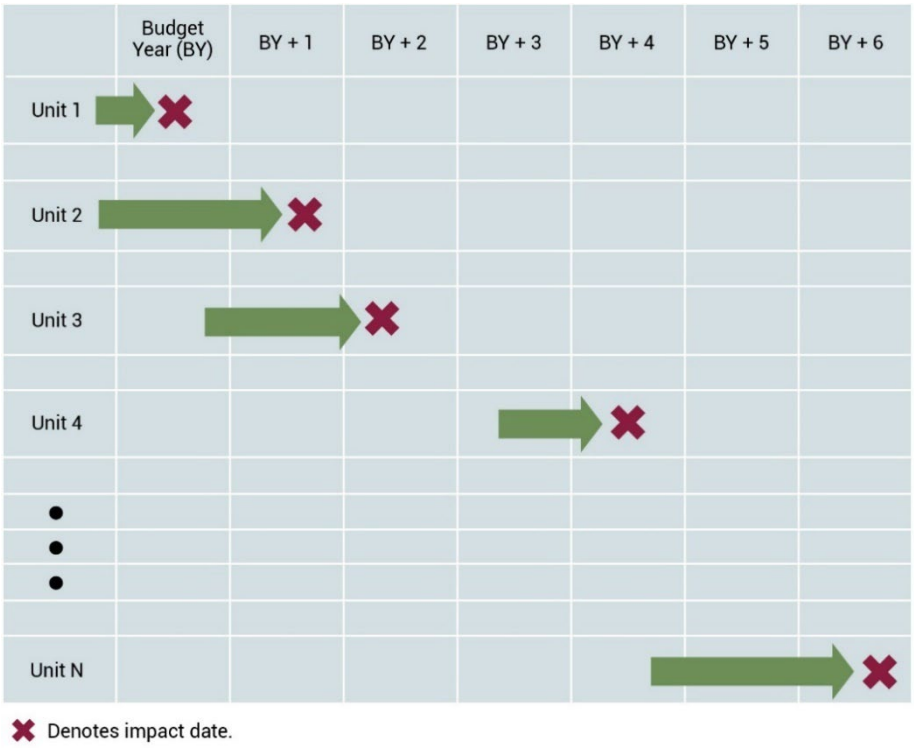
- ***Costs and timing associated with potential, future DMSMS issues***. This building block pertains to items where there is no known issue at the current time, but an issue could manifest itself at some point in the program/budget timeframe. In some cases, the issues may be anticipated based on expected life as projected by predictive tools, vendor surveys, or other research. In other instances, the issues could be completely unanticipated.

N.1.1.3.1 Estimating Program/Budget Requirements for Known, Current DMSMS Issues

The resolution-based health assessment technique (as described in Section 5.3.1 and Appendix L) is a best practice for making program/budget estimates for the first building block—resolutions for known, current DMSMS issues. Such an assessment determines the dates of impact (i.e., the point in time that a program office can no longer support the system) generally at the unit (e.g., box, WRA, LRU, software application) or sometimes assembly (SRA/LRU) level for the entire system. The impact date is a function of demand rate and stock on hand.

Figure 36 notionally depicts the results of a health assessment. The figure shows the expected DMSMS impact date as the point at which the implementation should be complete for various units in the system. The green arrows indicate when funds should be programmed/budgeted to resolve the issue before impact. Determining the appropriate lead time should consider technical content, non-recurring engineering, installation, and take into account backlogged issues, workload capacity, and administrative lead time.

Figure 36. Notional Depiction of Impact Date and Timing of Funding Requirements



At any point in time, implementation of some resolutions may be underway. In Figure 36, this would apply to units 1 and 2. In the case of unit 3 for example, funding should be put in place in the budget year to avoid an impact 2 years after that. For unit 4, funds should be programmed for three years after the budget year.

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The health assessment provides information regarding impact. Once that is determined, resolutions are evaluated in an AoA and the preferred resolution is selected. In many cases, the impact can be mitigated for a short time until a redesign can be planned and funded. Resolution cost should be estimated in the normal way by SMEs as described in Section 6 (the *Analyze* step of DMSMS management).

There is a potential alternative approach within this class of cost estimating techniques that simplifies the calculations. The basis for this modified approach is the empirical observation that most DMSMS resolution costs are associated with redesigns. An analysis of the 2014 Department of Commerce survey introduced in Section 6.3 determined that based on 4,562 DMSMS cases, and 95% of the costs, were associated with three redesign resolution types:

- Development of a new item or source,[302]
- Redesign—NHA, and
- Redesign—complex/system replacement.

A similar examination of thousands of other cases from two different data sets indicated that 80% and 96% of the resolution costs were associated with redesign.

Using an approach focused on redesigns simplifies the calculations because it should significantly reduce the number of issues where individual cost estimates would be needed. Once the redesigns are identified and their cost estimated (see Section 6.3), budgets and programs could be constructed that apply funding with the appropriate lead time before impact. Every non-redesign resolution identified by the health assessment could be priced using an average cost factor, preferably tailored to the individual program office. For these simpler resolutions, a program/budget lead time of one year should be sufficient.

N.1.1.3.2 Estimating Program/Budget Requirements for Potential, Future DMSMS Issues

There is a great deal of uncertainty associated with the second building block—projecting the time when an item that is currently being produced and sold will no longer be available for purchase. While health assessments for non-obsolete items have value for targeted monitoring, their use for programming/ budgeting is usually not practical. It may require a great deal of effort to project an end-of-sale date. That date may be hard to obtain and is often subject to large uncertainty, even in cases where the date is obtained via a vendor survey. In addition, the impact (as calculated in a health assessment) may be delayed by an unknown number of years because of an LON buy.

Utilizing technology refreshment planning (also known as design refresh planning) based on technology segments for estimating program/budget requirements for potential, future DMSMS issues is a best practice. As indicated in Section 4.6, the goal of technology refreshment is to resolve potential DMSMS issues before they materialize.[303] Because the DMSMS management community is not responsible for developing technology refreshment plans, this document does not describe methodological approaches. The DMSMS management community does provide input to the process (additional information is provided in Section 4.6). For example, health assessments for non-obsolete items may have an impact on refreshment planning and therefore should be providers to the technology refreshment planners.

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[302] This resolution type has been divided into development of a new source, design refreshment, and redevelop the item.

[303] The University of Maryland's Center for Advanced Life Cycle Engineering's Mitigation of Obsolescence Cost Analysis (MOCA) tools use technology segments to analyze a related problem. MOCA supports design refresh planning that optimizes either individual refreshment dates or (in a simplified application) periodic refreshment dates associated with the specific content and maintenance requirements of a system. A recent article on this subject provides a good list of references on this subject. See Peter Sandborn, "Design for Obsolescence Risk Management," *Procedia CIRP* 11 (2013): 15–22, available at https://doi.org/10.1016/j.procir.2013.07.073.

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N.1.1.3.3 Combining the Building Blocks

It is a best practice to combine the programs/budgets for the building blocks described in the previous two sections. Any duplicate redesigns (e.g., one from technology refreshment planning and the other from known DMSMS items that can no longer be procured) should be eliminated. The combined values should be compared on a year-by-year basis with a historical extrapolation (as discussed earlier) and the larger of the two should be selected for the final program/budget. Regardless of which value is greater, the details behind the building blocks should be included in the budget justification.

It is usually a mistake to base programs/budgets solely on resolutions from known, currently obsolete items if program offices do not have a funded technology refreshment plan. In this situation, it is a best practice to compare the program/budget for the currently obsolete items to historical extrapolations on a year-by-year basis and select the larger of the two. The details behind the resolutions for currently obsolete items should be included in budget justification.

N.1.2 Determine DMSMS Resolution Programming and Budgeting Requirements

This section discusses the enabling best practice for circumstances where DMSMS issues may be funded with resources obtained outside of programming and budgeting for DMSMS resolutions by the program office. When this is the case, funding requirements discussed previously may be decreased accordingly.

Some DMSMS resolutions during sustainment will be financed[304] (either in part or in entirety) by the organizations that manage WCFs, assuming that those organizations have sufficient lead time to award contracts to implement the resolution.[305] (See Appendix N.3 for a discussion of some of these programs.) There are, however, uncertainties associated with this process:

- The process varies by service. There are periodic project calls with different criteria for the project to meet. There are also processes to submit projects out-of-cycle.
- Organizations that manage WCFs will of course only consider contributing to resolutions for items that they manage.
- There may be limitations on the amount of money available for projects. Consequently, a project proposal may be rejected, or only partially funded, on that basis.
- The WCF corpus can only be used to support projects that provide an F3 replacement. Deliberate increases in capability are not allowed. However, there may be incidental improvement to capability from using newer technology.
- WCF dollars cannot be used to pay for fielding the changes. The obsolete items will be replaced by attrition (i.e., when the higher-level assembly cannot be repaired).
- There may be constraints on the extent of WCF resources available. For example, high-cost projects such as a redesign at a higher level of assembly may or may not be funded.
- The program office may be able to fund projects in conjunction with WCF projects. For example, the program office may fund costs associated with production, while WCFs would be used to finance costs associated with sustainment. More complex cost-sharing relationships may also be feasible. It is possible for joint funded or sequential projects. It is also possible to fund over several years.

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[304] The term *finance* is used instead of *fund* when referring to WCFs. Since WCFs are required to break even by including all their operating expenses in a cost recovery rate applied to all sales, WCFs technically do not fund anything. The cost of resolutions (among other things) is spread among all WCF customers in anticipation of future sales.

[305] Some programs are entirely CLS with no WCF involvement.

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Appendix N.4 lists other centralized, non-program office resources that may be used to resolve DMSMS issues. Generally, many projects compete for funding from these sources.

As was the case for DMSMS management operations, programming and budgeting for DMSMS resolutions may not be needed if a larger contract (e.g., a procurement contract, a modification contract, or a logistics support contract) is in place to fund them.[306] The program office should be aware of the level of funding allocated within the contract for DMSMS resolutions to ensure that there is alignment with the internal program office estimate. Separate programming and budgeting for resolutions applies only to DMSMS issues not funded within the contract. For resolutions not covered on contract, the program office may need to modify the contract to implement these resolutions.[307] Potentially, a separate contract may be used, depending on which course of action is preferable.

The program office must recognize that resolutions handled by the contractor generally apply only through the end of the contract period of performance and the government needs to understand that another resolution may be needed before a new contract is negotiated to extend beyond the period of performance of the initial contract.[308] During production, the contractor is responsible for delivering the units on contract, but programming and budgeting procurement appropriations may be necessary to implement a resolution before the next contract option is exercised. Depending on the timing, the resolution may be included in the funding for the follow-on contract. That resolution may be applied to already fielded units as the need arises. Even in sustainment with a PBL contract, programming and budgeting for DMSMS resolutions may need to consider programming and budgeting for the redesigns necessary to award the next PBL contract.

N.1.3 Consider the Appropriations Available

This enabling best practice involves taking the type of appropriation into account. As was described under programming and budgeting for DMSMS management operations, different appropriations are available to program offices as a function of where they are in the life cycle. During initial design and development, as well as design and development associated with modification, the non-recurring engineering and testing cost of DMSMS resolutions associated with the subsystems involved in the effort will normally be funded with the research, design, test, and evaluation (RDT&E) appropriation. Similarly, DMSMS issues associated with production, will normally be funded with one of the procurement appropriations.

The situation with fielded systems is more complicated, as follows:

- In some situations, when the system provides a service (e.g., airlift services), customers who pay for the service may be called upon to transfer additional amounts of their own funds (normally an O&M appropriation) to the service provider for resolutions.
- A program office may build a POM and budget for a redesign resolution using RDT&E appropriations to develop and test the resolution. For immediate fielding, depending on the item cost, procurement or O&M funding would be used to procure the item. O&M appropriations are normally used to install it. This could be treated as an independent modification effort. For fielding by attrition, no DMSMS resolution funds would be needed for procurement and installation; they would be covered through customer funds.

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[306] This is often the case for non-class 1 changes. Class 1 changes imply significant impact to functional and physical interchangeability or supportability. There is often a change to F3 or interface.

[307] For example, the DMSMS issue may apply to an item in a subsystem that is not being changed by a modification contract. Another theoretical example is one in which the contractor is required to fund a resolution that only applies until the end of the contract but the program office would rather fund a long-term resolution immediately and not wait until the period of performance has elapsed.

[308] There are examples of contractors making LON buys that go beyond the end of the contract if there is a business case for them to do so.

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- O&M funds may be used to fund the non-recurring engineering and testing for most low-cost resolutions that do not involve redesign. There are different categories of O&M funding that may apply. For example, there is sustaining engineering funding for both hardware and software, as well as corresponding logistics-related efforts.

N.1.4 Establish a Dedicated POM/Budget Line

As an enabling best practice, program offices should establish dedicated POM/budget lines for funding DMSMS resolutions. All the reasons described in the fourth enabling best practice for DMSMS management operations (see Appendix H.4) apply here. In addition, the DMSMS management community should not perpetuate the myth that it cannot project DMSMS resolution costs well enough to have a POM/budget line item. Some form of the projection methodologies from Section 5.3.1 and Appendix N.1.1 are used by many program offices. Of course, these methodologies only provide estimates, and these estimates could be too high or too low. But there is no downside to either if there was no deliberate attempt to bias the estimate for some other purpose. If it happened that the estimate is too high, program office management will welcome extra funding to apply to some other unfunded requirement. If the estimate is too low, Appendix N.2 of this document covers some ideas for obtaining extra resources during budget execution.

If there is no POM/budget line for DMSMS resolutions, the DMT will have to seek ad hoc funding every time an issue arises. That leads to an erroneous impression that the DMSMS management community is a disruption to normal program office financial operations when the opposite is true. The proactive DMSMS management community forewarns program office management about problems before they cause disruptions.

Ad hoc funding is *never* a best practice; it's what you do when all else fails. Furthermore, an ad hoc process can negatively impact program office performance, schedule, and cost. An ad hoc environment is likely to be reactive, consequently, there will be an implementation delay as compared to having a dedicated POM/budget line item to rely upon. Delays may occur in 1) determining what resolution should be pursued if it has not been considered earlier and 2) in identifying the funding necessary to implement the approved approach. These delays could increase cost because the window of opportunity to resolve them will be shorter. The delays could also contribute to disruptions to operations (e.g., reductions to mission capability, cannibalization) and schedule. Costs may also increase because trying to initiate modifications in a hurried way may suboptimize vendor selection, contracting strategy, and contract clauses and sacrifice economies of scale.

Although it may be necessary to treat an unanticipated DMSMS issue in an ad hoc way even when a formal programming and budgeting process was used, a dedicated POM/budget line item will reduce the frequency of obtaining ad hoc funding. In addition, obtaining funding in an ad hoc situation implies diverting resources from already funded activities. Those previously funded activities will generally be negatively affected. Lastly, ad hoc processes usually demand much more time and attention of people at all levels in the program office. This can take time away from other important activities and lower morale.

Having a POM/budget line item improves planning because it provides a basis for tradeoffs based on the funding profile available. As in the case of DMSMS management operations, a POM/budget line item gives visibility to the reality of obsolescence and the importance placed on addressing that obsolescence. It informs program office management that expenses must be incurred while illustrating that planning has been done for those expenses. It is also an opportunity to justify and defend the funds requested to all the organizations that review a program office's DMSMS resolution POM or budget. This is especially important when large DMSMS resolution funding requirements are expected.

Implementing DMSMS resolutions is not an optional activity—readiness, schedule, and cost impacts will occur otherwise. Showing the amount needed explicitly may serve to insulate this money from external

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and internal reductions. A POM/budget line item for DMSMS resolutions may be just internal to the program office or it may be displayed on external budget submissions. Visibility may help protect DMSMS-related funding; hiding it may have the opposite effect.

Naturally there should be a spend plan associated with the POM/budget line item for DMSMS resolutions unless it is part of a larger contract for design, development, production, or logistics support. As stated in Section 3.4.1, procedures for obtaining resources are Component dependent, and even within a single Component, those procedures may not be the same across all program offices. Regardless of Component, including DMSMS issue resolution requirements in the budgets of other activities, such as PM&P management, reliability and maintainability, or supportability activities, is often a successful tactic.

N.1.5 Prepare Programming and Budgeting Justification

Program offices should prepare persuasive programming and budgeting justification materials to support the need for funding for DMSMS resolutions as an enabling best practice. Justify resolution costs for known and anticipated issues separately because stronger justification can be established for currently obsolete items. If for some reason, the financial (or any other stakeholder) community questions the funding justification, it is important to discourage arbitrary reductions to the maximum extent possible by providing as much specific information as possible. Justification for anticipated issues will be based on the methodology used to estimate their cost.

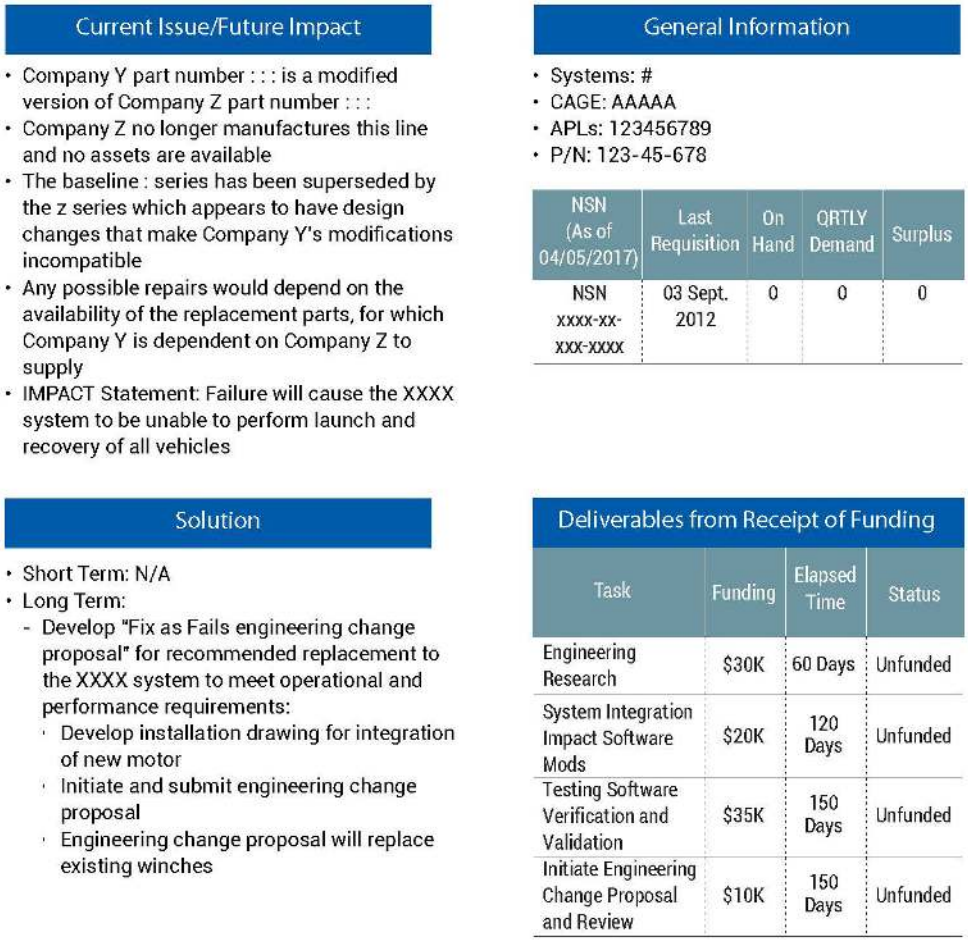
The use of a quad chart mechanism (or something analogous to it) can be an effective way to justify programming and budgeting requests for DMSMS resolutions. Quad charts should be socialized in advance with all key stakeholders. The following information for a DMSMS issue and its recommended resolution should be conveyed:

- Provide pertinent general information on the obsolete items, e.g., part number, manufacturer, supplier if appropriate, expected demand, number on-hand, consumption rate, and so forth.
- Describe the issue, the impact that will occur if the issue is not resolved, and the expected date that the impact will begin to materialize.
- Describe the recommended approach for resolving the issue, any risks associated with the approach, and how those risks will be managed.
- Show the funding profile needed to resolve the issue over time, the tasks to be funded, and estimated completion dates.

Figure 37 is an illustrative, generic example of a quad chart.

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Figure 37. Notional Quad Chart



N.1.6 Ensure Stakeholders Recognize the Importance of Funding Resolutions

Appendix H.4 covered this enabling best practice for programming and budgeting for DMSMS management operations. Some of the material applies to both management operations and resolutions and therefore is repeated here.

DMSMS management practitioners do not solely control the effectiveness and efficiency of DMSMS management–related activities in program offices. Everyone in a program office contributes. Programming and budgeting can only be successful if that is the case. Outreach to key stakeholders is the first essential step. While the need for outreach is mentioned elsewhere in this document, that material is organized by best practice. This section is organized by the stakeholder community interacting inside and outside the program office. It summarizes the information that should be conveyed and how the stakeholder community can help with DMSMS resolution-related programming and budgeting.

N.1.6.1 Program Office Management

The PM is the ultimate decision maker. To a large extent, the technical aspects of the programming and budgeting request should be coordinated and supported by the chief engineer and the PSM.

Program office management needs to understand the value of separate programming and budgeting line items for DMSMS resolutions (for both known and anticipated DMSMS issues) in terms of their

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contributions to proactivity and efficient program office operations. Program office management should also understand that lead times should be built into DMSMS resolution fund requests. Resources may be needed before the issue is expected to impact the system (perhaps several years before) because it takes time to develop and procure the necessary items. Finally, program office management should be aware that when unanticipated DMSMS issues occur, additional funding (not included in the budget) could be needed during execution because program office management will be called upon to help identify potential sources of funding to resolve those previously unanticipated DMSMS issues.[309]

N.1.6.2 Contracting

The contracting community should recognize that new contracting actions could be required during budget execution to resolve previously unanticipated DMSMS issues. In some cases, especially for LON buys, these contracting actions for DMSMS resolutions must be exercised quickly. Contracting should also understand that to avoid a negative impact on the system, putting some resolutions on contract could be time sensitive. In the case of an LON buy, the window for final purchase may be very brief. In addition, the contracting community should explain what it needs from the DMSMS management community to satisfy the contracting community's requests.

Another key interface with the contracting community relates to the different ways that resolution funding may be implemented. The prime contractor/OEM may be contractually obligated, under a fixed-price contract line item number (CLIN), to resolve all, some, or none of the DMSMS issues during development, production, or sustainment. This has an effect on how the total program and budget estimates are split between such fixed-price contract amounts and later contract modifications. For the fixed-price contract situations, the contractor resolution may only resolve the issue until the end of the contract, e.g., the contractor will buy additional items to resolve a DMSMS issue to satisfy demands up to the end of the contract period of performance.[310] The government should consider funding the portion of the costs that are beyond the contract end date. Otherwise, the government can expect to incur redesign costs before a new contract can be signed or as part of the new contract. This could be complicated further if the government does not have appropriate technical data rights to accomplish the redesign.

N.1.6.3 Financial Management

Along with program office management stakeholders, the financial management community also must appreciate the importance of having separate line items for DMSMS resolution-related programming and budgeting. In addition, these stakeholders need an understanding of and degree of comfort with how DMSMS resolution–related funding requests are determined. They must be aware of and support the estimation of the funding required to resolve both known and anticipated issues. The unanticipated element is very important since it is nearly a certainty that such issues will occur but the specific details are not certain. The financial community must recognize the validity of the estimates for resolving unanticipated DMSMS issues and not create barriers because of any uncertainty in the details because not resolving those issues could negatively impact cost, schedule, and readiness.

The financial management community is also a source of key information for DMSMS resolution–related programming and budgeting. They can provide information about the process and the types of appropriations that can be used for DMSMS activities and help make the justifications for the funding as

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[309] Although the budget should contain funding for unanticipated issues, it may be the case that the requirements exceed the funding available.

[310] In one sense, a prime contractor should never develop a design for the government for which parts are not available for manufacturing, either LRIP or for all the options of the first production contract. That is not to say that there will not be any obsolescence. The prime contractor should stockpile enough of the obsolete items to meet the terms of these contracts (i.e., to produce prototypes, engineering design models, LRIP units, units for the first production contract, and initial spares for the units produced for the first production contract). That is not always the case in practice.

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persuasive as possible. Financial management stakeholders should be aware that the cost of resolving unanticipated DMSMS issues could exceed the amount budgeted for that purpose. In that case, additional funding (not included in the budget) could be needed during execution. Since this community will have knowledge of budget execution rates, it also may be useful in identifying sources of funding for unanticipated issues that arise. Because there may be extensive approvals required for LON buys that exceed a certain quantity, the financial management community should know about the calculation of LON quantities and the fact that these quantities may be required to last well into the future. The financial management community can therefore help with the justification process. Finally, the financial management community should be aware of color of money repercussions because resolving DMSMS issues may increase capability.

N.1.6.4 WCF Organizations

The organizations that manage WCFs also pay for certain resolutions during sustainment. In some cases, these efforts are driven by requests from the program office, but in other cases, the requests may have been generated in the WCF organizations themselves. These resolutions include, but are not limited to, LON buys. Two-way communications on all these issues are essential not only to ensure that everyone knows what is happening but also to enable alignment with program office modification efforts which are often not known by WCF organizations.

N.1.6.5 IPTs

This is a generic heading because every program office may have a different IPT structure. IPT stakeholders include those organizations that are concerned with both readiness and system development and modification. DMT interfaces with various IPTs should be made on those IPTs' own terms. For example, it is generally more effective for DMSMS SMEs to attend the meetings of other IPTs than to invite various IPTs to attend the program office's DMT meetings. The DMSMS management community should convey how it can help the various IPTs and then what the IPTs need to do to better enable that support. This could eventually lead to a situation where the various IPTs contact the DMT to provide key information.

The appropriate IPTs should be informed of the relationship between DMSMS management and readiness. The "readiness" IPT (or the IPT most concerned with the subject, whatever it is called) can help support DMSMS resolution programming and budgeting requests. Multiple IPTs usually support the program office's modification and development efforts. It is important for planned modifications to be considered in the development and AoA to enable the DMT to determine the length of time that a short-term resolution should cover based on existing plans and/or to ask about whether/how existing plans could be revised. Without such information, there could be inefficiencies or wasted effort on the part of the DMT. The DMT may be trying to come up with a resolution without knowing the appropriate time horizon. In theory, if communications are poor, the wrong LON quantity may be purchased.

DMSMS health assessments should be communicated to the appropriate IPTs as an input to their modification planning. The appropriate IPTs need to know about planned DMSMS resolutions because they may impact modification plans. Efforts should be made to improve receptivity to the DMSMS information. When unanticipated issues arise during budget execution, certain IPTs may be able to provide funding if the DMSMS resolution–related execution budget line was not sufficient to resolve the issue. There also may be a relationship between certain IPTs and the engineering authorities external to the program office that technically approves a DMSMS resolution.

N.1.6.6 Engineering

This interface is important because technical approval of resolutions must be made by the appropriate engineering authority. The engineering authority may be in different organizations depending on how the service is organized—it may be at the HQ, field command, or program office level. The program office is

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of course always involved regardless of whether it is the technical decision maker. Depending on when the impact is projected for a DMSMS issue, technical approval may be preliminary. Nevertheless, the technical authority must be consulted to provide the best assessment of how the issue will be resolved to inform the resolution cost estimate, even if final approval has not been given.

N.2 Considerations for Funding DMSMS Resolutions in the Year of Execution

As stated in Section 7.1, reliance on obtaining funding for DMSMS issues in execution year is not a best practice. Despite this, financial issues may still arise during budget execution. This section discusses several budget execution considerations.

N.2.1 Potential Funding Sources during Budget Execution

During budget execution, unanticipated DMSMS issues may be discovered or the cost of implementing a resolution could be significantly higher than expected. As a result, the funds budgeted for DMSMS resolutions in the year of execution may be too low and it may become necessary to identify other sources of funding to resolve those issues quickly to prevent a specific impact (e.g., cost, schedule, and/or readiness) on the program office. The following are some potential approaches for obtaining additional resources given a high enough priority for initiating a resolution for the DMSMS issue during budget execution.

- ***Obtaining funds from other budget allocations***. During budget execution, unused funds may emerge. For a variety of reasons, appropriated budgets may not be obligated for their intended purposes and unless obligated for some purpose, those funds will expire at the end of the budget execution year.[311] This type of situation may be faced internal to a program office (e.g., budgets allocated to initial spares) or external to a program office (e.g., at a higher echelon of command such as the PEO or other organizations such as operating units). When that occurs, organizations often seek potential projects that can obligate those expiring funds (called sweep-up funds) just before the end of the fiscal year.[312]

  A centralized asset manager should be aware of the execution status of all aspects of the program office's budget. Likewise, asset managers for organizations external to a program office will be aware of the execution status of its funds. If implementation plans for DMSMS issues are already prepared, technically approved, and can be quickly put on contract, then these efforts would compete well for sweep-up funds. If unobligated funding is found, there may be the opportunity to cost share using some of those funds to resolve a DMSMS issue. Operating forces using a system with a DMSMS issue may be willing to provide funding to resolve the issue if the program office cannot find another alternative. For example, budgets for replenishment spares held by operating sources could be considered.
- ***Leveraging FMS***. DMSMS issues that affect DoD systems will affect similar systems sold to foreign partners if they are using the same items. Resolution funding should be equitably cost-shared among all affected partners, including DoD (see Section 3.5.1.2).
- ***Reallocating funds from ongoing projects where there is sufficient flexibility to slow the expenditure rate***. The execution of system modifications or even other DMSMS resolutions may be able to be delayed without severe negative repercussions in the current budget execution year assuming that funding is restored in the following budget year when

[311] For example, unexpected technical issue could delay the signing of follow-on contracts or contract negotiations and/or administrative lead time could be longer than planned.

[312] Some organizations do this both in the middle and at the end of the year.

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additional resources become available. There should be a process for adjusting spend plans rapidly if and when the need arises.

A key to success in overcoming budget execution shortfalls is outreach and communication. The DMT should be aware of calls for sweep-up fund projects, ongoing modification programs, potential funding sources, FMS, and centralized financial managers so it can act quickly when the need arises. Similarly, key points of contact in all these activities should be made aware of the uncertainties in DMSMS resolution budgets and recognize the roles they can play in minimizing the potential impacts of these uncertainties.

N.2.2 Contracting during Execution

There are two contracting-related considerations for funding DMSMS resolutions that may be of particular importance if a program is seeking to fund a DMSMS resolution during the budget execution year:

- ***Finding a contracting vehicle that is feasible to use***. Finding an appropriate contract vehicle for DMSMS resolutions can sometimes be difficult. A contract must be in place with the organization that will implement the resolution (e.g., the organization performing the non-recurring engineering or the organization that will sell the items). This is often a problem when a system is in sustainment with only organic support for the obsolete item. In this situation, even if the program office has funds to develop a resolution, there may not be a readily available contract to use that money. If the item is supported by a WCF, the program office should attempt to initiate a WCF project to develop the resolution, including an LON buy. Otherwise, the program office should identify a usable contract vehicle or issue an RFP to create a new contract vehicle. The problem with the latter is that it will require significant lead time before a new contract can be put in place. This is especially challenging for an LON buy as there may only be a very limited amount of time to purchase the needed quantities.
- ***Having an appropriate funding appropriation to use on that contract***. Restrictions exist on the use of all appropriations. In some cases, procurement funds are necessary to buy the new item; in others, RDT&E funding is generally required for non-recurring engineering and testing associated with redesign. O&M funding may be needed to pay for the installation of the item. There will be knowledgeable people in the program office that can provide advice on this subject.

  When a system is in production, a program office may not have any or only very limited O&M funds. When a DMSMS issue occurs, production money can be used to develop a resolution, which would likely be applicable to units in production as well as fielded units. Installation of the item in fielded units requires O&M funds and is generally done by attrition. In rare cases where there is a need to install the new item in fielded units rapidly (i.e., not through attrition), there could be a problem because installation if O&M funds are not available to support rapid installation. A much more likely problem is the situation of funding a resolution that applies to both in-production and fielded units using an LON buy. It is difficult to justify the use of procurement appropriations to buy LON quantities of an item to meet the demand of the fielded units. There is no best practice for this situation. Programs must strongly justify the requirement and the impact of not procuring the full amount (i.e., to meet the item demand for production and fielded units) and then work with their leadership to obtain approval. Another approach is to work with organizations managing WCFs to finance the LON buy, when the item in question is managed by that WCF.

N.2.3 Executing LON Buys

There are two limitations that can result in suboptimal LON purchases.

- ***Limitations on being able to buy to the estimated requirement***. According to 31 U.S.C. § 1502(a), the balance of an appropriation or fund, limited for obligation to a definite period, is available only for payment of expenses properly incurred during the period of availability or to

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complete contracts properly made within that period of availability and obligated consistent with 31 U.S.C. § 1501. The appropriation of funds is not available for expenditure for a period beyond the period otherwise authorized by law. To justify an LON purchase to include demand for the item beyond the expenditure period for the funds used to purchase those items, a "bona fide need" statement must be documented for the General Counsel's office. That statement should explain the DMSMS issue and the risk posed to the system if the items to meet demand outside the expenditure period, as well as describe how and why the resolution option was determined.

- ***Limitation on the acquisition of excess supplies***. 10 U.S.C. § 3070 may also be an issue. That section of public law applies only to WCFs. It also provides a basis for exceptions to the limitation. The interpretation of these statutes varies throughout DoD. There have been situations in DoD where the quantity was limited to one year, two years, three years, the number of years remaining in a production contract, and as determined on a case-by-case basis. Some program offices routinely obtain exceptions and others do not. Program offices should work with their associated financial communities and potentially service DMSMS leads to establish a routine process for justifying their entire estimated requirement.[313]

N.3 Leveraging WCFs to Fund DMSMS Resolutions

The Air Force and the Army each have comprehensive WCF programs to resolve DMSMS issues for WCF-managed items. The Navy has nothing equivalent. Under very limited circumstances, the Logistics Engineering Change Proposal program may be used to fund DMSMS resolutions. No further information has been provided. Program offices interested in learning more should refer to their principal point of contact for DMSMS matters.

The purpose of this appendix is to describe the extent to which the Air Force Working Capital Fund (AFWCF) and Army Working Capital Fund (AWCF) will finance resolutions to DMSMS issues.[314] Implementing or financing a resolution implies that the resolution has been determined, the non-recurring effort to develop and test the resolution to the problem has been funded and completed, and the replacement item/assembly can be obtained from the supply system.

N.3.1 Air Force Sustaining Engineering Program

N.3.1.1 Name and Purpose of Primary Mechanism for Using the WCF to Finance a Resolution to a DMSMS Issue

- Consolidated Sustainment Activity Group–Supply (CSAG-S)/General Support Division (GSD) Sustaining Engineering Program.
  - Purpose. Solicit and finance projects that mitigate obsolescence of DLRs (as part of CSAG-S) and consumables managed by the AFWCF and consumables managed by the

[313] One program office overcame the problem by adding the following words to its DMSMS resolution POM/budget line item: "Obsolescence above encompasses mitigation activities that protect the system and ensure a producible technical data package. This preserves an affordable future product cost with an acceptable production schedule. Examples of mitigation activities include component replacement parts, materials, qualification, alternative source/parts qualification, and piece part/material bridge buys to support subsequent year's production lots."

[314] This document generally uses the term *finance* instead of *fund*. Since WCFs are required to break even by including all their operating expenses in a cost recovery rate applied to all sales, WCFs technically do not fund anything. The cost of resolutions (among other things) is spread among all WCF customers in anticipation of future sales.

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Defense Working Capital Fund (DWCF) where the Air Force is the cognizant engineering authority (as part of GSD); recover expenditures in the AFWCF cost recovery rate.[315]

Unless otherwise noted, the remainder of this section is primarily focused on the CSAG-S/GSD Systems Engineering Program, its call for projects, and the rules by which projects can be approved and financed.

N.3.1.2 Service Policy Documents Governing the Implementation of the CSAG-S/GSD Sustaining Engineering Program

- Air Force Sustainment Center Instruction (AFSCI) 63-101, *CSAG/GSD Sustaining Engineering Program*, February 22, 2023.
- AFSCI 21-101, *Maintenance, Test Program Set*, July 13, 2021.
- Air Force Instruction (AFI) 23-101, *Materiel Management, Materiel Management Policy*, October 22, 2020, Air Force Material Command (AFMC) Supplement (Guidance Memorandum)—Improved Item Replacement Program (IIRP) verbiage.
- Department of the Air Force Instruction (DAFI) 65-601v1, *Financial Management, Budget Guidance and Procedures*, June 22, 2022.

N.3.1.3 Type of Obsolescence-Related Projects Allowed

- A project is allowable if its primary purpose is obsolescence mitigation for an allowable item and any product improvement that enhances capability is incidental. SOWs and project narratives must clearly not intend to enhance capability because that effort then becomes a modification.
- Support documentation must include the following:
  - Proof of obsolescence indicating that an item can no longer be purchased (does not include functional or technological obsolescence) and
  - A determination of the date that the obsolescence will impact operational availability.
- In the case of a redesign, a project must develop an F3 interface replacement.[316] It usually is for a single new item to replace a single obsolete item. Obsolete items may be grouped together such that an obsolete assembly or subassembly is replaced by a new set of items if it is cost effective to do so. LRUs and SRUs must be budget code 8 (repairable) or budget code 9 (consumable) before a project can be funded by AFWCF dollars.
- Other projects include those to reverse engineer an item to gain complete TDPs so that competition can result in future buys and repairs.
- Another type of project could be one to re-establish repair capability as a result of a related sustaining engineering project as data is the deliverable and not equipment. That equipment is the responsibility of the program office or the organic depot itself.
- Cost sharing between appropriated and WCF dollars is not allowed with one exception.[317] The WCF cannot be used to supplement appropriated funds.
- It is also possible to use AFWCF resources to investigate an issue to determine a resolution. Such efforts are not associated with the sustaining engineering program. The results could

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[315] Projects to improve reliability, availability, or maintainability; to correct safety of flight deficiencies; to develop repair procedures; to reestablish an organic repair capability following a redesign project; or to obtain needed technical data are also allowed. In addition, projects can maintain test program sets or units under test capability to isolate and resolve software discrepancy reports. Such non-DMSMS–related projects are not discussed further in this document.

[316] This is usually preferable from a program management perspective since it simplifies logistics in that there is no proliferation of the number of configurations to be supported.

[317] The sole exception to this is the case when it is not clear whether a deficiency is item related. A study to decide whether the deficiency is item related must be made with appropriated dollars. If the study concludes that the deficiency is item related, then the AFWCF may be used to finance the resolution.

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lead to either a sustaining engineering project or a program office deciding to deliberately increase capability to correct the deficiency and use appropriated dollars.

N.3.1.4 Allowable Costs

- Non-recurring engineering costs for redesign or reengineering.
- Prototype design and fabrication.
- Testing at the item level (and at higher levels of assembly) to ensure the item's requirements are met.
- Software costs when the software is embedded in hardware that is being redesigned or reverse engineered.
- Reverse engineering to develop a TDP if it is not available.
- Purchase of technical data associated with a project's engineering efforts, and the rights to use that data. Other technical data can be purchased with non-sustaining engineering AFWCF funds.
- Associated changes to technical manuals.

N.3.1.5 Exclusions

- AFWCF funds cannot follow appropriated procurement or RDT&E funds.
- Studies that determine what to do cannot be supported through the sustaining engineering program, but they can be financed in the AFWCF using the Contract Services program.
- While initial spares are not funded by sustaining engineering, they may be purchased with AFWCF because of a requirement established elsewhere.
- New test or support equipment.
- Production or implementation of kit.
- Projects related to software only.
- System level testing.
- Development projects: Project must be at Technology Readiness Level (TRL) 6 (demonstrated in a relevant environment) or higher.

N.3.1.6 Types of DMSMS Resolutions That Can Be Funded

- Approved part: Yes.
- LON buy: An LON buy cannot be executed through the project call, but it can be executed using the AFWCF (see Appendix N.3.3).
- Repair, refurbishment: Yes.
- Extension of production or support: Yes.
- Simple substitute: Yes.
- Complex substitute: Yes.
- Development of a new item or source: Yes.
- Redesign—NHA: Only if the NHA is managed by AFWCF, which means that it has been provisioned.
- Redesign—Complex or (sub)system replacement: Only if the subsystem is managed by AFWCF, which means that it has been provisioned.

N.3.1.7 Project Call Procedures

- Issuer/Point of contact: Systems Engineering Program Office.
- Approval authority: AFMC.

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- Frequency and timing: In-cycle data calls are issued December–March annually. If a need arises, projects may also be submitted out-of-cycle.
- Ranking criteria examples:
  - Whether the project is a continuation of a previous project;
  - Mission impact;
  - Severity of the impact;
  - Mission degradation date;
  - Mission item essentiality code;
  - Affected weapon systems;
  - Reliability, availability, maintainability impact;
  - ROI (to the Air Force as a whole); and
  - Impact on safety.

N.3.1.8 Limitations on the Dollar Amount That the WCF Can Finance

- There is no explicit limitation.
- The total dollar value of all projects from FY16 to FY19 has been $370 million, $118 million, $118 million, and $241 million, respectively. Obsolescence mitigation projects represent approximately 75% of the total.

N.3.1.9 Funding for Fielding New Items

- Appropriated funds can be used for scheduled installations across operating units if fielding is considered urgent.
- Installation by attrition is a second option.
- The IIRP is a mechanism for using the AFWCF to finance some fielding of replacement items to correct deficiencies, replace obsolete assets, and/or introduce, through technology insertion, state-of-the-art components that are stocked, stored, and issued as assets of supply.
  - There are no known funding limits. High priority candidate items for IIRP funding include those correcting a safety of flight issue, those providing a specialized mission capability, those whose fielding would lead to an ROI through improved reliability, and obsolete items (electronic as an example).
  - In the case of obsolete items, the rationale is that by the time an attempt is made to purchase the redesigned item, it may be obsolete because residual stock of the original item could last longer than the market life of the replacement electronics. In that situation, the cost of the original redesign project would have been wasted because another redesign will be required.

N.3.1.10 Reasons a Program Might Not Want to Try to Use WCF Resources on an Allowable Project

- PM might not want a solution at the part level but prefer a solution at the assembly level and potentially improve capability at the same time.

N.3.1.11 Other Mechanisms for Financing a DMSMS Resolution

- Outside of sustaining engineering, the AFWCF, as part of its normal supply operations, may execute an LON buy for either the DLRs or the consumables that it manages. Unless something significantly changes with the reliability or condemnation rate for a DLR, additional procurements are rarely needed. Even when one of those changes occurs, a DLR LON buy would not be required if it is possible to develop F3 replacements to obsolete components

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within the DLR or to make LON buys of the component itself. Furthermore, if there were issues at the component level, there may be no ability to buy the DLR itself. The AFWCF will execute an LON buy option for an item which will no longer be produced only when all other more economical and logistically acceptable alternatives to a material shortage or manufacturing discontinuance have been exhausted. While 10 U.S.C. § 3070 limits the ability of WCFs to buy stock beyond two years of need, it also establishes a mechanism for exceptions to be granted within a WCF organization itself. The likelihood of approval is heavily dependent on whether the purchase will impact AFWCF cash flow since the sales to reimburse the AFWCF for the purchase will be several years in the future.

- LON buys on items managed by DLA are executed by DLA. They are usually the result of a discontinuance, counterfeit, or change notice and are processed within the AFWCF for identification of all Air Force–managed next higher assemblies that will be impacted and computation of total projected requirements through the life of the program.
- To some extent, the AFWCF uses CLS and PBL contracts to repair some of the items that it manages. Those contracts may have requirements to resolve certain DMSMS issues. The funding for such contracts is not part of the project call. It is however possible that those contracts are modified to perform a funded project.

N.3.2 Army Obsolescence Mitigation Program

N.3.2.1 Name and Purpose of Primary Mechanism for Using the WCF to Finance a Resolution to a DMSMS Issue

- AWCF Obsolescence Mitigation Program.
- Purpose. Solicit and finance projects that mitigate obsolescence of DLRs and consumables managed by the AWCF.[318] Recover expenditures in the AWCF cost recovery rate.

Unless otherwise noted, the remainder of this paper is primarily focused on the AWCF Obsolescence Mitigation Program, its call for projects, and the rules by which projects may be approved and financed.

N.3.2.2 Service Policy Documents Governing the Implementation of the AWCF Obsolescence Mitigation Program

- U.S. Army Materiel Command (AMC) Memorandum, Subject: Supply Management, Army. AWCF Procedures of Submission of Secondary Item Obsolescence Requirements, October 1, 2020, updated annually.
- Annual Funds Release Execution Guidance.

N.3.2.3 Type of Obsolescence-Related Projects Allowed

- A project is allowable if its primary purpose is obsolescence mitigation for an AWCF-managed item and any product improvement that enhances capability is incidental. Support documentation must include the following:
  - Proof of obsolescence indicating that an item can no longer be purchased (does not include functional or technological obsolescence) and
  - A determination of the date that the obsolescence will impact operational availability.

[318] There are three other programs included in the project: Operating and Support Cost Reduction (OSCR), Reliability Improvement Program (RIP), and Cost-Wise Readiness. Since they are not focused on DMSMS issues, they are not discussed further in this document.

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- In the case of a redesign, a project must develop an F3 interface replacement.[319] It usually is for a single new item to replace a single obsolete item. Obsolete items may be grouped together such that an obsolete assembly or subassembly is replaced by a new set of items if it is cost effective to do so.
- A project whose primary purpose is product improvement to enhance capability would not be funded by the AWCF. However, if a single project includes both obsolescence mitigation and deliberate product improvement, a shared cost arrangement can be negotiated (if an equitable way of splitting the cost has been agreed upon and there are separate CLINs).[320]

N.3.2.4 Allowable Costs

- Non-recurring engineering costs for redesign or reengineering,
- Prototype development,
- Testing at the item level (and at higher levels of assembly) to ensure the item's requirements are met,
- Software costs when the software is embedded in hardware,
- Reverse engineering to develop a TDP if it is not available, and
- Purchase of technical data related to the project and the rights to use that data.

N.3.2.5 Exclusions

- Initial spares when needed are normally funded by the PM with appropriated funds.
- Production or implementation of kits.
- Projects related to software only.
- Studies that determine what to do are not part of the project call. However, such studies can be funded with AWCF logistics operations money.
- System level testing.
- Development projects: Project must be at TRL 6 (demonstrated in a relevant environment) or higher.
- New test equipment unless the old test equipment were AWCF managed.
- While changes to technical manuals and revisions to provisioning data are funded with appropriated resources, there are no programming and budgeting implications to the program office. Such changes are centrally funded with non-program office appropriated funds.
- Projects unique to special operations since they involve no AWCF parts.

N.3.2.6 Types of DMSMS Resolutions That Can Be Funded

- Approved Part: Yes.
- LON buy: An LON buy cannot be executed through the project call, but it can be executed using the AWCF (see Appendix N.3.1.11).
- Repair, refurbishment: Yes.
- Extension of production or support: Yes.
- Simple substitute: Yes.
- Complex substitute: Yes.

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[319] This is usually preferable from a program management perspective since it simplifies logistics in that there is no proliferation of the number of configurations to be supported.

[320] In some cases, it is difficult to make the distinction. There have been instances where the AWCF financed hardware changes and appropriated dollars paid for software changes.

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- Development of a new item or source: Yes.
- Redesign—NHA: Only if the NHA is managed by AWCF, which means that it has been provisioned.
- Redesign—Complex or (sub)system replacement: Only if the subsystem is managed by AWCF, which means that it has been provisioned.

N.3.2.7 Project Call Procedures

- Issuer/Point of contact: Army's Aviation and Missiles Command (AMCOM), Communications and Electronics Command (CECOM), and Tank and Automotive Command (TACOM).
- Approval authority: AMC.
- Frequency and timing: Data calls issued typically in April and November. If a need arises, projects may also be submitted out-of-cycle.
- Ranking criteria examples:
  - Amount of time until no items are available because of obsolescence,
  - Amount of reliability improvement that would be attained,
  - Stakeholder priority,
  - ROI (to the Army as a whole),
  - Technical risk and merit,
  - Impact on safety, and
  - Impact on the IPS elements.[321]

N.3.2.8 Limitations on the Total Dollar Amount That the WCF Can Finance

- Two and one-half percent of predicted AWCF sales allocated by systems command (AMCOM, CECOM, and TACOM).
- From 2008 through 2010, AMCOM was not allocated sufficient AWCF funds to meet its obsolescence mitigation requirements.[322] From 2011 through 2018, AMCOM has not spent all its allocation.[323] The average funding spent on obsolescence mitigation from 2008 through 2018 is $31.78 million per year with approximately $15.93 million spent on other supportability improvement projects.
- Since FY14, excepting FY18, CECOM has not been allocated sufficient AWCF funds to meet its obsolescence mitigation requirements. During that period, CECOM only submitted AWCF projects that were obsolescence related. Figure 38 shows data call requests made by CECOM Integrated Logistics Support Command and the corresponding amount funded by AMC by fiscal year.

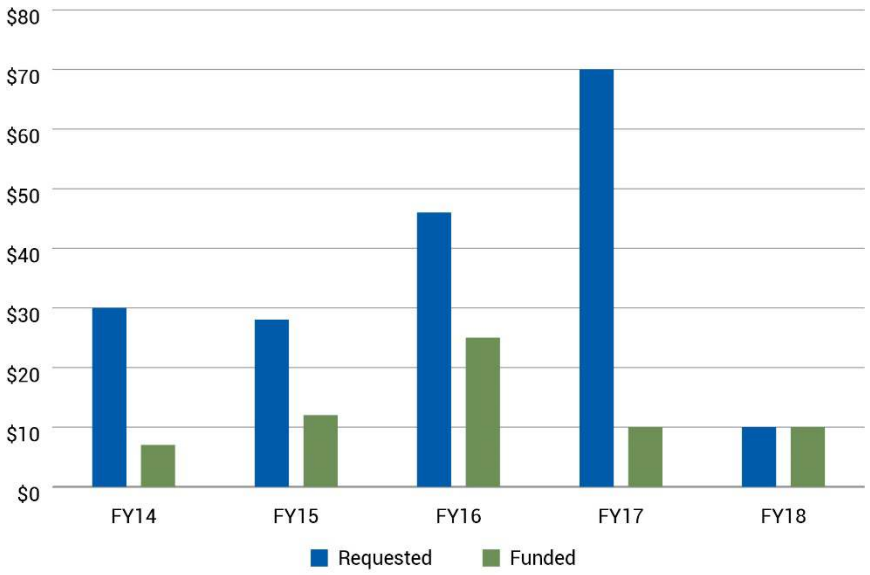
[321] DAU, *Integrated Product Support (IPS) Elements Guidebook*, March 2021, available at https://www.dau.edu/tools/integrated-product-support-ips-element-guidebook.

[322] In 2010 AMCOM had urgent obsolescence mitigation needs and was granted extra allocation from AMC.

[323] Because 2013 was a sequestration year, no "new starts" were approved by AMC. Only ongoing projects were funded in 2013.

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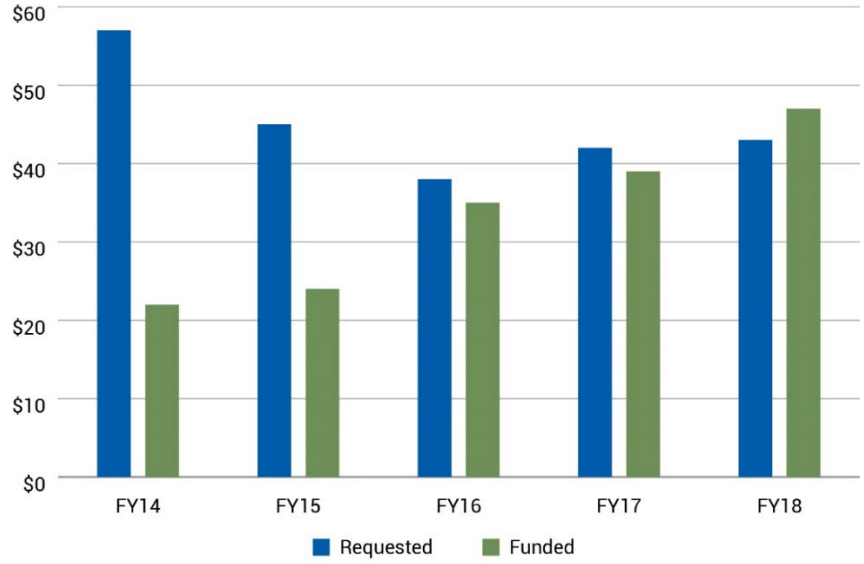
Figure 38. CECOM Requests and Funding for AWCF DMSMS-Related Projects



Note: Dollar amounts are in millions of dollars.

- Figure 39 shows TACOM requirements and funding for AWCF projects. Data was not available to differentiate DMSMS issue–related projects from other types of projects. All requirements from FY14 through FY17 were not funded. In FY18, requirements were fully funded and some additional resources were provided to deal with emergent problems.

Figure 39. TACOM Requests and Funding for AWCF Projects



Note: Dollar amounts are in millions of dollars.

N.3.2.9 Funding for Fielding New Items

- Appropriated funds can be used for scheduled installations across operating units if fielding is considered urgent.
- Otherwise, installation by attrition.

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N.3.2.10 Reasons a Program Might Not Want to Try to Use WCF Resources on an Allowable Project

- PM might not want a solution at the part level but prefer a solution at the assembly level and potentially improve capability at the same time.

N.3.2.11 Other Mechanisms for Financing a DMSMS Resolution

- While an LON buy cannot be executed through the project call, the AWCF could finance an LON buy for items that it manages as part of its normal supply functions. The decision would be made by the item manager and organization. Perhaps the more difficult aspect of this is the determination of whether an LON buy is the most cost-effective resolution over the life cycle of the system. The AWCF will execute an LON buy option for an item which will no longer be produced only when all other more economical and logistically acceptable alternatives to a material shortage or manufacturing discontinuance have been exhausted. Such an action is unusual for depot-level reparables themselves since DMSMS issues would normally be associated with the components used for repair and would be mitigated at that level. Furthermore, many of the components used for repair are DLA managed and LON buys would be funded by the DWCF (see below). When the AWCF does identify an LON as the preferred resolution, 10 U.S.C. § 3070 limits the ability of WCFs to buy stock beyond two years of need. However, Section 3070 also establishes a mechanism for exceptions to be granted within a WCF organization itself. The likelihood of approval is heavily dependent on whether the purchase will impact AWCF cash flow since the sales to reimburse the AWCF for the purchase will be several years in the future.
- It is unlikely that the AWCF would be used to supplement an inadequate DLA-funded LON buy for a DLA-managed item for which the Army is the technical authority. In such a situation, the Army is likely to try to use senior level influence to persuade DLA to buy additional items.
- To some extent, the AWCF uses CLS and PBL contracts to repair some of the items that it manages. Those contracts may have requirements to resolve certain DMSMS issues. The funding for such contracts is not part of the project call. It is however possible that those contracts are modified to perform a funded project.

The Army has three other WCF programs that theoretically could also be used to resolve a DMSMS issue.[324] Each program has different project acceptance criteria:

- The OSCR program is designed to “save the field money” by reducing secondary item acquisition costs, extending the life of the item, and reducing the number of events (removals or repairs) and the cost per event. OSCR promotes life cycle cost savings and avoidance in the field by redesigning, prototyping, and testing spare parts for fielded systems. OSCR projects involve an individual item or assembly of items, prototype, or test. The program will not fund production or implementation of kits, nor will it fund studies. Eligibility for the program requires a validated economic analysis.
- The RIP is a continuous process to look for opportunities to decrease demand, improve operations, and improve reliability. Projects must provide immediate help to the soldier and must show an ROI. This program will not fund production and studies.
- The Product Improvement Pilot program provides funding for product improvements such as improving reliability and maintainability, extending useful life, enhancing safety, and lowering maintenance costs. This program cannot be used to significantly change the performance envelope of an end item, and individual item costs may not exceed $1 million.

[324] For more information, an organization should contact its manager for each of the programs.

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N.3.3 WCFs and LON Buys

There are circumstances where WCFs finance LON buys for the items they manage. Because Service WCFs generally manage more expensive reparable items, LON buys are usually not the most cost-effective resolution to DMSMS issues and are often not feasible. That is one reason why the programs described earlier in this appendix were established. On the other hand, the DLA-operated DWCF manages less expensive consumable items where LON buys are not unusual.

Therefore, the purpose of this section of the appendix is to describe how DLA uses the DWCF to finance resolutions to DMSMS issues. This appendix also provides best practices that the Services should follow to optimize DLA support in obtaining such DWCF assistance. Answers are provided to the following questions:

1. What is the process by which the DWCF will finance an LON buy (either in whole or in part)?[325]
2. How much of the actual cost to make the LON buy or implement some other resolution will the DWCF finance?

DLA defines and distinguishes between diminishing manufacturing sources (DMS) and obsolescence in its own unique way. When an item can no longer be procured (i.e., it is non-procurable), it is declared by DLA as obsolete. An item is coded DMS by DLA when a PDN or equivalent has been received and there remains some opportunity to buy the item. A PDN ideally precedes DLA's definition of obsolescence, providing warning of part production termination, but not always.

The first section of this appendix primarily focuses on the single DLA office responsible for all LON buy processes across all DLA supply chains. It is in the DLA Land and Maritime DMSMS Office (DMSMS Office). The next section gives a brief description of other processes for accomplishing an LON buy or mitigating obsolescence in some other way. Mechanisms to revisit LON buy assumptions are addressed in the third section. Finally, the fourth section provides advice to the DMSMS management community on navigating all these DLA processes.

N.3.3.1 Primary DLA LON Buy Processes

The DMSMS Office deals exclusively with DMS items and it is the most likely organization to pursue an LON buy. Figure 40 depicts the DMSMS Office's processes that can result in either an LON buy or the issuance/funding of a request for engineering support for non-procurable (obsolete) items. Both the technical engineering support process and the logistics requirements function of executing an LON buy are conducted, in part, external to the DMSMS Office. The letters shown in Figure 40 represent labels used to reference different parts of the figure.

From the perspective of a program office trying to program and budget to resolve DMSMS issues,[326] a specific DMSMS situation or case[327] may lead to five potential outcomes:

- The resolution is an existing "approved item," with no programming and budgeting implications on the affected program offices.

[325] DLA normally uses the term *life-of-type buy*.

[326] An operating unit's perspective is that it will have to buy parts to maintain its equipment. It is possible that the price paid for parts will change or the reliability of parts is increased and therefore fewer will need to be purchased. Programming and budgeting implications of this are out of scope for this document.

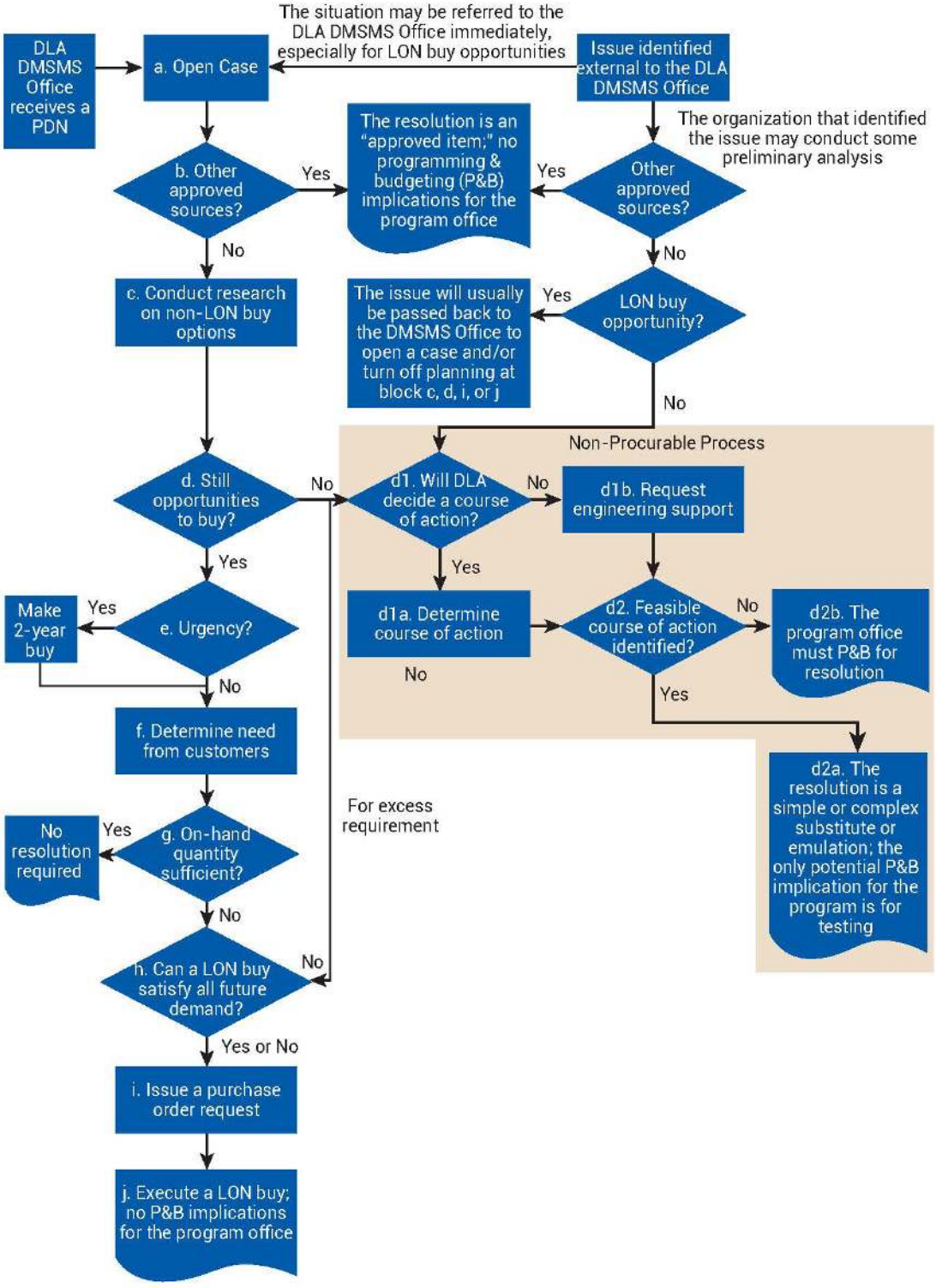
[327] DLA's definition of a case is different from the concept of a case within the context of a case management system as discussed in this document.

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- The resolution is a “simple substitute,” a “complex substitute,” or “a new item or source” specifically through emulation. There may be no programming and budgeting implications on the affected program offices or there could conceivably be some testing cost implications.
- The resolution is an “LON buy,” with no programming and budgeting implications on the affected program offices.
- No resolution is required.
- None of the above resolutions was feasible. An affected program office will need to develop and fund its own resolution.

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Figure 40. Primary DLA DMSMS-Related Processes



The following are brief descriptions of the principal blocks in the process depicted in Figure 40. The letters depicted below correspond to the letter labels in the principal Figure 40 boxes. It will be useful to refer to Figure 40 to locate the box being discussed.

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a. Open case.

This step happens very quickly. The DMSMS Office resolution process is usually initiated when it receives a PDN or equivalent thereof either from the GIDEP or directly from industry. The process may also be initiated upon notification along with evidence of the discontinuation (e.g., EOL notice from the manufacturer or an email from the manufacturer), from other sources. Possible sources include a program office,[328] another Service entity (e.g., a depot), or some other DLA organization (e.g., from a product specialist or material manager because of a no bid to an attempted order or direct notification from a Service). A DLA case (or cases) will be opened on item(s) that DLA manages (i.e., on a stock listed item with an NSN). A case may be opened at other points in the overall process as indicated in Appendix N.3.3.2.

b. Other approved sources?

After opening a case, the first consideration is whether there are any already identified approved sources. If there are, and those sources are willing to produce the item in question, the case is closed and DLA databases are updated accordingly. The resolution is an "approved item;" there are no DMSMS programming and budgeting implications on the affected program offices.

c. Conduct research on non-LON buy options.

This step may take a few days to complete if no approved options are found. The DMSMS Office codes all the items associated with the PDN as DMS. No DLA entity other than the DMSMS Office can change the DMS code for an item. When the code is set, DLA planning is stopped, no automatic requests to purchase the item can be issued, and DLA's process for identifying excess inventory no longer considers that item. In addition, the technical side of the DMSMS Office researches each of the DLA-managed items in a PDN to identify potential replacement items, new sources, complex substitutes, and/or the possibility of emulation, but that research is not immediately acted upon. The technical side of the DMSMS Office forwards the information to the logistics side of the DMSMS Office to investigate the viability of an LON buy and to issue a request for engineering support as needed.

The next decision in the flowchart splits the process into two major branches, depending on whether there are opportunities to buy an item after a PDN has been issued. While there normally are opportunities to buy an item after a PDN or equivalent has been received, DLA may not have received the notice in a timely manner and the period of time where the item could be purchased may have expired. When there are no opportunities to buy, DLA initiates its non-procurable item process where engineering support is sought from the appropriate Service engineering authority to determine another item or source. When there are opportunities to buy, DLA's LON buy process is continued. The following paragraphs describe both branches after the "still opportunities to buy" decision block. The non-procurable process is discussed first since the requests for engineering support could occur in the other branch.

d. Still opportunities to buy?

The first step in determining the viability of an LON buy is determining whether the original source is still willing to fulfill orders.

If there ARE NO opportunities to buy:

1) Will DLA decide a course of action?

The DMSMS Office usually is not involved in this branch of Figure 40. This branch is for DLA-defined obsolete or otherwise non-procurable items and would normally be led by the material planners or product specialists as discussed in Appendix N.3.3.2. The branch is however described in this section because the DMSMS Office often utilizes the same processes where there is an LON opportunity since the LON buy would not be DLA's resolution preference if another item or source can be approved for use instead of the DMS item.

[328] Program offices and other service organizations can initiate a case if they have access to DLA's shared data warehouse gateway. This is rare and probably not a best practice because direct contact with the DMSMS Office will begin more quickly if that office creates the case.

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DoDI 4140.69 establishes the criteria whereby DLA must obtain approval from the appropriate engineering support activity (ESA) for the use of a replacement item or another supplier.[329] The criteria are based on Federal Logistics Information System (FLIS) data elements for criticality and weapon system essentiality. DLA is the decision-making authority if the criticality code is blank or it indicates the item does not have a nuclear hardened feature or any other critical feature such as tolerance, fit restriction, or application (a value of X) and the essentiality code indicates that the failure of the part will not render the end item inoperable. Otherwise, the approval needs to come from the ESA. Even when DLA is the decision-making authority, it may still request ESA support to determine the appropriate action, typically a replacement item or source. This usually occurs when DLA believes that the item is used on weapon systems even though the FLIS record indicates otherwise.

a) If Yes, DLA Determines Action.

If DLA does not request engineering support, it determines the course of action to be taken using, in part, the information collected in the previous research step. This effort often involves both the DMSMS Office, the material manager, and the product specialist.

b) If No, DLA Requests Engineering Support.

If DLA is not the one to decide upon the course of action, it initiates a request for engineering support with the creation of DLA Form 339 incorporating the research results from c) above. After being reviewed within DLA, the form is sent to the appropriate Service ESA(s) by the designated DLA ESA focal point. Typically, the 339 (and any subsequent revisions) may request approval of replacement item or a new source of supply, identification of a replacement item or new source, or use of a waiver/deviation associated with replacement item or source. The information discovered in the previous research step is included in the request.

The required Service response time may be as low as 15 days for situations urgently impacting readiness. Otherwise, the response time requirement will vary between 30 to 90 days based on engineering support performance-based agreements (PBAs) between DLA and the Services.[330]

The DWCF pays for the non-recurring engineering efforts performed by the Service ESA on a fixed price basis as established in the PBA. What is paid may be different than the actual expenses incurred by the Service ESA. 339s may be revised in complex situations, but no additional funds are provided to cover any additional work related to such a revision. In some instances, the funding provided for Service ESA non-recurring engineering efforts will completely pay for the implementation of the resolution, but in many cases it will not.

DLA generally does not increase the amount paid to the ESA to perform testing because it asserts that in the determination of the resolution, the Service ESA should have already been assured that the resolution is technically acceptable. The only testing that DLA funds (outside of the 339 process) is a first article test (FAT) or a production lot test (PLT). Such tests would be requested by the Service ESA and paid to the performing organization (i.e., a contractor or a government laboratory) if it is put in the purchase order that DLA issues. Since the purpose of these tests is to ensure that the contractor can and does furnish a product that conforms to all specifications, DLA should have a TDP that contains those specifications so that it can judge the test results.[331] When there is no TDP, DLA occasionally may ask for a FAT or a PLT, but in such cases, the contractor can simply assert that the test was successful with providing DLA the technical data needed to verify the result.

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[329] DoDI 4140.69, "Engineering Support Instructions for Items Supplied by Defense Logistics Agency (DLA)," September 30, 2016.

[330] In some instances, iterations between DLA and the ESA extend the time for this process to six months.

[331] This implies that the acquisition management suffix code is "G'" indicating that the government has unlimited rights to the technical data and the data package is complete. Valid acquisition method codes are "1" which implies suitable for competitive acquisition, or "2" which is suitable for competitive acquisition for the first time.

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2) Feasible course of action identified?

a) If yes, Resolution is simple or complex substitute or emulation.

The only potential programming and budgeting implication for the program office is testing. DLA databases are updated accordingly.

b) If no, Program office must program and budget for resolution.

If no replacement item or source has been determined by the ESA or by DLA, then the users must determine and fund the resolution.

If there ARE opportunities to buy:

e. Urgency?

If an LON buy is feasible, the logistics side of the DMSMS Office determines the urgency. For high-demand items where the demand is forecastable (as indicated by an AAC of D[332]), the DMSMS Office usually recommends that a buy (not officially designated as an LON buy) be initiated for typically a two-year quantity based on historical demand as experienced by DLA. Such a purchase is unusual because high-demand items do not typically become obsolete.

f. Determine need from customers.

Regardless of whether an urgent buy occurred, the DMSMS Office sends out an "alert" about the DMS situation to customer focal points associated with the weapon systems linked to the item in question. The alert indicates that a PDN has been received and requests that information on the life-time demand for the item be provided back to the DMSMS Office within 30 days or less if the EOL date is less than 30 days away. The DMSMS Office requests monthly demand estimates for as long as the end item is expected to be in service.

When the logistics side of the DMSMS Office has received demand requirements from the customer focal points representing the weapons systems associated with the item, it will compare, in consultation with the DLA material planner and demand planner for that item, those demand requirements to historical demand. It will also perform its own research on how long the end item is expected to remain in service. If the demand requirements from the customer focal points are not consistent with DLA's historical demand or represent a high cost, the logistics side of the DMSMS Office will request further explanation, justification, and tangible evidence (e.g., purchase records) for the quantity of the item required from the customer focal points.[333] Future production needs are not an acceptable justification because DLA provides items for sustainment only. If the demand cannot be justified to DLA's satisfaction, DLA will use its own calculations made jointly by the DMSMS Office, the material planner, the demand planner, and the product specialist. In general, DLA forecasts approximately 60% of the items it manages. About 68% of those forecasts are close to actual demand on average. However, DLA internal data indicates their demand history is usually closer to actual demand than what the Services state their future requirement estimates will be. Statistically, collaborative input on future requirements provides a better forecast about 25% of the time.

If there are no current customer requirements, no backorders, and no current demands, the logistics side of the DMSMS Office is likely to recommend no further action regarding the item at that time.[334]

If there is any controversy, the decision on the LON buy quantity is made by a senior DLA decision maker.

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[332] AAC D is defined as follows: Issue, transfer, or shipment is not subject to specialized controls other than those imposed by the Integrated Materiel Manager/Service supply policy. 1) The item is centrally managed, stocked, and issued. 2) Requisitions must contain the fund citation required to acquire the item. Requisitions will be submitted IAW Integrated Materiel Manager/Service requisitioning procedures.

[333] Reasons for inconsistency include situations where customers buy the item commercially (bypassing DLA), cannibalization, and the items being bought commercially by a commercial support provider even though the Services are required to submit a Demand History Accounting when a materiel requirement for a DLA-managed part is satisfied outside the normal supply process.

[334] DLA will also try to locate customers by examining previous orders if there was no response to the alert.

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g. On-hand quantity sufficient?

This decision is a simple test of whether there is sufficient on-hand inventory to meet all future requirements for the item. If that is the case, no resolution or further action is required. If the answer is no, the LON buy is executed.

h. Can an LON buy satisfy all future demand?

An LON buy is executed to satisfy the requirement. There are however situations where the combination of on-hand stock and the maximum amount feasible to purchase will not satisfy all future needs. When that occurs, the maximum LON buy is executed and the remainder of the requirement is satisfied by the non-procurable process discussed above.

i. Issue a purchase request.

This step normally takes one to three days. The logistics side of the DMSMS Office recommends a certain quantity LON buy of the item to the appropriate DLA material planner responsible for the item. Sometimes, the DMSMS Office may relinquish control of the process and have the material planner execute this step.[335] Recommendations are prioritized based on the EOL date. The material planners or their supervisors (for higher dollar level buys) are the ones who approve an LON buy. When approved, the material manager issues a purchase request and sends it to the DLA buyer.[336] Since the determination of how many to buy is usually done jointly, the quantity requirement for the purchase normally does not deviate from what was originally determined. In some instances, when the material planner has not been involved, there may be a purchase quantity adjustment if the material planner has some additional specific knowledge about the item. In other instances, the Services have interacted directly with the material planner to change the size of the buy if there is a disagreement about the quantity.[337]

There are legal restrictions to the amount purchased as discussed in Section 7.2. Although 31 U.S.C. § 1502 does not apply, 10 U.S.C. § 3070 explicitly restricts the quantity of an item that can be purchased by a stock fund to two years of supply. In the case of a DMSMS item, there are national security reasons for allowing an exception—there is a validated future demand and the item will not be available for purchase in the future. The head of DLA's Contracting Agency can make such a determination and is asked to do so when the quantity exceeds two years of supply.

j. Execute the LON buy.

This step can take anywhere from 30–90 days or longer depending on the circumstances for the item being purchased. The DLA buyer orders the LON buy quantity from the OEM and pays for the purchased items using funds from the DLA DWCF. The LON buy quantity is then available through the DLA supply systems. The logistics side of the DMSMS Office will continue to leave the case open for that item until the LON buy stock for that item is delivered. There are no programming and budgeting implications for the program office if the quantity purchased by DLA is sufficient to meet the demand requirements of the program office and all other known DLA customers for that item.

N.3.3.2 Other Processes

When another organization in DLA (normally the material manager and/or the product specialist) discovers a DMS or obsolescence situation first, several things could happen:

- For a DMSMS situation, the information could be passed immediately to the DMSMS Office and the process occurs exactly as depicted in Figure 40. In some instances, some of the

[335] DLA material planners reside within the DLA supply chains; therefore, if an item is unique to a particular supply chain, the recommendation for an LON buy from the logistics side of the DMSMS Office will go to the appropriate DLA material planners in that supply chain. For example, if the item in question is aviation-unique, the recommendation for an LON buy will go to a material manager in the Aviation supply chain. Of note, the Land and Maritime supply chains are the only supply chains that have material managers with special DMSMS expertise. Those material planners manage a certain set of items associated with a particular federal supply class, as well as work with demand planners.

[336] The buyer is responsible for acquisition, the product specialist is responsible for the technical quality of the items delivered.

[337] This usually occurs outside of the Land and Maritime supply chain.

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beginning steps of the Figure 40 process could be conducted outside of the DMSMS Office and then it is turned over to the DMSMS Office and the remainder of the Figure 40 process proceeds. The point at which the turnover occurs varies.

- Sometimes for a DMSMS situation, the entire Figure 40 process is conducted outside of the DMSMS Office. Normally, when this happens, there is something time sensitive about the situation, e.g., a very short time until the item can no longer be purchased. Ultimately, the DMSMS Office should be informed to update the data elements in FLIS to turn off planning and exempt the item from being considered in excess inventory reporting. No case is opened.
- Regardless of whether the DMSMS Office is or is not involved, the branch of Figure 40 associated with no opportunity to buy an item (Appendix N.3.3.1.d1 and Appendix N.3.3.1.d2) would always be followed. In the aviation supply chain, a 339 is sometimes not issued because of the length of time it may take to obtain a final response. A pilot program has been established to form a "sourcing team" of very experienced people to conduct in-depth market research and to negotiate with the source of the item and the ESA to determine a resolution.
- For an item that is truly obsolete (non-procurable and a PDN was not received), the DMSMS Office is usually not involved and no case is opened.

N.3.3.3 Revisiting the LON Decision

The DLA DMSMS Program monitors DMSMS items annually and flags items that have a Stock Impact date of less than two years. DLA will open the case and determine the intended date of the original LON buy and coordinate with the program if this number falls short of the stock requirement.[338] If DLA's inventory becomes exhausted and there is no longer an ability to purchase the item, then the process in Figure 40 for no opportunities to buy (Appendix N.3.3.1.d1 and Appendix N.3.3.1.d2) is again followed.

There are rare instances where additional demand becomes known, and it is possible to expand the size of the original LON buy if the purchased quantity has not been fully delivered. For example, sometimes a back order will occur for a DMS item. When this happens, the material planner should contact the logistics side of the DMSMS Office. If the timeframe for making LON buy purchases of the item in question is still open, then the LON process would continue. If the timeframe for making LON buy purchases of the item has ended, this should trigger the no opportunity to buy process.[339]

N.3.3.4 DMSMS Management Take-Aways

Several key take-aways emerge for the DMSMS management community:

- Program offices, whether proactive or reactive, identify DMSMS issues on a continuing basis; some of these issues impact DLA-managed items. When it appears that an LON buy is a cost-effective resolution, program offices should contact both the DLA DMSMS Office and the cognizant material planner to expedite the LON buy process.
- The DMSMS Office relies, in part, on the user community, to determine LON buy demand quantities. It generally reaches out to specific Service identified points of contact to obtain that information. The Services should ensure that their internal communication processes rapidly provide well-prepared demand inputs from all concerned users.
- Service consumption of DLA-managed items may differ from DLA sales because some items are purchased commercially. The Services should be able to provide sufficient, compelling justification for the demand requirements for the item in question, as well as what the impacts would be if the LON buy purchase were not approved and executed. Services therefore

[338] The formula DLA uses to determine the Stock Impact Date is "Stock on Hand/Annual Demand Quantity." For example, an item with an Annual Demand Quantity of 2, with 10 Stock on Hand, has a stock impact date of five years. When reaching two years, DLA will research the case and see what the LON (if any) was planned for. If it is less than two years, no action is required. If more than two years, DLA will contact the program to notify and work a solution.

[339] DLA checks whether additional procurement is feasible after the EOL date has expired.

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should ensure that thorough records are kept to persuasively justify future support requirements.

- DLA will not include future production requirements in determining an LON buy quantity. Program offices should program and budget for LON buys associated with production.
- DLA might not agree with Service LON buy requirements. Affected program offices, in an integrated manner at least at the Service level, should monitor the LON buy process to understand how many DLA was willing to purchase as compared to the total requirements for all customers of that item. Senior Service representatives may be able to reverse the situation by communicating directly with DLA senior managers.
- Apart from a FAT or a PLT, the DWCF will not fund any testing when a replacement item or a new source is the resolution to be implemented beyond the amount of money on a 339 request. Program offices should program and budget for any additional testing necessary.
- Even when the LON buy quantity matches that indicated by the sum of all associated weapon systems' demand requirements for that item, the stock of that item in the supply system is available on a first-come, first-served basis. Any registered user can purchase the item.[340] For this reason, and because the assumptions used to determine the LON buy quantity may be incorrect, a program office that has demand for the item, should monitor the stock levels closely.

N.4 Other Resources That May Be Available to Finance DMSMS Resolutions

With the exception of the first initiative shown below, this appendix lists several external funding sources at the DoD and the Service levels.[341] These funding sources represent other potential resource options for DMSMS resolutions.[342] Most of these funding sources have a periodic project solicitation, but some do not. Projects may or may not be accepted off cycle. In some cases, the solicitation is directed at government program offices; in other cases, the solicitation is directed at industry. The focus areas for the project solicitations are defined by the funding initiatives themselves, based on their understanding of DoD needs. Although a program office with DMSMS issues can communicate its needs to the funding initiatives, the proposed DMSMS resolution must be aligned with the mission, requirements, restrictions, and goals of the funding initiatives to obtain funding. Key initiatives follow.

N.4.1 DoD's VE Program

There are no separate resources associated with DoD's VE Program. VE provides a systematic approach to analyze the functions of systems, equipment, facilities, services, and supplies to ensure they achieve essential functions at the lowest life cycle cost, consistent with required performance, reliability, quality, and safety.[343] Typically, the implementation of the VE process increases performance, reliability, quality,

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[340] One exception exists and that pertains to FMS programs that are users of the item for which an LON buy was implemented. FMS programs can identify their demand requirements for the item at the time that the case is being worked and then the FMS programs fund and take possession of their portion of the LON buy quantity purchased at the time that they are delivered.

[341] Air Force, Army, and Navy WCF initiatives are discussed in Appendix N.3.

[342] DMSMS management practitioners should also be aware of congressionally established programs that are not included in the DoD Presidential budget, for example, the Industrial Base Innovation Fund, the Rapid Innovation Fund, and the Defense Rapid Innovation Program. Such congressional programs are not discussed in this document.

[343] Office of Management and Budget, Circular A-131, Value Engineering, January 2013, available at https://www.whitehouse.gov/wp-content/uploads/legacy_drupal_files/omb/circulars/A131/a131-122013.pdf.

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safety, durability, effectiveness, or other desirable characteristics. VE has been used to mitigate DMSMS issues from two perspectives: funding and methodological.[344]

From a funding perspective, a value engineering incentive (VEI) clause is included in most supply/service contracts when the contract price exceeds $150,000. A value engineering change proposal (VECP) is a proposal submitted to the government by the contractor IAW the VEI clause. A VECP proposes a change to the contract that, if accepted and implemented, provides an eventual, overall cost savings to the government with a substantial share of that savings contributing to the contractor's profit. It provides a vehicle to reduce acquisition and operating costs, while increasing the contractor's rate of return. Typically, the contractor pays the non-recurring costs associated with the VECP and is reimbursed from the savings. To ensure that savings can be shared, the VECP must meet two primary requirements:

- It must require a change to the current contract under which it is submitted.
- It must provide an overall cost savings to the government after being accepted and implemented. (A VECP could result in increased unit cost but reduced O&S cost. Thus, there would be an overall savings to DoD.)

The key takeaway associated with the funding perspective is that the contractor has a profit-based incentive to resolve obsolescence issues earlier. Without this incentive, proactive issue identification may not occur and, consequently, resolutions may be more expensive. These concepts could be applied to an LON buy if the contractor would use its own resources to buy items and then sell them back to the government when needed. This could be especially valuable if the government were not able to obtain the necessary resources in a timely manner.[345]

From a methodological perspective, the VE process can augment the *Analyze* step in DMSMS risk management, as illustrated in the following example:

> *Obsolescence issues emerged for the Theater High Altitude Air Defense missile. The issues involved multiple subcontractors and various components. The major and minor redesign efforts recommended to address the obsolescence problems would have resulted in high costs and negative schedule impacts for the program office. The DMT used VE to evaluate each redesign proposal and determine if other mitigation efforts could be employed to overcome the obsolescence issues. A VECP was implemented to mitigate the obsolescence and minimize redesign cost without adverse schedule impacts. The total three-year cost avoidance by being proactive was calculated to be $21.2 million.*

N.4.2 DoD and Service ManTech Program

The ManTech program is codified in 10 U.S.C. § 4841 as a requirement for each Military Service and Component.[346] This is DoD's primary program for investing in next-generation manufacturing processes, materials, or technologies, but it also has the mission for the "development and application of advanced manufacturing technologies and processes that will reduce the acquisition and supportability costs of defense weapon systems and reduce manufacturing and repair cycle times across the life cycles of such systems."[347]

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[344] See the SD-24, *Value Engineering: A Guidebook of Best Practices and Tools*, June 2011, available at https://quicksearch.dla.mil/qsDocDetails.aspx?ident_number=277777, Chapter 8.

[345] When a PDN is anticipated but has not been released, it may be possible to adapt this approach to critical items that are expensive to redesign and test. Specific arrangements to do this would have to be negotiated in advance with the program office, the contracting officer, and industry.

[346] For more information, see https://www.dodmantech.mil/.

[347] Ibid.

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Thus, DMSMS resolutions that require producibility improvements or a new manufacturing capability can seek funding through the ManTech program, particularly if repair cycle time and support costs can be reduced. The DoD ManTech program is a joint service R&D program with appropriations in OSD, Army, Navy, Air Force, and DLA. Each Military Department or agency programs its investments separately but plans jointly through the Joint Defense Manufacturing Technology Panel. Annual solicitations from OSD and each Military Department or agency are released, and all proposals must contain clear transition paths and have Military Department support from the transition target PM.

N.4.3 Defense Production Act Title III Program

This program's mission is "designed to create, maintain, protect, expand, or restore domestic industrial base capabilities."[348] Production capabilities that would otherwise be inadequate are transformed to support the material requirements of defense programs in a timely and affordable manner. Title III focuses on materials and items that could be used across a broad spectrum of defense systems. The capabilities of defense systems depend upon the availability of materials and technologies.

The program can respond to material shortages using unique authorities that apply to three focus areas:

- "Sustain[ing] Critical Production,"
- "Commercializing R&D Investments," and
- "Scal[ing] Emerging Technologies."[349]

Title III authorities incentivize the creation, expansion, or preservation of domestic manufacturing capabilities for technologies, items, and materials needed to meet national security requirements.[350] The goal is not the production of materials or items themselves, but the creation or expansion of the industrial capacity to produce these items and materials. Title III mechanisms can include the following:

- Grants,
- Purchases and purchase commitments (not commonly used),
- Installation of production equipment,
- Development of substitutes (most commonly used via R&D contracts), or
- Loans and loan guarantees (not used since 1992 by memorandum of understanding with the DoD General Counsel).

N.4.4 Industrial Base Analysis and Sustainment Projects

The Industrial Base Analysis and Sustainment (IBAS) program is part of DoD's efforts to fortify the industrial base, particularly regarding DoD's supply chain. To that end, the mission of the program is to "Strengthen the competitive posture of the U.S. Defense Industrial Base (DIB) in the era of great powers and global competition."[351] "IBAS primarily invested in projects to preserve existing suppliers and capabilities. These projects focused on maintaining workforce capabilities such as engineers at a supplier's production facility whose specialized skills were at risk of atrophying due to a reduction in DoD's demand."[352]

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[348] For more information, see DoD Office of Industrial Policy, "Defense Production Act (DPA) Title III," available at https://www.businessdefense.gov/ibr/mceip/dpai/dpat3/index.html.

[349] Ibid.

[350] Ibid.

[351] DoD Office of Industrial Policy, "IBAS Opportunities," available at https://cornerstone.army.mil/.

[352] DoD National Defense Industrial Strategy, 2023, available at https://www.businessdefense.gov/docs/ndis/2023-NDIS.pdf.

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N.4.5 Foreign Comparative Testing Program

The Foreign Comparative Testing (FCT) Program's mission is to test high-TRL items and technologies from foreign allies to determine whether the items could satisfy U.S. military requirements or address mission-area shortcomings and could do so more quickly and economically than would otherwise be possible.[353] The program has resulted in "substantial savings by avoiding R&D costs, lowering procurement costs, reducing risk for major acquisition programs, and accelerating the fielding of equipment critical to the readiness and safety of U.S. operating forces."[354] Sponsoring organizations within the Department identify foreign items for inclusion in the program.[355] Through this program, "the OSD Comparative Technology Office funds testing and evaluation; the Services fund all procurements that result from a successful test."[356] DMSMS resolutions that have foreign involvement can use this program to qualify technology or items for procurement.

N.4.6 DLA's Reverse Engineering Program

DLA's reverse engineering program is designed to lower the price that DLA pays for an item and to reduce the cost that DLA customers incur. According to a DLA Land and Maritime website,[357] "Reverse Engineering is a process used to gather required data to adequately determine how an item is manufactured. This process involves a variety of engineering disciplines to evaluate structural, dimensional, and functional properties of a component or technology. This allows DLA to better understand a system's F3. DLA parts are examined and analyzed to determine how they are manufactured for the development of a complete TDP[358] to make the item more competitive and provide better value to the warfighter."[359]

Reverse engineering projects are usually identified by the DLA program office based on analyses of sources and sales; program offices are generally not involved. Projects are typically associated with sole source or hard to obtain items where competition is more likely to lead to a reduction in the purchase price of such items. The reverse engineering itself is often performed by in-house DLA engineers.

There are however instances where one of the DLA reverse engineering offices establishes an agreement with an ESA to perform reverse engineering. When that occurs, there can be an opportunity for the ESA to suggest reverse engineering candidates based on obsolescence. As a result, there have been DWCF reverse engineering projects approved to mitigate an obsolescence issue.

While there have been situations where reverse engineering projects have been undertaken for obsolete items, DLA's reverse engineering program is not designed to provide wide ranging support to the DMSMS management community, especially not directly to program offices. In special circumstances, it can provide limited obsolescence mitigation support to ESAs.

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[353] FCT Program, "OSD CTO Foreign Comparative Testing," https://ac.cto.mil/pe/fct/.

[354] Ibid.

[355] Ibid.

[356] Ibid.

[357] DLA, Land and Maritime, available at http://www.dla.mil/LandandMaritime/Offers/Services/TechnicalSupport/ValueMgtDiv.aspx.

[358] The TDP created by reverse engineering may evolve over time to enable companies to compete.

[359] DLA, Land and Maritime, available at http://www.dla.mil/LandandMaritime/Offers/Services/TechnicalSupport/ValueMgtDiv.aspx.

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N.4.7 DLA's Sustaining Engineering Program

A sustaining engineering program will generally result in a design refreshment or redevelopment of the item. Sustaining engineering proposals are evaluated when received. Explanatory material for the program states that projects should generate an ROI of 10:1 to DLA (at a minimum) and are expected to deliver other positive impacts. Although this ROI is calculated as a ratio, the definition of the term *ROI* is different from the typical economic one. The denominator is the DWCF investment only.[360] The numerator includes life cycle reduction in DWCF outlays (which is not DWCF life cycle savings).[361] The numerator may include other life cycle savings for the weapon system at the discretion of the sustaining engineering program. Considerations for savings for the numerator of the ratio for an obsolescence resolution project include:

- Reductions to DWCF outlays for the new item because of smaller demand, lower unit price, improved reliability, or maintainability. Projects are not limited to F3 replacements and may provide more than incidental improved performance because of new technology.
- The amount that DWCF would pay for the existing or obsolete item may increase as availability decreases.
- Cost that DLA would incur associated with extra testing that may be required to ensure that the supply for the existing or obsolete item is not counterfeit.
- Additional cost because of cannibalization to obtain the existing or obsolete item becoming more frequent.
- Cost associated with sidelining an expensive, depreciating end item (i.e., a weapon system or platform) due to its material condition (i.e., critical item unavailable due to obsolescence).

Beyond ROI, the sustaining engineering office also evaluates project proposals in terms of other positive impacts such as improving operational readiness, reducing lead time (e.g., administrative and/or production lead time), decreasing the number of field and depot maintenance actions necessary, and improving the competitive position of the government with regard to the item by ensuring availability of the technical data with the appropriate rights to be able to add sources of supply, if desired.

The sustaining engineering program office works with the weapon system program office that submits a proposal for consideration. If necessary, it will help the weapon system program office improve and revise a proposal to improve the likelihood of its approval. For those project proposals that are deemed "good candidates," the sustaining engineering program office will develop a sustaining engineering project approval document. A sustaining engineering board will be scheduled and held to decide whether the proposed project is worthy of funding within the sustaining engineering program. If this is the case, then the sustaining engineering program office will develop a service agreement between DLA and the weapon system program office regarding the project. With the service agreement in place, the sustaining engineering program office will also ensure that there are DWCF funds available to support the proposed project and execute the steps necessary to transfer the funds to the weapon systems program office.

After the weapon systems program office has received the funds, it is responsible for implementing the project. The weapon systems program office submits progress reports throughout the course of the project to process any of its invoices against the funds provided.

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[360] The sustaining engineering program may be willing to split the funding needed for the project with the Service whose weapons system program proposed the project to achieve the 10:1 ratio on DLA DWCF investments.

[361] Outlay reduction is measured by a decrease in the price that the DWCF pays for the item and/or a reduction in demand as compared to what would have been the case had the item not been replaced (typically because the replacement item has improved reliability over the original item).

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While it has never been used to resolve an obsolescence problem, the descriptive literature for the program indicates that such an application is feasible. The nature of obsolescence issues means that a sustaining engineering project is unlikely to be approved, but program offices should not dismiss this potential resolution path. There is no downside to exploring possibilities.

N.4.8 DLA's GEM Program

According to the GEM website,[362] the GEM technology was developed by DLA to provide a continuing solution to the microelectronics diminishing manufacturing sources problem. The program was initiated by DLA in 1987 and deployed in 1997 as an on-demand type contract. The program is currently managed by the DLA Land and Maritime with SRI International as prime contractor.

Utilizing a set of gate arrays and single line processing technology, SRI International's onsite flexible wafer foundry manufactures a continuing source of form, fit, and function replacements for non-procurable microcircuits. These GEM devices are supplied to existing system documentation and meet the physical, electrical, and environmental specifications of the original devices.

Non-recurring engineering costs to develop GEM devices may be subsidized by DLA. The program office normally pays for only the manufacturing costs of the devices it buys.

N.4.9 Aviation Component Improvement Program

The Aviation Component Improvement Program (AvCIP) applies to the Navy and Air Force. Within NAVAIR, AvCIP deals with common and unique avionics on in-production and fielded systems. It can fund non-recurring engineering activities such as redesign or modification, prototype development, test and evaluation, integration, and technical documentation in partnership with the Naval Supply Systems Command for repairable items and with DLA for consumable items. To qualify for funding, a project must address a critical near-term issue concerning reliability, maintainability, or DMSMS; must result in cost avoidance by being proactive; and must achieve significant gains in warfighting capability or readiness.[363] (No evidence exists to indicate that AvCIP will be used by the Air Force for a DMSMS issue.)

[362] See https://gemes.com/about-gem/.

[363] For more information, contact the NAVAIR AvCIP program manager.

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Appendix O. Provisioning

The DoD provisioning process is part of supply chain management and is the process of determining and acquiring the range and quantity of support items necessary to operate and maintain an end item of materiel for an initial period of service.[364] This appendix describes the provisioning process, when provisioning is conducted, and when a program is required to perform updates.

Provisioning planning begins with program initiation for the planning and acquisition of initial spares to support a new or existing weapon system or major end item and continues through the system acquisition.[365]

Material managers and Primary Inventory Control Activities are responsible for conducting provisioning data management to provide program managers with provisioning data requirements and deliverables for incorporation into end item acquisition solicitation documents. Provisioning is primarily applied to new major acquisition systems and uses four major provisioning methods to identify the data requirements:

- ***Conference Team Method***. The most common provisioning method used. The formal technical review is conducted by a team of government and industry experts.
- ***In-house/Desktop Method***. Used when a system is not complex and uses a committee of supporting activity SMEs to complete the technical reviews at the government facility.
- ***Resident Provisioning Team***. Used for major systems and consists of a small group of personnel assigned on a permanent basis to the contractor facility.
- ***Contractor Provisioning***. Conducted by the prime contractor that is responsible for supporting the weapon system during its initial period of service.[366]

Each of the methods has its own advantages and disadvantages. The method used depends on the nature of the item being provisioned and the type of weapon system supported.

In addition to provisioning actions associated with new major acquisition systems, provisioning is initiated for items that enter the supply system as replacements for existing items and those that enter the supply system when transitioned from CLS to organic support.[367]

Provisioning updates may be required if any data deficiencies are discovered after the contract for the procurement of the end item expires. Provisioning updates may also be required when unnecessary duplicate items are discovered in the supply system. In these situations, it is best to identify and correct these data deficiencies during the provisioning review process. ECP is another acquisition event that may require a change to provisioning documentation and other key data that influenced the initial provisioning decision.[368]

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[364] DoDM 4140.01, DoD Supply Chain Materiel Management Procedures: Demand and Supply Planning, Volume 2, November 9, 2018.

[365] DoDM 4140.01, DoD Supply Chain Materiel Management Procedures: Operational Requirements, Volume 1, December 13, 2018.

[366] DAU Acquipedia: Provisioning.

[367] DoDM 4140.01, Volume 1.

[368] MIL-HDBK-61, Revision B, April 7, 2020, ASSIST, ASSIST-QuickSearch Document Details (dla.mil).

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Appendix P. Abbreviations

2M	miniature/microminiature electronics repair
AAC	Acquisition Advice Code
AD-DSL	Aerospace and Defense Declarable Substance List
AEC	Allowance Equipage Code
AFI	Air Force Instruction
AFMC	Air Force Materiel Command
AFSCI	Air Force Sustainment Center
AFWCF	Air Force Working Capital Fund
AMC	U.S. Army Materiel Command
AMCOM	Aviation and Missiles Command
AME	Advanced Microcircuit Emulation
AMSC	Acquisition Method Suffix Code
AoA	analysis of alternatives
APL	Allowance Parts List
AS	acquisition strategy
ASD(S)	Assistant Secretary of Defense for Sustainment
ASIC	application-specific integrated circuit
ASR	Alternative Systems Review
ASSIST	Acquisition Streamlining and Standardization Information System
AvCIP	Aviation Component Improvement Program
AWCF	Army Working Capital Fund
BCA	business case analysis
BOA	Back Orders Avoided
BOM	bill of materials
BY	Budget Year
C.F.R.	Code of Federal Regulations
CAGE	Commercial and Government Entity
CCA	circuit card assembly
CCB	configuration control board
CDR	Critical Design Review
CDRL	Contract Data Requirements List
CECOM	Communications and Electronics Command
CER	cost estimating relationship
CLIN	contract line item number
CLS	contractor logistics support
CM	configuration management

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CMRMP	Chemical and Material Risk Management Program
COG	Cognizance (code)
COTS	commercial off-the-shelf
CPI	critical program information
CSAG-S	Consolidated Sustainment Activity Group–Supply
DAFI	Department of the Air Force Instruction
DAU	Defense Acquisition University
DAWIA	Defense Acquisition Workforce Improvement Act
DEF	defense exportability features
DBS	Defense Business Systems
DFARS	Defense Federal Acquisition Regulation Supplement
DI	developmental item
DID	data item description
DLA	Defense Logistics Agency
DLR	depot level reparable
DMAIC	define, measure, analyze, improve, control
DMP	DMSMS management plan
DMPO	DMSMS management performing organization
DMS	diminishing manufacturing sources
DMSMS	diminishing manufacturing sources and material shortages
DMT	DMSMS management team
DoD	Department of Defense
DoDI	Department of Defense Instruction
DoDM	Department of Defense Manual
DPA	Defense Production Act
DSIA	days of supply impact avoided
DSP	Defense Standardization Program
DSPO	Defense Standardization Program Office
DWCF	Defense Working Capital Fund
EBOM	engineering BOM
EC	emerging chemical
ECP	engineering change proposal
EEE	electrical, electronic, and electromechanical
EMD	Engineering & Manufacturing Development
EOL	end of life
ESA	engineering support activity
F3	form/fit/function
F3I	form, fit, function, or interface
FAR	Federal Acquisition Regulation

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FAT	first article test
FCT	Foreign Comparative Testing
FIIN	Find Item or Index Number
FLIS	Federal Logistics Information System
FMS	foreign military sales
FOC	full operational capability
FRP	full rate production
FY	fiscal year
GEM	Generalized Emulation of Microcircuits
GIDEP	Government Industry Data Exchange Program
GOTS	government off-the-shelf
GSD	General Support Division
HMTL	Hazardous Material Target List
HQ	Headquarters
IAEG	International Aerospace Environmental Group
IAW	in accordance with
IBAS	Industrial Base Analysis and Sustainment
ICA	Industrial Capability Assessment
IIRP	Improved Item Replacement Program
ILA	independent logistics assessment
IMP	integrated master plan
IMS	integrated master schedule
IOC	initial operational capability
IP	intellectual property
IPS	Integrated Product Support
IPT	integrated product team
LCC	life cycle cost
LCSP	Life Cycle Sustainment Plan
LOA	Letter of Offer and Acceptance
LON	life of need
LRIP	low rate initial production
LRFS	Logistics Requirements and Funding Summary
LRU	line replaceable unit
LSS	Lean Six Sigma
ManTech	Manufacturing Technology
MaSME	materials and structural, mechanical, and electrical
MCA	Major Capability Acquisition
MDA	Missile Defense Agency
MDD	Materiel Development Decision

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MICAP	mission capable
MilSpec	military specification
MIL-STD	Military Standard
MOCA	Mitigation of Obsolescence Cost Analysis
MTA	Middle Tier of Acquisition
MTR	Module Test and Repair
NAS	National Aerospace Standard
NASA	National Aeronautics and Space Administration
NAVAIR	Naval Air Systems Command
NAVSUP WSS	Naval Supply System Weapon Systems Support
NDI	non-developmental item
NHA	next higher assembly
NMCS	non-mission capable supply
NPV	net present value
NSN	national stock number
O&M	operations and maintenance
O&S	operations and support
OASD(IBP)	Office of the Assistant Secretary of Defense (Industrial Base Policy)
OASD(L&MR)	Office of the Assistant Secretary of Defense for Logistics and Materiel Readiness
OCM	original component manufacturer
ODASD(PS)	Office of the Deputy Assistant Secretary of Defense for Product Support
ODASD(SE)	Office of the Deputy Assistant Secretary of Defense for Systems Engineering
OEM	original equipment manufacturer
OSCR	Operating and Support Cost Reduction
OSD	Office of the Secretary of Defense
OUSD(R&E)	Office of the Under Secretary of Defense for Research and Engineering
P&B	programming & budgeting
Pb	lead
PBA	performance-based agreement
PBL	performance-based logistics
PCN	product change notice
PDC	priority designator code
PDN	product discontinuance notice
PDR	Preliminary Design Review
PEO	Program Executive Officer
PLT	production lot test
PM&P	parts, materials, and processes
PM	program office manager
PMR	Provisioning Master Record

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POM	Program Objective Memorandum
PPA	physical product audit
PPP	Program Protection Plan
PRR	Production Readiness Review
PSM	product support manager
PSS	product support strategy
QML	Qualified Manufacturers List
QMS	Quality Management System
QPL	Qualified Products List
R&D	research and development
RACI	responsible, accountable, consulted, and informed
RCM	Reliability Centered Maintenance
RDD	required delivery date
RDT&E	research, design, test, and evaluation
REACH	Registration, Evaluation, Authorisation, and Restriction of Chemicals
RFP	request for proposal
RIC	Repairable Identification Code
RIP	Reliability Improvement Program
RNVC	Reference Number Variation Code
RoHS	Restriction of Hazardous Substances
ROI	return on investment
RRAC	Regulatory Risk Analysis and Communication
S&T	Science and Technology
SCM	strategic and critical materials
SCRM	supply chain risk management
SD	Standardization-related Document
SDD	System Development and Demonstration
SE	systems engineering
SEP	Systems Engineering Plan
SETR	systems engineering technical review
SFR	System Functional Review
SME	subject matter expert
SOW	statement of work
SPC	Special Procedures Code
SR	sustainment review
SRA	shop replaceable assembly
SRAM	static random-access memory
SRR	Systems Requirements Review
SRU	shop replaceable unit

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SSE	system security engineer
STPP	Science and Technology Program Protection
SWA	Software Acquisition
SYSPARS	Systems Planning and Requirements Software
TACOM	Tank and Automotive Command
TDP	technical data package
THAAD	Terminal High Altitude Area Defense
TRA	Technology Readiness Assessment
TRL	Technology Readiness Level
U.S.C.	United States Code
UCA	Urgent Capability Acquisition
USD(A&S)	Under Secretary of Defense for Acquisition and Sustainment
USD(R&E)	Under Secretary of Defense for Research and Engineering
UWDP	Undersea Wildlife Detection Program
VE	value engineering
VECP	value engineering change proposal
VEI	value engineering incentive
VHDL	VHSIC Hardware Description Language
VHSIC	Very High-Speed Integrated Circuit
VSM	value stream map
WCF	working capital fund
WRA	weapon replaceable assembly

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Appendix Q. References

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Appendix R. Definitions

***Analyze* step**. The step in the DMSMS management process (detailed in the SD-22) that deals with analyzing alternative approaches for resolving DMSMS issues and identifying the preferred alternative.

approved item. An item already approved on a system's (or lower level of assembly's) drawing that is still in production but typically is not currently used on the system.

***Assess* step**. The step in the DMSMS management process (detailed in the SD-22) that examines the potential effects that a DMSMS issue, at any level of a system, may have on cost, schedule, availability, and readiness bill of material.

capability modification. An activity that alters the form, fit, function, or interface (F3I) in a manner that requires a change to the existing system, performance, or technical specification of the asset. Such modifications are accomplished to add a new capability or function to a system or component, or to enhance existing technical performance or operational effectiveness. See also *modification*.

closed DMSMS case. A DMSMS case where all implementation activities are complete, including changes to drawings and technical manuals.

complex substitute. A replacement item that has different specifications but requires no modification of the source product or the NHA, is researched and validated. The substitute may be the result of a redefined military requirement.

cost avoidance. The difference between the cost of the resolution implemented when a DMSMS issue is found proactively and the cost of the resolution had the DMSMS issue been found reactively. Cost avoidance is calculated by subtracting the cost of the proactive resolution from the estimated cost of the reactive resolution.

critical materials analysis. The investigation of critical materials in an item's supply chain (those not identified in a BOM) or its manufacturing process for the purpose of determining future item availability.

design refreshment. Replacing an item with a new item developed using existing technical data and without affecting the NHA. A design refreshment requires first-article testing along with any necessary testing to ensure the new item functionally meets all requirements when installed in the system. The manufacturing source for the new item may be the original manufacturer or a new source.

development of a new source. Establishing a new manufacturing or production source for an item using technical data without affecting the NHA. First-article testing is required along with any necessary testing to ensure that the new item functionally meets all requirements when installed in the system.

diminishing manufacturing sources and material shortages. The loss or impending loss of manufacturers of items or suppliers of items or the loss or impending loss of raw materials used to manufacture items of supply. Such losses may cause material shortages that endanger a weapon system's or equipment's development, production, or post-production support capability.[369]

[369] DAU, *Glossary*, July 21, 2020, available at https://www.dau.edu/glossary/Pages/Glossary.aspx#!both|A|26790.

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DMSMS case. Documents that enable the tracking and management of a DMSMS issue from its initial identification to the implementation of the selected resolution.[370]

DMSMS issue. The loss, or impending loss, of manufacturers or suppliers of items, raw materials, or software. See also *diminishing manufacturing sources and material shortages*.[371]

DMSMS management. A multidisciplinary process to identify risks resulting from obsolescence, loss of manufacturing sources, or material shortages; assess the potential for negative impacts on schedule or readiness; to analyze potential mitigations; and implement the most cost-effective resolution.[372]

DMSMS management operations. The activities performed by the DMPO to manage DMSMS, including determining how DMSMS management efforts will be integrated with other strategies, plans, and reviews (e.g. product support strategies, the systems engineering design and review process, and modification planning) for the system in question.

DMSMS management plan. A document describing proactive, risk-based DMSMS management processes and team structures necessary to identify DMSMS risks in all life cycle phases as described in the SD-22, assess the potential for negative impacts on schedule or readiness, analyze potential mitigations, and implement optimal resolutions.

DMSMS management team. The team established by a program office to oversee the execution of a DMP.

DMSMS management team lead. The spokesperson for the DMT who oversees and has the authority to control DMSMS management operations, such as coordinating DMT meetings and managing corresponding action items, identifying potential sources of funding and funding availability, requesting funding and other resources to support the DMSMS management program, overseeing the DMSMS management support contracts and agreements, interfacing with the CCB, and reporting on DMSMS risks at technical, logistics, and programmatic reviews. The DMT lead is also the DMSMS champion for the program office and, as a best practice, should be a full-time position. This role may sometimes be filled by the DMSMS SME.

DMSMS monitoring and surveillance. Recurring tasks performed by the DMT such as data clean-up, vendor surveys, item research, and the use of predictive DMSMS tools.

DMSMS operational processes. The five operational processes a DMT must perform to effectively manage DMSMS. These are: secure resources for DMSMS management operations; prioritize systems; identify and procure monitoring and surveillance tools; collect and prepare item data; and analyze item availability.

DMSMS predictive tool. Tools that collect information on parts, their life cycles, and manufacturers and use that data to forecast future obsolescence. Program offices can use these tools by loading their system BOMs.

DMSMS program capability levels. A framework of four levels indicating progressively more proactive and strategic approaches to DMSMS management. Practices associated with each level have been identified in the SD-22 (Appendix D, Table 26).

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[370] DoDI 4245.15, "Diminishing Manufacturing Sources and Material Shortages Management," November 5, 2020, available at https://www.esd.whs.mil/Portals/54/Documents/DD/issuances/dodi/424515p.pdf.

[371] Ibid.

[372] Ibid.

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DMSMS record keeping. Records collected to describe program office DMSMS management practices and outcomes. Records can be divided into two general categories: those associated with the costs of DMSMS resolutions and management operations and those associated with the efficiency of DMSMS management operations.

DMSMS resilience and/or DMSMS-resilient design. The use of design techniques that reduce the likelihood of near-term DMSMS issues and increase the probability of a quick recovery when issues do occur. DMSMS resilience is incorporated into a design by applying a modular, open system approach along with other supportability-related design considerations. This must be done in conjunction with part selection procedures that choose items with significant time left in their life cycle and with viable replacement options whenever possible.[373]

DMSMS resolution. A course of action that will bring a DMSMS case to a close. Resolutions fall into three broad categories: use of existing material; substitutes; and redesigns.

DMSMS risk. The likelihood of a DMSMS issue occurring within a particular timeframe coupled with the severity of the consequences of that issue. Items are considered a high DMSMS risk if it is likely that the DMSMS issue will occur during the life cycle of the system and that the DMSMS issue will negatively impact production, sustainment, or readiness.[374]

extension of production or support. A DMSMS resolution option where a supplier is incentivized to continue providing the obsolete items or services. This may involve long-term agreements to procure specific quantities of items, or long-term software licensing and/or support agreements.

first-order hardware or software changes. Changes made to either hardware or software to resolve first-order obsolescence.

first-order obsolescence. Obsolescence which is directly driven by market factors and regulation changes that affect DoD's ability to buy, license, obtain support for, or use hardware and software.

health assessment. A projection over time of item demand for a subsystem and the number available in inventory to meet item demand thereby showing the expected time when items will no longer be available due to obsolescence.

hierarchy of system items. A way to describe a system from the lowest levels of the system to successive higher levels comprising the whole system. If a four-level hierarchy is used, it contains: components (or "piece parts," or "devices"); assemblies (or "cards," "shop replaceable units (SRUs)," or "shop replaceable assemblies (SRAs)"); units (or "boxes," "weapon replaceable assemblies (WRAs)," "line replaceable units (LRUs)," or "software applications"); and systems and subsystems.

***Identify* step**. The step in the DMSMS management process (detailed in the SD-22) where, through near-term monitoring and surveillance and longer-term technology forecasting, a program office finds items with obsolescence risks.

***Implement* step**. The final step in the DMSMS management process (detailed in the SD-22) wherein the preferred resolution option for a DMSMS case found via the previous steps is executed.

indentured bill of material. A BOM that shows the relationships (generally in a top-down breakout format) of components to assemblies to units to the system.

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[373] Ibid.

[374] Ibid.

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independent subject matter experts. A U.S. government or commercial entity or person, external to the DMPO and absent other contractual relationship to the item being managed, that a DMPO uses to perform specified DMSMS management functions.

item. An all-inclusive term that can refer to a material, parts, equipment, assemblies, components, material, and software.[375]

life of need buy. A DMSMS resolution option where sufficient quantity of the item is purchased to sustain the product until its next technology refresh or the discontinuance of the host assembly.

lower-order functional obsolescence. Obsolescence caused by hardware or software changes made to resolve first-order obsolescence issues, perform a proactive upgrade, or respond to a requirement change.

market research. A process for gathering data on product characteristics, suppliers' capabilities, and the business practices that surround them. Includes the analysis of that data to inform acquisition decisions. There are two types of market research, strategic market research and tactical market research.

modification. A configuration change to the F3I of an in-service, configuration-managed or produced Configuration Item. Modifications are defined by their purpose.

modification planning. The proactive work to determine what and when configuration changes to the F3I should be implemented.

obsolescence. From a DMSMS management dictionary perspective, an item is obsolete if it is no longer current and its function is superseded by something new. An obsolete item is usually, but not always, out-of-production.

obsolescence issue. See *DMSMS issue*. From the perspective of this document, the process for resolving an obsolescence issue is the same as that for resolving a DMSMS issue.

open DMSMS case. A DMSMS case for which all resolution implementation activities have yet to be completed, including changes to drawings and technical manuals. See also *closed DMSMS case*.

original component manufacturer item number. The item number assigned by the original manufacturer of the item being reported.

original equipment manufacturer item number. The item number assigned by the OEM of the parent assembly of the item. If the referenced item is software or firmware, then the OEM item number refers to the item that the software or firmware is used on.

parts list. A list of all parts used in design or construction of the military system or equipment. Initially, it contains those items designed into the system. Upon production, it contains those items that are incorporated into the actual units produced.

***Prepare* step**. The step in the DMSMS management process (detailed in the SD-22) that establishes the foundations for robust DMSMS management, develops a DMP, forms a properly trained DMT to carry out all DMSMS activities, and establishes DMSMS operational processes.

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[375] Ibid.

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proactive DMSMS approach. An approach to DMSMS management that attempts to identify DMSMS issues before there is an unfulfillable demand for the item and extend the window of opportunity for resolving the DMSMS issue before it can have a negative effect on a system's production or sustainment.

product change notice. A notice from a manufacturer or supplier to inform customers about a change to a product or its manufacturing process.

product discontinuation notice. A notice from a manufacturer that it will no longer produce a product. Product discontinuation notices are generally issued in advance of the discontinuation, with a specified date the product will cease to be produced.

product improvement roadmap. A roadmap outlining changes in product capability to address a new threat, safety deficiencies, information assurance, or the introduction of new statutory or regulatory requirements.

product roadmap. A high-level visual summary that maps out the vision and direction of product offerings over time. It describes the goals and features of each software iteration and increment.

product supportability roadmap. A roadmap that encompasses improvements in the ability to maintain or sustain the product, increase its reliability, or extend its service life.

reactive DMSMS approach. An approach to DMSMS management that identifies DMSMS issues when there is an unfulfillable demand for an item. This generally occurs when no attempt was made to apply a proactive DMSMS management approach or an opportunity to identify DMSMS issues was missed before there is an unfillable demand for an item.

redesign—complex/system replacement. A DMSMS resolution option where a major assembly redesign affects assemblies beyond the obsolete item's NHA and may require that higher level assemblies, software, and interfaces be changed.

redesign—next higher assembly. A DMSMS resolution option where the affected item's NHA must be modified. Only the NHA is affected, and the new design will not affect anything at a higher level in the system.

redevelop the item. A DMSMS resolution option where the original item is replaced with a new item developed without the benefit of existing technical data and without affecting the NHA. The new item may be developed by emulating, reverse engineering, designing a replacement based on the original manufacturing designs and processes, or designing a different item based on the original or new requirements. The manufacturing source for the new item may be the original manufacturer or a new source.

repair, refurbishment, or reclamation. A DMSMS resolution option where the obsolescence issue is resolved by doing one of the following: instituting a repair or refurbishment program for the existing item or assembly, whether through a depot repair, a repair contract with the original manufacturer, or support from a third party; instituting a reclamation program to reclaim items from marginal, out-of-service, or surplus materiel; obtaining access to the software source code, development tools, and human resource skills necessary to change it to ensure continued support.

roadmap. Time-phased plan to identify the need for, determine, and implement resolutions to issues before those issues can have a negative impact and improve forecasts of future issues. See also *product improvement roadmap*, *product supportability roadmap*, *supportability roadmap*, and *technology roadmap*.

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simple substitute. A DMSMS resolution option where the item is replaced with an existing item that meets all requirements without modification to either the item or its NHA and requires only minimal qualification. Typically, this implies use of a commercial item or NDI that is an F3 substitute. This is sometimes referred to as an alternate.

system security engineering. A specialty engineering field that applies scientific engineering and information assurance principles to deliver trustworthy systems that meet stakeholder requirements within their established risk tolerance.[376]

supportability roadmap. See *product supportability roadmap*.

sustainment modification. Alters the F3I of an asset in a manner that does not change the existing system, performance, or technical specification of the asset. Such modifications correct product quality deficiencies or bring the asset in compliance with established technical or performance specifications associated with the asset. Sustainment modifications may improve the reliability, availability, maintainability, or supportability to reduce its ownership costs. See also *modification*.

technology insertion. A type of technology-related change that inserts or integrates technologies to improve the performance of the system.

technology management. Establishing overall program office (or another entity's) goals for technology development and acquiring the resources to perform the activities necessary to achieve those goals including integrating the technology into a system.

technology refresh. The periodic replacement of COTS items to assure continued supportability of that system through an indefinite service life.

technology roadmap. A high-level, visual plan that communicates a system's technology strategy, showing when technologies are mature enough for application on the system and the approximate timing for technology insertion or refreshment.

vendor surveys. A form of surveillance whereby a DMSMS manager contacts the vendor of a part to learn pertinent information, such as planned discontinuation dates for the part (or the lack of any plans for discontinuation).

watch list. A list of items to be monitored closely for DMSMS issues. An item can be placed on a watch list for any reason DMSMS managers see fit, but typical reasons include the item's criticality; a predictive tool indicating a high probability of obsolescence; elevated foreign ownership, control or influence from an unfriendly nation; or post-DMSMS case monitoring to ensure the assumptions of a closed DMSMS case for the item hold.

[376] DoDI 5200.44, *Protection of Mission Critical Functions to Achieve Trusted Systems and Networks*, February 16, 2024, available at https://www.esd.whs.mil/Portals/54/Documents/DD/issuances/dodi/520044p.pdf.

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